

● General Information

Considerations for ALL Exams:

-
- created by Center for Disease Control; AKA Standard Precautions; used to prevent the spread of bloodborne diseases; includes best practices and personal protective equipment
- ALARA Principle - As Low As Reasonably Achievable; use the lowest power settings and limit exam times to reduce ultrasound exposure
- Probe sanitization after each patient:
 - Cavicide Sani-wipes for most probes/cords are sufficient
 - Cidex should be used for endocavity probes
 - Iodine or alcohol based cleaners can damage the matching layer of the probe and degrade the adhesives used to construct the probe
- Transducers cannot be thermally sterilized; heating the transducer above the Curie point will destroy the ultrasound propagation principles of the piezoelectric elements

Occupation Safety and Health Administration (OSHA):

- Created by Congress
- A part of the Department of Labor
- Purpose is to assure safe and healthful working conditions by setting and enforcing standards and by providing training, outreach, education, and assistance

HIPAA Compliance:

- **Health Insurance Portability and Accountability Act**
- Provides the ability to transfer and continue health insurance coverage when you change or lose your job
- Reduces health care fraud and abuse
- Mandates industry-wide standards for health care information on electronic billing and other processes
- Requires the protection and confidential handling of protected health information

- Health care providers and organizations must develop and follow procedures that ensure the confidentiality and security of **protected health information (PHI)** when it is transferred, received, handled, or shared; Applies to all forms of PHI, including paper, oral, and electronic
- Only the minimum health information necessary to conduct business should be used or shared
- Most facilities prohibit posting of exam images on social media, even without any identifying information
- Always research your employer's policy on social media before posting anything
- HIPAA allows the discussion of results with the patient and their physician, but NOT with anyone else unless the patient supplies written permission
- Anonymous images are permitted for use in research and educational settings

Patient Scheduling:

- Using a computer can assist with data collection, sorting, and organization
- Necessary data includes patient name, DOB, ordering physician name, exam type, reason for the exam and any history of prior exams performed at the facility
- Additional helpful information includes follow up date or special assistance needed
- Always schedule **NPO** exams in the morning whenever possible, especially for diabetic patients and those requiring medication early in the morning that must be taken with food
- Renal artery Doppler, mesenteric Doppler, portal Doppler and most abdominal vasculature evaluations require NPO patient

Patient Infection Control and Isolation:

- Standard precautions should also be applied in all of the situations below
- **PPE =Personal protective equipment** used (**gowns, goggles, masks, gloves** etc)

Standard Precautions:

- Used for all patients
- Includes personal protective equipment
- Hand hygiene - hands should be washed
 - Before touching a patient, even if gloves will be used
 - Before exiting the patient's room after touching the patient or something in their room
 - After contact with blood, body fluids or excretions, or wound dressings
 - After glove removal
- Gloves
 - Wear gloves when there is potential contact with blood (e.g., during phlebotomy), body fluids, mucous membranes, nonintact skin or contaminated equipment
 - NEVER wear the same pair of gloves for the care of more than one patient

- Discard used gloves and never wash gloves for the purpose of reuse
- Perform hand hygiene before and immediately after removing gloves
- To remove gloves - the first glove, pull the wrist section of the glove toward the fingers, turning the glove inside out as it is removed; place the removed glove in the hand still wearing a glove; pull the second glove off like the first and you should have both gloves inside out and rolled together for disposal
- Gowns
 - Wear a gown to protect skin and clothing during times where contact with blood or body fluids is possible
 - Never wear the same gown for the care of more than one patient
 - Remove gown - pull sides together in the front to fold the exposed sides together, slide off the arms, roll gown up with exposed areas all facing inward and discard
 - Perform hand hygiene before leaving the exam room
- Facemasks
 - Wear a facemask to protect nose and mouth during times where contact with respiratory secretions and sprays of blood or body fluids is possible
 - Remove mask - untie the bottom and then the top, discard without touching the front of the mask
- Goggles or Face Shields
 - Wear eye protection during times where contact with respiratory secretions and sprays of blood or body fluids is possible
 - Personal eyeglasses and contact lenses are NOT considered adequate eye protection
 - Remove the goggles - touch the sides of the goggles to remove and discard, do not touch the front of the goggles

Isolation Precautions:

- Contact Precautions
 - Used to protect others from germs in patients that have stool incontinence, draining wounds and/or skin lesions that cannot be covered, or uncontrolled secretions
 - Enteric - difficile, rotavirus, or norovirus
 - Wound and Skin - MRSA, VRE, Scabies, Impetigo
 - Wear gloves when touching the patient and items in the patient's immediate environment
 - Wear a gown if substantial contact with the patient or their environment is expected
 - Perform hand hygiene after removal of PPE
- Droplet Precautions
 - Used to protect others from germs in the patient's nose, mouth, throat, and lungs

- Patients with suspected infection and that exhibits sneezing, coughing or spitting
- Includes patients with pneumonia, influenza, adenovirus, mumps, rubella, meningitis, bordetella pertussis
- Wear a facemask (surgical mask) for close contact with the patient
- Wear gloves when touching the patient and items in the patient's immediate environment
- Wear a gown and/or goggles if excessive spraying of respiratory fluids is anticipated
- Perform hand hygiene after removal of PPE
- Patient should wear a facemask when exiting the exam room
- Airborne Precautions
 - Strict isolation
 - Separation from others to prevent the spread, by contact or airborne transmission, of highly contagious diseases
 - Patients suspected to be infected with a pathogen that can be transmitted by airborne route
 - Tuberculosis, measles, chickenpox (varicella), shingles
 - Wear a fit-tested N-95 or higher level disposable respirator when in the patient's room
 - Put the respirator on before entering the room and remove it after exiting
 - Wear a gown and/or goggles if excessive spraying of respiratory fluids is anticipated
 - Perform hand hygiene after removal of PPE
 - Patient should wear a facemask when exiting the exam room
- Protective Environment Precautions
 - Reverse protective isolation
 - Protects patients from the germs of others
 - It is used with patients who have extremely impaired immune systems or recent organ transplant
 - Wear gloves and mask
 - Equipment should be cleaned and disinfected before and after leaving the patient room
 - Perform hand hygiene after removal of PPE

Invasive Procedures:

- Intraoperative ultrasound, intravascular ultrasound, angioplasty, and endovenous ablation procedures require the use of sterile technique
- PT and INR levels should be assessed prior to most invasive procedures; used to assess clotting ability of blood
- Normal INR 0.9 - 1.3, Normal INR for patient on blood thinners 2-3

- Informed consent form required for all invasive procedures
 - Lists the procedure and potential associated risks and complications
 - Must be signed PRIOR to the procedure AFTER physician answers any questions/concerns
 - By signing the form, the patient indicates that they have an understanding of the procedure, complications, and expected outcomes
 - The patient must be of sound mind and have a full understanding of the information presented when giving permission for a procedure by signing the form
 - A “Time Out” period should be performed before the procedure with all persons in the room agreeing to proceed

Absolute Contraindication:

- A situation which makes a particular treatment or procedure absolutely inadvisable
- These include a known allergy to the injected material (ablation, steroid injection) , lack of appropriate equipment or skill to complete the procedure, uncooperative patient (uncontrolled movement during the procedure can result in injury, laceration or hemorrhage)

Relative Contraindication:

- A condition which makes a particular treatment or procedure possibly inadvisable
- Changes can be made to make the procedure safer or easier to perform
- These include coagulopathy and the use of blood thinners, aspirin or antibiotics
- If permitted by their physician, the patient can discontinue their blood thinner medication until their PT and INR values return to normal levels and then proceed with the invasive procedure
- Warfarin, Coumadin, Jantoven, Xarelto, Pradaxa or Lovenox, Aspirin, Ibuprofen, NSAIDs should be stopped at least 7 days before a procedure and resumed 7 days after the procedure

Other relative contraindications include:

Unsafe access route - must avoid large arteries and adjacent nerves

Local infection, rash, or skin breakdown

Sterile Field Tips:

- The Sonographer should prepare the supplies for the biopsy
- All items and personnel in a sterile field must be sterile
- When in doubt about sterility, discard the potentially contaminated item and begin again
- Items should not be used that have expired sterilization dates

- Always monitor the sterile field during the entire procedure; sterile fields become non-sterile if they are left unmonitored
- Tables are considered sterile only at and above the level of the table
- Sterile packages or fields are opened or created as close as possible to time of actual use
- The sterile tray cover can be opened without wearing gloves but the far corner of the top should be pulled toward you to expose the products facing away from you; This prevents your arm from hovering over the supplies to prevent any debris from your person from falling into the tray
- Sterile packages can be opened with non-sterile gloves but if any items inside the tray will be touched, sterile gloves must be used
- Sterile gloves must be used to remove items for the tray to prepare them (lidocaine, needle, syringe, biopsy instrument, catheter)
- Items should not be used that have expired sterilization dates
- Edges of sterile areas or fields (generally the outer inch) are not considered sterile
- Moist areas are not considered sterile
- Contaminated items must be removed immediately from the sterile field
- Sterile gowns are considered sterile only in the front, from chest to waist and from the hands to slightly above the elbow; cuffs of the gown are considered non-sterile
- Sterile individuals must pass each other back to back (non-sterile side to non-sterile side) when moving around the procedure room
- Tables are considered sterile only at or above the level of the table
- Personnel with colds should avoid working while ill or apply a double mask; no coughing/sneezing over the sterile field

Management of Emergency Situations:

- American Heart Association CPR guidelines - 30 compressions/ 2 breaths adult and child single rescuer protocol; see chart below
- Choking - Heimlich maneuver, abdominal thrusts
- Choking patients who are late in pregnancy or obese - use chest thrusts instead of abdominal thrusts
- Vasovagal reaction - heart rate and blood pressure drop; can be caused by excessive standing, stress, trauma; place patient in supine position to prevent injury because it commonly leads to syncope
- Syncope - place patient in supine position and elevate legs
- Anaphylactic reaction - hives, feeling flush, nausea, vomiting, shortness of breath; physician may administer antihistamine

CPR

American Heart Association

Patient	Adult	Child	Infant
Recognition of Distress/Cardiac Arrest	Unresponsive No breathing or gasping for air No pulse for 10 seconds		
Order of Events C-A-B	Compressions Airway Breathing		
Compression Depths	At least 2 inches	1/3 AP diameter of chest About 2 inches	1/3 AP diameter of chest About 1.5 inches
Compression to Breath Ratio	1 or 2 rescuers 30:2	Single rescuer 30:2 Two rescuers 15:2	
AED	Minimize interruptions in chest compressions <10 seconds Deliver single shock for patients in ventricular fibrillation Resume CPR immediately after delivering a shock		

Choking – Heimlich Maneuver

American Heart Association

Conscious Adult	Unconscious Adult	Infant
- Stand behind the patient and tip them slightly forward - Wrap your arms around their waist - Make a fist with one hand and clasp the other hand around it - Place your hands just above the naval and press in and up 5 times - Assess the patient for the dislodged object - Repeat if necessary ***Chest thrusts are used in patients late in pregnancy and obese patients	- Lay the patient on their back - Perform a visual inspection of the mouth and remove any objects identified (no blind finger sweep) - Begin CPR compressions - Check the mouth periodically for the object	- Hold the infant face down and lie them across your forearm for support - Be sure to support the head and neck - Deliver 5 back blows - Turn infant over and assess for object in mouth - Hold the infant face up and lie them across your forearm for support - Deliver 5 chest thrusts - Repeat until object is dislodged

Setting Up For An Examination:

- Enter the patient information into the system; include first and last name, date of birth, medical record number, gender
- State and federal regulations prohibit the use of a patient's social security number as an identifier on imaging exams (and other types of medical records); the medical record number is created by each facility to identify the specific patient
- Referring physician, reading physician and exam indication may also be required information to enter into the system at some facilities
- Some facilities also require the height and weight are entered into the US system to calculate the body mass index (BMI)
- Select the appropriate transducer for the exam; higher frequencies used for thinner patients and superficial vessels
 - Extremity and cerebrovascular exams- linear array
 - Abdominal and large habitus lower extremity exams- phased array, curved array, vector array
- Activate the appropriate preset for the exam; inappropriate presets can degrade the exam and extend exam time; EXAMPLES
 - If the venous preset is selected for an arterial exam, aliasing of the arterial signal will most likely occur. Venous flow velocities are much lower than arterial velocities and this will require

- multiple changes to system settings including reduction of color and PW Doppler scale, adjusting baseline and sweep speed
 - If the carotid preset is selected for a lower extremity insufficiency exam, this will require multiple changes to system settings including imaging depth, focal zones, probe frequency, Doppler settings, calculations package
- Position the machine to allow your arm to be extended and abducted to a minimal degree with the wrist extended (NOT flexed)
- Adjust the table height to allow for the proper position listed above
- Never place the transducer cable around your neck or shoulders; loosely wrap cord around your wrist or **use the probe holders on the equipment to support the weight of the cord**
- Be careful of other medical equipment connected to patient; oxygen, IV therapy, monitors; NEVER alter settings or alter other equipment in use; consult the patient's nurse if adjustment is necessary

Preparing the Patient for the Exam:

Communication:

- ALWAYS verify that you have the proper patient for the requested exam
 - Ask the patient to confirm their name and date of birth prior to starting the exam
 - When performing exams on inpatients, use the patient wrist band to verify the patient name and medical record number
- Explain any preparation required for the upcoming exam or procedure
- Optimal images on any type of abdominal vasculature exam require the patient to be **NPO 6-8hrs** prior to the exam
- NPO status helps with the following:
 - Reduces bowel gas that could obscure adjacent structures
 - Allows gallbladder to fill with bile for better visualization
 - Decreases peristalsis in the GI tract to improve visualization of adjacent structures that could be obscured by a part of the stomach or intestine
- Explain the exam or procedure and answer questions related to how the exam is performed
 - Always consider the patient's mental state, age and other factors that may affect their ability to understand the exam explanation
 - A child (and their parent) should be given an easy to understand explanation and demonstration of the probe, gel etc.
 - A patient with dementia or Alzheimer's disease may be more comfortable with a family member or something familiar in their room; check with the patient's nurse for tips on communicating with the patient
 - A patient that appears unconscious should still be offered a short explanation of the exam, especially when they have family members in the room
 - Avoid compromising the exam quality and scanning environment whenever possible
- Obtain pertinent history related to indication for current exam
 - Discuss when the patient can expect the results and that as a Sonographer you cannot provide any results for the exam
 - Tell the patient they need to contact the physician that ordered the exam to obtain the results

- Ask the patient if they have additional testing scheduled after this exam; offer to assist them with getting to their next exam

Patient Preparation:

- Position the patient for appropriate imaging of the area of interest
 - Transcranial Doppler=TCD
 - Supine
 - The technologist usually sits at the head of the bed for easier access to the cranial windows
 - Carotid
 - Supine with neck slightly extended and head turned slightly away from the side being scanned
 - If the patient is unable to turn their head, a positioning wedge can be used to partially oblique the patient
 - Lower Extremity Arterial
 - Supine, patient may be semi-erect if there are breathing problems, vertigo or back pain that prevents them from laying supine
 - Knee should be bent slightly and leg abducted for better access to the medial aspect of the leg and popliteal fossa
 - Decubitus or lateral position may be necessary for accessing the popliteal fossa and posterior calf
 - Lower Extremity Venous for DVT
 - Supine, patient may be semi-erect if there are breathing problems, vertigo or back pain issues that prevent them from laying supine
 - The semi-erect position also will assist with increasing intra-abdominal pressures causing decreased venous outflow in the legs and the veins will be more easily visualized
 - Knee should be bent slightly and leg abducted for better access to the medial aspect of the leg and popliteal fossa
 - Decubitus or lateral position may be necessary for accessing the popliteal fossa and posterior calf
 - Lower Extremity Venous for Insufficiency
 - After performing an evaluation for DVT the patient will need to stand next to the bed to evaluate the GSV, preferably on a stool
 - The leg of interest should be slightly rotated outward to allow easier access to the medial aspect of the leg
 - The SSV can be evaluated with the patient seated at the edge of the bed, legs dependent
 - Abdominal Doppler
 - Supine
 - Renal Doppler exams usually benefit from placing the patient in the oblique or decubitus position

- Obstructive clothing, bandages, dressings and devices
 - The patient should remove any clothing in the area to be scanned and if they are wearing anything tight that could constrict flow more proximally, this should also be removed
 - If the patient is wearing compression stockings, ask them to remove them and walk around a few minutes and then stand for a few minutes to normalize flow
 - If bandages are obstructing areas that need to be directly evaluated, discuss removing them with the patient and/or the nurse prior to removing them
 - Always attempt to complete a thorough exam without removing the bandages whenever possible
 - Always use a sterile sheath and sterile gel if it is necessary to scan directly over a wound
 - If bandages/dressings are removed for the exam, they need to be replaced before the patient leaves
- Ask the patient if the room feels too cold or too hot and adjust room temperature accordingly
 - Changes in the patient temperature can mask pathology or cause a false positive exam
 - If the patient is cold, vasoconstriction can occur which will increase resistance in arterial flow and decrease venous filling
 - If the patient is too warm, vasodilation can occur which will decrease arterial resistance and increase venous filling
- Anxiety can induce vasoconstriction
 - Adjusting the lighting and turning on relaxing music can help reduce patient anxiety

Post-Examination Procedures:

Demographic Data:

- Vascular disease statistics are calculated using patient demographic information
- Age, gender, ethnicity, and location are commonly used to calculate vascular disease statistics

Technologist Findings:

- Most vascular labs use computers to generate a preliminary report of their findings for the physician
- Technologists should include the velocity measurements and Doppler angles used on the exam; the same angles should be used on the next exam for the best follow up evaluation
- Logging patient exam data in the computer allows for easier comparison of serial exams, tracking of patient disease progression and calculating disease statistics
- Always include any exam limitations in the technologist findings
- Preliminary reports should include pertinent velocity and/or pressure information, location and extent of obstruction, related findings

Communicating with the Radiologist/Vascular Physician:

- Technologist findings and impression
- Exam difficulties or limitations study = reason ex: pt wasn't fasting, pt is on oxygen
- Comparison of prior exams

- Anytime the procedure could be detrimental to the patient, you should clarify the order with the radiologist and/or referring physician; Also clarify the order when the suspected diagnosis is not able to be evaluated by the exam that is ordered
- Indicate any potentially critical findings that require immediate attention
- **STAT= emergency / immediate attention required**

Preliminary Reports and Critical Findings:

- Required for any finding that requires immediate intervention
- Study should be read by physician as soon as possible
- Technologist may be responsible for preliminary results in some facilities
- Always include patient name and ID#, DOB, intended recipient name, type of exam and reason for the exam
- Include any exam limitations and any pertinent anatomy that was not evaluated
- Location and extent of findings
- Document who received the preliminary report and write the information in the patient chart
- Always contact the physician on call to review the exam for a stat reading

Examples of Critical Findings:

- AAA >5.5cm diameter
- Popliteal aneurysm >2.0cm
- Splenic artery aneurysm
- Aneurysm rupture
- Pseudoaneurysm or AV fistula
- Acute arterial occlusion; chronic obstruction usually allows time for collateral formation so it is NOT a critical finding
- 70-99% carotid stenosis
- Lower extremity DVT
- Free floating tail - piece of thrombus that is not adhered to the vein wall
- Upper extremity venous thrombosis
- IVC thrombus
- Hepatic vein thrombus
- Renal vein thrombus

Professional Compliance:

- RVT status requires 30 CME credits in each triennium (every 3 years)= ARDMS
- RVS status requires 36 CME credits in each triennium=CCI
- Facility accreditation requires at least one certified vascular technologist at each site while some insurance carriers require all vascular exams to be performed by a registered technologist
- IAC Vascular Testing accreditation requires peer review, exam/report variability assessment, result correlation, report audits and quarterly staff meetings; requires 15 CME credits in topics related to vascular ultrasound

- ACR accreditation requires physician peer review and Sonographer compliance with ARDMS or CCI. ARRT requirements

Hemodynamics

- Refers to the study of blood and how/why it moves
- Arterial and venous hemodynamics are very different due to structure and purpose

Systemic VS Pulmonary Circulation:

Systemic:

Left ventricle – aorta – arteries – arterioles – capillaries – venules – veins – IVC/SVC -- right atrium

Pulmonary:

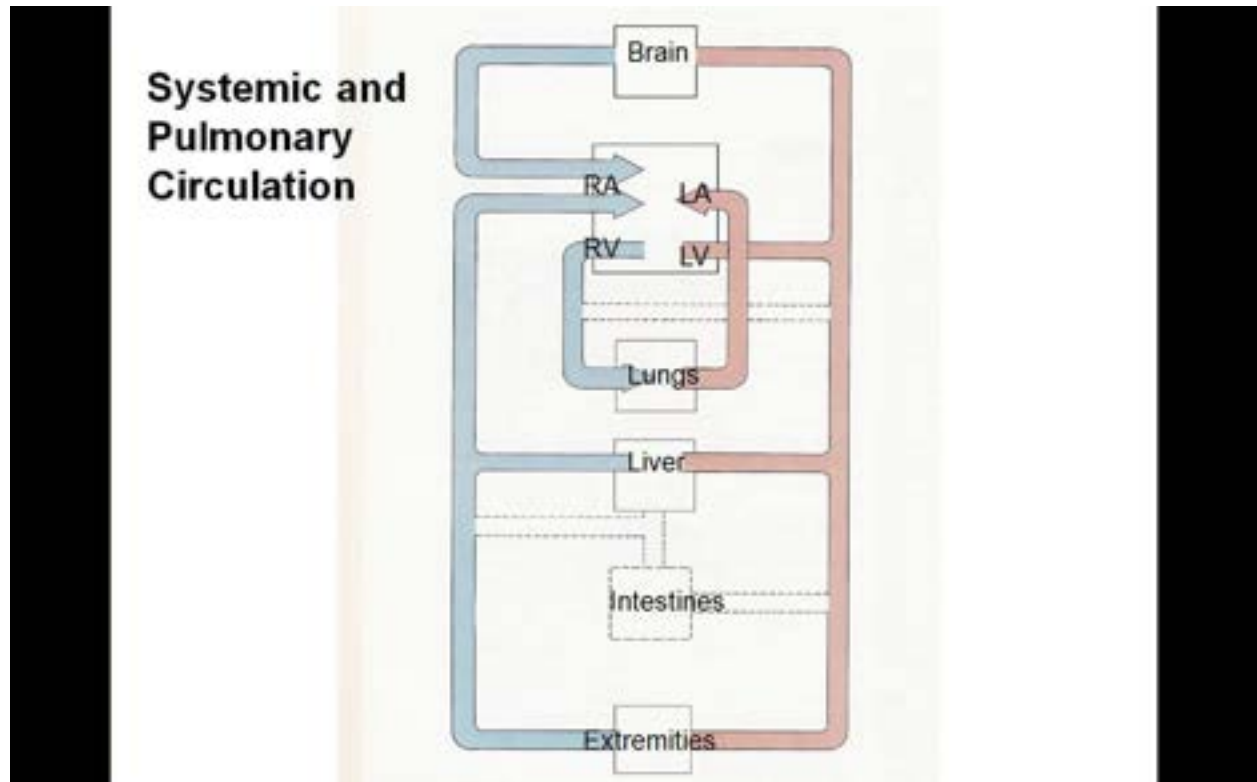
Right ventricle – pulmonary artery – lungs – pulmonary veins – left atrium

- Oxygen saturation is highest in the pulmonary veins and lowest in the IVC/SVC
- Pulmonary arteries are the only arteries to carry deoxygenated blood
- Pulmonary veins are the only veins to carry oxygenated blood

Cardiac Pump:

- Systole - LV pressures increase rapidly to exceed aortic pressure and push a bolus of blood into the arteries; contraction of the heart causes expansion of the arteries
- Diastole - LV pressure decreases rapidly and the LV refills with blood from the LA; relaxation of the heart allows for arterial recoil to normal dimension
- Normal heart rate 60 - 100bpm; Bradycardia <60bpm; Tachycardia >100bpm
- Pumps average of 70 ml of blood into aorta with each contraction
- The primary mechanisms that control arterial flow volume changes during the cardiac cycle are cardiac output and peripheral resistance
- Increased heart rate leads to increased cardiac output
- Cardiac function is directly related to the amount of blood entering the arteries
- Peripheral resistance and arterial pressure affect the amount of blood leaving the heart
- Systemic pressure is greatest at the heart and it gradually decreases as it moves distally through the system
- Low peripheral resistance allows for increased rate of flow
- Arterioles are responsible for regulating resistance in the tissues
 - Nerve fibers and adrenal glands release norepinephrine and vasoconstriction occurs

- Adrenaline is a vasodilator that is produced in the adrenal glands
- Capillaries are only composed of intimal tissues to allow perfusion of the surrounding tissues and the exchange of nutrients and waste products between blood and tissue



Energy Types:

- Kinetic: energy of motion, varies with mass and velocity squared; greater mass = more kinetic energy = greater velocity
- Potential: stored energy of motion; such as a sling shot; primary form of energy driving blood flow; related to the pressure distending the vessels
- Gravitational: stored energy related to elevated position; such as roller coaster (Hydrostatic pressure)

Kinetic energy reduced in blood vessels due to :

- Viscosity – thicker blood flows more slowly, increased hematocrit = decreased flow
- Friction – layers of blood sliding against each other; energy lost in the form of heat due to friction between layers and between blood and vessel wall
- Inertia – energy is lost when the speed of blood changes with systole and diastole of the heart

Energy Types

Gravitational
(and Kinetic)



Potential



Kinetic



Viscosity



The greater the viscosity, the slower the blood flow. Decreased viscosity will cause elevated flow velocities.



Arterial Blood Flow:

- Movement of blood requires a pressure difference and vessel to carry it
- Blood flow is related to kinetic and potential energy
- The moving blood represents kinetic energy

- The pressure build up caused by cardiac contraction represents the potential energy
- Total fluid energy is the sum of the gravitational, kinetic and potential components

Factors Affecting Blood Flow:

- Cardiac Function
 - Decreased function = decreased output = reduction in peak arterial flow velocities
 - Congestive Heart Failure (CHF)
 - Coronary artery disease and reduced ejection fraction (EF%) <50%
 - Diastolic dysfunction of the heart - impaired ability of ventricles to relax during diastole, less blood fills ventricles and systolic ejection is reduced
 - Cardiomyopathy - disease of the cardiac muscle that can reduce EF%
 - Pericarditis - inflammation of the sac around the heart that limits its ability to expand and contract
 - Increased function = increased output = increase in peak arterial flow velocities
 - Compensatory heart beats with arrhythmias
 - Volume overload in ventricles with normal function
 - Liver disease, such as cirrhosis, can lead to increased blood volume
 - Kidney disease can lead to sodium retention that results in varying degrees of volume overload
 - Pregnancy and obesity can cause volume overload
- Peripheral Resistance
 - Primary factor that determines the resistance characteristics of the arterial signal displayed by PW Doppler is the part it is supplying
 - Determines the rate of flow in the arteries and is regulated by the arterioles
- Vessel Compliance
 - Stiffer or more calcific vessels = higher resistance flow
- Tone of vascular musculature
 - More muscular the patient = less compliant vessels = higher resistance flow
- Pattern of branches or collaterals
- Vasoconstriction
 - Tightening of vessel walls due to stimuli; cold, anxiety, smoking, norepinephrine
- Vasodilation
 - Expansion of vessel walls due to stimuli; heat, HTN medication, proximal stenosis
- Viscosity
 - Thickness of the blood
 - Refers to internal friction between the adjacent layers of a fluid
 - Energy lost in the form of heat
 - The greater the viscosity, the greater the resistance to flow
 - Causes of increased viscosity:
 - Viscosity is directly related to hematocrit levels in blood

- Increased hematocrit levels = increased resistance = decreased velocity of flow
- Sickle cell anemia is a genetic disorder that causes the body to produce abnormally shaped red blood cells; It can lead to increased viscosity, thrombosis and ischemia due to clumping of abnormal cells
- Polycythemia vera is a rare condition causing erythrocytosis (increased number of red blood cells in the bloodstream); It can lead to increased blood viscosity, decreased flow velocities throughout the arterial system
- Reduced temperature can increase blood viscosity
- Smoking can increase blood viscosity
- Exercise
 - Induces vasodilation in the large vessels supplying the skeletal muscles
- Autoregulation
 - Vascular beds alter resistance to flow to maintain the flow levels needed for normal function

Factors Affecting Doppler Evaluation
Reduced Systolic Velocity in Arteries
Congestive Heart Failure (CHF)
Coronary artery disease
Diastolic dysfunction
Significant Aortic Stenosis
Significant Mitral Stenosis
Dilated Cardiomyopathy
Pericarditis
Increased blood viscosity (polycythemia vera, sickle cell anemia, leukemia, smoking)
Cold exam room
Reduced heart rate
Hypocalcemia
****Mitral regurgitation normally does NOT affect the systolic velocities in the arterial system

Factors Affecting Doppler Evaluation
Elevated Systolic Velocity in Arteries
Compensatory heart beats with arrhythmias
Volume overload in normal ventricles (Liver disease, renal disease, pregnancy, obesity)
Systemic HTN
Significant Aortic Regurgitation
Hypertrophic Cardiomyopathy
Reduced blood viscosity (thrombocytopenia, chronic anemia)
Warm exam room
Increased heart rate
Hypercalcemia
****Mitral regurgitation normally does NOT affect the systolic velocities in the arterial system

Resistance:

- Determines the rate of flow in the arteries and is regulated by the arterioles
- "Push-back" from vascular beds against forward flow of blood during the cardiac cycle
- Varies with metabolic demand of tissue being fed, fluid properties, and characteristics of the vessel carrying the fluid
- Increased viscosity = increased resistance
- Increased vessel length = increased resistance
- Increased vessel radius = decreased resistance
- Changes in vessel radius have the greatest effect on flow resistance
- Extremities/muscles have much higher resistance to blood flow than the organs
- As resistance increases, blood flow decreases with no change in pressure
- If resistance increases, pressure must increase to maintain constant blood flow
- Stenosis causes resistance to increase proximal to a stenosis
- Resistance decreases distal to a stenosis

Arterial Resistance

$$\text{Resistance} = \frac{8nL}{\pi r^4}$$

n = blood viscosity

L = length of blood vessel

r = vessel radius

Resistance

↑ resistance = ↓ flow

↑ resistance requires an ↑ in pressure to maintain constant flow

Causes of ↑ resistance:

- Stenosis
- Vasoconstriction
- Organ supplied not functioning properly

High Resistance	Low Resistance
Sharp upstroke	Slow upstroke
Rapid deceleration	Slow deceleration
Little to no antegrade diastolic flow	Antegrade diastolic flow
Diastolic flow reversal	No diastolic flow reversal
Feeds tissues that do not require constant flow	Feeds tissues that require constant flow
Muscles, extremities	Brain, Kidneys, Spleen, Liver

High Resistance Flow



Low Resistance Flow



Poiseuille's Equation:

- Demonstrates relationship between pressure, flow volume and resistance
- Used to describe how much fluid (blood) moves through a pathway (artery); related to Ohm's Law of current moving in an electrical circuit
- Small changes in vessel radius cause most significant changes in flow
- Vessel length and viscosity are relatively constant; increases in either one will cause increased resistance

Poiseuille Equation

$$\text{Flow Volume (Q)} = \frac{(P_1 - P_2) \pi r^4}{8nL}$$

P = pressure

n = blood viscosity

L = length of blood vessel

r = vessel radius

Simplified Poiseuille Equation

$$\text{Flow Volume (Q)} = \frac{(P_1 - P_2)}{R}$$

P = pressure

R = resistance

Ohm's Law of Electrical Current

$$\text{Electrical Current (I)} = \frac{(V1 - V2)}{R}$$

V = voltage

R = resistance in the circuit

Bernoulli Effect:

- Total fluid energy is the sum of the gravitational, kinetic and potential components
- When fluid flows at a constant velocity from one point to another, the total energy remains constant
- If the pressure changes, the velocity will also change to maintain total fluid energy
- Slow moving blood will exert more pressure than faster moving blood
- There is a reduction in pressure seen with an increased blood velocity at a stenosis; decreased pressure = increased velocity
- After the stenosis, the velocity decreases causing an increase in pressure
- Explains the decrease in pressure seen with increased blood flow velocity
- Also describes the effects of turbulence and layers of flow separation within the carotid bulb

Bernoulli Equation

$$\Delta \text{ Pressure} = 4V^2$$

Sample:

$$\Delta P = 4 (3\text{m/s})^2$$

$$\Delta P = 36\text{mmHg}$$

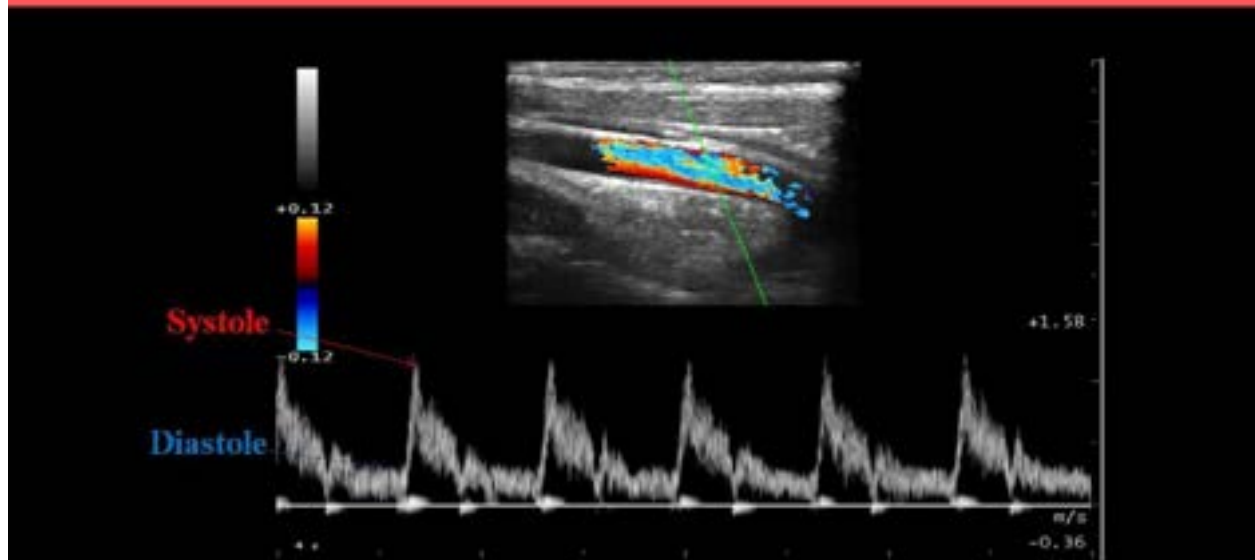
Pressure:

- Highest pressure in the system is in the left ventricle, average 120mmHg
- Increases with systole, decreases in diastole
- As peripheral pressure meets and exceeds ventricular pressure, diastole occurs
- Average pressure in the veins 2 mmHg, average arterial pressure 100 mmHg
- Arterial stenosis leads to decreased pressure as the velocity increases with the stenosis

Flow Volume:

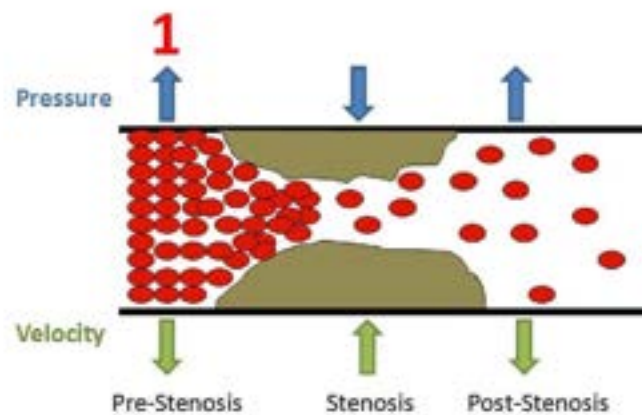
- Always proportional to blood pressure
- Increased volume = increased blood pressure

High Resistance Flow



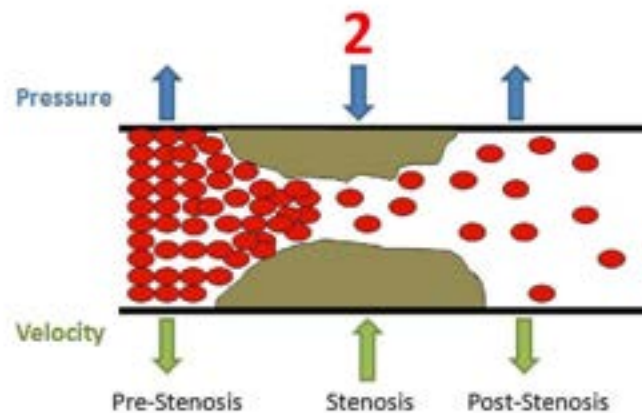
Bernoulli Effect

1. Pressure is higher proximal to the stenosis causing lower velocity.



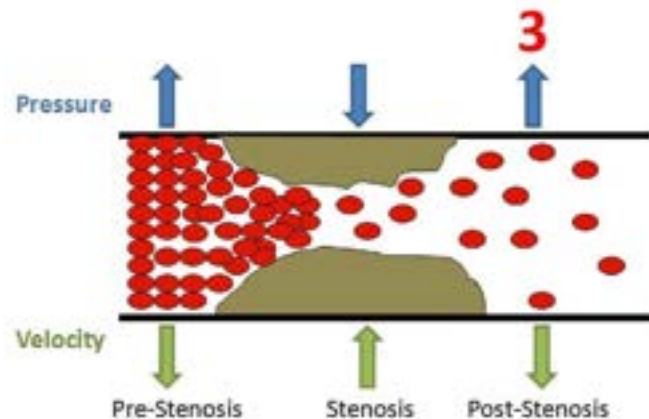
Bernoulli Effect

2. Pressure drops at the stenosis causing a large gradient and high velocity.



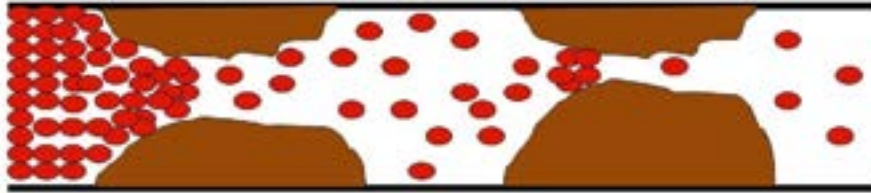
Bernoulli Effect

3. Pressure increases mildly just distal to the stenosis causing drop in velocity.



Pressure

- Multiple lesions affect flow more than a single lesion
- Flow will decrease significantly distal to multiple lesions



- Collaterals can mask disease by allowing blood to course to distal locations through alternate pathways

Arterial Flow Characteristics

Laminar Flow:

- Concentric layers of flow each with a slight difference in velocity
- Center layers have the highest velocity
- Outer layers are slower due to energy loss from friction
- Parabolic flow is identified in medium sized vessels and the layers of flow have a narrower range of velocities than laminar flow

Plug Flow:

- Identified during systole in larger vessels (aorta) and all layers of flow are moving at relatively the same speed
- Also seen at arterial branch origins and proximal to a stenosis
- Also called blunt flow

Arterial waveforms are determined by where the artery arises and what it is feeding

Systole: Cardiac contraction pushes blood forward through the arterial system

Diastole: The heart relaxes, refills with deoxygenated blood from the body on the right and oxygenated blood from the lungs on the left; forward flow continues in low resistance arteries

Pulsatility: Continuous throbbing or beating, flow in the arteries pulsates due to continuous cardiac contractions, venous flow is not considered pulsatile because it is not moved by cardiac contraction

Arterial Pulsatility:

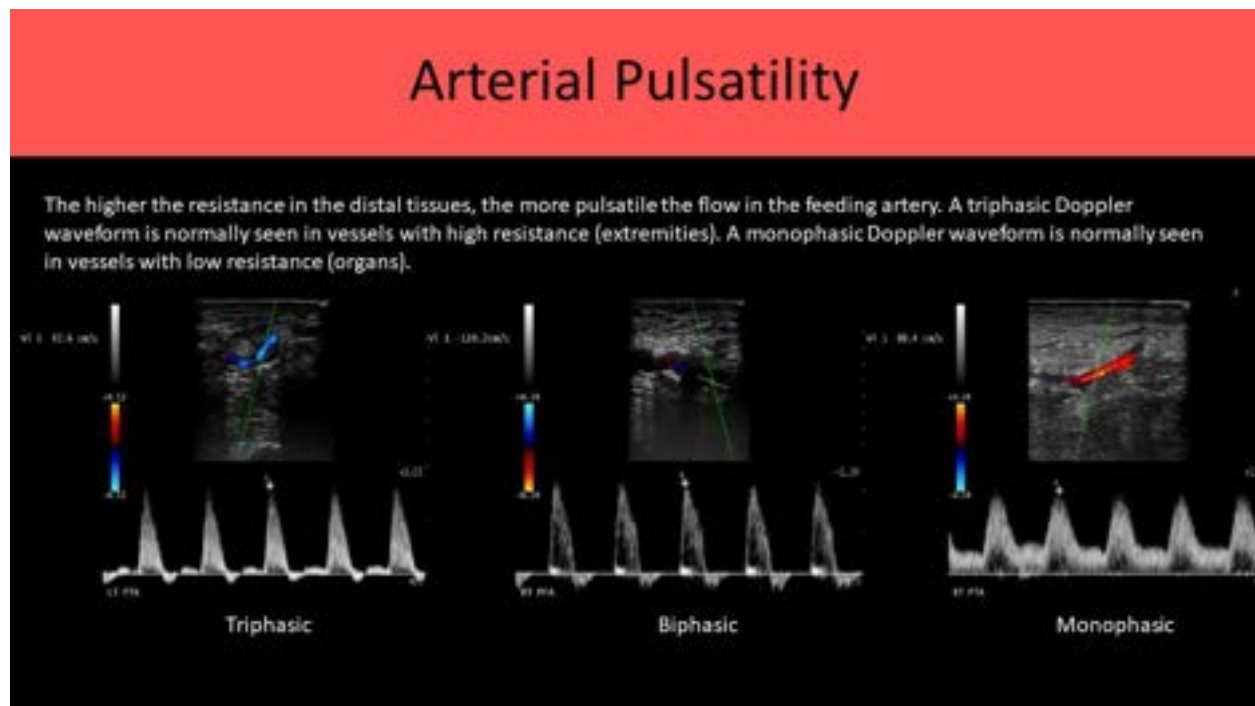
- Related to the number of changes in flow direction during one cardiac cycle
- Primary factor that determines the resistance characteristics of the arterial signal displayed by PW Doppler is the part it is supplying
- Vasoconstriction in the extremities leads to increased pulsatility in arteries and decreased pulsatility in capillaries and arterioles

- Vasodilation and stenosis in the extremities lead to decreased pulsatility in arteries and increased pulsatility in capillaries and arterioles

Monophasic - antegrade flow continues through the entire cardiac cycle; vessels that feed low resistance vascular beds

Biphasic - Flow during systole is antegrade and some flow is moving in a retrograde fashion during diastole; vessels feeding medium to high resistance vascular beds

Triphasic - Flow during systole is antegrade and during diastole initially some flow moves in a retrograde direction followed by a small amount of forward flow in end diastole



Blood Flow Terminology:

1. *Antegrade*: moving forward in the normal direction of blood flow
2. *Retrograde*: backward flow or filling, or against the normal direction of flow
3. *Turbulence*: disrupted flow caused by a stenosis, tortuosity or bifurcation; appears at the exit point of a stenosis
4. *Murmur* – abnormal blood flow sound in the heart, usually valvular regurgitation
5. *Bruit* – abnormal blood flow sound in a blood vessel; can be due to stenosis; also seen with vessel branching or tortuosity
6. *Thrill* – abnormal blood flow sensation in a blood vessel (vibration), can be due to stenosis; also seen with pseudoaneurysm and hemodialysis grafts

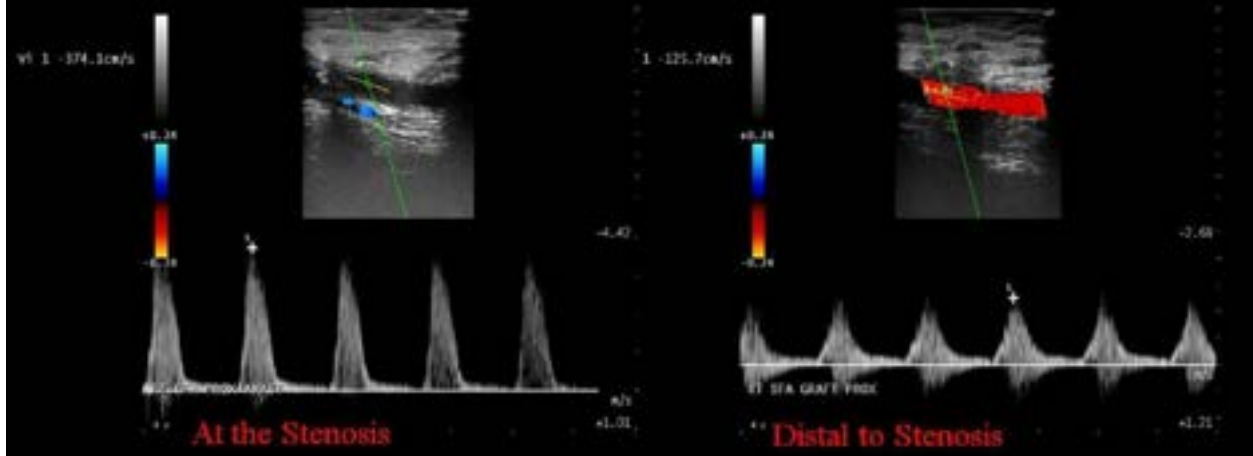
Extremity Arterial Hemodynamics

Arterial stenosis causes several key changes to the PW Doppler waveform. The example demonstrates a "normal" extremity waveform with high resistance flow proximal to the stenosis. At the area of the stenosis, the velocity of the flow increases significantly and spectral broadening occurs due to turbulent flow. Blood flow in the vessel distal to the stenosis will demonstrate a monophasic, low resistance waveform with increased diastolic flow.



Arterial Hemodynamics

Proximal disease decreases the strength and velocity of the flow distally. Arterial resistance will drop to aid in compensating the distal vascular bed for the loss of flow volume due to the stenosis.



Turbulence	
Anatomic Causes	Acquired Causes
Bifurcation/Branching	Atherosclerosis formation
Tortuous vessel course	Stent placement
Kinking of vessel	Bypass graft placement
Coiling of vessel	Myointimal hyperplasia
Eccentric change in vessel course	Aneurysm formation

Stenosis:

- Narrowing of a vessel lumen
- Factors that determine the hemodynamic significance of a stenosis:
 - shape and degree of the stenosis
 - diameter of the stenosis
 - length of the stenosis
 - contour of the stenosis (smooth vs. rough)
 - distal peripheral resistance
 - pressure gradient
 - presence/absence of collaterals
 - compounding factors from other diseases
- 50% diameter stenosis and 75% area stenosis are considered hemodynamically significant in most arteries
- Leads to increased resistance proximal to site
- Causes increased peak systolic and end diastolic velocities at the site of stenosis, due to the body trying to maintain blood volume
- The most common reason for underestimation of arterial stenosis is improper sample volume location
- The cursor should be placed initially at the narrowest point of the stenosis
- Increased velocities can also be detected immediately distal to the stenotic area
- It is important to move the Doppler cursor through the stenotic area and out the other side to find the highest velocity

- Distal to the stenosis vasodilation occurs in the capillaries causing lower velocity, low resistance flow with an increase in antegrade diastolic flow as the vascular beds try to "encourage" more flow distally
- Turbulent, swirling blood is identified just distal to a focal stenosis; referred to as eddy currents or vortices
- Smaller potential collateral vessels usually have higher resistance flow than their native vessels, due to decreased vessel diameter and increased length
- As stenosis increases in the larger artery, resistance to flow increases
- When the resistance in the larger vessel exceeds that in the smaller collateral vessels, flow enters the collateral pathways
- Collateral flow occurs when the smaller vessels have lower resistance than the primary pathway that has the stenosis/obstruction
- Collaterals have limited ability for vasodilation and do not respond the same as native arteries do to exercise

Significant Stenosis Leads To:

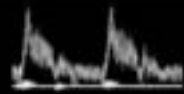
1. Proximal to the stenosis, flow velocity will be dampened with increased resistance
2. Increased velocity as the blood flows through and escapes the stenotic area
3. Increased velocities can also be detected immediately proximal and distal to the stenotic area
4. Post Stenotic Turbulence AKA eddy currents AKA flow vortices refers to areas of flow separation distal to the stenosis
5. Drop in pressure at the stenosis = greater the velocity at the stenosis
6. Drop in flow volume at stenosis
7. Decreased resistance and velocity distal to stenosis

Doppler Interpretation:

- Arteries carrying blood to organs normally exhibit low resistance waveforms with antegrade flow throughout the cardiac cycle
- Arteries carrying blood to extremities/muscles normally exhibit high resistance waveforms with retrograde flow during diastole
- Anytime a monophasic waveform is seen in an extremity artery = ABNORMAL
- Anytime a biphasic/triphasic waveform is seen in an artery supplying an organ = ABNORMAL
- Vasoconstriction leads to increased resistance to flow in the vessels - can change a monophasic waveform to biphasic
- Vasodilation causes decreased resistance to flow in the vessels - can change a triphasic waveform to monophasic
- Clenching the ipsilateral fist during an upper extremity Doppler evaluation can increase distal resistance and pulsatility in the proximal arteries

- Upon release of the clenched fist, the distal resistance drops significantly, pulsatility is reduced and flow increases to the hand

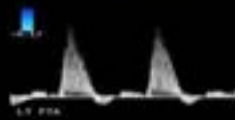
Doppler Interpretation



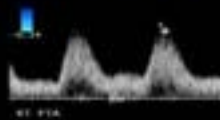
CCAW/ NL ICA



CCAW/ ICA OCCLUSION



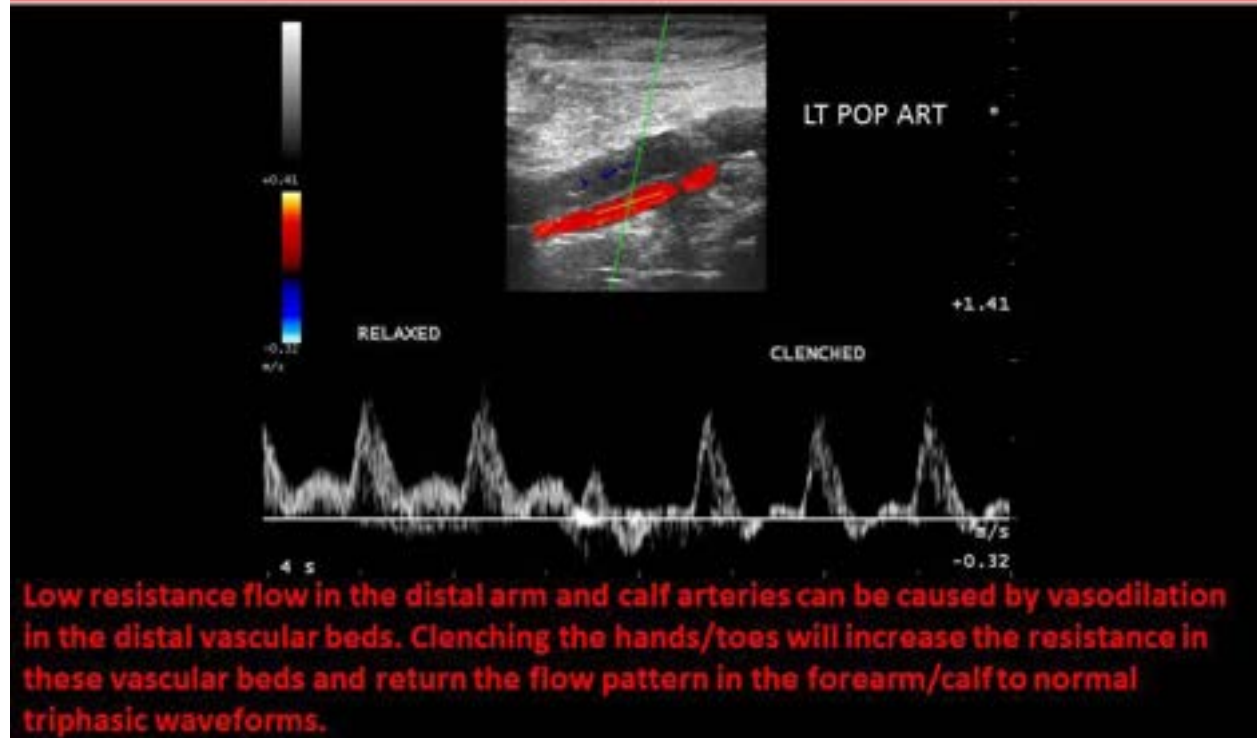
PTA NL



PTA W/ PROXIMAL DISEASE

Changes in phasicity indicate diseased area

Changing Resistance in Distal Vascular Beds



Factors Affecting Doppler Evaluation

Reduced Systolic Velocity in Arteries

Congestive Heart Failure (CHF)

Coronary artery disease

Diastolic dysfunction

Significant Aortic Stenosis

Significant Mitral Stenosis

Dilated Cardiomyopathy

Pericarditis

Increased blood viscosity
(polycythemia vera, sickle cell anemia, leukemia, smoking)

Cold exam room

Reduced heart rate

Hypocalcemia

****Mitral regurgitation normally does NOT affect the systolic velocities in the arterial system

Factors Affecting Doppler Evaluation
Elevated Systolic Velocity in Arteries
Compensatory heart beats with arrhythmias
Volume overload in normal ventricles (Liver disease, renal disease, pregnancy, obesity)
Systemic HTN
Significant Aortic Regurgitation
Hypertrophic Cardiomyopathy
Reduced blood viscosity (thrombocytopenia, chronic anemia)
Warm exam room
Increased heart rate
Hypercalcemia
****Mitral regurgitation normally does NOT affect the systolic velocities in the arterial system

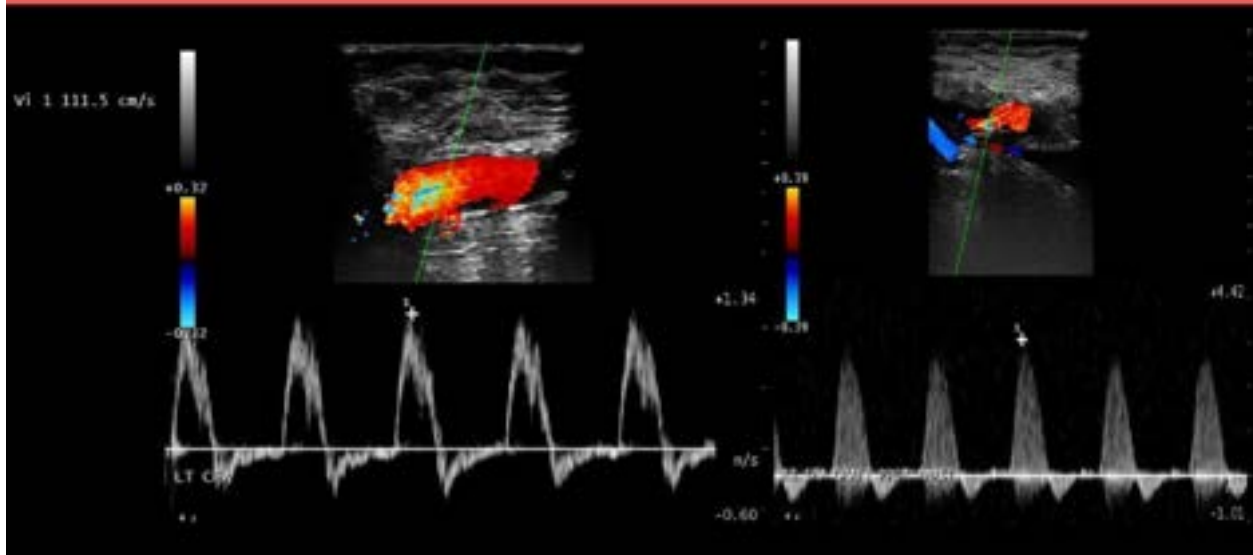
Spectral Broadening:

- Widening of the spectral waveform with filling in of the spectral window
- Bandwidth refers to the difference between the highest and lowest frequencies in the pulse
- Laminar flow has a narrow bandwidth of reflected signals because the blood cells are all moving at relatively the same speed
- This allows for the display of a spectral window on the tracing
- With turbulent flow, the blood cells are moving at many different velocities
- This produces a wide bandwidth of reflected signals and the spectral window fills in
- Causes: stenosis, tortuosity, improper sample location closer to the wall, increased sample size, and bifurcations

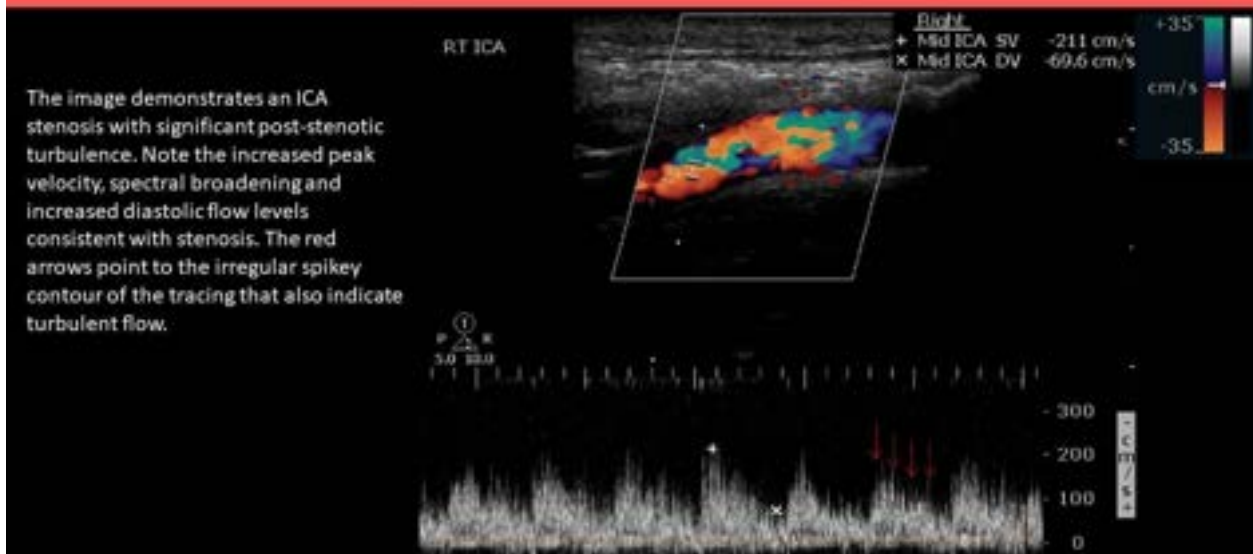
Post Stenotic Turbulence:

- Eddy currents or vortices
- Flow spreads out into the larger vessel area distal to stenotic area
- Wide range of flow velocities
- Causes aliasing on Color Doppler and spectral broadening on PW Doppler

Spectral Broadening



Post-Stenotic Turbulence



Acceleration Time :

- Measured to differentiate inflow from outflow disease in the lower extremity
- Time from onset of systole to point of maximum systolic peak

- Increased sweep speed makes measurement of the AT easier due to wider waveforms on the Doppler display
- AT increases in extremities as stenosis increases; takes a longer amount of time to reach peak systole due to narrowed lumen
- Decreased cardiac output can increase the AT in bilateral extremity arteries
- In normal lower resistance vessels like the ICA and cerebral vessels, the AT is increased (longer) when compared to a higher resistance vessel (extremity)
- AT increases (longer time) distal to a significant stenosis due to decreased resistance

Resistive Index:

- Measurement of vascular resistance
- Compares the difference between the systolic and diastolic velocities to the maximum velocity of the vessel segment; $200\text{cm/s} - 60\text{cm/s} / 200\text{cm/s} = \text{RI } 0.7$
- If there is no diastolic flow (EDV 0 cm/s) then the RI will be 1.0; $200\text{cm/s} - 0\text{cm/s} / 200\text{cm/s} = 1.0$
- Used to evaluate tissue resistance to blood flow (e.g. renal parenchyma, liver parenchyma, transplant evaluation)
- Can also be used to demonstrate changes in resistance caused by a stenosis/occlusion
- A distal occlusion will cause increased resistance proximal to the obstruction = high RI

Pulsatility Index:

- Degree of resistance in a vessel
- Measures the variability in blood velocity during the cardiac cycle
- Compares the difference between the systolic and diastolic velocities to the average velocity of the vessel segment
- Used to distinguish inflow disease from outflow disease
- Values increase as you evaluate vessels further from the aorta; Normal CFA >5.0 and popliteal artery >8.0
- Higher in extremities than organs, normally; organs usually <1.5 and brain <1.0
- Damping factor
 - Describes the attenuation of the Doppler signal with a stenosis
 - Compare the PI proximal to the stenosis to the PI distal to the stenosis
 - A significant stenosis has a higher damping factor than a mild stenosis

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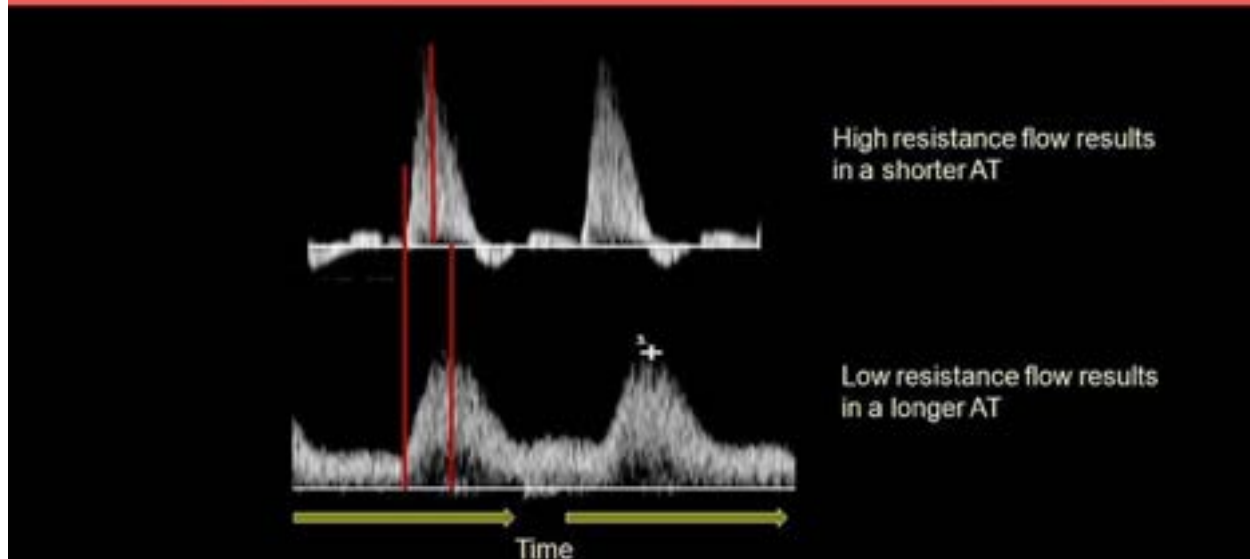
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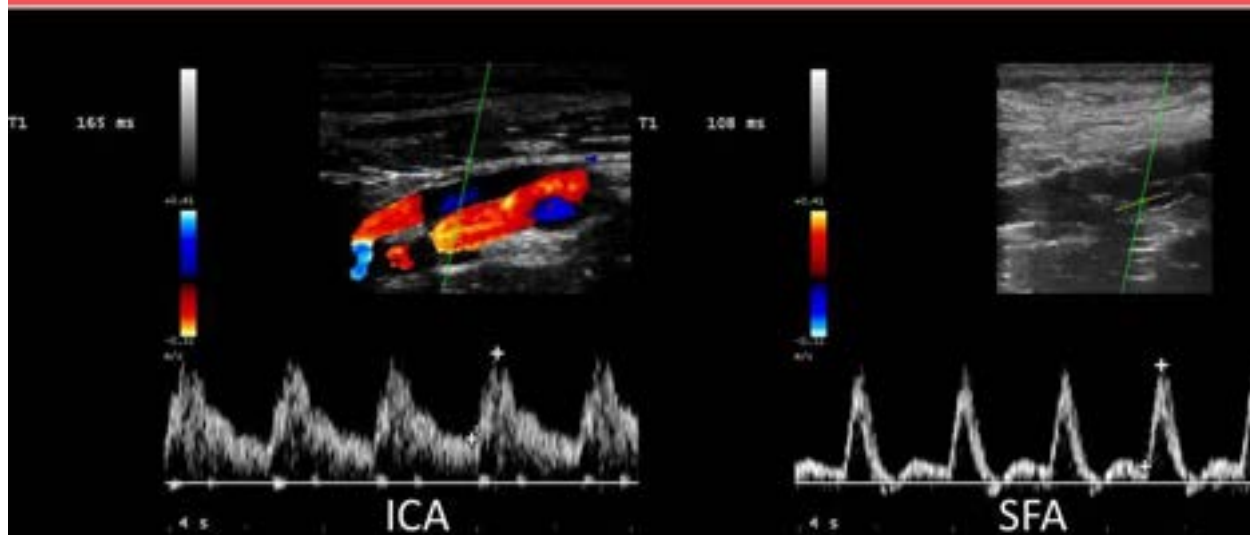
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Acceleration Time (AT)



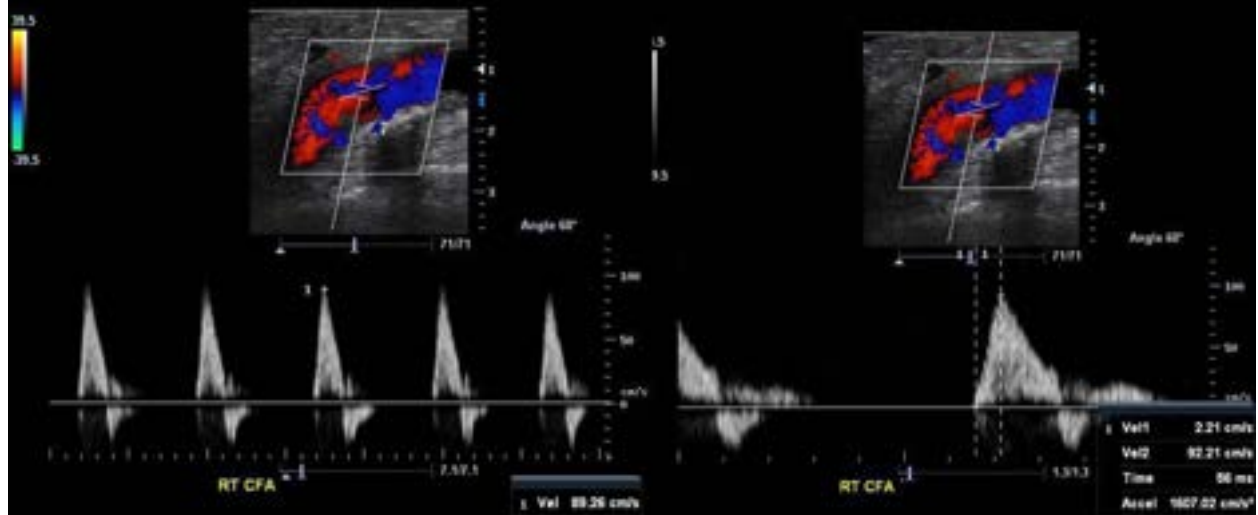
Acceleration Time (AT)



Extremities normally offer higher resistance to flow compared to organs = shorter AT. If the AT in the femoral artery is increased >140ms, inflow disease is suspected.

Organs like the brain offer low resistance to flow compared to extremities = longer acceleration times. If the AT in the carotid system is reduced, there is a distal obstruction.

Measuring Acceleration Time



Increasing the sweep speed will spread out the waveform to help in obtaining a more accurate AT measurement.

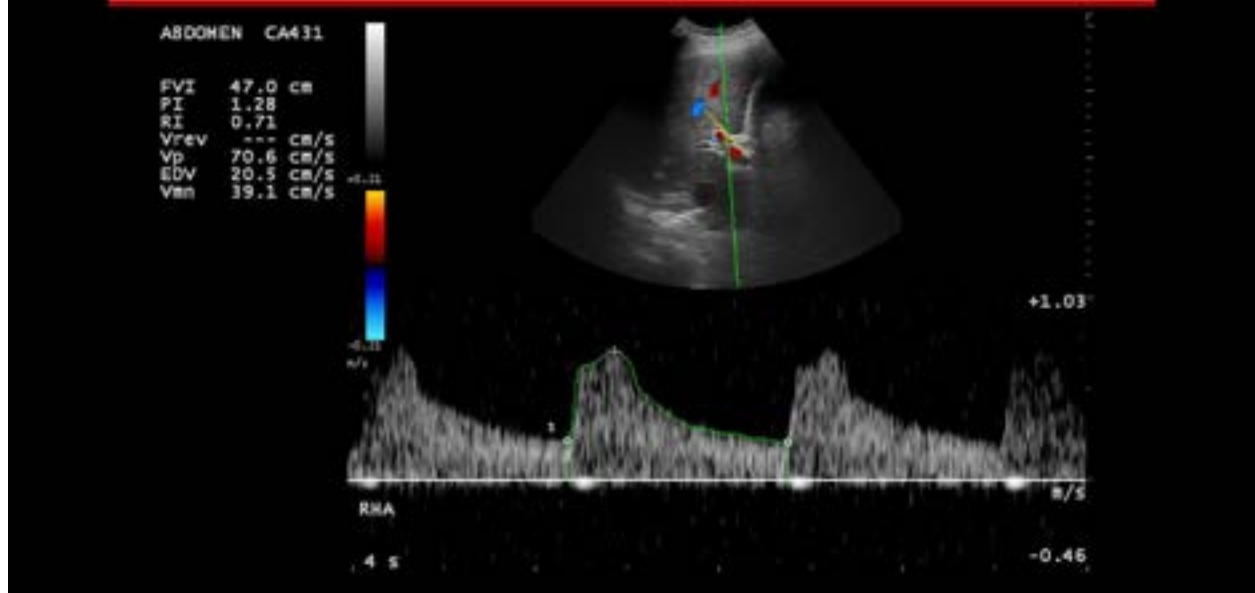
Pulsatility Index (PI)

$$PI = \frac{PSV - EDV}{\left(\frac{PSV + EDV}{2} \right)}$$

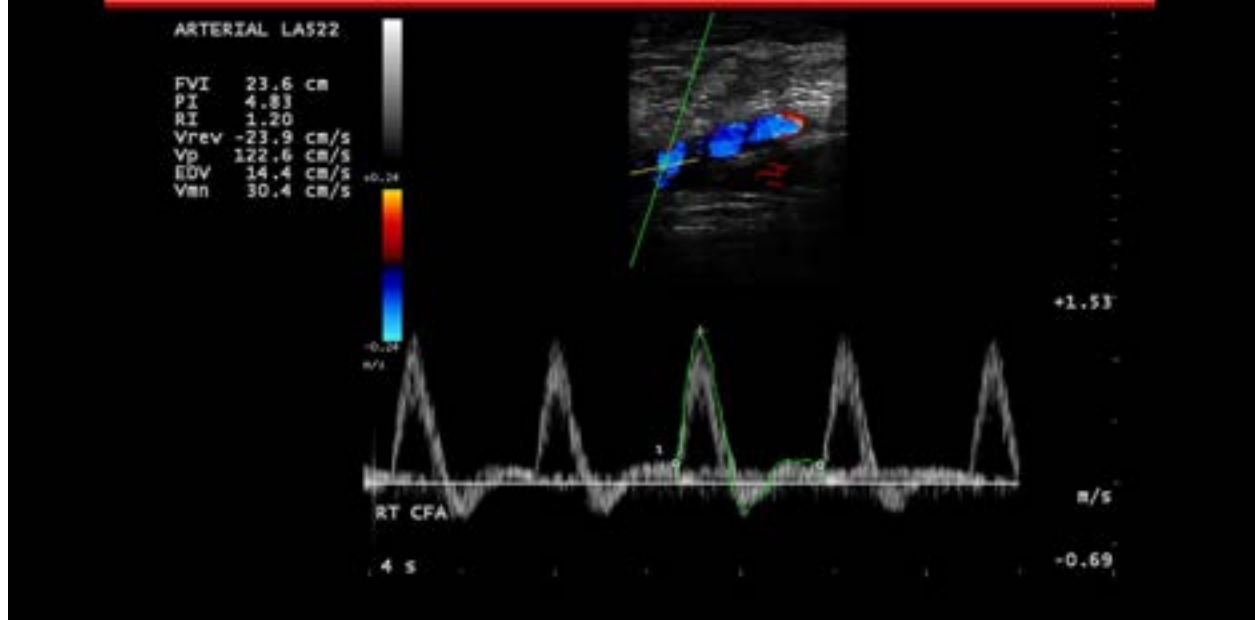
Resistive Index (RI)

$$RI = \frac{PSV - EDV}{PSV}$$

Organ PI and RI



Extremity PI and RI



Effects of Exercise on Arterial Flow:

- Exercise normally increases flow throughout the body to 3-5 times the resting flow

- Exercise should increase flow and the pressure gradient in normal arteries
- As muscles require more blood flow, the arterioles open and the resistance is reduced to increase flow to the muscles
- In normal patients, ankle and brachial pressures after exercise will be the same as the resting pressures or mildly increased
- When stenosis is present, the arterioles are already open wider with lower resistance to flow because the vascular bed is not receiving enough flow at rest
- When exercise is performed in a patient with stenosis, the muscles demand more flow but the arterioles cannot reduce the resistance any further
- The reduced inflow due to increased demand causes the pressure at the ankles to drop
- In patients with arterial disease, the ankle pressures drop while brachial pressures stay the same or increase
- The drop in ankle pressure leads to a reduced ABI calculation
- Ischemic symptoms occur in the muscles due to the reduced flow to the region

Collateral Formation:

- Numerous, smaller vessels that provide an alternative pathway to perfuse an organ/extremity
- Blood vessels that dilate in response to increased flow in an alternate pathway due to stenosis or obstruction in the primary pathway
- As stenosis increases, resistance to flow increases
- When the resistance in the larger vessel exceeds that in the smaller collateral vessels, flow enters the collateral pathways
- Collateral flow occurs when the smaller vessels have lower resistance than the primary pathway that has the stenosis/obstruction
- Can mask symptoms that would be caused by the stenosis of the primary pathway
- Associated with chronic disease, NOT seen with acute obstruction

Venous Flow Characteristics

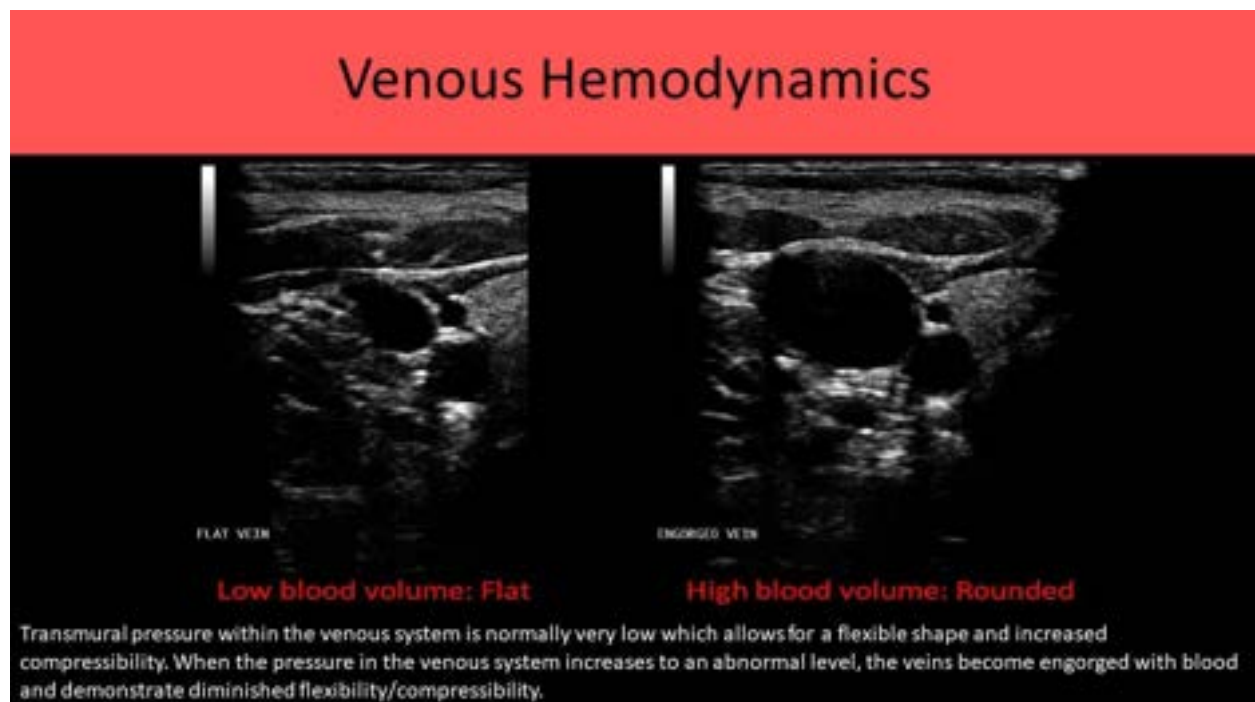
Hydrostatic Pressure:

- Weight of the column of blood inside the vessels
- Directly proportional to height

- Calculated by multiplying fluid density X gravity X height of the column
- Applies to arteries and veins
- Equal to zero at the level of the heart
- Changes with variation in patient position; usually constant at the level of the heart and diaphragm
- Supine: HP = 0mmHg
- Average Patient Standing:
 - HP = 0mmHg at heart
 - HP = 100mmHg at ankle
 - HP = -30mmHg at head
- When standing, the pressure in the arteries in the ankle is the ankle BP + 100mmHg

Ambulatory Venous Pressure:

- Defined as the lowest pressure level in the limb during exercise
- Normal levels are approximately 30mmHg and will increase when reflux is present.
- Walking causes the pressure in the calf veins to drop to about 20-30mmHg in normal patient
- In a patient with incompetent valves, the walking pressure in the veins will remain at nearly the same level as when standing still or it will be higher due to increased arterial inflow and little to no outflow



Transmural Pressure:

- Veins hold 2/3 of all blood in the body
- Normal venous volume <150mL, increases with insufficiency
- Pressure within the veins pushing outward
- Normally low
- Increases with venous volume
- Decreases with ambulation and veins refill after 3-5 minutes of standing
- Determines the shape of the vein; flat or circular
- Flattened veins with low volume have low transmural pressure which offer more resistance to flow and less resistance to compression
- Circular veins with higher volume and higher transmural pressure offer less resistance to flow and more resistance to compression

Calf Muscle Pump:

- Movement of venous blood occurs due to respiration and the calf muscle pump
- Calf veins act as reservoirs for the blood
- Muscle contraction causes the deep system valves to open and flow moves through deep venous system toward heart
- Muscle relaxation causes proximal valves to close due to hydrostatic pressure and distal valves remain open to refill the deep system
- Flow moves from superficial to deep system through the calf perforators during muscle relaxation
- Competent valves normally allow unidirectional flow toward the heart AND from superficial to deep system
- Effective calf pump reduces blood stasis/pooling, decreases venous pressure
- Ineffective calf pump leads to increased stasis, venous volume and pressure; reflux is a contributing factor
- Causes of pump failure include muscle wasting, obstruction and incompetent valves; venous pressure will decrease only slightly with ambulation and refill time is significantly shortened

Doppler Flow Characteristics of Veins:***Spontaneous Flow:***

- Flow is detected in the vein without flow augmentation
- Calf veins and saphenous veins normally do not demonstrate spontaneity
- All other veins should demonstrate spontaneity

Respiratory Phasicity:

- Flow moves from areas of high pressure to low pressure

Inspiration:

- Decreased intrathoracic pressure with increased intra-abdominal pressure
- Increased flow in upper extremities toward the heart
- Decreased flow in lower extremities

Expiration:

- Increase in intrathoracic pressure with decreased intra-abdominal pressure
- Decreased flow in upper extremities
- Increased flow in lower extremities toward the heart

Cardiac Pulsatility:

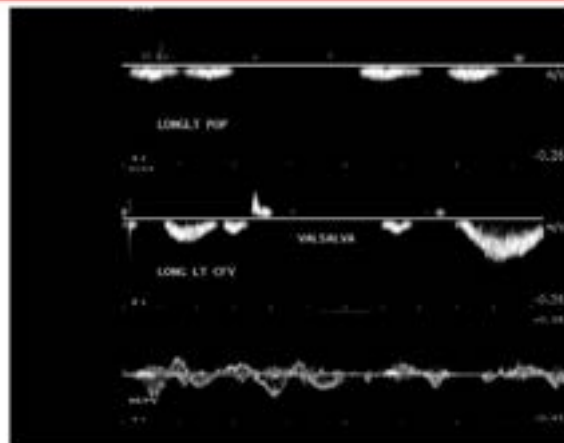
- "Ripples" of motion caused by adjacent cardiac contraction
- Causes turbulent flow in the upper vena cava, upper arms, and hepatic veins
- Abnormal if seen in lower extremities = CHF

Venous Waveforms

Respiratory Phasicity

Valsalva

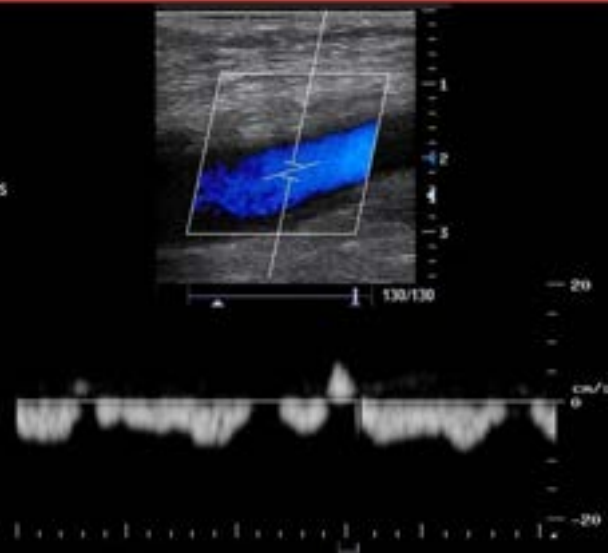
Cardiac Pulsatility



****the further from the heart you get the cardiac pulsatility decreases and respiratory phasicity increases*

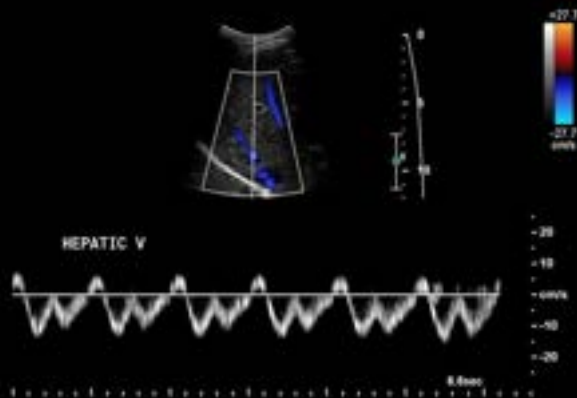
Spontaneous Flow & Respiratory Phasicity

Normal venous flow is spontaneous, demonstrates respiratory phasicity and responds to augmentation maneuvers properly.

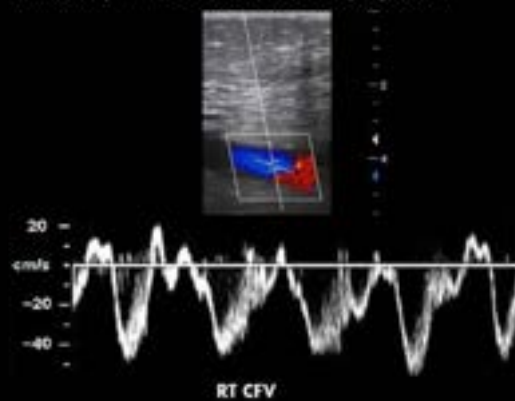


Cardiac Pulsatility

Normal in veins adjacent to the heart

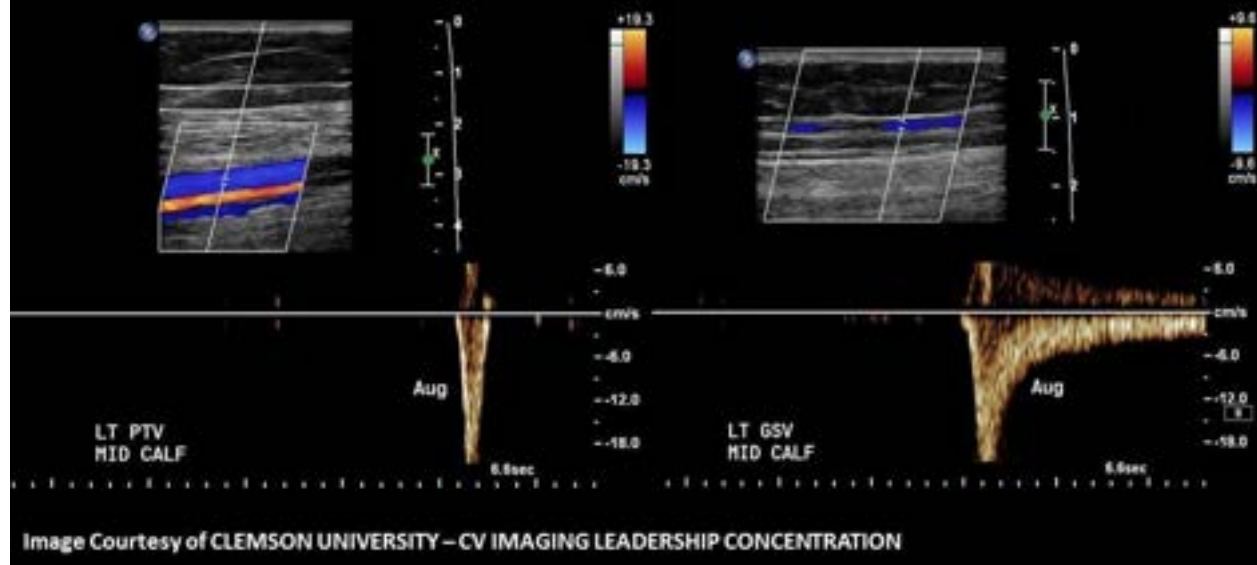


Abnormal in the lower extremity veins - Indicates elevated right atrial pressure; e.g. CHF, Pulmonary HTN, severe tricuspid regurgitation



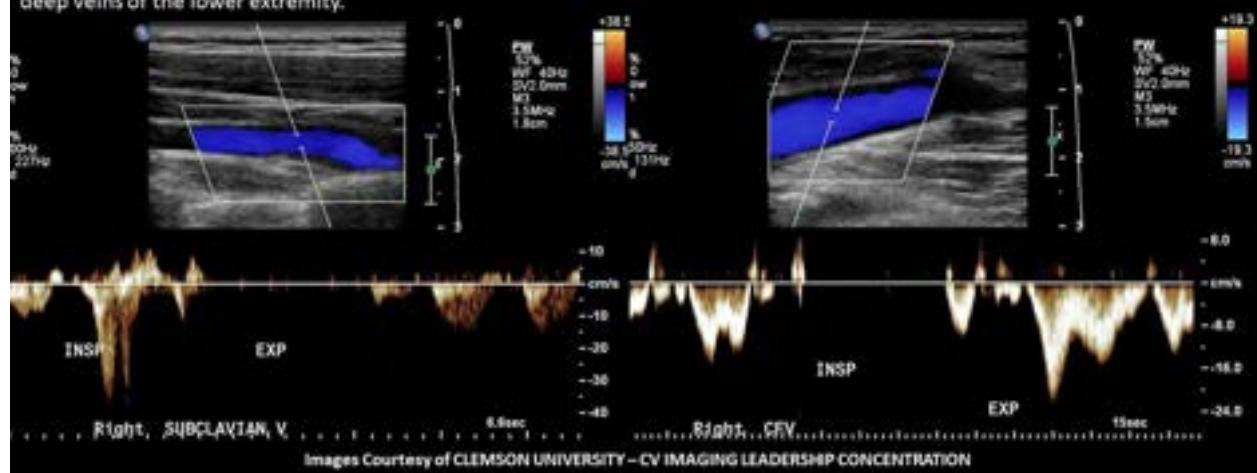
Non-Spontaneous Flow

The saphenous and calf veins normally demonstrate non-spontaneous flow. Distal augmentation is used to assess patency.



Respiratory Phasicity

Note that inspiration increases flow toward the heart in the deep veins of the upper extremity, but flow ceases in the deep veins of the lower extremity. Expiration stops flow in the deep veins of the upper extremity, but increases flow toward the heart in the deep veins of the lower extremity.



Pressure Changes

	Intrathoracic Pressure	Intra-abdominal Pressure
Inspiration	↓	↑
Expiration	↑	↓
Valsalva Strain	↑	↑

Factors Affecting Doppler Evaluation
Increased Pulsatility of Venous Flow
Congestive Heart Failure (CHF)
Significant Tricuspid Regurgitation
Significant Tricuspid Stenosis
Significant Pulmonic Regurgitation
Significant Pulmonic Stenosis
Diminished/Absence of Respiratory Phasicity Venous Flow
Unilateral – proximal venous obstruction in the ipsilateral extremity
Bilateral – central venous obstruction (thrombus, tumor, extrinsic compression)

The respiratory response to venous flow in the deep veins in the lower extremities is the opposite of the response that occurs in the upper extremity deep veins. As the patient takes in a breath, the diaphragm moves inferiorly to increase the pressure in the abdomen while decreasing the pressure in the chest cavity. Flow in the lower extremities will decrease as the diaphragm moves inferiorly and flow in the arms will increase toward the heart. When the patient lets the breath out, the diaphragm moves superiorly to decrease the intra-abdominal pressure while increasing the intra-thoracic pressure. Expiration will allow an increase in lower extremity flow while the upper extremities will decrease their flow volume toward the heart.

Distal Augmentation:

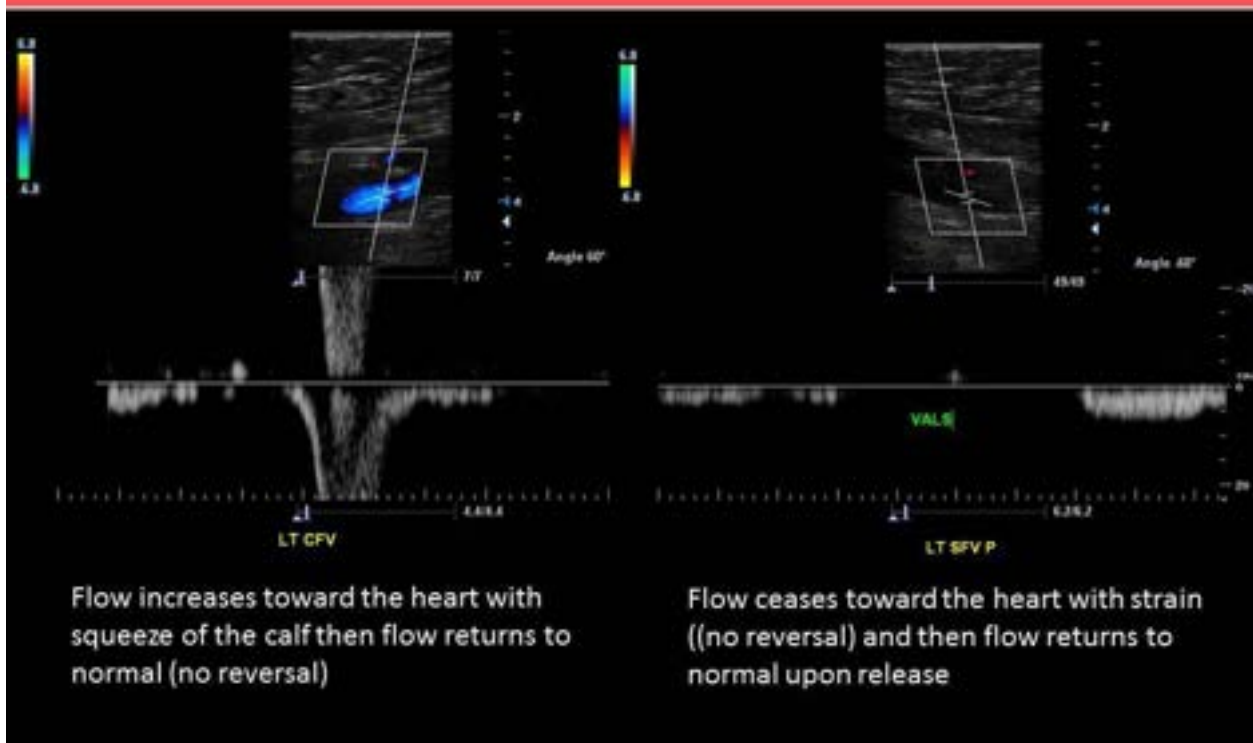
- Lightly squeeze muscular portion of patient's calf/forearm or have patient flex their foot/hand quickly
- Normally increases flow toward heart followed by a lack of flow reversal and flow normalization toward heart
- Decreased augment = obstruction between probe and augmentation location
- Valvular insufficiency will demonstrate an increase in flow toward the heart followed by retrograde flow toward distal extremity

- Length of time reflux occurs is directly related to the severity of valvular insufficiency

Valsalva Maneuver:

- Stressing increases intrabdominal pressure; simulate stressing during bowel movement
- Evaluates extremity veins, iliac veins and IVC
- Normally the maneuver performed, flow toward heart should stop with no flow reversal
- Augmentation of signal toward the heart upon **release**
- Augmented flow reversal at onset of Valsalva = venous valvular incompetence

Distal Augmentation vs. Valsalva Maneuver



Sonography Principles & Instrumentation

2D Evaluation:

Interaction of Sound with Tissue:

1. Reflection - return of the ultrasound wave to the transducer from a smooth surface; greatest amount of reflection occurs when incident beam is perpendicular to structure (specular reflection)
 2. Scatter - deflection of ultrasound waves from small structures; red blood cells scatter sound
 3. Refraction - deflection of ultrasound waves due to media with varied acoustic impedance
 4. Attenuation - loss of ultrasound signal due to absorption in tissues as it travels; higher frequencies have higher attenuation rates
- In order to transmit sound into the patient, an acoustic coupling agent is required; ultrasonic gel is used to transmit the sound from the transducer into the patient; Air between the probe and the patient will cause the sound to scatter and no image will be produced

Production of Ultrasound:

Pulser:

- Generates the voltage that excites the piezoelectric element so that it produces a sound pulse
- Determines the amplitude of the voltage spike applied to the element
- Determines the PRF of the transducer
- Can be adjusted by the Sonographer using the power output
- If the image is too DARK, always increase the gain before increasing the voltage to the pulser (ALARA principle)
- If the image is too BRIGHT, always decrease the output power first, before decreasing gain, to reduce patient exposure

Transducers:

- Single Element - requires mechanical focusing and steering; with only one element, electronic focusing and steering is not possible
- Phased Array - multiple elements used to create an image by varying timing of the electrical pulses (firing) to the different elements in the array (electronic steering and focusing)
- Transducer frequency is inversely proportional to penetration and directly related to resolution; high frequency probes have low penetration and high resolution
- Beam width, spatial pulse length and pulse duration are inversely proportional to the transmitted frequency; decrease with increased transmit frequency

Annular Array:

- Usually constructed of 5 concentric, ring-shaped piezoelectric elements that produce a cylindrical beam
- Sector shaped image
- Mechanical wobbler can use annular array instead of single element
- Requires mechanical steering, if needed; electronic not possible with circular elements
- Can be electronically focused by timing voltages to different elements (phased annular array)
- Circular elements allow for electronic focusing in all planes at all depths to provide optimal lateral resolution at all depths on the image
- Central elements used to image more superficial structures and outer elements used to image deeper structures

Sequential Linear Array:

- AKA switched array
- Contains approximately 200 rectangular elements placed in a line across the face of the transducer
- Parallel acoustic lines are transmitted sequentially
- Each acoustic line a single sound pulse created by the transducer element
- Same size field-of-view in near and far fields (no trapezoid FOV)
- Uses mechanical focusing (curved lens or crystal)
- Lateral resolution is improved by firing a few elements at exactly the same time to create a narrow beam
- No beam steering
- Damage to a single element will only cause the loss of the portion of the image produced by that element

Linear Phased Array:

- Contains approximately 200 elements

- Elements fired at varied times to allow for electronic focusing and beam steering; approximately 10 nanosecond delays between firing
- Phasing used to create trapezoidal FOV, steer and focus beam (electronic beam focusing and steering)
- Electronic steering allows a linear probe to produce a trapezoidal image
- Trapezoidal linear display leads to degraded lateral resolution in the far field; contrast, axial and elevational resolution are not affected
- Large footprint makes it difficult to obtain images in areas with small imaging windows
- Used for abdominal, OB, small parts, vascular, and musculoskeletal imaging

Curved Sequential Array:

- Similar construction and function as a sequential linear array
- Blunted sector FOV
- Requires mechanical focusing
- No beam steering

Curved Linear Phased Array:

- Contains a large number of linearly arranged piezoelectric elements (120-250)
- Sequentially pulsed transducer elements
- Produces curved/blunted sector image with a larger near field footprint
- Uses electronic focusing and steering
- AKA curvilinear array or convex array
- Can be used for abdominal, obstetrical, gynecological, and small parts imaging

Sector Phased Array:

- Contains a large number of linearly arranged rectangular piezoelectric elements
- Small footprint with 100-300 elements
- Wedge-shaped sections; Pie-shaped
- Acoustic lines are transmitted at an angle that is different from that of the previous acoustic line
- Distance between individual acoustic lines in the far field is greater than in the near field
- Allows multiple focal points at varied depths (variable focusing)
- Triangle or sector FOV created by electronic steering
- Most limited field-of-view in the near field compared to all transducer types
- Greater field-of-view in the far field than a linear image
- Electronically focused and steered by firing groups of elements to produce one pulse at a time (Transmit Focusing)
- Used for echocardiography, abdominal, pelvic, vascular, transcranial, and neonatal brain imaging

Vector Array:

- Combine sequenced linear array and phased array techniques to provide a trapezoidal imaging format
- Slightly larger footprint than a phased array transducer
- Sector image has a flat top with a wider field-of-view in the near field
- Employs electronic steering and focusing
- Used for echocardiography, abdominal, pelvic, vascular, transcranial, and neonatal brain imaging

Harmonic Imaging:

- A low frequency beam is transmitted into the patient but the system displays the received information as double the frequency; EX: transmitted 5MHZ, received/displayed 10MHZ
- Allows deeper penetration with lower transmitted frequency and improved lateral resolution with increased received frequency
- Provides narrower beam width to reduce side lobe/grating lobe artifact

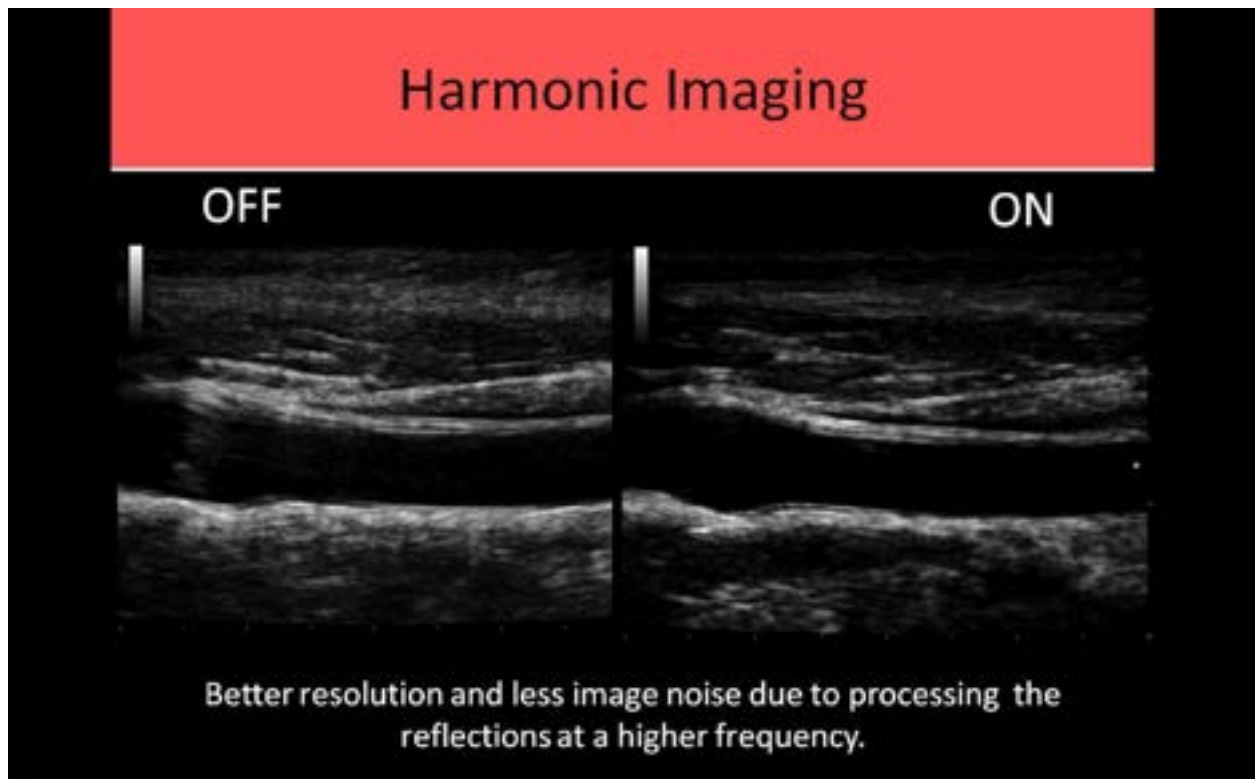


Image Resolution:

1. Axial Resolution

- Type of spatial resolution
- Ability to resolve two separate structures that lie parallel to the ultrasound beam; useful in evaluating wall thickness of a vessel
- Improves with higher transducer frequency, wider bandwidths and shorter pulse lengths
- Usually 1mm for 3MHz probe and 0.5mm for 7.5MHz probe

1. *Lateral Resolution*

- Type of spatial resolution
- Ability to resolve two separate structures that lie perpendicular to the ultrasound beam; useful for differentiating two atheroma that are side by side in the vessel
- Improves with higher transducer frequency, beam focusing, wider bandwidths and more superficial structures

1. *Contrast Resolution*

- Ability to differentiate two structures with varied echogenicity
- Compression will change the contrast resolution on the image
- Increasing compression/dynamic range = more shades of gray = improved contrast resolution
- Monitor contrast setting can also adjust image contrast

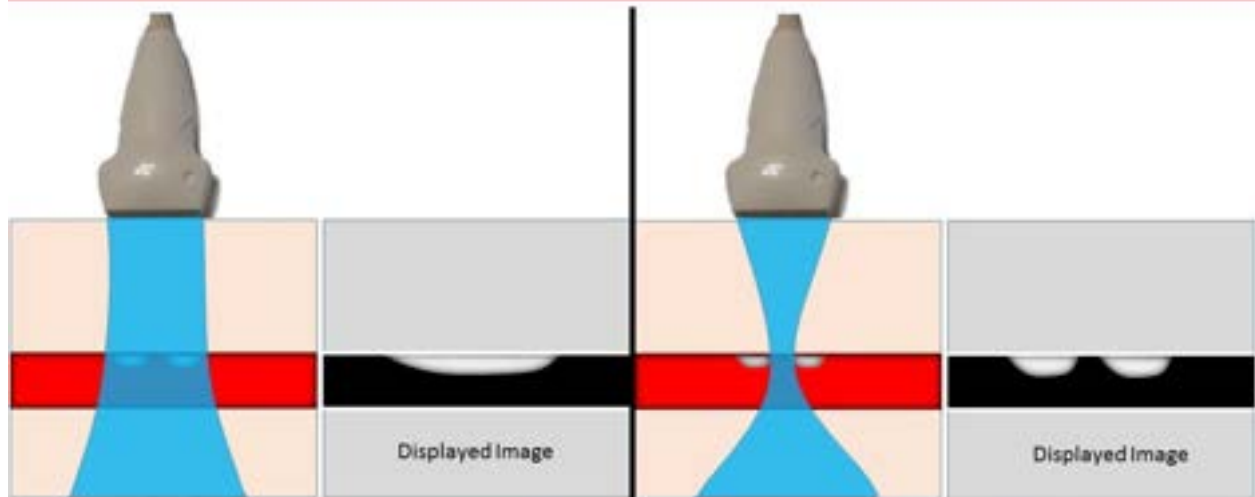
1. *Elevational Resolution*

- AKA slice thickness resolution
- Improves with thinner image slices
- Most difficult to adjust
- Slight improvement with increased transducer frequency and focusing

1. *Temporal Resolution*

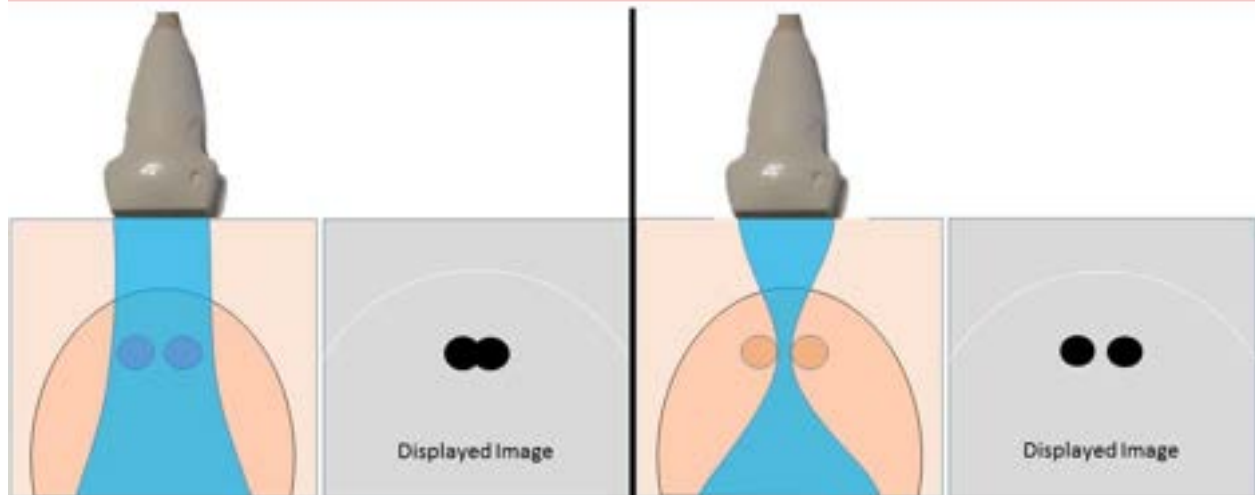
- The ability to detect that an object has moved over time
- Decreased temporal resolution will cause the image to "drag" with a visible difference in the motion of the probe and the display of the structure
- Improves with decreased number of focal zones, smaller field of view, decreased image depth and decreased line density

Lateral Resolution



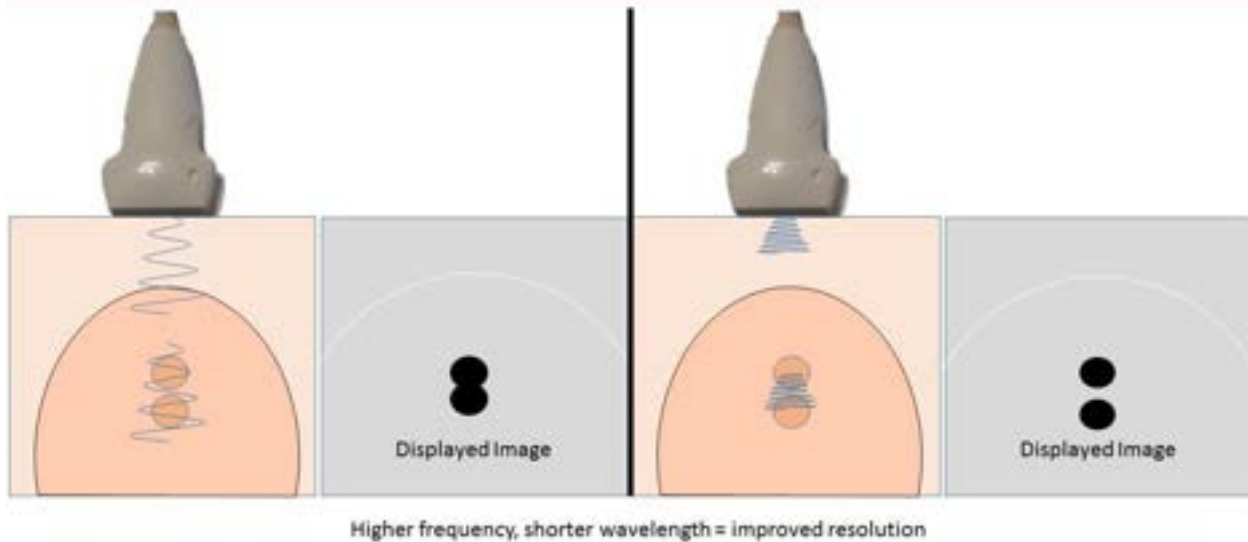
Beam width must be less than the distance between two objects for them to be viewed separately

Lateral Resolution

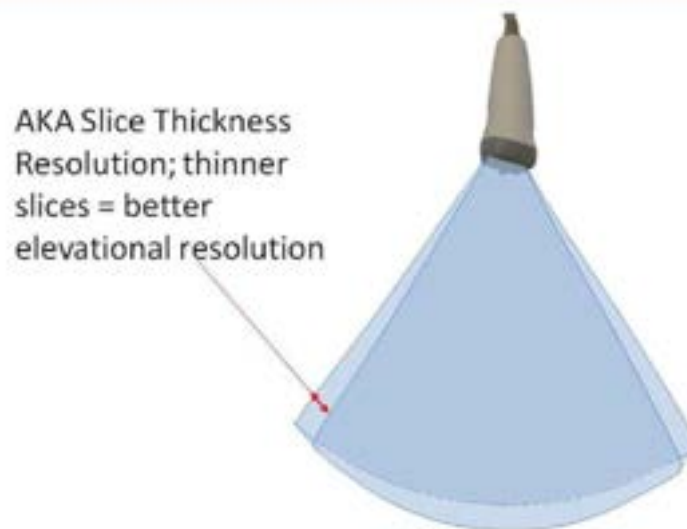


Beam width must be less than the distance between two objects for them to be viewed separately

Axial Resolution



Elevational Resolution



External or Mechanical Focusing - lens placed inside transducer in front of element to focus the beam before it exits the transducer

Internal or Electronic Focusing - varied timing of the pulser voltage used to vary the firing of the elements in an array; used to focus the beam

Electronic Beam Steering - varied timing of the pulser voltage used to vary the firing of the elements in an array; used to steer the beam; helps to visualize needles during procedures

STORAGE DEVICES:

Hard Copy Storage:

- Any kind of film
- Thermal paper
- Kodak glossy printers
- Inkjet printers
- Videotapes
- Disadvantages: Images degrade over time; can be damaged by moisture or heat exposure; requires large storage area for films/tapes; videotapes have limited resolution and can be erased

Hardware, Software and Digital Media:

- DICOM images = digital imaging and communication in medicine; standard image format for the medical community
- PACS (Picture Archival and Communication System) is the best form of storage for DICOM images; computer servers (hardware) that store and display DICOM images; easy to back up info to prevent corruption or damage to all files
- NAS (Network Attached Storage) computer memory devices that are part of the PACS system and is used to store the original study and the back-up files in separate locations; hardware
- RIS (Radiology Information System) or HIS (Hospital Information System) refers to a software program that stores medical records, appointments and other patient information
- Examples of digital media - CD, DVD, MO Disk, hard drives, flash drives
- Advantages: large amounts of exams can be stored in a small space; image resolution is consistent with the resolution of the original image
- Disadvantages: requires DICOM viewer; file size is limited to storage media size; files can be deleted; CDs and MO disks can be broken or cracked

2D IMAGING:

- Grayscale images
- Used to assess chamber size, structure and motion, valvular structure and motion
- Allows mainly subjective assessment of valvular motion
- Reflections are processed and displayed as varied levels of gray of the image
- The amplitude of the reflection is directly proportional to the brightness of the pixel on the display

Image Depth and the Range Equation:

- Image depth is inversely proportional to pulse repetition frequency
- As the pulse travels deeper into the patient, it takes a longer time to reach the intended structure; less pulses per second can be emitted due to the increased distance and travel time for the pulses
- Range equation used to calculate the time of flight or go-return time; the time sound travels TO and FROM the transducer
- Used to calculate distance to reflector
- US system uses this information to locate pixel placement on the screen
- Range Equation: used to calculate depth of a structure for pixel placement on the display;
distance to reflector = $1/2 \text{ speed of sound} \times \text{time of flight}$

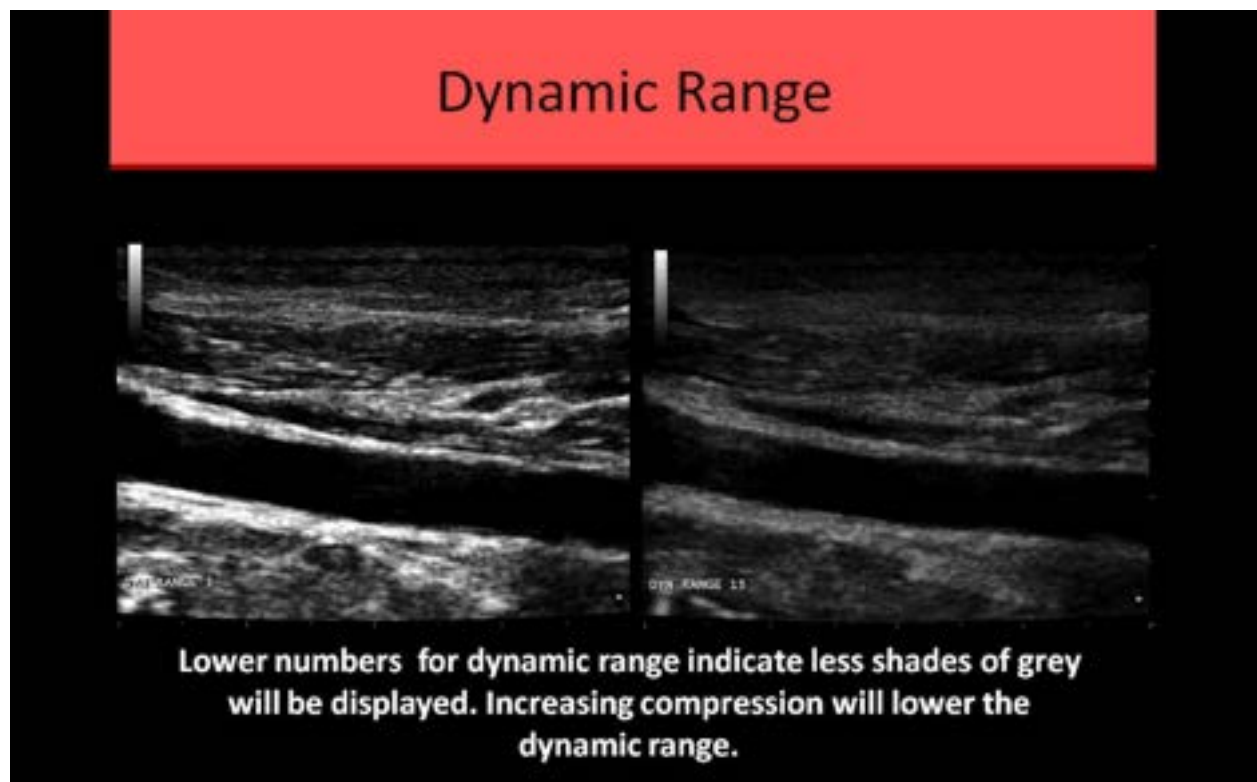
2D Operator Controls:

1. Power Output - adjusts the electrical voltage applied to the PE elements in the transducer; changes the energy of the beam; use as a last resort to increase penetration and reflected sound wave amplitude
2. Overall Gain - adjusts the displayed amplitude of the reflected sound waves on the entire image
3. Time Gain Compensation - allows varied levels of amplification of the reflected sound waves across the image; near field gain is usually set lower to suppress reflections and the far field gain is usually set higher to enhance reflections; compensates for beam attenuation
4. Depth - the deeper the structure being evaluated, the lower the PRF and frame rate (PRF = pulse repetition frequency and refers to the number of ultrasound pulses emitted per second); axial resolution is constant at a set depth but the lateral resolution varies with shape of the beam
5. Dynamic Range/Compression - adjusts the range of grays displayed on the image; changes the contrast level of the image; increased compression/larger dynamic range = decreased image contrast (more black and white image)
6. Focal Zone - used to improve lateral resolution; increased focal zone number causes decreased temporal resolution; activating beam focusing will increase the intensity of the beam at the focal zone
7. Gray Map - changes the shades of gray assigned to the reflected echo amplitudes; used to adjust spatial and contrast resolution; change map before image is frozen = pre-processing; change map on stored image = post-processing

2D Image Optimization Hints:

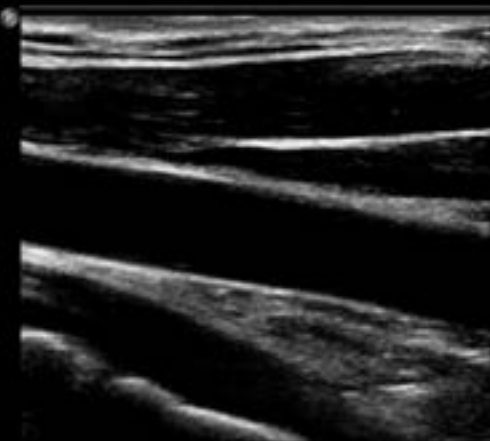
- Adjust probe position to obtain images of the vessel with the beam perpendicular to the vessel walls; 2D imaging improves with perpendicular incidence

- Adjust compression to vary the "black & white" appearance of the image to improve visualization of vessel walls and to see soft atheroma formation; increasing compression = more shades of gray
- If the reverberation artifact is prevalent in superficial vessels, adjust probe position to create more distance between the probe face and the vessel
- Adjust the gray scale map (pre- and post-processing) to improve lumen characteristics and identify plaque/thrombus
- Adjust gain settings, monitor brightness and contrast



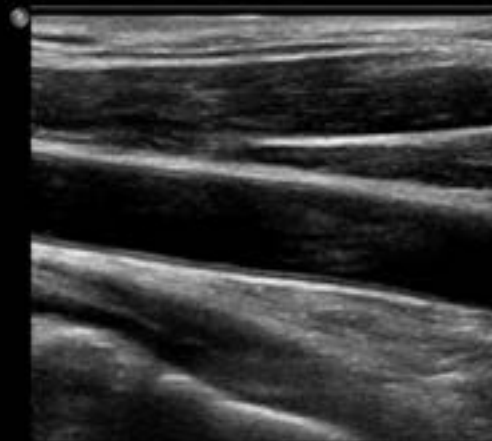
Dynamic Range

Note the image is more black and white with the reduction in the number of shades of grey displayed on the image



Low or Narrow Dynamic Range

Note the increase in the number of shades of grey displayed on the image



High or Wide Dynamic Range

Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Focal Zone Placement



Harmonic Imaging

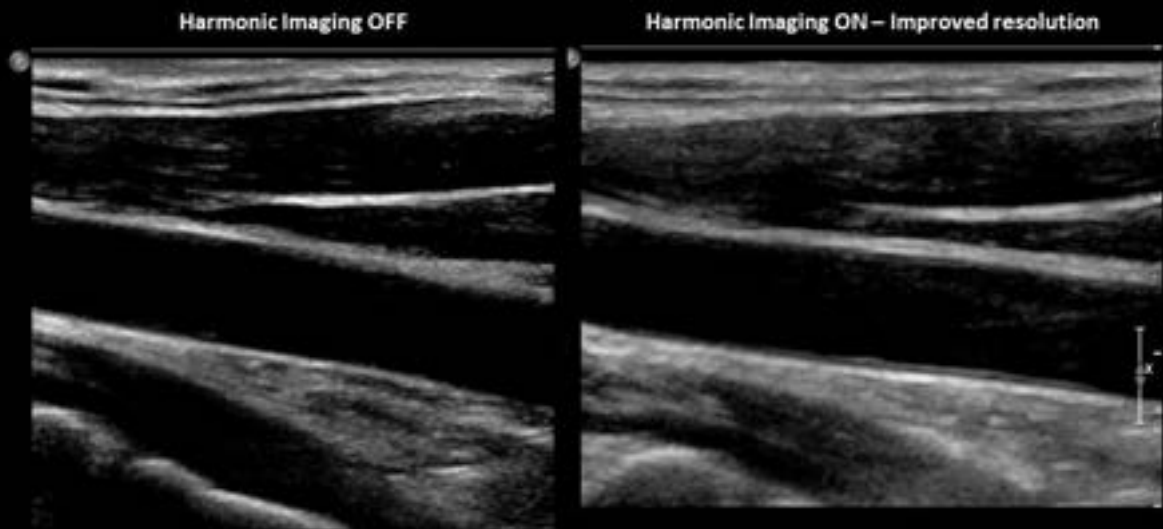


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Image Depth

Optimal depth settings: Reduce the image depth to center the vessel on the image

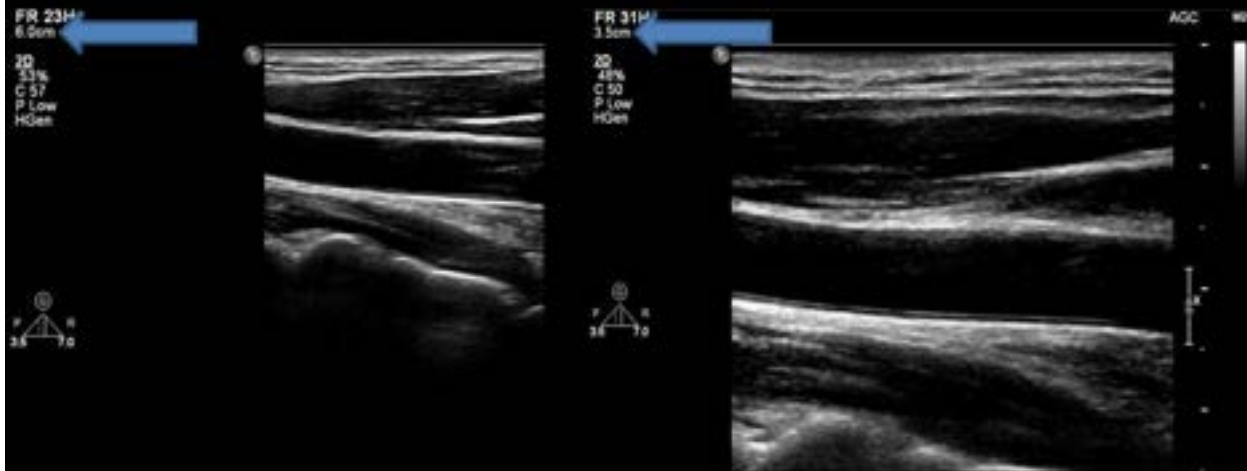
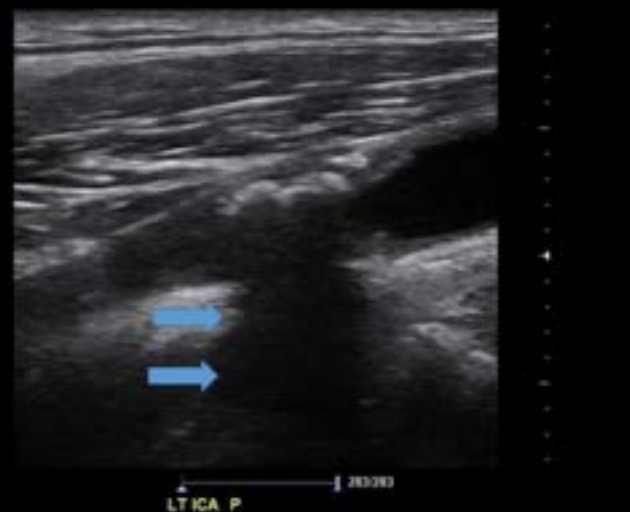


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

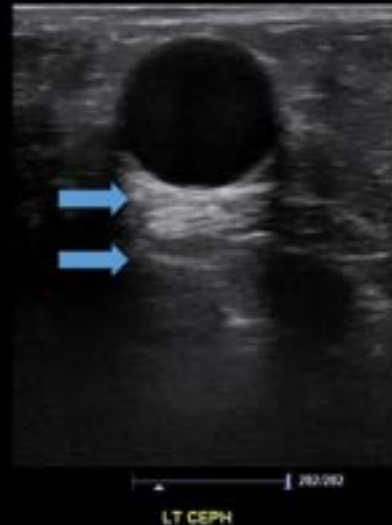
Posterior Shadowing

Occurs posterior to highly attenuating structures. Plaque in the arterial wall is highly reflective. The posterior shadowing can limit color and PW Doppler evaluation of the arterial segment posterior to the plaque.



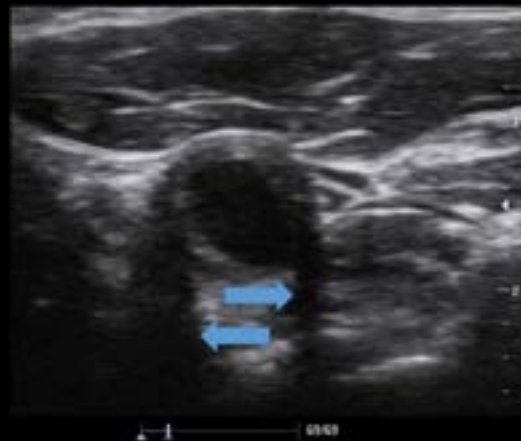
Posterior Enhancement

Occurs posterior to poorly attenuating structures. Fluid filled structures are non-reflective. The structures located deep to the fluid filled structure will demonstrate increased echogenicity.



Edge Shadowing/Refractive Shadowing

When the ultrasound beam strikes a rounded structure, sound waves are scattered and refracted, leading to energy loss and the formation of a shadow. It causes linear posterior shadowing on each side of the structure.



Mirror Image Artifact

Occurs when there is a highly reflective surface in the path of the primary beam. Places a second copy of the structure deeper on the image.



Microbubble Contrast Imaging:

- Contrast-enhanced ultrasound (CEUS) for vascular imaging indications has increased over the last few years
- Gas-filled microbubbles are injected into the bloodstream and increase the reflectivity of blood on ultrasound
- Used to evaluate the carotid arteries, the abdominal aorta, renal arteries and the kidneys
- Improves the delineation of the vessel wall
- Aids in detection of endoleaks following endovascular treatment of abdominal aortic aneurysms
- Used to assess the tissue perfusion in native and transplanted kidneys

Contrast Evaluation

IV Contrast can be used to improve the visualization of the PW Doppler signal. The contrast material makes the blood more reflective, therefore the PW tracing is brighter.

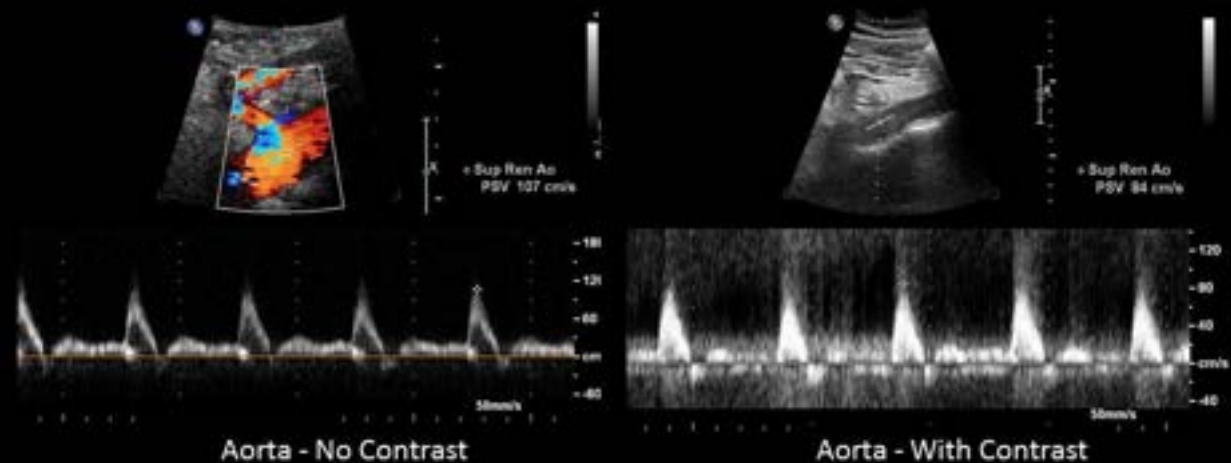


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

DOPPLER FLOW EVALUATION TECHNIQUES

Doppler Ultrasound Physics

- Johann Christian Doppler first identified the physical principles involved in Doppler evaluation
- Doppler shift: the mathematical difference between the transmitted and received frequencies, occurs when structures are moving toward or away from the "listener"
- Train whistle blows as it moves toward you at the station, as it gets closer to the station the approaching train whistle has a higher pitch (frequency); the departing train whistle has a different, lower pitch (frequency)
- Tissues and other non-mobile structures do not exhibit a Doppler shift
- Reflected sound has the same frequency as the transmitted sound if the blood is stationary
- If the blood is moving away from the transducer, the reflected sound has a lower frequency
- If the blood is moving toward the transducer, the reflected sound has a higher frequency
- Fast fourier transformer processes Doppler shifts into Doppler tracing displayed on screen

- Doppler shift frequencies are displayed on a tracing as a function of time; x axis = time, y axis = Doppler shift

The ultrasound system calculates VELOCITY, not SPEED!

Speed – measure of magnitude, any distance divided by unit time, 30 m/s

Velocity – measure of magnitude & direction, positive or negative direction related to starting point, + 30m/s or - 30m/s

Doppler Calculations:

- The Doppler shift is directly proportional to the transducer frequency and blood flow velocity
- As blood flow velocity increases, the Doppler shift increases
- As the transducer frequency increases, the Doppler shift increases BUT the calculated velocity is unchanged because the ultrasound system is programmed to "disregard" the change in transducer frequency
- Increasing the cursor angle from 45 to 60 degrees = decreased cosine value = increased calculated velocity
- Decreasing the cursor angle from 60 to 45 degrees = increased cosine value = decreased Doppler velocity
- Overestimating the angle between the beam and the blood flow = overestimated flow velocity
- Underestimating the angle between the beam and the blood flow = underestimated flow velocity
- The greater the Doppler angle, the greater the amount of error in the calculated velocity

Doppler Equation

Used to calculate Doppler frequency shift and Doppler velocities that are displayed during a PW/CW exam

$$D_f = \frac{2 T_f V \cos \phi}{c}$$

D_f = Doppler shift frequency

T_f = Transducer frequency

V = Velocity of the moving reflectors

ϕ = Doppler Angle

c = Speed of Sound in Tissue

Doppler Equation

Solved for velocity

$$V = \frac{D_f c}{2 T_f \cos \phi}$$

The Doppler shift frequency and Speed of Sound in Tissue are directly related to the calculated velocity

D_f = Doppler shift frequency

T_f = Transducer frequency

V = Velocity

ϕ = Doppler Angle

c = Speed of Sound in Tissue

Doppler Equation

Doppler shift = 2000Hz
 Transducer = 3MHz
 Velocity = ??? m/s
 $\phi = 60^\circ$
 $C = 1540 \text{ m/s}$

$$V = \frac{2000\text{Hz} \times 1540\text{m/s}}{2 (3,000,000\text{Hz}) \cos 60^\circ}$$

$$V = \frac{3,080,000}{6,000,000\text{Hz} (0.5)}$$

$$V = 1.0\text{m/s}$$

What if the angle decreases to 45 degrees?

$$V = \frac{2000\text{Hz} \times 1540\text{m/s}}{2 (3,000,000\text{Hz}) \cos 45^\circ}$$

$$V = \frac{3,080,000}{6,000,000\text{Hz} (0.7)}$$

$$V = 0.7\text{m/s}$$

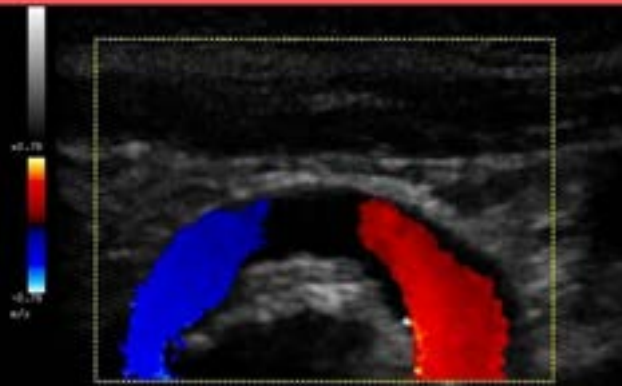
What if the angle increases to 75 degrees?

$$V = \frac{2000\text{Hz} \times 1540\text{m/s}}{2 (3,000,000\text{Hz}) \cos 75^\circ}$$

$$V = \frac{3,080,000}{6,000,000\text{Hz} (0.3)}$$

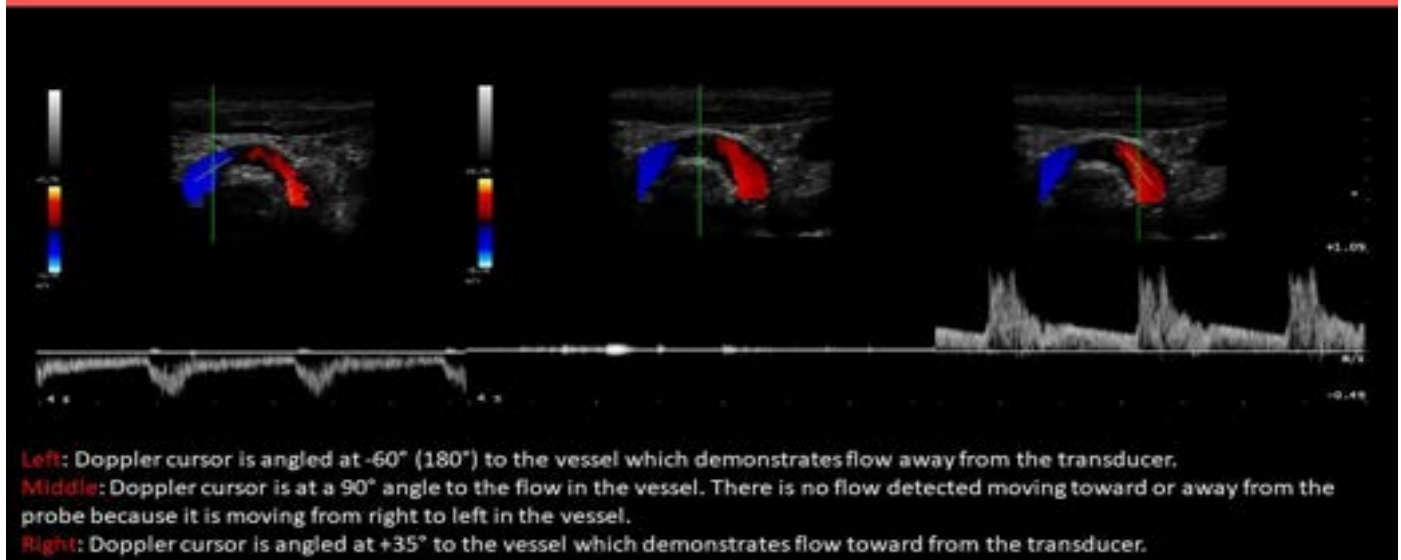
$$V = 1.7\text{m/s}$$

Doppler Incident Angle

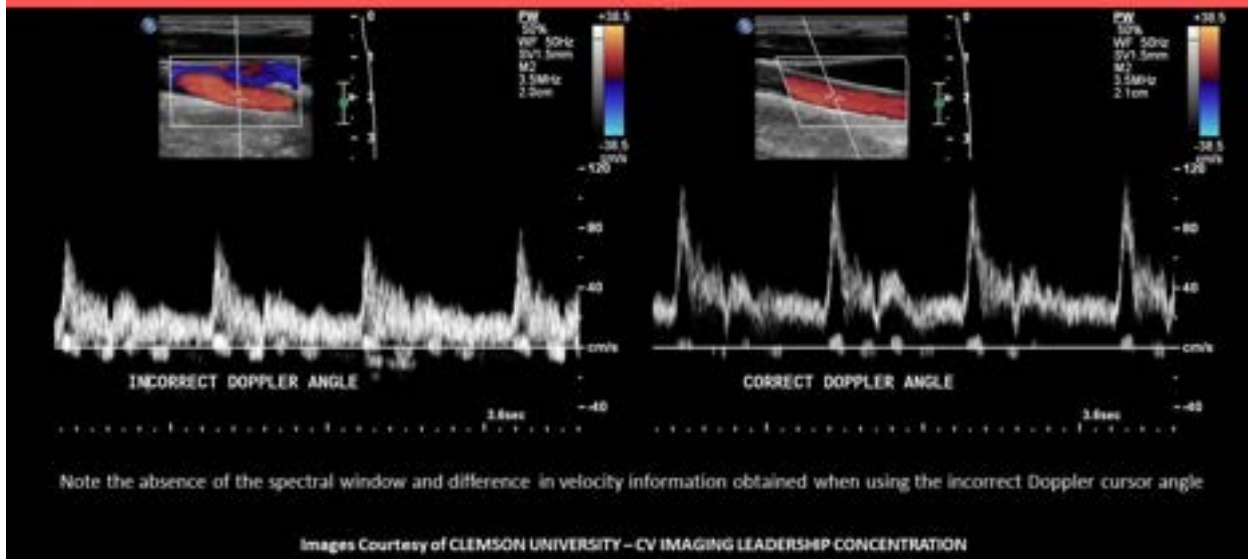


Blood flow toward the probe will demonstrate a positive change in frequency which will be displayed on the top side of the Doppler baseline. Blood flow away from the probe will demonstrate a negative change in frequency which will be displayed on the bottom side of the Doppler baseline.

Doppler Incident Angle



Doppler Angle



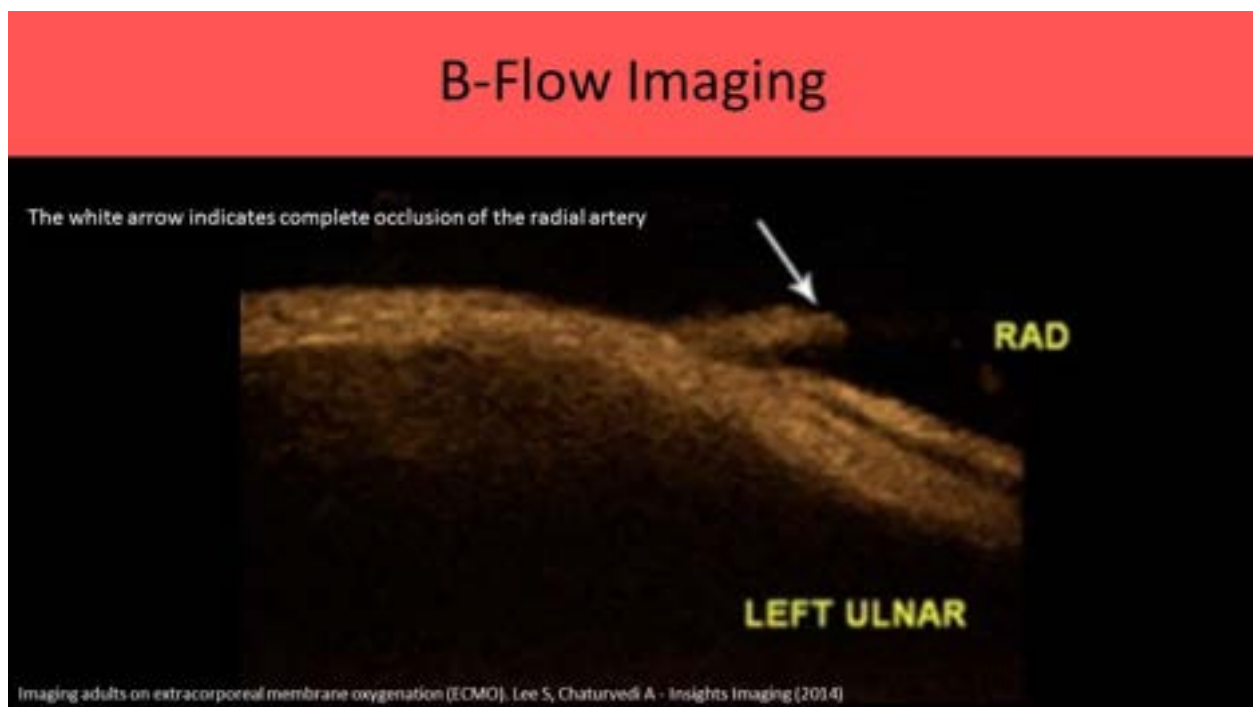
Blood Flow Ultrasound Evaluation Techniques:

- B Mode Flow
- Color Doppler
- Power Doppler

- Spectral Doppler; Pulsed Wave Doppler and Continuous Wave Doppler
- Analog Doppler
- Tissue Doppler

B Mode Flow Imaging:

- Recent technology
- Maintains the available reflection details in the individual echoes received from the RBCs
- Main benefit is that this technique provides more hemodynamic information and increased visual differentiation between true blood flow and wall motion artifacts when compared to color Doppler



Color Doppler:

- Detecting Doppler shifts from hundreds of locations
- Colors are encoded to display varying hues
- Superimposed over 2D image
- Threshold control determines which shades of grey are covered by color pixels; lower settings reduce the color display over the 2D image
- Indicates the direction and the average velocity of blood flow
- Thousands of samples to process
- Autocorrelation is used to manipulate data

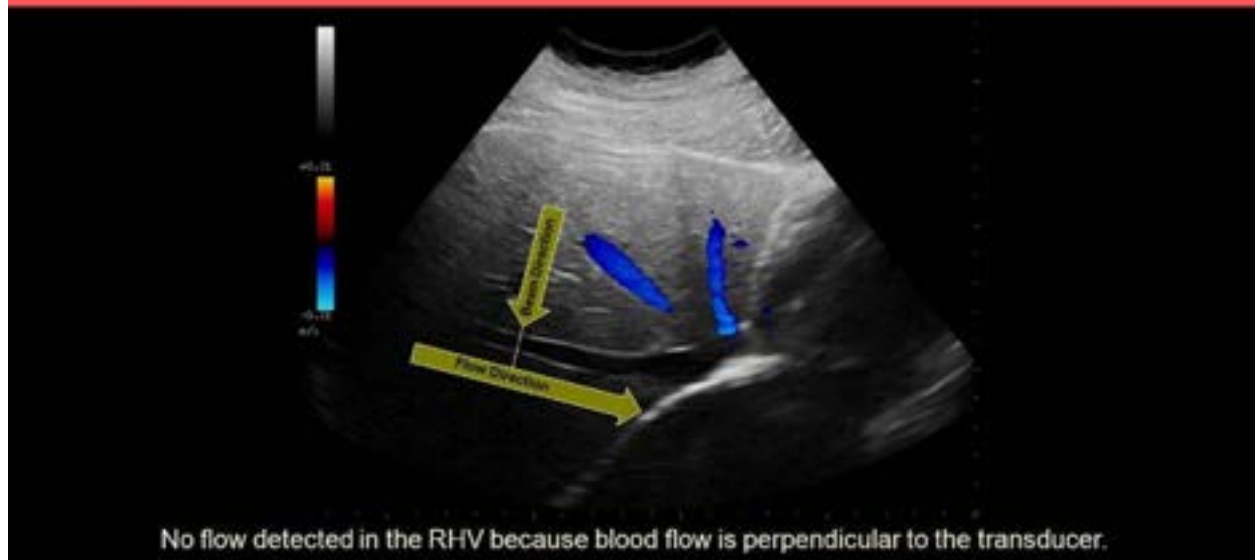
Packet Size:

- Helps to find a balance between the accuracy of the color displayed and the temporal resolution of the image
- 2D = 1 pulse per scan line
- Color Doppler uses from 3-32 pulses per scan line = larger packet size
- Larger packet sizes cause decreased frame rates but produce more accurate flow velocity information
- Larger packet sizes are better able to identify low velocity flow more accurately
- Parallel processing can be used to improve frame rate and temporal resolution

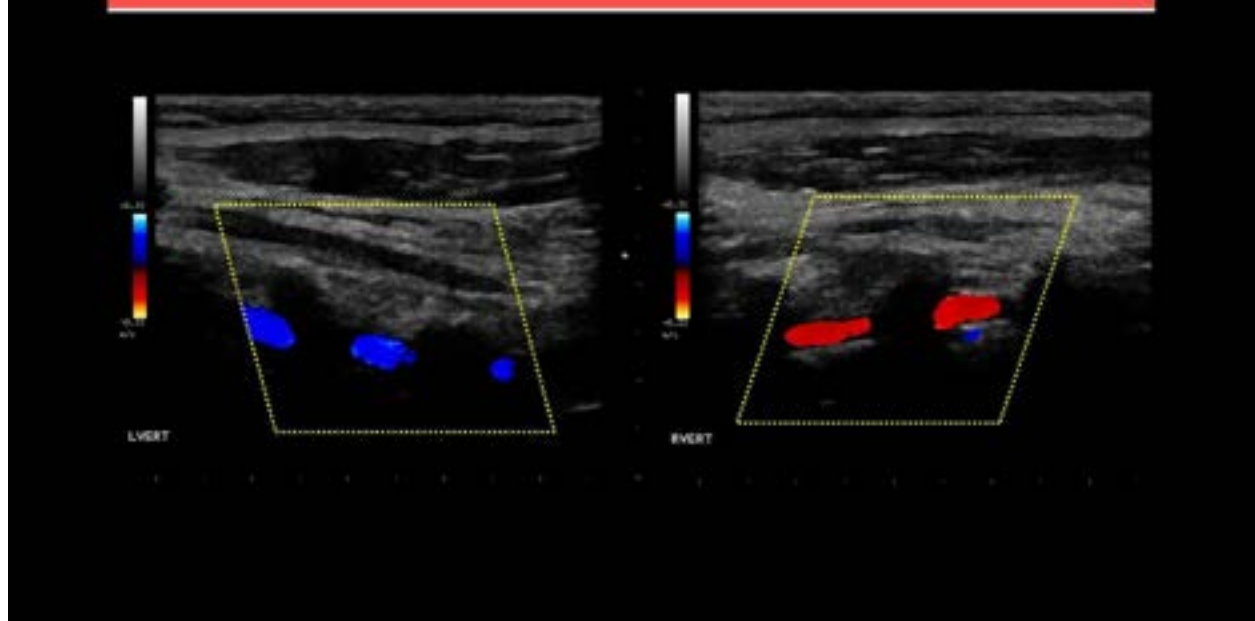
Color Doppler Operator Controls:

1. Color Gain - adjusts the displayed amplitude of the reflected sound waves from the flowing blood; over-gaining can lead to noise artifact
2. Color Sample Angle - the optimum angle between the sample box and the blood flow is 0 degrees or parallel to the flow; the steering function is used to make the sample box more parallel to the flow
3. Sample Volume Width - adjusts the amount of flow evaluated in a specific location; smaller sample sizes produce more accurate color Doppler information with faster frame rates and improved temporal resolution
4. Sample Volume Depth - used to evaluate flow in specific locations; more superficial structures provide better color display with faster frame rates and improved temporal resolution
5. Color Packet Size - refers to the number of samples taken during evaluation; larger packets have higher line density but lower frame rates and decreased temporal resolution
6. Wall Filter - eliminates low frequency Doppler shifts from display; used to reduce color ghosting
7. Baseline - used to demonstrate positive and negative shifts appropriately without aliasing; can be used to eliminate aliasing
8. Velocity Scale or PRF - increasing the scale allows the display of higher velocities without aliasing
9. Color Map - varied color assignments to Doppler shifts; pre- or post-processing function
10. Variance - degree of variability of mean velocity at different depths in the sample; typically displayed as a GREEN scale superimposed on the red/blue display; used to highlight flow turbulence
11. Color Priority - determines if a pixel is displayed as shade of black and white or as a color; lower threshold settings display less color (lower color priority); higher threshold settings display more color (higher color priority)

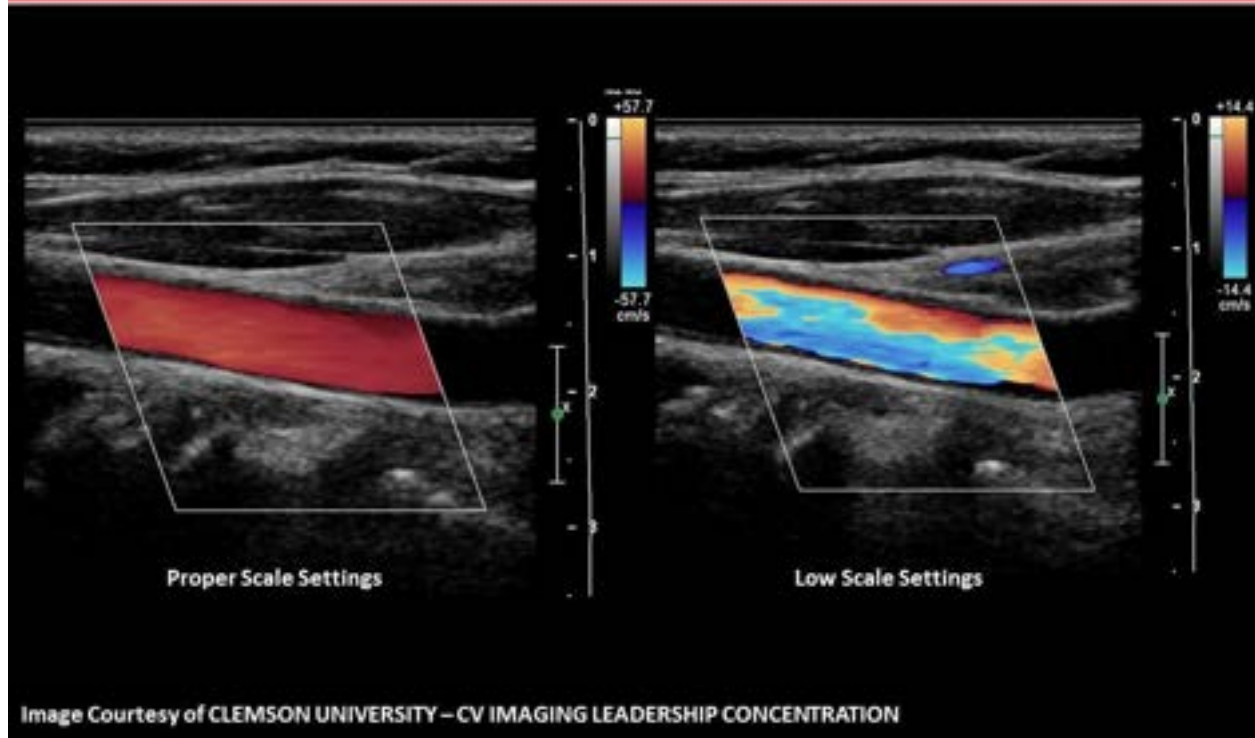
Color Doppler Evaluation



Color Doppler Display



Color Velocity Scale (PRF)

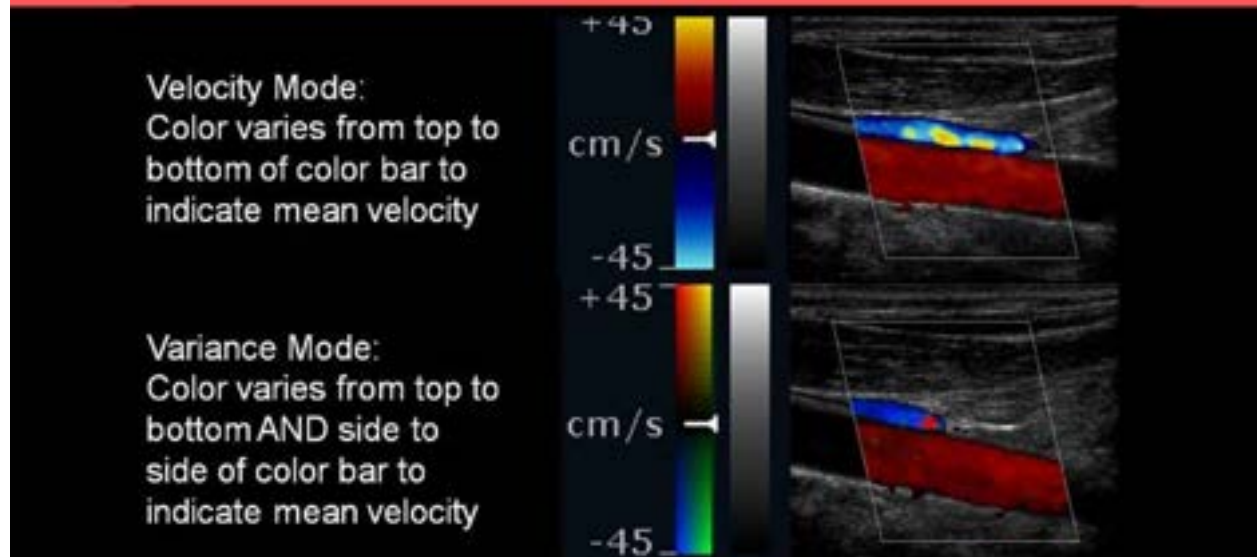


Velocity vs Variance Mode:

Velocity – colors on top of the stripe indicate blood flow toward the probe and colors on the bottom of the stripe indicate blood flow away from the probe; color change with velocity variation demonstrated as varied colors up and down the stripe; aliasing causes flow to be displayed incorrectly; high velocity antegrade flow that exceeds the Nyquist limit will be incorrectly displayed as the color on the bottom of the color stripe and shades of white and yellow (depends on the map); color image looks the same as variance map with laminar flow; display of turbulent flow demonstrates a different color image than with the variance map

Variance – used to determine turbulence and represent velocity; Turbulent flow displayed as GREEN color; colors on top of the stripe indicate blood flow toward the probe and colors on the bottom of the stripe indicate blood flow away from the probe AND colors on the left side of the stripe represent laminar flow and colors on the right side of the stripe indicate turbulent flow; color change with velocity variation always seen up and down the stripe AND from left to right; color image looks the same as velocity map with laminar flow; display of turbulent flow demonstrates a different color image than with the velocity map

Velocity vs Variance Color Map



Autocorrelation:

- Used to process Doppler shift information for color Doppler
- Automatically correlates data from multiple sampling sites
- At least three pulses per scan line necessary to generate a single frame of color Doppler information
- Displays average Doppler shift frequencies as varied hues
- Velocity measurement accuracy is limited because angle correction is not utilized

Advantages of Color Doppler:

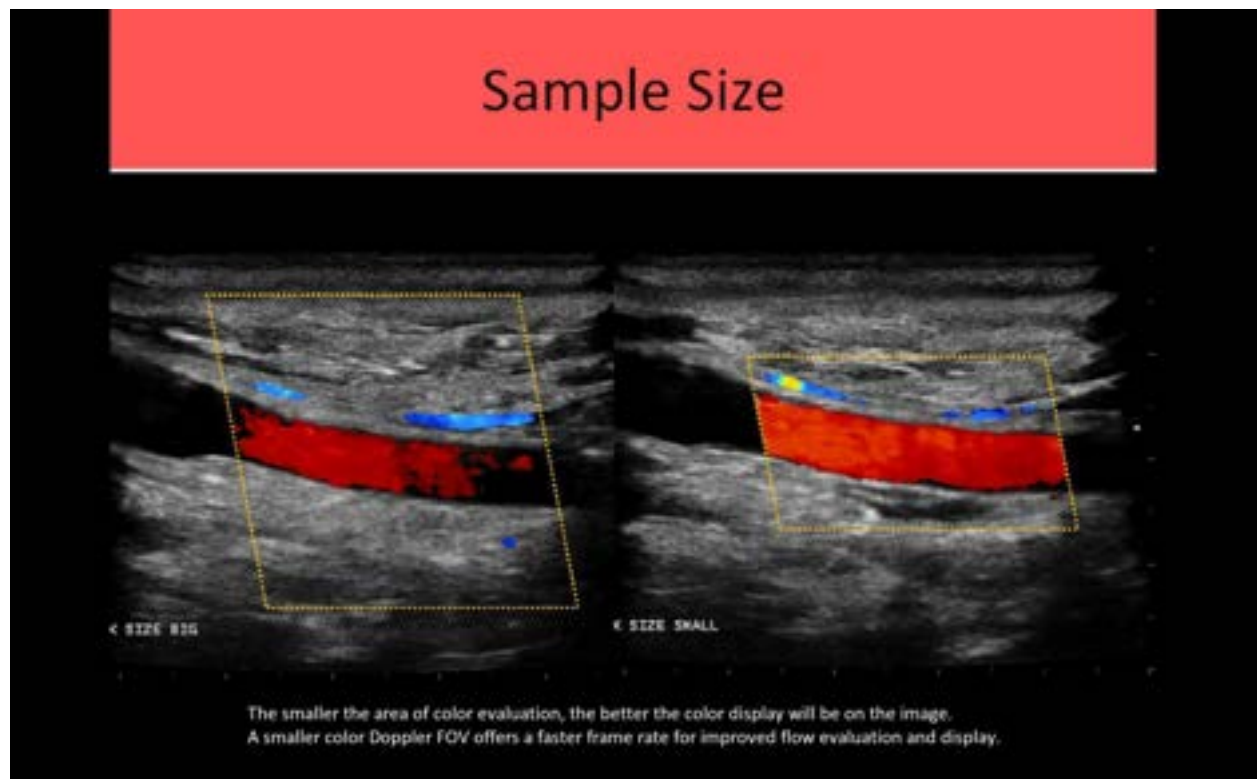
- “Triplex” capabilities
- Permits instantaneous identification of areas of abnormal blood flow
- Easier placement of the single line cursor that is used for spectral Doppler

Potential Errors:

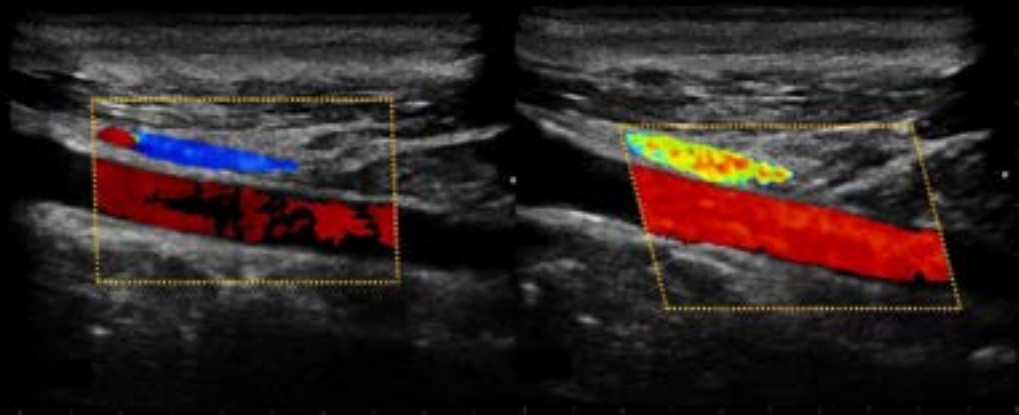
- Box position
- Box size
- Gain too low/high
- Filter too high/low
- Scale set too low/high

Color Doppler Hints:

- Adjust scale, gain, and baseline
- Adjust color threshold
- Decrease size of sample for improved flow visualization
- Create angle of vessel across the screen, NO flow will be detected when it moves at 90° to the US beam



Electronic Steering



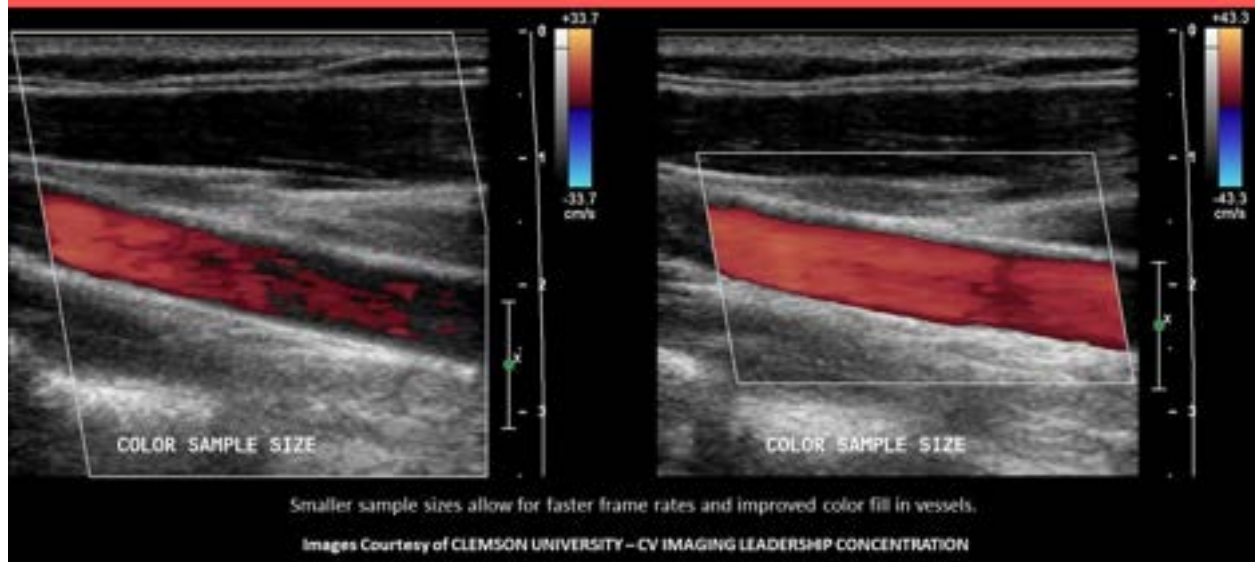
The transducer will detect blood from that is moving toward or away from the probe. Steering the color box will make the flow in the vessel less perpendicular to the US beam, therefore improving the detection and display of the flow.

Color Steering



Images Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Color Doppler Sample Size



Reynold's Number:

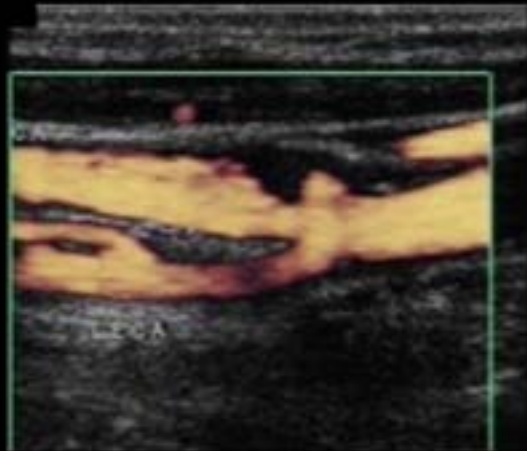
- # expressing the balance of inertial and viscous forces acting on the blood flowing
- Used to discuss color Doppler turbulence
- Average flow speed X vessel diameter X density
- < 1500 for laminar flow
- > 2000 abnormal

Power Doppler Imaging:

- AKA Color Angio or Energy Mode
- Displays an approximation of lower echo amplitudes
- More sensitive to lower Doppler shifts than normal color-flow Doppler
- Demonstrates very low velocity blood flow with no aliasing
- Disadvantages: Low frame rates and more susceptible to tissue-motion artifacts

Ulceration

Power Doppler is much more sensitive to low velocity flow movement than color Doppler but the direction of flow is not displayed with Power Doppler.



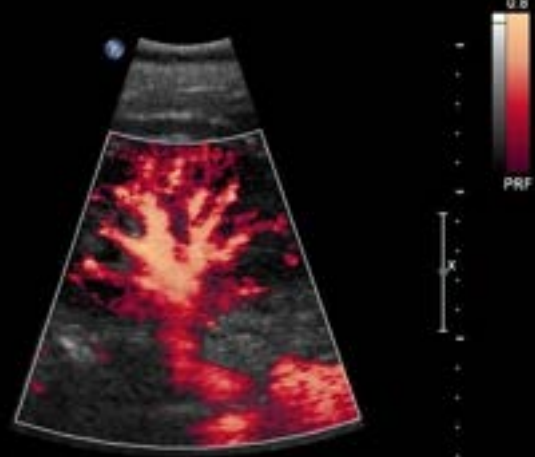
www.qualityvascular.com

Color Doppler vs. Power Doppler



Right KIDNEY

+15.4
-15.4
cm/s



Right KIDNEY RRA

0.8
PRF

Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Analog Doppler:

- Graphical recording of pulsatile Doppler in a non-spectral (in the form of compressed, thin-line analog tracings) or a strip chart recording
- Uses pencil style transducer
- Uses zero crossing detector - counts each time the reflected signal crosses through the baseline over time
- Does NOT display a range of frequency shifts, displays average shift as single line; no demonstration of turbulence or spectral broadening
- Uses strip chart recorder to display waveform
- Underestimates high velocities and overestimates low velocities; cannot detect flow velocities less than 6cm/s

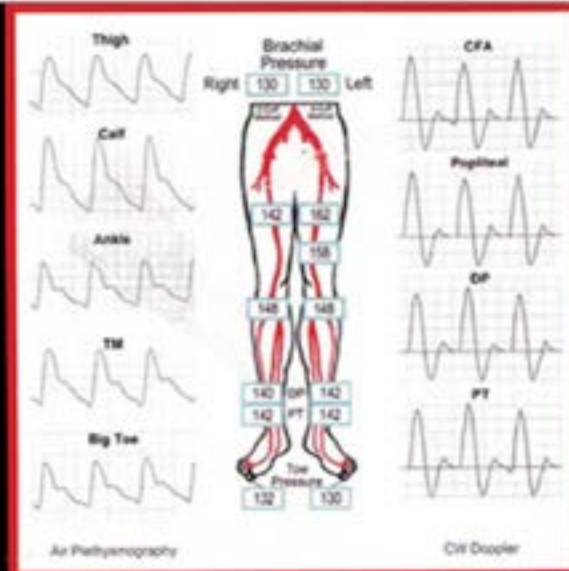
Limitations of Lower Extremity CW Doppler Exam:

- Cannot differentiate stenosis from occlusion
- Cannot precisely localize an area of obstruction
- Cannot discriminate between iliac and common femoral disease
- Limited value in the presence of multi-level disease and collateral formation
- Decreased cardiac output can cause reduced arterial velocity
- Improper angles of insonation can degrade recorded waveforms

Analog Doppler Display

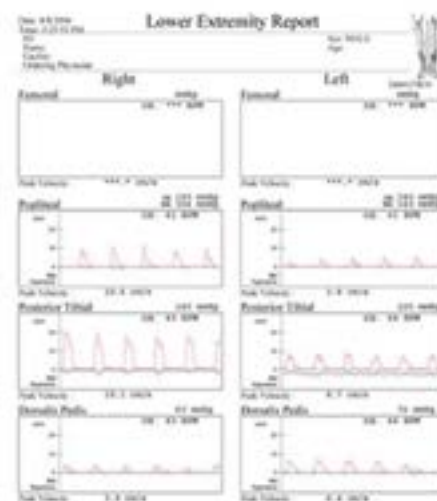
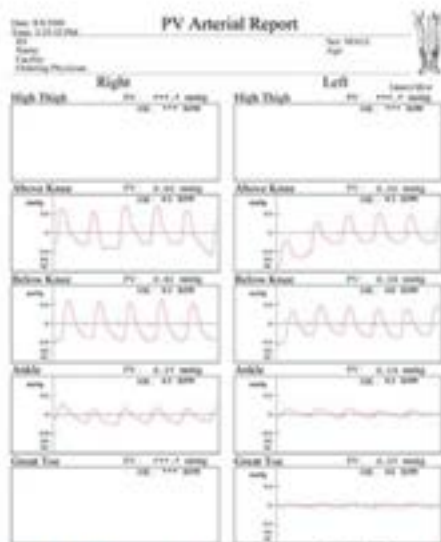
PVR

Doppler



Courtesy of Dr. Paul Kasselman
www.kovon.com

Abnormal Lower Extremity Exam



Courtesy of Dr. Paul Kasselman
www.kovon.com

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Spectral Doppler:

- Doppler shifts are detected along a single line and used to produce a graphic representation of the blood velocity
- Spectral Doppler systems contain spectrum analyzers to produce the graphic displays
- Multiple frequencies are displayed over time
- Displayed on the monitor NOT strip chart recorder; big advantage over analog Doppler

Spectrum Analyzer:

- Contained within the Doppler system
- Produces the graphic displays
- Blood velocity is calculated, NOT measured
- Calculates peak and mean Doppler shift frequencies
- Uses the mathematical process Fast Fourier Transform (FFT) to process data and provide it for graphic display
- Displays positive Doppler shifts above the zero baseline
- Displays negative Doppler shifts below the zero baseline
- Magnitude of the shift directly related to the amplitude of the tracing

Quadrature Phase Detection:

- Methods used to determine the direction of blood flow for bidirectional Doppler systems
- Requires two speakers to allow one for flow toward and one for flow away

PW Doppler:

- AKA gated Doppler; 1-1.5mm gate size recommended; small sample sizes demonstrate a more specific location of the velocity detected
- Uses controlled time intervals between transmitted and received sounds to collect information using a single PZT element
- Depth selected by technologist
- Gate size or sample volume used to select location and amount of blood sampled (range resolution or free of range ambiguity)
- Aliasing is the biggest disadvantage, limiting velocities that can be evaluated
- Nyquist Limit indicates the highest velocity that can be displayed without aliasing
- Aliasing can be corrected by using a lower frequency transducer, increasing PRF or scale and adjusting baseline
- Proper setting of the angle correction cursor = accurate velocity calculations
- Increased transmitted frequency will cause an increase in the Doppler shift detected; Incident frequency is directly related to the Doppler shift
- Doppler angles increase, calculated velocity decreases = Stenosis can be underestimated

- Doppler angles decrease, calculated velocity increases = Stenosis can be overestimated
- As the angle approaches 90 degrees, the velocity will decrease until it reaches 0
- Using different angles for different exams or for different vessels on the same exam can cause a variation in the velocities and false negative or false positive results
- Number of reflected frequency shifts correlates with the number of different velocities of the blood cells; laminar flow has blood cells that move at similar speeds which produces similar reflected frequency shifts(spectral window); turbulent flow will produce a wide variety of frequency shifts due to the significant variation in the speed of the different blood cells (no spectral window)

Doppler Operator Controls:

1. Doppler Gain - adjusts the displayed amplitude of the reflected sound waves from the flowing blood
2. Cursor Angle - any exam that requires measurement of peak flow velocities requires angle correction; the optimum angle between the cursor and the blood flow is 0-degrees or parallel to the flow; standard setting for most vascular evaluations is a 60-degree cursor angle; changing the cursor angle changes the velocity calculated from the frequency shift detected
3. Sample Volume Size - adjusts the amount of flow evaluated in a specific location; smaller sample sizes produce more accurate Doppler information with improved evaluation of laminar flow and appearance of the spectral window; average size of 1-1.5mm for arterial evaluations; larger sample sizes helpful for evaluating venous flow and performing volume flow calculations for dialysis access grafts
4. Sample Volume Depth - used to evaluate flow in specific locations; necessary for evaluating arterial stenosis; deeper vessels require lower frequency transducers to obtain samples at increased depths
5. Wall Filter - AKA high pass filter; eliminates low frequency Doppler shifts and high amplitude signals from display; lower settings used for venous evaluation compared to arterial evaluations
6. Baseline - used to demonstrate positive and negative shifts appropriately without aliasing
7. Velocity Scale or PRF - increasing the scale allows the display of higher velocities without aliasing; lower settings used for venous evaluation compared to arterial evaluations
8. Sweep Speed - speed of the spectral Doppler display; 50mm/s is the standard setting; High sweep speed (100mm/s) is recommended for precise measurements of acceleration time; Slower sweep speeds (25mm/s) can be useful for evaluating venous reflux time

Potential Errors:

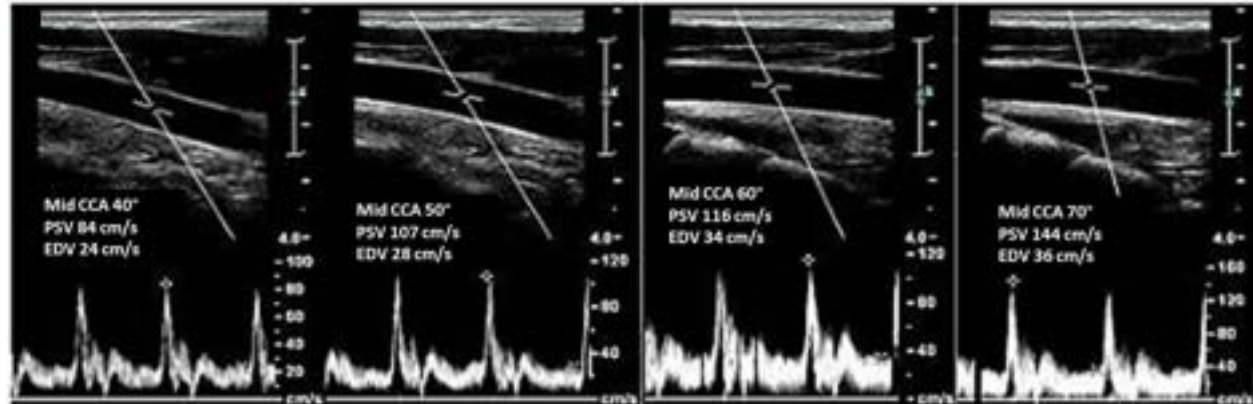
- Improper sample location #1 reason missing a stenosis
- Improper probe position
- Improper angle of incidence

- Large sample size
- Excessive probe pressure
- Improper filter settings
- Insufficient gel
- Aliasing due to structure depth

PW Doppler Hints:

- Always sample the center of the vessel EXCEPT when atheroma/stenosis is present
- Take multiple samples in the area of the stenosis in case an eccentric jet is present
- Always use $<60^\circ$ angle on cursor with any exam where velocity measurements are needed
- Adjust scale, baseline, and gain to properly demonstrate waveforms
- Smaller sample sizes give more accurate waveforms at specific depths
- Cursor angle must be as parallel to flow as possible, can be less than 60° in order to stay parallel

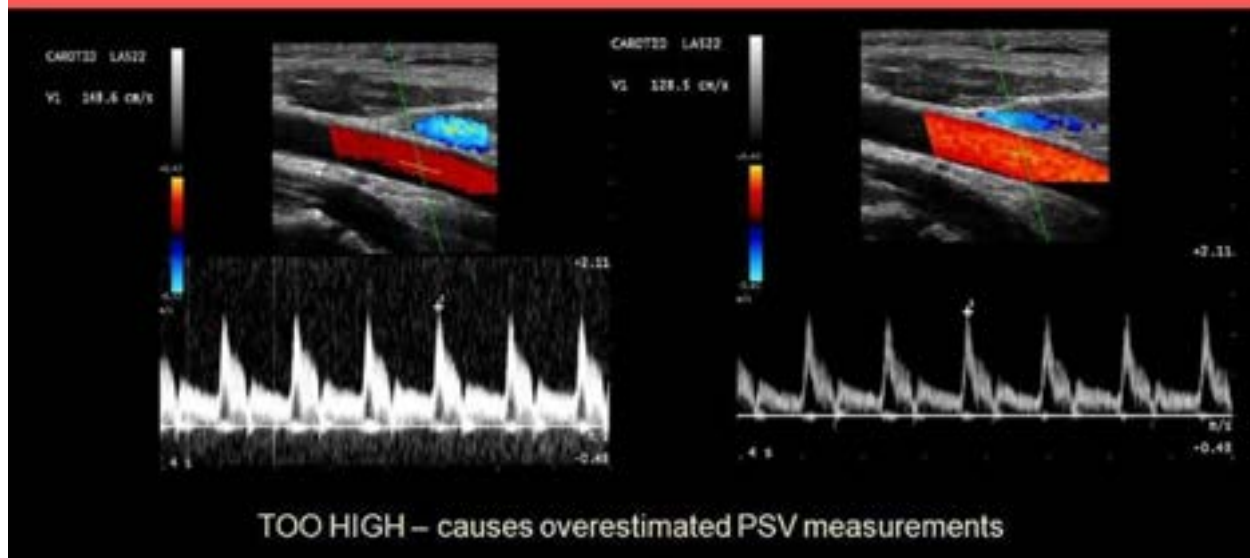
Effect of Cursor Angle on Velocity



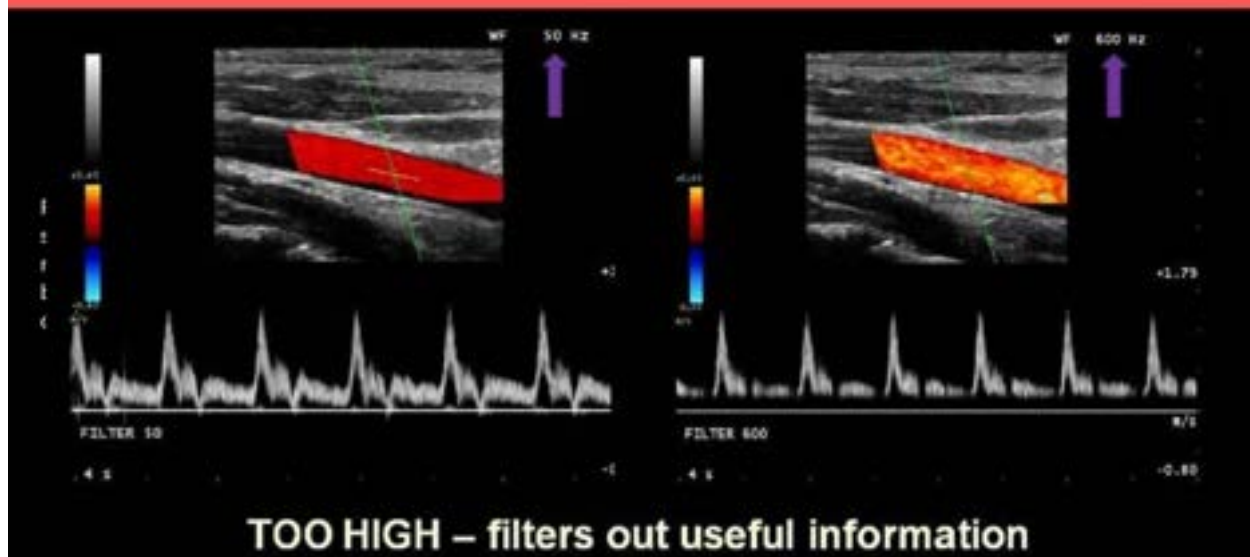
It is important to use a consistent cursor angle setting when performing serial exams. Note the difference in velocities obtained from the same vessel using different Doppler angles. 60° is recommended for carotid exams.

Adapted from: Standardized ultrasound evaluation of carotid stenosis for clinical trials: University of Washington Ultrasound Reading Center. Beach KW, Bergelin RO, Leotta DF, Primorich JF, Sevareid PM, Stutzman ET, Zierler RE - Cardiovasc Ultrasound (2010)

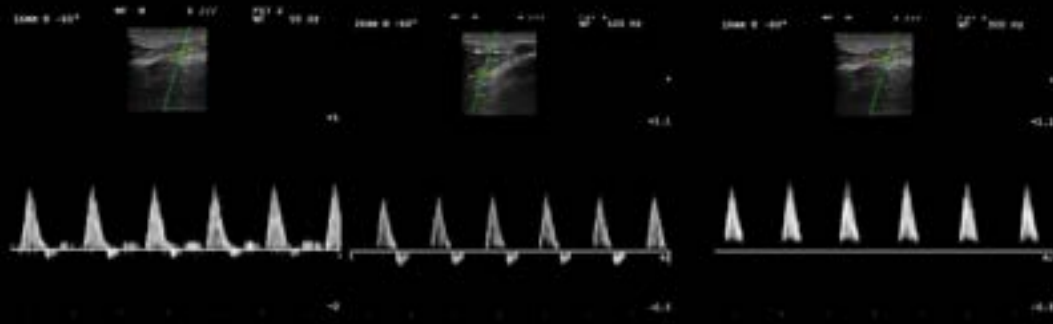
Effect of Cursor Angle on Velocity



Wall Filter Settings



Incorrect Filter Settings



Wall filter settings can negatively affect the evaluation of low velocity flow and arterial pulsatility. All three Doppler tracings are from the same vessel but with increasing filter settings. Note the loss of the diastolic flow components on the display as the filter is increased.

Doppler Cursor Angle

No Cursor Angle: non-parallel insonation provides a poor quality waveform

Correct Cursor Angle: parallel insonation provides a clear waveform with accurate velocity calculations

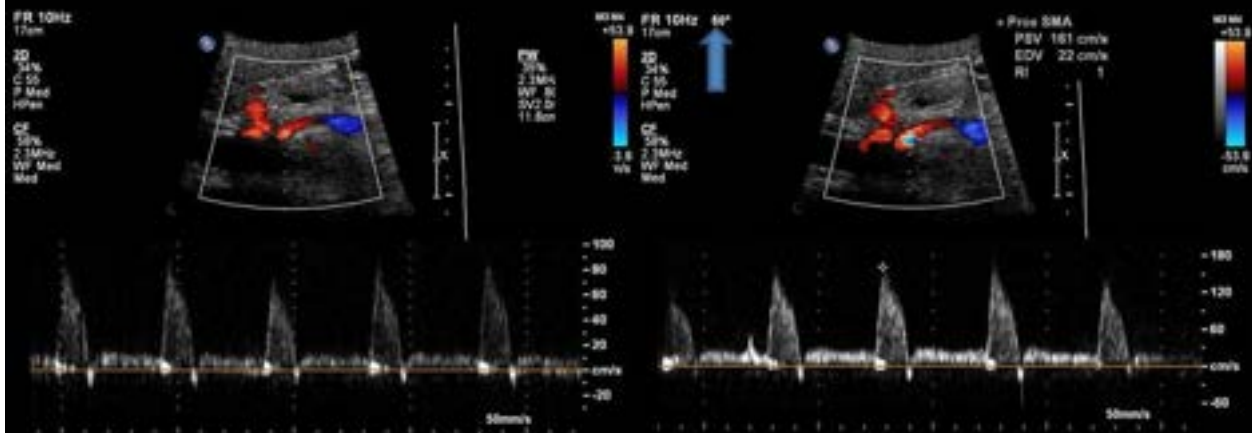
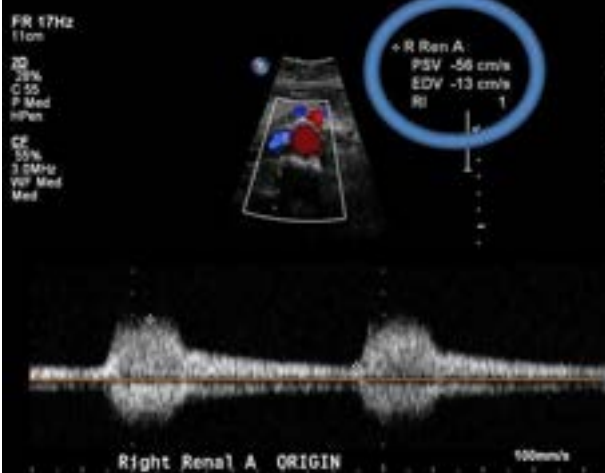


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Doppler Cursor Angle

No Cursor Angle: non-parallel insonation provides a poor quality waveform



Correct Cursor Angle: parallel insonation provides a clear waveform with accurate velocity calculations

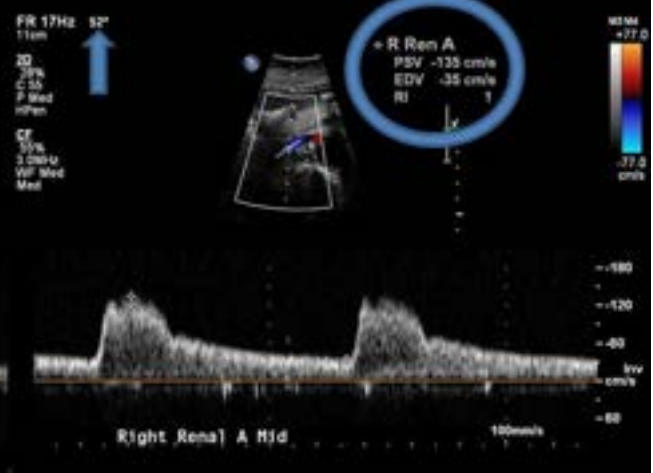
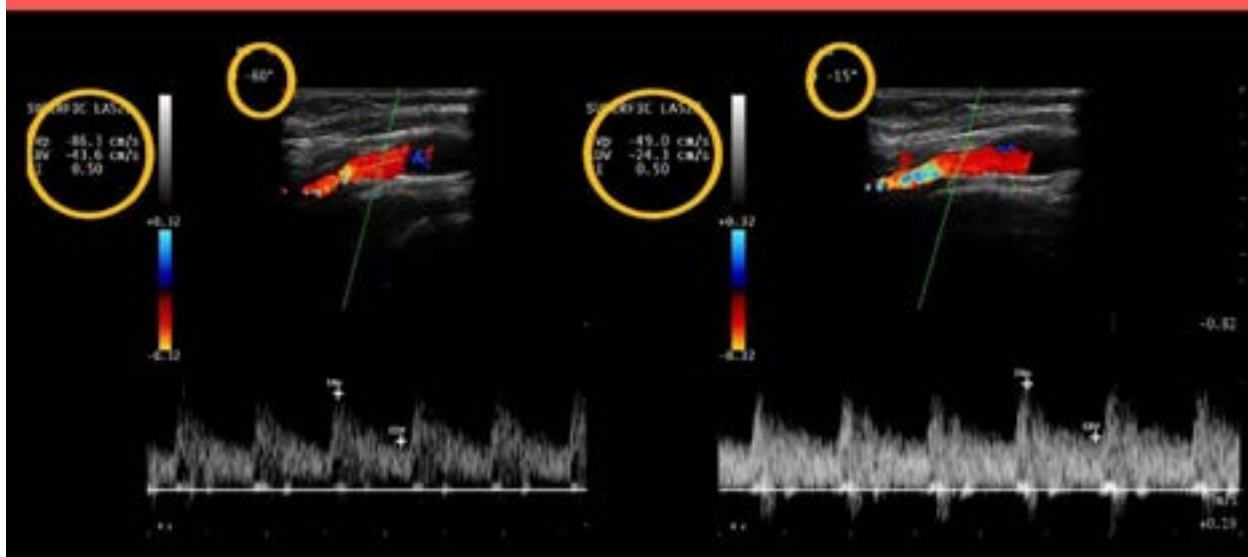
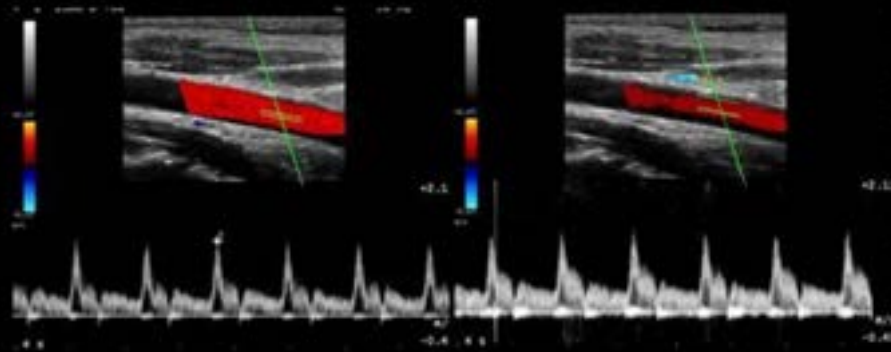


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Proper vs Improper Angle Correction



Sample Size



TOO LARGE – causes spectral broadening

CW Doppler:

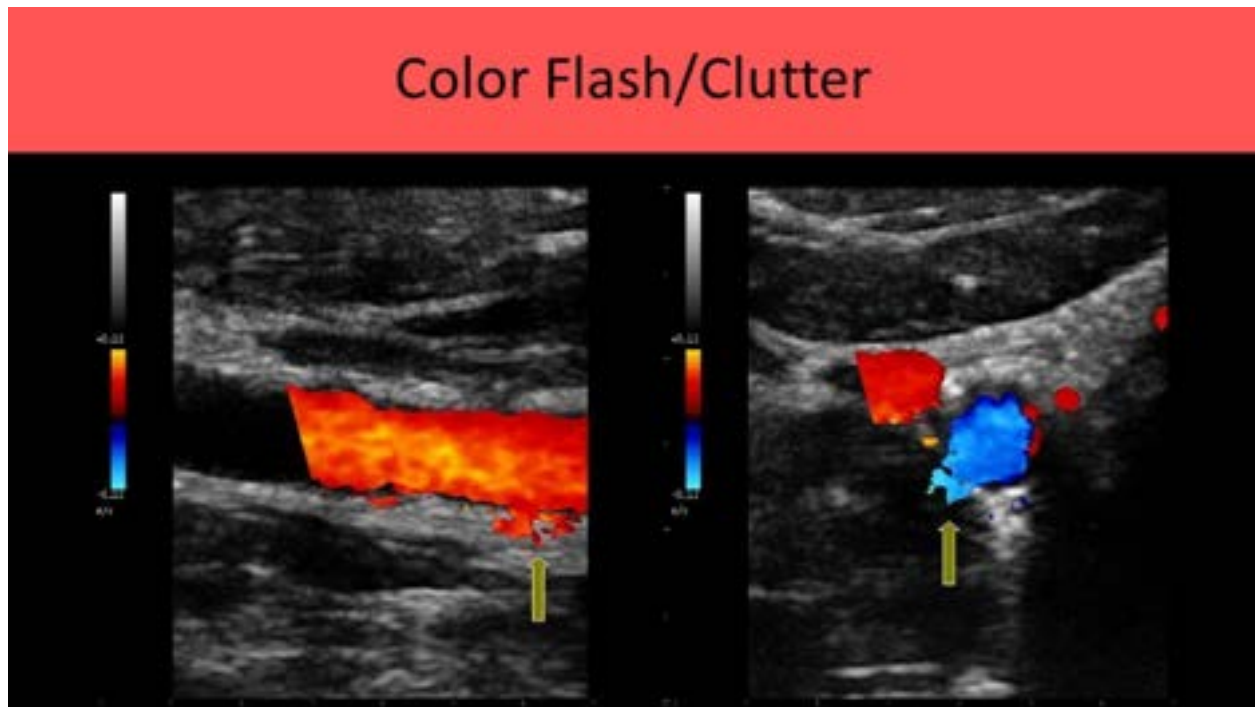
- Dedicated
- CW probes do not use B-mode imaging
- Separate transmit and receive PE elements
- Biggest advantage is it can be used to calculate blood flow at HIGH velocities
- Dedicated CW Doppler probes have increased sensitivity to small Doppler shifts
- Must place probe at a 45-60-degree angle to the skin/vessel; as the incident angle of the probe increases toward perpendicular incidence, frequency shift will decrease
- Cannot determine location of highest velocity which is called Range Ambiguity; EX: cannot differentiate external iliac stenosis from common iliac stenosis
- Provides Doppler shift information for all vessels in its path, no depth control
- Insonation of multiple vessels is a significant limitation

DOPPLER ARTIFACTS:

Clutter:

- Unwanted Doppler display patterns typically caused by a vessel's wall motion
- Called color flash for color Doppler
- Filters used to correct the issue

1. Filter: an electronic circuit designed to allow signals to pass while stopping signals of other frequencies
2. High-pass: a filter that allows high, but not low, frequencies to pass through
3. Low-pass: a filter that allows low, but not high, frequencies to pass through



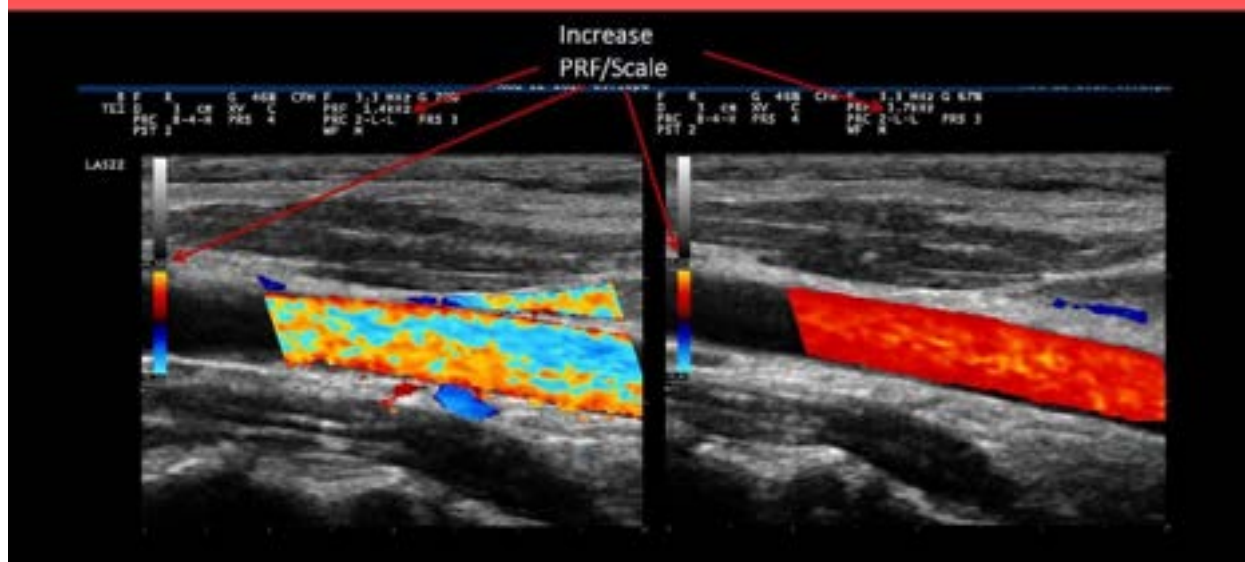
Aliasing:

- Misrepresentation of the Doppler signal due to PRF limitations
- Causes colors/waveform to “wrap around” from top to bottom on the baseline

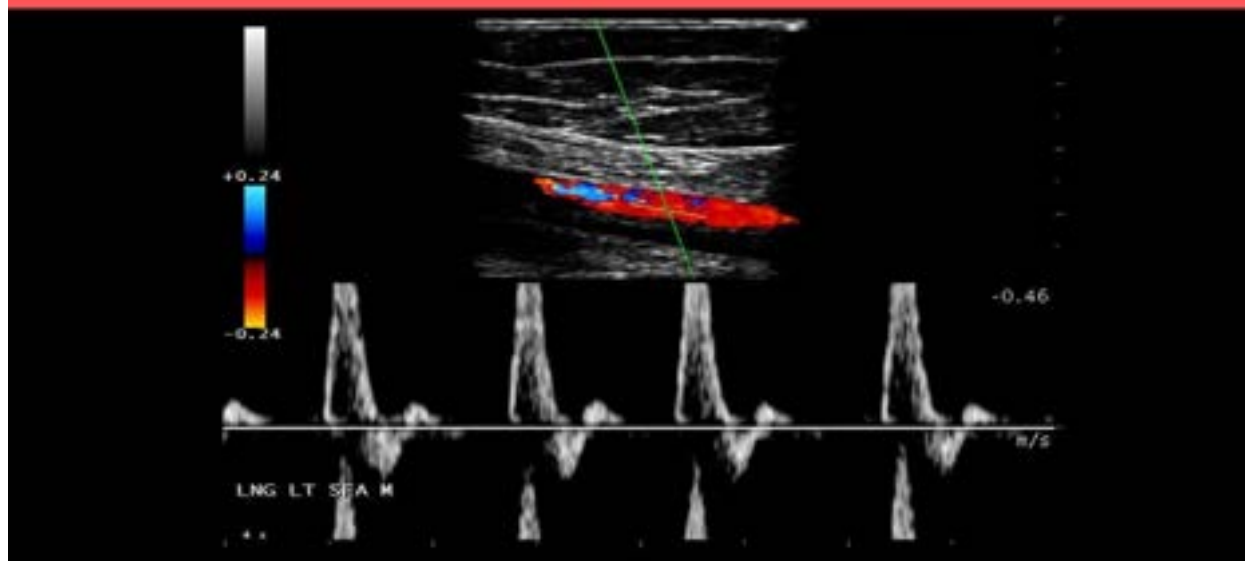
SOLUTIONS

1. Increase scale
2. Lower freq. probe
3. Change angle°
4. Adjust approach to decrease distance to vessel
5. Switch to CW

Color Doppler Aliasing



Spectral Doppler Aliasing



Cross Talk:

- Type of mirror imaging
- Caused by adjacent highly reflective structures, receiver gain too high or incident Doppler angle near 90°

Mirror Image Artifact

The lung apex (blue arrows) is located immediately posterior to the subclavian artery and acts as a highly reflective acoustic interface that causes the artifact.

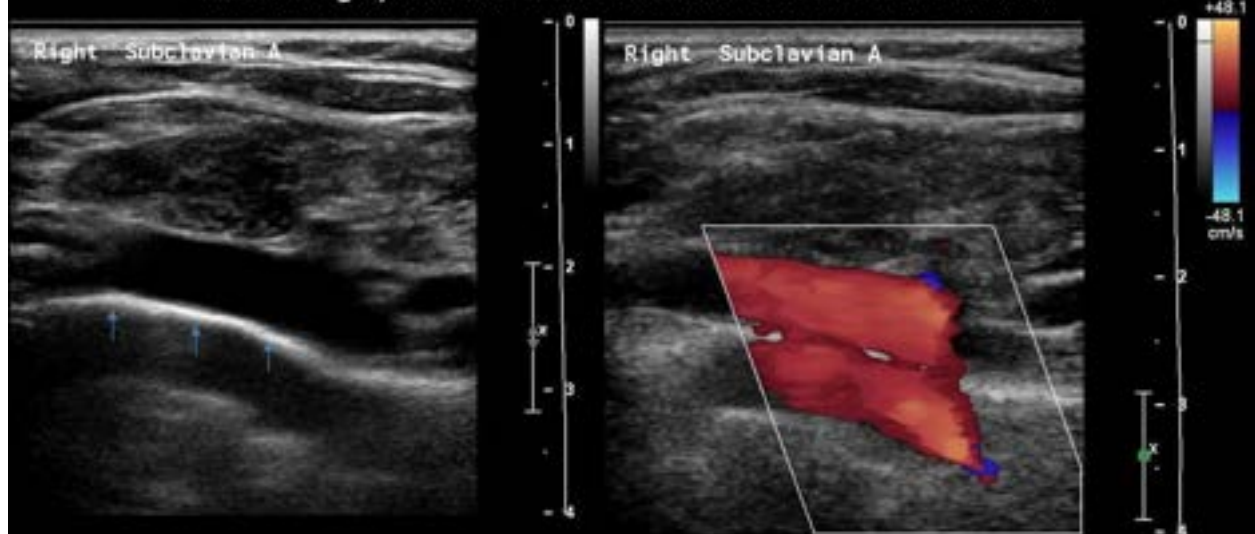


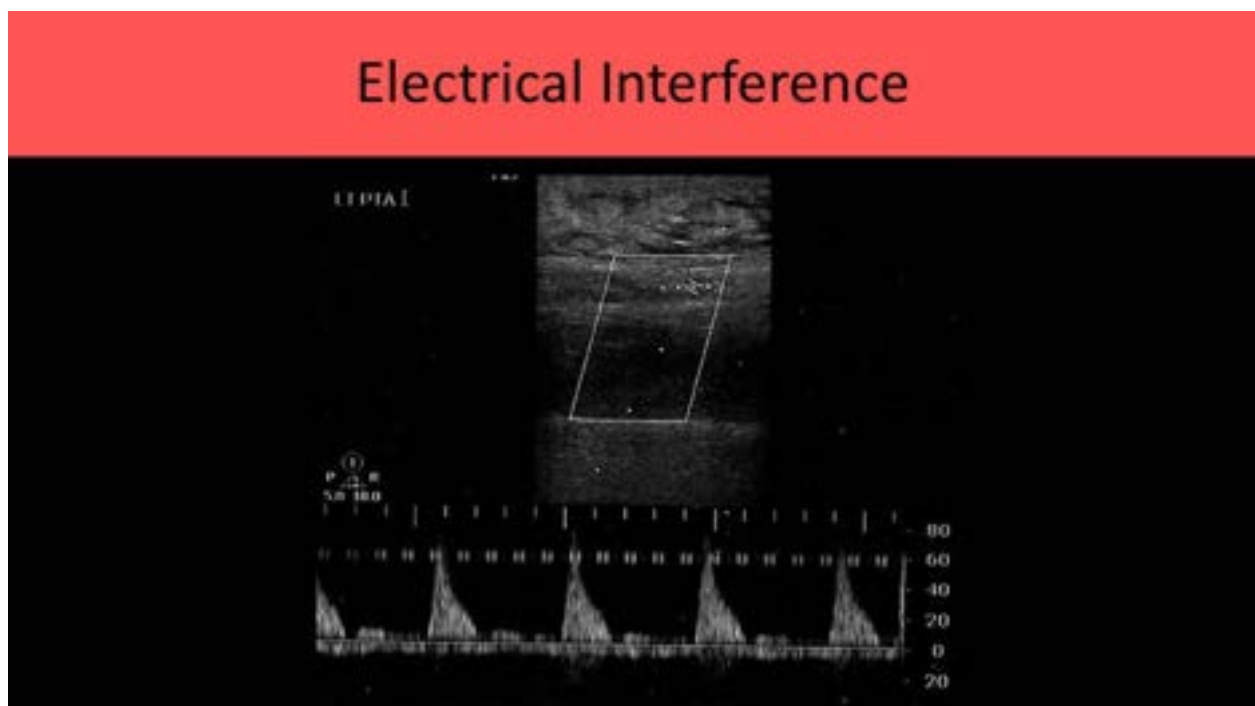
Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Cross Talk Artifact



Electrical Interference/Noise:

- Appears as repetitive, consistent artifactual echoes demonstrated on the Doppler tracing
- The frequency of the electrical signal is directly related to the number of artifact echoes demonstrated in one second
- Reduced by decreasing Doppler gain, increasing wall filter settings, try another wall outlet



Phantom Testing:

- A phantom is used as the standard for testing US system functionality
- Required for system calibration and to satisfy medical/legal needs for the facility

Uniformity:

- Evaluating the compensation controls
- Image taken with TGC set to display an even echogenicity from the top to bottom of the image
- TGC is then turned off to compare the images

- Normal results demonstrate a reduction in amplitude in the echoes in the far field on the second image

Dead Zone:

- Area closest to the transducer that cannot produce an accurate image due to reverberation
- Caused by the time it takes for the PE crystal and receiver to switch from transmit to receive
- Measured from the top of the phantom to the depth that uniform tissue is identified
- The dead zone depth should be as thin as possible with recommendations for normal depth set by manufacturer
- In normal scanning, a standoff pad is used to eliminate the dead zone and visualize superficial structures

Sensitivity:

- Ability to detect low level echoes accurately
- Turn the gain all the way down and then slowly increase it until the far field rods are visible on the image = minimum sensitivity
- Normal overall sensitivity is indicated when all rods are displayed on the image
- Number of visible rods on the image should be consistent between semi-annual assessments
- If less rods are visible on the current sensitivity test compared to the last sensitivity test = decreased system sensitivity, system service is needed

Registration Accuracy:

- Accuracy of the placement of the echoes on the screen as related to the structure placement in the body when the structure is scanned from different windows

Vertical Depth Accuracy:

- AKA range accuracy
- Reported in cm
- Accuracy of depth assessment of structures within the phantom
- A reflector is placed at a specific depth in the phantom
- A transducer is placed on the phantom and the distance to the reflector is measured
- The depth measured should correlate with the known depth of the reflector

Horizontal Calibration:

- Accuracy of the placement of echoes in the correct position along the horizontal axis of the image
- Evaluates the position of echoes along a line that is perpendicular to the beam
- Reported in cm

Spatial Resolution:

- Evaluates the minimum resolution capability of the system
- Determines minimum axial and lateral resolution capability
- Reported in mm
- Axial resolution refers to the smallest distance between two pins located parallel to the beam that are properly displayed as two separate echoes
 - 6 or more rods are placed in a row at different millimeter intervals from top to bottom in the phantom
 - The ability to resolve the pins as separate structures and the minimum distance visible between those pins is used to assess the minimum axial resolution
 - Axial resolution should be constant across the image
- Lateral resolution refers to the smallest distance between two pins located side by side (perpendicular to the beam) that are properly displayed as two separate echoes on the image
 - 6 or more rods are placed at specific depths on the image
 - The vertical depth calibration rods can be used to assess the width of the reflector rods
 - Rod width is constant and they are placed at specified depths within the phantom
 - The transducer is tested for proper display of an expected width measurement of the rods at each depth
 - Lateral resolution is degraded as the beam gets farther from the transducer and the rods appear widest in the far field

Contrast Resolution:

- Structures of different densities are placed within the phantom
- Tissue background is the reference point at 0dB
- Negative values indicate a less echogenic or anechoic structure
- Positive values indicate a more echogenic structure

QUALITY ASSURANCE

Note: Click any image to enlarge.

Quality Assurance

- Required due to medical and legal reasons
- Necessary to evaluate equipment for consistency in performance and quality of image production
- Testing should be performed under predetermined, well defined guidelines with constant settings
- Phantoms should provide measurable data
- System assessment can detect gradual changes in system performance
- Preventive maintenance helps to minimize downtime
- Protocols assist in maintaining the exam
- and reduce number of repeat exams

Four Components of a QA Program:

- Preventative maintenance
- Repairs
- System assessment of all components
- Record keeping

Preventive Maintenance:

- Performed by the manufacturer every 6 months, included in most service contracts
- Sonographers should vacuum filters, wipe down entire machine on a biweekly/monthly basis
- Sonographers should wipe screen, cords on a weekly basis
- Always clean keyboard, trackball, probe, and handles/levers used for equipment maneuvering between exams

Record Keeping:

- Another important part of preventive maintenance/QA program

- Current log of QA results
- Warranty contract for equipment
- System manual for equipment
- Maintenance Log with downtime recorded

SYSTEM ASSESSMENT

- Requires the tests performed under specific and standardized conditions
- Requires constant or standard instrument settings
- Should use a phantom with measurable and reportable characteristics
- Imaging of the phantom for each assessment should be performed in a consistent environment

Element Damage:

- If a Piezoelectric (PE) element is not functioning, data will be missing from the image
- If the probe is a single element (mechanical) probe, the entire image will not be displayed
- An element array will demonstrate missing info related to just the element that is damaged
- A linear, phased, or vector array will demonstrate the area of no reflected info as a vertical black stripe; can also affect ability to focus/steer beam
- An annular array will demonstrate an anechoic horizontal arc of missing data
- A curved array will demonstrate a vertical line of missing data along the radius of the curve
- Being unable to steer the color sample box in both directions is also a sign of element damage



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- Area closest to the transducer that cannot produce an accurate image due to reverberation
- Caused by the time it takes for the PE crystal and receiver to switch from transmit to receive
- Measured from the top of the phantom to the depth that uniform tissue is identified
- The dead zone depth should be as thin as possible with recommendations for normal depth set by manufacturer
- In normal scanning, a standoff pad is used to eliminate the dead zone and visualize superficial structures

Sensitivity:

- Ability to detect low level echoes in the far field
- Turn the gain all the way down and then slowly increase it until the far field rods are visible on the image = minimum sensitivity
- Normal overall sensitivity is indicated when all rods are displayed on the image
- Number of visible rods on the image should be consistent between semi-annual assessments
- If less rods are visible on the current sensitivity test compared to the last sensitivity test = decreased system sensitivity, system service is needed

Registration Accuracy:

- Accuracy of the placement of the echoes on the screen as related to the structure placement in the body when the structure is scanned from different windows

Vertical Depth Accuracy:

- AKA range accuracy
- Reported in cm
- Accuracy of depth assessment of structures within the phantom
- It is assumed that the speed of sound in the tissue in the phantom is 1540m/s
- If tissue propagation velocity is greater than 1540m/s, reflectors will be placed more superficial on the image than their actual depth
- If tissue propagation velocity is below 1540m/s, reflectors will be placed deeper on the image than their actual depth

Horizontal Calibration:

- Accuracy of the placement of echoes in the correct position along the horizontal axis of the image
- Evaluates the position of echoes along a line that is perpendicular to the beam
- Reported in cm

Spatial Resolution:

- Evaluates the minimum resolution capability of the system
- Determines minimum axial and lateral resolution capability
- Reported in mm
- Axial resolution refers to the smallest distance between two pins located parallel to the beam that are properly displayed as two separate echoes

- 6 or more rods are placed in a row at different millimeter intervals from top to bottom in the phantom
- The ability to resolve the pins as separate structures and the minimum distance visible between those pins is used to assess the minimum axial resolution
- Axial resolution should be constant across the image
- Lateral resolution refers to the smallest distance between two pins located side by side (perpendicular to the beam) that are properly displayed as two separate echoes on the image
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 - The vertical depth calibration rods can be used to assess the width of the reflector rods
 - Rod width is constant and they are placed at specified depths within the phantom
 - The transducer is tested for proper display of an expected width measurement of the rods at each depth
 - Lateral resolution is degraded as the beam gets farther from the transducer and the rods appear widest in the far field

Contrast Resolution:

- Structures of different densities are placed within the phantom
- Tissue background is the reference point at 0dB
- Negative values indicate a less echogenic or anechoic structure
- Positive values indicate a more echogenic structure

Dynamic Range:

- The number of shades of black, white and grey displayed on the image

Slice Thickness Phantom:

- Evaluates elevational resolution
- Elevational resolution improves with thinner image slices

Tissue Equivalent Phantom:

- Used for B-mode quality control
- Contains a tissue-mimicking water based material that simulates the echotexture of a normal liver

- Material has same propagation speed (1540m/s) and attenuation characteristics of tissue (1540m/s)
- May also contain wires, fluid filled and solid objects
- Measures dead zone, resolution, penetration, distance accuracy, and gray scale characteristics

AIUM Recommendations for Ultrasound Safety:

- Do not perform studies without reason (prudent use)
- The potential benefit of the ultrasound exam should always outweigh the risk of ultrasound exposure
- Do not prolong studies without reason
- Minimize exposure time for a complete diagnostic exam
- Always use minimum output power and highest receiver gain required to produce optimal images (ALARA principle)

AIUM Test Objects:

- Line Density Targets
- Phased out since the tissue phantoms were created
- Rods made of steel at 0.75mm thick
- Plastic case holds fluid within that has the same propagation speed as soft tissue but NOT the same attenuation as tissue
- Evaluates:
 1. Dead zone: area closest to transducer face, imaging not possible in this area, scanned from TOP of phantom viewing Row 1
 2. Axial resolution: scanned from TOP of phantom viewing Row 2, pins are spaced at different millimeter intervals
 3. Lateral resolution: scanned from the RIGHT SIDE of the phantom viewing Row 3, pins are spaced at different millimeter intervals
 4. Distance calibration: scanned from the TOP of phantom viewing Row 5, Can scan from either side viewing Row 3 and 4 instead
 5. Sensitivity: measures the weakest signal the system can detect
 6. Registration Accuracy - system's ability to place echoes in the correct position when scanning the same structure from multiple windows; phantom scanned from at least two sides (preferred 3 sides) to evaluate the placement of a specific rod on the image
 7. Horizontal Calibration - system's ability to place echoes in the correct position along a line that is perpendicular to the US beam

Doppler Phantoms:

- Flow phantom or Fluid-Filled phantom- simulates moving blood flow with specific frequency shifts in static vessels in the phantom
- Evaluates the accuracy of detected frequency shifts
- Color Doppler is also evaluated with the fluid filled phantom
- Temporal resolution is evaluated using a fluid filled phantom because it requires moving structures
- Vibrating string phantom is another type of phantom used for Doppler evaluation

Beam Profiler:

- Records a three-dimensional evaluation of the reflected amplitudes
- Tank equipped with pulser, transducer and receiver with metal rods
- Hydrophone is another type of beam profiler that uses an oscilloscope
- Schlieren method
 - Transmitting US beam through water to assess the beam pattern by displacement of water molecules
 - Uses the refraction properties of water to evaluate the beam pattern and measure intensity

Hydrophone:

- Small needle with a PE crystal attached to the end is placed into the beam
- Evaluates amplitude, period, pulse duration, pulse repetition period
- Measures temporal peak or pulse average intensity
- Evaluates acoustic pressure field and beam intensity profile
- Important for assessment of cavitation risks

Calorimeter:

- Converts acoustic energy into heat (type of transducer)
- Used to calculate the total power of the US beam
- Important for evaluation of thermal effects of sound

Intensity Ratios:

- from lowest to highest
 1. SATA
 2. SPTA
 3. SATP or SAPA

4. SPTP or SPPA

- For Pulsed instruments, peak measurements are always greater than average measurements
- $SP > SA$ and $TP > TA$
- SPTA values are used to assess patient exposure
- PW Doppler uses highest intensity
- Average range of Doppler intensities is 0.2 - 400 mW/cm²
- 2D imaging average range 0.1 - 300mW/cm²

Why do we need to be aware of beam intensity?

Bioeffects:

- The biological effects of ultrasound on tissue
- FDA studies ultrasound bioeffects; AIUM makes recommendations related to the FDA data
- Safety Indices developed by the AIUM to predict potential adverse effects from ultrasound exposure (MI and TI)
- Bioeffects have not been confirmed for Non-focused SPTA < 100mW/cm² OR Focused SPTA < 1W/cm²
- If you are unsure if your transducer is focused, use the guidelines for non-focused intensities

ALARA Principle:

- As Low As Reasonably Achievable
- When using diagnostic ultrasound, the prudent course of action is to use ultrasound when the expected benefit outweighs the potential risk of exposure.
- Reduce exam time, whenever possible
- Reduce orbital exposure, reduce power output to lowest possible setting

Thermal (Heat):

- Friction produced from the vibration of sound in tissue
- Negligible when using pulsed ultrasound for diagnostic exams
- Can be harmful to a fetus if the exam increases the temperature in the womb above 41° C
- Fetal tissue that lies adjacent to bone is at the greatest risk for the thermal effects of ultrasound
- Unfocused transducers have a greater thermal effect than focused transducers
- PW Doppler, CW Doppler and M-mode have a greater thermal effect than 2D or color Doppler
- Higher duty factors increase the risk of thermal bioeffects

- Much greater effect when ultrasound used for therapeutic purposes

Thermal Index (TI):

- Used to monitor patient exposure
- Ratio of power produced to the power needed to increase temperature in tissue by 1 degree
- TI = 1 means the output level is high enough to cause up to a 1-degree temperature increase in the tissues in a stationary transducer
- TI above 1 will cause temperature increase in bone first because it has the greatest stiffness
- Reduce exposure time to keep TI below 1
- A TI setting of 1.5 indicates the potential to increase the temperature in the tissues by 1.5 degrees
- TIS – soft tissue index
- TIB – bone index
- NO THERMAL BIOEFFECTS HAVE CAUSED ADVERSE EFFECTS WHEN THE TISSUE TEMPERATURE HAS INCREASED LESS THAN 2° C (TI <2.0)

Mechanical (Microcavitation):

- Production of gas bubbles in the tissue
- Seen with higher intensities than normal diagnostic ultrasound
- Stable – seen at lower MI values; bubbles vibrate without bursting
- Transient - seen at higher MI values; causes the microbubbles to collapse and/or burst; has potential for significant cellular damage
- 2D and color Doppler have greater risk of cavitation than PW/CW Doppler and M-mode
- Higher pulser voltages increase the risk of cavitation

Mechanical Index (MI):

- Predicts the likelihood of cavitation
- Lower frequencies produce higher MI and greater risk of cavitation
- Used to control intensity on contrast exams
- Adjusting the power output will adjust the MI

EVALUATING TEST RESULTS:

Gold Standard Testing:

- Refers to an accepted test that is assumed to be able to determine the true disease state of a patient, regardless of positive or negative test findings or sensitivities or specificities of other diagnostic tests used
- As new diagnostic methods become available, the "gold standard" test may change over time
- For example, an aortogram used to be the gold standard test for the diagnosis of aortic dissection, but today MRA has become the new "gold standard" test for aortic dissection because it has the highest sensitivity and specificity levels

1. Sensitivity-The ability of a test to detect disease when it IS present. The number of truly positive tests performed in the current modality is compared to the positive tests diagnosed by the gold standard test

True Positive Tests / Total # Positive Tests proven positive by the gold standard test

2. Specificity-The ability of a test to rule out disease when it IS NOT present. The number of truly negative tests performed in the current modality is compared to the negative tests diagnosed by the gold standard test

True Negative Tests / Total # Negative Tests proven negative by the gold standard test

3. Positive Predictive Value-Calculates how often a positive study is correctly diagnosed; ability of an exam to predict the presence of disease

True Positive Tests / Total # Positive Tests

4. Negative Predictive Value-Calculates how often a negative study is correctly diagnosed; ability of the exam to predict the absence of the disease

True Negative Tests / Total # Negative Tests

5. Accuracy-ability of test to correctly identify disease or lack of disease; total number of correct diagnoses compared to the total number of tests;

True Positive Tests + # True Negative Tests / Total # Tests Performed

Overall accuracy value is always between sensitivity and specificity AND between PPV and NPV

EX: sensitivity 80%, specificity 92%, PPV 94%, NPV 88%

Accuracy must be between 80 - 92% AND between 88 - 94%

To meet all requirements the accuracy must be between 89 -91%

89% is greater than 80% sensitivity and greater than 88% NPV

91% is less than 92% specificity and less than 94% PPV

Appropriate Use Criteria For Vascular Exams

"An appropriate imaging study is one in which the expected incremental information, combined with clinical judgment, exceeds the expected negative consequence by a sufficiently wide margin for a specific indication that the procedure is generally considered acceptable care and a reasonable approach for the indication."

ACCF/ACR/AIUM/ASE/ASN/ICAVL/SCAI/SCCT/SIR/SVM/SVS 2012 Appropriate Use Criteria for Peripheral Vascular Ultrasound and Physiological Testing Part I: Arterial Ultrasound and Physiological Testing.

ACCF/ACR/AIUM/ASE/IA/SCAI/SCVS/SIR/SVM/SVS/SVU 2013 Appropriate Use Criteria for Peripheral Vascular Ultrasound and Physiological Testing Part II: Testing for Venous Disease and Evaluation of Hemodialysis Access. See [references](#) for link to full report.

The information below is our summary of the Appropriate Use Criteria for vascular ultrasound and physiologic testing that was developed through the collaboration of several vascular ultrasound and medical societies. Our primary focus is on the generally appropriate and inappropriate categories. To learn more about the indications that "may be appropriate", please visit the original article.

- The most common indications given for vascular ultrasound and physiologic testing were scored for appropriate use
- Not all exam indications were evaluated
- The indications were scored while considering vascular ultrasound or physiologic testing by itself as a reasonable exam choice
- Three categories used to classify indications
 - 7-9 points: Generally appropriate indication

- 4-6 points: May be appropriate or reasonable indication; situational consideration
- 1-3 points: Inappropriate or unreasonable indication
- If there is more than one indication for an exam, the indication with the highest score should be used to determine appropriate use

General Recommendations for Performance of Arterial and Physiologic Testing:

- Use a standard protocol for duplex and physiologic exams, including stress testing
- Results evaluated with standard diagnostic criteria
- A Doppler angle of 60 degrees or less should be used for all exams and the Doppler angle must be documented on the report
- Testing should be performed by a registered Sonographer and at an accredited facility
- Repeat examinations should be performed at the same facility as the initial exam
- Screening examination: testing used to determine the presence or absence of disease in an asymptomatic patient
- Baseline examination: testing that leads to the initial diagnosis or the initial exam following vascular intervention/surgery that is usually done within first month after the procedure
- Surveillance examination: testing used to monitor known disease progression; follow up exam timing is based on the time since initial diagnosis or revascularization

Extracranial Evaluation:

- MRA or CTA is recommended as the initial imaging test for suspected vertebral artery stenosis and carotid dissection due to trauma
- Carotid ultrasound is not recommended to evaluate suspected carotid dissection due to trauma because a distal dissection of the internal carotid artery may not be detected
- Framingham risk score - gender specific criteria for estimation of 10 year risk of cardiovascular disease
- Low scores indicate screening exams are unnecessary
- High scores may be appropriate for screening in some cases

Venous Evaluation:

- For all unilateral extremity venous exams, the flow in the contralateral extremity should be documented for comparison
- Common femoral vein for the lower extremities and subclavian or internal jugular vein for the upper extremities

Appropriate Use Criteria – Cerebrovascular Duplex Exam

ALWAYS APPROPRIATE

- New or worsening hemispheric neurological symptoms
- Evaluation of patient with TIA or stroke
- Hollenhorst plaque identified on retinal examination
- Lightheadedness/impaired vision with upper extremity exertion
- Suspected subclavian–vertebral steal
- Suspected symptomatic vertebrobasilar occlusive disease
- Cervical bruit in patient with no prior CV duplex exam
- Atherosclerotic disease in other vascular areas (PAD, CAD, AAA)
- Follow up exam 6 months after initial diagnosis on patient with $\geq 70\%$ ICA stenosis
- Follow up exam every 6-12 months after first year following diagnosis of $\geq 70\%$ ICA stenosis
- Follow up exam annually after initial diagnosis on patient with 50-69% ICA stenosis
- 1 month follow up post-intervention (baseline)
- 6 month follow up after baseline in first year post-intervention
- 12 month follow up on stable patients after first year post-intervention

Appropriate Use Criteria – Cerebrovascular Duplex Exam

INAPPROPRIATE

- Follow up exam within 12 months on a patient with a normal prior examination (no plaque, no stenosis)
- Follow up exam within 12 months on a patient with plaque, but no significant stenosis of the ICA
- Follow up exam within 12 months on a patient with plaque, but $<50\%$ ICA stenosis
- Screening exam on a patient with low Framingham risk scoring and no other assessment
- Screening exam on a patient with low to intermediate Framingham risk scoring and normal alternative risk assessment (carotid IMT thickness or coronary calcium scoring)

Appropriate Use Criteria – Upper Extremity Arterial Duplex/Physiologic Exam

ALWAYS APPROPRIATE

- Claudication
- Finger discoloration
- Ulceration or gangrene
- Unilateral cold hand
- Suspected positional arterial obstruction
- Pre-op evaluation of the radial artery for coronary bypass
- Suspected vascular injury due to trauma
- Pulsatile mass or hand ischemia following vascular access
- Bruit at intervention access site
- 1 month follow up baseline exam post-intervention
- Post-intervention trauma
- New or worsening symptoms after intervention
- 6-8 month follow up on stable venous bypass graft in first year (<6 months deemed inappropriate)
- After first year post-intervention, annual follow up exam on stable patient (vein or prosthetic graft)

Appropriate Use Criteria – Lower Extremity Arterial Duplex/Physiologic Exam

ALWAYS APPROPRIATE

- Claudication
- Rest pain
- Ulceration or gangrene
- Leg infection with non-palpable pulses
- Clinical signs of embolism (eg blue toes)
- New symptoms in patients with previously normal or abnormal duplex or ABI exam
- 1 month follow up baseline exam post-revascularization
- Follow up exam post-revascularization with new or worsening symptoms
- 6-8 month follow up on stable patients after angioplasty, placement of stent, vein bypass graft, or prosthetic bypass graft
- After first year post-revascularization, annual follow up on stable patients after angioplasty, placement of stent, vein bypass graft, or prosthetic bypass graft
- Any of the following post-intervention symptoms: pulsatile groin mass, thrill or bruit at groin, severe pain or prominent hematoma formation
- Screening ABI exam on patients with femoral bruit or diminished pulses
- Screening ABI exam on patients >50yrs with diabetes or smoking history
- Screening ABI exam on patients >70yrs

Appropriate Use Criteria – Lower Extremity Arterial Duplex/Physiologic Exam

INAPPROPRIATE

- Nocturnal pain or cramping with normal pulses on clinical exam
- Normal pulses, but no leg/foot hair
- Lower extremity swelling with normal pulses
- Screening exam on a patient with low Framingham risk score
- Diabetic neuropathy with normal pulses
- Follow up exam on stable patients with normal ABI on baseline exam

Appropriate Use Criteria – Aortic Duplex Exam

ALWAYS APPROPRIATE

- Lower extremity claudication
- Decreased or absent femoral pulse
- Ulceration or gangrene
- Abdominal or femoral bruit
- Pulsatile abdominal mass
- Aneurysmal pulse in femoral or popliteal arteries
- Clinical signs of embolism (e.g. blue toes)
- Physiologic testing that indicates aortoiliac disease
- Abdominal x-ray that suggests AAA
- Diagnosis of femoral, popliteal or thoracic aorta aneurysm
- New or worsening symptoms in patient with AAA

Appropriate Use Criteria – Aortic Duplex Exam

ALWAYS APPROPRIATE

- Lower extremity claudication
- Decreased or absent femoral pulse
- Ulceration or gangrene
- Abdominal or femoral bruit
- Pulsatile abdominal mass
- Aneurysmal pulse in femoral or popliteal arteries
- Clinical signs of embolism (e.g. blue toes)
- Physiologic testing that indicates aortoiliac disease
- Abdominal x-ray that suggests AAA
- Diagnosis of femoral, popliteal or thoracic aorta aneurysm
- New or worsening symptoms in patient with AAA

Appropriate Use Criteria – Aortic Duplex Exam

ALWAYS APPROPRIATE

Aortic Screening:

- Patient >60yrs with first degree relative with an abdominal aortic aneurysm
- Patient >65yrs with current or previous history of smoking

AAA Surveillance in First Year After Diagnosis (stable patients):

- 3 - 3.9 cm diameter = follow up 9 – 12 months
- 4 – 5.4 cm diameter = follow up 6 – 12 months
- 5.5 cm or greater diameter = follow up 3 – 9 months

AAA Surveillance after First Year with No or Slow Progression:

- 3 - 3.9 cm diameter = follow up every 12 months or greater
- 4 – 5.4 cm diameter = follow up every 12 months
- 5.5 cm or greater diameter = follow up every 6 – 12 months

AAA Surveillance after First Year with Rapid Progression:

- 3 - 3.9 cm diameter = follow up every 6 – 12 months
- 4 – 5.4 cm diameter = follow up every 6 – 12 months
- 5.5 cm or greater diameter = follow up every 6 months

Appropriate Use Criteria – Aortic Duplex Exam

ALWAYS APPROPRIATE

- 1 month follow up baseline exam post-revascularization (endograft or stents)
- Follow up exam post-revascularization with new or worsening symptoms
- Aortic graft with endoleak and increasing sac diameter = follow up exam 6 – 12 months in first year
- Aortic graft with endoleak and increasing sac diameter = follow up exam every 6 – 12 months after first year
- Aortic graft with endoleak and decreasing sac diameter = follow up exam every 12 months after first year

Appropriate Use Criteria – Aortic Duplex Exam

INAPPROPRIATE

- Nonspecific lower extremity discomfort
- Lower extremity swelling
- Systemic Hypertension
- Fever of unknown origin
- Aortic screening on patient <65yrs with no history of smoking
- 6 month follow up exam on stable patients with 3 – 3.9cm diameter AAA
- Surveillance performed < 6 months after the post-vascularization baseline exam for a stable patient
- Surveillance performed at 6 month intervals after the first year post-vascularization for a stable patient

Appropriate Use Criteria – Renal Artery Duplex Exam

ALWAYS APPROPRIATE

- Hypertension in young person (age <35 years)
- Resistant hypertension (>140/90 mm Hg on ≥3 meds)
- Malignant hypertension (>160/80 mm Hg)
- Blood pressure becoming less manageable over time
- Unexplained size discrepancy between kidneys
- Unknown cause of azotemia
- Elevated creatinine levels
- Epigastric bruit
- Acute renal failure with aortic dissection

Appropriate Use Criteria – Renal Artery Duplex Exam

ALWAYS APPROPRIATE

- Refractory heart failure
- (Symptoms of CHF at rest or even with optimal drug therapy)
- “Flash” pulmonary edema
- (Respiratory distress due to the rapid accumulation of fluid within the lung tissues with heart failure)
- Follow up known renal artery stenosis
- 1 month follow up baseline exam post-revascularization
- Follow up exam post-revascularization with new or worsening symptoms
- Follow up exam every 12 months in stable patient

INAPPROPRIATE

- Atherosclerotic vascular disease in other areas (e.g., PAD, CAD) and well controlled hypertension

Appropriate Use Criteria – Mesenteric Duplex Exam

ALWAYS APPROPRIATE	INAPPROPRIATE
Post prandial pain or weight loss not otherwise explained by GI evaluation	Evaluate for acute abdominal pain 'out of proportion to exam'
Follow up known mesenteric artery stenosis	Chronic constipation or diarrhea with no prior GI evaluation
1 month follow up baseline exam post-revascularization	Surveillance < 6 months after baseline exam post-revascularization
Follow up exam post-revascularization with new or worsening symptoms	
Follow up exam every 12 months in stable patient post-revascularization	Follow up exam every 6 months after first year of post-revascularization

Appropriate Use Criteria – Lower Extremity Venous Duplex Exam (Thrombosis)

ALWAYS APPROPRIATE

Unilateral and bilateral acute swelling
Unilateral chronic swelling
Bilateral chronic swelling with no other potential explanation (CHF, renal failure, hypoalbuminemia, lymphedema)
Palpable, painful cord in the lower extremity
Pain in the lower extremity, not related to the hip, knee or ankle joints
Suspected or diagnosed pulmonary embolism
New or worsening symptoms in a patient being treated for venous thrombosis
New or worsening symptoms in a patient not being treated for venous thrombosis
Evaluation of proximal propagation of calf vein thrombosis in a patient with contraindication to anticoagulation therapy
Vein mapping for vein harvesting for bypass surgery
Lower extremity swelling or pain after endovenous ablation
Follow up exam within 10 days of endovenous ablation
Physiologic testing that indicates venous obstruction

Appropriate Use Criteria – Extremity Venous Duplex Exam (Thrombosis)

INAPPROPRIATE

- Prolonged ICU stay (e.g., >4 days)
- Patients with acquired or inherited risks of thrombosis
- Positive D-dimer test
- Screening after recent orthopedic surgery

Appropriate Use Criteria – Lower Extremity Venous Duplex Exam (Insufficiency)

ALWAYS APPROPRIATE

- Hyperpigmentation and skin changes seen with venous insufficiency, no visible varicose veins
- Visible varicose veins with associated symptoms (leg heaviness, discoloration, ulceration)
- Active or healed venous ulcer
- Vein mapping for endovenous ablation
- New or worsening symptoms in a patient previously treated with endovenous ablation
- Suspected or diagnosed pulmonary embolism

INAPPROPRIATE

- Telangiectasia (spider veins)
- Prior endovenous saphenous ablation procedure with no residual symptoms

Appropriate Use Criteria – Extremity Venous Physiologic Exams

ALWAYS APPROPRIATE

- Hyperpigmentation and skin changes seen with venous insufficiency, no visible varicose veins
- Visible varicose veins with associated symptoms (leg heaviness, discoloration, ulceration)
- Active venous ulcer

INAPPROPRIATE

- Unilateral acute swelling
- Suspected DVT
- Possible pulmonary embolism

Appropriate Use Criteria – Lower Extremity Venous Duplex Exam (Insufficiency)

ALWAYS APPROPRIATE

- Hyperpigmentation and skin changes seen with venous insufficiency, no visible varicose veins
- Visible varicose veins with associated symptoms (leg heaviness, discoloration, ulceration)
- Active or healed venous ulcer
- Vein mapping for endovenous ablation
- New or worsening symptoms in a patient previously treated with endovenous ablation
- Suspected or diagnosed pulmonary embolism

INAPPROPRIATE

- Telangiectasia (spider veins)
- Prior endovenous saphenous ablation procedure with no residual symptoms

Appropriate Use Criteria – Extremity Venous Physiologic Exams

ALWAYS APPROPRIATE

- Hyperpigmentation and skin changes seen with venous insufficiency, no visible varicose veins
- Visible varicose veins with associated symptoms (leg heaviness, discoloration, ulceration)
- Active venous ulcer

INAPPROPRIATE

- Unilateral acute swelling
- Suspected DVT
- Possible pulmonary embolism

Appropriate Use Criteria – Upper Extremity Venous Duplex Exam (Thrombosis)

ALWAYS APPROPRIATE

- Unilateral acute swelling
- Bilateral acute swelling with suspected central venous obstruction
- Unilateral chronic swelling
- Bilateral chronic swelling with no other potential explanation (CHF, renal failure, hypoalbuminemia)
- Pain in the upper extremity (not related to joints) with indwelling upper extremity venous catheter
- Palpable, painful cord in the upper extremity
- New or worsening symptoms in a patient being treated for venous thrombosis
- New or worsening symptoms in a patient not being treated for venous thrombosis
- Vein mapping for upper extremity vein harvesting due to lack of appropriate lower extremity vein

Appropriate Use Criteria – Upper Extremity Venous Duplex Exam (Thrombosis)

INAPPROPRIATE

- Fever of unknown origin in patient with no indwelling upper extremity venous catheter
- Shortness of breath in a patient with previously diagnosed upper extremity DVT
- Prolonged ICU stay (e.g., >4 days) with or without indwelling upper extremity venous catheter
- Prior to pacemaker or cardiac defibrillator placement procedure
- Patients with acquired or inherited risks of thrombosis
- Patients at risk of thrombosis due to hypercoagulable state
- Positive D-dimer test

Appropriate Use Criteria – IVC/Iliac Vein Duplex Exam

ALWAYS APPROPRIATE

- Lower extremity swelling with proximal extremity thrombosis, unilateral or bilateral
- Abnormal flow pattern in the common femoral vein, unilateral or bilateral

INAPPROPRIATE

- When extremity thrombosis or pulmonary embolus is suspected and an extremity venous evaluation has not been performed
- Abdominal pain or bruit
- Fever of unknown origin

Appropriate Use Criteria – Renal Vein Duplex Exam

ALWAYS APPROPRIATE

- No indications are listed as always appropriate

INAPPROPRIATE

- Gross or microscopic hematuria
- Pulmonary symptoms/Suspected embolism
- Drug resistant HTN
- Epigastric bruit
- Fever of unknown origin

Appropriate Use Criteria – Hepatic and Portal Venous Duplex Exam

ALWAYS APPROPRIATE

- Portal HTN
- Cirrhosis
- Hepatomegaly/Splenomegaly
- TIPS surveillance

INAPPROPRIATE

- Jaundice that has not been otherwise evaluated
- Pulmonary symptoms/Suspected embolism
- Fever of unknown origin

Appropriate Use Criteria – Hemodialysis Vascular Access Duplex Ultrasound	
ALWAYS APPROPRIATE	
•	Vein mapping < 3 months before hemodialysis graft placement
•	Suspected failure to mature on clinical exam performed > 6weeks after placement
•	Graft dysfunction during dialysis
•	Inability to cannulate at dialysis (multiple personnel)
•	Palpable mass at the graft/fistula
•	Absence of palpable thrill at the graft/fistula
•	Arm swelling
•	Signs of hand ischemia (pain, pallor, ulceration)
INAPPROPRIATE	
•	Cool extremity with no other symptoms
•	Vein mapping > 3 months before hemodialysis graft placement
•	Routine surveillance

Abdominal Aorta: Sonographic Evaluation and Disease

Blood Vessel Anatomy:

- Tunica Intima-inner layer endothelial cells
- Tunica Media-middle layer, muscle, thicker and more organized layer in arteries
- Tunica Adventitia/Externa-outer layer, epithelial cells, connective tissue

- Veins have thinner media layer to allow flexibility of lumen size with varied flow volume

AORTA

Anatomy:

- Trunk artery
- Ascending aorta arises from left ventricle
- First branches of the ascending aorta are the coronary arteries
- Aortic arch starts at the level of the innominate artery and extends distal to the left subclavian artery
- Three branches of the arch in order of origin: innominate artery, left common carotid artery, left subclavian artery
- Bovine Arch
 - The most common anatomic variation of the arch anatomy
 - Two branch arch formation: the innominate and left CCA origin are combined and the left subclavian artery is the second branch
 - Although "bovine" indicates a cow, this arch formation is NOT related to the arch formation in cattle.
- Descending aorta extends through the chest and abdomen until it bifurcates at the level of the umbilicus
- Descending thoracic aorta - above the diaphragm
- Descending abdominal aorta - below the diaphragm
- Courses anterior to the spine and to the left of the IVC
- Posterior to the IVC until the level of the umbilicus
- Distributes blood to organs and limbs
- Lower resistance in the upper abdominal aorta due to low resistance branches; higher resistance in the distal aorta due to high resistance branches
- In most cases the branch is named after the organ it is feeding

Paired Branches:

- Suprarenal arteries
- Renal arteries
- Gonadal arteries
- Lumbar arteries
- Common Iliac arteries

Unpaired Branches:

- Celiac Axis
- SMA

- IMA

Anterior Branches:

- Celiac Axis
- SMA
- IMA
- Gonadal arteries

Lateral Branches:

- Renal arteries
- Common Iliac arteries

COMMON ILIAC ARTERIES: (CIA)

- Originate at the aortic bifurcation at L3 or L4, umbilicus level
- Supply legs and pelvis
- Bifurcate into the internal and external iliac arteries
- High resistance flow
- Above the umbilicus, arteries course posterior to veins
- Below the umbilicus, the arteries course anterior to the veins; common iliac arteries are anterior to the common iliac veins
- Evaluated for arterial pathology in longitudinal and transverse planes

INTERNAL ILIAC ARTERY: (IIA)

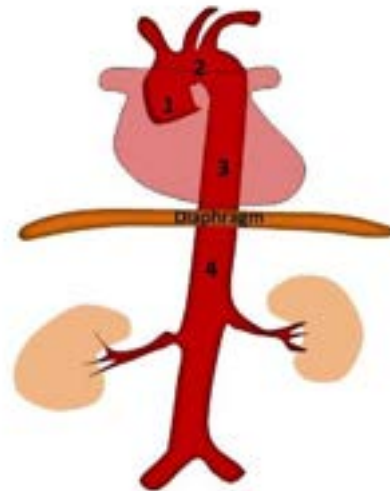
- AKA hypogastric artery
- Travels medially and supplies the pelvic organs, NOT GONADS
- Triphasic or biphasic, lower resistance than the common iliac

EXTERNAL ILIAC ARTERY: (EIA)

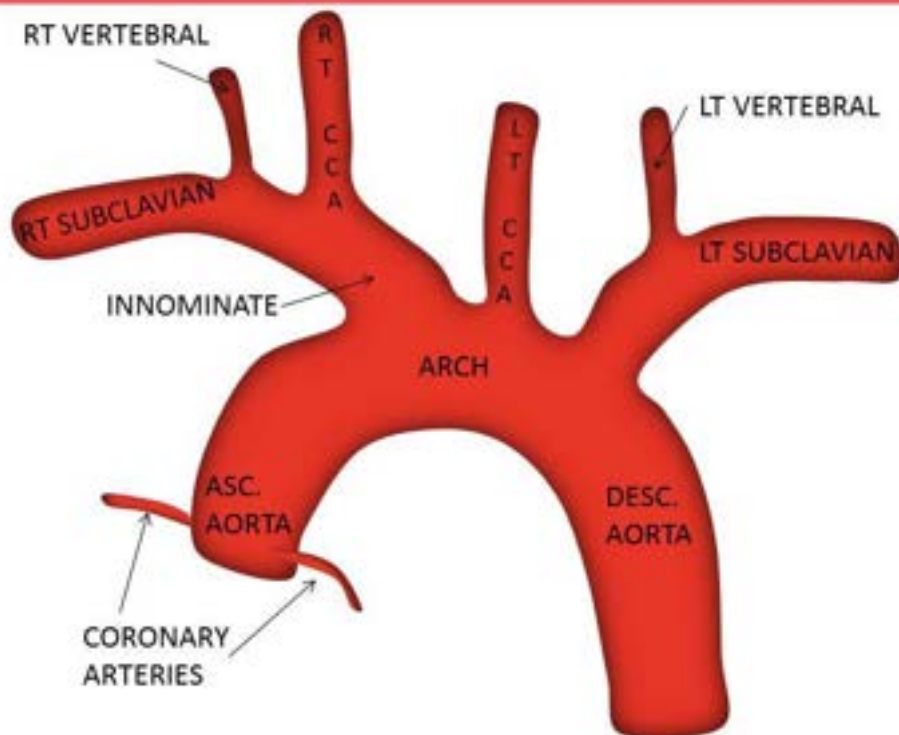
- Travels laterally to pass under the inguinal ligament to become the common femoral artery
- Courses along the medial side of the psoas muscle
- Supplies the extremities with blood
- Triphasic, higher resistance than the common iliac

Aorta

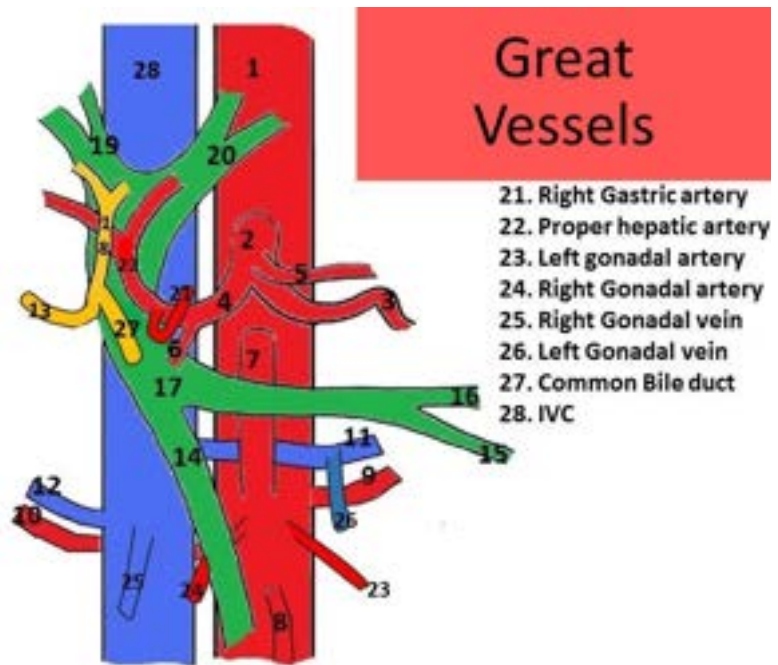
1. Ascending
2. Arch
3. Descending Thoracic
4. Descending Abdominal



Aortic Arch



1. Aorta
2. Celiac
3. Splenic artery
4. Common hepatic artery
5. Left Gastric artery
6. Gastroduodenal artery
7. SMA
8. IMA
9. LRA
10. RRA
11. LRV
12. RRV
13. Cystic duct
14. SMV
15. IMV
16. Splenic Vein
17. Portal confluence
18. Common Hepatic duct
19. RPV
20. LPV



Sonographic Evaluation:

Indications:

- Abdominal pain
- Pulsatile mass
- AAA on plain film
- F/U AAA
- Trauma
- Decreased pedal pulses

Physical Exam:

- Palpation:
 - The distal aorta can usually be palpated by pressing on the abdomen at the level of the iliac crest
 - An AAA can cause a pulsatile mass that can be palpated
- Auscultation:
 - Place a stethoscope in multiple locations on the abdomen
 - Ask the patient to hold their breath while listening for the flow sounds
 - LUQ bruit can be associated with splenic artery stenosis or left renal artery stenosis
 - RUQ bruit can be associated with right renal artery stenosis

- A bruit can also be associated with celiac artery and SMA stenosis, usually heard more centrally

Exam Technique:

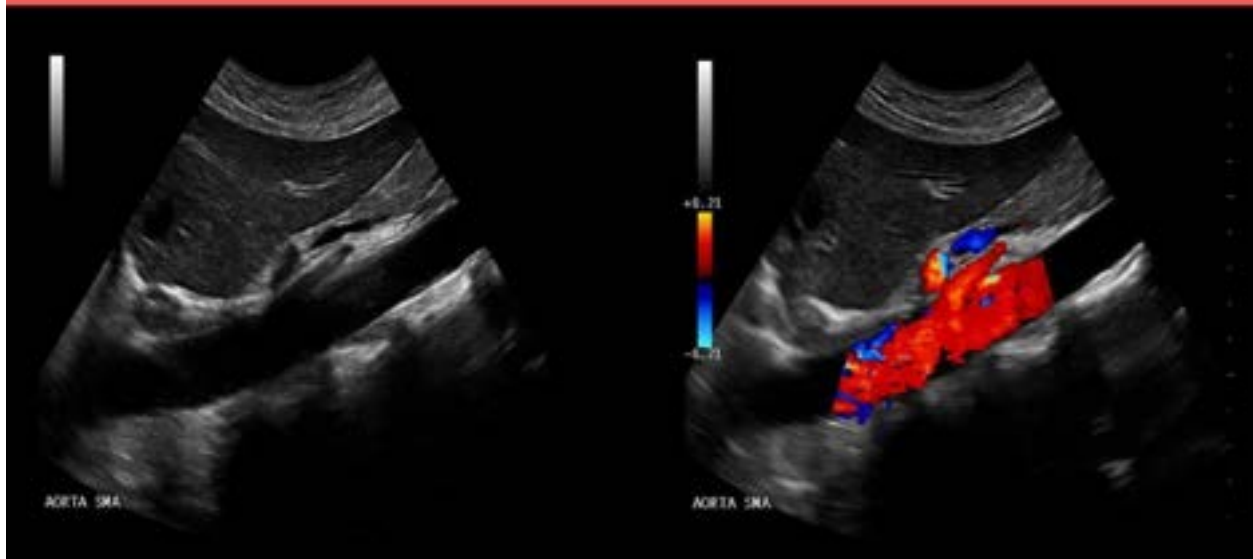
- 2.5MHz to 4MHz curvilinear array - adult
- 4MHz to 8MHz curvilinear array - pediatric
- Phased array may be used for larger patients
- Patient must be NPO to best visualize the abdominal vasculature
- Supine, oblique and decubitus positions may be necessary
- 2D, Color and Doppler evaluation of proximal, mid, and distal segments with iliac arteries at bifurcation

US of the Aorta:

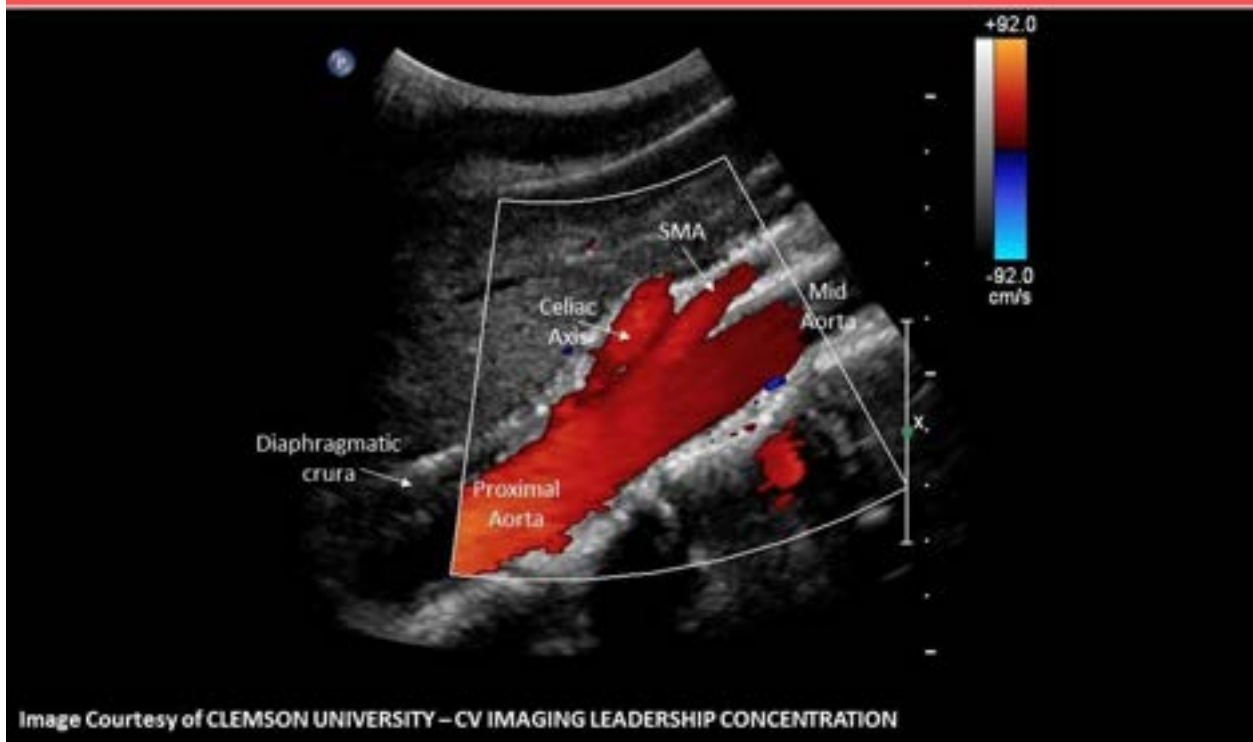
- Best anatomic landmark for abdominal structures
- Longitudinal - anechoic, hollow tube anterior to spine
- Transverse - anechoic, circular structure to the left of the IVC
- Coronal view can demonstrate both the aorta and IVC on same image
- AP and transverse measurements should be obtained of the aorta and the proximal iliac arteries
- AP dimensions are obtained in longitudinal plane, from outer wall to outer wall of the vessel
- Width dimensions are obtained in the transverse plane, from outer wall to outer wall of the vessel
- Gradual distal tapering should be identified in the normal aorta
- Becomes more anterior distally
- The proximal, mid and distal aortic segments are documented, along with both common iliac arteries
- The common iliac artery ends at the bifurcation into the external and internal iliac arteries
- Average Normal Measurements:

- “ Proximal-2.0-2.6 cm; portion between diaphragm and celiac axis/SMA
- “ Mid-1.6-2.4 cm; portion from celiac axis/SMA to just below renal artery origins
- “ Distal-1.1-2.0 cm; portion from below renal arteries to iliac bifurcation
- “ Iliacs-0.6-1.4 cm

Aorta

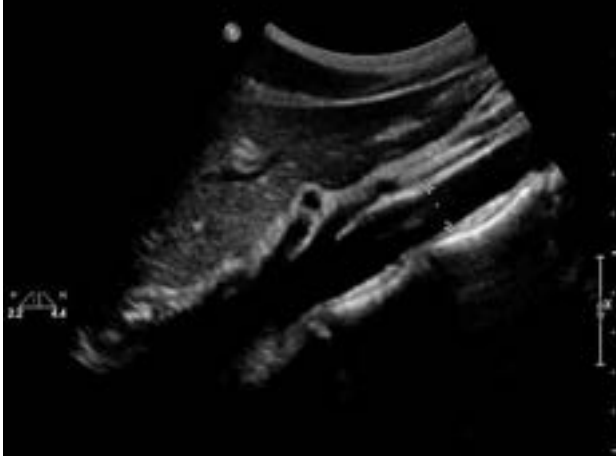


Aorta



Aortic Measurements

INCORRECT Measurements – Inner wall to inner wall



CORRECT Measurements – Outer wall to outer wall

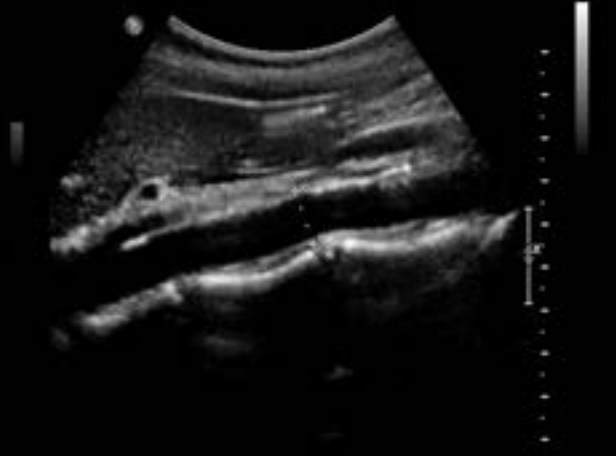


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Proximal Abdominal Aorta

Segment between the diaphragm and the celiac origin.

Measure the diameter above the celiac origins.

The diaphragmatic crura can be seen anterior to the proximal aorta.

+ Prox Ao Long 2.1 cm

Diaphragmatic
crura



+ Prox Ao AP 2.1 cm
= Prox Ao Trnv 2.2 cm

Diaphragmatic
crura



Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Mid Abdominal Aorta

Segment from celiac origin to just distal to the renal artery origins.

Measure the diameter just below the SMA origin.

The SMA should be demonstrated anterior to the aorta in the transverse image.

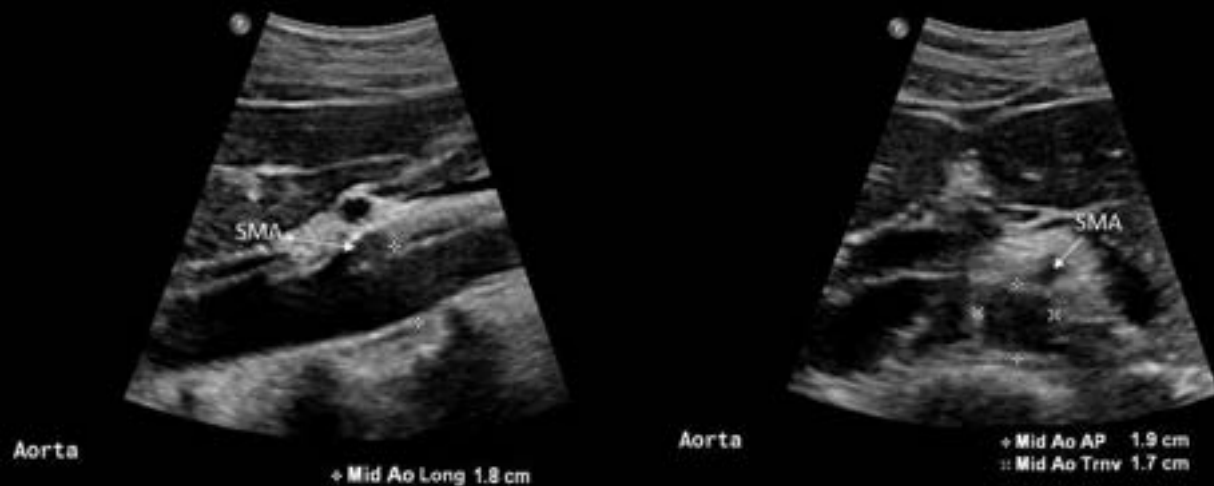


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Distal Abdominal Aorta

Segment from distal to the renal origins to the bifurcation.
Measure the diameter just above the bifurcation, before it tapers into the iliac arteries
Aortic diameters should decrease slightly as the vessel courses distally.

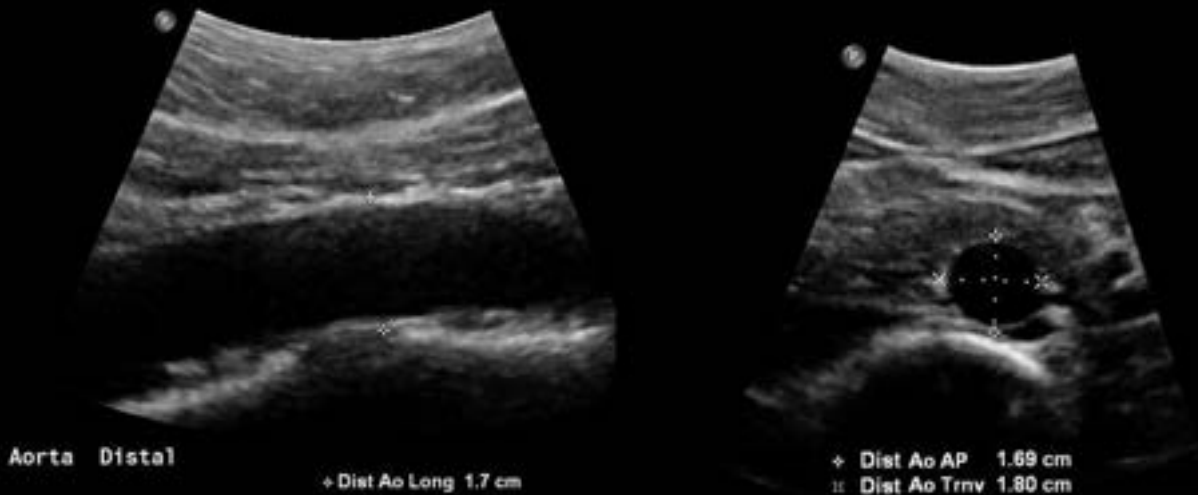
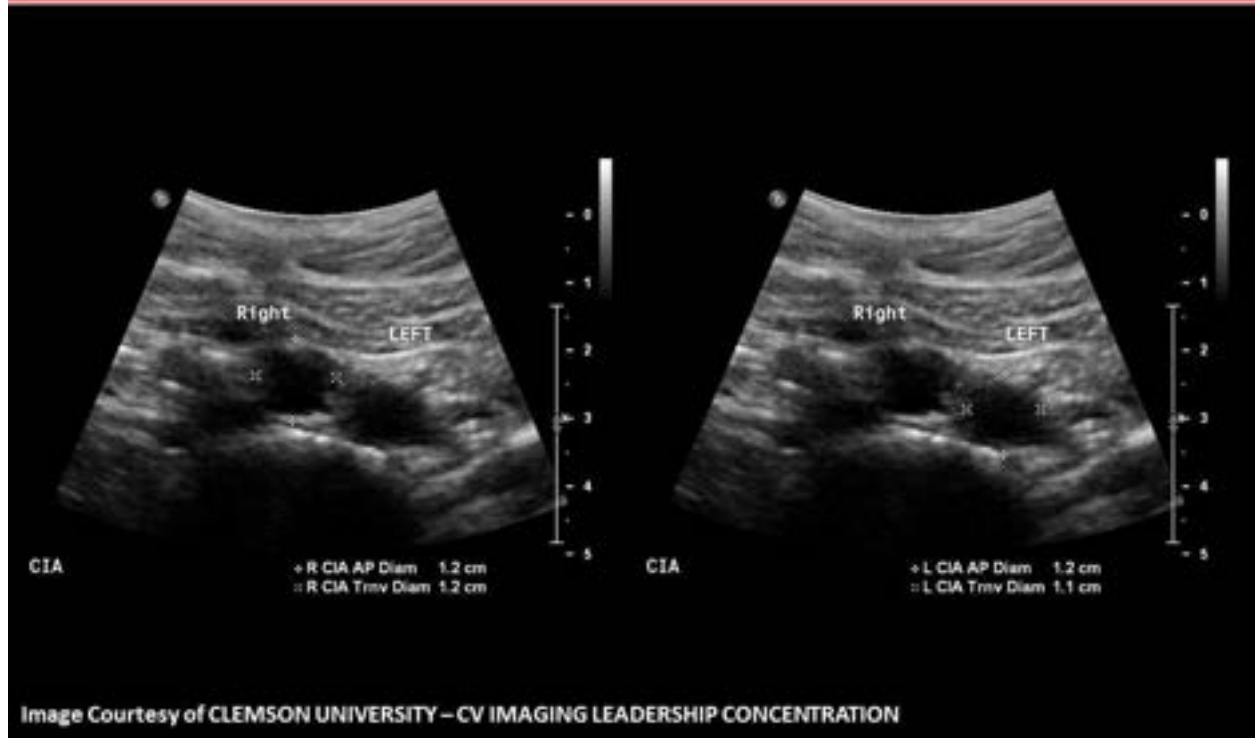
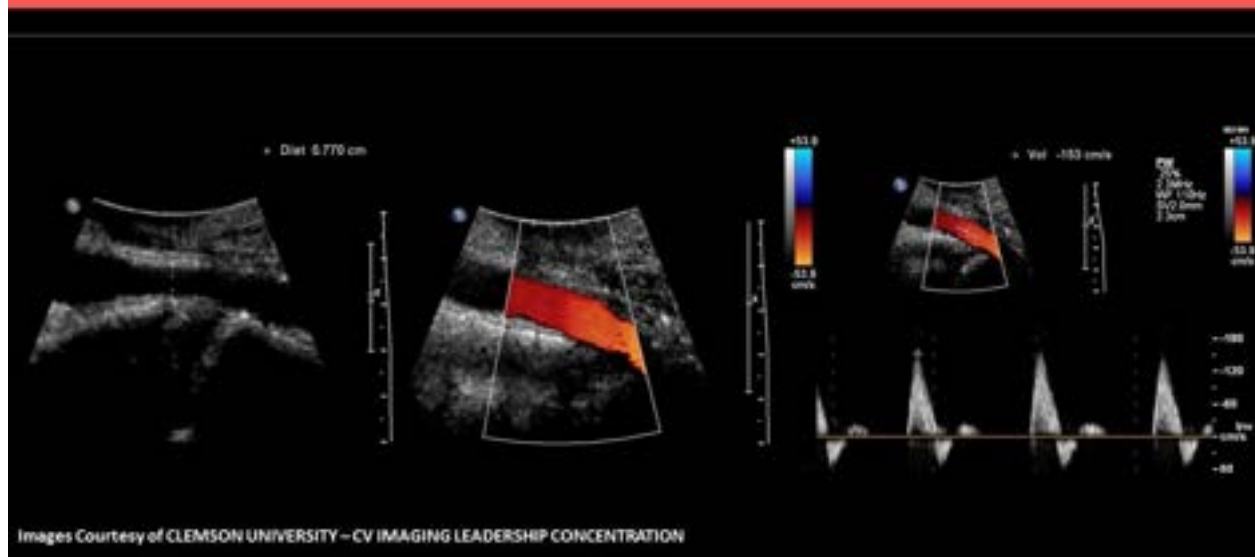


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

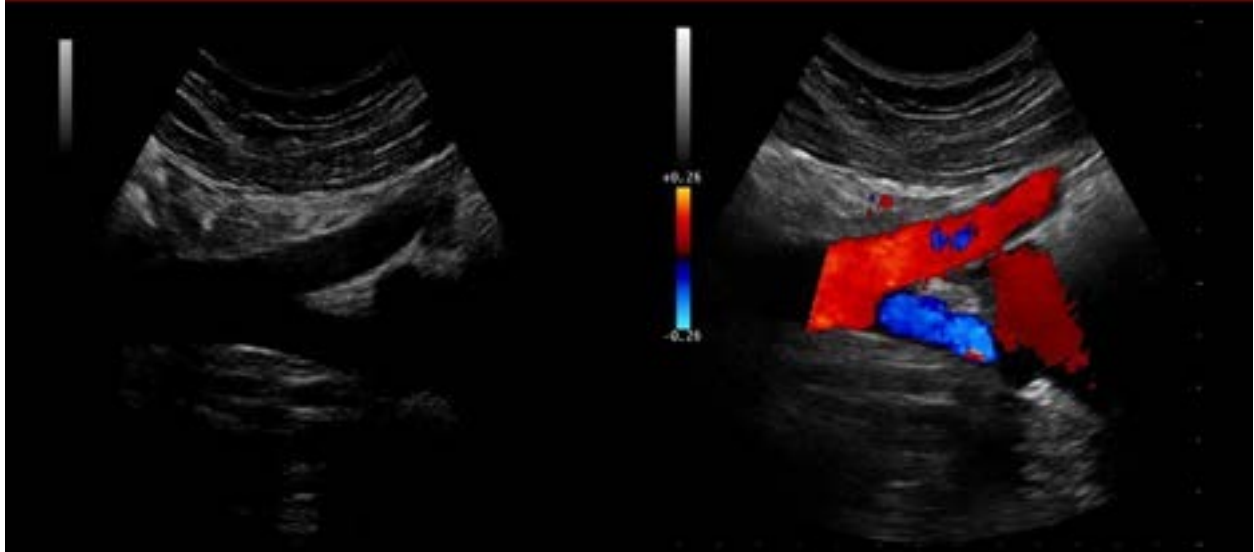
Aortic Bifurcation – Iliac Arteries



Common Iliac Artery



Aortic Bifurcation



To obtain a view of both iliac arteries at the bifurcation, use a lateral or coronal approach. It may also be helpful to place the patient in the decubitus position.

Doppler of the Aorta:

- High resistance, plug flow
- Average velocity 70 to 100 cm/sec
- Clean spectral window
- Biphasic above renal arteries due to numerous low resistance branches (organ arteries)
- Resistance in the aorta increases as flow moves distally
- Triphasic below renal arteries due to high resistance branches (iliacs)
- When evaluated as part of a mesenteric, iliac or renal artery exam the aorta is usually assessed with Doppler just proximal to the branch of interest to compare flow patterns
- Occlusive aortic disease may change resistance and pulsatility of waveform; increased velocity and waveform changes will be transmitted into branches
- Aneurysmal disease may show increased turbulence and non-laminar flow, especially with color Doppler; decreased velocity and waveform changes will be transmitted into branches

Aortic Doppler

Note the reduction in antegrade diastolic flow in the distal aorta. The proximal aortic segment supplies several low resistance vascular beds (liver, spleen, kidneys) that require continuous antegrade flow throughout the cardiac cycle.

The distal aortic segment supplies mostly high resistance vascular beds (extremities).

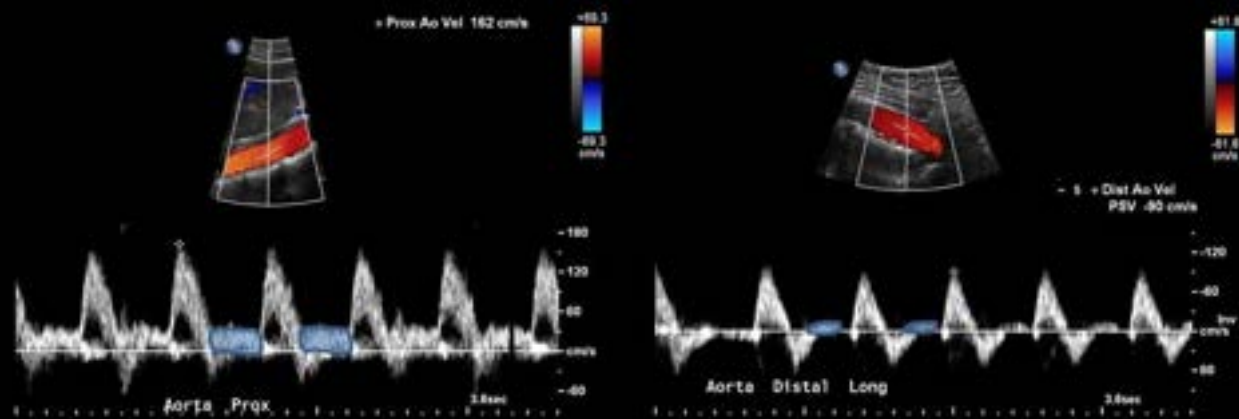


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Limitations:

- Bowel gas
- Patient not NPO
- Obesity
- Abdomen pain with probe pressure
- Patient unable to follow breathing instructions

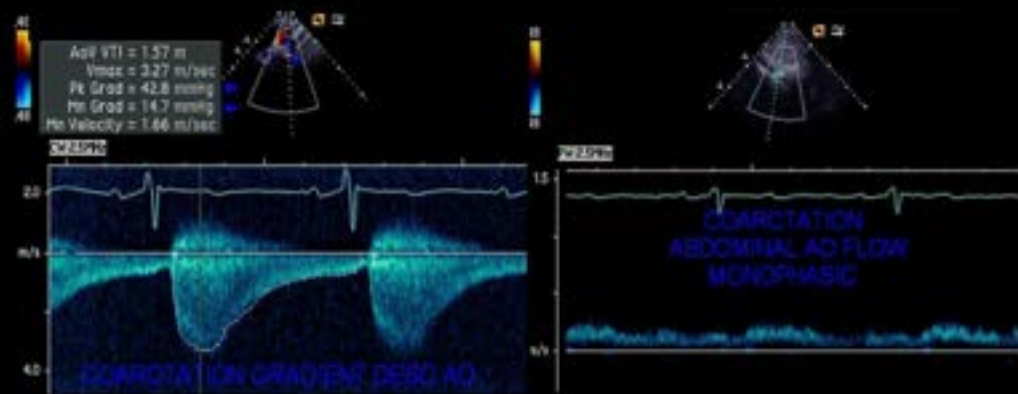
AORTIC PATHOLOGY

Coarctation:

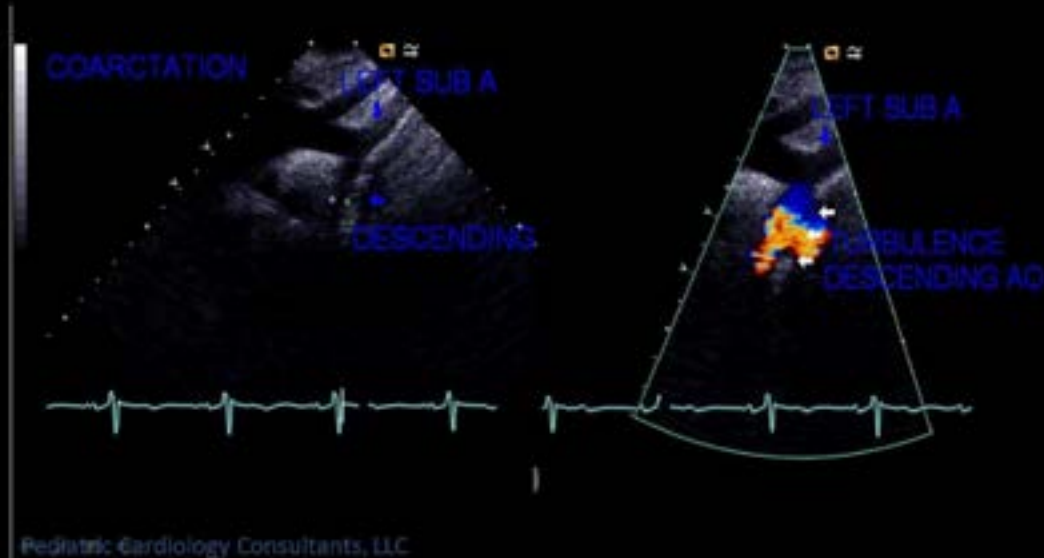
- Congenital narrowing of the aorta
- Usually occurs with other heart defects and is a component of Shone complex
- Most commonly occurs distal to the origin of the left subclavian
- Decreased flow distal to the obstruction
- Monophasic flow with continuous flow in diastole identified in the abdominal aorta
- Causes lower extremity ischemia

- Decreased bilateral pedal pulses
- Decreased bilateral ankle pressures; brachial pressures may be elevated if renal inflow reduced
- Because the arch is proximal to the obstruction, flow to the head and neck is usually normal
- Systemic HTN is usually present because the coarct causes renal ischemia and the renin-angiotensin system is activated causing increased systemic pressure
- If aortic coarctation occurs just distal to the origin of the left common carotid artery, BP will be reduced in the left arm and increased in the right arm

Coarctation



Coarctation



Takayasu Arteritis:

- Presents as hypertrophic area(s) of inflamed tissue usually found in the aorta and its branches
- Associated with supralvalvular aortic stenosis and aortic coarctation
- Commonly affects the subclavian and carotid arteries
- Can also affect the renal arteries
- Most commonly seen in young females

Aortic Ectasia:

- Lack of tapering of the aorta as it travels distally, size remains constant from proximal to distal portions
- Can be a precursor to aneurysm formation

Abdominal Aortic Aneurysm: (AAA)

- True aneurysm
- Most commonly caused by degenerative disease
- Caused by a breakdown of the media and adventitia layer of the wall
- Symptoms include pulsatile abdominal mass, low back pain, abdominal bruit

- Family history of AAA = increased risk 4X
- If the family member is in your immediate family, risk increases 12X
- Associated with Ehler-Danlos syndrome (reduced and abnormal collagen formation)
- >60% of patients with a popliteal aneurysm will have an aortic aneurysm
- Ruptured AAA can cause hypotension, severe abdominal pain, shock, death
- Most commonly forms in the infrarenal aortic segment

Types:

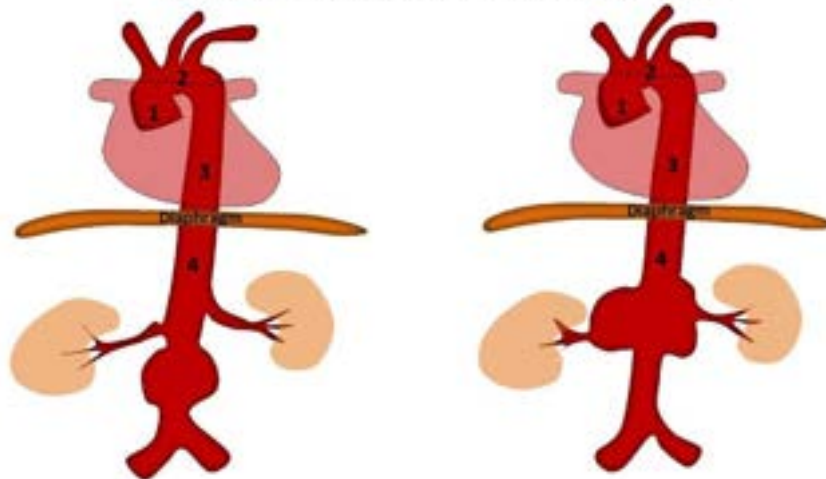
1. Fusiform - vessel wall stretches in a circumferential manner; all three wall layers intact
2. Saccular - formed with a stalk connecting dilated wall portion to main vessel; usually associated with wall dissection
3. Berry - tiny out pouching, usually found in cerebrum/circle of Willis; usually congenital
4. Mycotic - due to infection in the artery and the arterial wall; syphilis most common
5. Vasculitis - due to inflammatory process of vasa vasorum and tunica adventitia
 - Focal dilatation of the aorta >3cm; AP diameter is the most reliable measurement for assessing AAA
 - Another method used to identify an aneurysm is if there is a 50% increase in the diameter of an aortic segment compared to the adjacent unaffected segment
 - Above the Renal Arteries - Surgical intervention asap
 - Below the Renal Arteries - Most common; surgical intervention at a diameter >5.5cm
 - Serial exams will be scheduled more frequently as the aneurysm size increases; one exam per year 3-4cm size, exam every 3 months 5cm in size
 - Aneurysm diameter change of >4 mm on a yearly follow up exam, the next follow-up should be at 3 months
 - Arteriography normally performed prior to surgery; CTA also commonly performed to obtain most accurate diameters of the aorta
 - Iliac artery considered aneurysmal when diameter exceeds 1.5cm or when the diameter increases by 50% as compared to the adjacent segment
 - Stent or graft placement usual treatment

Complications:

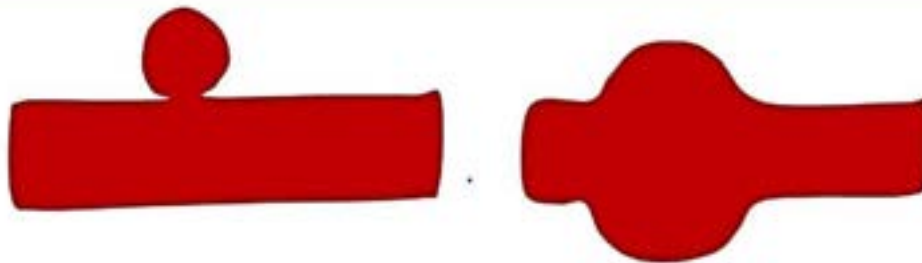
- Rupture is the most common and most critical complication
- Decreased flow to extremities
- Blue Toe Syndrome with thrombus accumulation and embolization to the toes; acute ischemia causes color changes in the toes
- Affects renal circulation and systemic BP if proximal to renal artery origins, if kidneys are ischemic they raise BP to increase flow

Aorta

Any aneurysm above the renal arteries (Suprarenal) or at the level of the renal arteries (Juxtarenal) is considered critical and treated immediately.



Saccular vs Fusiform



***Increased risk of rupture due to increased pressure in the neck of the aneurysm.

***A saccular cerebral aneurysm is termed a berry aneurysm. This is a critical finding with immediate surgical repair required.

***Most common type of AAA

***Infrarenal type monitored until surgical intervention required at > 5.5cm

Sonographic Appearance:

- Noting AAA location in relation to the renal arteries is very important for treatment considerations
- Measure outer wall to outer wall

- The Anterior-Posterior (AP) measurement is considered more reliable than the transverse diameter
- Aneurysmal >3cm diameter
- Possible thrombus accumulation causes increased echogenicity in the lumen
- Turbulence seen with color and PW Doppler evaluation, Yin Yang sign on color shows swirling flow
- Aortic rupture can demonstrate free fluid in the perirenal space between the kidney and psoas muscle; may see debris due to blood content

Endovascular Aortic Repair (EVAR):

- Used to improve and streamline flow through a stenosis or AAA
- Endovascular graft easier and better for patient than external graft or surgical resection of a AAA
- Kissing stents used for stenosis at the origin of the common iliac arteries
 - Two stents are deployed, one is placed into each iliac artery
 - Both stents are inflated with the catheter balloon
 - The proximal ends of the stents are touching in the distal aorta (kissing)

Intravascular Ultrasound (IVUS):

- Can be used to guide the graft during placement
- Physician uses an ultrasound catheter to visualize the lumen prior to the procedure and evaluate the best location for the endoluminal graft placement

Advantages of IVUS over Angiography during Angioplasty:

- No need to cut the aorta to insert the graft
- No radiation
- No contrast required
- Can provide reconstructed 2D and 3D images of the vessel and lumen

Limitations of IVUS:

- Catheter size limits the vessels that can be evaluated
- Introduction of the catheter can lead to arterial spasm
- Significant atherosclerosis can blur the image
- Images suffer from ringdown artifact caused by dead space

- Cost of the disposable ultrasound catheter; cannot be heated/sterilized because the piezoelectric elements will lose their US properties

Intraoperative Duplex Ultrasound:

- Can be used to assess the flow changes during and after the procedure
- Surgeon places transducer directly on the open wound or exposed vessel; requires sterile technique
- Sonographer may be asked to assess distal flow at the ankles during and after the procedure (outside of sterile field)
- Recovery room monitoring includes pressure cuffs, PVR, PPG, PW Doppler or CW Doppler evaluation of the ankle/feet/toes
- Ankle pressures should demonstrate a mild increase after the graft is placed

Complications:

- Stenosis
- Thrombosis
- Endoleak
 - blood leakage outside the graft but inside the aneurysm being treated by the graft
 - increased aneurysm size after graft placement indicates a possible endoleak

Four Types of Endoleaks:

Type I : leak at the site of attachment

- Type Ia = leak at proximal attachment site, Type Ib = leak at distal attachment site
- Flow pattern resembles the flow pattern within the graft

Type II : leak at the branch vessels

- Will see retrograde flow into the aneurysm sac and increase in AAA diameter
- Inferior mesenteric artery, internal iliac, lumbar arteries and accessory renal arteries are potential sources for the leak
- Usually demonstrates a bidirectional flow pattern, Doppler waveform morphology reflects the end source resistance
- If flow moves from a branch vessel into the sac, flow will be bidirectional

Type III : leak caused by a tear in the graft or separation of the main graft from the iliac limbs

Type IV : transgraft leak through porous material of the graft (no tear)

***Some resources acknowledge a Type V: endotension- no leak present but there is continued expansion of the aneurysm sac greater than 5 mm

- Type II is most common and usually does not require intervention
- Type I is most dangerous and requires surgical intervention

Post-Surgical Evaluation:

- Performed at 1 month post-op, then at 3, 6, 9, 12 and 18 months, followed by annual assessment after the first year
- Be sure to review the details of the endovascular repair prior to beginning the exam
- Measure the outer wall to outer wall diameter of the AAA to compare to prior exam (should see NO increase in size)
- When assessing the aortic sac for enlargement on serial exams, be sure you are comparing measurements taken from the same aortic segment on both exams
- >0.5cm increase in diameter indicates a possible endoleak
- Evaluate the structure and location of the graft for compression or tortuous segments
- Document the proximal attachment site and confirm it has not migrated distally
- Use color Doppler to evaluate the graft for potential endoleak
- The color box should be large enough to encompass the AAA sac, but not too large to stimulate artifact
- No color flow should be seen in the sac
- Any color flow demonstrated within the sac should also be evaluated by PW Doppler
- Color Doppler alone cannot diagnose an endoleak
- Because low velocity and filter settings are used for the exam, color artifact can complicate the exam
 - Calcification of the aortic wall can cause twinkle artifact
 - Patient movement and respiration can cause flash artifact
- A suspected leak on color Doppler must be confirmed by PW Doppler
- Evaluate flow velocities in the aorta proximal to the graft, at the proximal attachment, proximal graft, mid graft, distal graft and each graft limb with their distal anastomosis sites
- PSV and EDV measurements should be recorded
- A true endoleak will have reproducible arterial waveforms that are easily differentiated from the graft flow
- Potential signs of an unstable AAA sac
 - Echolucent areas within the sac
 - Changes in aneurysm shape and/or sac size
 - Increased pulsatility of the sac

Dissection:

- Intimal layer tears and allows flow between the intima and media layers into a blind pocket (false lumen)
- Remaining lumen (true lumen) is decreased in size due to flap and flow beneath
- Causes weakened vessel wall, risk of vessel rupture
- Thrombosis can occur in the false lumen causing a significant stenosis/occlusion of the true lumen leading to distal ischemia
- Most commonly found in the ascending aorta, the second most common site is just distal to the left subclavian artery
- A significant aneurysm of the ascending aorta aneurysm (>5cm) can lead to dissection that enter the branches on the arch - critical finding, can be fatal
- Increased risk with connective tissue disorders like Marfan syndrome and Ehler-Danlos syndrome
- CTA is preferred for diagnosis
- Surgical intervention required

Sonographic Appearance:

- Linear echogenicity seen in the lumen of the vessel, separating it into two channels
- Color flow demonstrates two lumens, both with turbulence
- Bidirectional flow seen in false lumen
- Intimal flap can cause stenosis in the true lumen
- Stanford Classification:
 - Type A: dissections that involve the ascending aorta and arch
 - Type B: dissections that involve the aorta from the left subclavian artery to the iliac bifurcation
- DeBakey Classification:
 - Type I: dissections that involve the ascending and descending aorta (= Stanford A)
 - Type II: dissections that involve only the ascending aorta (= Stanford A)
 - Type III: dissections that involve the descending aorta only, starting after the origin of the left subclavian artery (= Stanford B)

Aortic Stenosis:

- Most commonly caused by atherosclerotic changes
- Can also be related to AAA thrombosis or arteritis
- Resistance will increase proximal to the stenosis
- Causes increased velocity at the site of the stenosis
- Distal to the stenosis the flow will be dampened and possibly tardus parvus

- If the stenosis is superior to the renal artery origins, renal ischemia will activate the renin-angiotensin system causing systemic HTN
- Effects will be similar to coarctation with increased brachial pressures and decreased bilateral ankle pressures
- Treatment includes: Stent, balloon angioplasty, endarterectomy or bypass

Leriche Syndrome:

- AKA aortoiliac occlusive disease
- Due to occlusion of the abdominal aorta just above the site of its bifurcation
- Causes bilateral symptoms and flow changes in the lower extremities
- Symptoms include fatigue of both lower limbs, intermittent bilateral claudication with ischemic pain, absent or diminished femoral pulses and pallor or coldness of both lower extremities
- Doppler waveforms will demonstrate low resistance, post-stenotic flow changes throughout both legs

Retroperitoneal Fibrosis:

- AKA Ormond disease
- Most commonly occurs at the level of the aortic bifurcation and inferiorly in pelvis
- Idiopathic(usually) overgrowth of fibrous tissue around an atherosclerotic aorta
- Can be related to drugs, infection, malignancy or cancer therapy
- May lead to ureteral obstruction causing hydronephrosis
- May compress the IVC causing bilateral pedal edema
- May compress the gonadal veins causing scrotal swelling

Sonographic Appearance:

- Soft tissue mass surrounding great vessels
- Hypoechoic
- Smooth borders

Aorta “Most Commons”

Bovine Arch is the most common anatomic variant of the aortic arch
Coarctation most commonly occurs just distal to the origin of the left subclavian artery
Takayasu Arteritis most commonly occurs in females
AAA is most commonly caused by degenerative disease
AAA most commonly occurs in the infrarenal aortic segment
Syphilis is the most common cause of a mycotic aneurysm
Rupture is the most common and most critical complication of AAA
Type II endoleak is most common and usually does not require intervention
Type I endoleak is most dangerous and requires surgical intervention
Aortic dissection is most commonly found in the ascending aorta segment, while the second most common location is the descending aorta just below the isthmus
Connective tissue disorders are commonly associated with aortic aneurysm/dissection
Aortic stenosis is most commonly caused by atherosclerotic changes
Retroperitoneal fibrosis most commonly occurs at the level of the aortic bifurcation and extends inferior into pelvis

Mesenteric Vessels: Sonographic Evaluation and Disease

Indications to Scan the Mesenteric Vessels:

- Post-prandial pain
- Abdominal bruit
- Recent weight loss
- "fear" of food/eating (sitophobia)
- Diarrhea
- Nausea and vomiting

Limitations for the Exam:

- Open abdominal wounds, drains, sutures or staples
- Patient not NPO
- Overlying bowel gas - Simethicone may be administered to reduce gas
- Obesity
- Patient cooperation, inability to comply with holding breath

Vessels Included in Exam:

- Proximal abdominal aorta
- Celiac axis (CA)
- Common hepatic artery (CHA)
- Splenic artery (SA)
- Superior mesenteric artery (SMA)***
- Inferior mesenteric artery (IMA)***

***Proximal, Mid and Distal portions of each mesenteric artery should be evaluated with Doppler

CELIAC AXIS: (CA)***Anatomy:***

- Anterior branch of aorta
- Originates from the aorta within first 2 cm below diaphragm
- Normally 1-3cm in length
- Superior to the body of the pancreas
- Splits into the Common Hepatic Artery (CHA), Left Gastric Artery (LGA), Splenic Artery (SA)
- In some patients the celiac artery and the SMA may share a common trunk
- In some patients the hepatic artery may originate from the SMA
- In some patients the hepatic artery may originate directly from the aorta

***Exam Techniques:***

- 2.5MHz to 4MHz curvilinear array - adult
- 4MHz to 8MHz curvilinear array - pediatric probe
- Patient must be NPO to best visualize the abdominal vasculature
- 2D, Color and Doppler evaluation of celiac trunk and branches

US of Celiac Axis:

- Transverse-origin and bifurcation, looks like a "T", bird sign, seagull sign
- Longitudinal-profile for Doppler assessment

Doppler of the Celiac Axis:

- Low resistance waveform; feeds organs requiring constant flow
- Flow pattern does NOT normally change after eating
- Evaluated at the origin from the aorta, usually in the longitudinal plane
- Occlusive disease may cause increased velocity and resistance
- Extrinsic compression can also affect blood flow in the celiac axis
- NL PSV < 2.0 m/s
- ABNL PSV >2.0 m/s indicates 70-99% stenosis
- ABNL EDV >0.55 m/s indicates >50% stenosis
- Celiac occlusion leads to retrograde flow through the gastroduodenal artery to supply blood to the common hepatic artery to maintain blood supply to the liver and spleen
- The common hepatic trunk may also demonstrate retrograde flow as it sends blood to maintain blood supply to the spleen

BRANCHES OF CELIAC AXIS:

1. Left Gastric Artery (LGA):

- Smallest branch of the CA
- Travels anterior and cephalad
- Supplies the stomach and pylorus
- Not usually seen sonographically due to course and structures supplied

2. Common Hepatic Artery (CHA):

Anatomy:

- Right branch of celiac axis (right wing of bird sign)
- Supplies the liver, gallbladder and stomach
- Branches into the proper hepatic artery which enters the liver and the gastroduodenal artery that supplies the pancreas and duodenum
- Hepatic artery supplies 20-30% of the blood to the liver; oxygenated blood 95% O₂ saturation

Indications to Scan the Common Hepatic Artery:

- Chronic liver disease
- Liver transplant (pre- or post-)
- Trauma

Anatomic Variants:

- Hepatic artery originates from SMA (most common variant)
- Hepatic artery originates from aorta

Exam Techniques:

US of Common Hepatic Artery:

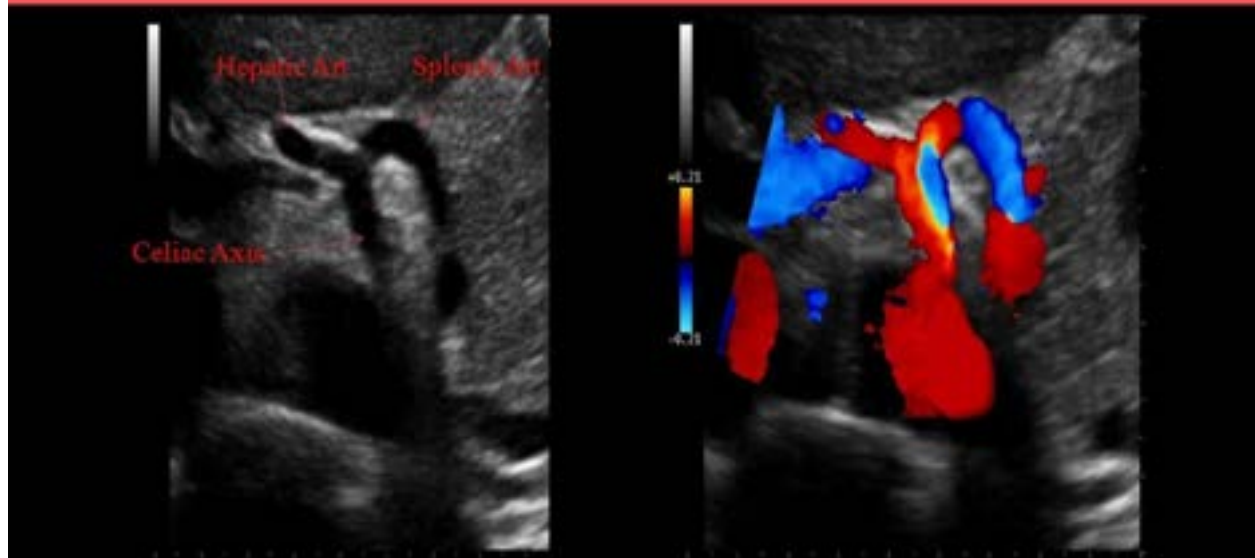
- Best visualized in the transverse plane, anterior approach to view the area around the porta hepatis
- Right wing of bird sign
- Travels along the superior border of the head of the pancreas

Doppler of Hepatic Artery:

- Low resistance (organ flow)
- Tortuous
- Trace waveform for Resistive Index and Peak Systolic Velocity

- Flow increases with cirrhosis, metastasis, lymphoma
- RI >0.78 implies portal HTN

Celiac Axis



Normal Hepatic Artery



3. Splenic Artery (SA):

Anatomy:

- Largest branch of the celiac axis
- Tortuous course posterior to body and proximal tail of pancreas
- MOST tortuous artery in body
- Supplies spleen, pancreas, and fundus of stomach
- Most common site of aneurysm of a visceral vessel; more common in females

Exam Techniques:

US of Splenic Artery:

- Origin and proximal portion best visualized in the transverse plane
- Distal segment best evaluated through the splenic hilum from a left lateral window
- Left wing of bird sign
- Travels posterior and superior to the pancreas body/tail

Doppler of Splenic Artery:

- Low resistance
- Tortuous, may normally find areas of high PSV
- Trace waveform for Resistive Index and Peak Systolic Velocity

SUPERIOR MESENTERIC ARTERY: (SMA)

Anatomy:

- Anterior branch of aorta originates about 1-2 cm below the celiac axis
- Runs parallel to the aorta posterior to the body of the pancreas
- Supplies blood to the jejunum, ileum, cecum, ascending colon, proximal 2/3 transverse colon, duodenum and pancreatic head
- Communicates with celiac artery via the pancreaticoduodenal arcade
- Terminates near the ileocecal valve

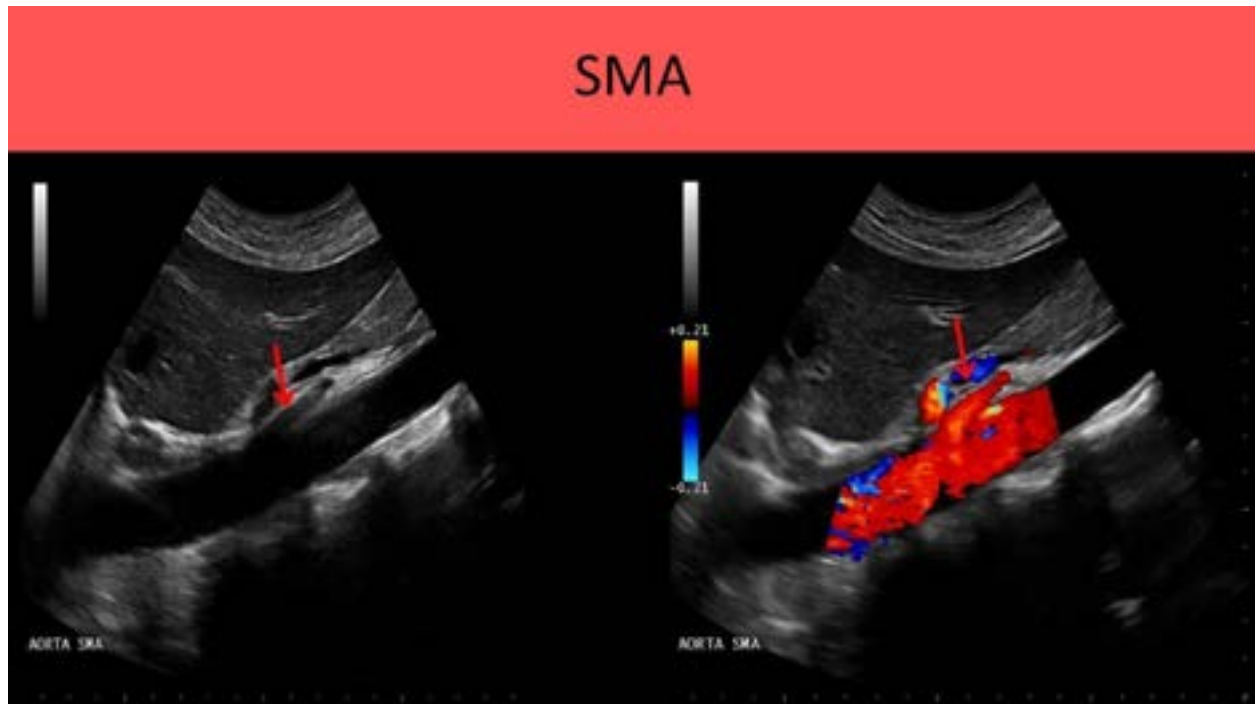
Exam Techniques:

- 2.5MHz to 4MHz curvilinear array - adult
- 4MHz to 8MHz curvilinear array - pediatric
- Patient must be NPO to best visualize the abdominal vasculature
- 2D, Color and Doppler evaluation
- Most patients should be NPO prior to the exam, but always ask the patient when they last ate

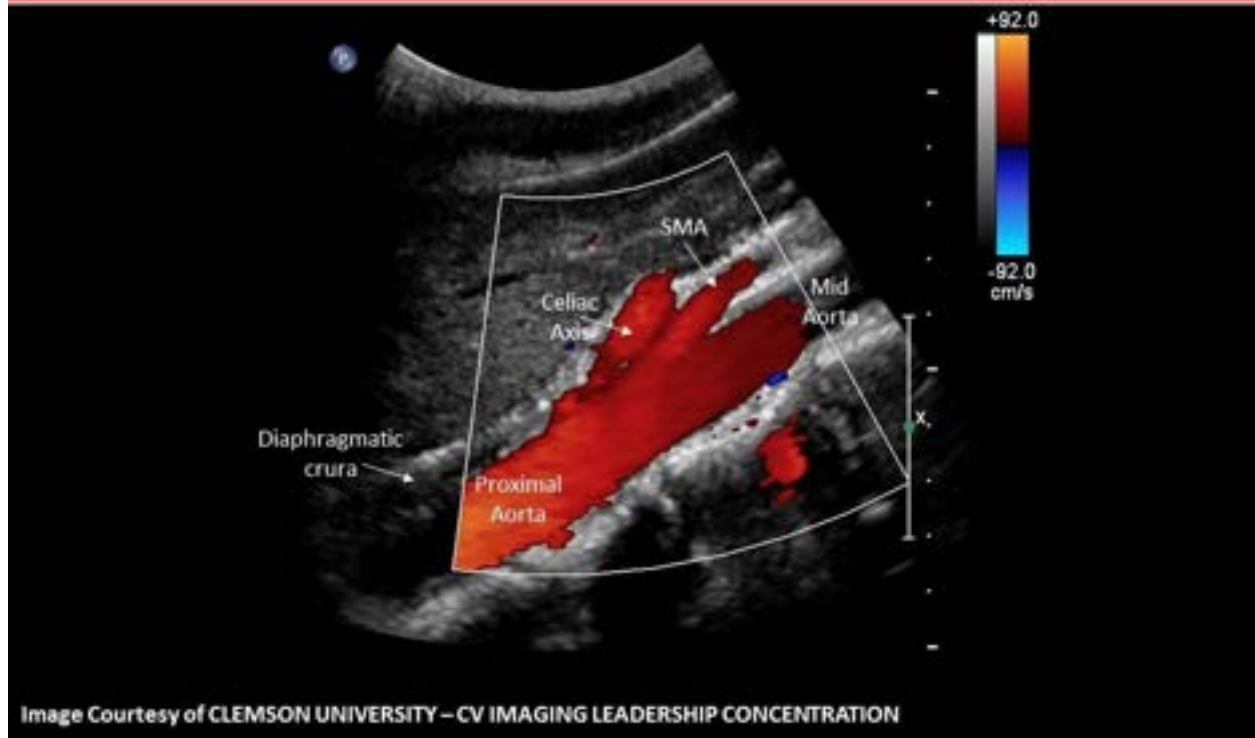
- The SMA waveform is different when evaluated pre- vs. post-prandial

US and Doppler of SMA:

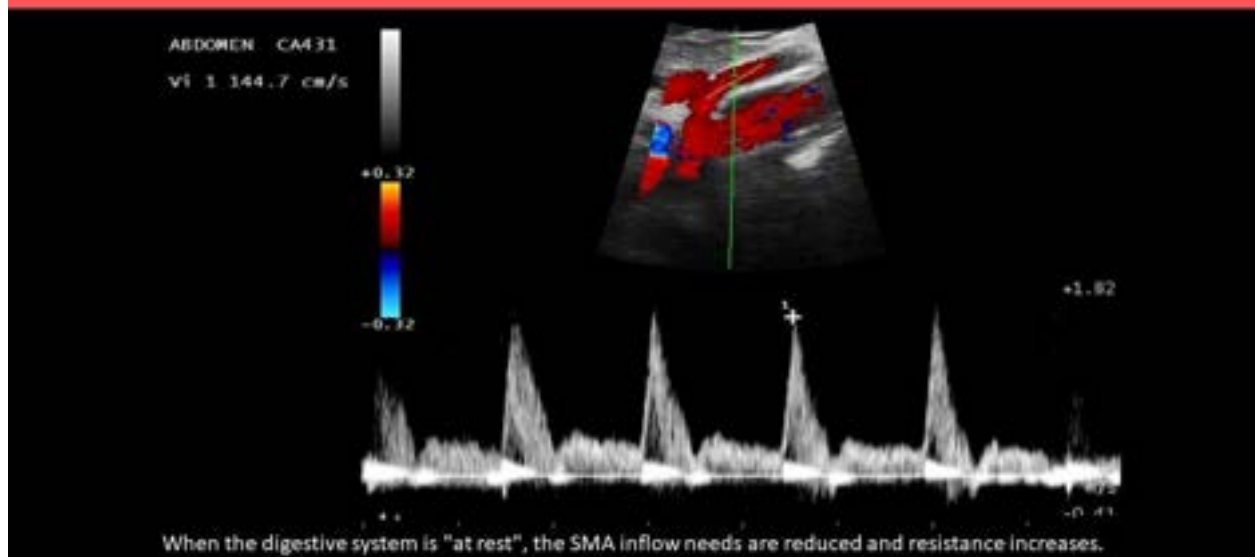
- Longitudinal - tubular structure coursing parallel to anterior aorta
- Transverse - circular structure anterior to aorta and posterior to pancreas body
- Pre-prandial - high resistance flow with minimal flow in late diastole; peaked waveform
- Post-prandial - low resistance flow with continuous flow in diastole and rounded waveform
- If no resistance changes occur stenosis is present
- PSV can also increase after eating
- Intestines need continuous flow while digestion occurs, if stenosis is present, intestinal angina can occur

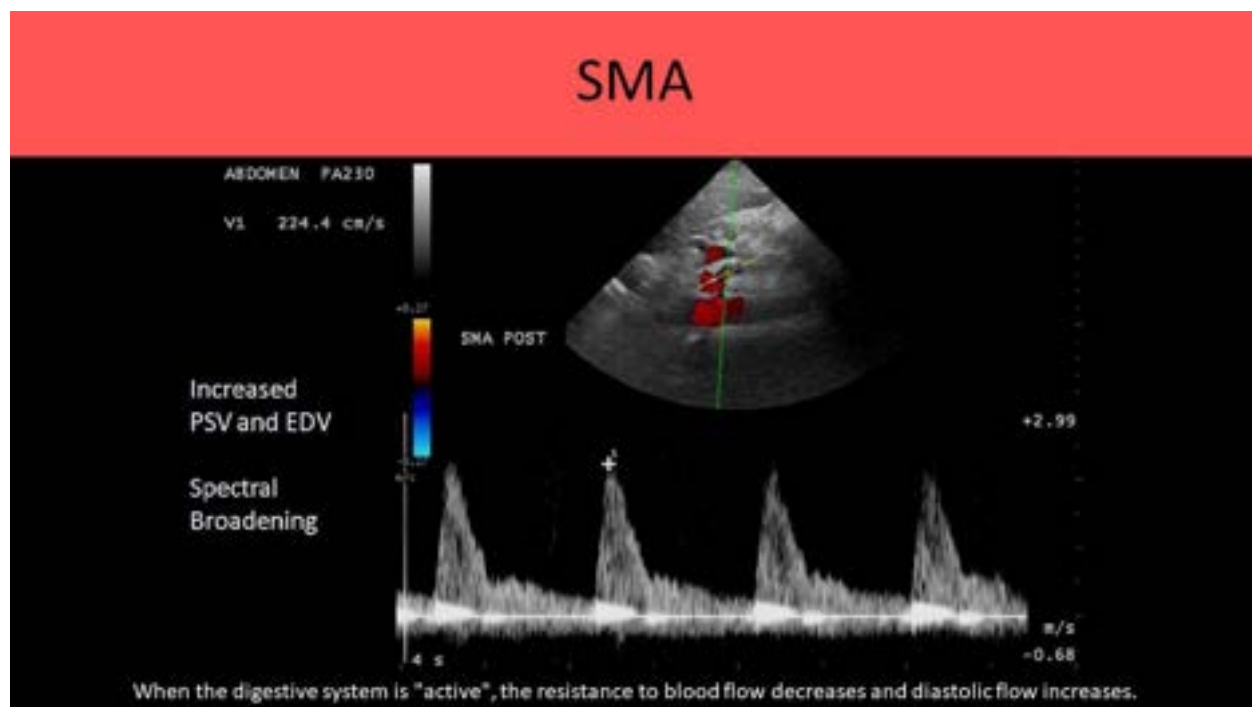


Aorta



SMA





INFERIOR MESENTERIC ARTERY: (IMA)

- Originates 3-4cm above the aortic bifurcation, on the left anterolateral aspect of the aorta and courses inferiorly to the left
- IMA provides blood supply to the distal colon and proximal rectum
- SMA communicates with the IMA through the arc of Rioloan and the marginal artery of Drummond; these vessels provide the potential for collateral flow with mesenteric arterial obstruction
- Courses inferiorly to supply the distal colon
- Gas artifact usually impedes visualization
- Can be identified using a transverse view and locating the renal arteries, then sliding inferiorly; IMA normally visualized originating from the left anterolateral aspect of the aorta
- IMA can also be found by identifying aortic bifurcation then sliding slightly superiorly
- Doppler waveform is usually high resistance with minimal late diastolic flow
- If the IMA is easily identified by ultrasound, SMA stenosis or occlusion should be suspected
- It usually means it is dilated due to compensatory flow related to an SMA stenosis

Inferior Mesenteric Artery

To locate the IMA, find the renal artery origins and slide inferior OR find the aortic bifurcation and slide superior



Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Normal Inferior Mesenteric Artery

Normal IMA flow demonstrates a triphasic or biphasic, high resistance Doppler waveform. Stenosis would cause increased velocity and reduced resistance with antegrade diastolic flow.



Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

ABNORMALITIES OF MESENTERIC ARTERIES:

Mesenteric Ischemia:

- Pre-prandial duplex evaluation of the celiac artery, splenic artery, common hepatic artery, proximal, mid and distal SMA, IMA and aorta
- Patient given a high calorie LIQUID meal to stimulate the digestive system, while minimizing gas production from digestion
- Post-prandial evaluation of the same group of arteries should be performed 20 - 30 minutes after the meal OR when the patient has symptoms
- The time between ingestion of the meal and Doppler evaluation should be documented, along with any patient symptoms experienced during the exam
- The celiac artery, hepatic artery and splenic artery waveforms should not demonstrate changes in flow after eating
- Normal SMA and IMA flow pre-prandial = high resistance, post-prandial = lower resistance with increased diastolic flow
- Abnormal SMA and IMA flow post-prandial = no change in resistance
- Intraluminal stenosis can occur due to atherosclerosis, arteritis or fibromuscular dysplasia
- Mesenteric ischemia is diagnosed with stenosis/occlusion of *two or more of the three major arteries* which supply the small bowel and colon with oxygenated blood (CA, SMA, IMA)
- The digestive tract requires more blood flow during the digestion process
- The patient experiences reproducible pain after eating due to bowel ischemia caused by stenosis
- May demonstrate an abdominal bruit
- Collateral pathways can form with chronic disease:
 - Pancreaticoduodenal arcade (gastrooduodenal artery to the pancreaticoduodenal arteries); most common
 - Arc of Riolo
 - Marginal artery of Drummond
- Portal venous gas may be identified in severe chronic cases; dirty shadowing from the portal structures
- Arteriography preferred for mesenteric, liver and splenic arterial evaluation
- CTA limited anatomy demonstrated and requires iodinated contrast
- MRA demonstrates necessary anatomy very well but requires long exam time and possible gadolinium injection

Doppler Evaluation:

- Mesenteric/Aortic Ratio is usually around 1.0
- Mesenteric/Aortic Ratio >3.0 indicates hemodynamically significant stenosis
- SMA

- .. PSV >2.75m/s > 70% stenosis (Moneta)
- .. EDV >0.45m/s >50% stenosis (Bowersox)
- .. Increased diastolic flow
 - Celiac Axis

- .. ABNL PSV >2.0 m/s indicates > 70% stenosis (Moneta)
- .. ABNL EDV >0.55 m/s indicates >50% stenosis
 - IMA

- .. ABNL PSV >2.0 m/s indicates stenosis (Pellerito)
- .. ABNL EDV >0.25 m/s indicates stenosis
 - If SMA and IMA are low resistance with increased diastolic flow even in a fasting patient, this indicates the capillary beds are constantly vasodilated due to ischemia
 - Celiac occlusion leads to retrograde flow through the gastroduodenal artery to supply blood to the common hepatic artery to maintain blood supply to the liver and spleen
 - The common hepatic trunk may also demonstrate retrograde flow as it sends blood to maintain blood supply to the spleen
 - If there is significant obstruction or occlusion of one mesenteric artery, but the other two main arteries are normal there can be associated changes in these normal vessels
 - If the SMA is occluded, the celiac artery may demonstrate flow with increased velocity and a normal spectral window (no post-stenotic turbulence)
 - If the SMA is occluded, the IMA can increase in size and demonstrate increased flow velocity without post-stenotic turbulence
 - If the celiac artery is occluded, the SMA can demonstrate increased flow velocity without post-stenotic turbulence
 - Keep in mind that compensatory flow can sometimes mimic stenosis
 - It is very important to evaluate the waveform characteristics with elevated systolic velocities
 - Compensatory flow = increased velocity + spectral window
 - Stenosis = increased velocity + post-stenotic turbulence

Treatment:

- Angioplasty
- Stent placement
- Bypass graft from aorta to location distal to stenosis in SMA or celiac
- Intravascular ultrasound can be performed during stent placement and for angioplasty guidance

- Flow resistance should increase and peak systolic and end diastolic velocities decrease postprocedure
- When evaluating mesenteric arteries that have been treated for stenosis, compare the velocities on the follow up exam to the pretreated velocity information
- The velocities should be reduced on the follow up exam, but may still be elevated compared to normal lab criteria for stenosis
- Stented visceral arteries will demonstrate increased velocity without the presence of stenosis
- The amount of reduction in velocity from the prestented values helps to determine the success of the procedure

Median Arcuate Ligament Syndrome:

- Can be mistaken for celiac stenosis with mesenteric stenosis
- Intermittent extrinsic compression of the celiac axis
- Caused by diaphragm moving superiorly and the median arcuate ligament pinching the celiac axis
- Rarely causes symptoms but may see pain with expiration

Sonographic Appearance:

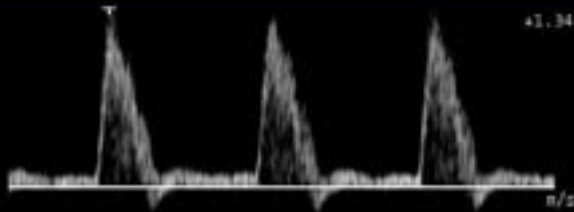
- Intermittent increased PSV in celiac axis with respiration maneuvers
- Velocities will increase with expiration and return to normal on inspiration

SMA Syndrome:

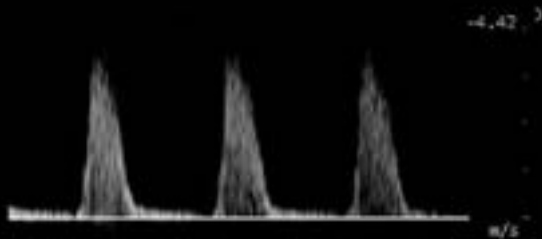
- AKA Wilkie syndrome
- If a patient suffers from severe weight loss, the fatty tissue around the SMA is diminished, causing acute angulation and reduction in the distance between the aorta and the superior mesenteric artery
- Acute angulation of SMA causes compression and obstruction of the third part of the duodenum
- Symptoms include: early satiety, recurrent episodes of abdominal pain, vomiting
- Related conditions include anorexia nervosa, malabsorption, malignancy, increased spinal lordosis, short ligament of Treitz, abnormally inferior SMA origin
- Usually requires surgical intervention

SMA - Abnormal

PRE-
Prandial



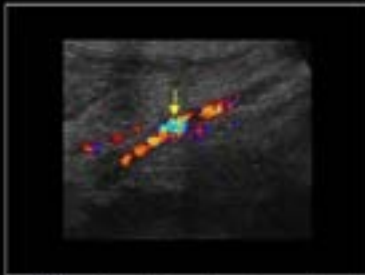
POST -
Prandial



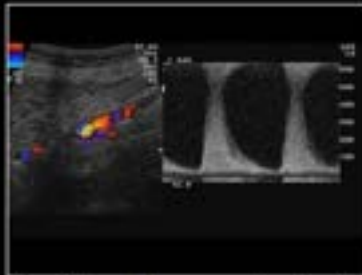
When the digestive system is "active" in a patient with functional SMA stenosis, the resistance to blood flow will remain high resistance after eating. Ischemia will lead to symptoms.

SMA - Stenosis

Significant SMA stenosis can present as a monophasic waveform with increased systolic and diastolic flow due to constant ischemia even pre-prandially. Note the significant increase in PSV along with the significant drop in resistance causing a large amount of flow to occur in diastole. The RI is also very low which indicates very low resistance in the mesenteric bed as it tries to compensate for the ischemia.



Aliasing at focal stenosis



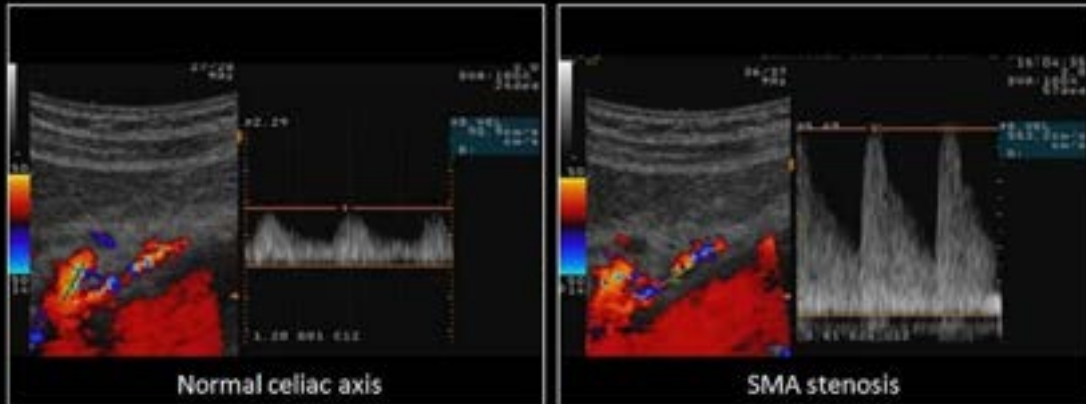
Significant increase in velocity
at focal stenosis



Distal to the focal stenosis

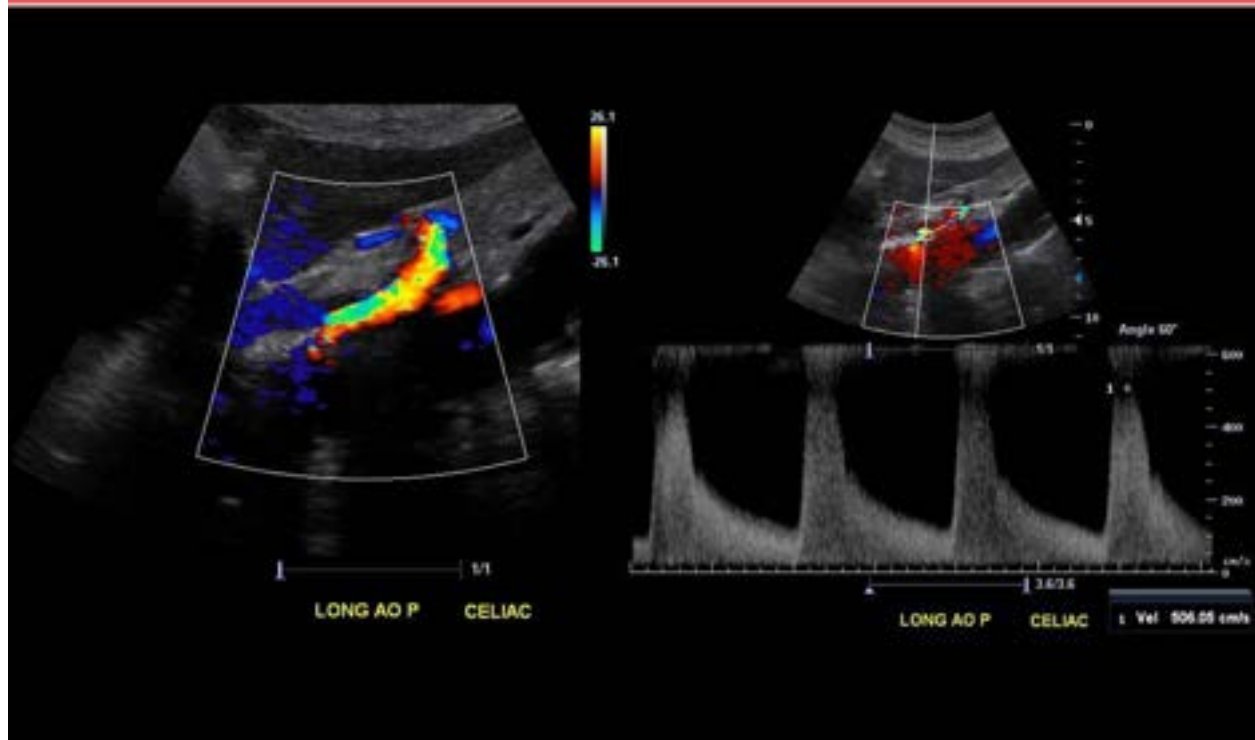
Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

SMA Stenosis



Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

Celiac Axis Stenosis



Renal Vessels: Sonographic Evaluation And Disease

Anatomy:

- Transverse or lateral branches of the aorta
- Originate 2-3 cm below SMA
- Supply the kidneys, adrenals, and ureters
- LRA originates from the posterolateral aspect of the aorta and courses anterior to the psoas muscle to reach the left kidney
- RRA originates from the anterolateral aspect of the aorta and courses posterior to IVC to reach the right kidney
- RRA origin usually slightly inferior compared to the LRA origin because of the location of the liver in the abdomen
- RRA is longer than LRA
- Renal arteries normally course posterior to the renal veins
- Locating the LRV coursing between the SMA and aorta is a landmark that can be used to identify the LRA
- Up to 30% of patients have multiple accessory renal arteries (most common anatomic variant)

Exam Techniques:

- 2MHz to 4MHz curvilinear probe - adult
- 4MHz to 8MHz curvilinear probe - pediatric
- Patient should be NPO 8-12hrs to best visualize the abdominal vasculature

- Simethicone may be administered to reduce bowel gas
- Place the patient in the supine position to evaluate the aorta and renal artery origins in the transverse plane
- Left lateral decubitus position is preferred for viewing the entire length of the right renal artery
- Right lateral decubitus position is preferred for viewing the entire length of the left renal artery
- 2D, Color and Doppler evaluation of the kidneys for renal artery stenosis, AV Fistula, renal vein thrombosis or other vascular abnormality
- Origin of renal arteries most common site of stenosis, most important to evaluate for stenosis

US of the Renal Arteries:

- Document renal length and width for both kidneys
 - <2cm difference in length is normal
 - >2cm difference in length can indicate renal disease or RAS in the smaller kidney
- Origin of the arteries is best seen in transverse view
- Find SMA and slide inferiorly
- RRA originates from the aorta at the 10 o'clock position
- LRA originates from the aorta at the 3-4 o'clock position
- Banana Peel view
 - Patient in left lateral decubitus position
 - Transducer is placed in the longitudinal position on the lateral right abdomen and angled through the abdomen to visualize the IVC and aorta on the same image
 - The liver serves as an acoustic window
 - Both renal arteries should be identified "peeling" off from the aorta
 - RRA is the anterior branch seen on the image and LRA is the posterior branch
- Decubitus position with a posterolateral transducer approach is preferred to evaluate the length of the artery
- RRA evaluated with the patient in the left lateral decubitus, LRA evaluated with the patient in the right lateral decubitus
- Use color Doppler to aid visualization

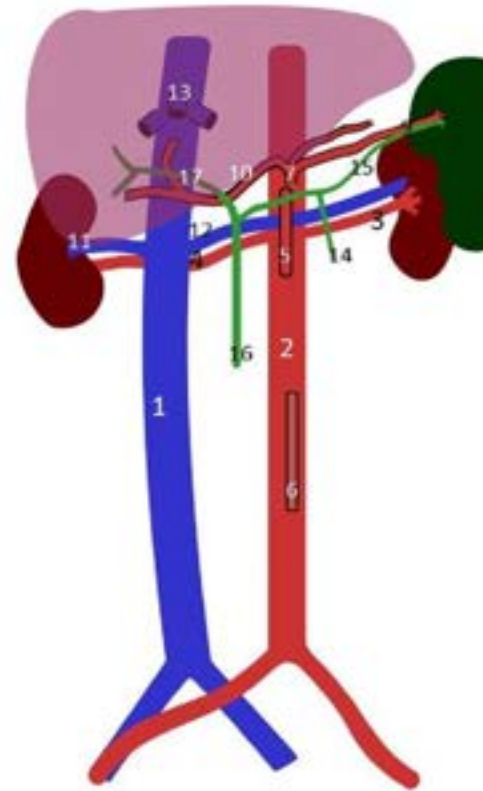
Doppler of the Renal Arteries:

- Use color to locate the RRA posterior to the IVC and LRA posterior to the LRV
- Use Doppler angles of 60 degrees or less to prevent artifactually increasing the PSV measurement
- Use Doppler sample size of 2mm for renal arteries
- Doppler at the origin, mid and distal artery, then trace for peak systolic velocity and resistive index
- Doppler interlobar (parenchymal) vessels of kidneys also; assess RI and AT
- Low resistance flow, rapid systolic upstroke with increased diastolic flow

- Normal flow demonstrates an early systolic peak (ESP)

Abdominal Vasculature

1. IVC
2. Aorta
3. Left renal artery
4. Right renal artery
5. Superior mesenteric artery
6. Inferior mesenteric artery
7. Celiac axis
8. Left gastric artery
9. Splenic artery
10. Hepatic artery
11. Right renal vein
12. Left renal vein
13. Hepatic veins
14. Inferior mesenteric vein
15. Splenic vein
16. Superior mesenteric vein
17. Portal vein



Renal Vasculature



Renal Artery Origins – Transverse View

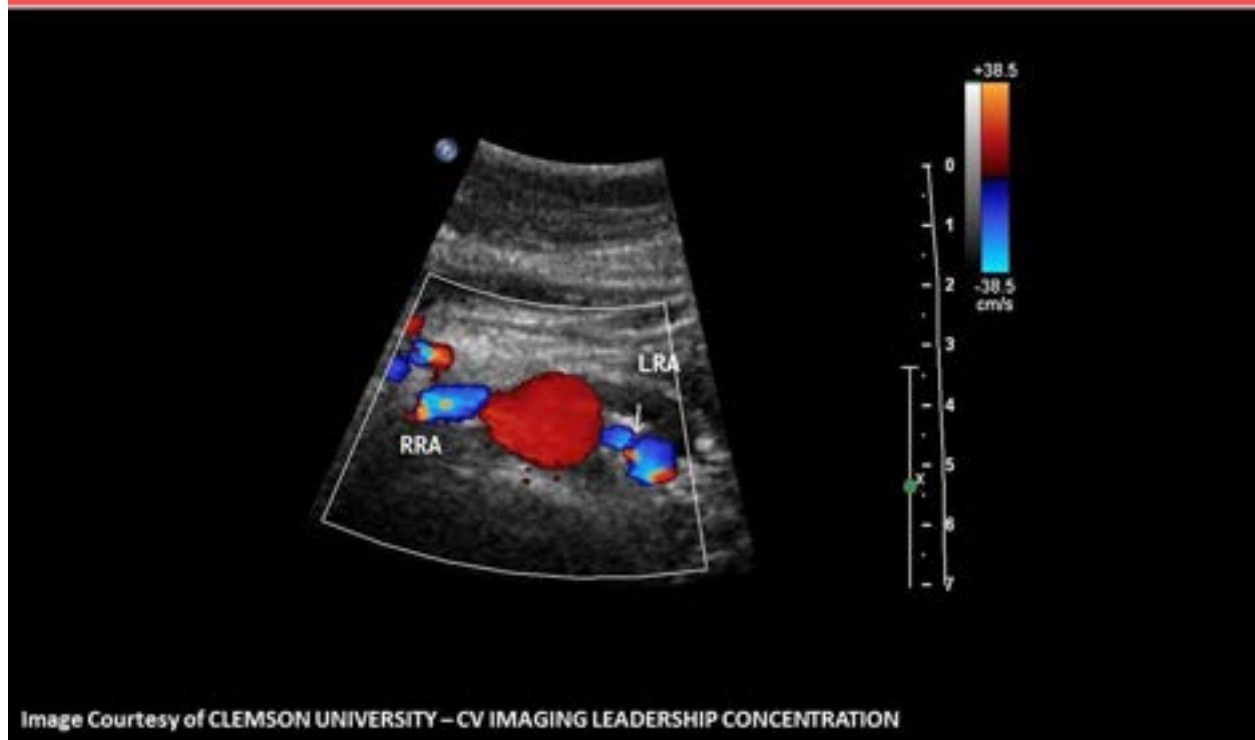
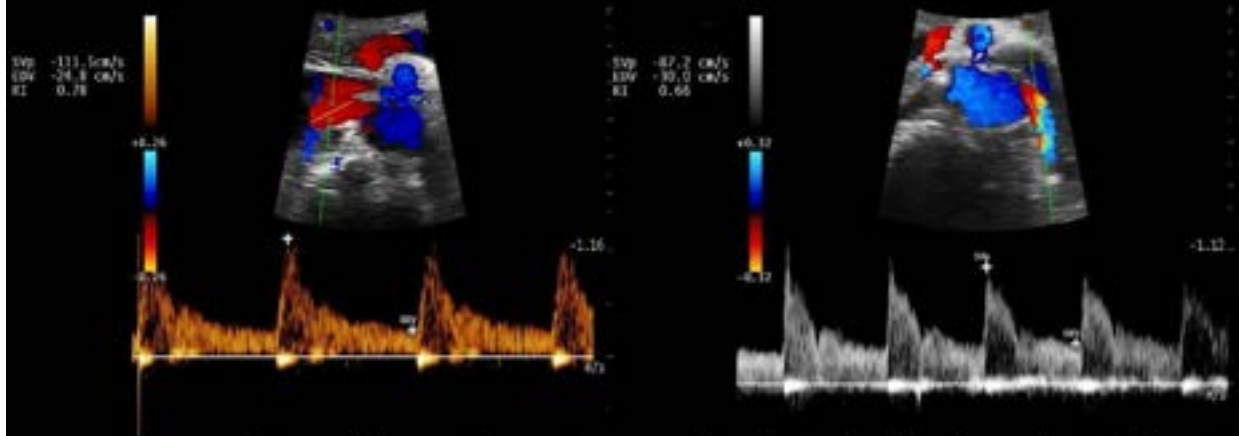


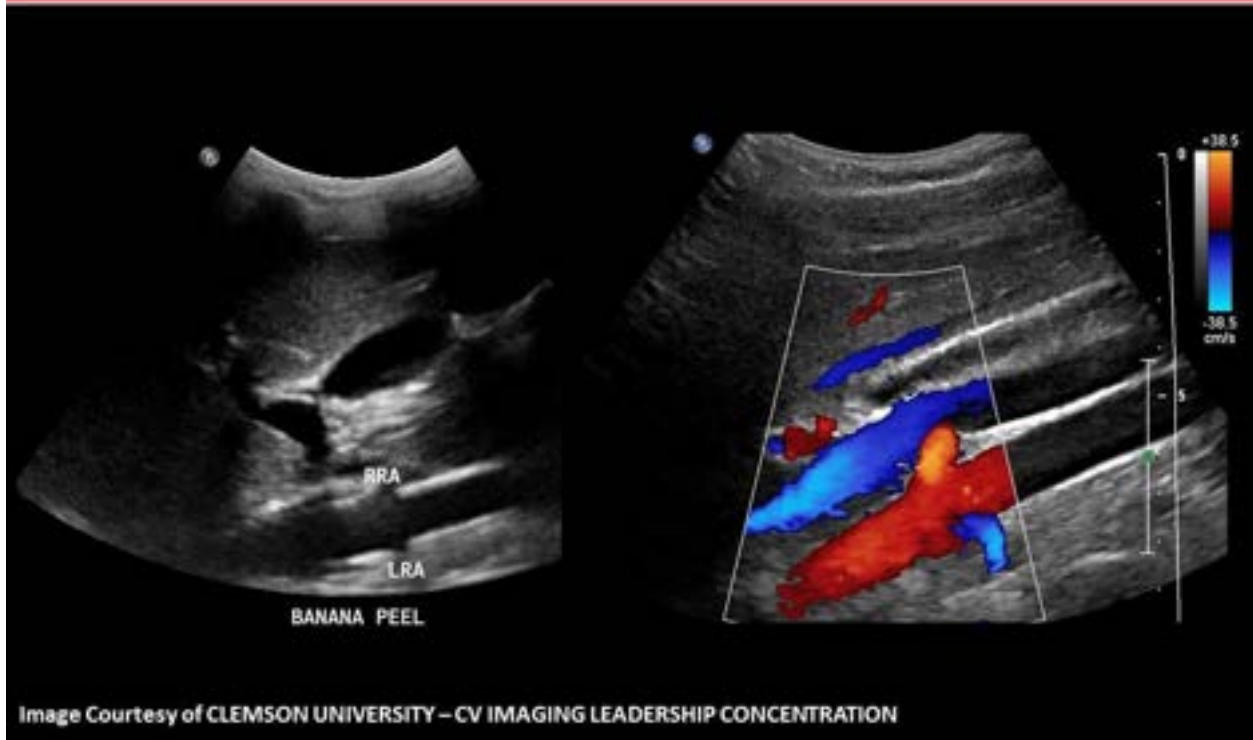
Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Renal Arteries

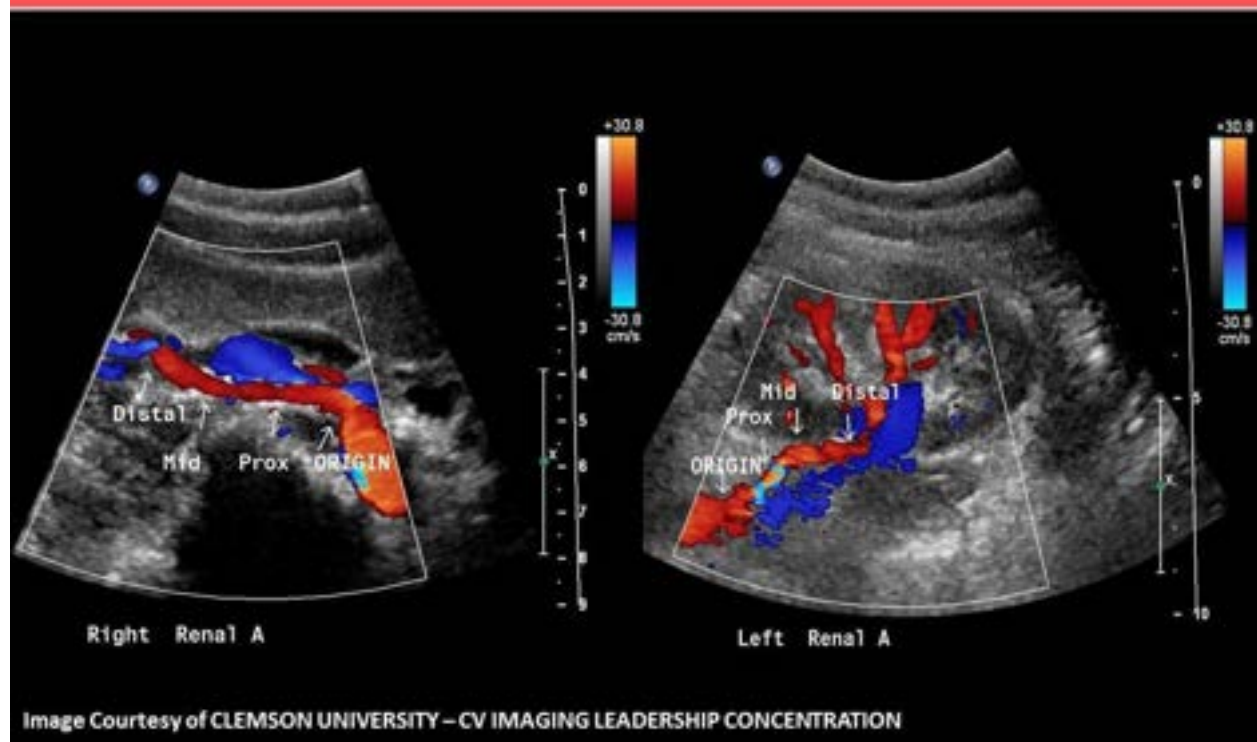


The renal artery origins are the most common site of stenosis with atherosclerotic disease. ¹

Renal Artery Origins Banana Peel View



Renal Artery Anatomy



Right Renal Artery



Duplicated Renal Artery



Renal Length

Kidney length should be assessed as part of a renal artery duplex exam.
<2cm difference is considered within normal limits

← Dist 11.0 cm



Right KIDNEY

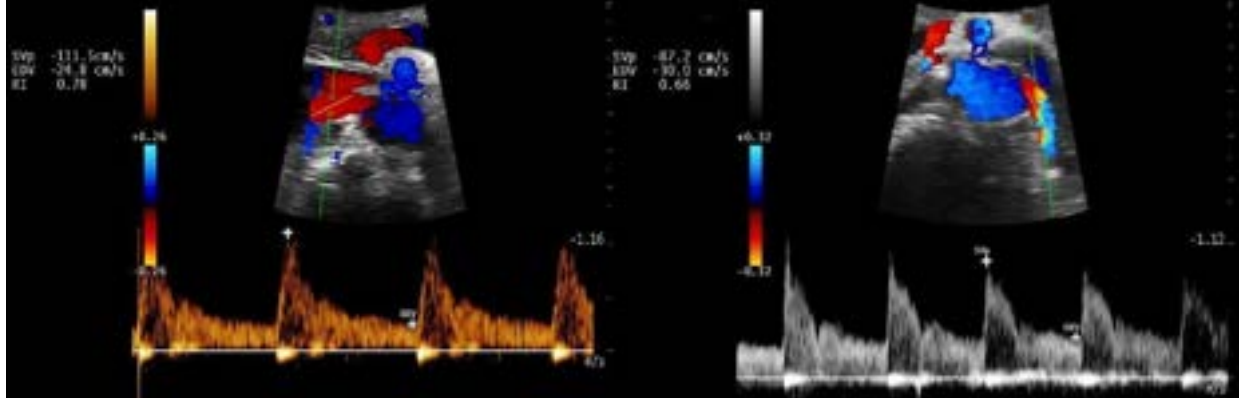
← Dist 12.3 cm



Left KIDNEY

Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Renal Arteries



The renal artery origins are the most common site of stenosis with atherosclerotic disease. ¹

Segmental Renal Artery

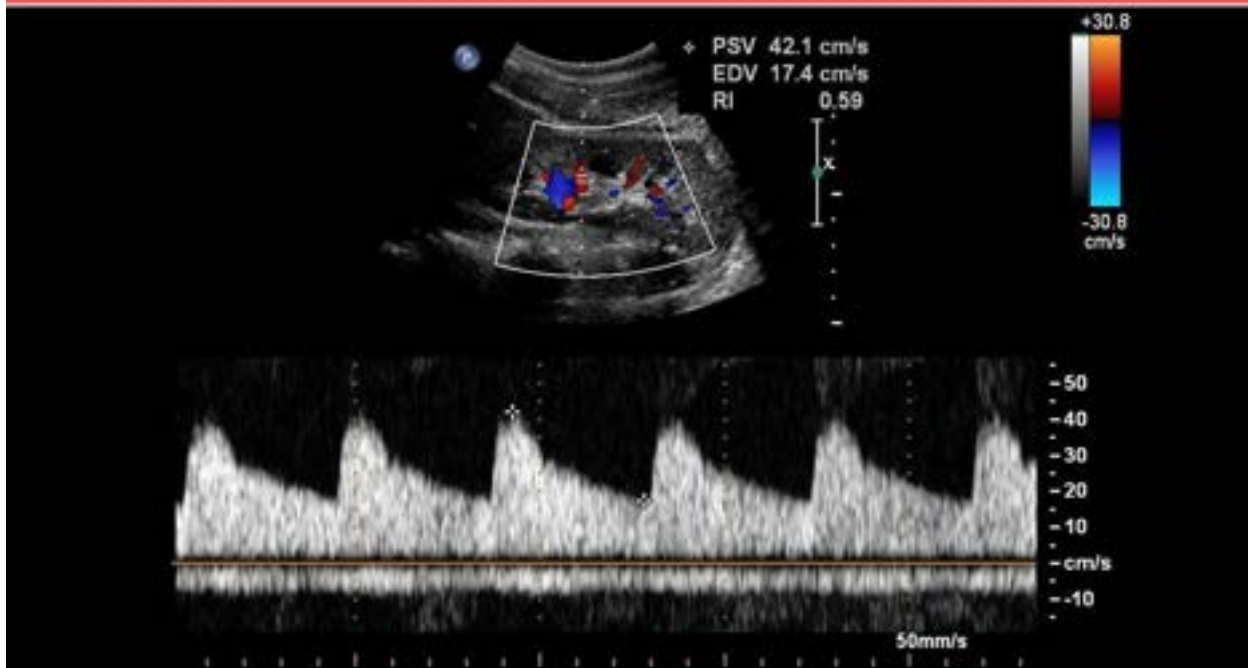


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Interlobar Renal Artery

The interlobar arteries course perpendicular to the renal capsule and flow moves parallel to the ultrasound beam. This provides the most accurate Doppler tracing of the parenchymal arterial flow. The arcuate arteries course parallel to the renal capsule and Doppler evaluation is difficult because flow is perpendicular to the ultrasound beam.

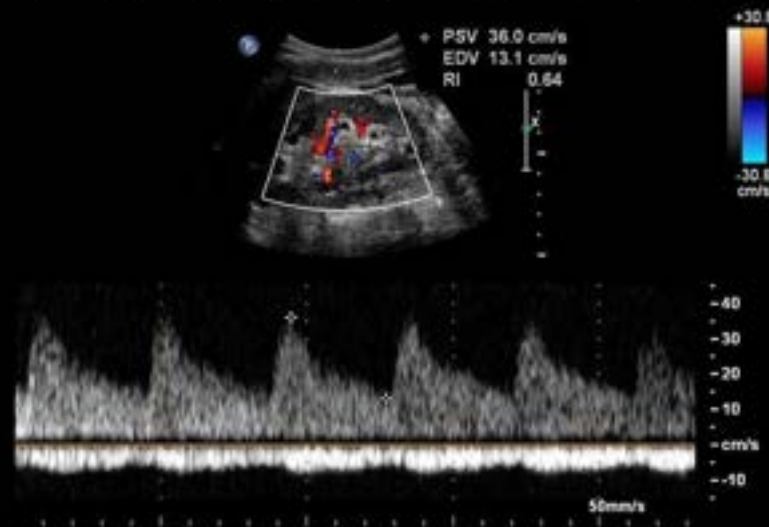
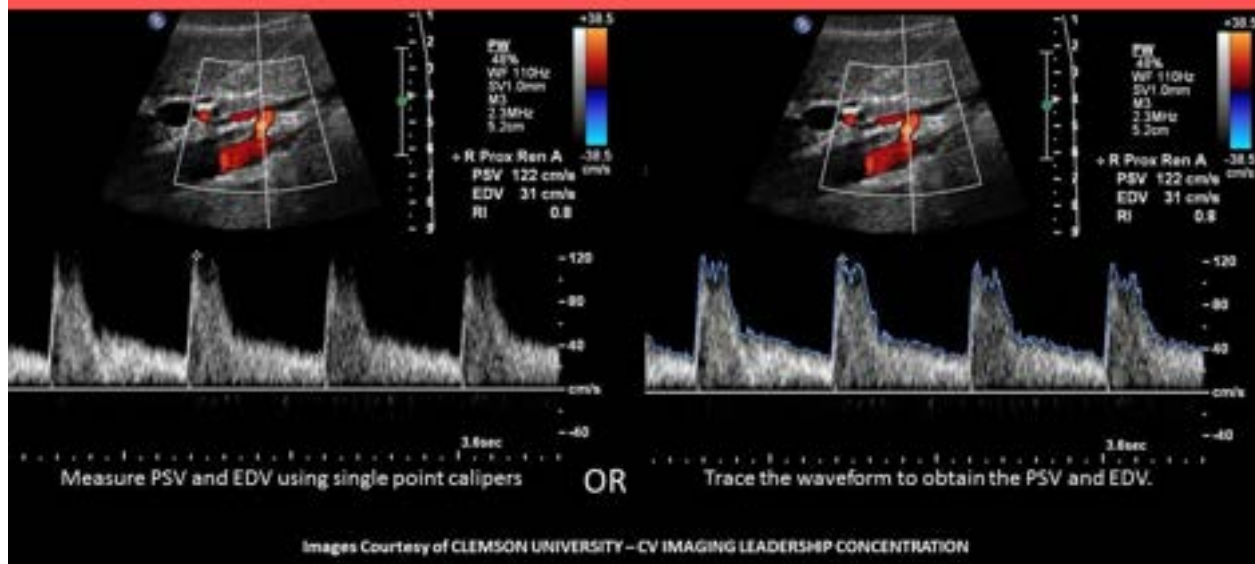


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Measuring Resistive Index



Renovascular HTN and Renal Stenosis:

- Most common cause of secondary HTN (elevated systemic pressure cause by another medical condition)
- Refers to HTN that is unresponsive to medications
- BP may be at very high levels and termed malignant HTN
- Serum BUN and creatinine levels will be increased with decreased renal perfusion/function
- Discrepant kidney sizes can be a sign of stenosis (stenosed artery of the smaller kidney)
- Kidneys receive less blood flow due to renal artery stenosis
- They perceive this as the body having an overall low blood pressure
- Renal response is the activation of the renin-angiotensin aldosterone system which raises the pressure to try to increase the flow to the kidneys, which leads to systemic hypertension
- Long standing stenosis can lead to parenchymal damage and renal failure
- RAS is most commonly caused by atherosclerosis
 - Most commonly occurs at the origin of the renal artery
- Fibromuscular dysplasia second most common cause
 - Most commonly affects the mid-distal renal arteries
 - If FMD is identified in the renal arteries, the carotid and mesenteric arteries should also be evaluated for FMD
- Ultrasound evaluation of the renal arteries can be limited by:
 - patient habitus
 - bowel gas
 - patient unable to follow breathing instructions

- presence of duplicated renal arteries
- course of the vessel (unable to obtain Doppler angle of 60 degrees or less)
- Angiography is the gold standard in diagnosis of renal stenosis but requires an invasive technique and iodinated contrast
- CTA offers detailed alternative but requires iodinated contrast
- MRA is limited by stent placement, tends to overestimate stenosis, and also may require contrast that has been associated with renal complications
- Doppler evaluation provides physiologic, as well as, anatomic information; benefits include inexpensive, noninvasive and does not require contrast material; limitations include technologist experience, patient habitus and depth of structures

Risk Factors:

- HTN
- Diabetes
- Smoking
- Family History
- Obesity
- High Cholesterol
- Age
- Gender
- History of scleroderma
- History of CREST syndrome (most commonly affects parenchymal renal arteries)
- History of fibromuscular dysplasia

Patients with Suspected Renal Artery Stenosis:

- Recent onset of uncontrollable HTN is the most common indication for renal duplex exam
- Young patients with significant hypertension
- Patients with malignant hypertension
- Patients with hypertension that is difficult to control with medication
- History of hypertension and deteriorating renal function
- Patients with renal insufficiency and discrepant kidney size
- Significant aortoiliac atherosclerotic disease

Sonographic Appearance:

- Aorta should be evaluated for atherosclerosis and aneurysm
- Aorta Doppler evaluation performed immediately proximal to the renal artery origin
- Kidney length should also be obtained bilaterally; <9cm length is abnormal
- A longitudinal image of the renal arteries should be obtained to best evaluate the full length of the artery for stenosis

- This usually requires decubitus position with probe positioned lateral to midline or use the liver/spleen for window
- 2D visualization of plaque very difficult in renal arteries
- Color demonstrates turbulence in stenosis at the main renal artery
- Doppler velocities can be artifactually diminished in patients with decreased cardiac output/cardiac disease
- RAR is invalid in patients with an associated AAA or aortic stenosis
- Direct evaluation - uses Doppler to evaluate the length of the renal artery (proximal, mid, distal segments); uses 60-degree Doppler angle; velocities used to assess RAS
- Indirect evaluation - intrarenal evaluation of the segmental/parenchymal arteries at the upper, mid and lower poles; uses 20-degree or less Doppler angle; waveform shape used to assess RAS; resistive index and acceleration time evaluated
- Parenchymal resistance ratio (PRR) or end diastolic ratio is used to evaluate parenchymal resistance (EDR)
- $EDR = EDV / PSV$
- >0.2 EDR is normal

Diagnostic Criteria – Renal Artery Stenosis

Characteristic	Normal	>60% Stenosis
Kidney Length	10-12cm length	Decreased overall size
Renal Artery PSV	<180 cm/s	>180 cm/s
Turbulence	Absent	Present
Renal/Aortic Ratio	<3.5	>3.5
Parenchymal Thickness	>10mm	<10mm
Parenchymal Resistance	Low Resistance RI<0.7	Increased Resistance RI >0.75
Systolic upstroke Acceleration Time	AT <100ms	Slower systolic upstroke AT >100ms
Parenchymal Resistance Ratio (EDV/PSV)	EDR >0.20	EDR <0.20

- A normal renal artery waveform demonstrates an early systolic peak (ESP); stenosis will cause the ESP to disappear and the waveform will be more rounded
- Increased resistance in the renal arteries can indicate distal stenosis or kidney disease
- Intrarenal arteries demonstrate damping with an acceleration time $>100\text{ms}$ when stenosis is present proximally
- Tardus Parvus waveform - absence of early systolic peak and prolonged acceleration time
- Damped PSV distal to site of stenosis

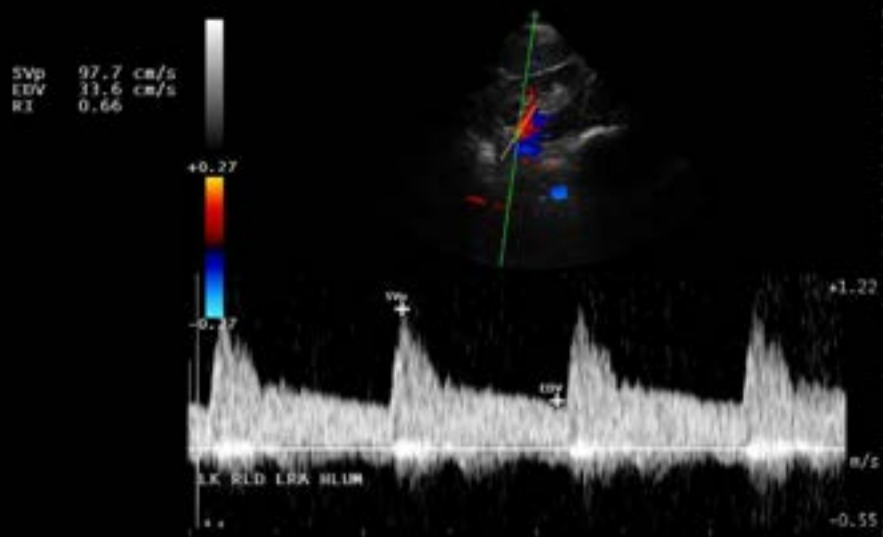
Renal Artery Occlusion:

- Absence of a visible main renal artery
- Kidney size $< 9\text{cm}$ in length
- Either absence of blood flow or very low amplitude
- Damped intrarenal flow signals (tardus parvus)

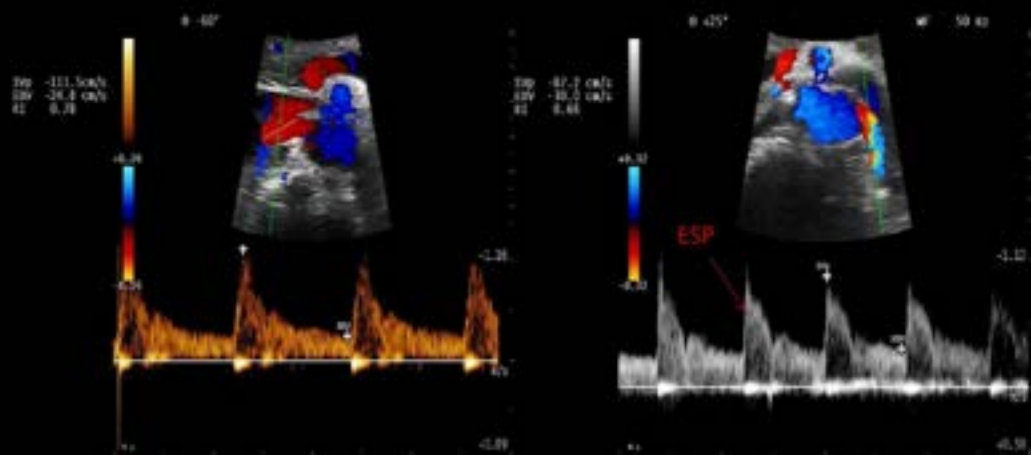
Treatment:

- Stent placement or angioplasty
- Intravascular ultrasound can be utilized to guide the procedure but very costly due to disposable imaging catheter
- Physician uses an ultrasound catheter made of multiple elements arranged in a circular array; average 20-40MHz frequency; placed over the guidewire and evaluates the vessel lumen for abnormalities

Early Systolic Peak

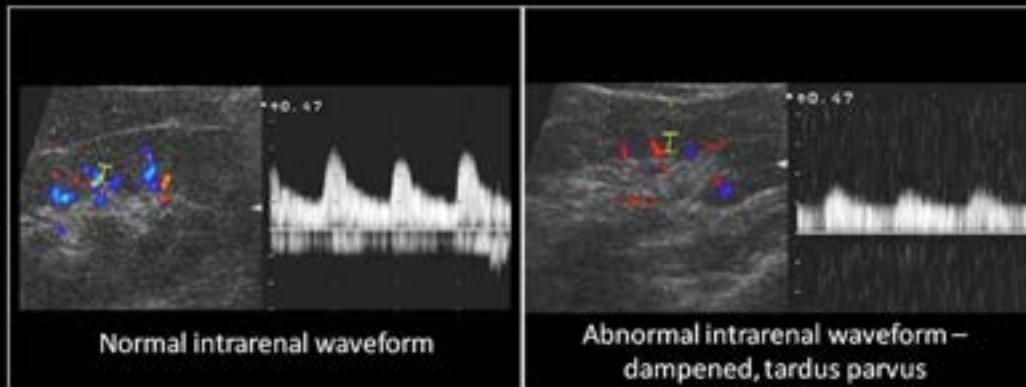


Normal Renal Artery



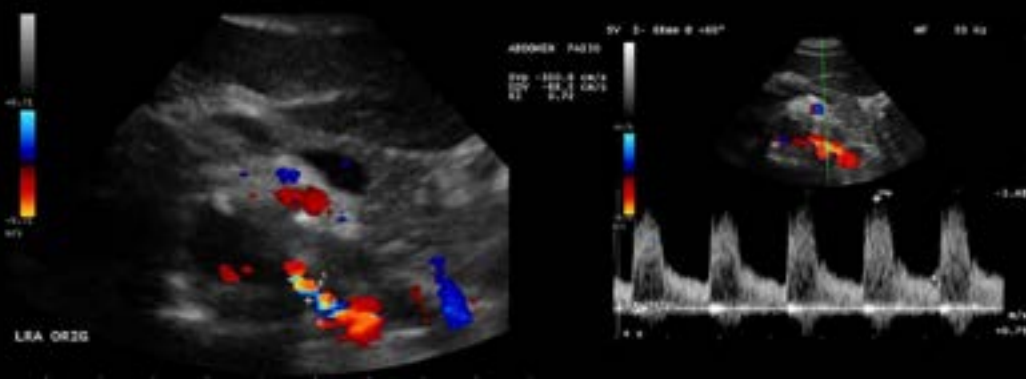
Note the presence of the low resistance waveform with rapid upstroke and peaked appearance. The early systolic peak (ESP) is visualized.

Indirect Evaluation of Renal Artery Stenosis



Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

Direct Evaluation of Renal Artery Stenosis



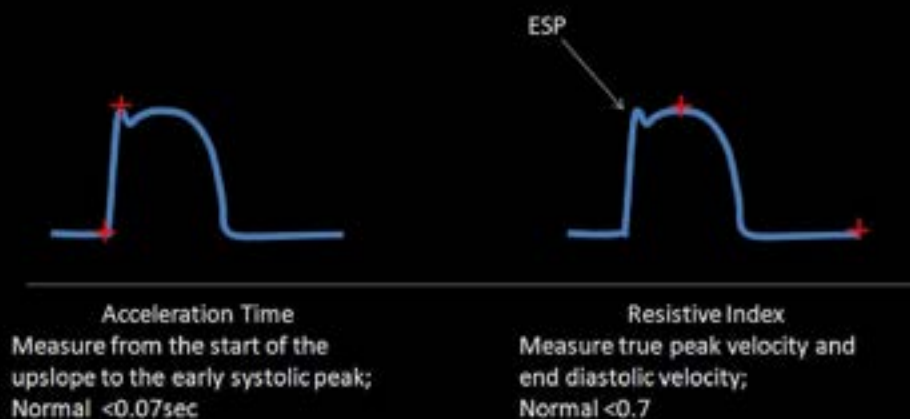
Note the absence of the early systolic peak (ESP).

Tardus Parvus Waveform

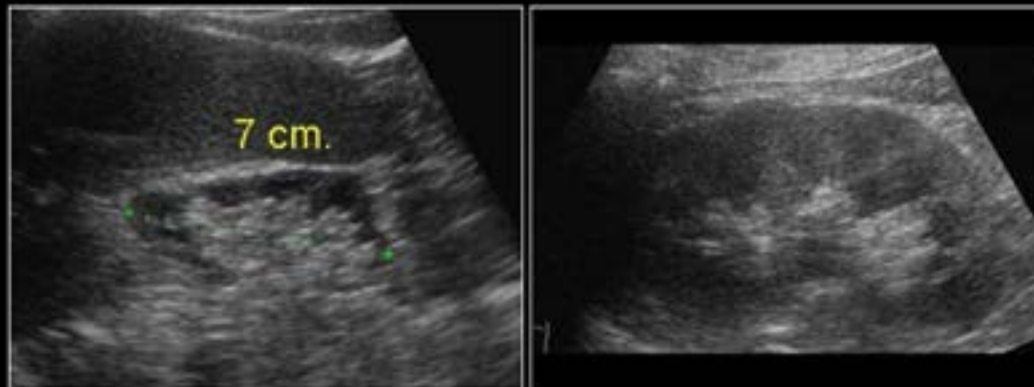


Courtesy of Dr. Taco Geertma
www.ultrasoundcases.info

Parenchymal Flow Evaluation

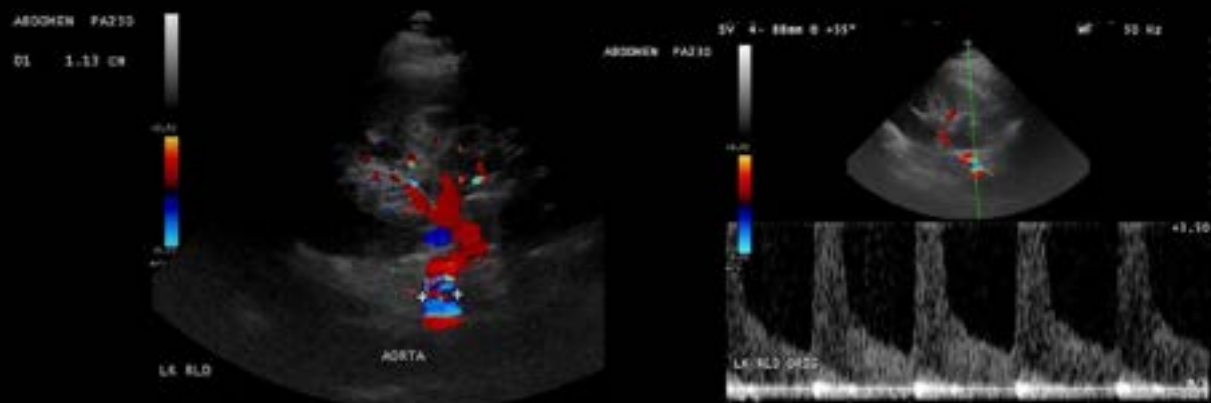


Renal Artery Stenosis



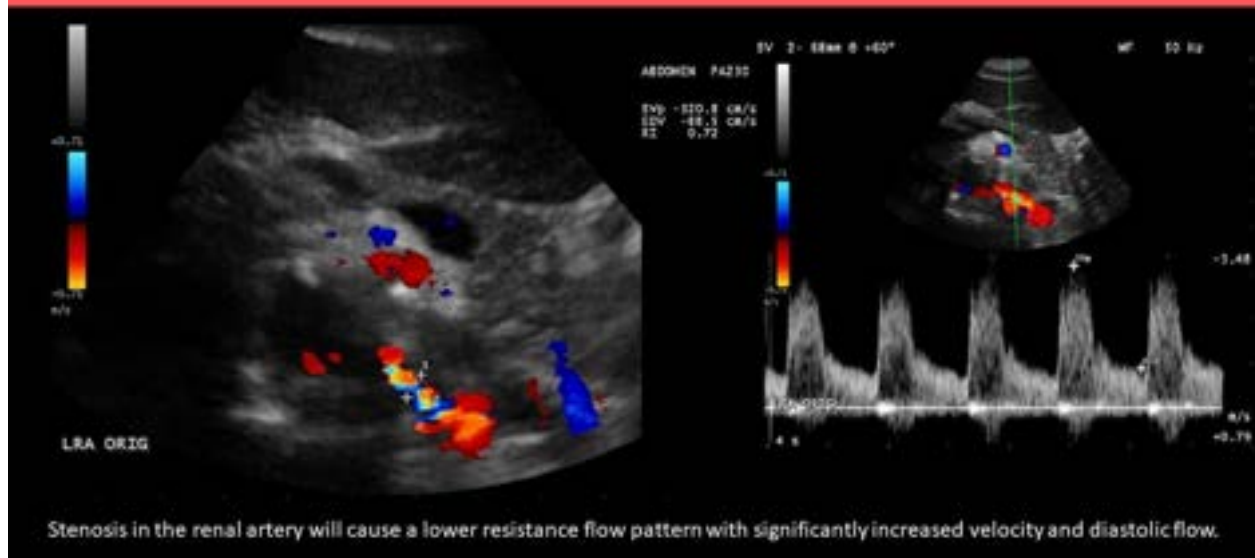
Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

Renal Artery Stenosis

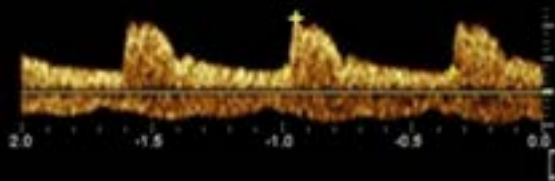


Stenosis in the renal artery will cause a lower resistance flow pattern with significantly increased velocity and diastolic flow.

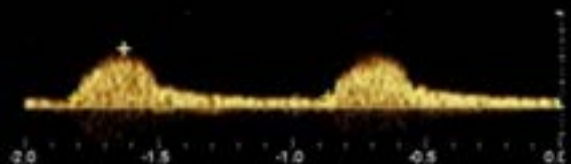
Renal Artery Stenosis



Normal vs Abnormal Renal Artery Waveforms



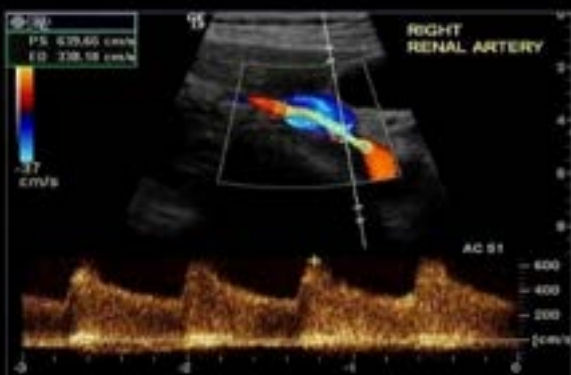
Normal renal artery tracing with an early systolic peak



Abnormal renal artery tracing with a late systolic peak (tardus parvus) Indicates proximal stenosis

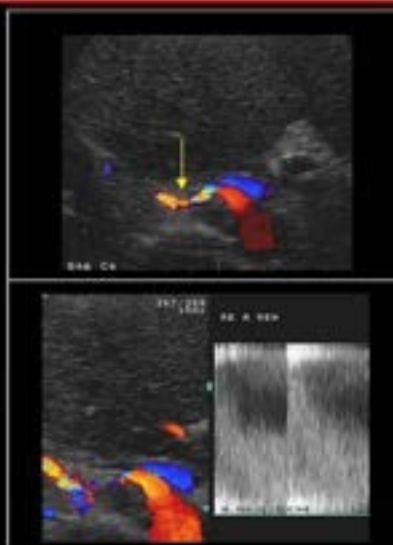
Adapted from P. Karmis, MD. <https://www.imagingrealm.com/vascular.html>

Renal Artery Stenosis – Tardus Parvus



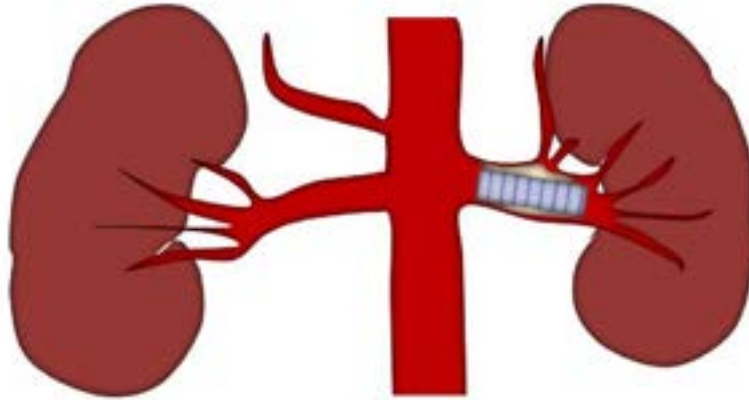
Fibromuscular Dysplasia

FMD is most commonly a diffuse process that can affect many arteries in the body. It is most commonly seen in the renal arteries. Consider this diagnosis when bilateral RAS is present, especially in younger patients. Angiography is the preferred modality to diagnose FMD.



Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

Renal Artery Stent



Renal Artery Stent



Renal Transplant:

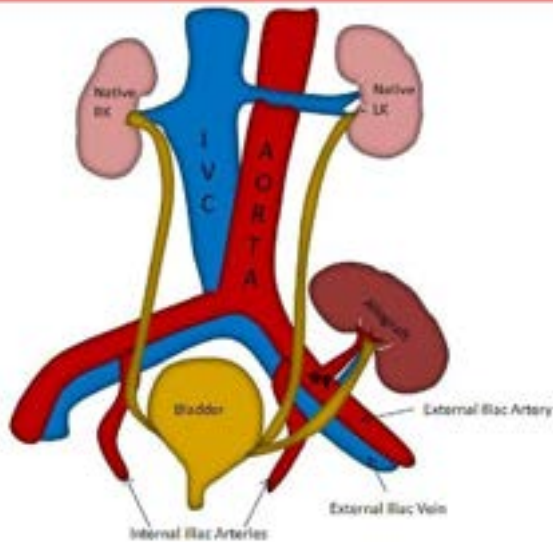
- Gold standard in treatment for patients with end-stage renal disease

Exam Technique:

- Curvilinear or phased array, 2MHz to 6.5MHz

- Native inflow and outflow vessels and transplant vessels evaluated for patency
- Iliac arterial and venous anastomosis sites also evaluated
- Right iliac fossa most common site for transplant
- External iliac artery most commonly used for arterial anastomosis (internal iliac artery is an alternative)
- External iliac vein is the most common site for venous anastomosis (internal iliac vein is an alternative)
- Allograft vessels are also evaluated
- Evaluate for rejection, venous thrombosis, arterial stenosis/occlusion, AV fistula
- Protocol:
 - Measure kidney length and width at upper, mid and lower poles
 - Measure cortical thickness
 - Evaluate kidney echogenicity
 - Document perinephric fluid collections and any hydronephrosis
 - Evaluate the bladder for signs of ureteral dilatation
 - Color and PW Doppler evaluation of:
 - External iliac artery proximal and distal to anastomosis of allograft artery
 - Main renal artery anastomosis, proximal, mid and distal vessel
 - Interlobar arteries in the upper, mid and lower pole
 - Document flow in the main renal vein of the allograft
 - Peak systolic velocity, end diastolic velocity and resistive index measured on each waveform
 - Acceleration time is also measured on the interlobar tracings
 - Calculate the velocity ratio: PSV of the main renal artery of the allograft divided by the iliac artery PSV

Renal Allograft



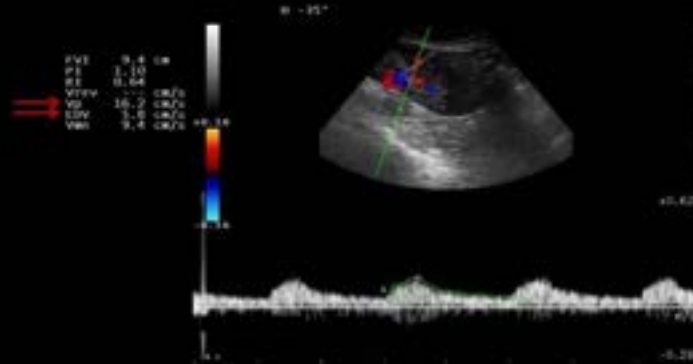
- The external iliac vessels are the most common vessels used for a renal transplant.

Renal Transplant Doppler

Parenchymal Resistance Ratio

PR = EDV/PSV NL > 0.2

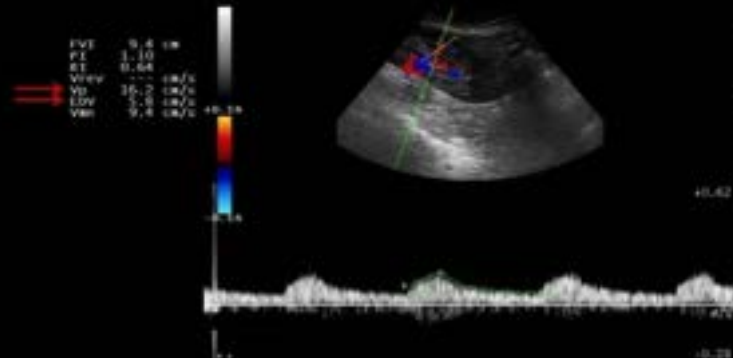
$$\frac{6 \text{ cm/s}}{16 \text{ cm/s}} = 0.375$$



Renal Transplant Doppler

$$\text{RESISTIVE INDEX} = \frac{\text{PSV} - \text{EDV}}{\text{PSV}}$$

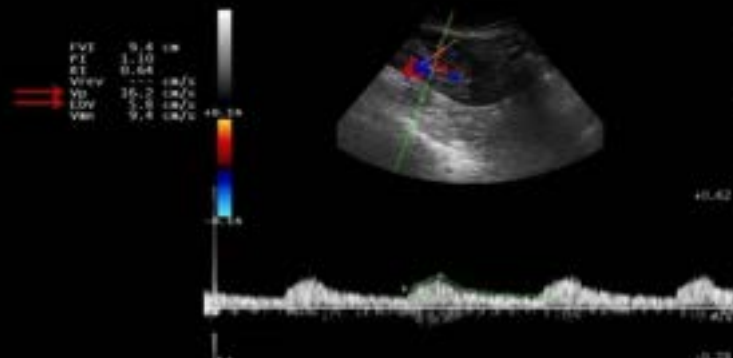
$$\frac{16 - 6}{16} = 0.63$$



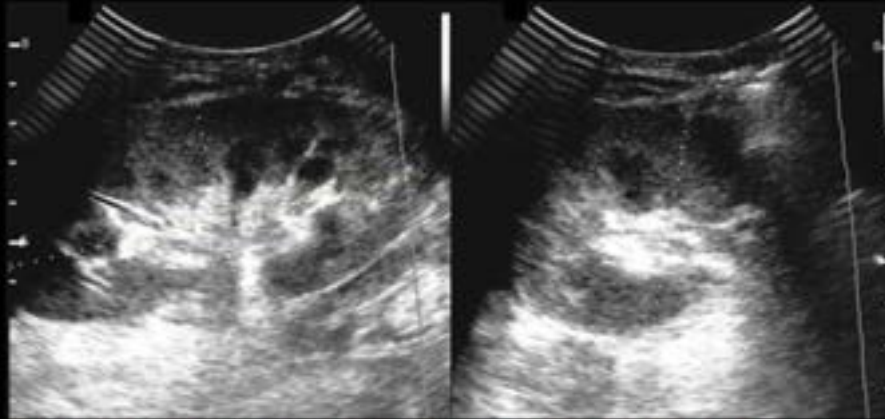
Renal Transplant Doppler

$$\text{RESISTIVE INDEX} = \frac{\text{PSV} - \text{EDV}}{\text{PSV}}$$

$$\frac{16 - 6}{16} = 0.63$$



Renal Transplant Evaluation



2D ultrasound is used to assess allograft length and cortical thickness.

Adapted from: Early ultrasound assessment of renal transplantation as the valuable biomarker of long lasting graft survival: a cross-sectional study. Enhesari A, Mardpour S, Makki Z, Mardpour S- Iran J Radiol (2014)

Renal Transplant Evaluation



Doppler evaluation of the interlobar artery to assess RI, PI and EDV.

Adapted from: Early ultrasound assessment of renal transplantation as the valuable biomarker of long lasting graft survival: a cross-sectional study. Enhesari A, Mardpour S, Makki Z, Mardpour S- Iran J Radiol (2014)

Renal Transplant Rejection:

- Allograft failure commonly caused by Acute Tubular Necrosis (#1 acute cause), rejection, drug toxicity
- Other causes include infection, vascular thrombosis/stenosis
- Signs include oliguria and elevated BUN and creatinine levels

- Acute rejection usually develops within 1-3 weeks to months after transplantation
- Chronic rejection occurs as a progressive decrease in renal function beginning at least 3 months after surgery
- Biopsy of the cortex is the most accurate way to determine the cause for rejection
- Perinephric fluid collections are very common after surgery
 - Hematoma - collection of blood, seen immediately after surgery and usually resolve on their own
 - Urinoma - collection of urine, form within 2 months after surgery, must be drained
 - Lymphocele - collection of lymphatic fluid, usually resolve on their own
 - Abscess - caused by infection and can be related to allograft failure; require immediate treatment to save allograft

Sonographic Appearance:

Normal:

- Parenchymal resistance ratio or end diastolic ratio $EDV / PSV = >0.2$
- Interlobar Resistive Index <0.7
- Normal levels of diastolic flow, continuous flow to the kidney throughout the cardiac cycle

Abnormal:

- Increased kidney size and echogenicity
- Increased parenchymal resistance $RI >0.7$
- $PRR <0.20$
- Loss of diastolic flow or diastolic flow reversal in renal transplant parenchymal arteries
- Anastomosis stenosis - increased PSV, decreased perfusion, most common complication
- Thrombosis of transplant artery
- Acute occlusion of the renal artery or supplying artery is a critical finding in a renal transplant patient
- Thrombosis of the transplant vein will cause increased resistance to flow in the renal arteries, reduced or absent diastolic flow, kidney will become enlarged with plump cortex
- AV fistula most commonly caused by biopsy, but can occur with transplant
- Fluid collections - urinoma, lymphocele, abscess
- Hydronephrosis with urinary tract obstruction

Acute Tubular Necrosis – Transplant Rejection



Increased size and echogenicity

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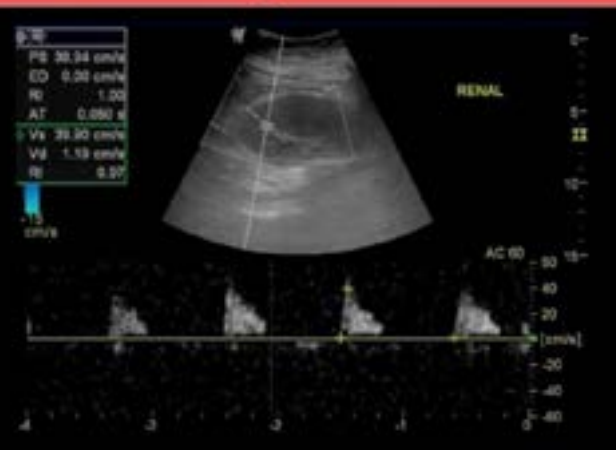
Loss of corticomedullary definition

Renal Transplant Arterial Stenosis vs Acute Rejection



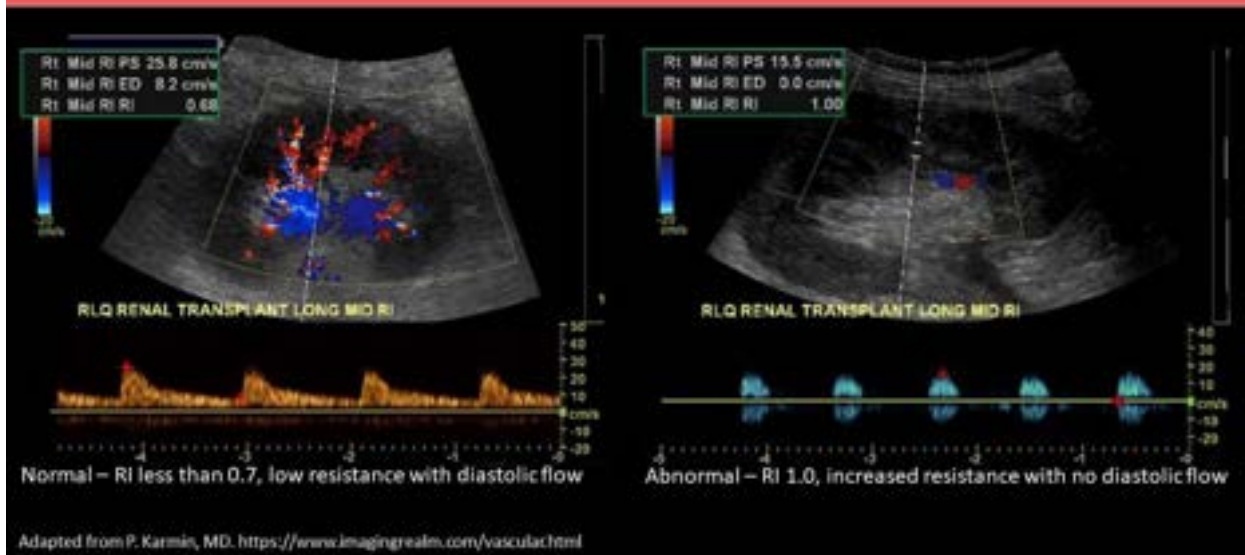
A tardus parvus waveform in the parenchyma indicates a proximal stenosis in the renal artery

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Increased parenchymal resistance with acute rejection leads to an RI 1.0, increased resistance with no diastolic flow

Renal Transplant Rejection



Renal “Most Commons”

The most common anatomic variant of the renal arteries is the presence of 2 or more renal arteries

Renal artery stenosis is most commonly caused by atherosclerosis

The most common site of atherosclerotic stenosis is at the origin of renal arteries

The most common site of stenosis caused by FMD is in the mid-distal renal arteries

Renovascular HTN and Renal Stenosis is the most common cause of secondary HTN

Recent onset of uncontrollable HTN is the most common indication for renal duplex exam

FMD is most commonly seen in the renal arteries of young women

Renal vein thrombosis is most commonly caused by primary renal disease

CREST syndrome most commonly affects the tiny parenchymal arteries in the renal circulation

Right iliac fossa most common site for renal transplant

External iliac artery most commonly used for arterial anastomosis

External iliac vein is the most common site for venous anastomosis

Renal transplant rejection is most commonly caused by Acute Tubular Necrosis (#1 acute cause)

Stenosis at the arterial anastomosis is the most common complication of renal transplant

Inferior Vena Cava: Sonographic Evaluation and Disease

Anatomy:

- Trunk vein formed by the confluence of the common iliac veins just below the aortic bifurcation (level of L5)
- Empties into the right atrium
- Has NO true venous valves
- A Eustachian valve is a flap of tissue at the junction of the IVC with the right atrium that is necessary for fetal circulation
- The Eustachian valve may persist in some adults, but it has no effect on circulation
- Courses to the right of the aorta within the abdomen
- Anterior to the aorta above the umbilicus, posterior to aorta below umbilicus
- Posterior to the duodenum and pancreas head
- Returns all blood from legs and non-digestive abdominal organs to heart
- Inspiration increases the diameter due to cessation of flow with increased intra-abdominal pressure
- Expiration decreases the diameter with decrease in intra-abdominal pressure
- Valsalva causes dilatation
- Abnormal dilatation >2.5 cm in upper abdominal segment; >2.1cm at junction with hepatic veins`

Tributaries:

- Common iliac veins

- Lumbar veins
- Gonadal veins - Right gonadal vein empties into IVC, Left gonadal vein empties into the left renal vein
- Renal veins
- Suprarenal veins
- Hepatic veins - largest tributaries

Indications to Scan IVC:

- Abdominal pain
- Palpable mass
- F/U filter placement
- History of renal tumor - carcinoma commonly invades the renal veins/IVC

Variants of IVC Anatomy

- Duplication - most commonly occurs below the level of the renal veins
- Transposition - IVC on the left side, can be associated with situs inversus
- Interruption - the intrahepatic segment of the IVC does not develop; the azygos vein drains the flow from the pelvis and lower extremities into the SVC; hepatic veins empty directly into the right atrium

Exam Techniques:

- 2.5MHz to 6.5MHz adult probe
- 4MHz to 8MHz pediatric probe
- Patient must be NPO to best visualize the abdominal vasculature
- 2D, Color and Doppler evaluation of length of vessel

US of the IVC:

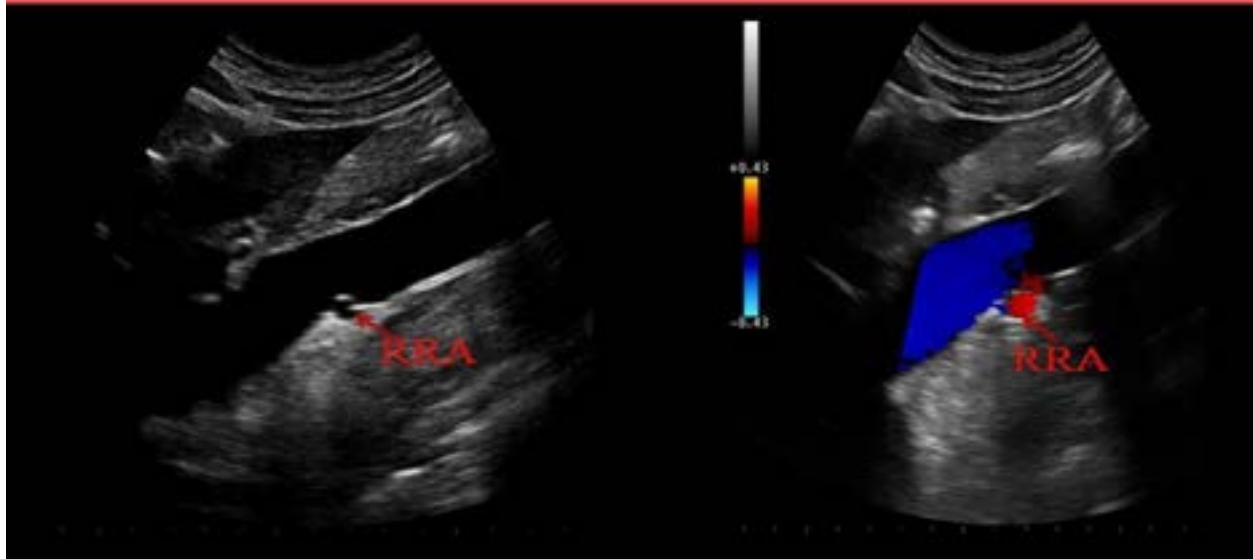
- Subxiphoid probe location to evaluate the suprahepatic, intrahepatic and infrahepatic segments
- The confluence of the iliac veins into the infrahepatic segment may be best visualized with the patient in the decubitus position and scanning from the right lateral abdomen
- Longitudinal-tubular structure to the right of the aorta
- Transverse-oval structure or dumbbell shaped
- Walls thinner than aorta
- Should change diameter with respiration (aorta does NOT change with respiration)

Doppler of the IVC:

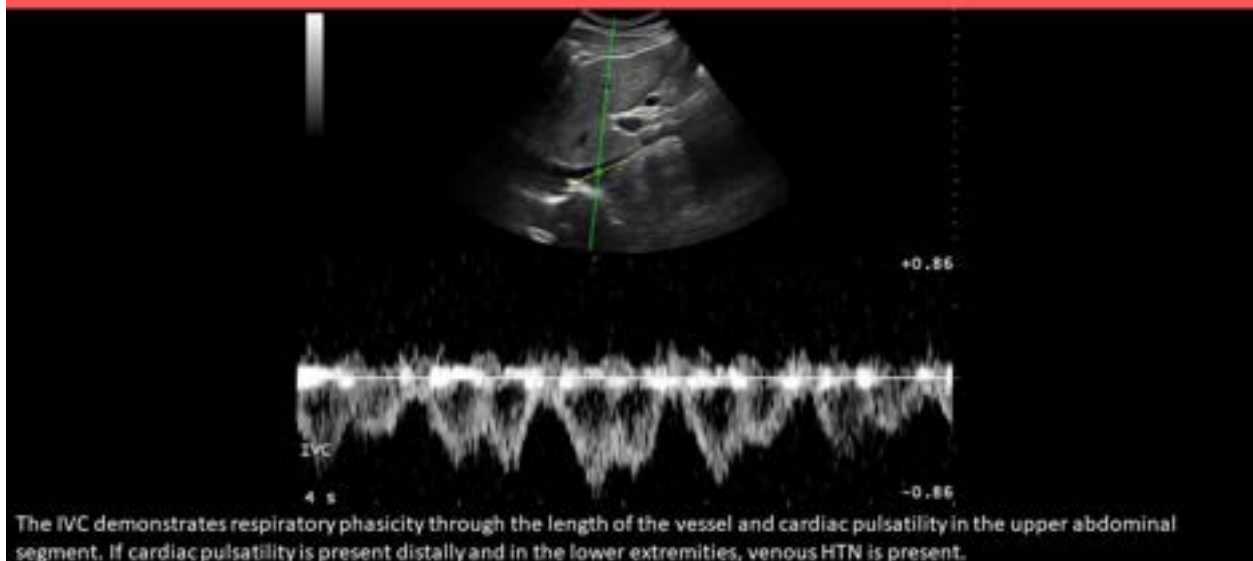
- Longitudinal plane
- Flow toward right atrium
- Cardiac pulsatility and respiratory changes
- Can see respiratory changes in 2D also

Aorta	IVC
Does not touch liver	Touches liver
Thicker, more echogenic walls	Thin walls
Multiple anterior branches	Hepatic veins are the only anterior branches
No vessels course posterior to it	RRA courses posterior to the IVC
No respiratory changes to flow	Respiratory phasicity
Unaffected by Valsalva maneuver	Diameter changes with Valsalva

IVC



IVC Doppler

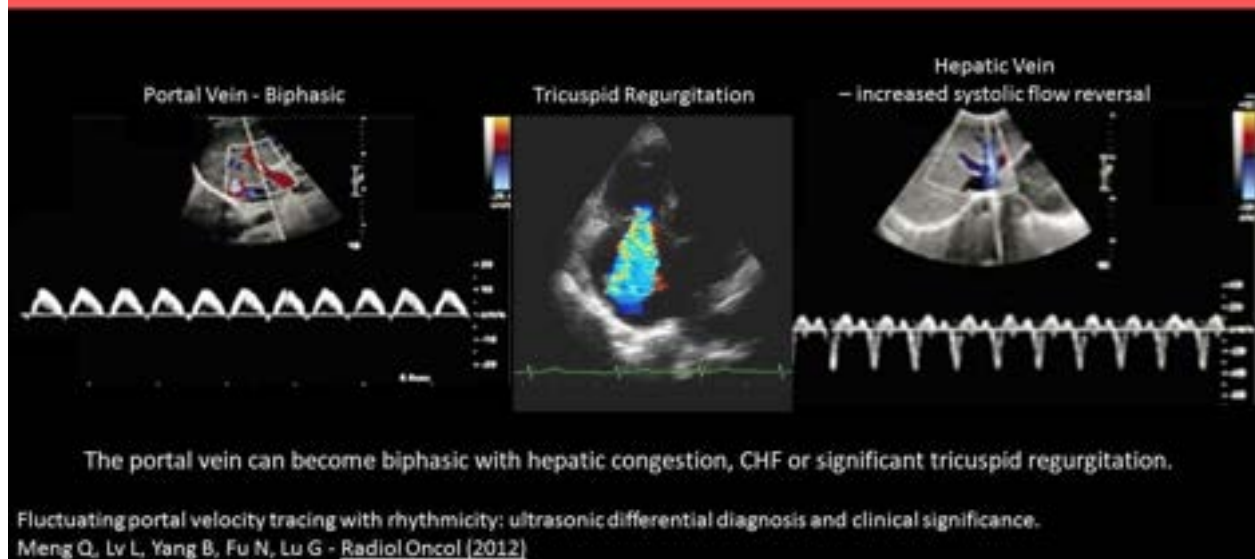


Dilated IVC:

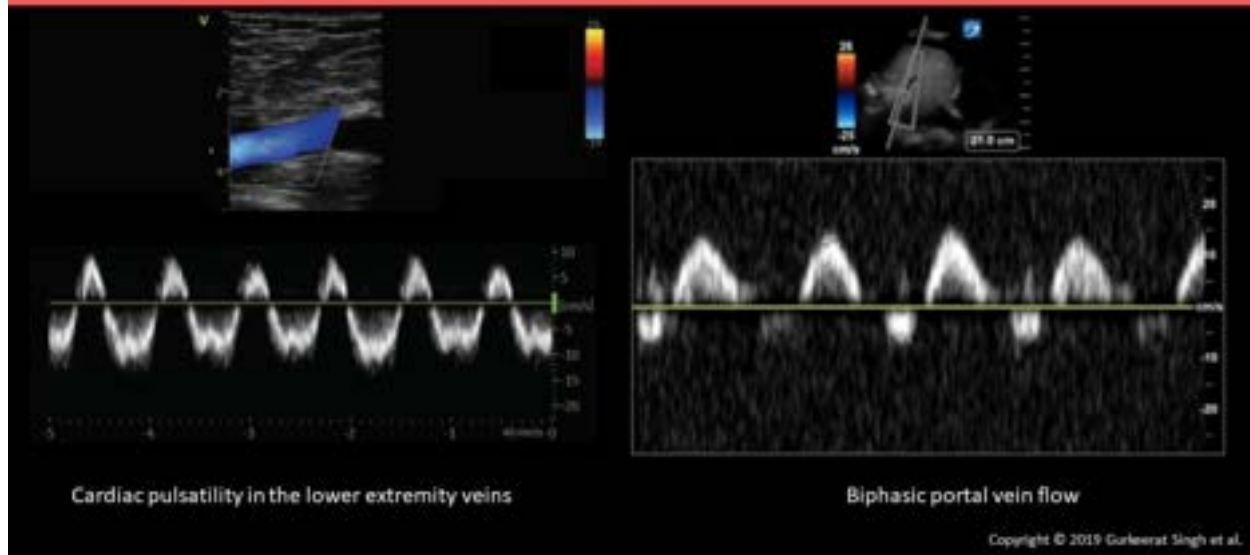
- The IVC will act as a reservoir for the blood if there are other circulatory issues
- Greater than 2.5cm in upper segment
- IVC should change in size with respiration

- IVC should collapse >50% when the patient performs a quick sniff
- Dilatation of the IVC and loss of respiratory phasicity can be caused by CHF, pulmonary HTN, decreased cardiac function, electrolyte imbalance, renal dysfunction
- Severe tricuspid regurgitation can lead to congestive heart failure (CHF) and IVC dilatation which will cause increased systolic flow reversal in the hepatic veins
- CHF will cause increased pulsatility throughout the venous system, including the lower extremity veins
- Biphasic flow in the portal veins is also associated with CHF and IVC dilatation
- Thrombus can form with blood stasis due to congestion within the IVC
- Obstruction of the IVC will lead to continuous flow in both iliac veins
- Tumor invasion of the IVC most commonly occurs with renal cell carcinoma; can also be seen with hepatic carcinoma
- Use color Doppler to evaluate the IVC blockage, tumor invasion should show internal vascularity while thrombus will not have any internal flow

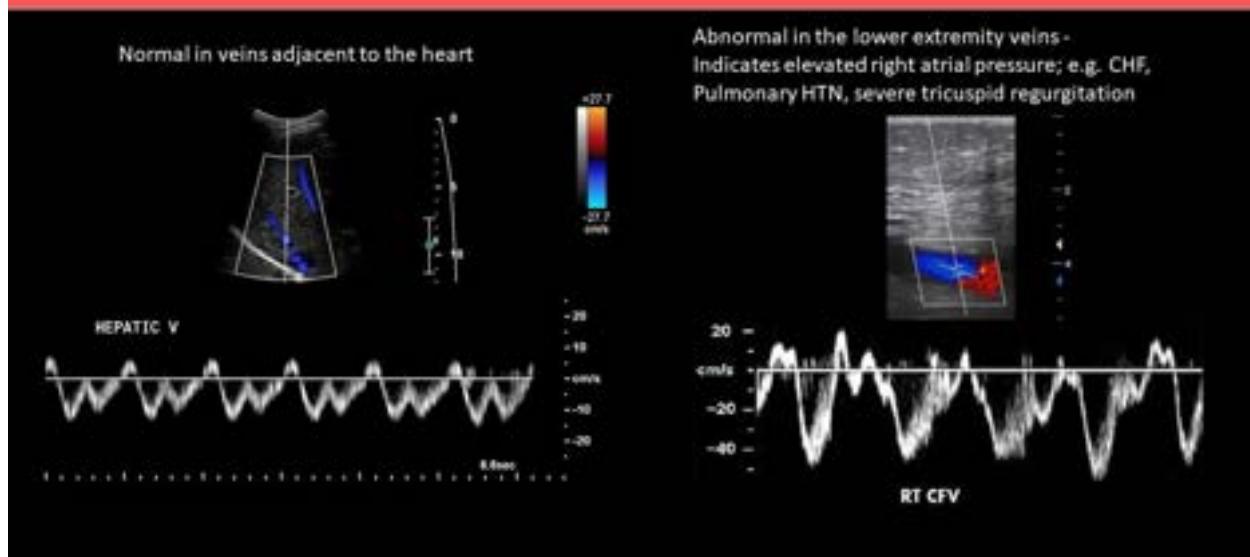
Tricuspid Regurgitation - CHF



Congestive Heart Failure – Distal Effects



Cardiac Pulsatility



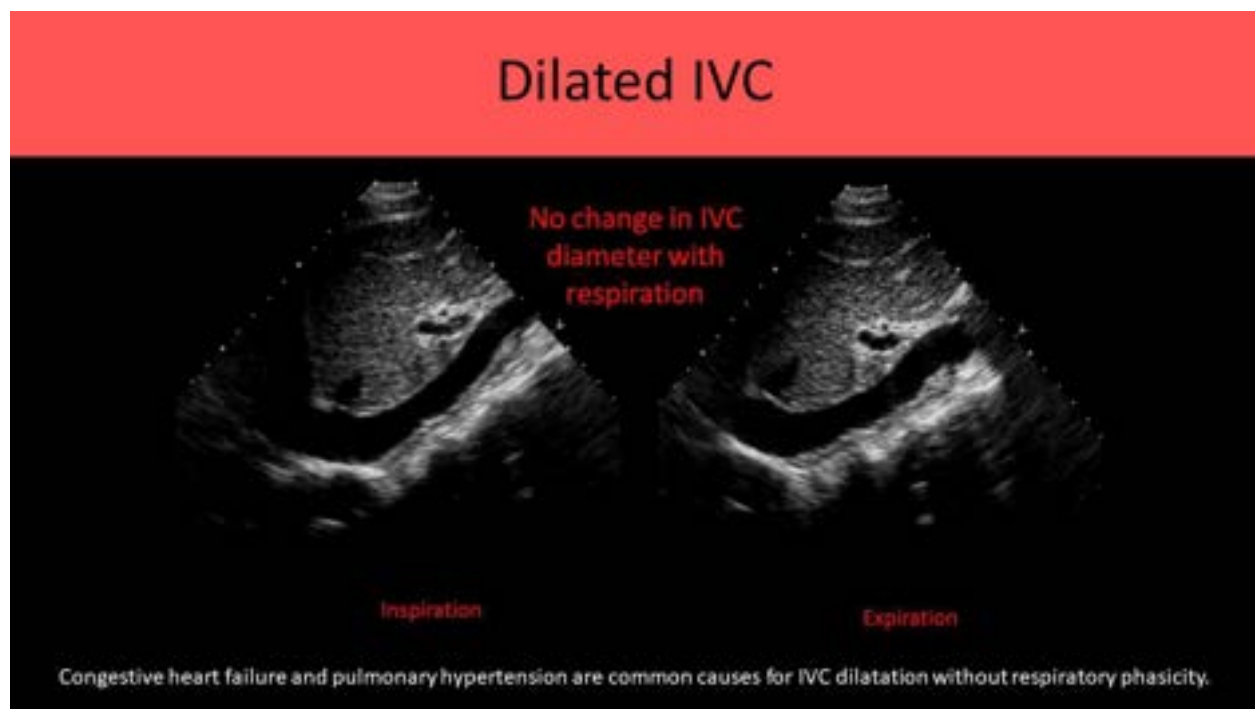
IVC Aneurysm

- Can be saccular or fusiform, but the saccular formation is slightly more common
- Aortic aneurysms are more commonly fusiform
- Anatomic classification system:
- Type I: occurs in the suprahepatic IVC without venous obstruction

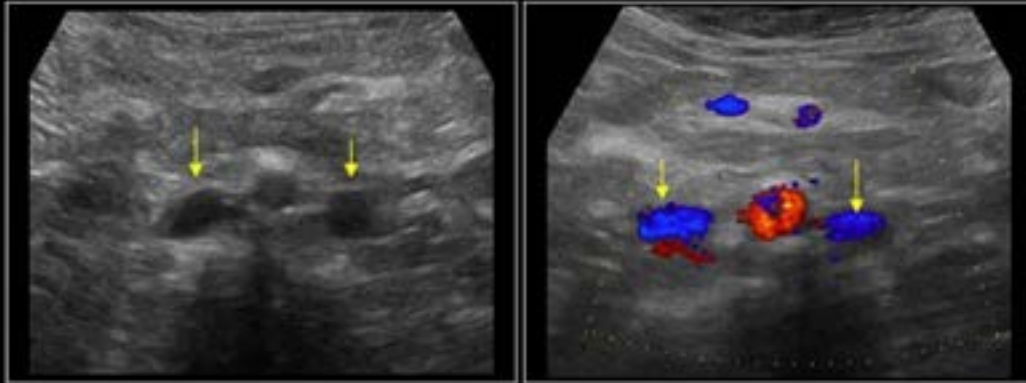
- Type II: associated with interruption of the IVC above or below the hepatic veins
- Type III: occurs in the infrarenal IVC without associated venous anomaly
- Type IV: a left-sided IVC with an aneurysm at the confluence of the left internal and external iliac veins
- Can lead to thrombus formation and increases risk of embolism

Congenital Abnormalities:

- Duplication - in most cases, each iliac vein extends separately into the abdomen; left IVC joins the left renal vein and then crosses midline to join the right IVC before it enters the chest to go to the right atrium
- Situs Inversus - organs of the abdomen and chest are reversed; IVC on left and aorta on the right
- Interrupted IVC - failure of the IVC to form correctly in the upper abdomen/liver; azygous and hemiazygos veins drain blood from the distal IVC segment; hepatic veins drain directly into the right atrium

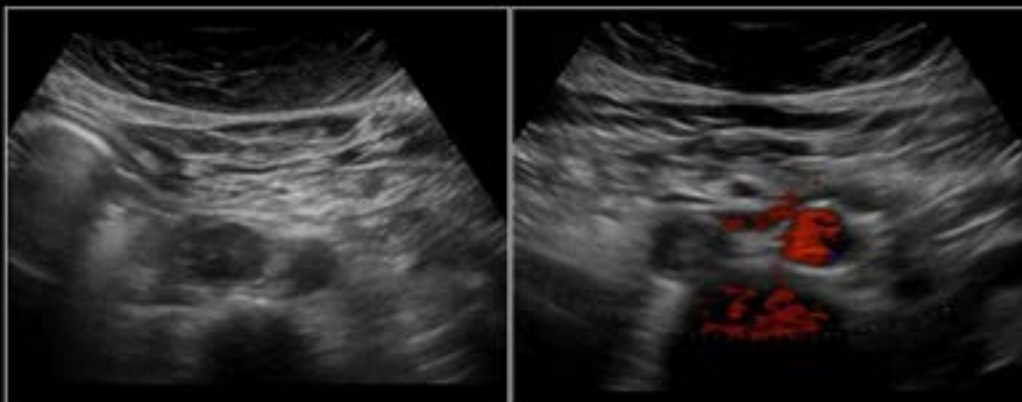


Duplicated IVC



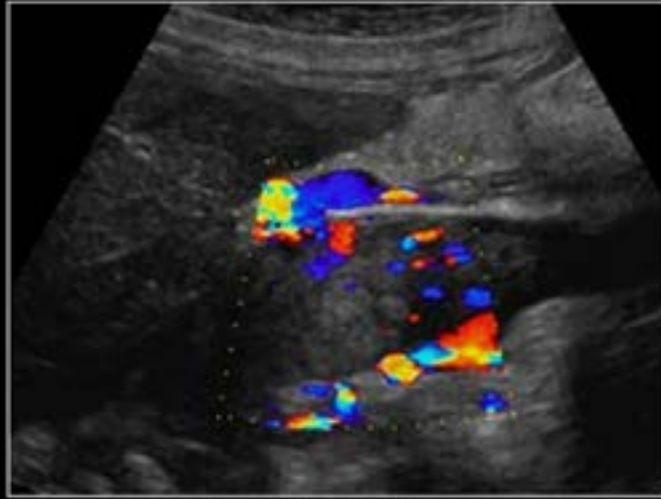
Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

IVC Thrombus



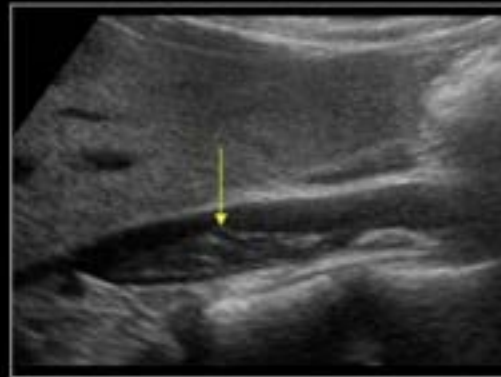
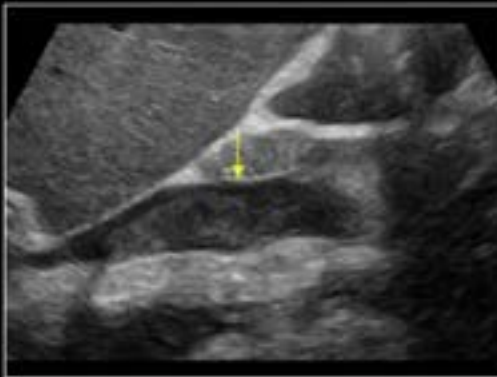
Courtesy of Dr. Taco Geertsma
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IVC Thrombus



Courtesy of Dr. Taco Geertsma
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IVC Thrombus



Courtesy of Dr. Taco Geertsma
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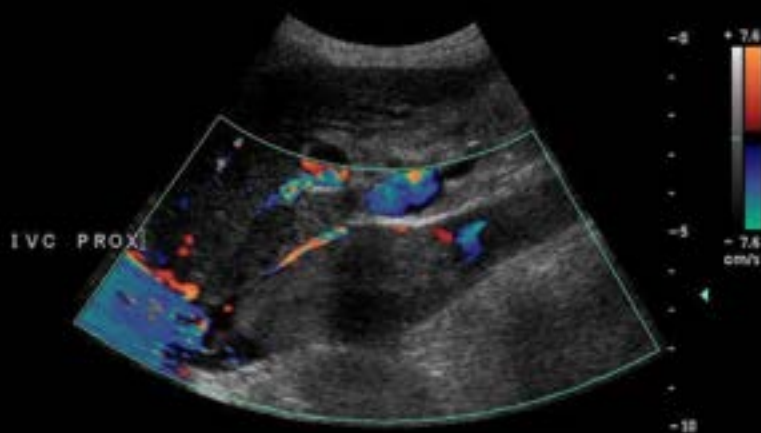
IVC Tumor Invasion



Quality Vascular Imaging, Inc.

www.qualityvascular.com

IVC Tumor Invasion



Quality Vascular Imaging, Inc.

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HEPATIC VEINS: (HV)

Anatomy:

- Largest tributaries of IVC
- Left, Right and Middle branches come together to join the IVC superior to the liver
- In most patients the left and middle hepatic veins come together into a common trunk before exiting the liver and entering the IVC
- RHV-drains right lobe, divides right lobe into anterior and posterior segments
- LHV-drains left lobe, divides left lobe into medial and lateral segments
- MHV-drains portions of right and left lobes, divides liver into right and left lobes
- Caudate lobe drained by the emissary veins that empty into the IVC; the separate venous drainage system allows this lobe to be spared from complications caused by hepatic venous outflow obstruction
- Thinner, less echogenic walls as compared to the portal veins
- Course inferior to superior through liver toward diaphragm to meet the IVC
- Increase in size as they near the liver periphery

Indications to Scan Hepatic Veins:

- CHF - can cause increased resistance to flow entering the right atrium
- Significant tricuspid regurgitation - can cause increased pulsatility and systolic flow reversal in the hepatic veins
- Thrombosis
- Liver tumor-mass effect
- Hepatic congestion
- Budd Chiari Syndrome
- Liver disease

US of Hepatic Veins:

- LHV and MHV best seen in transverse plane, subxiphoid approach
- RHV best evaluated in longitudinal plane
- Diameter and flow normally vary with respiration
- Increase in size as they near the IVC

Doppler of the Hepatic Veins:

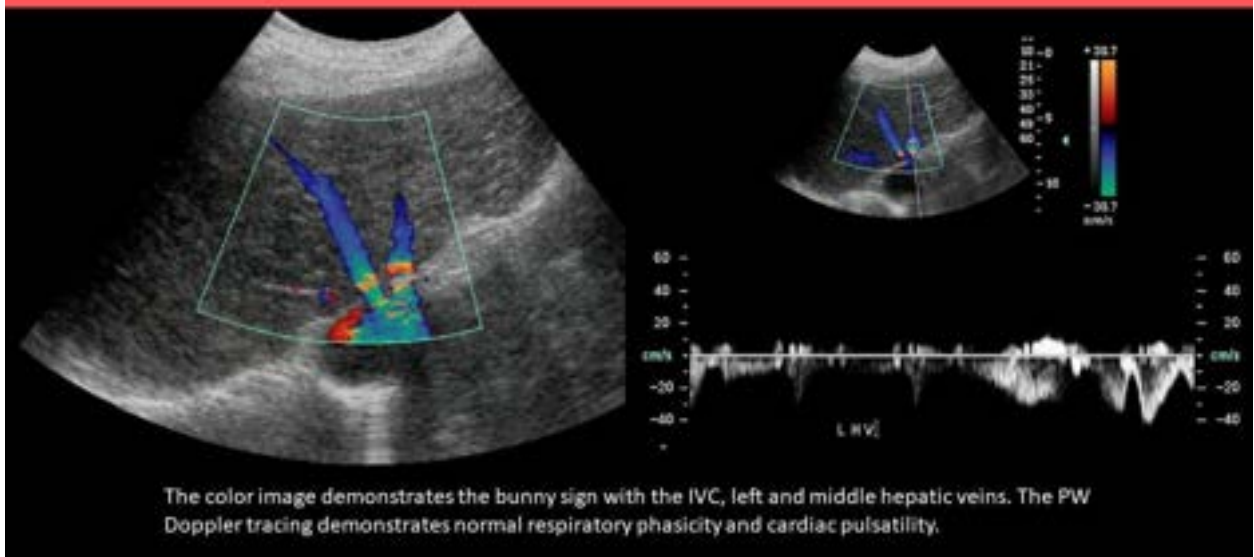
- Tracing obtained during quiet respiration
- 0 degree cursor angle because flow velocity is not evaluated
- MHV and LHV in transverse, RHV in longitudinal
- Normal blood flow is away from the liver (hepatofugal)
- Normal HV flow is considered triphasic

- The tracing normally demonstrates respiratory phasicity and cardiac pulsatility
- The majority of flow moves forward into the right atrium until the atrial "kick" occurs
- The Doppler tracing will demonstrate two large antegrade diastolic and systolic waves followed by a small retrograde component that corresponds with the atrial contraction
- Increased cardiac pulsatility seen with CHF and significant tricuspid regurgitation

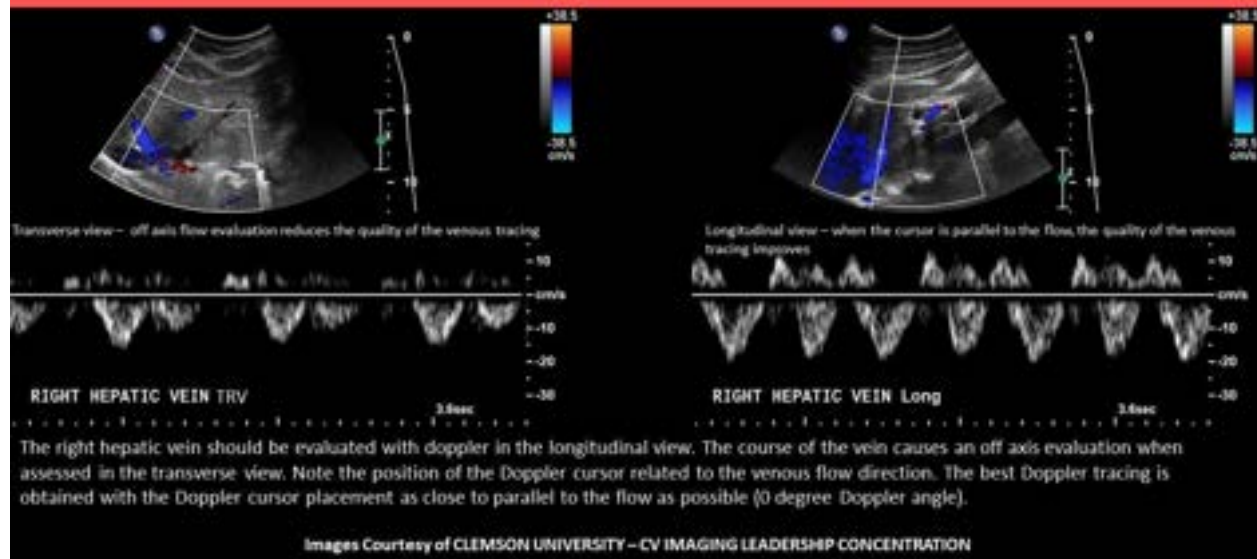
Hepatic Veins



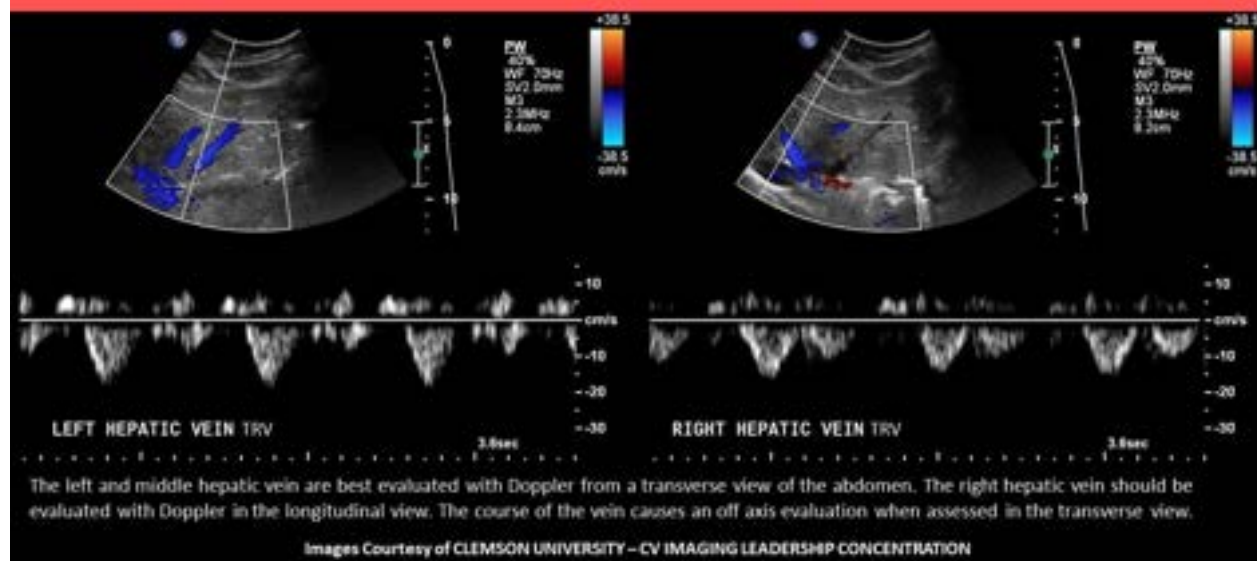
Hepatic Veins



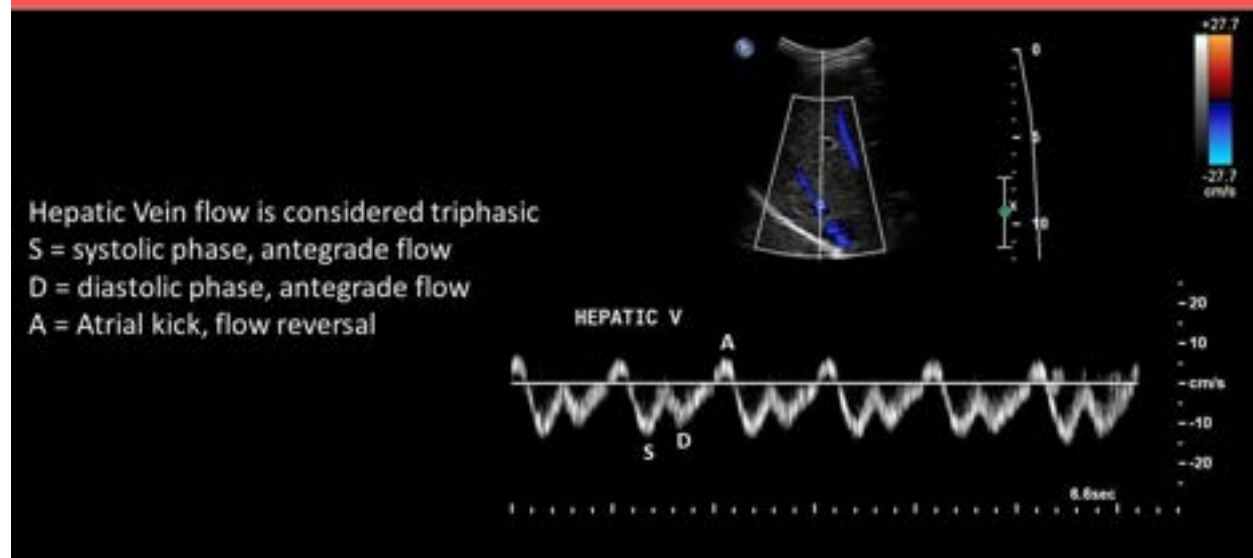
Hepatic Vein Doppler



Hepatic Vein Doppler



Normal Hepatic Vein Waveforms



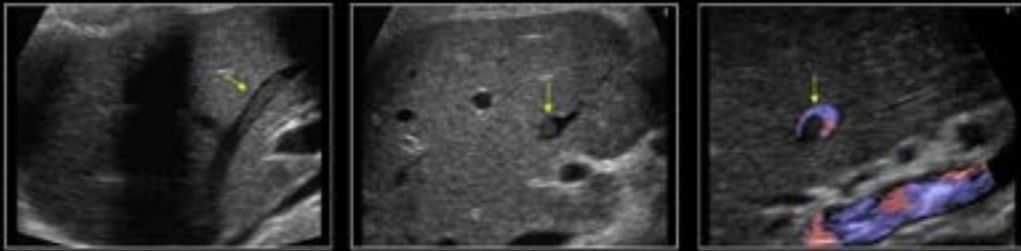
Budd-Chiari Syndrome:

- Obstruction of HV by thrombus (#1), stenosis, membrane, compression or tumor
- Causes include oral contraceptives (OCP), hepatocellular CA, renal CA, adrenal CA
- Pain, jaundice, hematemesis, ascites, hepatomegaly, portal HTN

Sonographic Appearance:

- Varies with degree of obstruction; echogenic material within veins
- Hepatomegaly (Acute)
- Caudate lobe hypertrophy (Acute)
- Marked ascites (Acute)
- Hepatic veins tough to see due to significant fluid accumulation (Acute)
- Right lobe atrophy and fibrosis (Chronic)
- Splenomegaly (Chronic)
- HV thick walled, difficult to see within the liver parenchyma (Chronic)
- Lack of flow within hepatic veins
- Portal vein flow slow or reversed, portal HTN

Thrombosed Hepatic Vein



Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

RENAL VEINS:

Anatomy:

- Drains the kidneys, ureters, and adrenals
- RRV courses medially and anterior to RRA
- LRV courses between the aorta and SMA; longer than the RRV

Indications to Scan Renal Veins:

- Renal Mass - renal cell carcinoma is the most common type of tumor to invade the renal vein of the ipsilateral kidney
- Transplant - evaluate for thrombosis
- Nutcracker Syndrome - SMA compresses left renal vein against the aorta; can lead to left renal vein thrombosis and left renal vascular congestion

Exam Techniques:

US of Renal Veins:

- RRV best seen in transverse view, anterior to renal artery
- LRV best seen in transverse view, anterior to the aorta and posterior to SMA; use the left renal artery as a landmark to find the vein

- If tumor invasion is suspected within the vein, evaluate the adjacent organs for mass formation; renal cell carcinoma is most common
- If thrombosis is suspected, evaluate the kidney and IVC for a potential cause

Doppler of Renal Veins:

- Use 3-4mm Doppler sample size
- At the origin, flow is mildly pulsatile with minimal respiratory phasicity
- Continuous flow with minimal respiratory phasicity seen near renal hilum
- CHF can cause bilaterally increased pulsatility
- Unilateral increase in pulsatility related to acute urinary obstruction or distal renal vein thrombosis

Renal Vein Thrombosis:

- Can cause hematuria and acute renal failure
- Increased risk for pulmonary embolism
- Most commonly caused by primary renal disease
- Caused by extrinsic compression, trauma, surgery and dehydration
- Considered a critical finding due to increased PE risk and potential kidney damage

Sonographic Appearance:

- Increased kidney size
- Hypoechoic renal cortex and decreased corticomedullary differentiation with mottled echogenicity
- Decreased or absent flow detected with Doppler in the renal veins
- Continuous PW Doppler signal with partial thrombosis
- Increased resistance in the renal arteries with reduced or absent diastolic flow
- Parenchymal RI >0.7
- Chronic thrombosis can lead to renal atrophy

Renal Veins

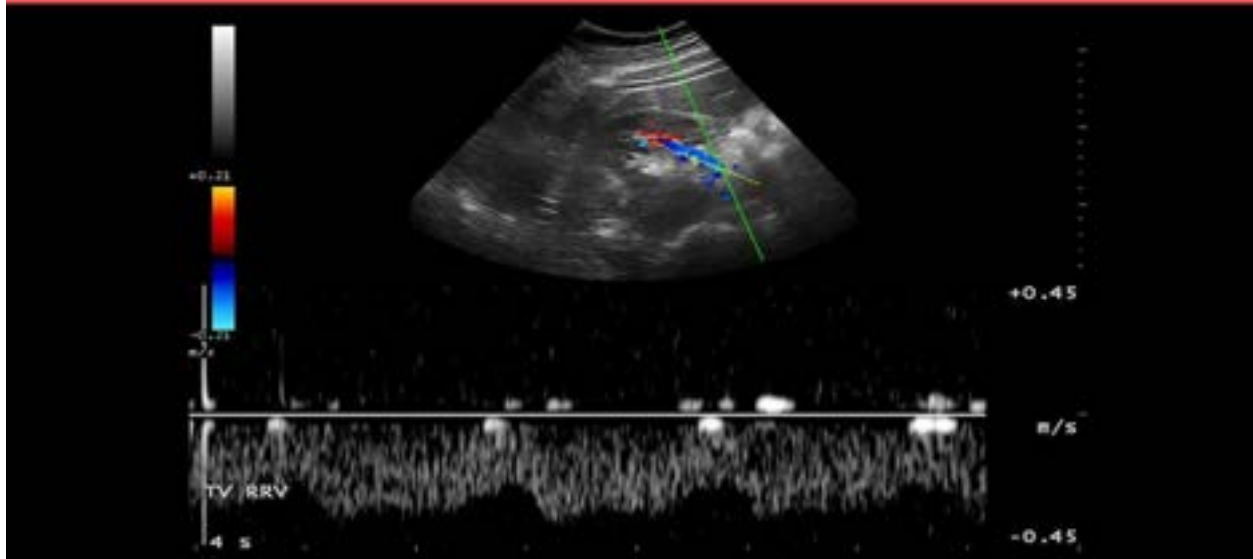


Nutcracker Syndrome



Courtesy of Dr. Taco Geertsma
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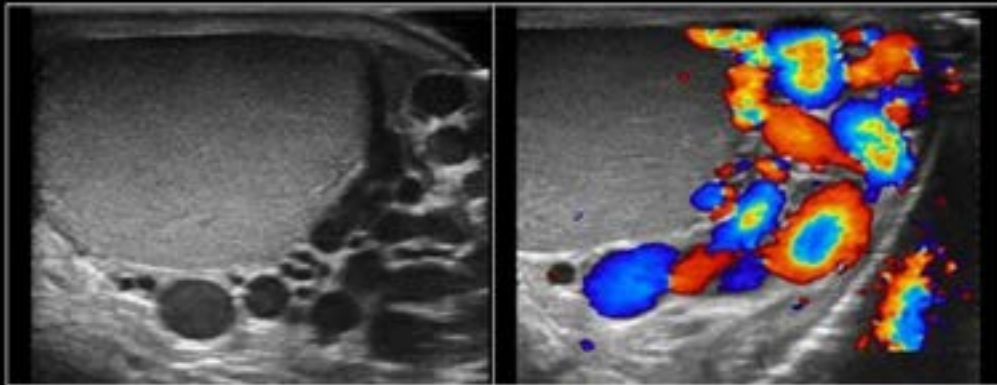
Renal Vein Doppler



GONADAL VEINS:

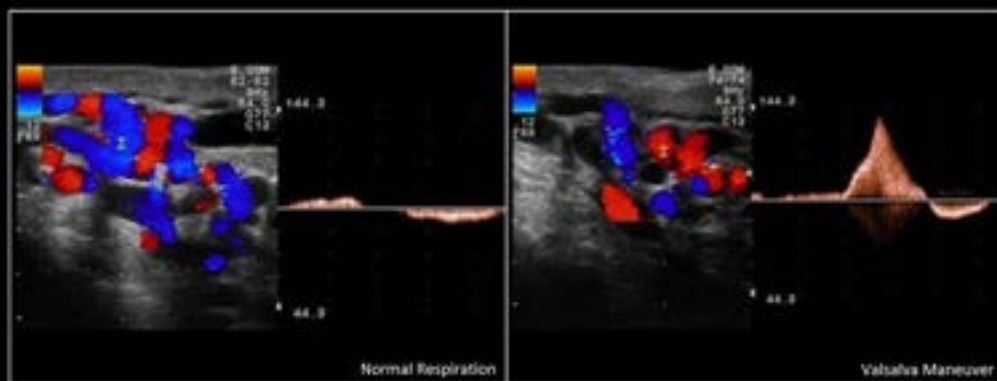
- Drain the blood from the testicles or ovaries
- The right gonadal vein empties directly into the IVC
- Varicocele
 - The left gonadal vein empties into the left renal vein and has a higher risk of left sided varicocele formation due to anatomic course
 - Distal end called pampiniform plexus just outside testicle; can dilate causing a varicocele; varicoceles have been linked to infertility
 - Valsalva maneuver used to increase varicocele vessel size and demonstrate reflux on Doppler
- Ovarian Vein Compression Syndrome
 - The ovarian vein normally crosses anterior to the ureter as it travels cephalad to join the IVC/left renal vein
 - A dilated ovarian vein can cause notching, dilatation, or obstruction of the ipsilateral ureter
 - Most commonly related to varicosities of the ovarian vein or ovarian vein thrombosis

Varicocele



Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

Varicocele – Reflux on Spectral doppler



Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

Portal Venous System: Sonographic Evaluation and Disease

Anatomy:

- Confluence of superior mesenteric vein and splenic vein posterior to the neck of the pancreas; inferior mesenteric vein joins splenic vein prior to splenic confluence with SMV
- Enters liver carrying blood drained from the digestive system
- Branches into:

“ Right Portal

- * Anterior branch - supplies anterior right lobe
- * Posterior branch - supplies posterior right lobe

“ Left Portal

- * Medial branch - supplies medial left lobe
- * Lateral branch - supplies lateral left lobe

US of Portal System:

- Thicker, echogenic walls as compared to the hepatic veins
- Course transversely through liver, decreasing in size as they near the diaphragm

Main Portal Vein: (MPV)

- Arises from junction of SV and SMV
- Anterior to the IVC
- Enters liver at porta hepatis and splits into right and left branches
- Drains GI tract and sends blood to liver for filtration
- Carries 70-80% of total blood flow into liver; deoxygenated blood from digestive tract 85% O₂ saturation
- Hepatic artery supplies 20-30% of the blood to the liver; oxygenated blood 95% O₂ saturation

US of MPV:

- Simethicone may be administered to reduce bowel gas
- The right coronal oblique *scanning plane* with an intercostal approach is very helpful in patients with large amounts of ascites and bowel gas
- The intercostal approach uses the liver as an acoustic window, reduces the AP distance between the probe and the vein, and allows for Doppler angles between 0 and 60 degrees
- To obtain right coronal oblique scanning plane, place the patient in the *LPO or LLD position*
- Position the transducer at a rib interspace and aim the ultrasound beam at the hepatic hilum
- Demonstrates a tubular structure coursing from the pancreas head to liver
- Diameter changes with respiration and Valsalva
- The portal vein should be assessed with the patient supine and with quiet respiration
- The inner wall to inner wall diameter measurement should be obtained at the point the portal vein crosses the IVC
- Normal diameter <13 mm

Doppler of MPV:

- Sample obtained during quiet respiration
- Evaluated in transverse/oblique plane
- Flow toward the liver (hepatopetal)
- Continuous flow with minimal respiratory phasicity in flow due to more "rigid" walls than hepatic veins
- Normal flow is low velocity 15-20cm/s
- If measuring the peak velocity of flow, angle correction must be used
- Thrombosis causes decreased respiratory and cardiac variation in flow or can occlude vessel
- Liver disease can cause increased resistance to flow and even flow reversal
- Right heart failure and tricuspid regurgitation may cause exaggerated portal vein pulsatility

Hepatic Veins	Portal Veins
Thin walls	Thick, echogenic walls
Get larger as they travel through the liver to meet IVC	Get smaller as travel through the liver
Vertical orientation	Horizontal orientation
Hepatofugal flow	Hepatopetal flow
Cardiac pulsatility and respiratory phasicity	Mild respiratory phasicity

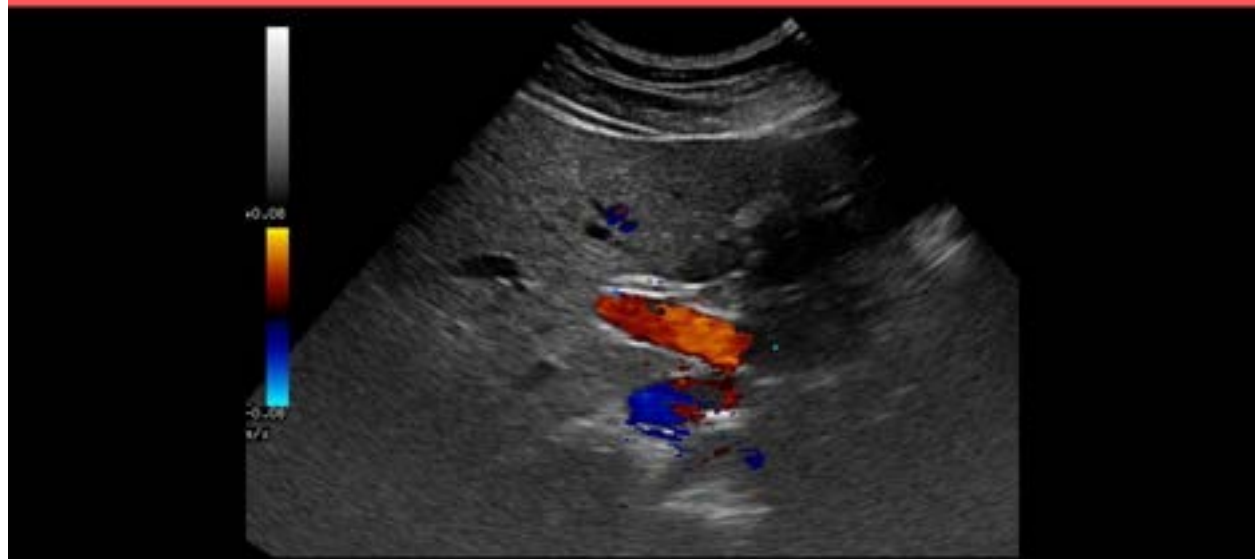
Main Portal Vein



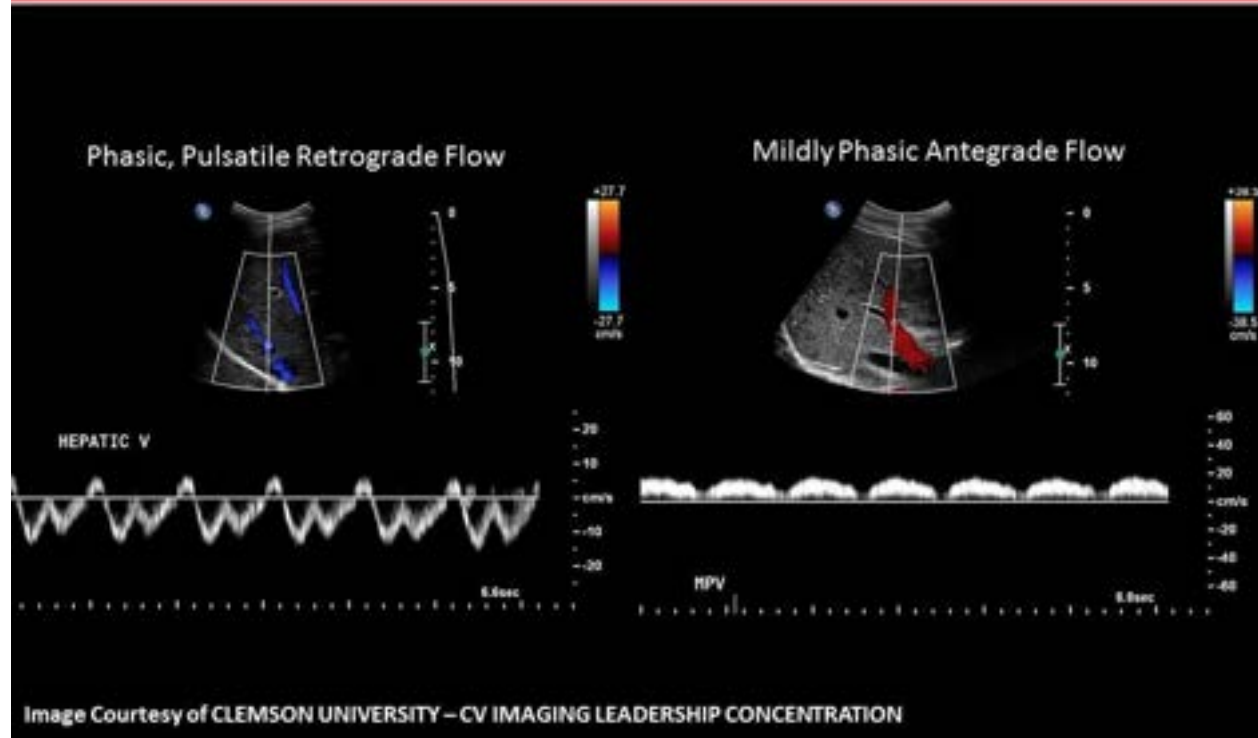
Main Portal Vein



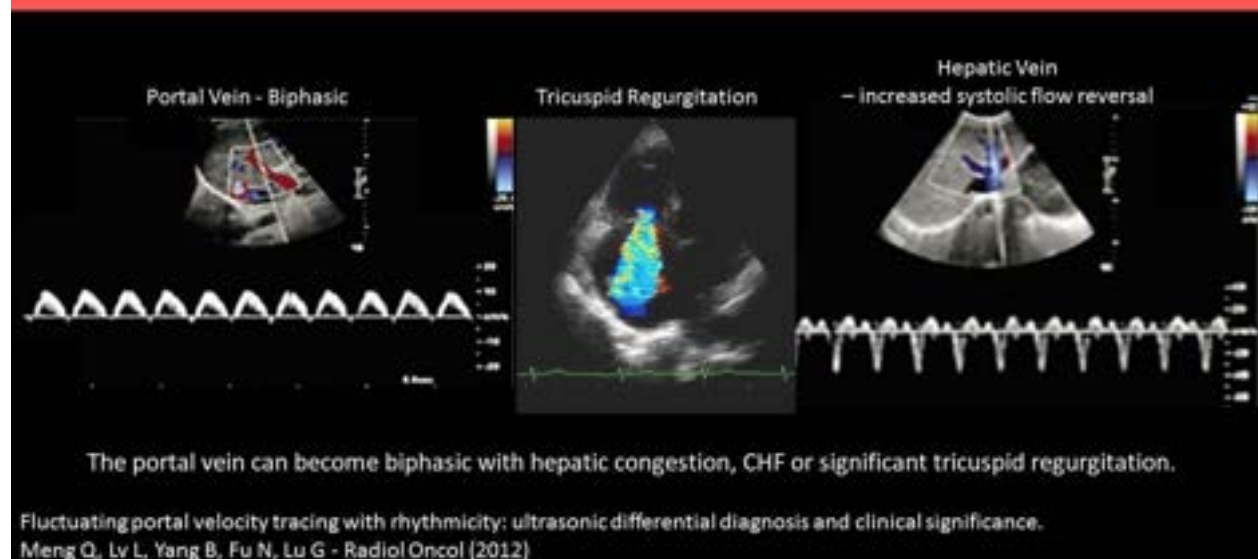
Main Portal Vein – Hepatopetal Flow



Hepatic Vein vs. Portal Vein



Tricuspid Regurgitation - CHF



How is the MPV Red on one pic and Blue on the other but both demonstrate normal flow direction??? Always be aware of your scanning approach of the portal vein. This will affect the Doppler evaluation. Rely on your color map for appropriate flow direction!!

SPLenic VEIN: (SV)

- Originates at the splenic hilum
- Course posterior to the pancreas
- Joins SMV to form MPV
- Drains the stomach, spleen, and pancreas

US and Doppler of the Splenic Vein:

- Seen in transverse as a tubular structure posterior to the body and tail of pancreas
- Flow toward the liver and away from spleen

SUPERIOR MESENTERIC VEIN: (SMV)

- Originates at the mesentery
- Joins splenic vein posterior to the neck of the pancreas to form MPV
- Drains the small intestine and proximal colon

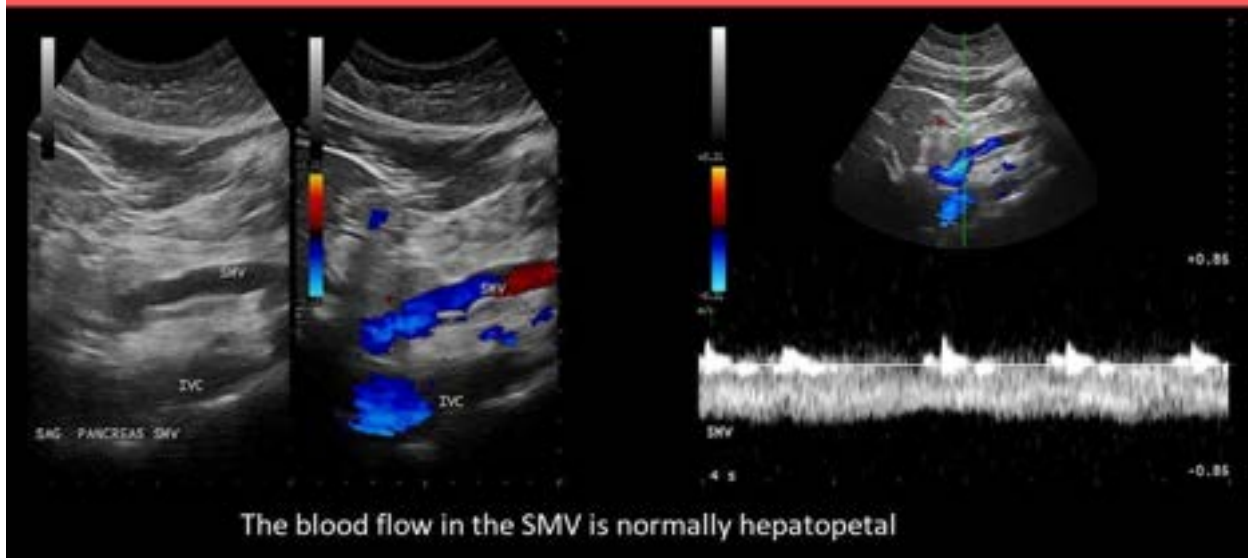
US and Doppler of the SMV:

- Best seen in longitudinal as tubular structure to the right of SMA
- In transverse seen as a circular structure
- Flow toward liver and away from mesentery

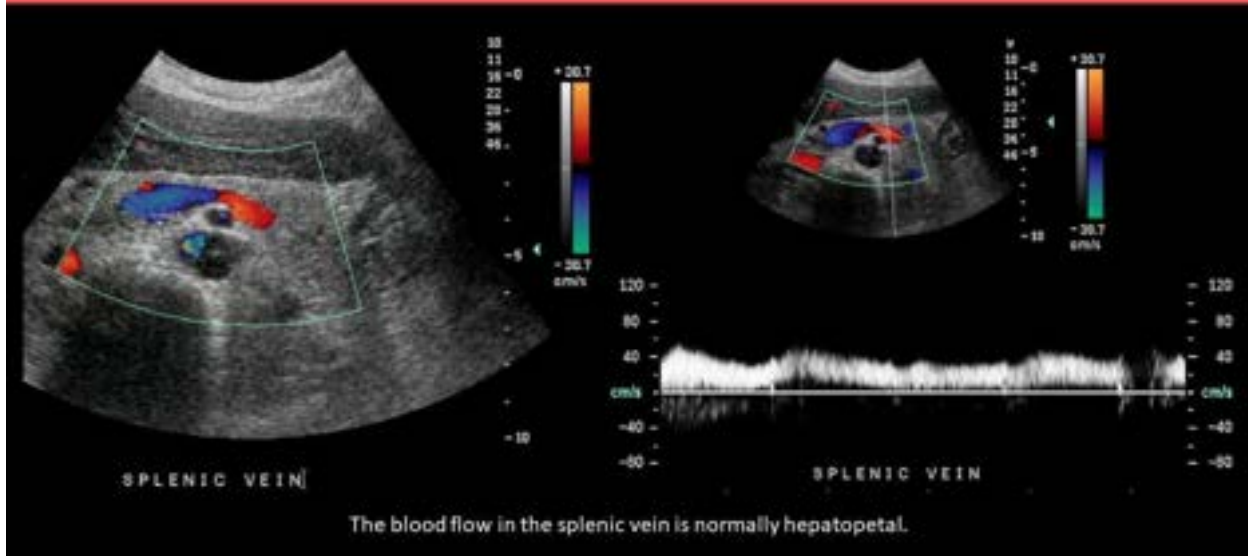
INFERIOR MESENTERIC VEIN: (IMV)

- Arises from colon
- Ascends to the left of the IMA
- Joins splenic vein posterior to the body of the pancreas
- Drains distal colon
- Usually not seen

Superior Mesenteric Vein



Splenic Vein



Hepatoportal Duplex Evaluation:

- The exam is performed to assess flow in and out of the liver
- Exam indications include portal HTN, history of cirrhosis, Budd Chiari syndrome
- 2D evaluation:
 - Measure the length of the liver and spleen

- Cirrhosis causes a reduction in liver size and an increase in spleen size
- Budd Chiari syndrome causes hepatomegaly in the acute phases and causes a reduction in liver size with an increase in spleen size with chronic obstruction
- Evaluate the echotexture of the liver and spleen, note any nodules or mass formation
- If cirrhosis is suspected, the liver surface should be evaluated with a high frequency LINEAR transducer
- Color and PW doppler evaluation should include:
 - Use angle correction in order to measure peak velocity of flow of any vessel
 - Main portal vein
 - Right and left portal vein
 - Splenic vein
 - Superior mesenteric vein
 - Right, left and middle hepatic veins
 - Right and left hepatic arteries
 - Any visible collaterals present

Limitations:

- Bowel gas
- Patient not NPO
- Obesity
- Abdomen pain with probe pressure
- Patient unable to follow breathing instructions

Portal Hypertension:

- Normal portal pressure is 5-10mmHg
- Elevated pressure in the portal system due to flow obstruction, can lead to flow reversal in the portal system
- Most commonly caused by cirrhosis
- Collateral formation occurs causing varices to form; gastroesophageal varices can lead to life threatening hemorrhage
- Esophageal varices are a common sign of portal HTN; cause hematemesis; form due to retrograde flow in the coronary (left gastric) vein
- Coronary vein is a branch of the portal vein, near the liver hilum; hepatofugal flow; normally drains into the splenic vein near the portal confluence
- Associated with recanalized umbilical vein; dilated tubular structure connected to left portal vein and extending toward the umbilicus; flow is hepatofugal; may cause the LPV to remain hepatopetal due to blood exiting the liver through the collateral pathway

- Caput Medusa - tortuous vessels around the umbilicus due to portal HTN
- Portal-Caval Shunts are the preferred treatment
- Extrahepatic presinusoidal HTN - PV Thrombosis
- Intra-hepatic presinusoidal HTN - Schistosomiasis
- Intra-hepatic HTN - Most common type; Cirrhosis
- Intrahepatic postsinusoidal HTN - Hepatic vein obstruction, Budd Chiari syndrome
- The liver surface should be evaluated with a higher frequency linear transducer for surface nodularity that is common with cirrhosis
- Diameter, flow direction and flow velocity of portal vein should be evaluated
- Evaluate the adjacent vascular changes = size of the spleen and collateral formation at the porta hepatis and splenic hilum; assess for umbilical recanalization
- Treated with portosystemic shunt

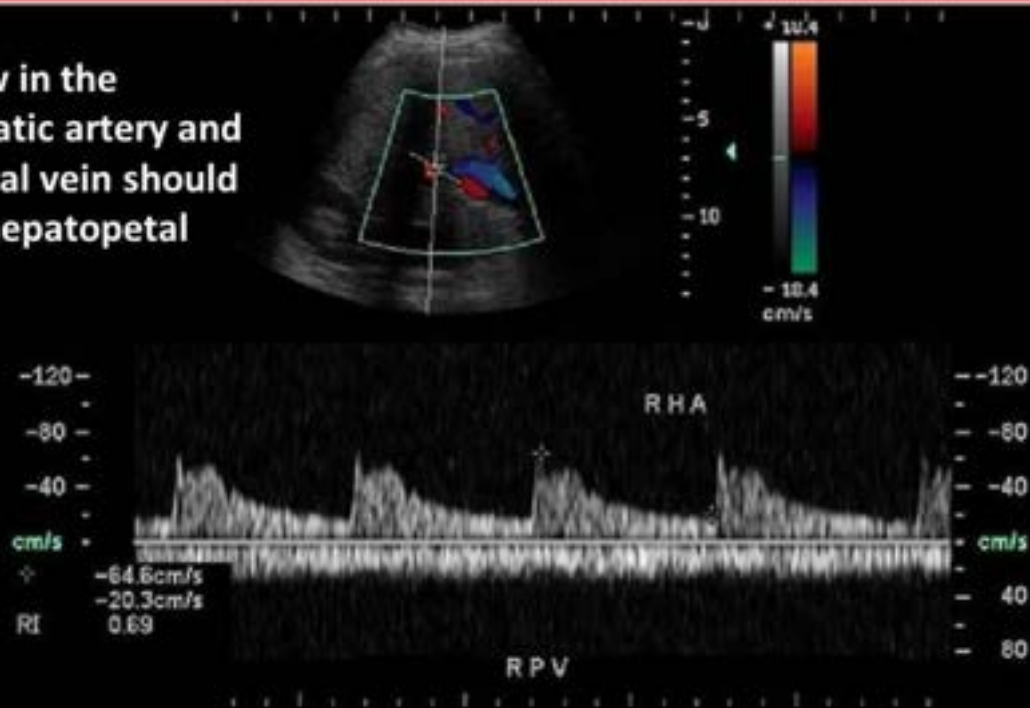
Sonographic Appearance:

- Splenomegaly >13cm
- Dilated PV >13mm
- Dilated splenic vein >10mm
- Slowed or reversed flow in portal system (hepatofugal flow)
- SMV and splenic vein significantly increase in size with deep inspiration
- Cavernous transformation - formation of venous varices/collaterals at porta hepatis
- As portal HTN increases, portal flow will become more continuous and lose all respiratory variation in flow
- Portal flow can reverse direction(hepatofugal) in severe cases
- Resistance in the hepatic artery will increase with the increase in resistance to flow in the cirrhotic liver tissue
- Can see associated portal vein thrombosis

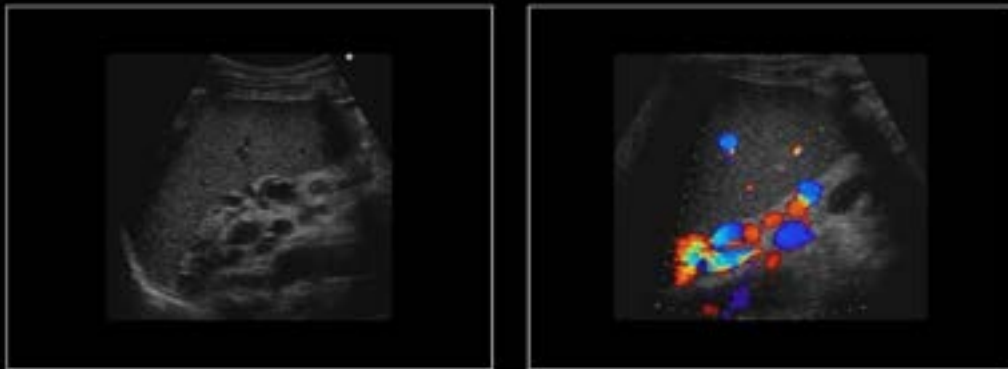
Doppler Hint!! Hepatic Artery and Portal Vein carry blood into the liver and should be demonstrated on the same side of the Doppler baseline in normal patients.

Hepatofugal Portal Vein Flow

Flow in the
hepatic artery and
portal vein should
be hepatopetal

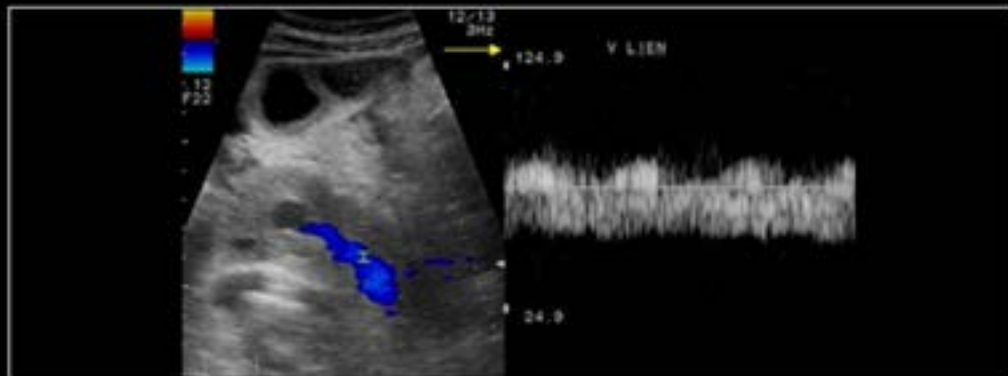


Splenic Varices



Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

Splenic Vein Flow Reversal



Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

Portal Vein Thrombosis:

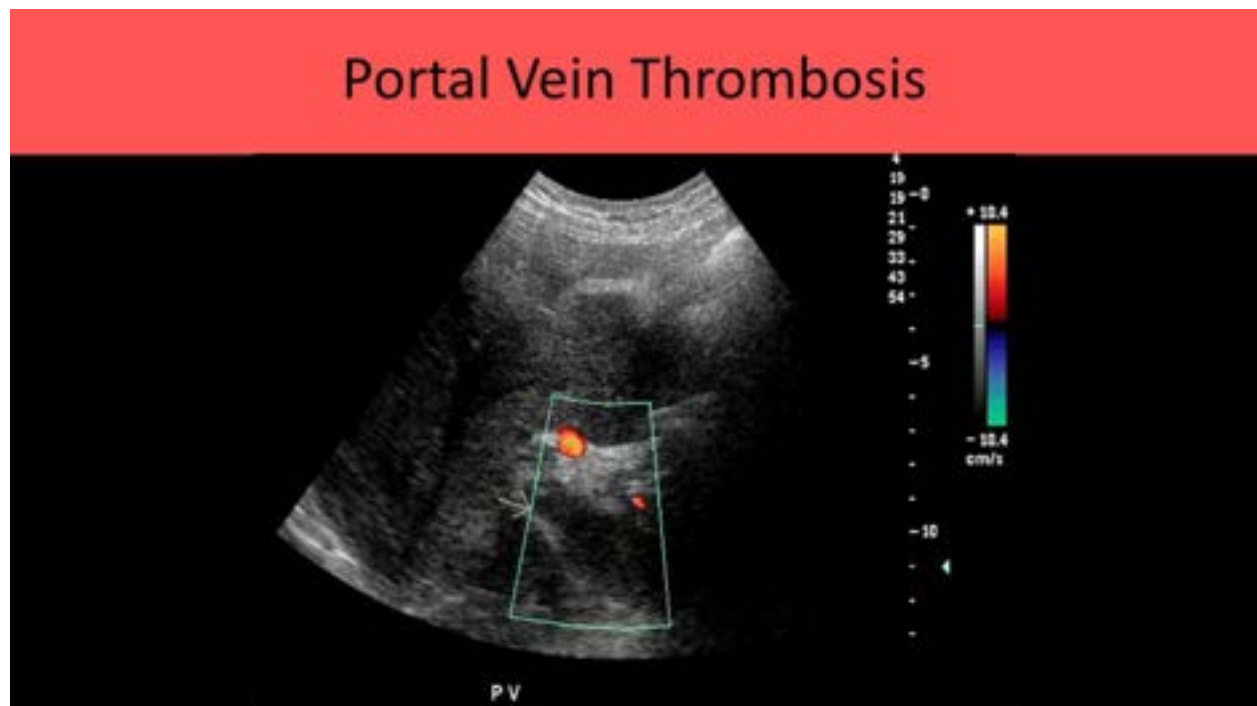
- Causes include portal HTN, inflammatory process, trauma, post-op complications, extrinsic compression
- Pain, fever, rigid abdomen, nausea, vomiting
- Increased LFTs (Liver Function Tests)
- Cavernous transformation occurs 1-3 weeks after PV occlusion by thrombus
- Care must be taken to distinguish the portal vein and the common bile duct when no flow is present in either structure

Sonographic Appearance:

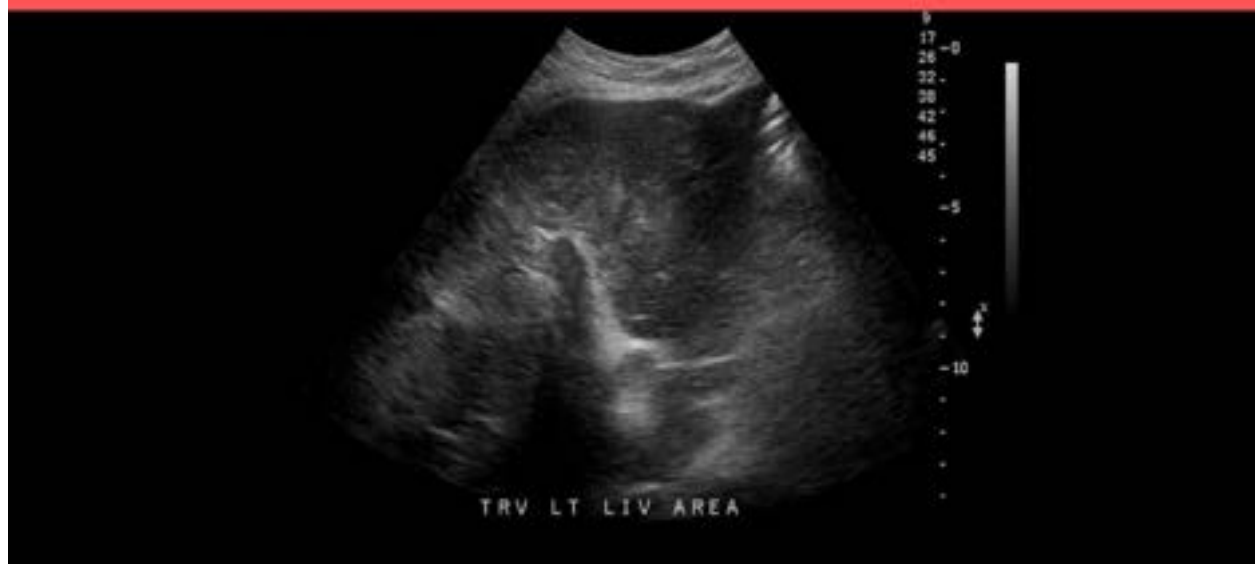
- Increased diameter of splenic vein and superior mesenteric vein
- Acute-echogenic debris in lumen, >13mm, absence of flow
- Chronic-more echogenic, <13mm, cavernous transformation, numerous collaterals, decreased flow

Treatment:

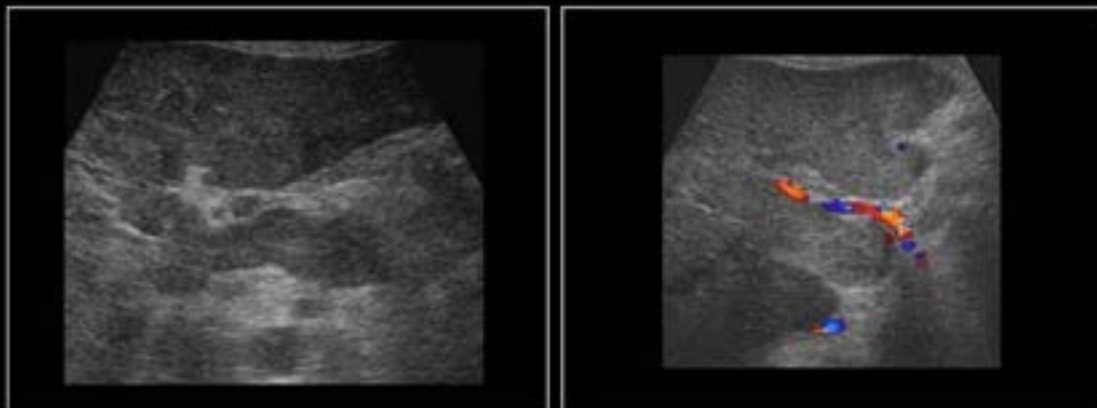
- Portal-Caval Shunts



Portal Vein Thrombosis



Portal Vein Thrombosis



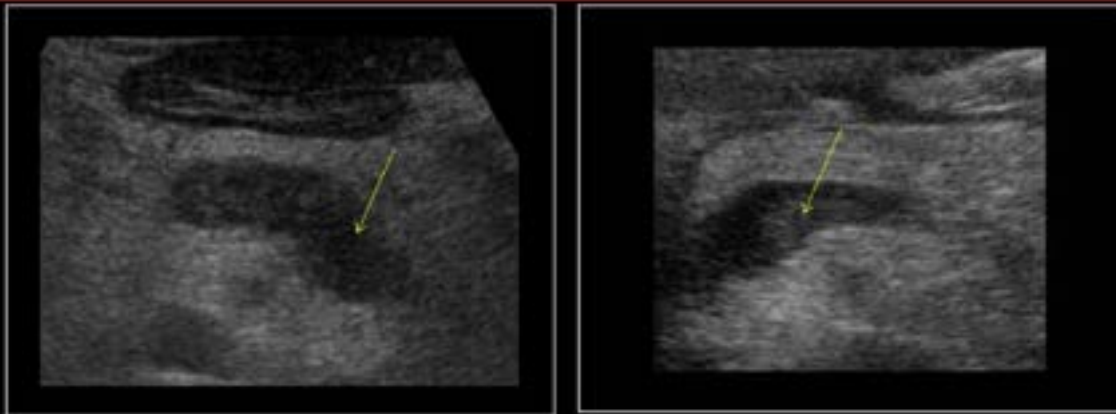
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Portal Vein Thrombosis

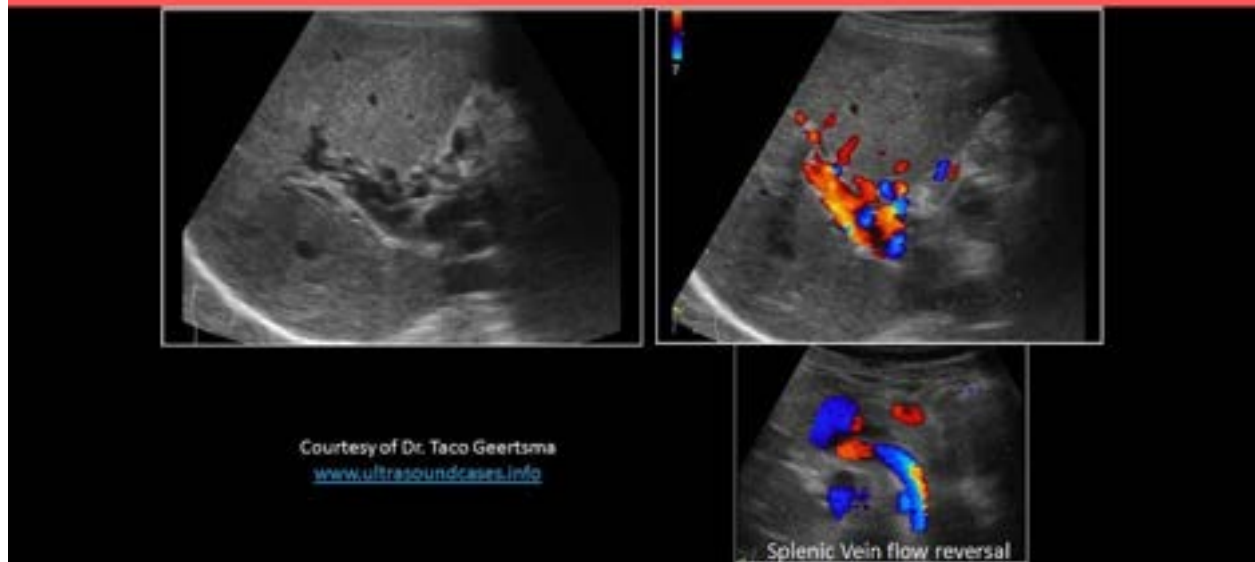


Splenic Vein Thrombosis



Courtesy of Dr. Taco Geertsma
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Cavernous Transformation



Portosystemic or Portal - Caval Shunt:

- Metallic device used to reduce pressure and increase flow in the extrahepatic portal system; strongly reflective material used for easy visualization sonographically
- Used to alleviate portal hypertension
- Catheter inserted into the IJV and advanced into the hepatic vein through the liver and into the portal system
- A shunt is placed between the portal system (usually MPV) and the hepatic vein (usually RHV)
- Blood flow bypasses the liver and goes from the digestive system straight to the IVC system
- PV connected to Hep V/IVC - Transjugular Intrahepatic Portosystemic Shunt (TIPS), most common
- Easiest to see with US, Doppler used to assess patency
- Flow should be hepatopetal in the proximal shunt and hepatofugal in the distal shunt
- Complications include infection, abscess, shunt stenosis and thrombosis
- Thrombosis is the most common cause of acute shunt stenosis/occlusion
- The most common site of TIPS stenosis (chronic occlusion) is at the hepatic vein anastomosis
- Neointimal hyperplasia is the most common cause of chronic graft stenosis/occlusion

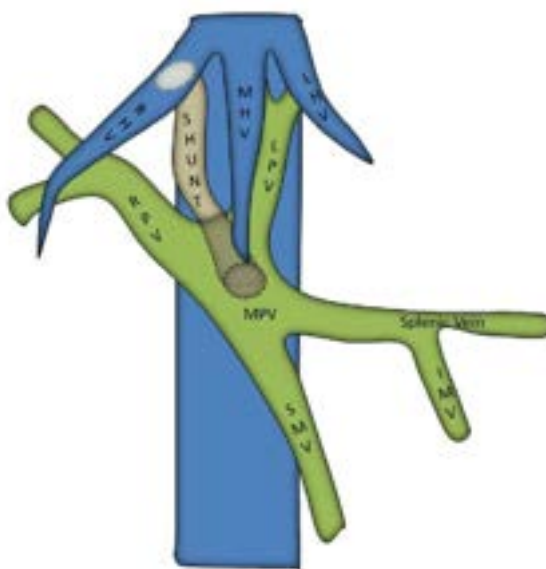
Sonographic Appearance:

- Pre-OP exam includes evaluation of flow in the portal vein, splenic vein, and SMV; liver should be evaluated for mass formation that will affect the path of the shunt

- Post-OP exam should be performed within the first 24 hours of shunt placement; document the patency of the shunt and establish baseline velocities
- Shadowing from air in the graft material immediately post-op can degrade the follow up exam when performed 48 - 72hrs after procedure
- Shunt walls are very echogenic with linear echoes in longitudinal and echogenic foci seen in the wall in the transverse plane
- Normal flow is from the portal system into the shunt then into the hepatic system
- Right and left portal veins will demonstrate hepatofugal flow (toward the shunt)
- Shunt flow is monophasic and slightly pulsatile with moderate spectral broadening
- Similar velocities should be identified at both ends of the shunt
- **MUST USE ANGLE CORRECTION WHEN EVALUATING THE FLOW VELOCITY IN A VESSEL** (not just arteries, any flow that velocity is being measured)
- Peak flow velocity at any point along the shunt should not fall below 50cm/s and should not exceed 200cm/s
- Hepatopetal portal vein flow <30cm/s indicates abnormal shunt flow and impending graft failure
- Hepatofugal flow in the main portal vein is abnormal
- Hepatopetal flow in the right and left portal veins is abnormal
- Hepatopetal flow in the associated hepatic vein is abnormal

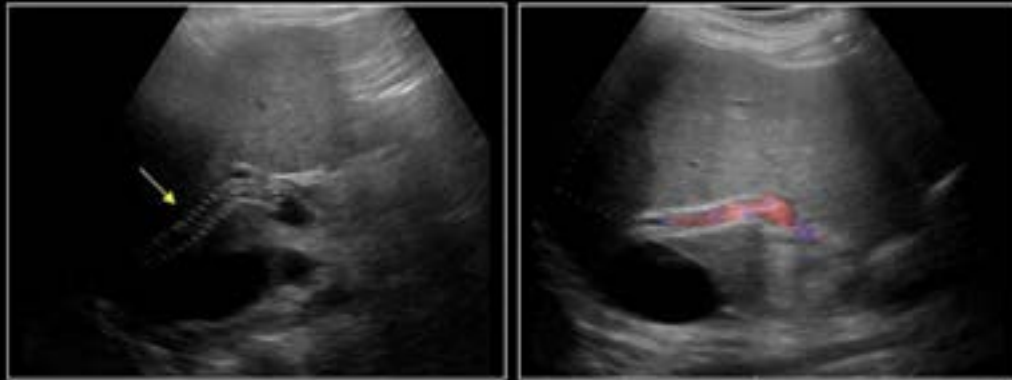
Transjugular Intrahepatic Portosystemic Shunt

In order to relieve severe portal HTN, a synthetic shunt is connected from the portal system through the liver tissue and into the hepatic veins (or IVC). It is usually introduced into the body by inserting a catheter into the jugular vein.



TIPS

Note the blood flow in the shunt is from the portal vein into the hepatic vein/IVC.

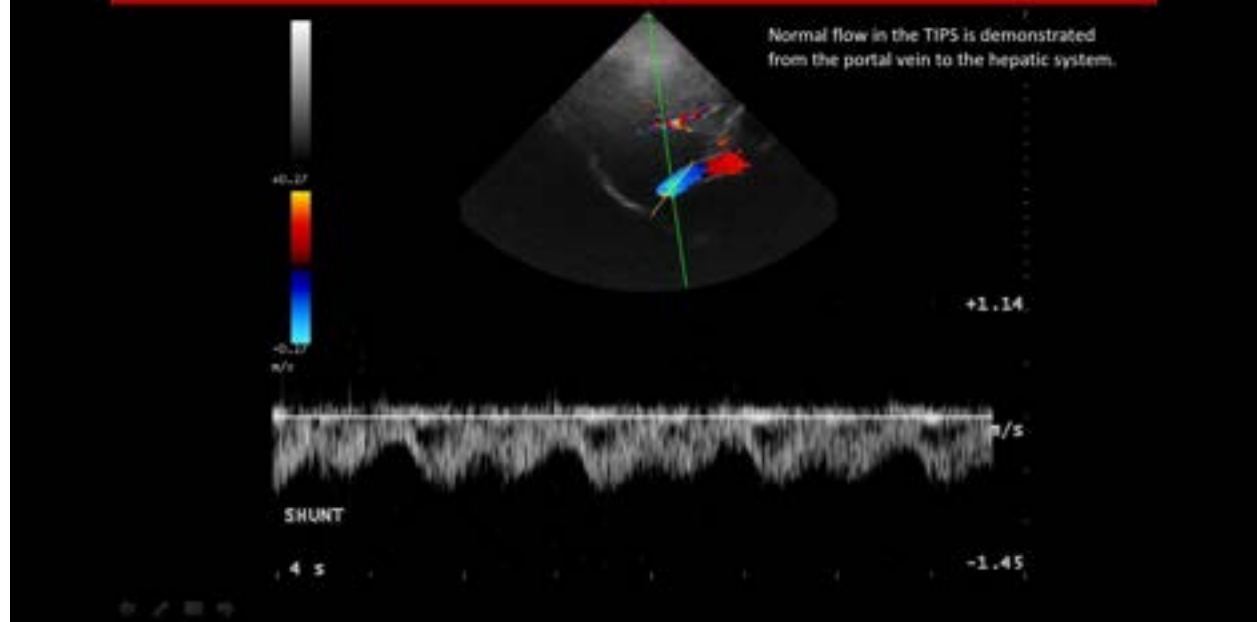


Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

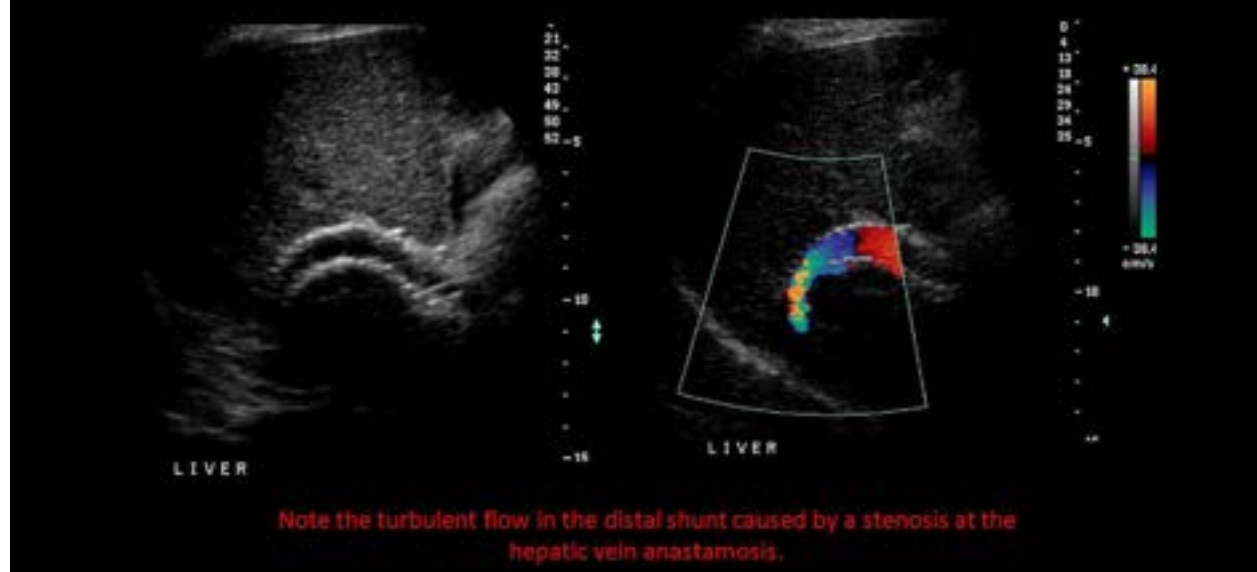
TIPS - Normal Color Flow



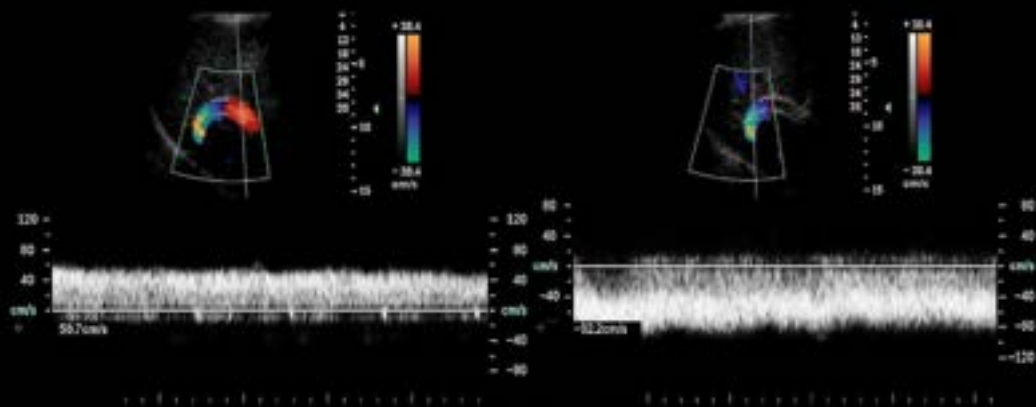
TIPS Normal PW Doppler



Abnormal Transjugular Intrahepatic Portosystemic Shunt



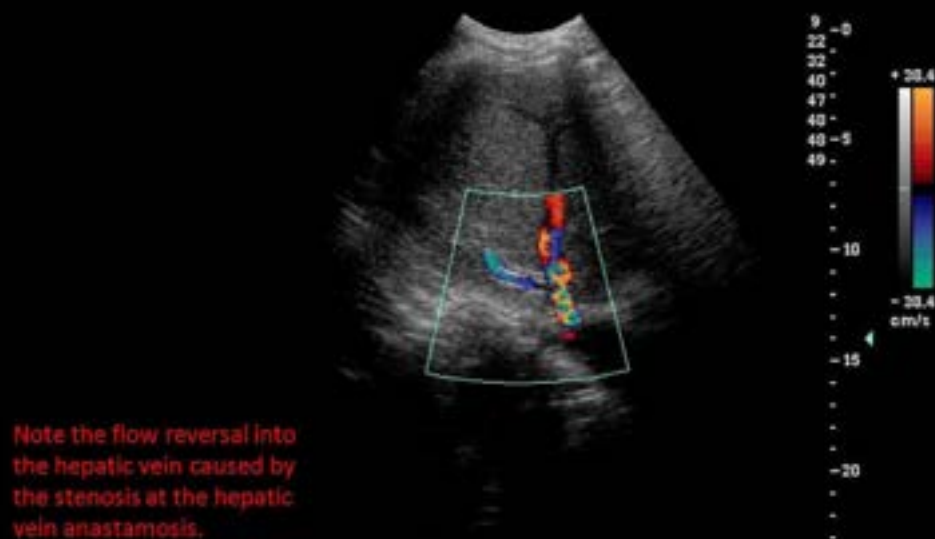
Abnormal Transjugular Intrahepatic Portosystemic Shunt



Note the increasing velocity of the flow within the shunt as it nears the hepatic vein anastomosis.

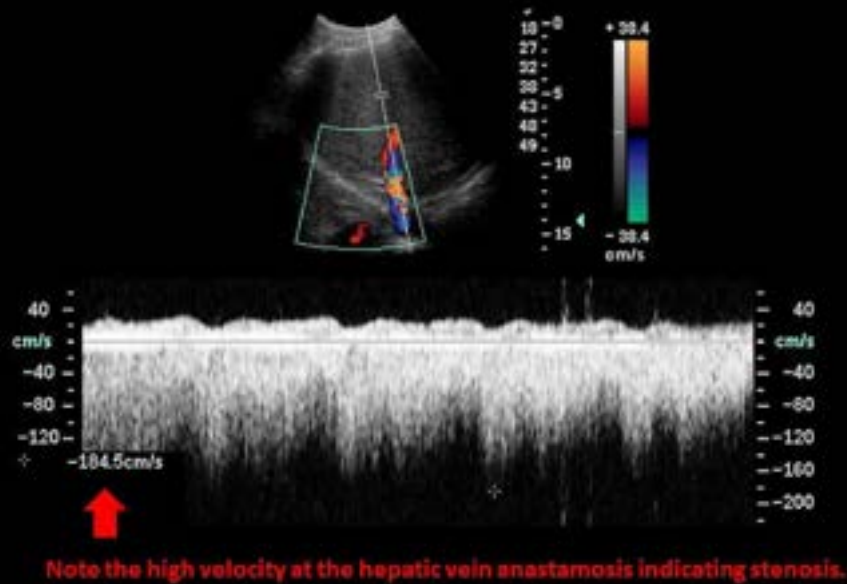
Normal velocities should not exceed 200 cm/s

Abnormal Transjugular Intrahepatic Portosystemic Shunt



Note the flow reversal into the hepatic vein caused by the stenosis at the hepatic vein anastomosis.

Abnormal Transjugular Intrahepatic Portosystemic Shunt



Hereditary Hemorrhagic Telangiectasia:

- AKA Osler-Weber-Rendu Disease
- Autosomal dominant disorder
- Affects vasculature of multiple organs
- Characterized by multiple arteriovenous malformations (AVMs)
- Most commonly the AVMs are superficially located in the skin, but can also occur in pulmonary, cerebral, GI tract, and liver vasculature
- Liver - AV malformations (AVM can be arteriovenous or portovenous), fibrosis, and cirrhosis
- Recurrent episodes of bleeding
- Contrast echocardiography can be used to diagnose a suspected pulmonary AVM
- Ultrasound is most helpful in evaluating liver AVMs

Sonographic Appearance:

- Dilated HA (up to 10mm)
- Multiple dilated tubular structures with color flow demonstrated
- Dilated HV
- Causes signs of portal HTN due to increased portal flow
- Increased resistance in the hepatic arterial flow with abnormal arterial connections
- Can cause mesenteric angina from steal phenomenon

Bean Syndrome:

- AKA Blue rubber bleb nevus syndrome
- Soft blue skin lesions
- Multiple bowel venous malformations that can cause lower GI bleeding
- Can also affect other organ systems

Liver Transplant:

- Most are orthotopic transplant of whole liver grafts
- Other types include segmental liver transplantation and split allografts(used for pediatric patients)
- Donor vessels connected to native vessels IF PATENT
- Hepatic artery and portal vein connected to the native vessels just proximal to the liver hilum
- If portal thrombosis is present, a bypass graft is used to alleviate the flow obstruction

Indications:

- Cirrhosis
- Hepatocellular carcinoma
- Chronic hepatitis B, chronic hepatitis C, autoimmune
- Acetaminophen overdose
- Primary biliary cirrhosis
- Primary sclerosing cholangitis
- Hemochromatosis
- Wilson's disease

Contraindications:

- Extrahepatic malignancy
- Hepatocellular carcinoma >5cm tumor or more than three tumors >3cm in diameter
- Diffuse infiltration of malignancy
- Complete thrombosis of the portal and superior mesenteric veins
- Active systemic infection
- Active substance or alcohol abuse
- Human Immunodeficiency Virus (HIV)
- Previous malignancy

Complications:

- Hepatic artery thrombosis (#1)
- Hepatic artery stenosis
- Pseudoaneurysm

- AV fistula
- Venous thrombosis
- Hematoma
- Biloma
- Infection/Abscess
- Biliary stenosis at the anastomosis site

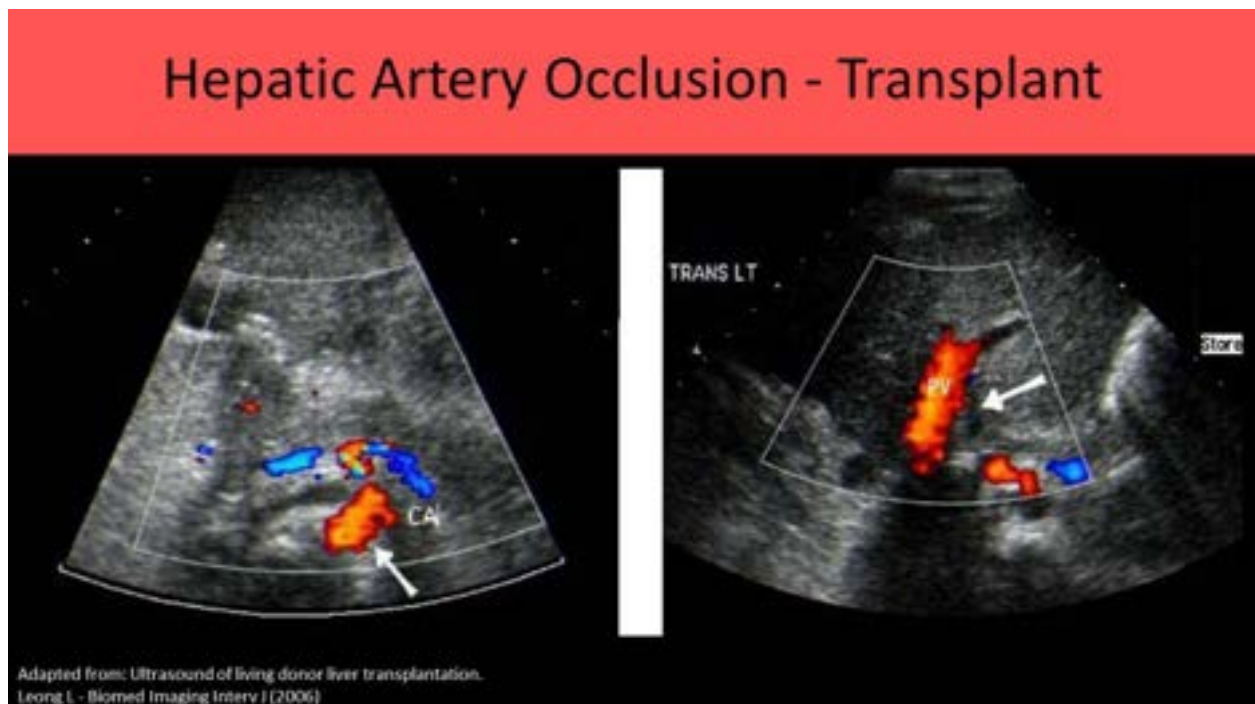
Exam Technique:

- 2D Ultrasound and Doppler evaluation preferred for assessment of allograft
- Curvilinear or phased array, 2MHz to 6.5MHz commonly used
- Pre-Op
 - Donor - assess hepatic vein number and size to guide resection
 - Recipient - assess vessel patency, portal, splenic and hepatic veins, hepatic artery
- Post-Op
- Before beginning the exam, the surgical notes should be reviewed to determine if the patient has a full or partial transplant, a piggyback or interposition anastomosis and note the anastomosis sites for all vasculature
- All native vessels and transplant vessels must be evaluated with Color and PW Doppler for patency
 - Hepatic artery
 - Native vessel
 - Anastomosis
 - Donor artery
 - Right and left hepatic arteries
 - Measure peak velocity, resistive index (RI) and acceleration time (AT)
 - Main portal vein and right and left portal branches
 - Angle-corrected peak velocity measurements proximal to, at, and distal to the main portal vein anastomosis
 - Hepatic veins and IVC
 - Determine the type of surgical anastomosis (piggyback versus interposition) before scanning
 - Piggyback - hepatic confluence of transplant liver attached to native IVC
 - Interposition - donor IVC segment replaces a segment of the native IVC
 - Doppler evaluation of right, middle, and left hepatic veins and the IVC in whole-liver transplants
 - Doppler evaluation of the existing hepatic veins and IVC in partial-liver transplants
 - Doppler evaluation of both the recipient IVC and the piggybacked hepatic vein confluence in patients with piggyback hepatic venous anastomosis
- Anastomosis sites common site for complication

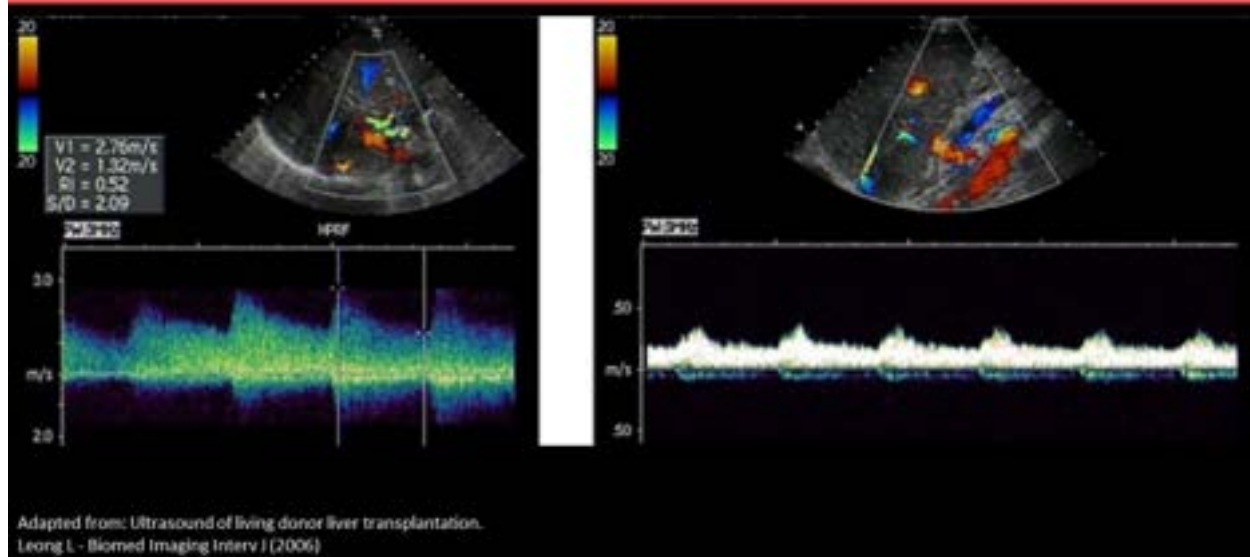
- GB is normally removed from the donor liver prior to the transplant
- NPO status assists in imaging
- Serial exams are required to assess for complications; Doppler US screening usually performed on a daily basis for at least the first 3 days post-op, then follow up exams at 1, 6, and 12 months

Sonographic Appearance:

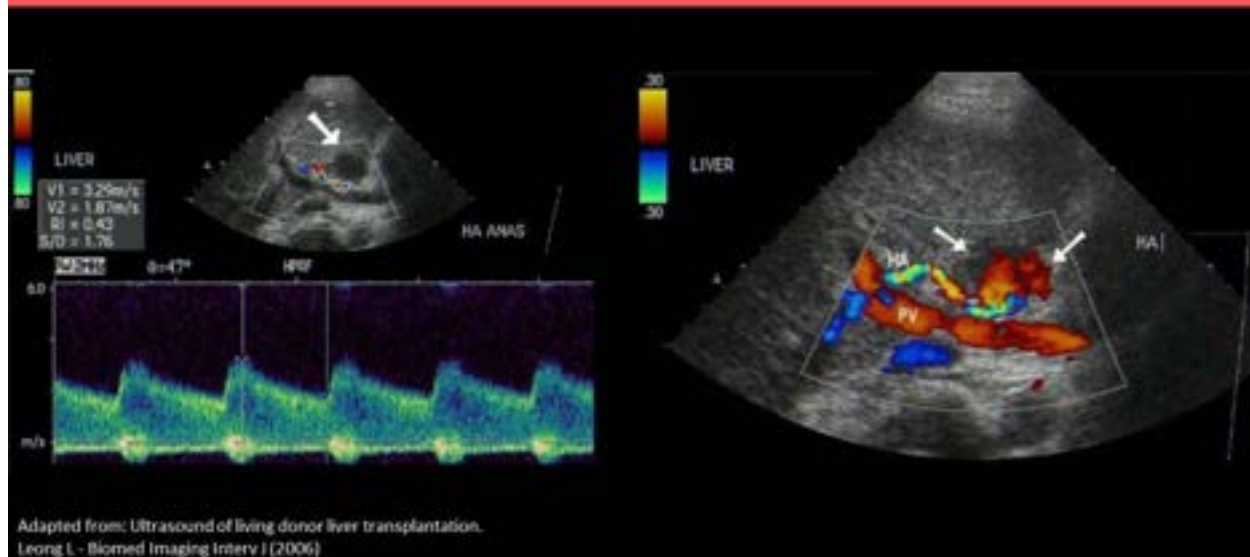
- Liver should have a homogenous texture and be of a medium echogenicity
- Hepatic artery PSV <200cm/s NL; Resistive Index (RI) >0.6-0.8 NL
- In the immediate post-op patient there can be increased hepatic pressure usually secondary to edema
- This can cause the RI values to be elevated
- Serial exams can be performed to monitor the expected increase in diastolic flow and decline in RI values
- Normal portal vein waveform is monophasic, continuous hepatopetal flow with slight respiratory phasicity
- Normal hepatic veins and IVC waveforms will be triphasic



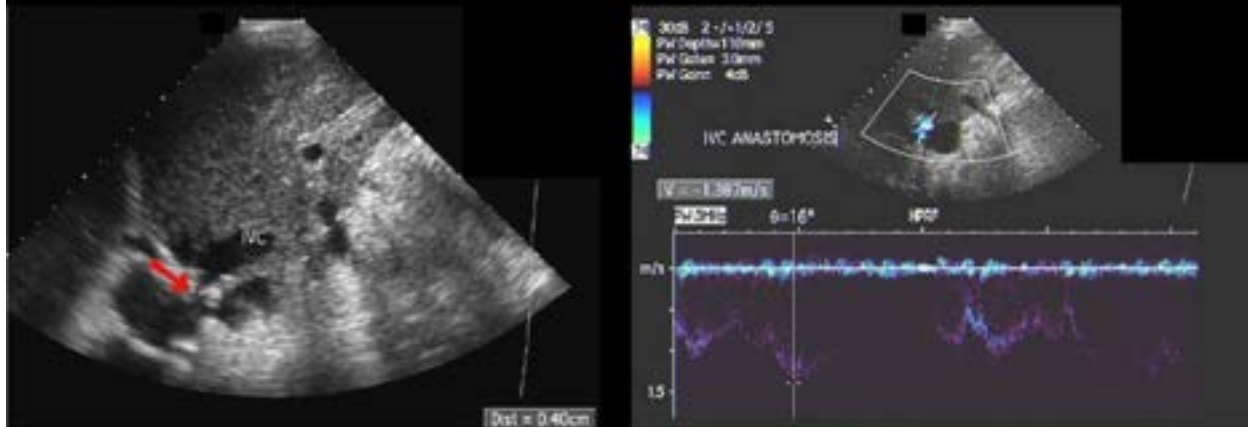
Hepatic Artery Stenosis - Transplant



Hepatic Artery Pseudoaneurysm - Transplant

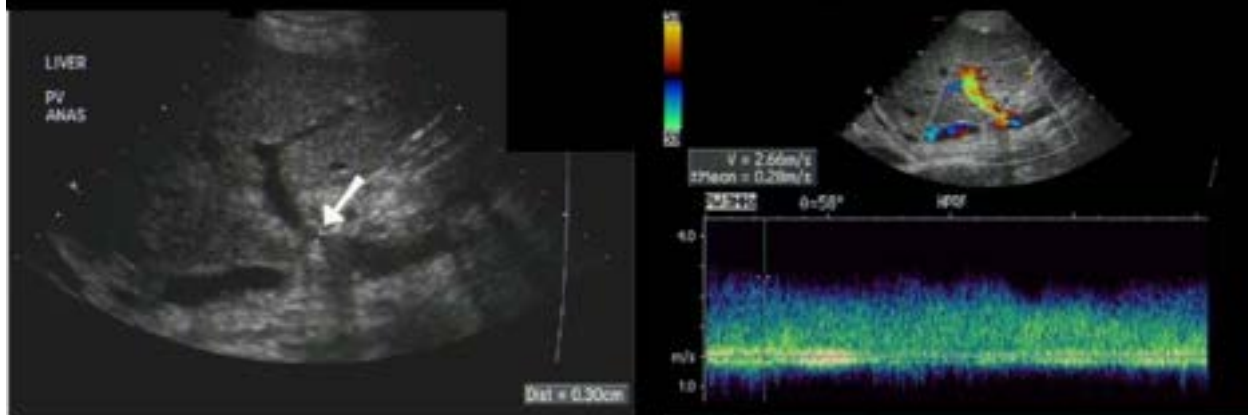


IVC Stenosis - Transplant



Adapted from: Ultrasound of living donor liver transplantation.
Leong L - Biomed Imaging Interv J (2006)

Hepatic Vein Stenosis - Transplant



Adapted from: Ultrasound of living donor liver transplantation.
Leong L - Biomed Imaging Interv J (2006)

Subphrenic Fluid Collection - Transplant



Portal/Mesenteric/IVC “The Most Commons”

Portal HTN is most commonly caused by cirrhosis

Transjugular Intrahepatic Portosystemic Shunt (TIPSS) is the most common portocaval shunt.

Thrombosis is the most common cause of acute shunt stenosis/occlusion

The most common site of TIPSS stenosis (chronic occlusion) is at the hepatic vein anastomosis

Neointimal hyperplasia is the most common cause of chronic graft stenosis/occlusion

The most common site for hepatic artery stenosis in a transplant is at the native hepatic artery anastomosis site located just outside the hilum of the liver

Most common variant of celiac axis anatomy is the hepatic artery originates from SMA, instead of the celiac axis

The splenic artery is the most common site of aneurysm of a visceral vessel and these aneurysms are more common in females

Tumor invasion of the IVC most commonly occurs with renal cell carcinoma

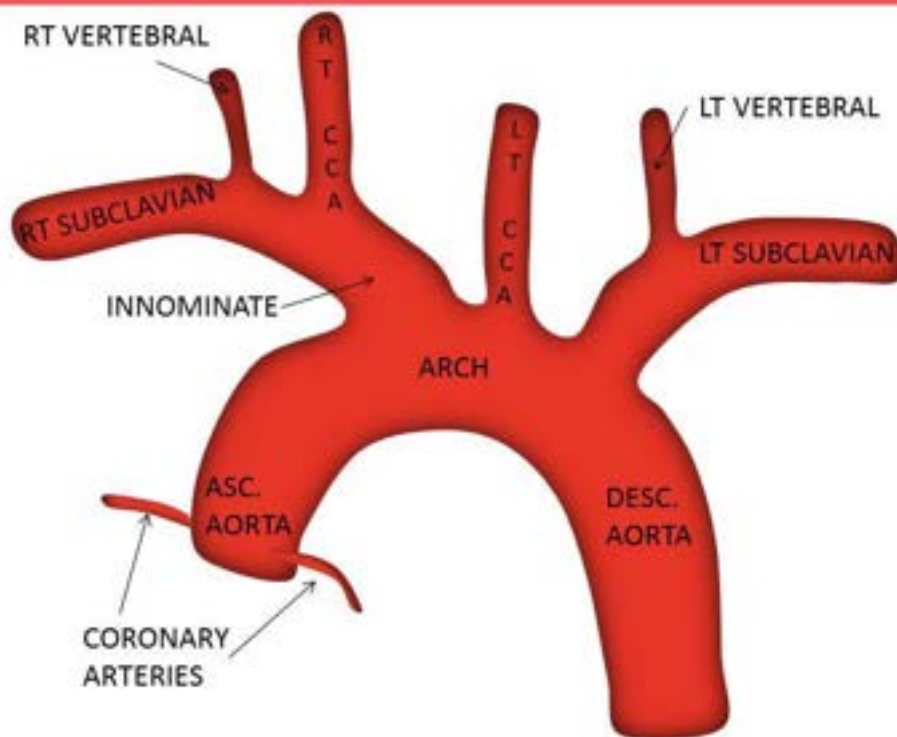
Renal vein thrombosis is most commonly caused by primary renal disease

Cerebrovascular Anatomy

Blood Vessel Anatomy:

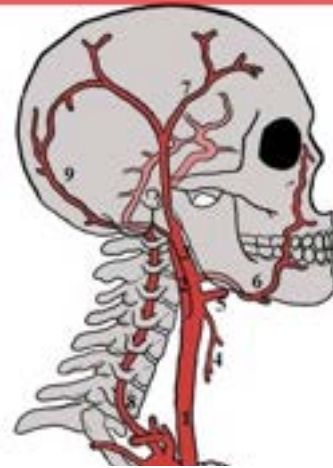
- Tunica Intima: inner layer endothelial cells
- Tunica Media: middle layer, muscle
- Tunica Adventitia/Externa: outer layer, epithelial cells, connective tissue

Aortic Arch



Cerebrovascular Anatomy

1. CCA
2. ICA
3. ECA
4. Superior Thyroidal Artery
5. Pharyngeal Artery
6. Facial Artery
7. Superficial Temporal Artery
8. Vertebral Artery
9. Occipital Artery



Innominate Artery (CCA):

- First major branch from the arch
- Also called the brachiocephalic artery or trunk
- Only ONE innominate artery, may be called the right innominate artery (NO left innominate artery)
- Courses upwards and rightward, anterior to the trachea
- Bifurcates posterior to the right sternoclavicular joint
- The right CCA and right subclavian artery are the only two branches

Common Carotid Artery (CCA):

- Right CCA originates from the innominate artery
- Left CCA originates from the aortic arch (second branch from arch)
- Courses cephalad on anterolateral neck
- Distal portion enlarges to form the carotid bulb and then branches into the internal and external carotid arteries
- Bulb contains nerve endings called baroreceptors and the carotid body lies behind the bulb and acts as a chemoreceptor that can alter the heart rate with changes to blood pH and oxygen levels
- Bifurcates at the level of the upper border of the thyroid cartilage
- In most patients the right CCA bifurcates at a lower level than the left CCA
- The bifurcation is the most common location of atherosclerosis due to the flow disturbances in this region

- Combination of ICA and ECA flow, higher resistance with good diastolic flow
- Bulb area normally exhibits areas of flow separation on color Doppler and short periods of systolic flow reversal; caused by swirling of the flow as it enters the enlarged segment
- CCA demonstrates low resistance waveform with mild diastolic flow
- May demonstrate a short duration of diastolic flow reversal at the end of systole
- Majority of the CCA blood enters the ICA due to the low resistance vascular bed distally

Common Carotid Artery

2D and color Doppler evaluation are necessary to determine atherosclerosis formation and areas of flow disturbance.

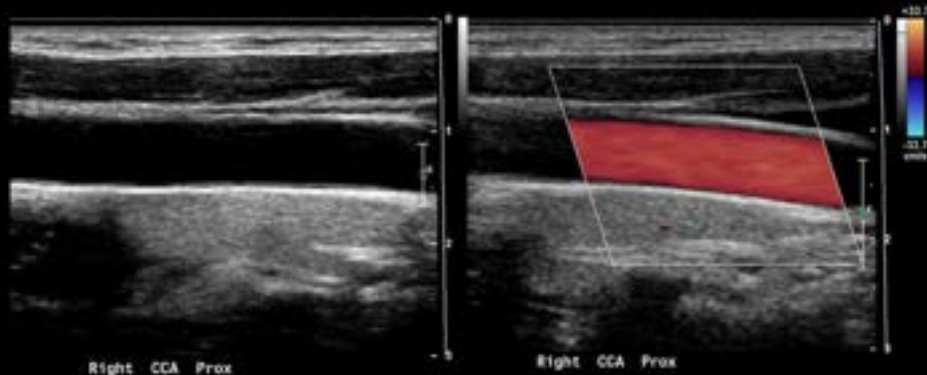
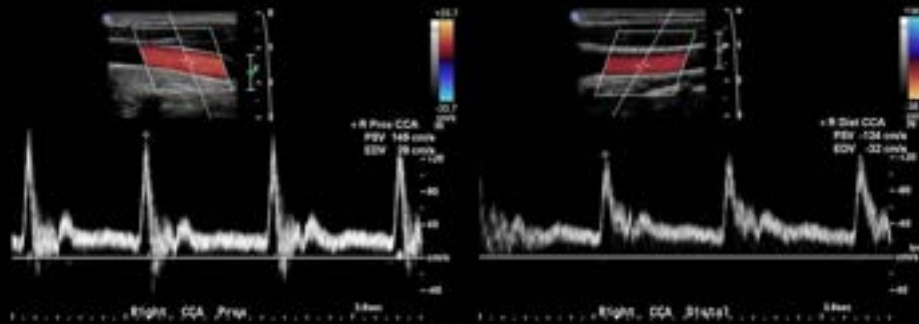


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Normal CCA Flow Pattern

Note the turbulence in the proximal CCA associated with the origin of the artery. The distal CCA demonstrates more laminar flow with a cleaner spectral window.



The CCA waveform demonstrates a combination of the ECA and ICA flow characteristics. The CCA typically demonstrates a higher resistance peaked waveform (ECA flow characteristic), but a mild amount of diastolic flow is present (ICA flow characteristic).

Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

External Carotid Artery (ECA):

- In most patients the ECA is the anterior medial branch of CCA
- There are several normal variants in ECA/ICA position, with the most common being the ECA lies posterior and lateral to the ICA
- Smaller diameter than ICA
- ECA branches supply neck, thyroid, face and scalp
- Terminates at the superficial temporal artery
- Branches:
 - Superior Thyroidal Artery - first branch courses inferior to thyroid, visible in most patients
 - Lingual Artery - anterior branch, supplies mouth/tongue
 - Ascending Pharyngeal Artery - posterior branch, supplies pharynx
 - Facial Artery - anterior branch, supplies facial muscles, gives off numerous small branches before it terminates into angular artery
 - Occipital Artery - posterior branch, branches connect to the branches of the distal vertebral artery
 - Posterior Auricular Artery - posterior branch, supplies the auricle and the scalp
 - Superficial Temporal Artery - terminal branch of ECA, gives rise to the maxillary artery, anterior and posterior branches
- ECA branches provide collateral pathways when stenosis is present in the intracranial system

- Superficial temporal artery branches connect to small branches of the ophthalmic artery
- Facial and maxillary artery branches connect to small branches of the orbital artery
- Ascending pharyngeal branches connect to small branches of the vertebral artery
- Superficial temporal artery used to perform the temporal tap; will demonstrate flow oscillations during diastole; branches connect to the supraorbital artery with potential for ICA collateralization
- High resistance, pulsatile flow with minimal to mild antegrade diastolic flow
- May demonstrate a short duration of diastolic flow reversal at the end of systole
- Normal PSV <150cm/s
- Rapid acceleration and deceleration of flow
- Demonstrates flow oscillations with temporal tap

External Carotid Artery

The ECA is usually the smaller, more anterior of the two CCA branches. Color Doppler evaluation can help identify visible extracranial branches that are an ECA characteristic

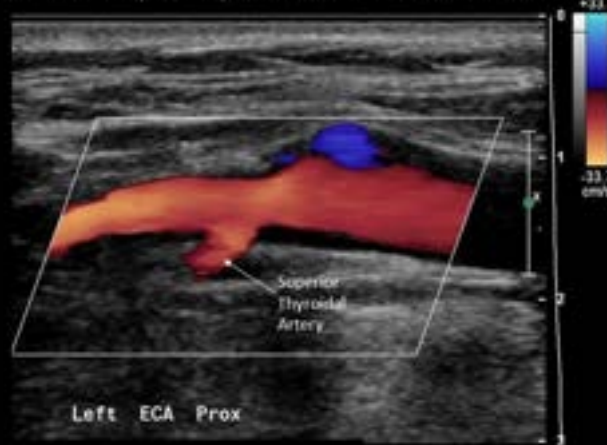
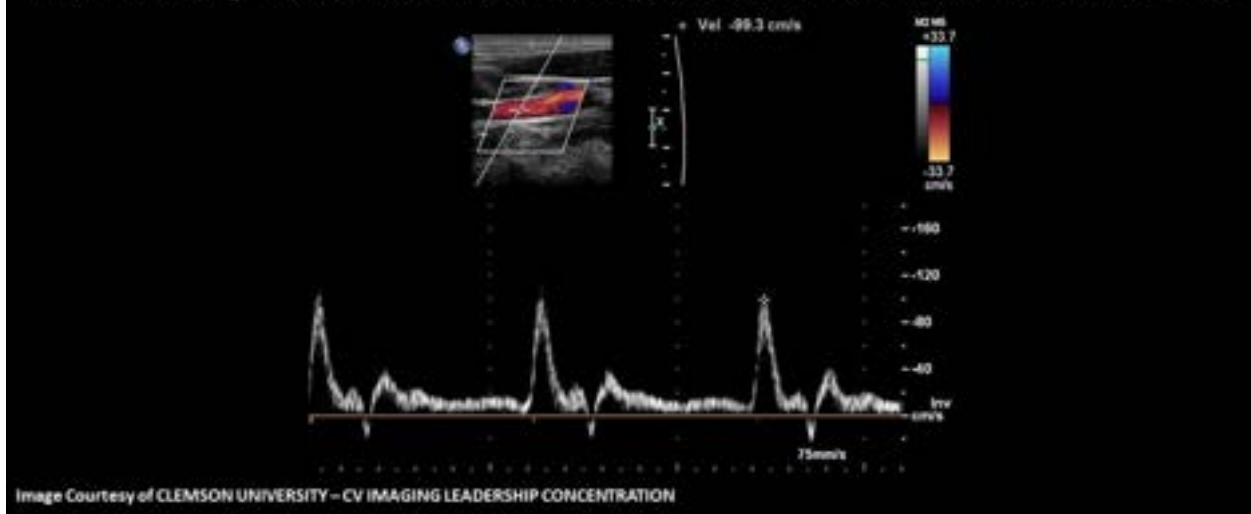


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Normal ECA Flow Pattern

ECA flow is usually high resistance due to the vascular beds it supplies. There is also less diastolic flow than in the CCA and ICA.



Internal Carotid Artery (ICA):

- Posterior/Lateral branch of CCA; courses cephalad to enter the cranium
- Larger diameter than ECA
- Siphon Area - distal ICA courses anterior then medial then posterior forming an "S"
- Siphon causes turbulence and has increased risk for flow reducing lesions such as atheroma and thrombus
- Can demonstrate a relatively straight course but can see
 - Tortuosity - S-shaped curve
 - Kinking - takes a sharp turn, most commonly associated with symptomatic cerebral ischemia
 - Coiling - vessel forms a circle during its course
- No visible branches extracranially
- Branches supply the eyes, nose, forehead, corpus callosum, frontal and parietal lobes
- Ophthalmic artery
 - First branch from siphon area inside the cranium, NOT visible on carotid evaluation
 - Supplies the eye
 - Branches into nasal, frontal & supraorbital arteries
- Anterior choroidal artery and posterior communicating artery are the other two main intracranial branches of the distal ICA
- Terminates at the bifurcation into the ACA and MCA, once it reaches the circle of Willis
- Supplies 75% of arterial blood to the brain

- Low resistance flow with increased antegrade diastolic flow; slow acceleration and deceleration of flow
- Does NOT respond to temporal tap

Internal Carotid Artery - Proximal

The ICA is usually the larger, more posterior of the two CCA branches.
Color Doppler evaluation can help identify visible extracranial branches that are an ECA characteristic.

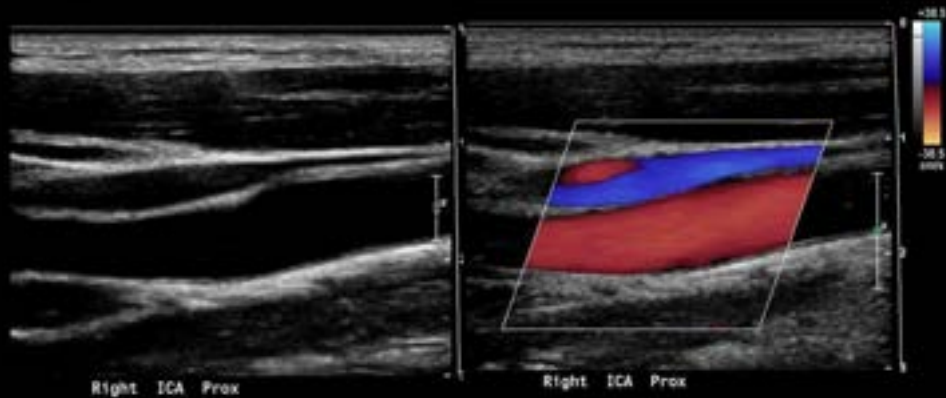


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Normal ICA Flow Pattern

ICA flow is usually low resistance due to the vascular bed (brain) it supplies.
There is also more diastolic flow than in the CCA and ECA.

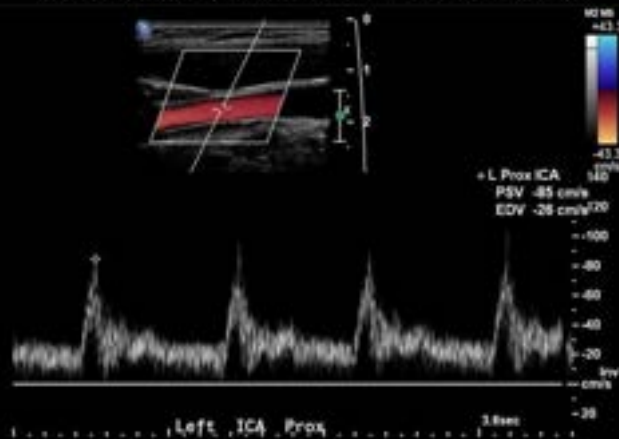


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Carotid Anatomy

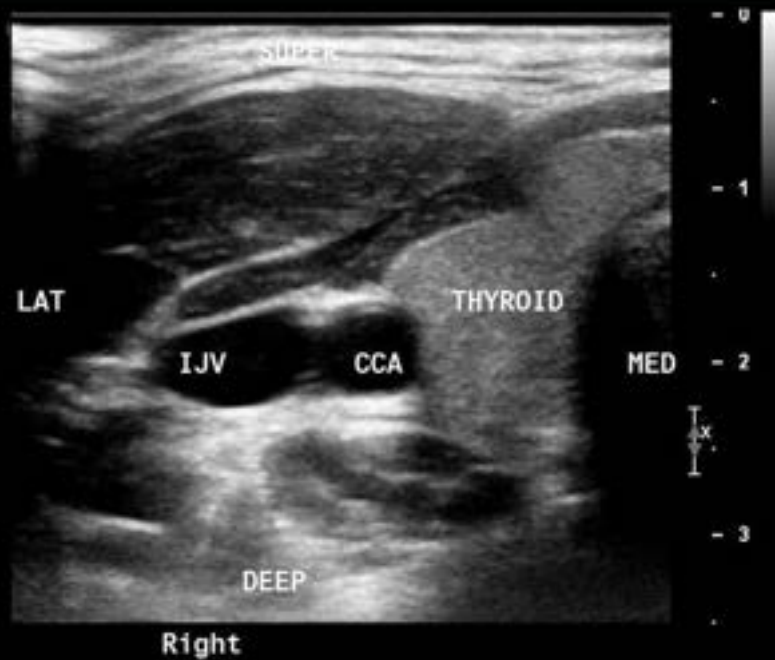
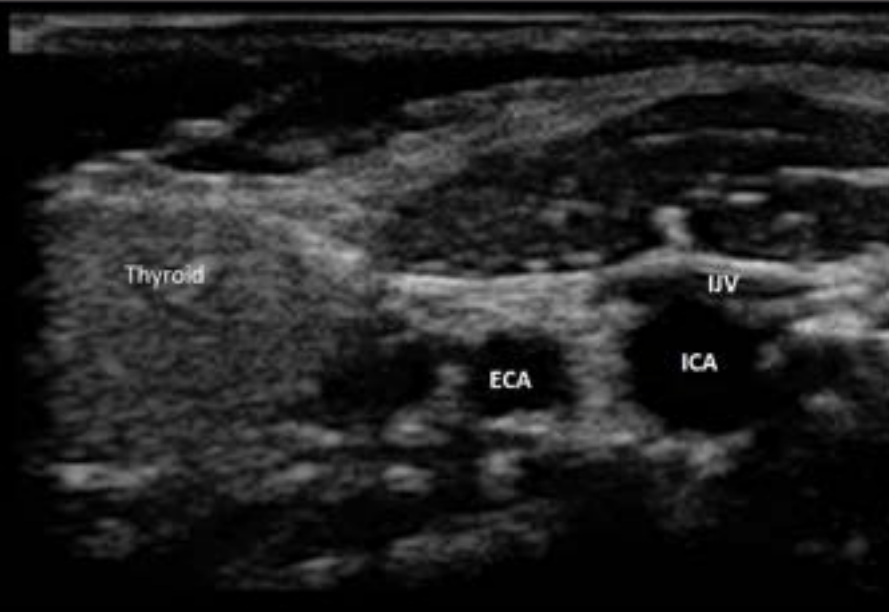


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Right Carotid Anatomy



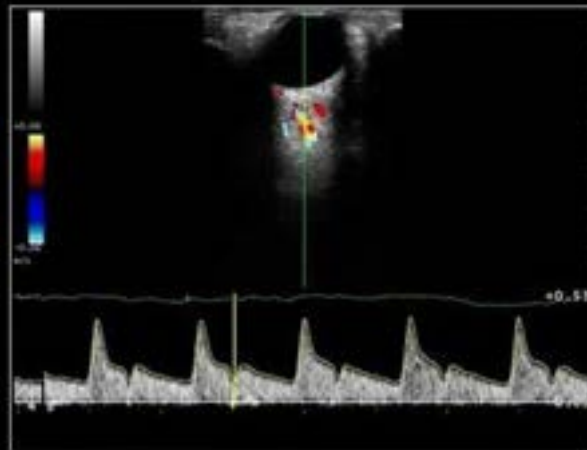
Left Carotid Anatomy



Periorbital Arteries:

- Supraorbital Artery - originates from the ophthalmic artery and joins the superficial temporal artery(branch of ECA)
- Frontal Artery - originates from ophthalmic artery; also called the supratrochlear artery; supplies nose with nasal artery
- Lacrimal Artery - originates from the ophthalmic artery
- All are potential collateral pathways for ECA to ICA collateral flow

Periorbital Doppler



Adapted from: Ultrasound entropy may be a new non-invasive measure of pre-clinical vascular damage in young hypertensive patients. Beakley C, McCann A, McClenaghan V, Hamilton PK, Millar A, Pumb R, Harbinson M, McWhig GE - Cardiovasc Ultrasound (2015)

Vertebral Arteries (VA):

- Supplies 10-25% of blood flow to cerebrum
- Course within foramina of the pedicles of the vertebra of the C6 through C1
- Pass through the foramen magnum and converge into the basilar artery after entering the posterior skull
- Left vertebral artery is usually larger in diameter
- Branches supply the medulla and inferior cerebellum and can provide collateral pathways for intracranial flow
- Right vertebral originates from the proximal right subclavian
- Left vertebral originates from the proximal left subclavian
- Carries 25% of blood flow to the brain
- Flow velocities normally <50cm/s
- Similar flow pattern to the ICA
- Slow acceleration and deceleration with increased diastolic antegrade flow

Normal Vertebral Artery & Vein

The normal vertebral arteries should always demonstrate antegrade flow. Always be mindful of your color map settings when evaluating flow direction.

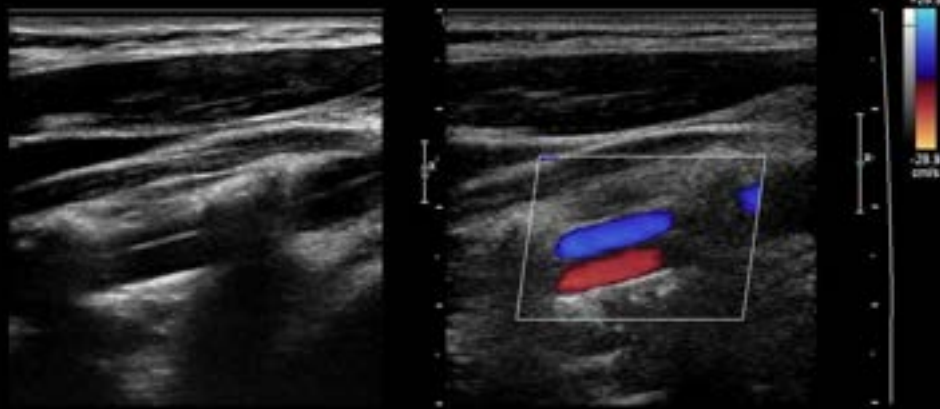


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Normal Vertebral Artery

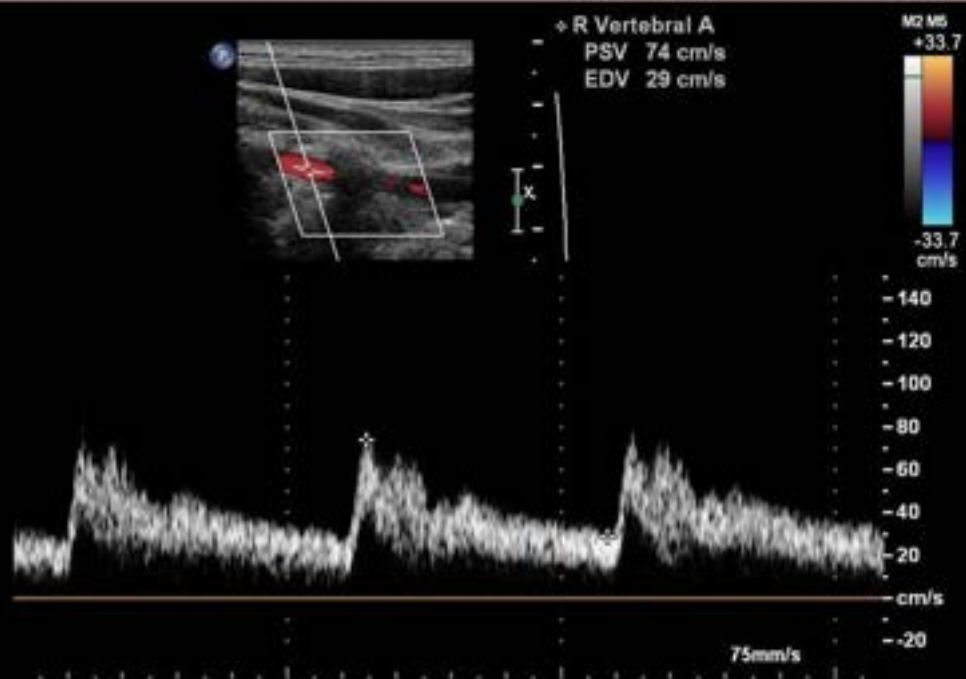
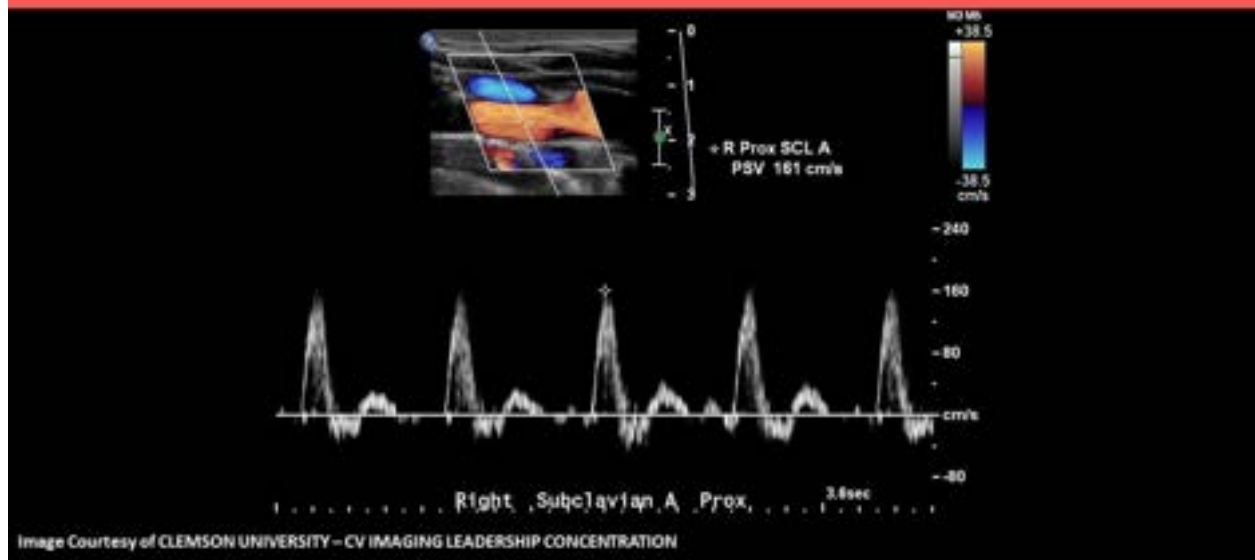


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Subclavian Artery:

- Some vascular labs include this artery in all carotid examinations, but it could be optional depending on suspected pathology
- Supplies blood to the vertebral artery and the arm
- Right subclavian artery originates from the innominate artery
- Aberrant right subclavian artery
 - In some patients the subclavian artery can be a direct branch of the aorta distal to the left subclavian origin
 - Called retroesophageal subclavian artery or aberrant subclavian artery
 - Usually originates from a dilated segment of the proximal descending aorta called the Kommerell diverticulum
 - Can cause dysphagia due to compression of the esophagus by the aberrant subclavian artery
 - Also related to Ortner syndrome - palsy of the recurrent laryngeal nerve
- Left subclavian originates at the aortic arch (3rd branch)
- Courses posterior to the anterior scalene muscle and clavicle
- Courses anterior to the apex of the lung
- Becomes the axillary artery at the level of the first rib
- Branches from subclavian
 - Vertebral artery
 - Internal mammary artery (AKA internal thoracic artery)
 - Thyrocervical and Costocervical Trunks - can be differentiated from the vertebral arteries by their many branches and lower end-diastolic flow velocities
- Triphasic waveform, high resistance
- Flow velocity varies
- Compare the velocities in both subclavian arteries; High peak systolic velocity can indicate stenosis in that artery
- Biphasic or monophasic flow patterns associated with obstruction

Normal Subclavian Artery Flow Pattern



Basilar Artery:

- The left and right vertebral arteries merge at the base of the skull to form a SINGLE basilar artery
- Branches supply the superior cerebellum and portions of the brain stem
- Bifurcates at the circle of Willis to form both posterior cerebral arteries

Circle of Willis:

- Network of arteries that are seated on the inferior cerebrum surface
- Supplies all lobes of the cerebrum
- Most common anatomy demonstrated below, but it is only seen in about 20% of patients
- The circle is usually complete in only 50% of patients
- Provides potential for collateral flow in the presence of disease

Anterior Cerebral Artery - supplies frontal and parietal lobes, corpus callosum and cavum septum pellucidum

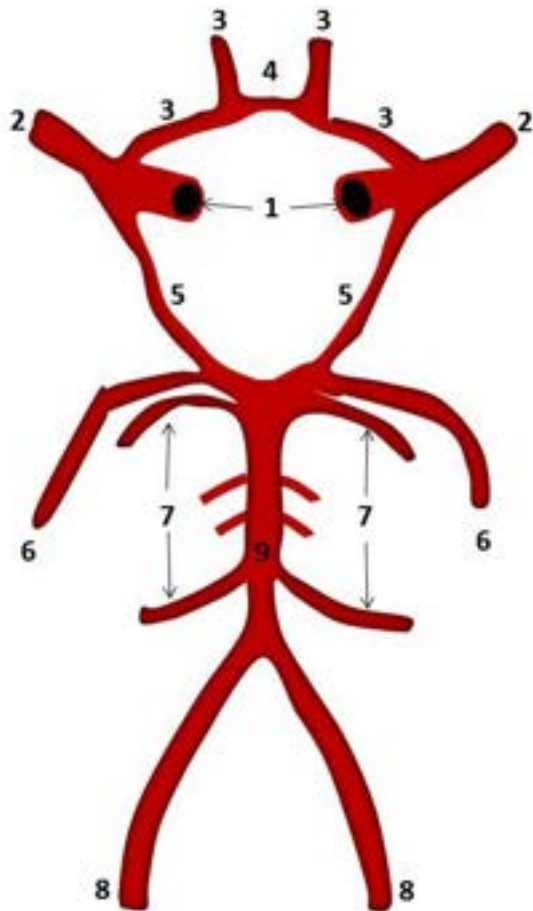
Middle Cerebral Artery - supplies frontal, parietal and temporal lobes

Posterior Cerebral Artery - supplies temporal and occipital lobes

Cerebral Circulation:

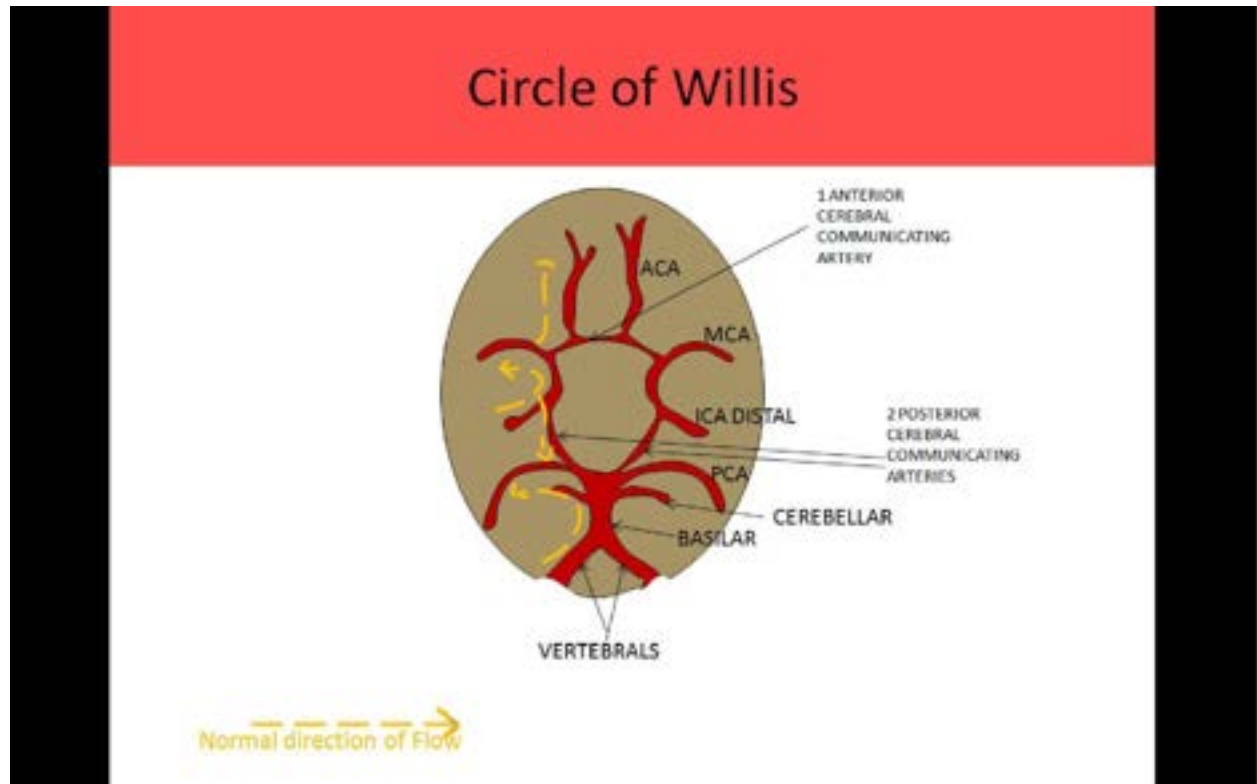
- Superficial branches spread over the surface of the brain EX: Circle of Willis

- Penetrating branches enter the brain tissue segments for nutrient exchange and neuronal function
- Intracranial venous sinuses are located between the dura mater and the periosteum drain cerebral blood into the internal jugular vein



Circle of Willis

1. Internal carotid arteries (ICA)
2. Middle cerebral arteries (MCA)
3. Anterior cerebral arteries (ACA)
4. Anterior communicating artery
5. Posterior communicating artery
6. Posterior cerebral arteries (PCA)
7. Cerebellar arteries
8. Vertebral arteries
9. Basilar artery



Signs & Symptoms of Cerebrovascular Disease

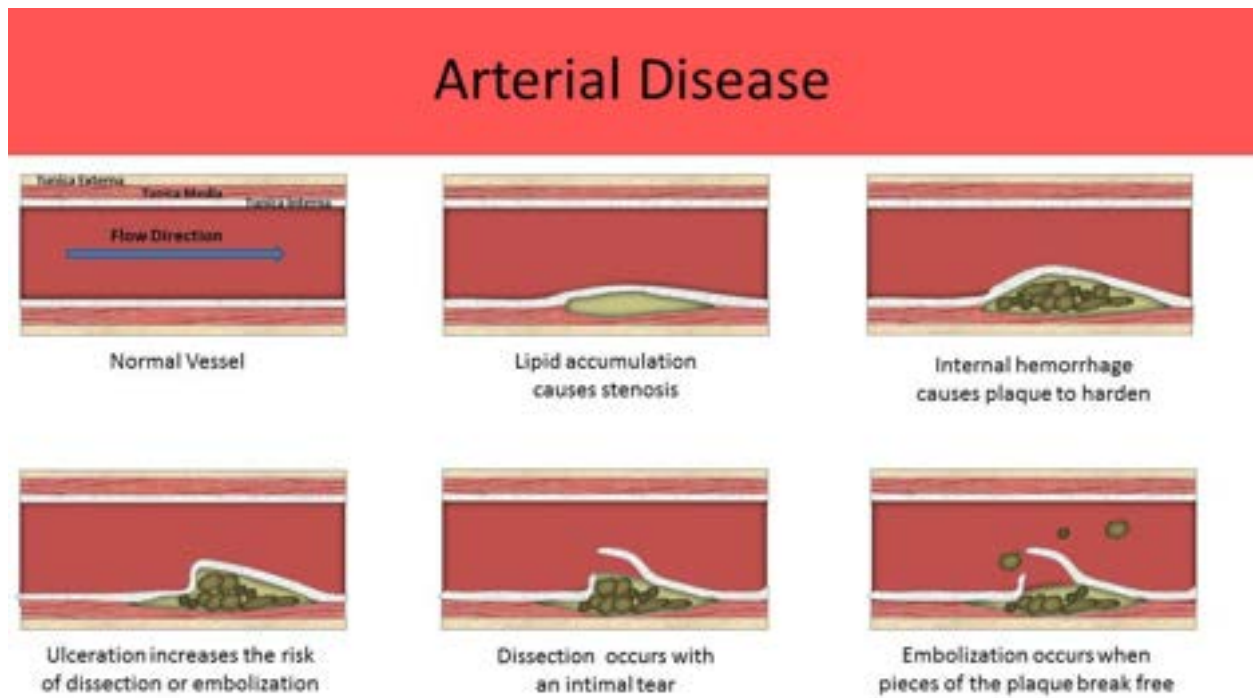
Mechanisms of Stroke:

- Ischemic stroke is the most common type of stroke
- Atherosclerosis causes stenosis which leads to decreased blood supply and cerebral ischemia
 - ICA stenosis presents the highest risk of TIA/stroke
- Embolization of atherosclerotic material which lodges in a distal artery and leads to cerebral ischemia

- Can be caused by cardiac embolus
- HTN - strongest risk factor for stroke; shearing forces of the blood can exert too much pressure on intracranial vessels leading to hemorrhage
- Dissection - intimal tear leads to thrombus/embolus formation
- Polycythemia Vera - thickening of the blood reduces flow and may lead to ischemia distally; increased risk of thrombus/embolus formation
- Vasospasm
- Intracranial hemorrhage

Atherosclerosis Formation:

- #1 cause of vascular disease
- Accumulation of atheroma, smooth muscle, collagen, fibrin and platelets
- Forms within and beneath the intima
- >1.2mm IMT indicates increased risk for stenosis
- Most common cerebrovascular site - CCA bifurcation
- 2nd most common cerebrovascular site - origin of the CCA



Modifiable Risk Factors for Stroke:

- These risk factors can be reduced or controlled to reduce the risk of stroke
- Hypertension - the main risk factor for stroke; treated with medication
- Diabetes - treated with medication and/or dietary changes
- Hyperlipidemia - treated with medication and/or dietary changes

- Smoking - quit
- Inactivity - get some exercise
- Obesity - add exercise and improve nutrition
- Poor nutrition - change eating habits

Other Major Risk Factors:

- Gender - strokes are more common in men, but women die from strokes more often
- Race/Ethnicity - African Americans have a higher incidence of stroke than Caucasians
- Age - risk increases with age
- Family history of TIA or stroke

Terminology:

- Transient Ischemic Attack (TIA) - Symptoms last <24hrs
- Cerebrovascular Accident (CVA) - Symptoms/damages last longer than 24hrs
- Brain Infarction-insufficient blood supply to brain due to stenosis or occlusion
- Subarachnoid Hemorrhage-rupture of intracranial vessels
- Hemispheric Symptoms-Specific symptoms are related to right and left brain with loss of functionality

“ Right brain controls Left side of the body - RICA stenosis = LT paresis, paresthesia etc.

“ Left brain controls Right side of the body - LICA stenosis = RT paresis, paresthesia etc.

***Stroke is the fourth leading cause of death in the US (CDC 2015)

***85% of strokes are caused by ischemia (other causes include hemorrhage, vasospasm)

***Atheroma in the carotid bifurcation is most common cause of an embolic stroke, cardiac emboli #2

Neurological Exam:

- To identify an abnormality in the nervous system
- To differentiate peripheral from central nervous system lesions
- General Appearance:
 - Level of consciousness - do they know basic facts? what year is it? what season is it? who is the President of the US?
 - Personal Hygiene and Dress - is the patient performing personal hygiene; is clothing appropriate for the season/temperature
 - Posture and Motor Activity - involuntary movements or tremors
 - Vital Signs
- Cognition - comprehension, attention and calculation, ability to follow simple commands and orientation

- Examination of cranial nerves - evaluates smell, sight, muscle movement in the face, speech, hearing
- Motor system evaluation - body positioning, involuntary movements, muscle tone and strength
- Sensory exam - pain sensation (pin prick), light touch sensation (brush), position sense, stereognosia (depth perception), graphesthesia (ability to recognize numbers/letters "written" on the skin), and extinction
- Reflex exam - reflex hammer used to assess deep tendon reflexes
- Coordination and Gait evaluation

Clinical Exam:

- Palpation:
 - Most arteries are palpated by placing two fingers over the artery
 - For carotid evaluation, the carotid pulses can be palpated
 - Bounding pedal pulses refers to normal, easily palpable pulses that are in sync with the heartbeat
- Auscultation:
 - Place a stethoscope over the at the level of the carotid bifurcation to identify a possible bruit
 - Ask the patient to hold their breath while listening for the flow sounds
 - A stenosis in the carotid system can cause a bruit
 - Aortic valve stenosis, low hematocrit and increased cardiac output can cause bilateral bruits
 - Bruits are described as 1+ mild, 2+ moderate or 3+ severe
 - A bruit that occurs throughout the cardiac cycle indicates possible severe stenosis
 - >90% stenosis does NOT usually cause a bruit due to the significant reduction in flow
 - A bruit can be missed if cardiac output is low
- Blood Pressure:
 - The blood pressures in the arms should not differ more than 20mmHg or a subclavian stenosis is suspected in the arm with the lower pressure

Symptoms of Cerebrovascular Disease:

Non-specific Symptoms:

- Carotid Bruit
 - the carotid artery is the most commonly assessed artery for the presence of a bruit

- noise heard with a stethoscope caused by turbulent flow
- carotid bruit heard through systole and diastole usually indicates a significant stenosis
- bilateral bruits heard proximally can indicate aortic valve stenosis
- Headache
- Syncope
- Dysarthria (disturbed speech)
- Dizziness

Carotid Sinus Massage:

- Used to assess the cause of syncope
- The carotid artery is massaged at the level of the cricoid cartilage for 5-10 seconds
- Will lead to bradycardia and hypotension if carotid sinus hypersensitivity is present
- Should NOT be performed on patients with an occluded carotid or recent TIA/CVA

ICA Disease Symptoms:

- Unilateral symptoms with contralateral side stenosis
- Paresthesia - tingling of skin
- Hemiparesis - weakness
- Paralysis - unable to move
- Broca Aphasia - AKA expressive aphasia; can understand others and follow directions but unable to speak
- Hollenhorst plaques - cholesterol emboli originate from atherosclerotic lesions in the ipsilateral carotid artery or aorta and lodge in the small branches of the retinal artery; usually identified by an optometrist on a standard vision exam because the patient is asymptomatic
- Homonymous Hemianopia - visual defect in half of the eye; more common with posterior circulatory issues and occipital lobe infarcts but can occur in the anterior circulation and parietal lobe infarcts
- Amaurosis Fugax
 - Temporary partial or total monocular blindness
 - Can be referred to as a TIA of the eye
 - Ophthalmic artery branches from ICA
 - Caused by ipsilateral stenosis of the ICA and/or ophthalmic artery
 - RICA stenosis leads to right eye amaurosis fugax

Symptoms of ICA Disease



Hemispheric Symptoms (Contralateral)
Left side disease = Right side symptoms
Right side disease = Left side symptoms



Broca's Aphasia - can understand others and follow directions but unable to speak



Unilateral Visual Symptoms (Ipsilateral)
Left side disease = Left side symptoms
Right side disease = Right side symptoms

MCA Disease Symptoms:

- Wernicke Aphasia - AKA receptive aphasia; totally incomprehensible, unable to understand speech, speak, or follow directions
- Dysphasia
- Behavioral changes
- Severe hemiparesis in contralateral face/arms

Symptoms of MCA Disease



Contralateral Face and Arm Symptoms
Left side disease = Right side symptoms
Right side disease = Left side symptoms

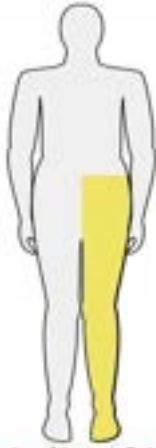


Wernicke's Aphasia - unable to
understand speech, speak, or follow
directions

ACA Disease Symptoms:

- Loss of coordination
- Incontinence
- Severe leg hemiparesis or hemiplegia in contralateral leg
- Facial drooping or asymmetry

Symptoms of ACA Disease



Contralateral Lower Extremity Symptoms
Left side disease = Right side symptoms
Right side disease = Left side symptoms



Facial Drooping

PCA Disease Symptoms:

- Dyslexia
- Coma

Vertebral/Basilar Disease Symptoms:

- Vertigo: unable to maintain equilibrium
- Syncope: fainting
- Ataxia: lack of coordination
- Bilateral visual blurring or double vision (diplopia)
- Bilateral homonymous hemianopia - medial or lateral half of the field of view disrupted in each eye; most commonly associated with occipital infarcts
- Bilateral paresthesia
- Drop Attack - falling to the ground without loss of consciousness
- Dysphagia - trouble swallowing

Symptoms of Vertebral Disease



Bilateral Symptoms (Contralateral)
Left side disease = Right side symptoms
Right side disease = Left side symptoms



Dizziness, vertigo and
ataxia



Bilateral Visual Symptoms
Bilateral homonymous hemianopia
Bilateral visual blurring or double vision

Ultrasound Evaluation of the Extracranial Arteries

Carotid Duplex Exam Techniques:

- Obtain bilateral brachial blood pressures
 - This helps to determine if a subclavian steal may be present
 - It alerts you to potential cardiac effects on carotid flow

- Patient is placed in the supine position with head tilted up and slightly away from the side being examined
- Use 5-10MHz linear probe
- Provide transverse 2D and color Doppler images of the CCA proximal, mid and distal, CCA bulb, and CCA bifurcation
 - Document plaque location and characteristics
 - Obtain area stenosis measurements, as needed
- Obtain longitudinal 2D images of the CCA, ECA and ICA to assess plaque formation
 - Document plaque location and characteristics
 - Obtain diameter stenosis measurements, as needed
 - Apply Color and PW Doppler to all vessels to demonstrate flow characteristics
 - CCA - proximal, mid, distal
 - ECA - demonstrate its origin from the CCA and assess the proximal ECA
 - ICA - demonstrate its origin from the CCA, proximal, mid and distal
 - Measure the peak systolic and end diastolic velocities on each waveform
- Vertebral arteries - document flow direction and velocity bilaterally
- Subclavian arteries - document flow characteristics (tri, bi, mono) and flow velocity
- When stenosis is present:
 - Sample with Doppler proximal to, at and distal to the stenosis
 - Move the Doppler cursor through the stenosis to find the highest velocity
 - Angle the cursor to keep it parallel to the blood flow; keep in mind that the flow is NOT always moving parallel to the vessel wall; the cursor should be aligned with any eccentric flow and not the wall
 - Calculate the ICA/CCA ratio
 - Use YOUR LAB'S Doppler criteria to assess the level of stenosis (Sample criteria: Carotid Stenosis page)

ICA US Characteristics	ECA US Characteristics
No extracranial branches seen	Extracranial branches seen
Larger caliber	Smaller caliber
Courses posterior and lateral	Courses anterior and medial
Low resistance flow	High resistance flow
No temporal tap response	Temporal tap response

Intimal-Media Thickness (IMT):

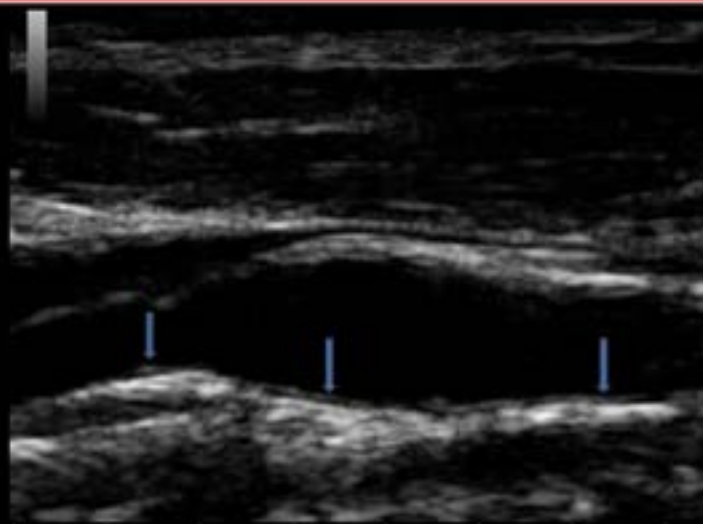
- Predictor of cardiovascular events
- Protocols usually require three measures of primary carotid disease: average wall intima-media thickness, a measurement of degree of focal plaque and the systolic velocity ratio of ICA and CCA
- Images of the right and left common carotid artery, carotid bulb, and the first centimeter of the internal carotid artery are obtained at end diastole
- The best IMT measurement is obtained with the US beam perpendicular to the vessel wall
- Due to varied course of the vessel, different probe angles are usually necessary to optimize the IMT measurement of each segment
- Distal CCA measurements are most commonly used to assess risk of cardiovascular events
- Normal values for intimal thickness range from 0.4-1.0mm
- Values normally increase with age

Vessel Wall Layers



Measurements of carotid intima media thickness in non-invasive high-frequency ultrasound images: the effect of dynamic range setting.
Gaarder M, Selerstad T - [Cardiovasc Ultrasound](#) (2015)

Intima-Media Thickness (IMT)



Intimal Thickening



2D Wall Abnormality Classifications:

- Fatty Streaks - thin layer of lipid accumulation; low level echoes, all ages
- Soft - lipids, fibrofatty
- Calcific - fibrous
- Complex - mixed fibrofatty and fibrous
- Hypoechoic Plaque - fibrofatty; low level reflections
- Hyperechoic Plaque - fibrous; medium level reflections with or without shadowing
- Calcific Plaque - bright echoes with shadowing; calcium deposits
- Homogeneous Plaque - uniform echoes
- Heterogeneous Plaque - varied echogenicity across plaque formation
- Surface Characteristics - smooth versus irregular (ulcerations)
- Ulcerated plaque has an increased risk of embolism due to shearing forces of the flowing blood hitting the irregular surface and can cause debris to break loose
- Intraplaque Hemorrhage - Sonolucent area within the plaque formation
- Thrombus - low level echoes, can attach to other surface irregularities

Plaque Formation

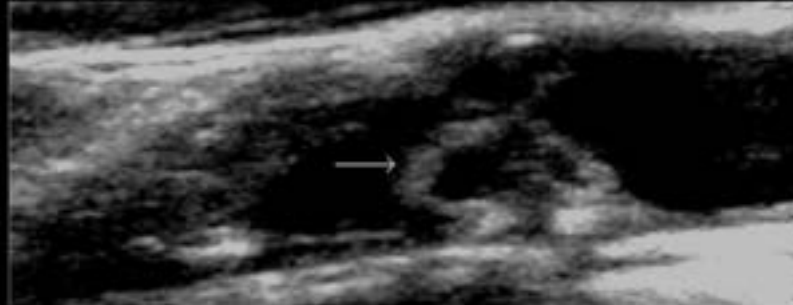


Homogeneous Smooth Plaque



Hemorrhagic Plaque

Note the anechoic area within the plaque formation of the carotid artery



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Area Stenosis



Area Stenosis:

$$\frac{\text{Vessel Area (+)} - \text{True Lumen Area (x)}}{\text{Vessel Area (+)}}$$

$$0.53 - 0.38 / 0.53 \times 100\% = 28\% \text{ Area stenosis}$$

$$50\% \text{ Diameter stenosis} = 75\% \text{ Area stenosis}$$

The transverse view is most accurate for evaluating area and diameter stenosis.

Diameter Stenosis



Diameter Stenosis:

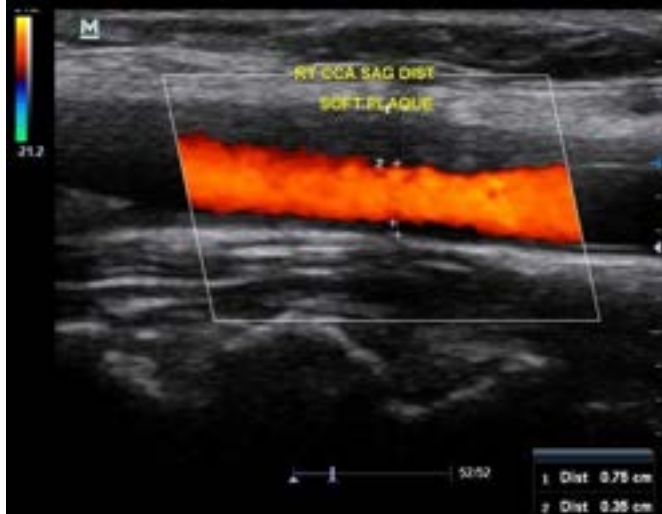
$$\frac{\text{Vessel Diameter (1)} - \text{True Lumen Diameter (2)}}{\text{Vessel Diameter (1)}}$$

$$0.79 - 0.39 / 0.79 \times 100\% = 51\% \text{ Diameter stenosis}$$

$$50\% \text{ Diameter stenosis} = 75\% \text{ Area stenosis}$$

The transverse view is most accurate for evaluating area and diameter stenosis.

Diameter Stenosis



Diameter Stenosis:

$$\frac{\text{Vessel Diameter (1)} - \text{True Lumen Diameter (2)}}{\text{Vessel Diameter (1)}}$$

$$0.75 - 0.35 / 0.75 \times 100\% = 53\% \text{ Diameter stenosis}$$

$$50\% \text{ Diameter stenosis} = 75\% \text{ Area stenosis}$$

The TRANSVERSE view is most accurate for evaluating area and diameter stenosis.

Diameter Stenosis



Blood Flow:

- Cardiac Function: decreased function = decreased flow
- Vessel Compliance: more stiff or calcified vessel = higher resistance flow
- Tone of vascular musculature: younger patients have better muscle tone = less compliant vessels = higher resistance flow
- Pattern of branches: more branches, decreases resistance
- Flow pattern is determined by where the vessel originates and where it is going
- 75% of blood flow to brain through ICA
- 25% of blood flow to brain through vertebral system
- High Resistance: extremities, muscles, pre-prandial digestive organs
- Low Resistance: liver, kidney, brain

High Resistance	Low Resistance
Sharp upstroke	Slow upstroke
Rapid deceleration	Slow deceleration
Little to no antegrade diastolic flow	Antegrade diastolic flow
Diastolic flow reversal	No diastolic flow reversal
Feeds tissues that do not require constant flow	Feeds tissues that require constant flow
Muscles, extremities	Brain, Kidneys, Spleen, Liver

Normal Doppler Evaluation:

- Use a 60 degree or less cursor angle to obtain the most accurate velocity
 - Cursor angle should be parallel to the flow in the vessel
 - Sample the center of the vessel unless stenosis is present
 - Multiple samples in different locations should be taken when stenosis is present
 - If the exam is a follow up to a prior exam, cursor angles used on current exam should be the same as those used on prior exam for best comparison
1. CCA - combination of ECA and ICA flow; higher resistance flow than the ICA and more diastolic flow than the ECA; normal flow demonstrates PSV <125 cm/s and EDV <40cm/s
 2. ECA - high resistance, minimal diastolic flow; may see diastolic flow reversal; normal flow demonstrates PSV <150cm/s
 3. ICA - low resistance, increased antegrade diastolic flow; normal flow demonstrates PSV <125 cm/s and EDV <40cm/s

Common Carotid Artery

2D and color Doppler evaluation are necessary to determine atherosclerosis formation and areas of flow disturbance.

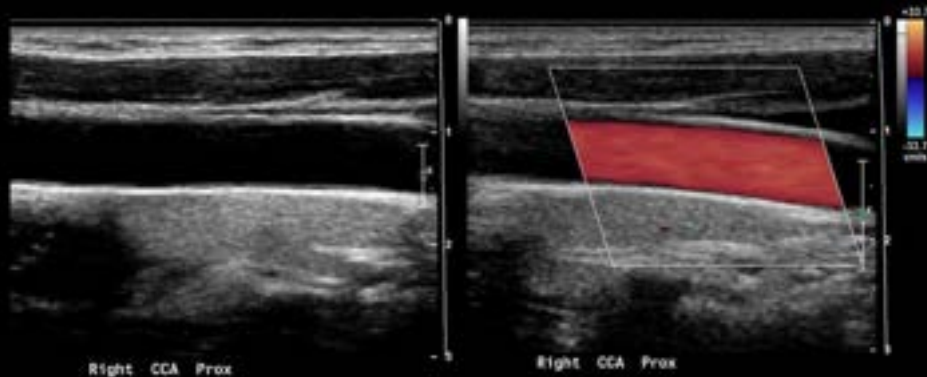
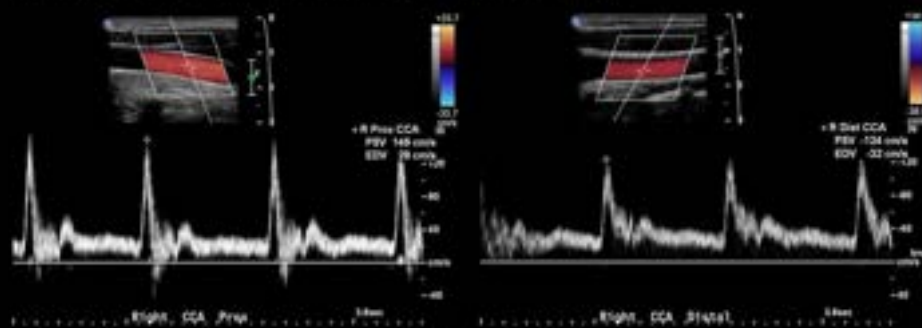


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Normal CCA Flow Pattern

Note the turbulence in the proximal CCA associated with the origin of the artery. The distal CCA demonstrates more laminar flow with a cleaner spectral window.



The CCA waveform demonstrates a combination of the ECA and ICA flow characteristics. The CCA typically demonstrates a higher resistance peaked waveform (ECA flow characteristic), but a mild amount of diastolic flow is present (ICA flow characteristic).

Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

External Carotid Artery

The ECA is usually the smaller, more anterior of the two CCA branches. Color Doppler evaluation can help identify visible extracranial branches that are an ECA characteristic.

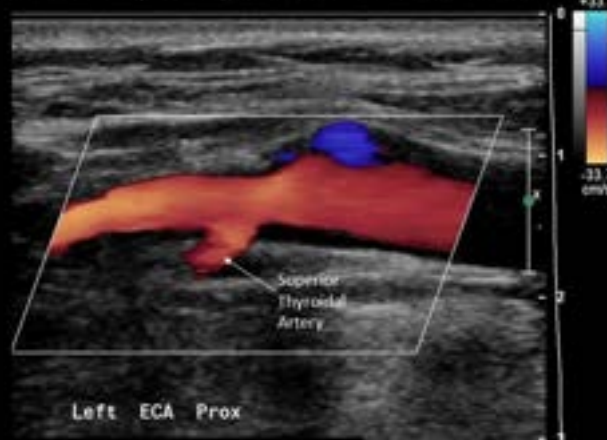


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Normal ECA Flow Pattern

ECA flow is usually high resistance due to the vascular beds it supplies. There is also less diastolic flow than in the CCA and ICA.

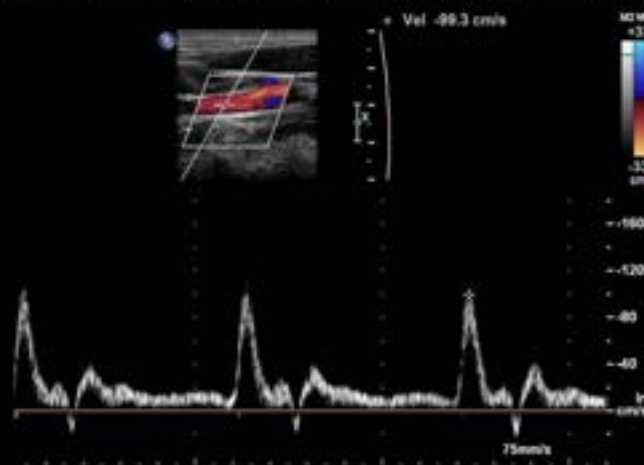
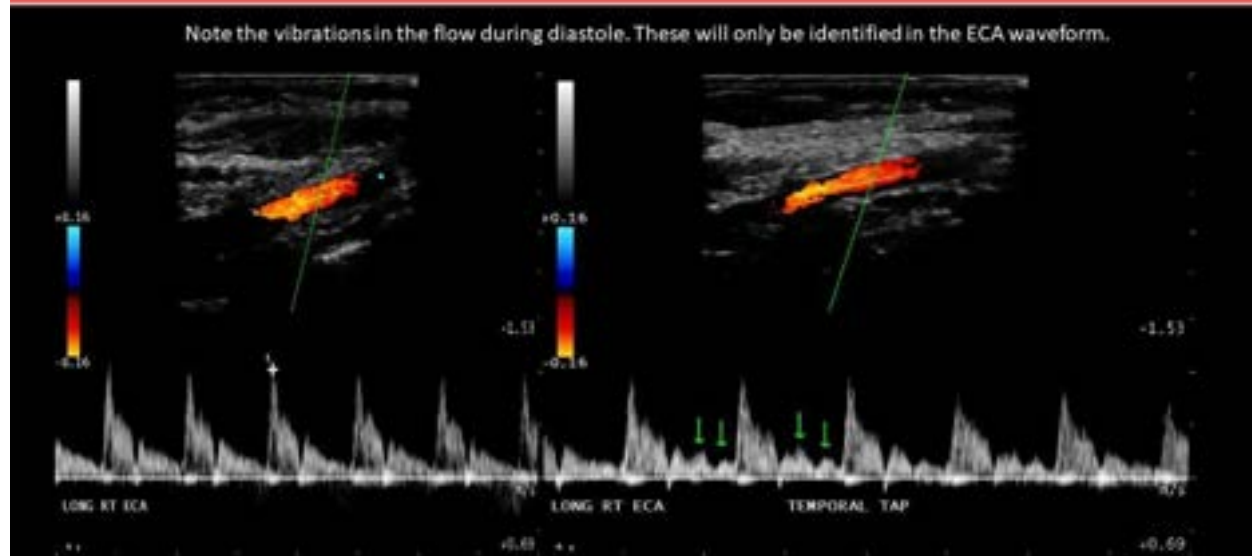


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

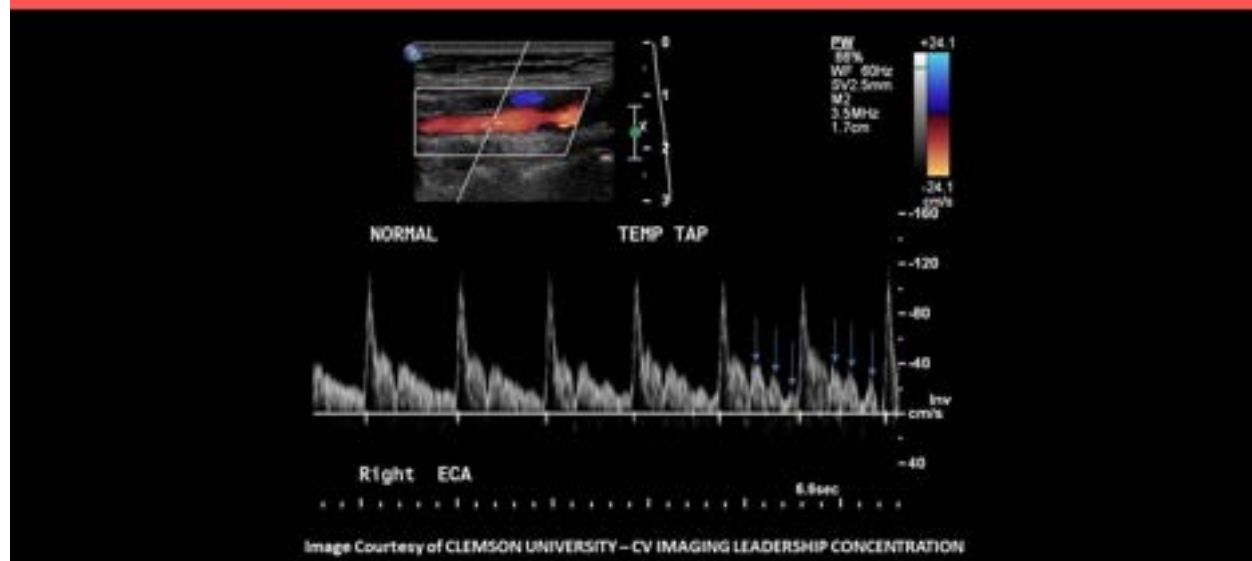
Temporal Tap:

- Assists in differentiating ECA from ICA
- Press index finger against patient's temple
- Tap superficial temporal artery repeatedly to demonstrate flow oscillations during diastole in ECA
- ICA will show NO response in most patients, temporal artery branches from ECA

Temporal Tap



Temporal Tap



Internal Carotid Artery - Proximal

The ICA is usually the larger, more posterior of the two CCA branches.
Color Doppler evaluation can help identify visible extracranial branches that are an ECA characteristic.

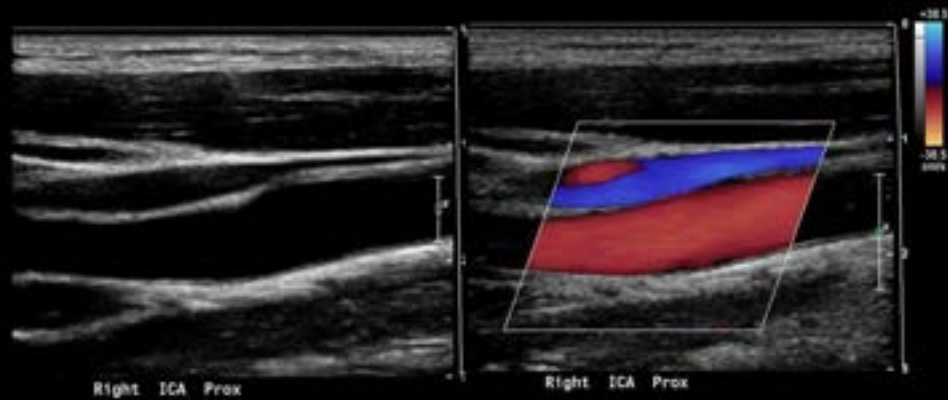


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Normal ICA Flow Pattern

ICA flow is usually low resistance due to the vascular bed (brain) it supplies. There is also more diastolic flow than in the CCA and ECA.

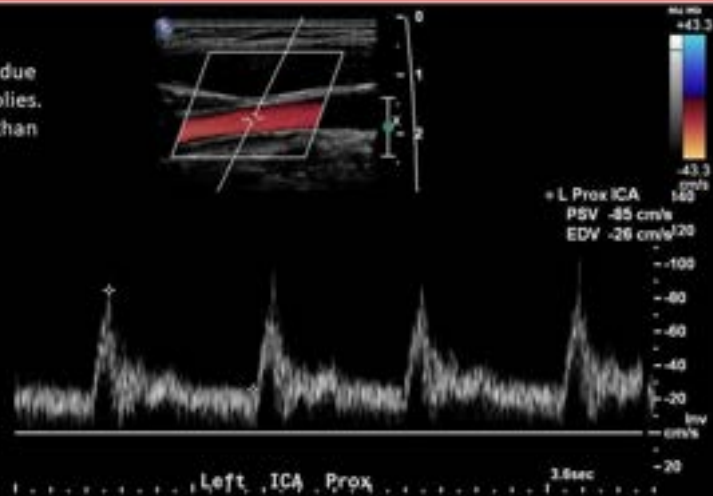


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Normal ICA Mid

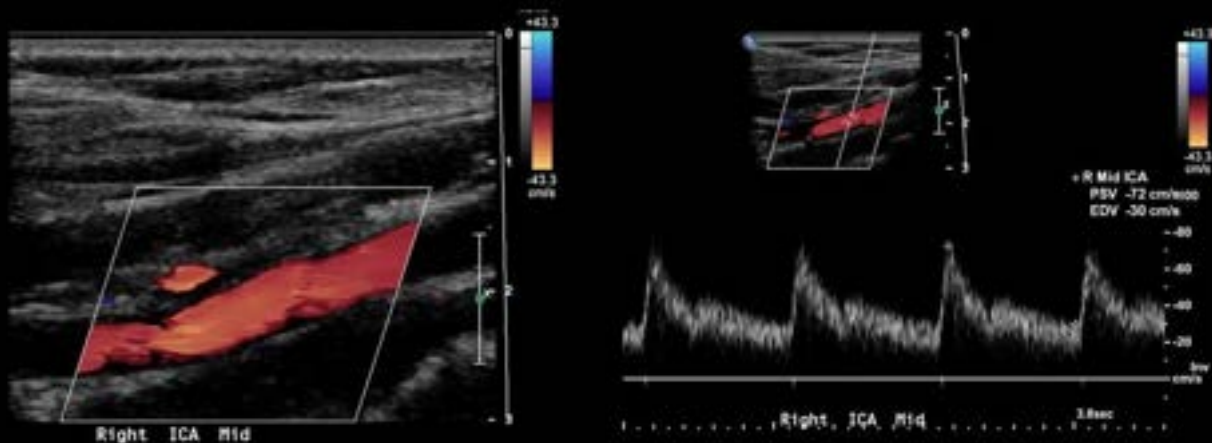
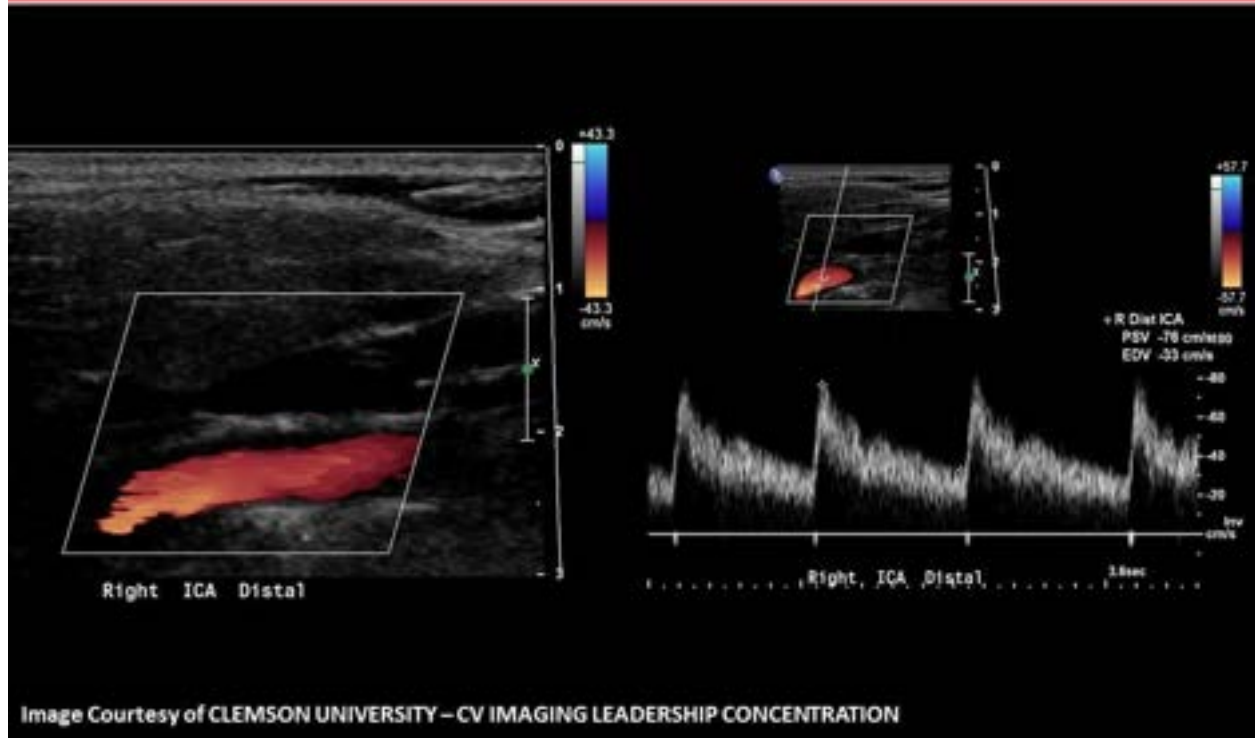


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Normal ICA Distal



Vertebral Arteries:

- Originate from the subclavian arteries bilaterally
- Travel through the vertebral bodies to merge inside cranium to form the basilar artery
- Locate the CCA and then angle posterior
- Vertebral arteries and veins seen between the vertebrae
- Velocity and flow direction should be compared between the right and left side
- Both arteries should demonstrate antegrade flow moving toward the brain
- Flow should be monophasic with mild diastolic flow
- Velocities should be similar

Normal Vertebral Artery & Vein

The normal vertebral arteries should always demonstrate antegrade flow. Always be mindful of your color map settings when evaluating flow direction.

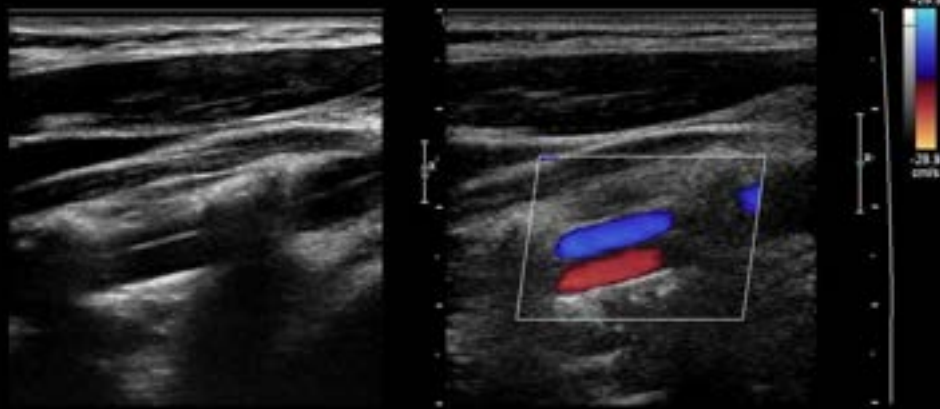


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Normal Vertebral Artery

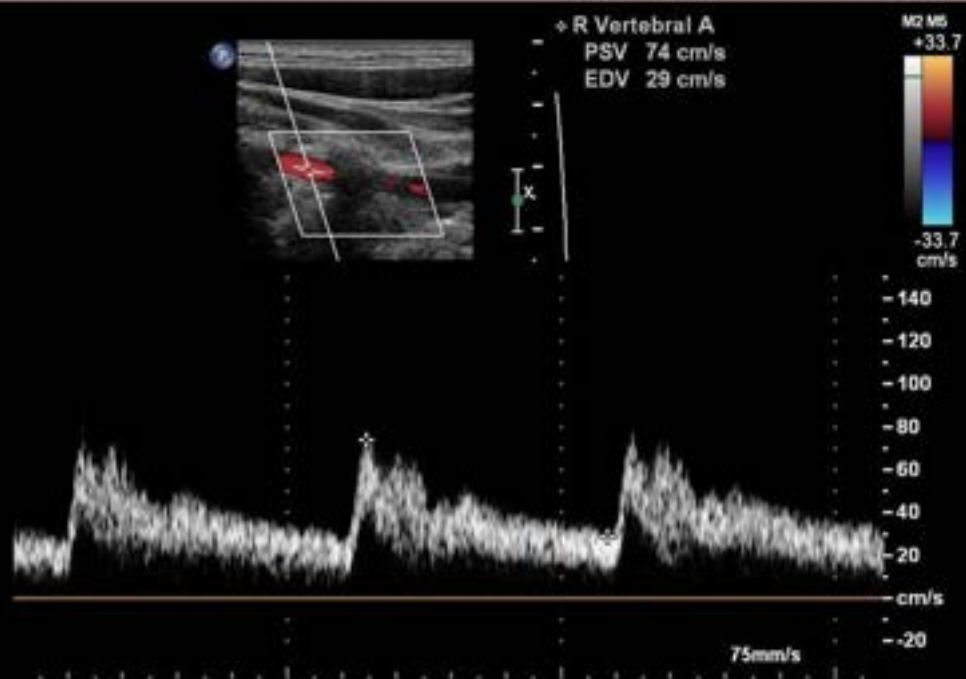


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Transcranial Ultrasound Evaluation

Transcranial Doppler (TCD):

- TCD - Transcranial Doppler:
 - Uses 2MHz non-imaging probe with good depth resolution to perform Doppler evaluation of the cerebral vessels (non-imaging), uses TAMV velocities
 - 0-degree angle of insonation is assumed
 - Vessel velocity, direction, pulsatility, and turbulence assessed
 - Time averaged maximum velocity (TAMV) most commonly used for TCD evaluation
 - Microbubble contrast can be used to improve exam; requires constant infusion of contrast and two transducers inserted into head gear that holds the transducers in place during constant monitoring
- TCCD - Transcranial Color Duplex Exam:
 - Uses 1.8 - 3.5MHz phased array to produce B-mode, color and Doppler images
 - OFFERS ABILITY FOR ANGLE CORRECTION = PSV MEASURED FOR DIAGNOSIS INSTEAD OF TAMV
 - Transcranial Imaging (TCI) is another way to describe TCCD
 - Peak systolic velocity (PSV) also can be used for TCCD evaluation
 - Vertebrobasilar system and Circle of Willis evaluated with PW Doppler
 - Evaluate bilateral ACA, PCA, MCA, siphon, vertebral and single basilar vessels
 - Can be used to detect collateral flow by evaluating flow direction and velocity in the orbital/ophthalmic artery, and basilar artery
 - Use a 5-10mm Doppler sample size
 - Never invert the PW Doppler baseline during the exam, demonstrate flow toward the probe as above the baseline, flow away from the probe as below the baseline
 - Use small field of view to increase frame rates for 2D and color Doppler
 - Always set the color Doppler map to demonstrate flow toward the probe as RED and flow away from the probe as BLUE

- To improve color Doppler visualization, set the color priority to higher levels to demonstrate more color over the 2D image

Intracranial Arterial Anatomy:

Internal Carotid Artery:

- Cervical ICA
 - Extracranial segment that originates at the carotid bifurcation and extends to the petrous portion of the temporal bone
 - Evaluated on extracranial Duplex examination
- Petrous ICA
 - Intracranial segment that courses along the petrous portion of the temporal bone
 - Not able to evaluate using 2D imaging, requires Doppler evaluation
- Cavernous ICA
 - Tortuous intracranial segment that extends from the petrous portion to the circle of Willis
 - Distal end of the cavernous segment is called the supraclinoid segment
 - The ophthalmic artery is the first branch of the ICA and originates from the cavernous segment
 - The ICA terminates when it splits into the MCA and ACA

Middle Cerebral Artery:

- The ICA terminates when it splits into the MCA and ACA
- MCA receives the majority of blood from the ICA and is a larger vessel than the ACA
- Supplies most of the temporal lobe, anterolateral frontal lobe, and parietal lobe
- Each MCA divided into 3 segments: M1, M2, M3
- M1 - sphenoidal or horizontal segment, first portion of the vessel that contains multiple penetrating branches
- M2 - Sylvian or insular segment, located in the sylvian fissure
- M3 - cortical segment, extends superiorly and inferiorly around the outer cerebrum and gives rise to numerous penetrating branches
- Terminal MCA branches connect to the terminal ACA and PCA branches
- Symptoms of MCA disease include dysphasia, behavioral changes, severe hemiparesis in contralateral face/arms
- Also related to Wernicke Aphasia - totally incomprehensible, unable to understand speech, speak, or follow directions

Anterior Cerebral Artery (ACA):

- The ICA terminates when it splits into the MCA and ACA
- Supplies the frontal and parietal lobes and the corpus callosum
- Each ACA is divided into two segments: A1 and A2
- A1 segment located between the ICA and the anterior communicating artery origin
- A2 segment located distal to the anterior communicating artery origin
- Symptoms of ACA disease include loss of coordination, incontinence, severe leg hemiparesis or hemiplegia in contralateral leg, facial drooping or asymmetry

Posterior Cerebral Artery (PCA):

- Originates from the top of the basilar artery
- Curve laterally, posteriorly, and superiorly around the midbrain
- Supply parts of the midbrain, thalamus, temporal lobe, and occipital lobe
- Each PCA is divided into two segments: P1 and P2
- P1 segment located between the basilar artery and the posterior communicating artery origin
- P2 segment located distal to the posterior communicating artery origin
- Symptoms of PCA disease include dyslexia and coma

Vertebral Arteries:

- Two vessels (right, left)
- Course from subclavian arteries through vertebral pedicles and enter cranium through the foramen magnum
- Merge together in the cranium to form the single basilar artery
- Proximal segments seen on extracranial evaluation, distal segments seen on TCD exam

Basilar Artery:

- Formed at the junction of the two vertebral arteries inside the cranium
- Supplies brainstem and cerebellum
- Bifurcates into the 2 PCA after the superior cerebellar arteries branch off of basilar artery
- Vertebrobasilar symptoms include vertigo, ataxia, bilateral visual blurring or double vision, bilateral homonymous hemianopia, bilateral paresthesia, drop attack

Anterior Communicating Artery:

- The most common congenital anomaly of the Circle of Willis is the absence of one or more communicating arteries
- Single vessel that connects the two ACA branches
- Potential for collateral flow between the right and left anterior circulatory systems
- Most common location for intracranial aneurysm

- Not normally directly evaluated in a TCD exam but can be indirectly assessed by evaluating the flow direction of the main arteries of the Circle of Willis

Posterior Communicating Artery:

- The most common congenital anomaly of the Circle of Willis is the absence of one or more communicating arteries
- Two vessels (right, left)
- Branches from the distal portion of the ICA before it terminates at the MCA/ACA bifurcation
- Connects the PCA and MCA on each side
- Potential for collateral flow
- Second most common location for intracranial aneurysm
- Not normally directly evaluated in a TCD exam but can be indirectly assessed by evaluating the flow direction of the main arteries of the Circle of Willis

Indications for TCD/TCCD:

- Most common indication is to assess vasospasm/vasoconstriction caused by subarachnoid hemorrhage
- Stroke - MCA is the most commonly occluded intracranial vessel with an acute stroke; Increased TAMV and PSV with turbulence
- TIA
- Intracranial stenosis/occlusion
- Collateral flow due to occlusion
- AV malformation - increased systolic and diastolic velocities (TCCD only)
- Vasospasm - increased peak systolic velocity
- Monitoring during/after surgical procedures (TCD only)
- Microemboli detection (TCD only)
- Right to left shunt detection, patent foramen ovale (TCD only)
- Pediatric flow evaluation due to sickle cell anemia, moyamoya disease
- Suspected brain death

Contraindications for TCD/TCCD:

- Recent eye surgery

Limitations of TCD/TCCD:

- Hyperostosis - thickening of the cranial bones, seen in women and older patients
- Requires extensive time from technologist and patient
- Incorrect vessel identification due to limited or absent vessel imaging
- Vasospasm can be mistaken for stenosis

- Cannot evaluate the cerebral circulation for aneurysm formation
- Patient cooperation/condition

Advantages of TCD: (Compared to CT, MRI and Angiography)

- Noninvasive
- Can detect rapid alterations in flow velocity in real time; angiography provides real time information on vessel patency as contrast fills vessels, but no velocity information is available
- Provides direct, immediate information on cerebral perfusion
- Does not require contrast administration
- No radiation exposure

Limitations of TCD:

- One of the most common limitations is inaccurate vessel identification due to non-imaging techniques
- No imaging capabilities = possible incorrect information with improper identification of the vessels
- Requires technologist skill and experience
- Requires patient cooperation = no movement or talking during the exam
- Difficulty penetrating the skull due to hyperostosis
 - Hyperostosis of the skull causes calvarial thickening
 - Diffuse involvement is related to Paget disease, metastatic disease, anemia, hyperparathyroidism

Standard TCD/TCCD Protocol:

Vessels Interrogated:

- Distal ICA segment at the entrance to the skull (when appropriate)
- Internal carotid artery (ICA) siphon
- Terminal internal carotid artery (TICA)
- Proximal M2 middle cerebral artery (MCA)
- A1 anterior cerebral artery (ACA)
- Ophthalmic artery (when appropriate)
- P1 or P2 posterior cerebral artery (PCA)
- Terminal vertebral artery (VA)
- Proximal and distal basilar artery
- Collateral flow via anterior communicating artery (when detectable)
- Collateral flow via posterior communicating artery (when detectable)

Standard Views:

- Sample depth, flow direction and flow velocity are used to identify a vessel on TCD exam
- Transtemporal, Ophthalmic, Submandibular and Suboccipital windows used to assess intracranial blood flow patterns
- Recent eye surgery is a contraindication for the transorbital view
- Use maximum power output for all windows except transorbital window; **MUST REDUCE POWER OUTPUT WHEN USING TRANSORBITAL WINDOW**
- All intracranial arteries demonstrate low resistance flow with spectral broadening
- There are no visible spectral windows on the tracings due to the small caliber of the vessels

Transtemporal Window:

- Patient in the supine position with head supported
- Place probe on cranium in temporal area, superior to the zygomatic arch, with notch toward patient's nose
- Used to evaluate the MCA, ACA, PCA and terminal ICA
- If stenosis and collateral flow are present, the communicating arteries can also be evaluated with this window
- Sphenoid bone used as landmark for MCA
- Midline probe placement with slight anterior angulation will demonstrate the MCA flow (at appropriate depths)
- MCA demonstrates flow toward the transducer
- From the MCA, angle more sharply anterior and adjust depth of cursor for the ACA
- ACA identified medial and anterior to the MCA
- ACA demonstrates flow away from the transducer
- From the MCA, angle posterior and adjust depth to assess P1 and P2
- Cerebral peduncles used as landmark for PCA
- P1 demonstrates flow toward the transducer, P2 demonstrates flow away from the transducer
- From the MCA, angle inferior to the origin of the MCA to locate the ICA
- Foramen lacrum used as landmark for the terminal ICA
- Terminal ICA identified slightly inferior to MCA/ACA level
- Flow direction in the terminal ICA varies due to varied course of the terminal ICA

Suboccipital Window:

- AKA Transoccipital
- Used to evaluate the vertebral and basilar arteries
- Place patient in decubitus position with hand placed under head for support
- If the patient cannot lie on their side, turn their head away from the side of interest and place the transducer to the left or right side of the foramen magnum

- If the patient cannot lie down, the patient can be seated, neck flexed and head supported in hands
- Place probe at the base of the neck and angle cephalad
- Foramen magnum used as landmark for vertebral arteries; both vertebral arteries demonstrate flow away from the transducer
- Basilar artery located at a depth >80mm and demonstrates flow away from the transducer

Transorbital Window:

- AKA Ophthalmic window
- Used to evaluate the ophthalmic artery and carotid siphon
- Patient in the supine position with head supported
- Reduce output setting to lowest setting possible that still permits acceptable imaging of the vessels
- Limit the time spent using the Transorbital window
- Place probe over the eye with sufficient gel to maintain contact of the probe face
- Notch should be medially oriented toward nose
- Demonstrates the ophthalmic artery ADJACENT to the optic nerve; flow should be toward the transducer; care must be taken not to confuse with the retinal artery that courses WITHIN the optic nerve
- Carotid siphon is a curved segment; flow direction varies with portion evaluated

Supplemental View:

Submandibular Window:

- Patient is supine with neck extended
- Probe placed under the jawline (mandible) toward the right or left side and angled cephalad
- Used to evaluate the distal ICA and siphon, if not evaluated in other view
- Normal flow moves away from the probe

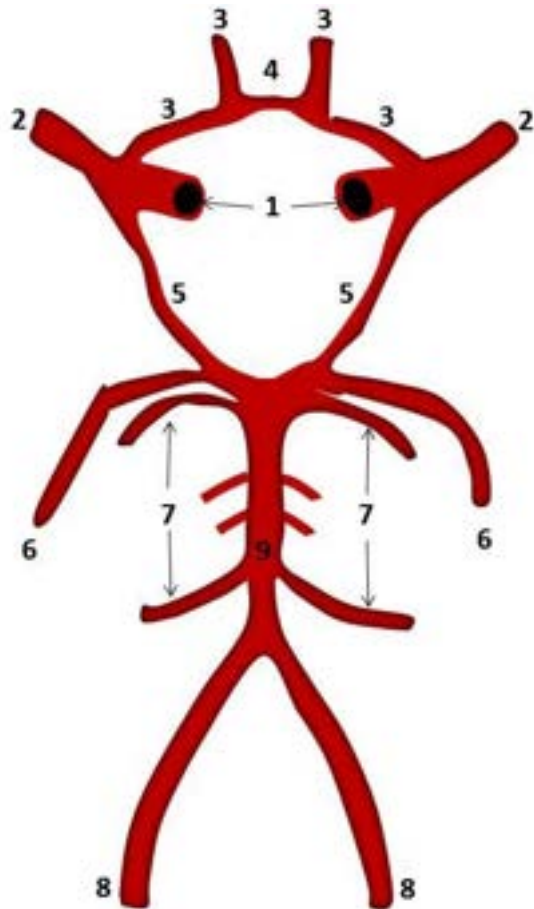
Normal Doppler Parameters:

- TAMV most commonly used to assess vessel stenosis in TCD, but peak systolic velocity should be reported on TCCD
- Mean velocity is more accurate because it is not affected by systemic factors
- Flow velocities normally decrease as the patient ages
 - Average MCA velocity age 10yrs = 70cm/s, age 70yrs = 41cm/s
- Velocity ratios are helpful to compare flow in different vessels
- Normal velocity ratios
 - ACA/MCA <1.2
 - MCA/ICA 1.1 - 3.0

- Basilar artery / extracranial vertebral artery <2.0
- Pulsatility Index refers to variability in systolic and diastolic flow velocities
- Normal PI 0.5 - 1.1
- PI increases proximal to an obstruction and decreases distal to an obstruction
- Flow velocities should be similar between the left and right side, a difference greater than 30% in the same vessel on opposite sides can indicate disease is present

Abnormal Doppler Parameters:

- Stenosis is most common in the MCA, siphon and terminal ICA
- Focal increase in TAMV and PSV; Refer to charts for specific parameters
- Post stenotic turbulence
- Tardus parvus waveform
- Increased resistance in flow proximal to an obstruction
- Abnormal velocity ratios
 - ACA/MCA >1.2
 - MCA/ICA >3.0
 - Basilar artery / extracranial vertebral artery >2.0 indicates basilar vasospasm



Circle of Willis

1. Internal carotid arteries (ICA)
2. Middle cerebral arteries (MCA)
3. Anterior cerebral arteries (ACA)
4. Anterior communicating artery
5. Posterior communicating artery
6. Posterior cerebral arteries (PCA)
7. Cerebellar arteries
8. Vertebral arteries
9. Basilar artery

Circle of Willis



Courtesy of:
Nevit Dilmen, MD.

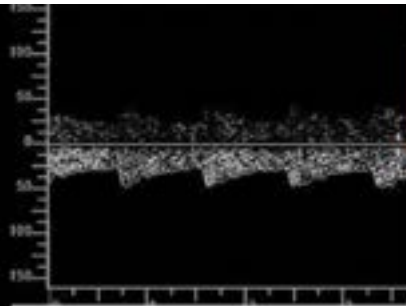
TCCD - Peak Systolic Velocity

Intracranial Artery	Normal PSV (Using Angle Correction for PW Doppler)
MCA	<220cm/s
ACA	<155cm/s
PCA	<145cm/s
Basilar	<140cm/s
Vertebral	<120cm/s

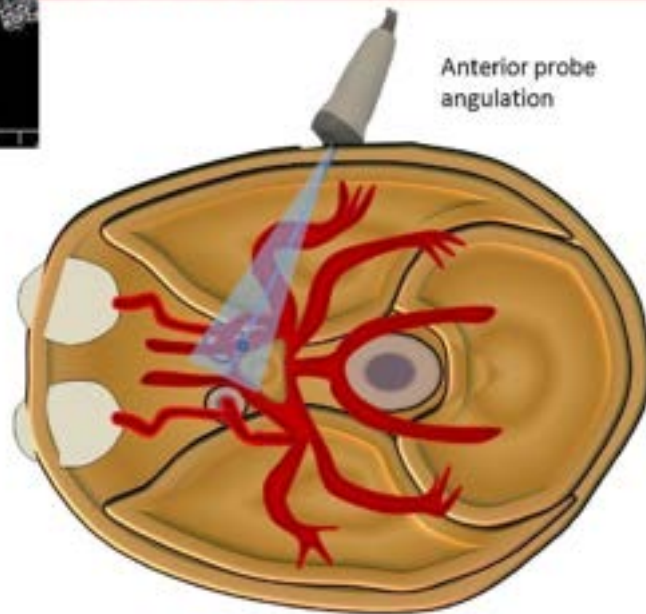


Transtemporal Window

Intracranial Artery	Average Depth Setting	Probe Angulation	Time Averaged Maximum Velocity	Direction of Flow
MCA (M1)	30 - 60mm	Anterior	55cm/s ± 12 cm/s	Toward
ACA	60 - 80mm	Anterior	50cm/s ± 11 cm/s	Away
PCA (P1)	60 - 70mm	Posterior and Caudal	39 cm/s ± 10 cm/s	Toward
PCA (P2)	60 - 70mm	Posterior	40cm/s ± 10 cm/s	Away
Distal ICA	55 - 65mm	Anterior and Caudal	39cm/s ± 9 cm/s	Bi-directional



Transtemporal Window – ACA;
Depth = 60-80mm



Average TAMV
50cm/s + or - 11

Transtemporal –ACA

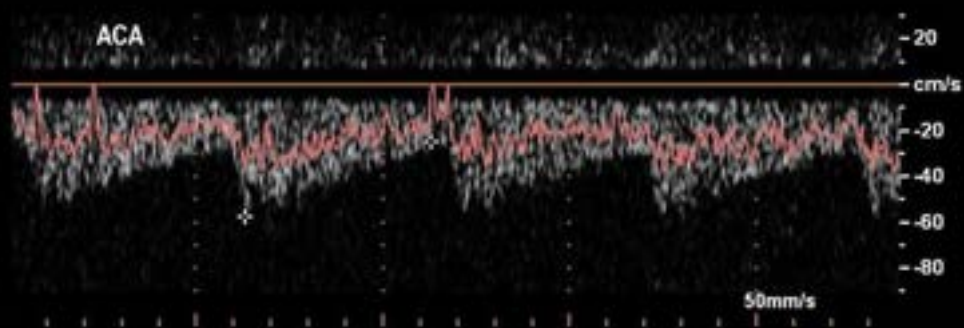
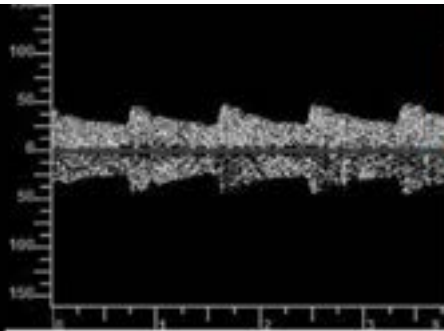
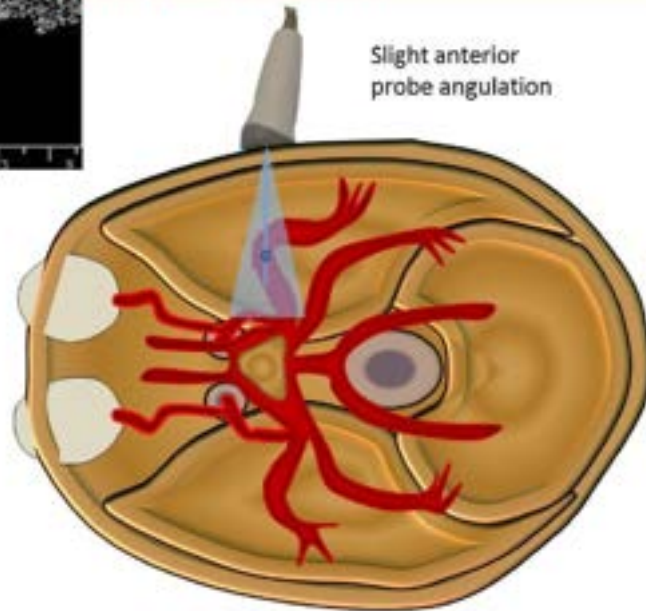


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

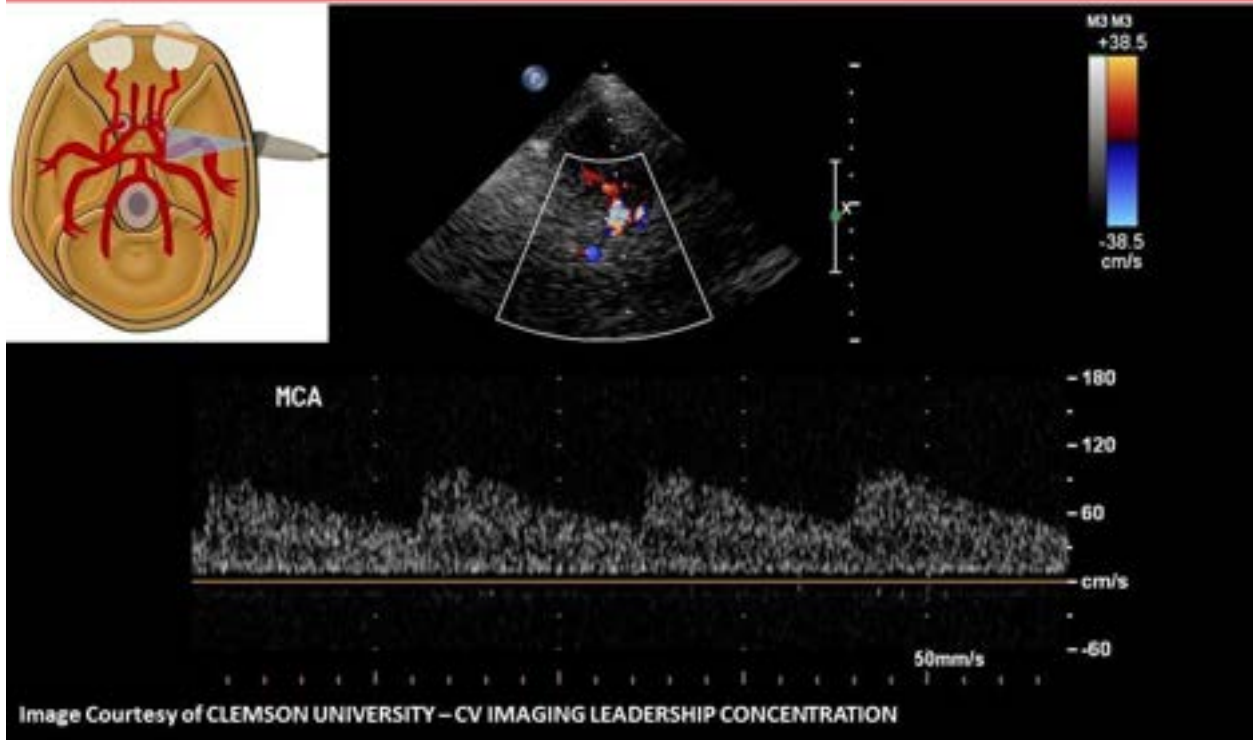


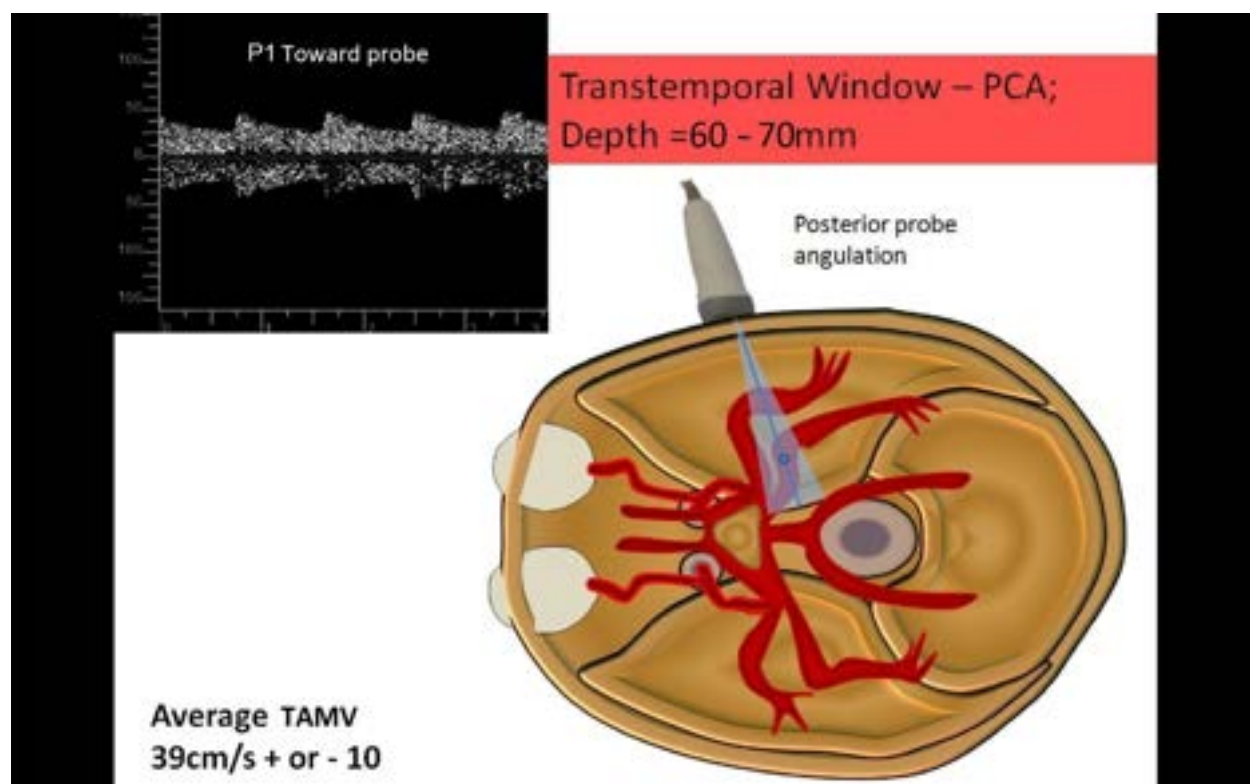
Transtemporal Window – MCA;
Depth = 30-60mm



Average TAMV
55cm/s + or - 12

Transtemporal – MCA





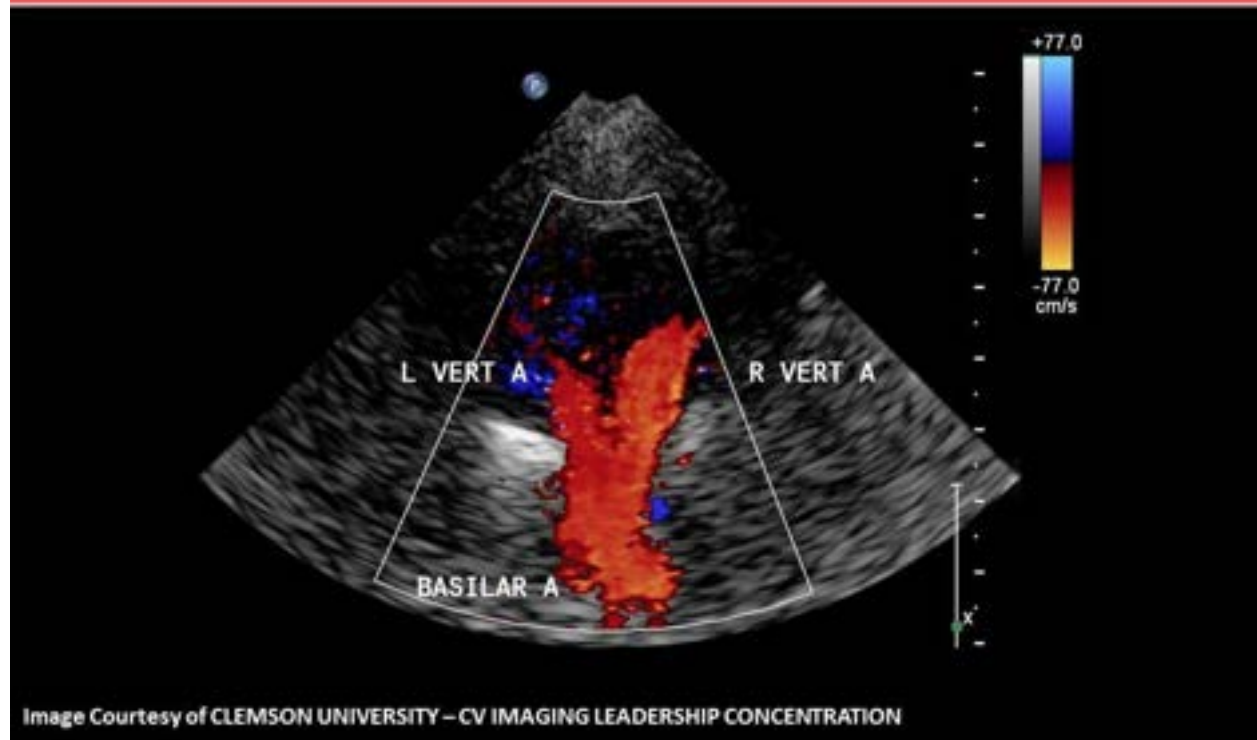
Suboccipital Window

Intracranial Artery	Average Depth Setting	Probe Angulation	Time Averaged Maximum Velocity	Direction of Flow
Vertebral	60 - 90mm	Lateral	34cm/s $\pm$ 8cm/s	Away
Basilar	80 - 120mm	Midline	41cm/s $\pm$ 10cm/s	Away

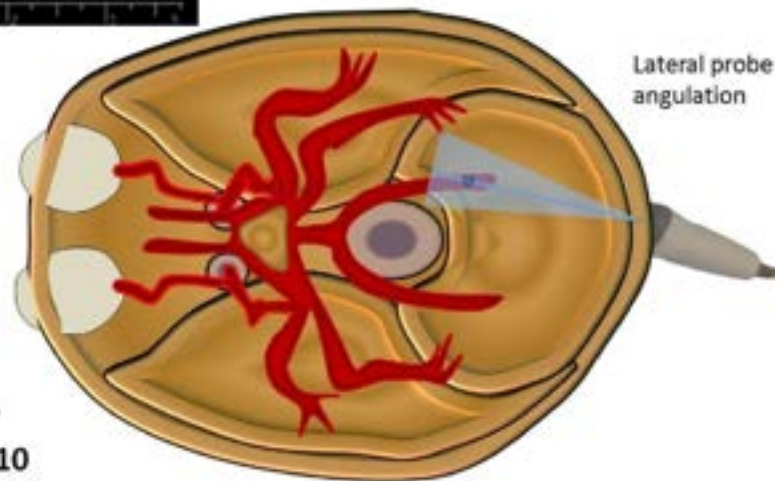
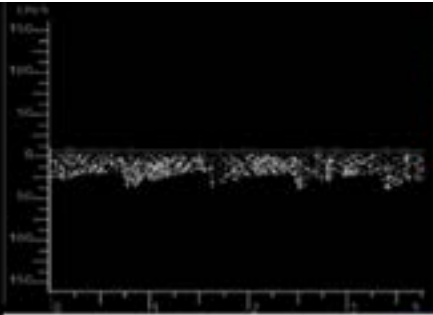
***PCA segments can also be evaluated in this view in some patients

- P1 away
- P2 toward

Suboccipital – Vertebrobasilar Arteries

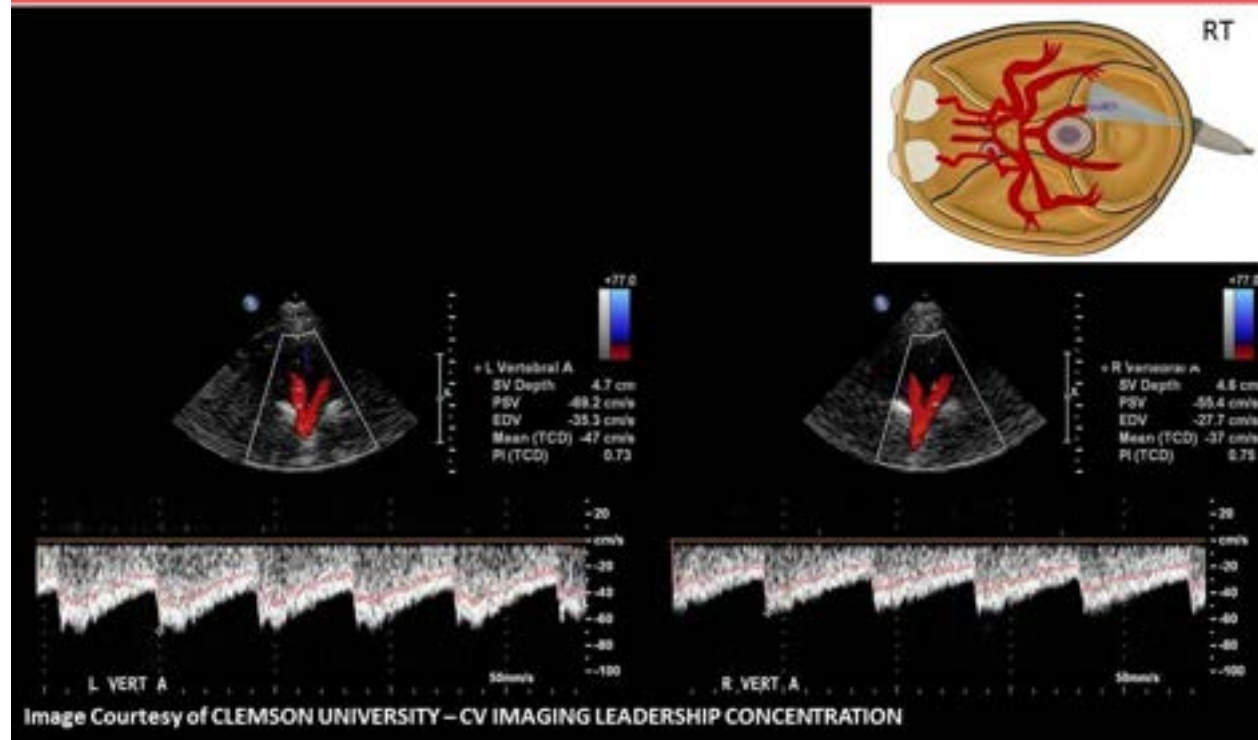


Suboccipital Window – Vertebral
Artery; Depth =60 - 90mm

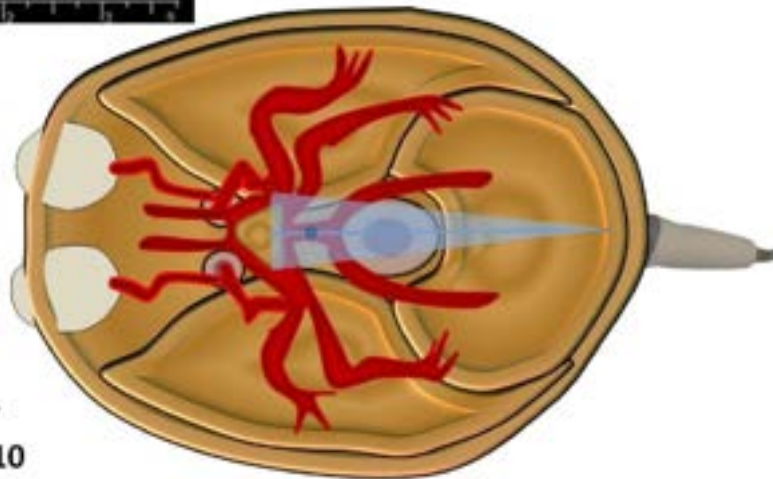
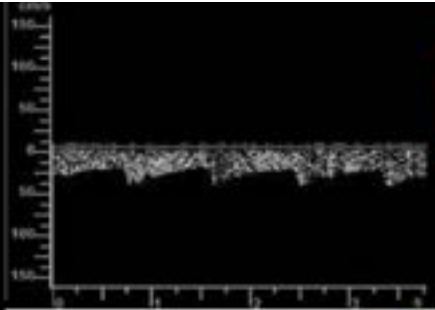


Average TAMV
38cm/s + or - 10

Suboccipital – Vertebral Arteries

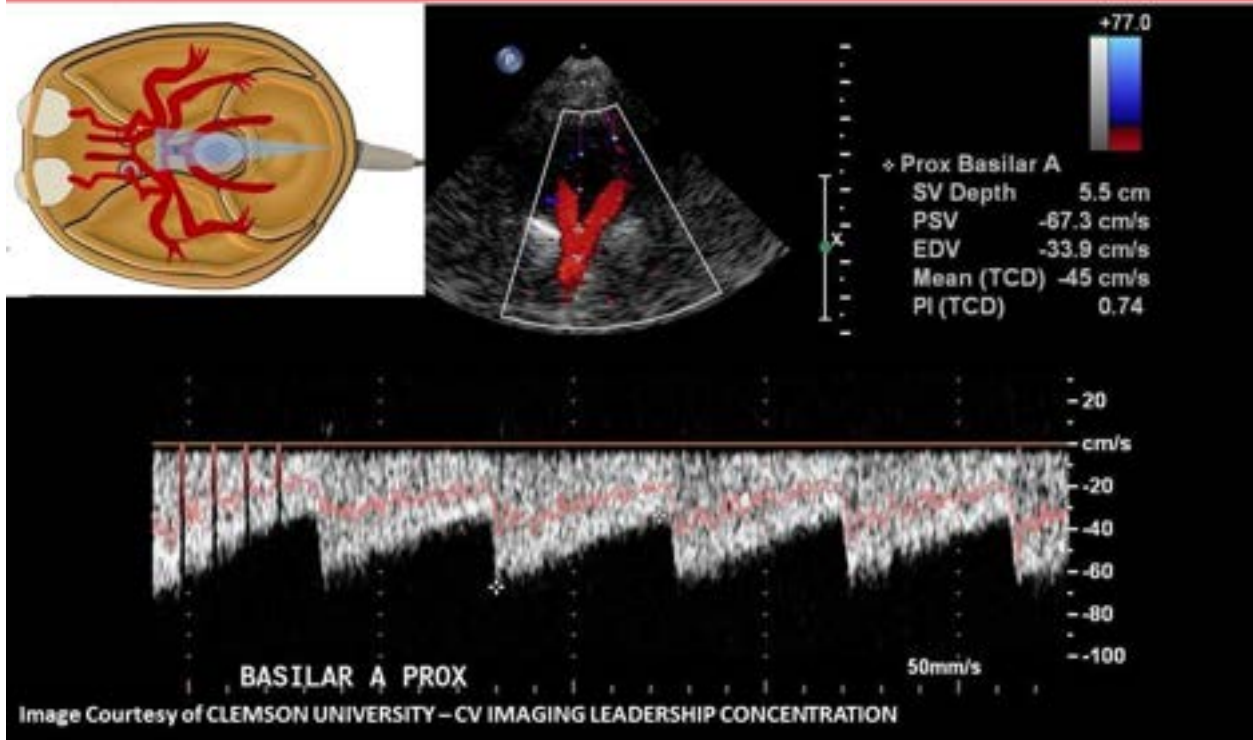


Suboccipital Window – Basilar
Artery; Depth = 80 - 120mm

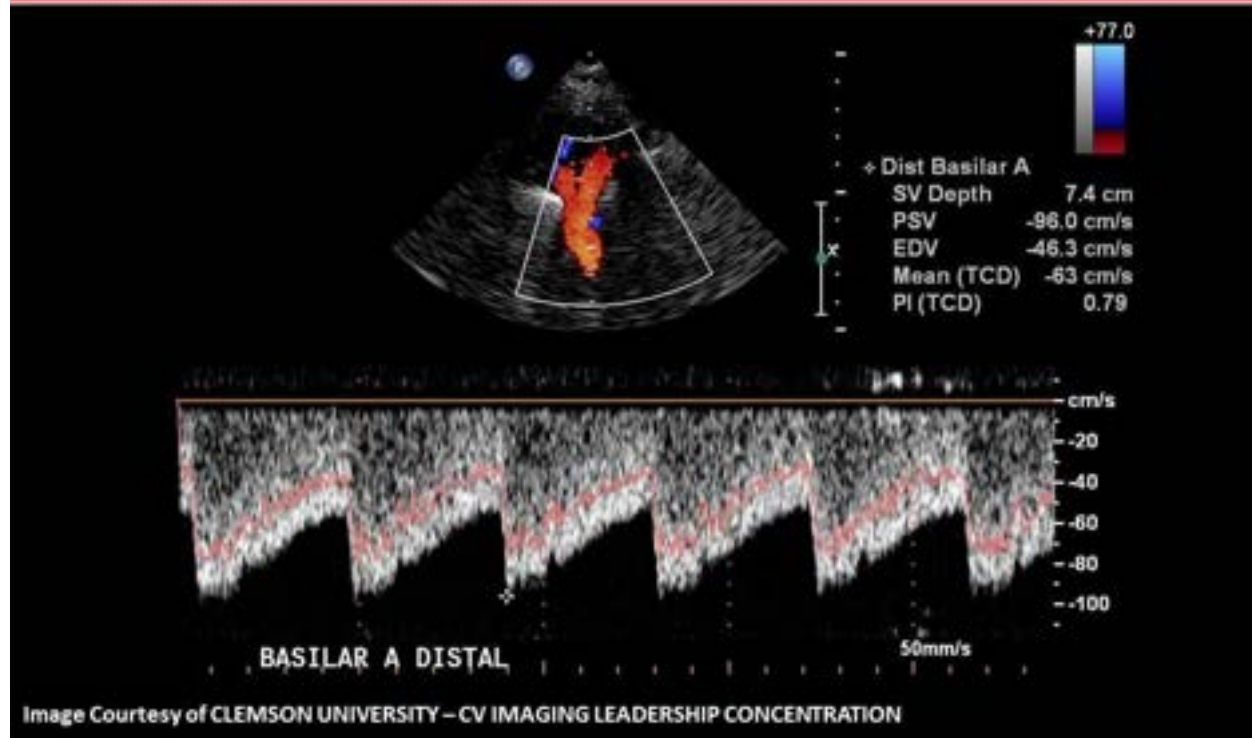


Average TAMV
41cm/s + or - 10

Suboccipital – Basilar Artery



Suboccipital – Distal Basilar Artery



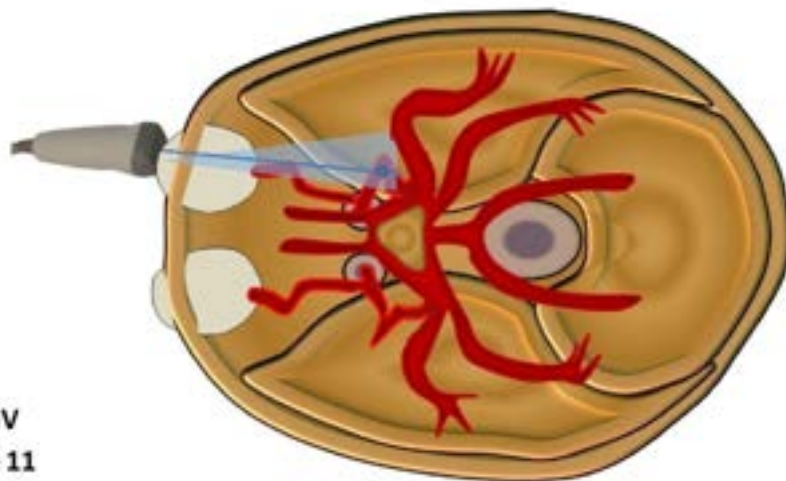
Transorbital Window

Intracranial Artery	Average Depth Setting	Probe Angulation	Time Averaged Maximum Velocity	Direction of Flow
Siphon	60 - 80mm	Slight Lateral	41cm/s $\pm$ 11cm/s	Away Can be bi-directional
Ophthalmic	40 - 60mm	Medial	21cm/s $\pm$ 5cm/s	Toward

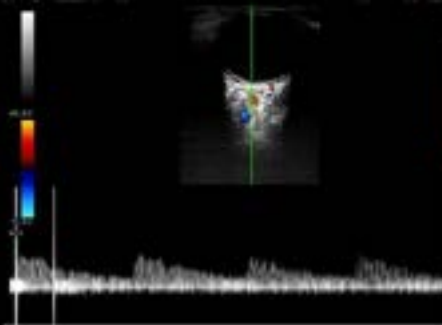
Reversal of the orbital artery indicates the presence of cerebral collateralization.

Ophthalmic Window Internal Carotid Siphon; Depth = 60 - 80mm

Flow will move away from the transducer or will be bidirectional, depending on the point of insonation of the siphon

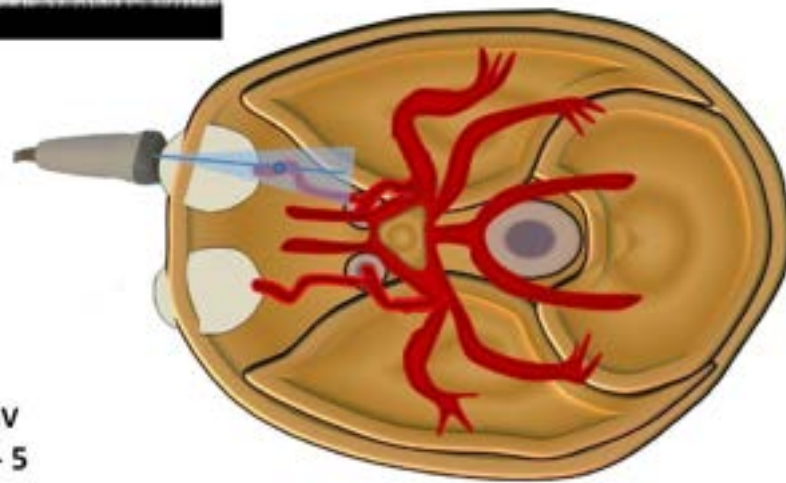


Average TAMV
41cm/s + or - 11



Ophthalmic Window – Ophthalmic Artery; Depth =40 - 60mm

***Always use lowest power settings when using the eye as an acoustic window.



Average TAMV
21cm/s + or - 5

Transorbital - Ophthalmic Artery

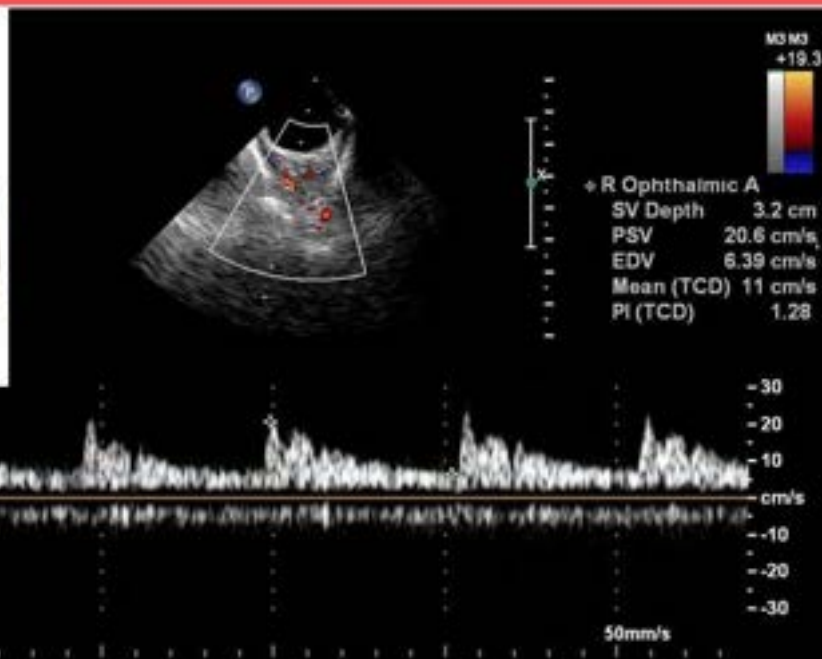
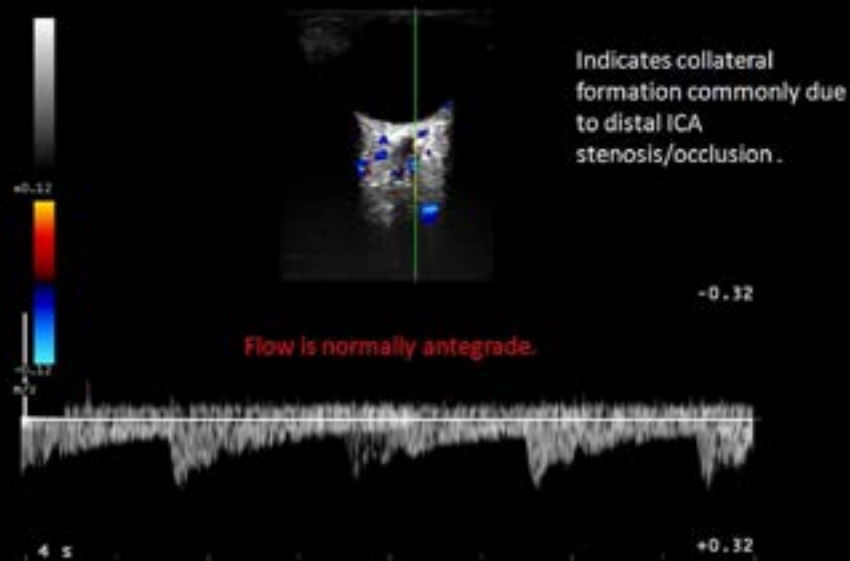


Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Abnormal Ophthalmic Artery Flow

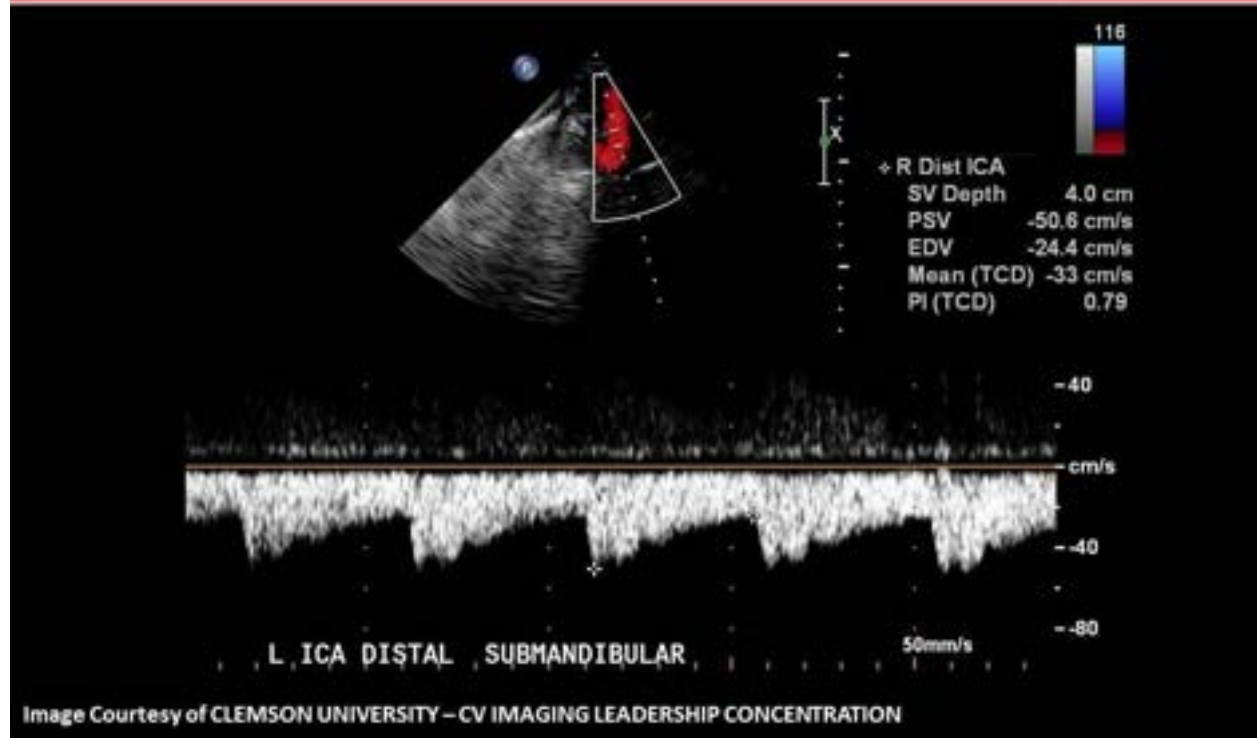


Reversal of flow in the orbital artery indicates the presence of cerebral collateralization.

Submandibular Window

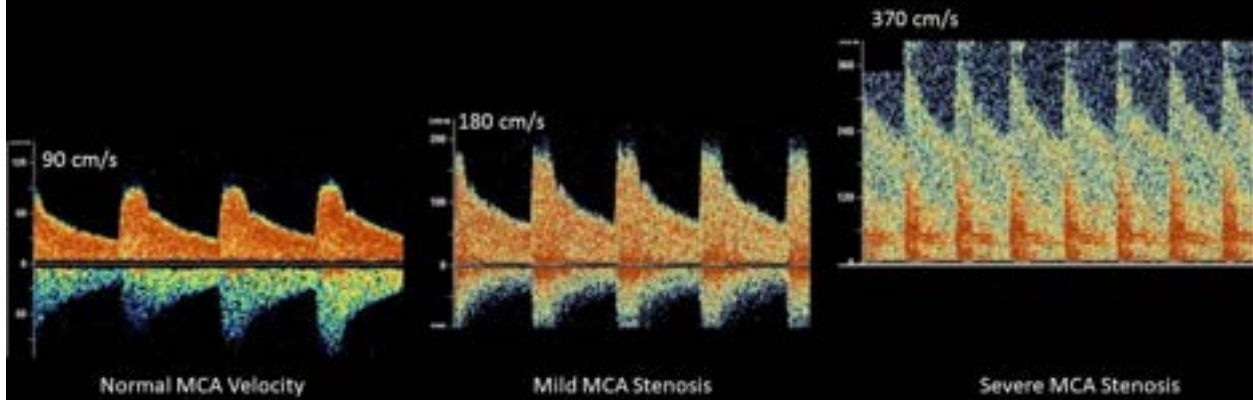
Intracranial Artery	Average Depth Setting	Probe Angulation	Time Averaged Maximum Velocity	Direction of Flow
Distal ICA	60 - 80mm	Medial	30cm/s ± 9cm/s	Away Can be bi-directional

Submandibular – Distal ICA



Intracranial Vessel	Abnormal PSV (Obtained with TCCD and Angle Correction)
MCA	$\geq 220 \text{ cm/s}$
ACA	$\geq 155 \text{ cm/s}$
PCA	$\geq 145 \text{ cm/s}$
Basilar	$\geq 140 \text{ cm/s}$
Vertebral	$\geq 120 \text{ cm/s}$

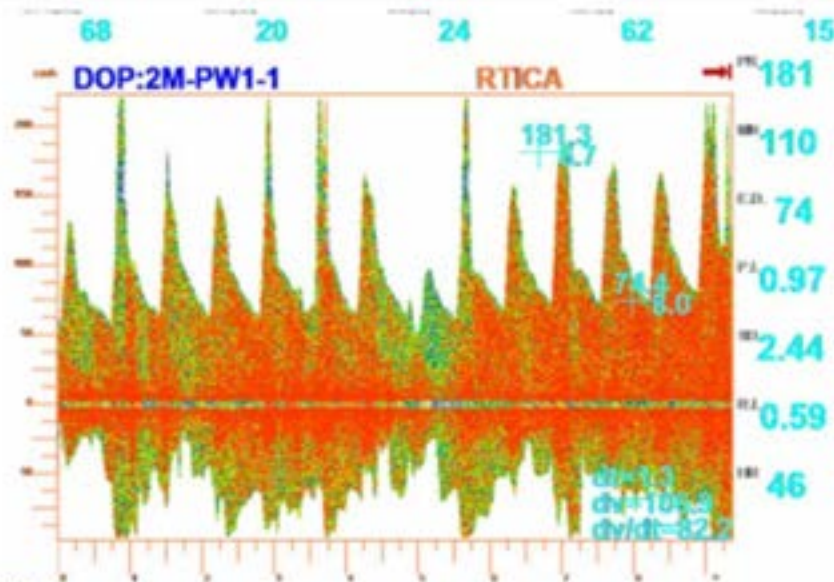
MCA Stenosis



Note the significant increase in systolic and diastolic velocities.

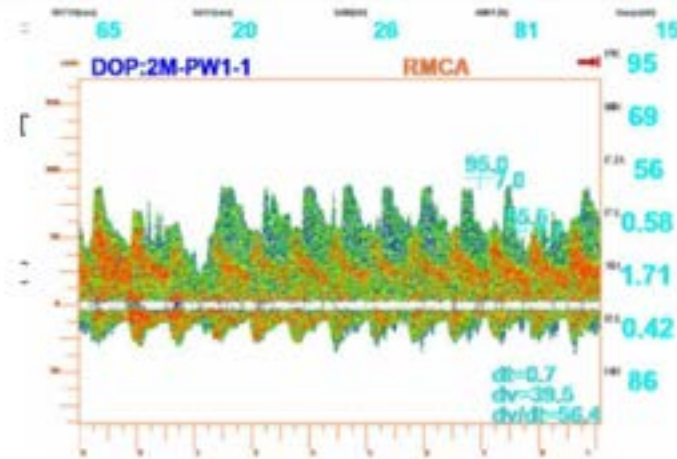
Adapted from: Microembolic signals detected with transcranial doppler sonography differ between symptomatic and asymptomatic middle cerebral artery stenoses in Northeast China. Wu X, Zhang H, Liu H, Xing Y, Liu K. PLoS ONE (2014)

ICA Siphon Stenosis



www.learntcd.com

Ipsilateral ICA Occlusion



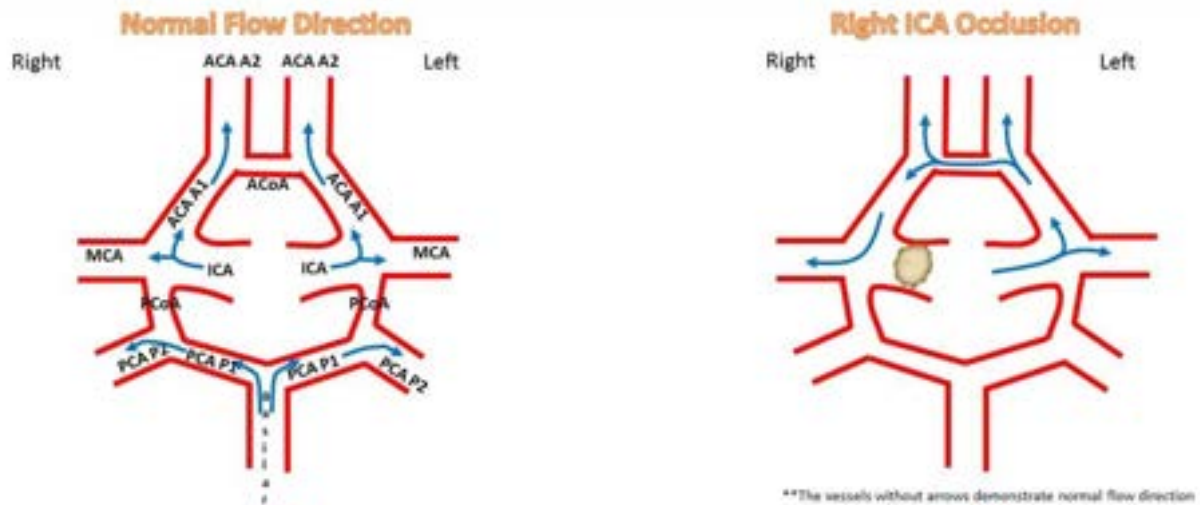
With the ipsilateral ICA occlusion, the normal MCA flow is seen with the smaller anterior waveform. The anterior communicating artery waveform is seen behind the MCA tracing. This vessel is "cross-filling" the right MCA with flow from the left sided cranial vessels.

www.learnTCD.com

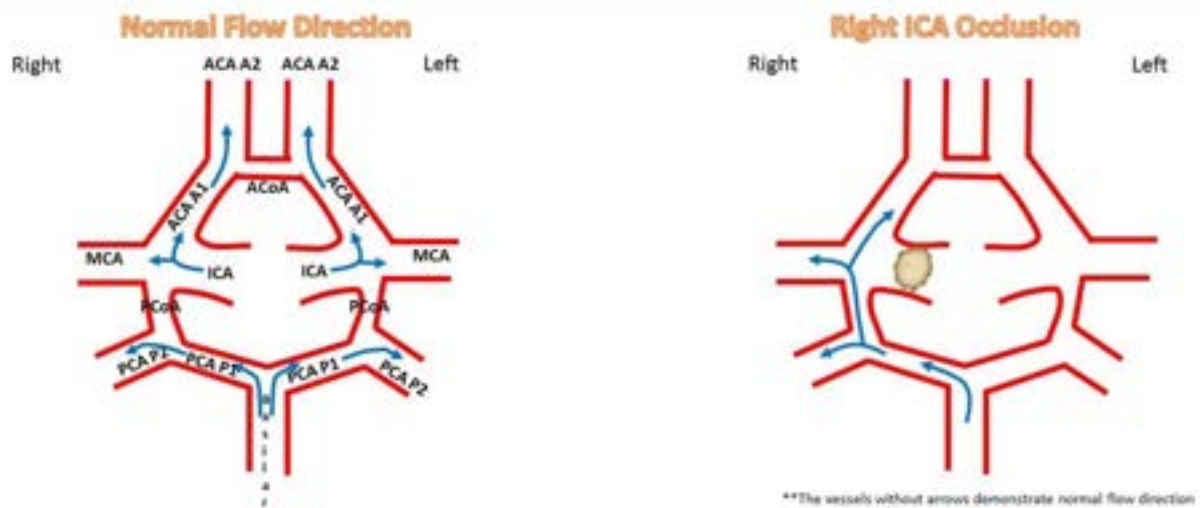
Intracranial Collaterals:

1. Anterior Communicating Artery (most common collateral pathway) provides collateral flow between the right and left anterior circulatory systems for high grade ipsilateral ACA or ICA stenosis or occlusion; contralateral ACA will demonstrate high velocities; flow is reversed in the A1 segment of the ipsilateral ACA
2. Posterior Communicating Artery provides collateral flow between the carotid and basilar arteries for ipsilateral ICA stenosis or occlusion; increased velocity in the ipsilateral P1 segment; decreased velocity in the ipsilateral P2 segment
3. ECA to ICA collateralization occurs with ipsilateral ICA stenosis/occlusion; reversal of flow in the ipsilateral ophthalmic artery; increased systolic and diastolic velocities in the contralateral ophthalmic artery
 - ECA > superficial temporal artery > supraorbital artery > ophthalmic artery > ICA
 - ECA > superficial temporal artery > frontal artery > ophthalmic artery > ICA
 - ECA > facial artery > angular artery > frontal artery > nasal artery

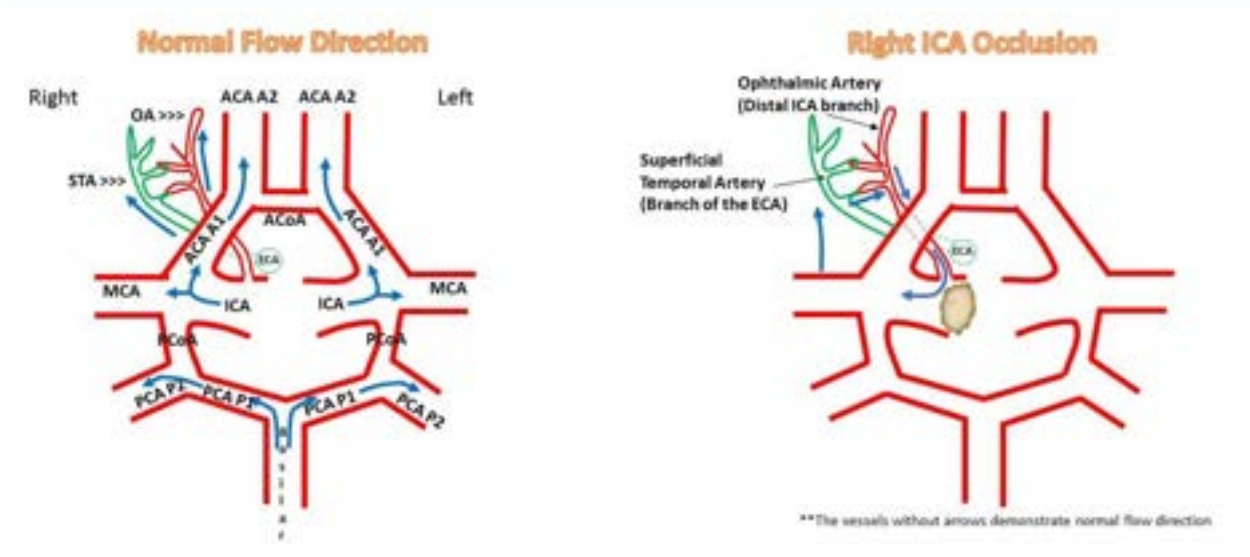
Cross - Over Collateralization



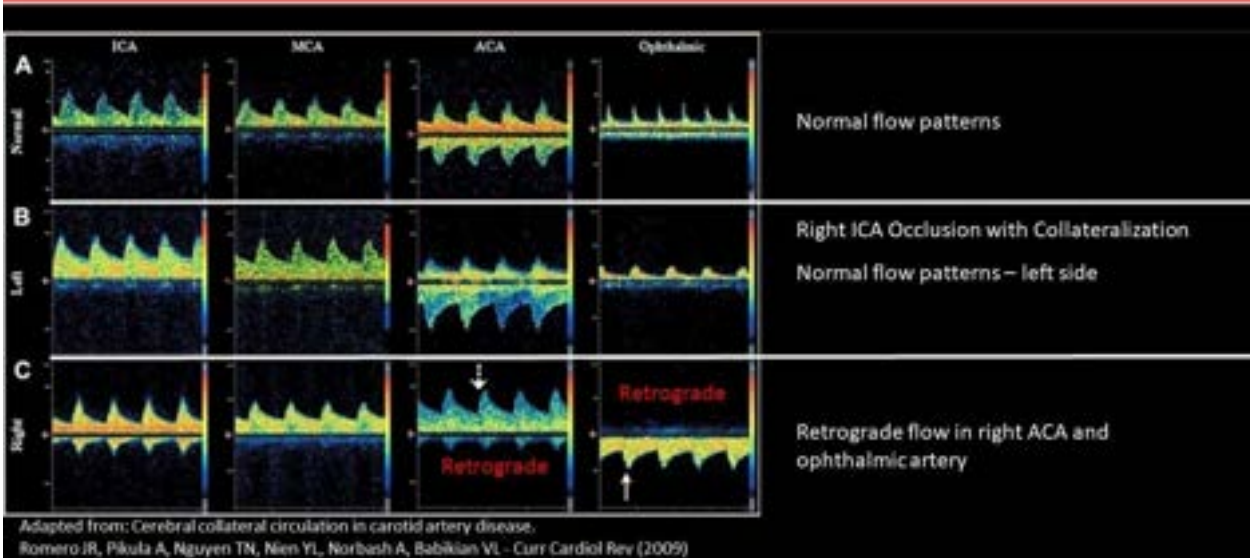
Posterior to Anterior Collateralization



ECA to ICA Collateralization



TCD – Collateral Formation



Periorbital Doppler:

- CW Doppler evaluation of the frontal artery (branch of the ophthalmic artery)
- Patient supine with sufficient head support to prevent motion

- 8-10MHz pencil probe placed near the inner canthus of the eye
- Normal flow direction will be toward the probe
- Flow moving away from the probe indicates a significant obstruction in the ipsilateral ICA
- Compression of the infraorbital, facial and superficial temporal arteries is performed on both sides of the head, while monitoring the frontal artery flow
- If compression of one of the above arteries causes decreased or reversed flow in the frontal artery, that vessel is being used as a collateral to fill the frontal artery (not the ICA)
- This method can only determine if there are patent collaterals being used to redirect flow, it cannot distinguish between a stenosis and an occlusion as the cause for the redirected flow
- Other limitations include patient motion, uncooperative patient, and technologist experience

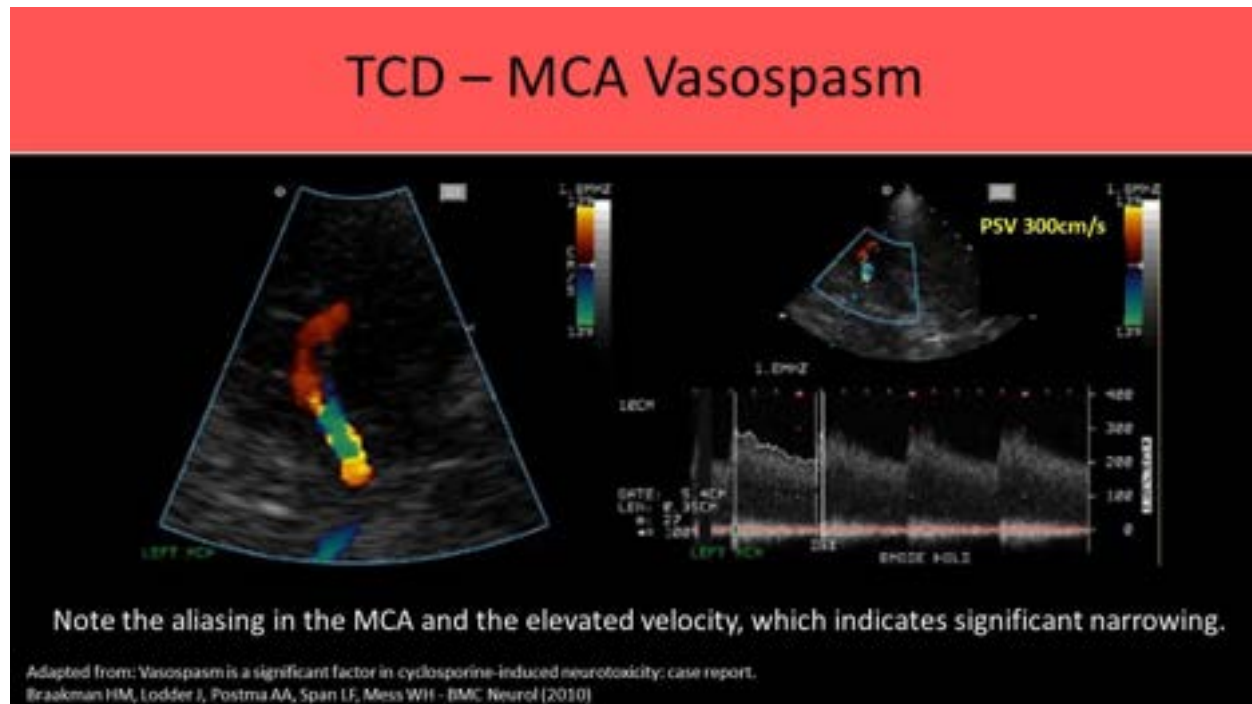
Intracranial Arteriovenous Malformations:

- Most brain AVMs emerge during fetal development
- Risk factors include gender (more common in males) and family history
- Headache or seizures
- Complications include subarachnoid hemorrhage, cerebral ischemia, cerebral aneurysm, brain damage or stroke
- Vein of Galen defect - a large AVM that is usually identified on fetal ultrasound exams or soon after birth
- Wyburn-Mason syndrome - typically presents with unilateral vascular AVMs that primarily involve the brain, orbits and facial structures
- Ultrasound is not typically used for evaluating cerebral AVMs, but it may be used during open neurosurgery for the purposes of AVM localization
- Treatment options include surgical resection, embolization and radiotherapy

TCD for Vasospasm:

- Vasoconstriction of intracranial arteries
- Complication in patients with subarachnoid hemorrhage, usually a slow bleed
- Vasospasm usually occurs within 14 days of the start of the hemorrhage
- Flow velocities increase in MCA, ACA and PCA with subarachnoid hemorrhage
- Serial evaluations are performed daily or every other day to record velocities in MCA
- Always perform the repeat examinations using the same depth settings used for the initial evaluation
- MCA mean velocity over 200cm/s and TAMV over 120cm/s indicate severe significant reduction in intracerebral flow
- Record the highest mean MCA velocity and the terminal ICA velocity
- Calculate the Hemispheric Index - divide MCA velocity by the distal/terminal ICA velocity; HI > 3 is mildly abnormal, HI > 6 indicates severely abnormal flow
- Pulsatility index >1.2 indicates vasospasm

- Can be treated with vasodilator, such as Papavarine



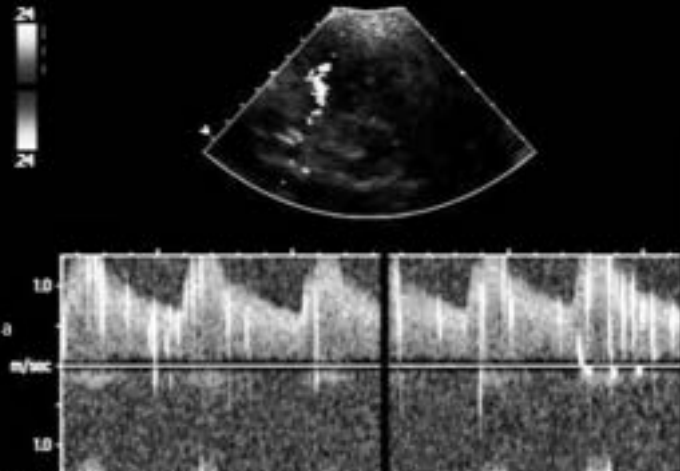
Patent Foramen Ovale:

- Opening in the interatrial septum fails to close after birth
- Patient is supine and will perform the Valsalva maneuver during the procedure
- A baseline MCA evaluation is performed
- Agitated saline injected into the venous system
- Microbubbles enter the lungs if no shunt is present
- If the foramen ovale is patent, bubbles will cross into the left heart and enter the arch, then cerebrovascular system
- If a shunt is present, emboli can be detected in the MCA within a few seconds after injection
- High Intensity Transient Signals (HITS) are visible as bright spikes on the Doppler tracing
- The HITS can also produce an audible signal that makes a "chirping" noise
- If the bubbles are identified in the intracranial arteries, there must be a right to left shunt in the heart to send the bubbles into the arterial system
- Valsalva maneuver may be used to push flow across the PFO into the left atrium
- The number of HITS recorded is directly related to the size of the shunt

MCA Doppler – PFO Study

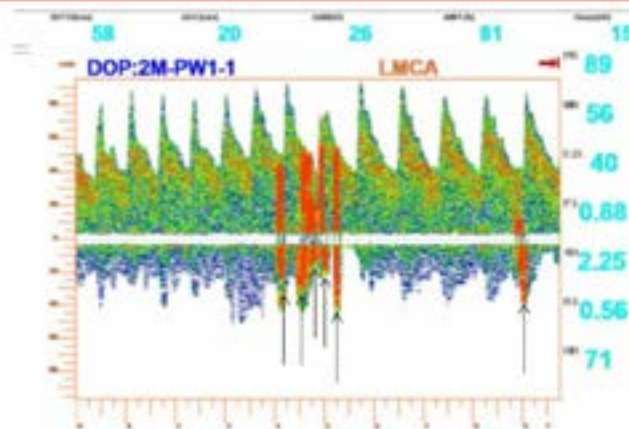
Air bubbles can be used to assess the presence of a patent foramen ovale (PFO) or septal defect. Microbubbles injected intravenously should not be demonstrated in the cerebral arterial evaluation normally.

Note the vertical spikes within the tracing. These represent microbubbles flowing through the MCA and a PFO/defect is present.



Adapted from: Percutaneous treatment of patients with heart diseases: selection, guidance and follow-up. A review. Costantini C, Lodi MA, Rapacciollo A, Prastaro M, Lombardi R, Parlo V, Parrella LS, Di Nardo C, Giannando A, Puglia R, Esposito G, Protoni E, Betocchi S - Cardiovasc Ultrasound (2012)

Positive PFO Exam



After an agitated saline injection mixed with 1cc of air, the patient must perform a 5-10 sec valsalva maneuver. If the exam is positive, emboli will be observed on the MCA tracing within a few seconds. The most common way the agitated saline can go from a vein to the MCA is the presence of a PFO. A pulmonary shunt is the only other way the bubbles can reach the MCA.

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Sickle Cell Anemia:

- Abnormally shaped RBCs produced by spleen

- Can cause increased PSV in cranial vessels and increased risk of stroke
- >200 cm/s is abnormal for terminal ICA and MCA and indicate increased risk of stroke in children

Intraoperative Monitoring:

- Most studies use depth of 50-55mm to evaluate M1
- Can be performed at short intervals or continuously using headband
- Evaluating the patient for microemboli from the procedure
- Advantages of TCD monitoring over Angiography during surgery include noninvasive, provides flow velocity information, no radiation exposure to patient or physician, does not require contrast, more cost effective

TCD Monitoring Used During:

1. Carotid endarterectomy
 - Carotid artery clamped during procedure and ipsilateral MCA evaluated for collateral flow
 - Used to monitor the presence of cerebral emboli and to help determine if a shunt is necessary during the procedure
 - Pre-op evaluation is done to find the appropriate transcranial "window" and the headgear is fitted for a trial
 - Unilateral monitoring of the ipsilateral MCA is performed
 - The Doppler evaluation begins just before the procedure starts to establish baseline flow
 - If MCA flow present >10cm/s after clamping, adequate collateral flow is present
 - Blood flow is monitored closely to identify and count any embolic signatures that may occur during the dissection, endarterectomy, and closing phase of the surgery, as well as, in the recovery room
 - Technique has helped to revise surgical techniques
1. Open heart surgery with bypass
 - Used to assess cerebral flow during extracorporeal bypass which can lead to hyperperfusion of the cerebral vessels
 - Bilateral MCA monitoring performed
 - Microemboli monitoring used to predict postoperative stroke
1. ICU Monitoring
 - Used to monitor critically ill patients; head injury, suspected intracranial hemorrhage, cervical artery dissection, hydrocephalus, intracranial and extracranial occlusive disease
 - Bilateral MCA monitoring performed
1. Brain Death
 - Used to evaluate the intracerebral flow to confirm brain death

- Waveforms demonstrate a characteristic to and fro pattern due to the significant increase in resistance of the cerebral tissues
- Peaked MCA waveform with moderate flow reversal in diastole
- 1. AV Fistula and Aneurysm
 - Requires TCCD for accurate identification
 - Color flow turbulence and visual assessment of Doppler cursor placement aid in evaluation of these defects

Cerebrovascular “Most Commons”		Page 3
The most common congenital anomaly of the Circle of Willis is the absence of one or more communicating arteries		
The most common indication for TCD is suspected vasospasm/vasoconstriction caused by subarachnoid hemorrhage		
Time averaged maximum velocity (TAMV) most commonly used for TCD evaluation		
Stenosis in the intracranial arteries is most common in the MCA, siphon and terminal ICA		
Anterior Communicating Artery is the most common collateral pathway for ICA disease		
ACoA is the most common location for intracranial aneurysm PCoA is the 2nd most common location for intracranial aneurysm		
Intracranial Arteriovenous Malformations are most common in men		

Other Cerebrovascular

Evaluation Techniques

Angiography:

- Considered the gold standard exam in arterial evaluation
- Radiography exam that uses iodinated contrast to evaluate the lumen characteristics of vessels
- Seldinger technique used to insert a catheter into the common femoral artery
- The brachial artery is an alternative access point
- The catheter is advanced through the aorta to the carotid arteries for carotid and circle of Willis evaluation and to the subclavian arteries for vertebrobasilar and circle of Willis evaluation
- Contrast opacifies the vessel lumen in normal vessels
- Areas with an absence of contrast indicate the presence of narrowing/obstruction
- Preferred method for the evaluation of fibromuscular dysplasia
- Used to estimate diameter stenosis and this value can be compared with diameter stenosis information obtained from ultrasound evaluation

Limitations:

- Invasive procedure causes increased risk to patient
- Contrast allergy
- Radiation exposure
- Single plane evaluation causes overestimation or underestimation of stenosis
- Can only estimate diameter stenosis (no area stenosis)
- Does not offer information on flow velocity
- Cannot demonstrate areas of turbulent flow

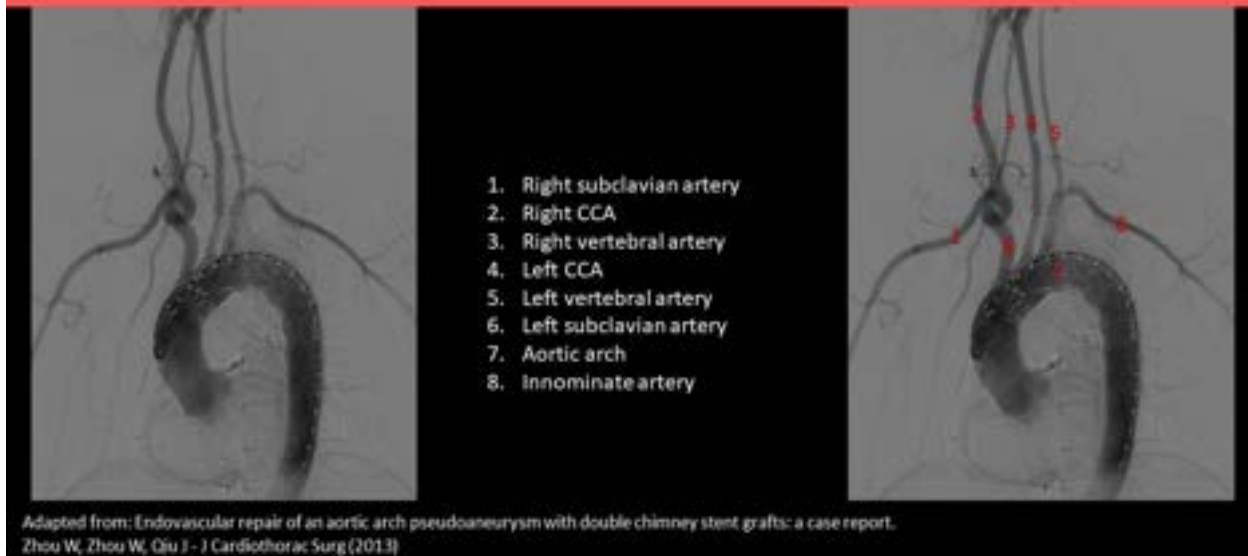
Magnetic Resonance Angiography (MRA):

- Uses radio frequency waves to produce images in multiple planes
- Quantifies blood flow, evaluates bones and tissues
- Tends to overestimate stenosis, may require contrast and costly
- US Advantage: Doppler flow evaluation

Computed Tomography Angiography (CTA):

- Uses radiation to evaluate cerebral infarcts, aneurysms, AV malformations and hemorrhage
- Can only image one plane, difficult to see stenosis
- NOTE: New technology allows for 3D reconstruction of the carotids following a contrast exam
- US Advantage: Doppler flow evaluation

Angiography



Angiography

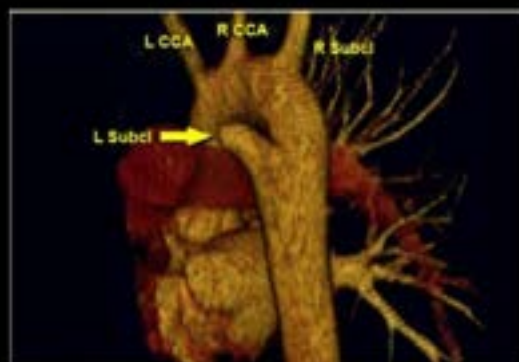


Note the stenosis at the origin of the left common carotid artery and the total obstruction of the subclavian artery (absent).

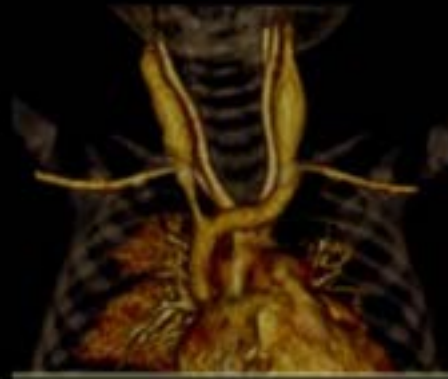
Adapted from: Bronchiectasis and hoarseness of voice in takayasu arteritis: a rare presentation.
Nid Perera G, C Jayasinghe A, D Dias L, Kulatunga A - BMC Res Notes (2012)

CT - 3D Reconstruction

Aortic Arch



Carotids, Arch, Jugular Veins and SVC



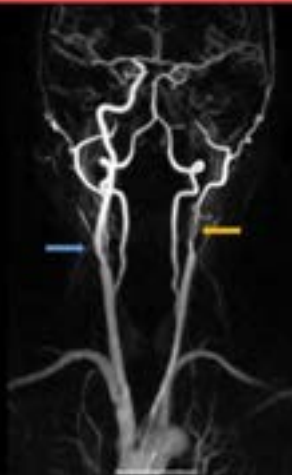
Marilyn J. Siegel and Robin Smithuis; www.radiologyassistant.nl

CT- 3D Reconstruction



Martin J. Siegel and Robin Smithius
www.radiologyassistant.nl

MR Angiography



Note the occlusion of the left internal carotid artery (yellow arrow) and there is a 41-59% stenosis of the right internal carotid artery (blue arrow).

Adapted from: Spontaneous recanalization of complete internal carotid artery: a clinical reminder.
Som S, Schanzer B - J Surg Tech Case Rep (2010)

MR Angiography



Note the stenosis in the right MCA and the right renal artery

Adapted from: Metabolic syndrome predicts vascular changes in whole body magnetic resonance imaging in patients with long standing diabetes mellitus. Findeisen HM, Weckbach S, Stark RG, Reiser MF, Schoenberg SO, Parhofer KG - Cardiovasc Diabetol (2010)

Carotid Artery Disease

Carotid Stenosis:

Doppler Evaluation/Characteristics:

- As area of vessel decreases, flow velocity increases
- Flow changes direction as it enters and leaves stenosed portion causing turbulence and spectral broadening
- The most common reason for underestimation of arterial stenosis is improper sample volume location
- Pre-Stenotic Zone
 - Increased resistance and resistive index
 - Flow velocity may be damped with loss of diastolic flow
 - Short acceleration time
- Stenotic Zone

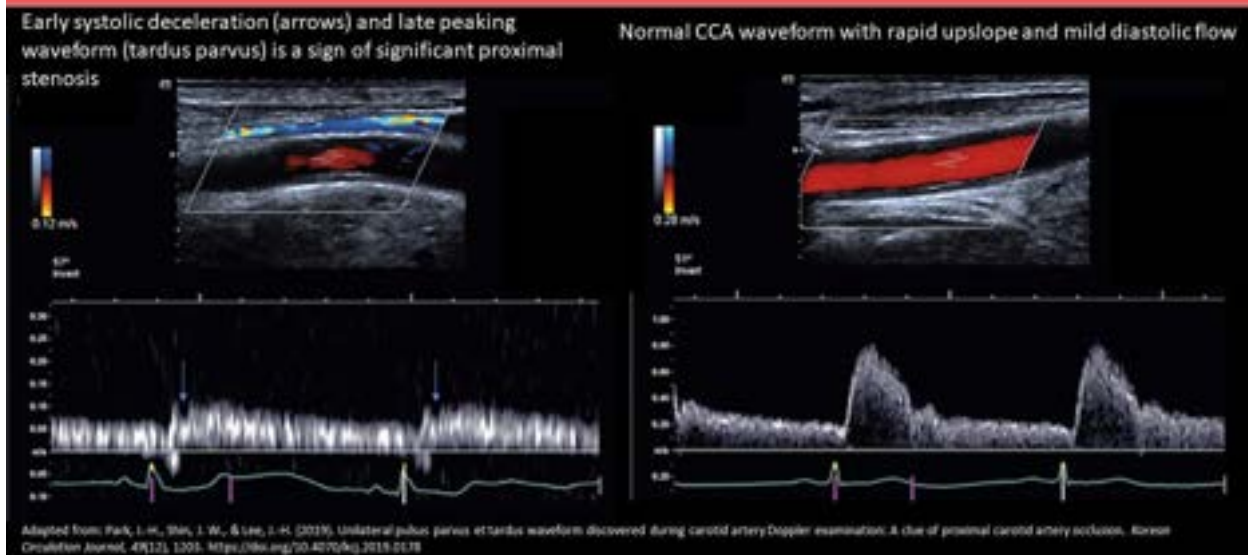
- High peak systolic velocity
- Increased diastolic flow
- Decreased pressure
- Spectral broadening
- The highest velocity can be detected within the stenosis or just distal to the stenosis
- Post-Stenotic Zone
 - Turbulence
 - Low resistance flow
 - Increased diastolic flow
- Distal to a *critical* ICA stenosis
 - Diastolic and systolic flow will be low velocity with a blunted waveform
 - Slow acceleration time
 - Tardus parvus waveform

CCA Waveforms:

- CCA normally demonstrates low resistance waveform with mild diastolic flow
- More than a 30cm/s difference in PSV between the right and left CCA can indicate an issue
 - Single CCA with reduced velocity, slower upstroke and holodiastolic flow = stenosis proximal to the point of insonation
 - Single CCA with reduced velocity, quick upstroke and minimal diastolic flow reversal = occlusion distal to the point of insonation
 - Single CCA with elevated velocity = tortuosity on that side or compensatory flow due to obstruction in the other carotid system
- Bilateral CCA waveforms should be symmetric with minimal difference in amplitude
 - Bilateral CCA flow abnormalities indicate a physiologic cause for changes
 - Increased velocity = overestimated carotid stenosis
 - Systemic HTN
 - Increased cardiac output
 - Decreased hematocrit (thin blood); can be due to anticoagulation medication; Heparin, Coumadin, Warfarin are blood thinners that will reduce the hematocrit (viscosity) levels of the blood
 - Mild to moderate aortic stenosis
 - Decreased velocity = underestimated carotid stenosis
 - Congestive heart failure (CHF)
 - Decreased cardiac ejection fraction
 - Severe aortic valve stenosis
 - Increased hematocrit (thick blood)
 - Polycythemia vera

- Bilateral CCA with increased diastolic flow *reversal* can indicate significant aortic valve regurgitation
- Cardiac Assist Devices can cause an unrecognizable arterial signal in the bilateral carotid arteries
 - Intra-aortic balloon pump (IABP)
 - IABP assisted waveforms have 2 systolic peaks for each pulse
 - The first peak indicates the left ventricular contraction
 - The second peak indicates the balloon inflation
 - EDV cannot be determined in the patients with an IABP
 - Flow reversal is commonly seen after the balloon deflation
 - If possible, turn off the IABP during carotid evaluation
 - If this is not possible, velocity from the first peak should be used as the PSV
 - Left Ventricular Assist Device (LVAD):
 - Doppler waveforms demonstrate low-velocity flow with delayed upstroke and loss of pulsatility
 - Carotid Doppler waveform morphology is highly variable among patients with LVADs and is very dependent on pump settings
 - It is very important to document the presence of a left ventricular assist device or intra-aortic balloon pump on the preliminary report
 - The waveform contour cannot be evaluated in these patient

Proximal CCA Stenosis



ECA Waveforms:

- Normally high resistance, pulsatile flow with minimal to mild antegrade diastolic flow
- May demonstrate a short duration of diastolic flow reversal at the end of systole
- Peak velocity normally less than 150cm/s
- >50% ECA stenosis = velocity at stenosis at least 2X the proximal ECA velocity
- Ipsilateral ICA occlusion can cause the ECA flow to become low resistance as it becomes involved in collateral pathways
- Ipsilateral CCA occlusion can cause the ECA flow to reverse so that it can supply the ICA with flow

ICA Waveforms:

- The most accurate Doppler predictor of ICA stenosis is the end diastolic velocity
- Normal peak systolic velocity <125cm/s and end diastolic velocity <40cm/s
- As stenosis increases = PSV and EDV will increase until the level of critical stenosis is reached
- Assess the waveform for spectral broadening
- Evaluate the peak systolic and end diastolic velocities
- Calculate the velocity ratios

Velocity Ratios:

- The highest ICA velocity is usually identified in the first 3cm of the vessel
- The highest CCA velocity is usually identified in the first 3cm of the vessel

- Systolic velocity ratio (SVR) - compares the highest ICA velocity to the distal CCA velocity
- Diastolic velocity ratio (DVR) - compares the highest end diastolic ICA velocity to the end diastolic velocity in the distal CCA
- Higher ratios indicate more significant stenosis
- When a stenosis is present in the CCA, use the highest normal CCA velocity in the ratios instead of always using the distal CCA

Indirect Doppler Criteria for Significant ICA Stenosis:

- Internalization of the ECA (becomes low resistance) with flow reversal in the ophthalmic artery
- Collateral flow noted in TCD exam such as cross filling in anterior or posterior communicating arteries
- Decreased, blunted flow in one MCA

Critical Stenosis:

- Near occlusion that should be treated as a critical finding to prevent full occlusion where treatment is not an option
- Significant stenosis that demonstrates a small amount of flow through the narrowed segment
- String flow sign - thin string of color Doppler demonstrated through the tiny opening
- Flow velocity is very low with decreased pulsatility
- A bruit is NOT usually present in these patients because of the minimal flow moving through the stenosis

Sample Doppler Criteria for ICA Stenosis: *Varies per lab preference*

ICA Stenosis

University of Washington (Strandness)

% Stenosis	PSV	EDV	Waveform Characteristics
1-15%	<125cm/s	---	Spectral broadening during deceleration phase of systole
16-49%	<125cm/s	---	Spectral broadening throughout systole
50-79%	>125cm/s	<140cm/s	Marked spectral broadening
80-99%		>140cm/s	Marked spectral broadening

***>70% = ICA/CCA ratio >4.0

Adapted from Moneta GL, Edwards JM, Papanicolaou G, Hatsukami T, Taylor LM Jr, Strandness DE Jr, Porter JM. Screening for asymptomatic internal carotid artery stenosis: duplex criteria for discriminating 60% to 99% stenosis. J Vasc Surg. 1995 Jun;21(6):989-94.

ICA Stenosis

Society of Radiologists in Ultrasound (SRU)

% Stenosis	% Plaque Visible	PSV	EDV	ICA/CCA Ratio
<50%	<50%	<125cm/s	<40cm/s	<2
50-69%	>50%	125-230cm/s	40-100cm/s	2-4
>70% (less than near occlusion)	>50	>230cm/s	>100cm/s	>4
Near Occlusion	Visible	High, low or undetectable	Variable	Variable

Adapted from Grant E G, Benson C B, Moneta G L, et al. Carotid artery stenosis: Gray-scale and Doppler US diagnosis—Society of Radiologists in Ultrasound Consensus Conference. Radiology 2003;229:340-346.

Velocity Ratios

Systolic Velocity Ratio

$$\frac{\text{PSV ICA}}{\text{PSV CCA}}$$

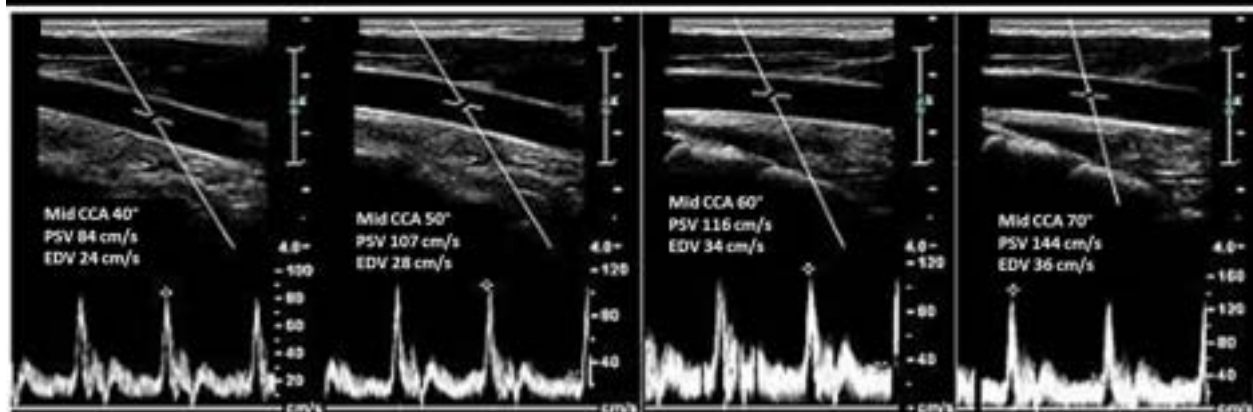
Diastolic Velocity Ratio

$$\frac{\text{EDV ICA}}{\text{EDV CCA}}$$

Estimating Carotid Stenosis

Overestimated	Underestimated
Systemic HTN	Reduced systemic blood pressure
Compensatory flow due to stenosis on other side	Proximal stenosis – significant aortic valve stenosis
Short area of focal stenosis	Long area of focal stenosis
Smaller vessel size	Tandem stenosis or eccentric stenosis
Elevated heart rate	Critical stenosis (string flow)
Measuring a compensatory heart beat with arrhythmias	Reduced cardiac output

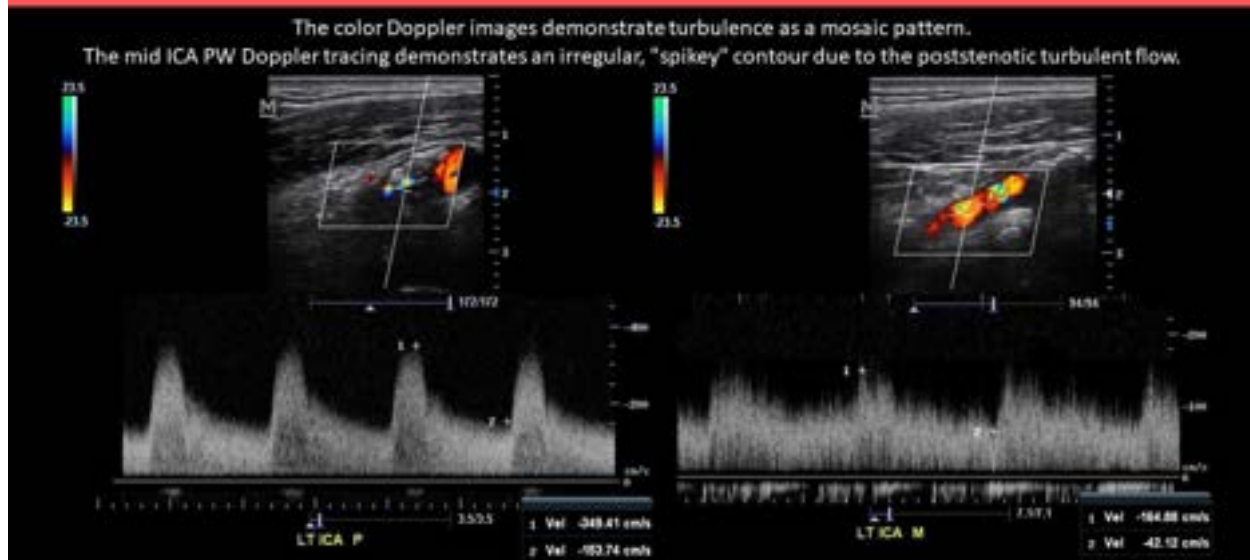
Effect of Cursor Angle on Velocity



It is important to use a consistent cursor angle setting when performing serial exams. Note the difference in velocities obtained from the same vessel using different Doppler angles. 60° is recommended for carotid exams.

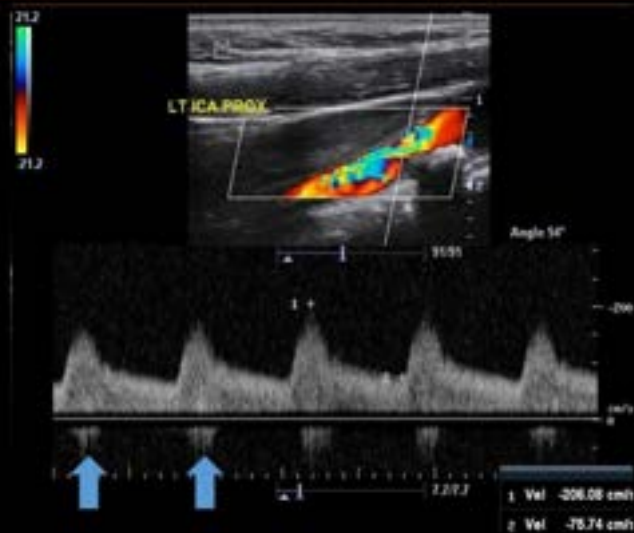
Adapted from: Standardized ultrasound evaluation of carotid stenosis for clinical trials: University of Washington Ultrasound Reading Center.
Beach KW, Bergelin RO, Leotta DF, Primozich JF, Sevareid PM, Stutzman ET, Zierler RE - Cardiovasc Ultrasound (2010)

Poststenotic Turbulence

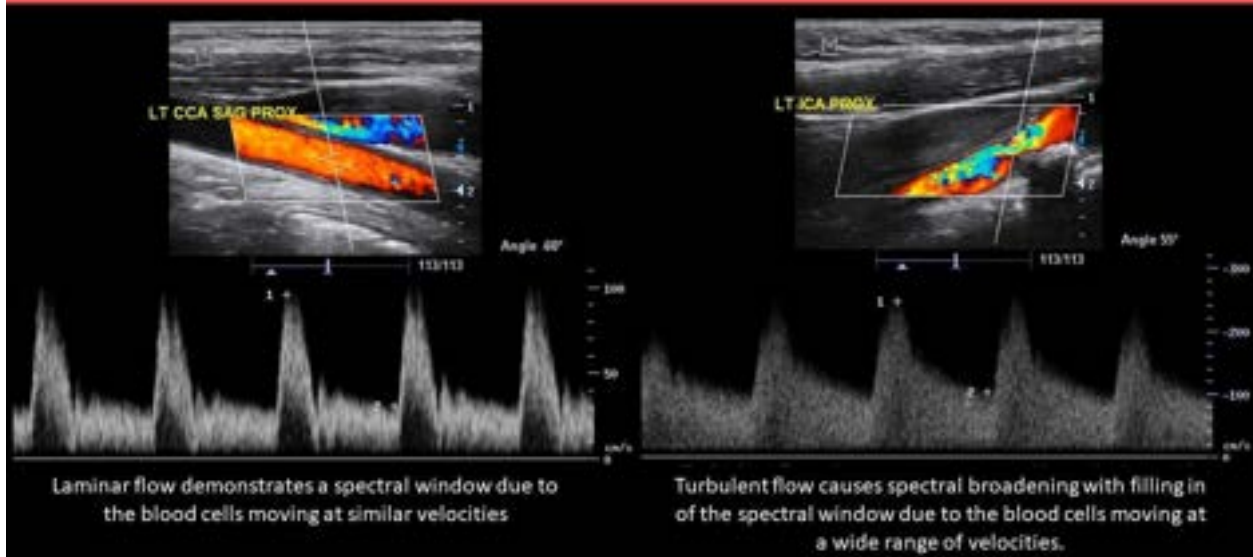


Post-Stenotic Turbulence

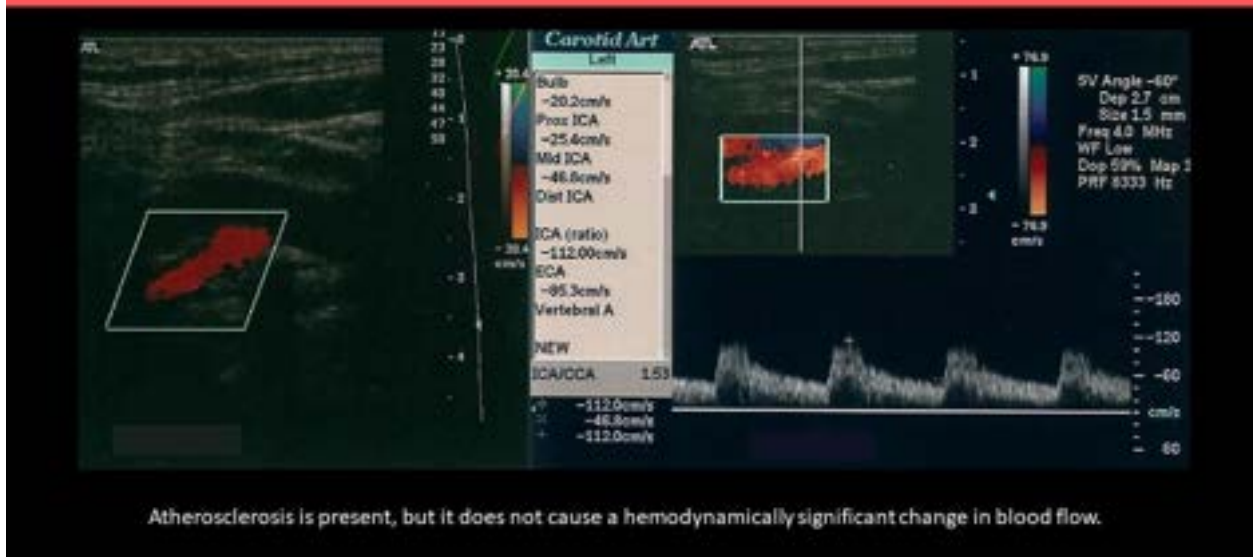
Post-stenotic turbulence can cause flow to be demonstrated below the baseline (blue arrows) in cases of significant stenosis.



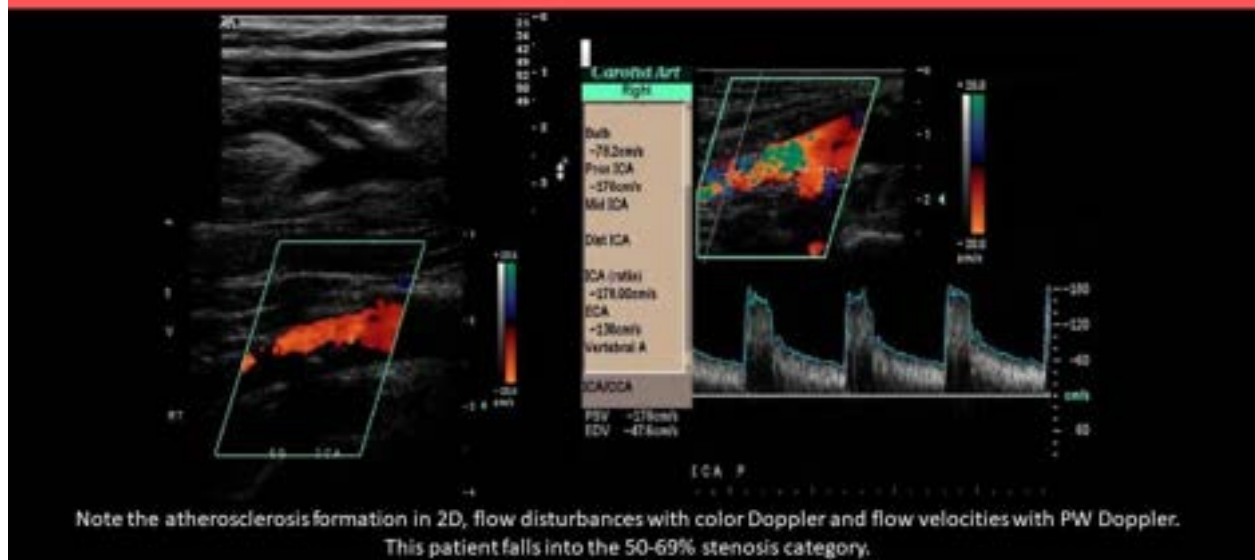
Spectral Broadening



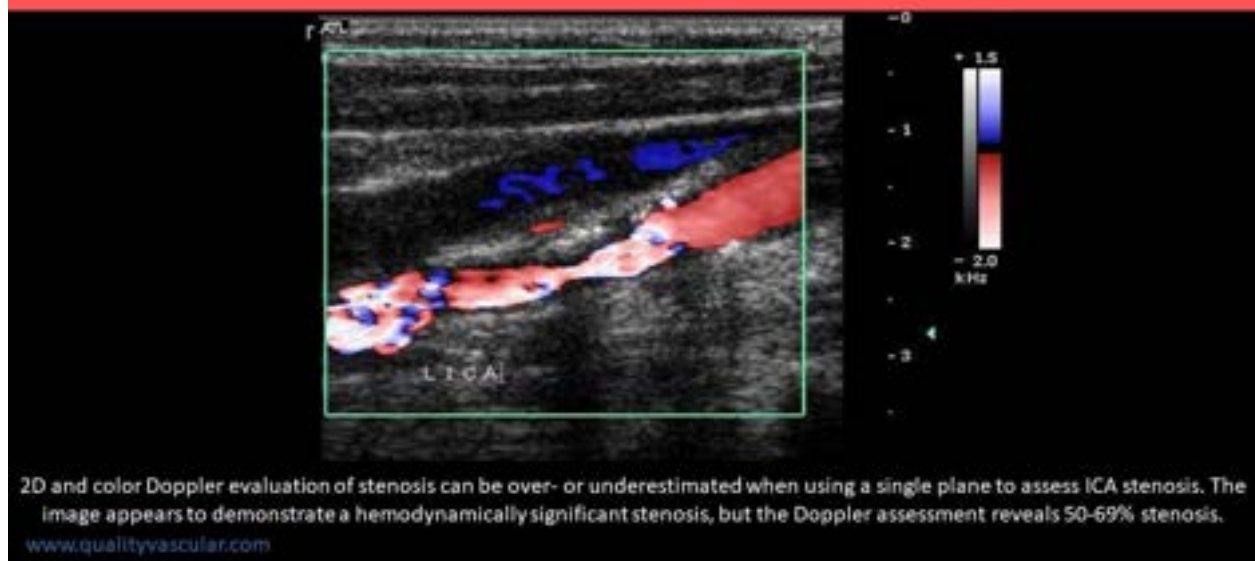
40 – 50% ICA Stenosis



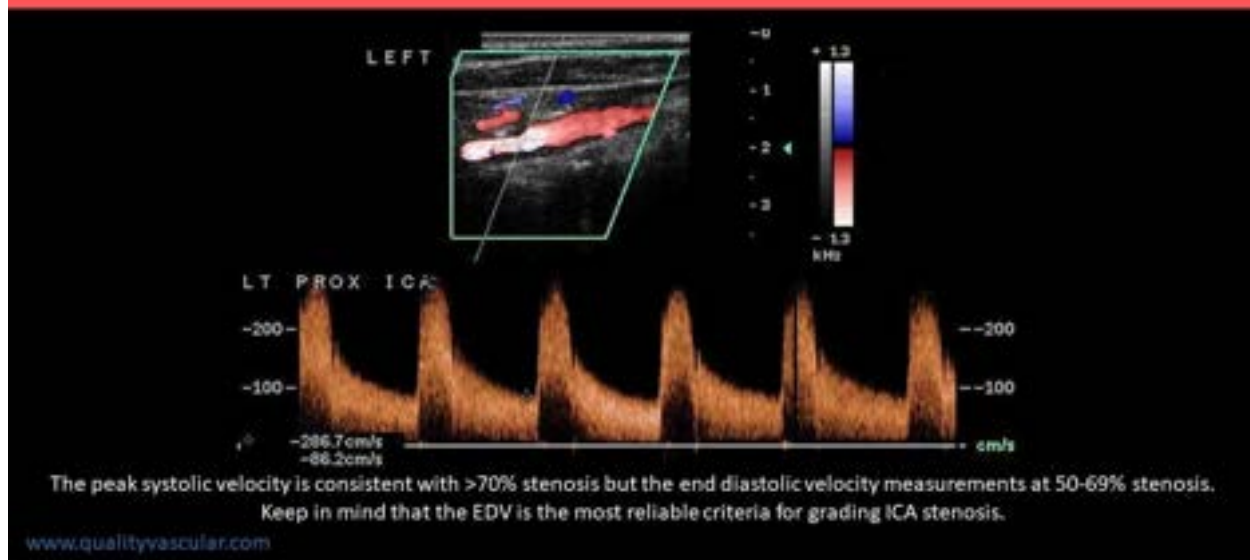
50 – 69% ICA Stenosis



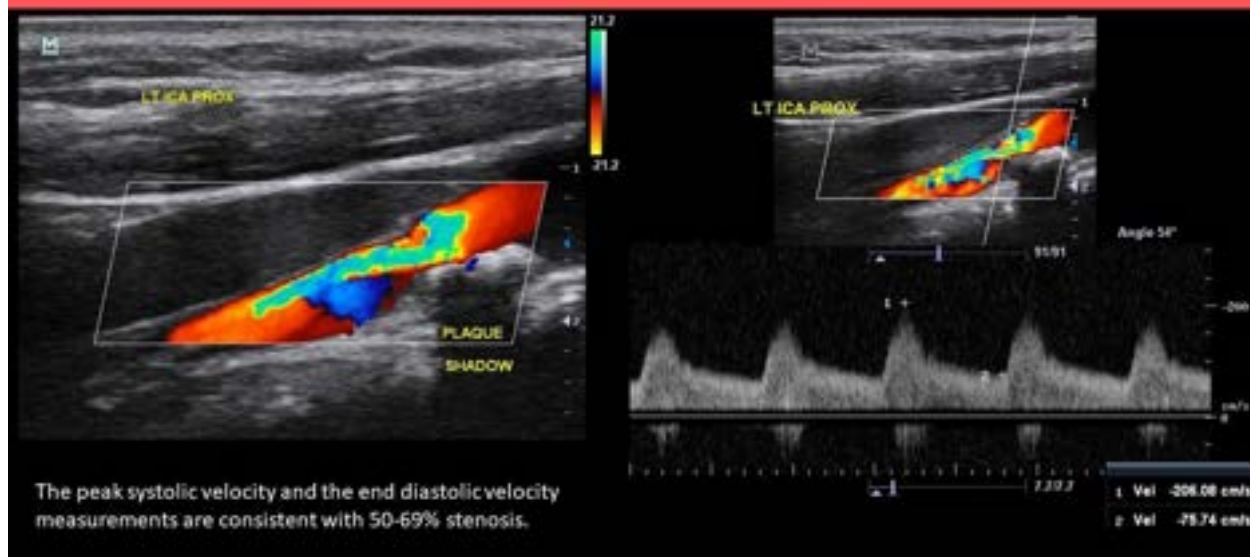
50 – 69% ICA Stenosis



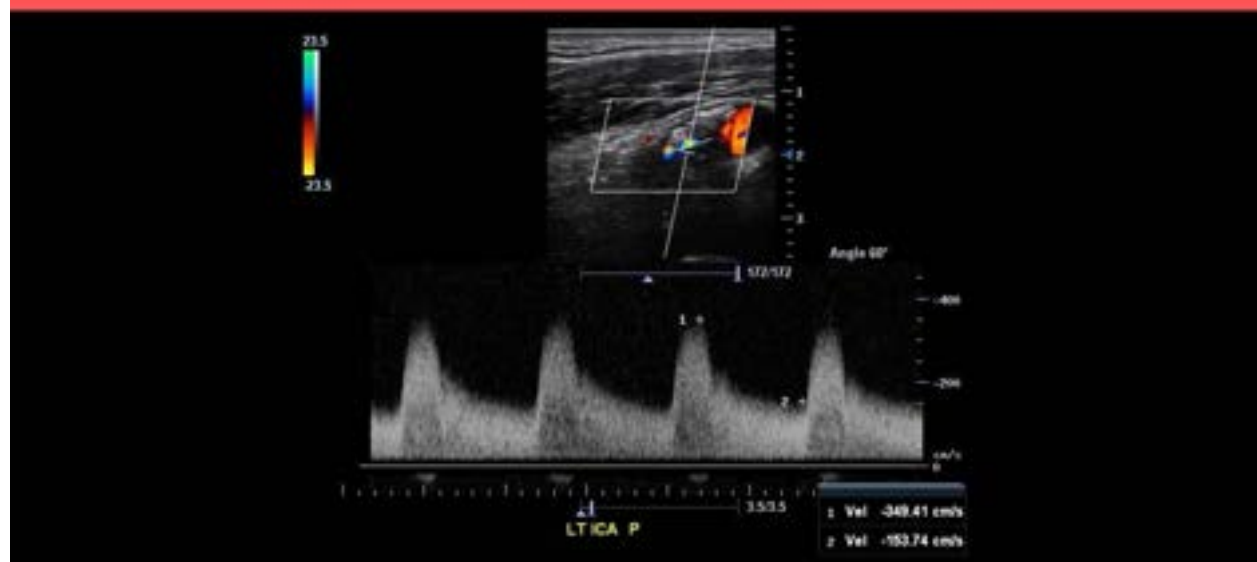
50 – 69% ICA Stenosis



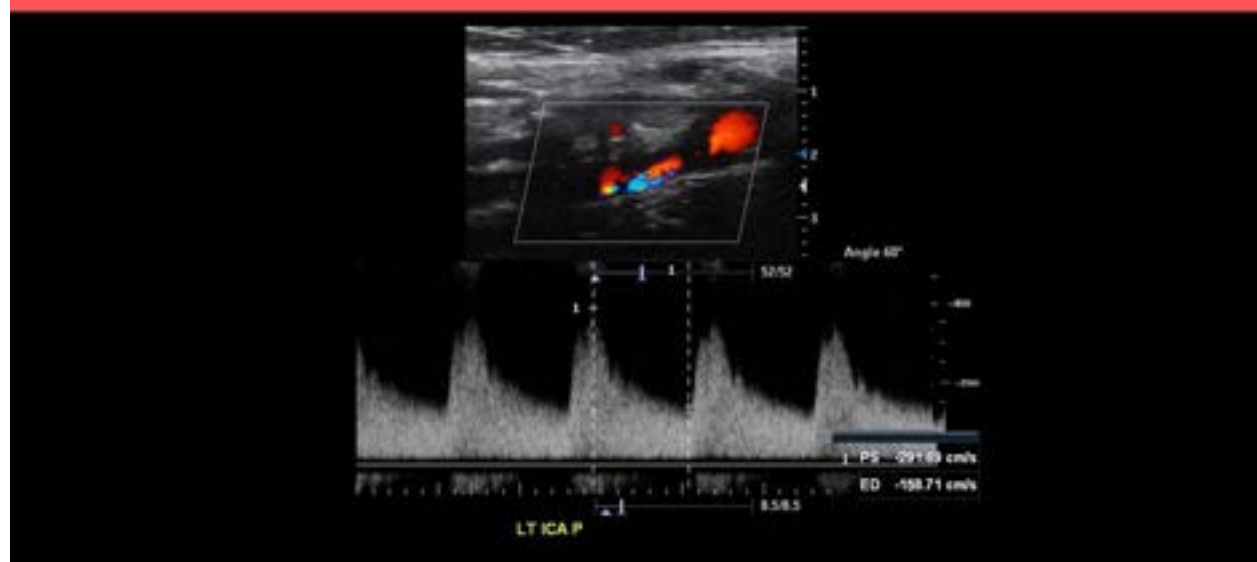
50 – 69% ICA Stenosis



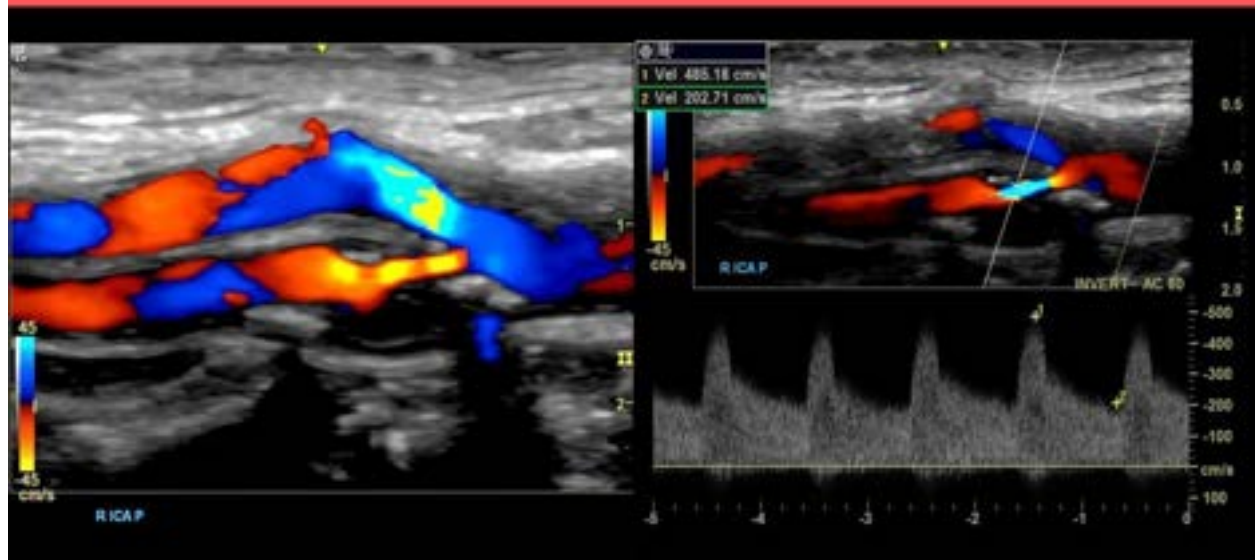
$\geq 70\%$ ICA Stenosis



$\geq 70\%$ ICA Stenosis

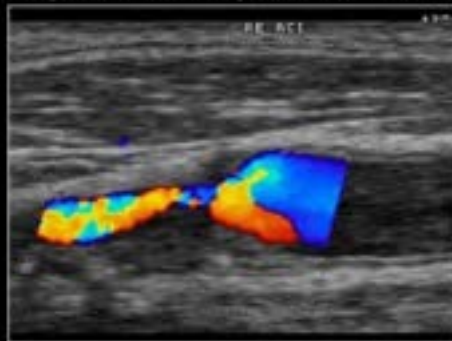


$\geq 70\%$ ICA Stenosis

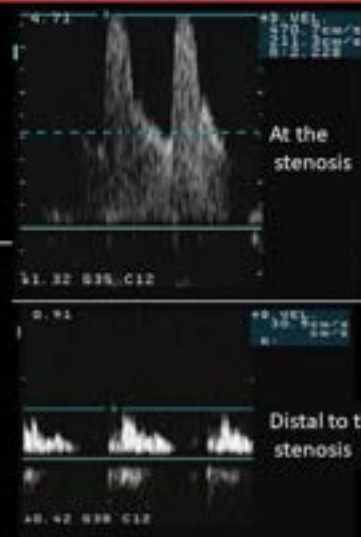


$\geq 70\%$ ICA Stenosis

Note the significant amount of diastolic flow at the stenosis. The flow distal to the stenosis is blunted with no diastolic flow. This is a key characteristic of a proximal critical stenosis.

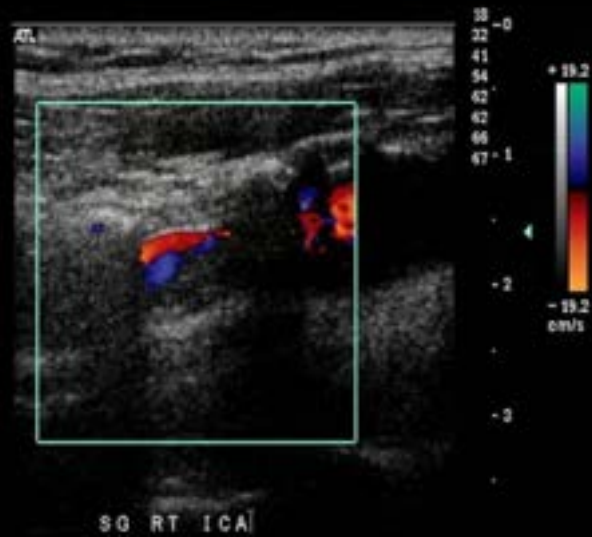


Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

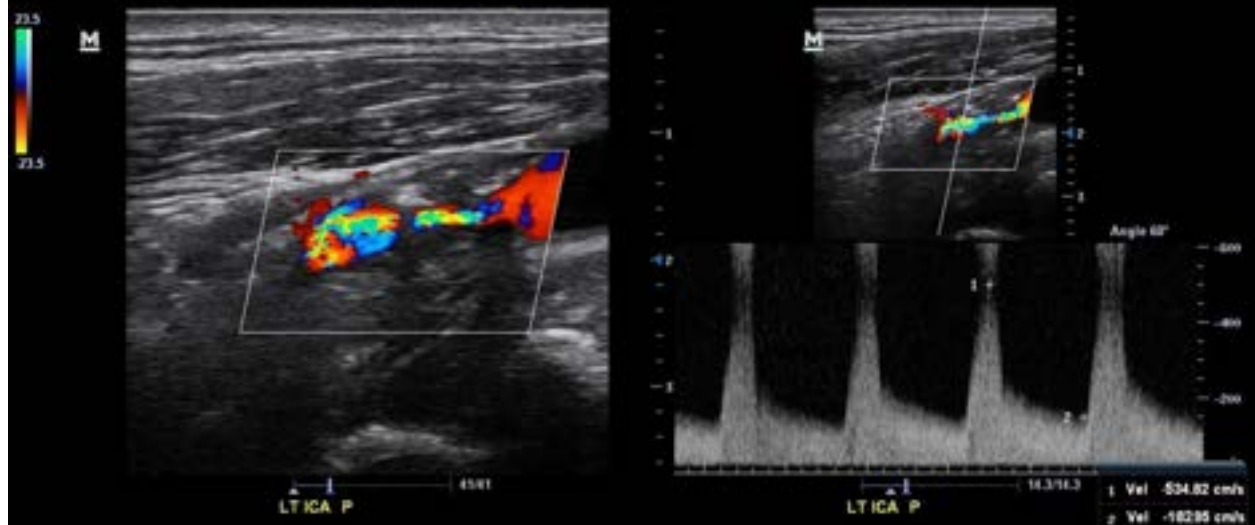


Critical ICA Stenosis

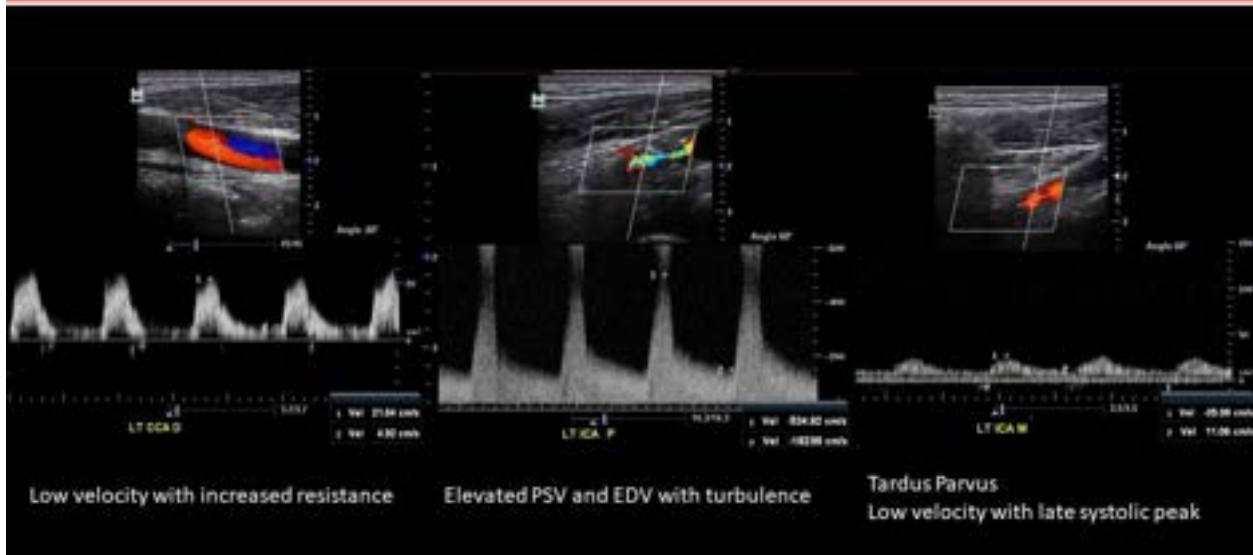
Note the significant atherosclerosis formation and the sliver of color. The string sign is associated with critical stenosis.



Critical ICA Stenosis



Critical ICA Stenosis



THERE IS AN IMPORTANT DIFFERENCE BETWEEN 50-70% STENOSIS AND 70-99% STENOSIS. A HEMODYNAMICALLY SIGNIFICANT STENOSIS REQUIRES INTERVENTION TO REDUCE THE RISK OF A STROKE, INFARCT OR EVEN DEATH. AT 70-99% STENOSIS, THE BENEFIT OF THE SURGERY OUTWEIGHS THE RISK OF PERFORMING IT.

Vertebral Artery Waveforms:

- Waveform assessed for flow direction, velocity and resistance
- Normally demonstrates a low to medium velocity, low resistance waveform
- Variations in flow velocity between the right and left side are normal, the left side is usually dominant
- Stenosis usually occurs in the proximal artery and this segment is not normally evaluated on a duplex exam
- Loss of diastolic flow indicates a distal occlusion
- Early systolic deceleration is a sign of subclavian artery stenosis

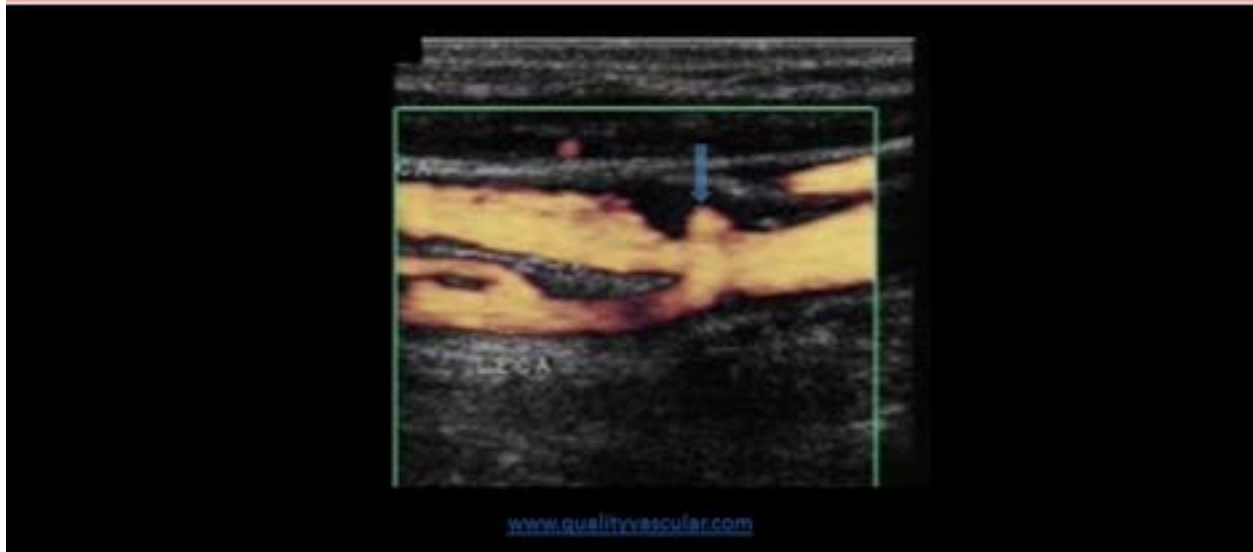
Subclavian Artery Waveforms:

- Normally a high resistance, triphasic waveform
- Right and left subclavian artery waveforms should be compared for expected similarity
- Biphasic or monophasic waveforms are associated with stenosis or occlusion

Ulceration of Plaque:

- Very difficult to accurately identify on sonography
- Formation of a crevice within the atheroma deposit
- Leads to the formation of a protruding edge of the atheroma that could be sheared off by the blood flow becoming an embolism
- Soft plaque has the greatest risk of embolization due to its composition and decreased adhesive levels
- Complex and calcific plaque becomes increasingly fibrous and more easily adheres to the vessel wall

Ulceration



Carotid Occlusion:

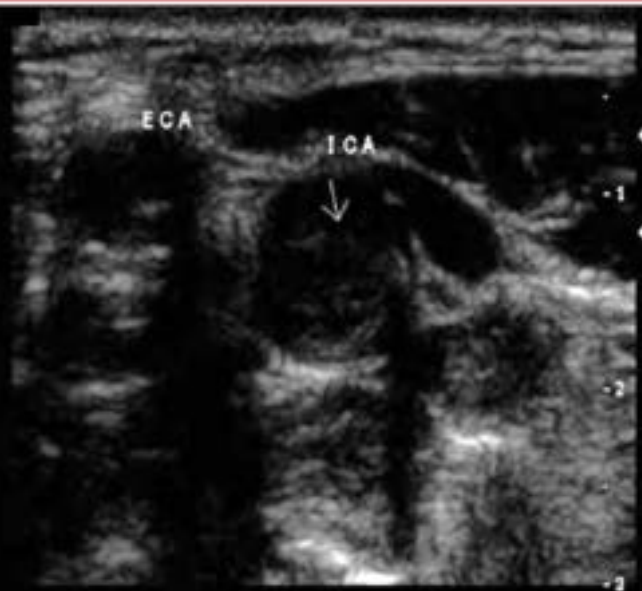
- Risk of a future stroke drops significantly when the artery is totally occluded
- Echogenic material fills the vessel lumen; mixture of atheroma and thrombus
- Horizontal motion of the vessel wall instead of wall pulsation with internal flow
- No flow identified with Color or PW Doppler techniques
- Increased resistance in the vessel proximal to the occlusion
- Immediately proximal to the occlusion, flow patterns are very low velocity and high resistance; called thump flow, stump flow or thud flow
- Use color Doppler with reduced scale, rejection and filter settings, increase Doppler gain
- Power Doppler should be used to assess the vessel for very low velocity flow
- PW Doppler - always move sample volume around within vessel, increase sample size and try different probe locations to confirm an occlusion is present

Diagnostic Criteria for Carotid Occlusion:

- Absent flow in the distal ICA
- Power Doppler is preferred for confirmation of occlusion due to its flow sensitivity
- High resistance, peaked waveform with absent diastolic flow or diastolic flow reversal proximal to occlusion
- Decreased, blunted flow in ipsilateral MCA
- Decreased resistance with increased diastolic flow in the ipsilateral ECA
- Evidence of collateralization

ICA Occlusion

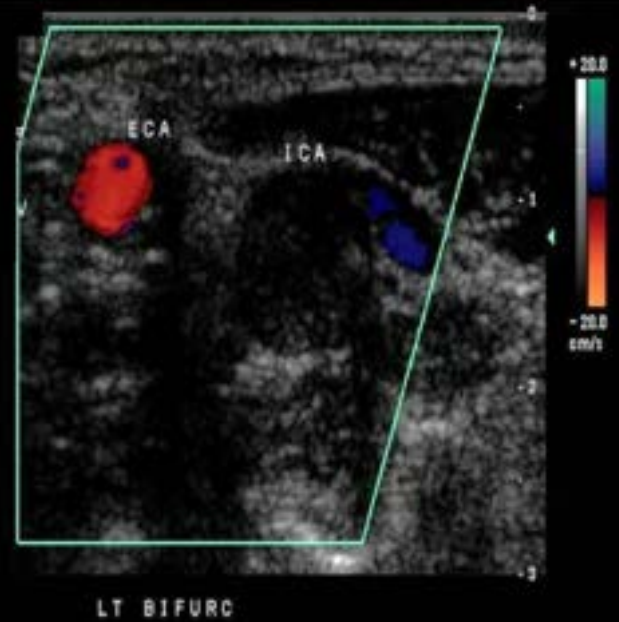
Note the echogenic material within the vessel lumen. The echogenicity of the plaque/thrombus and the overall atrophy of the vessel indicates a chronic occlusion.



LT BIFURC

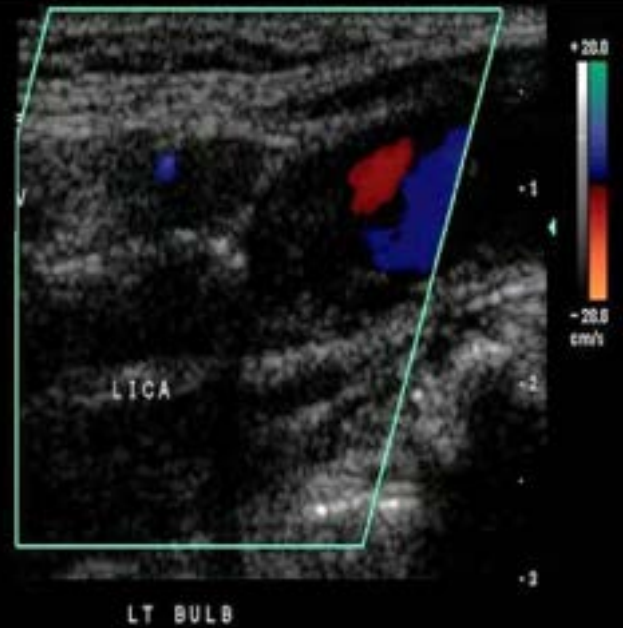
ICA Occlusion

Note the echogenic material within the vessel lumen. No flow is identified in the ICA using color Doppler imaging.



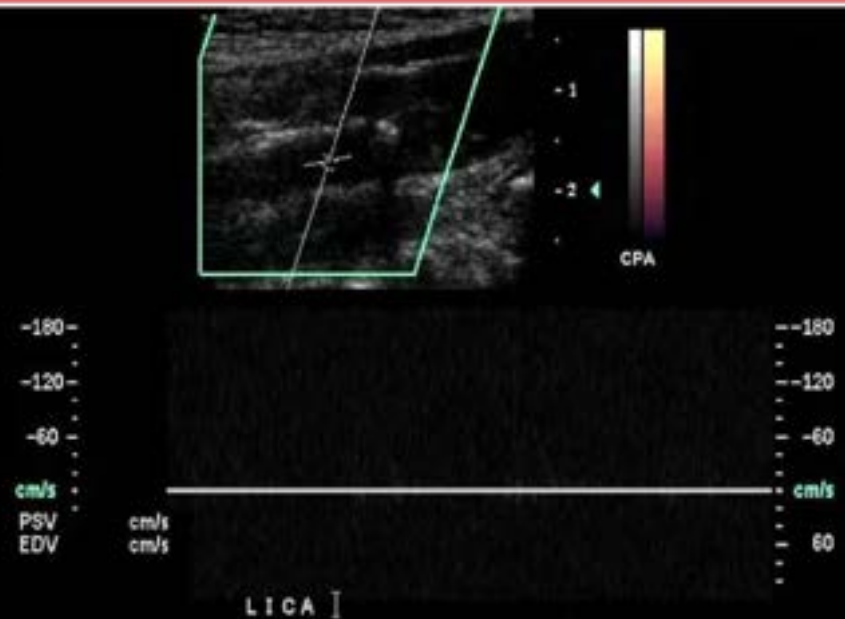
ICA Occlusion

Note the echogenic material within the vessel lumen. No flow is identified in the ICA using color Doppler imaging.



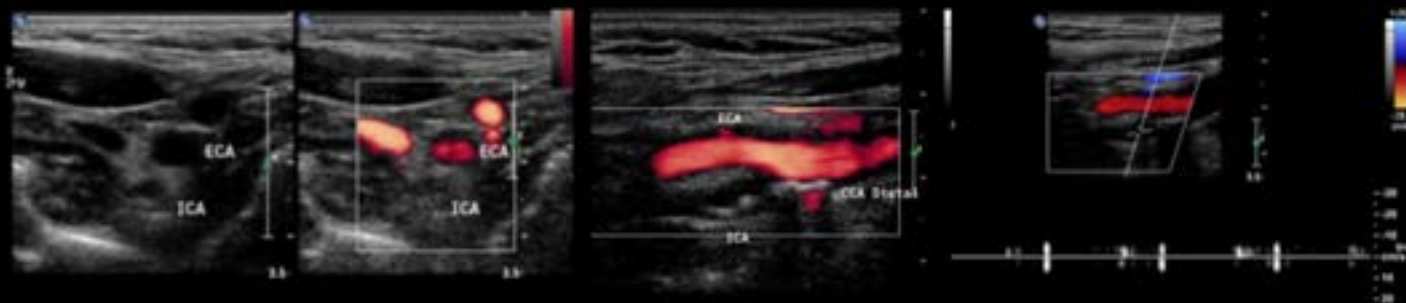
ICA Occlusion

Power Doppler and PW Doppler should also be used to document an occlusion.



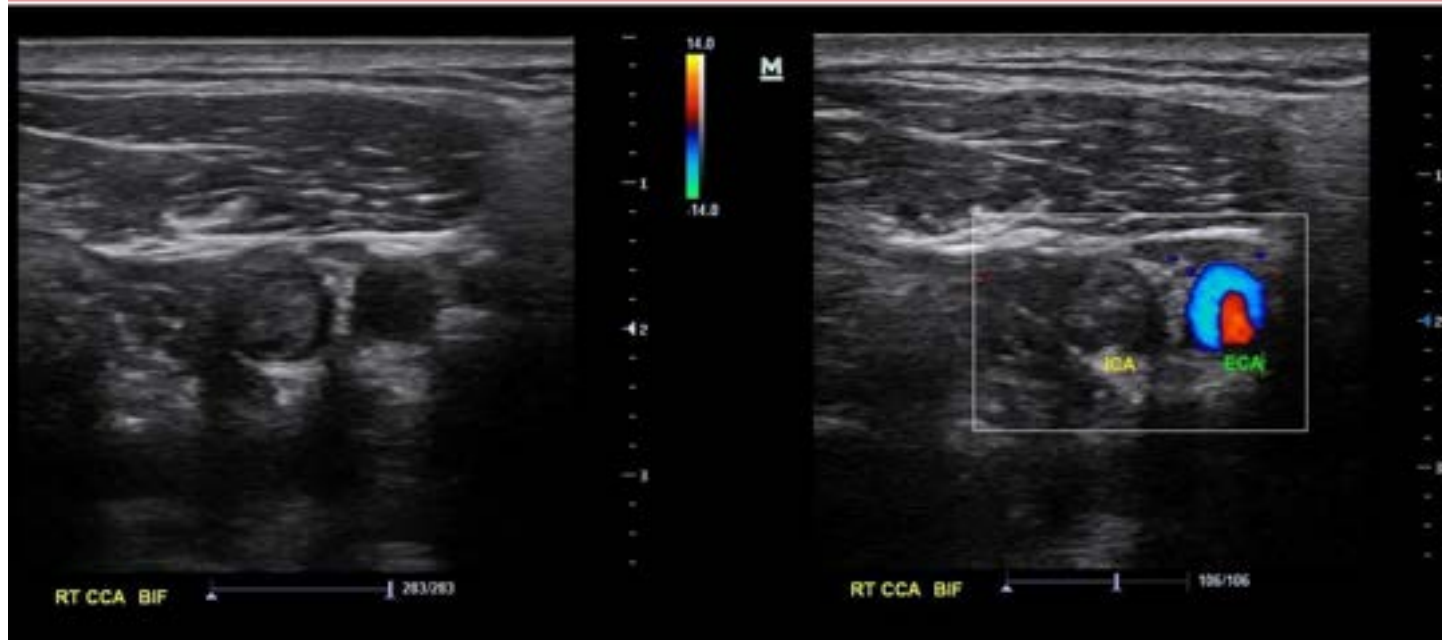
ICA Occlusion

When occlusion is suspected, use all available techniques to evaluate the artery for flow.



Power Doppler is more sensitive than color Doppler and should be used to look for signs of string flow.

ICA Occlusion

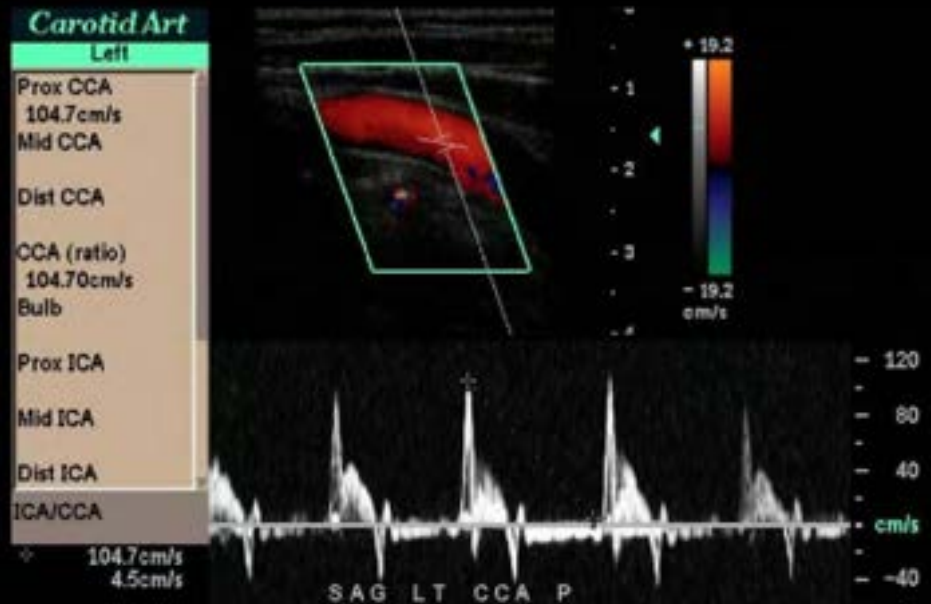


ICA Occlusion



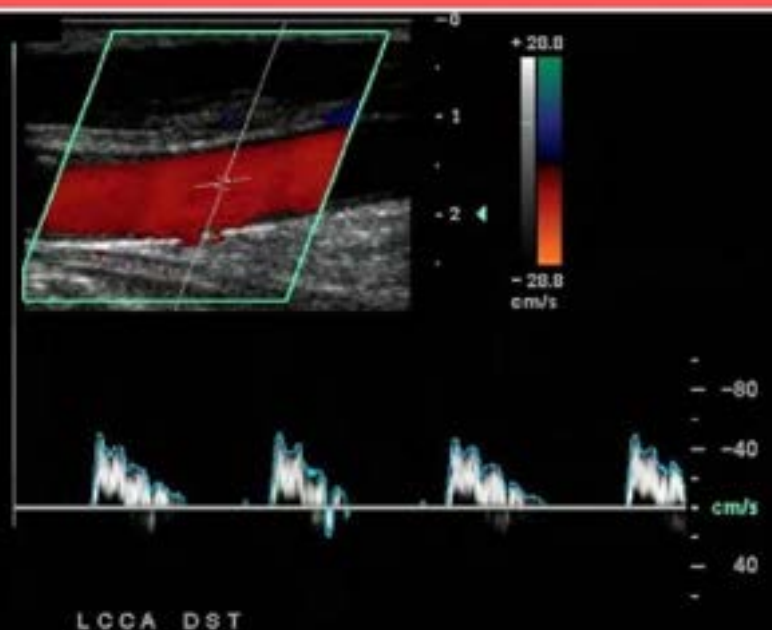
ICA Occlusion

The CCA flow will be affected by an ICA occlusion. Note the triphasic appearance of the CCA flow demonstrated. The flow will demonstrate an increase in resistance and, in most cases, flow reversal in diastole.

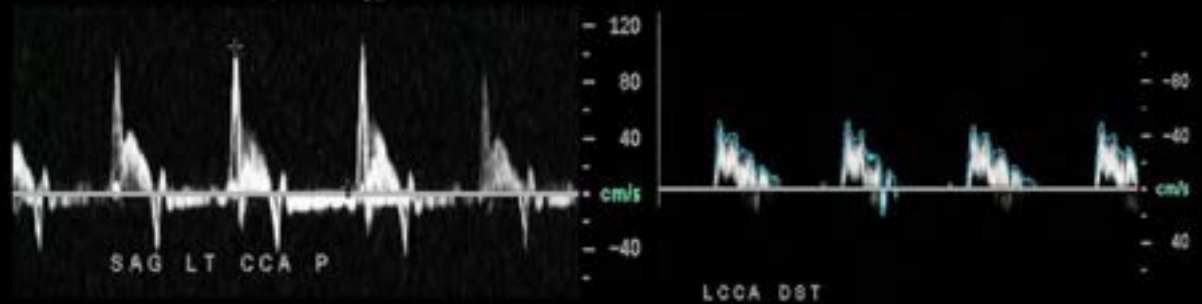


ICA Occlusion

Complete loss of diastolic flow in the CCA indicates a distal occlusion.

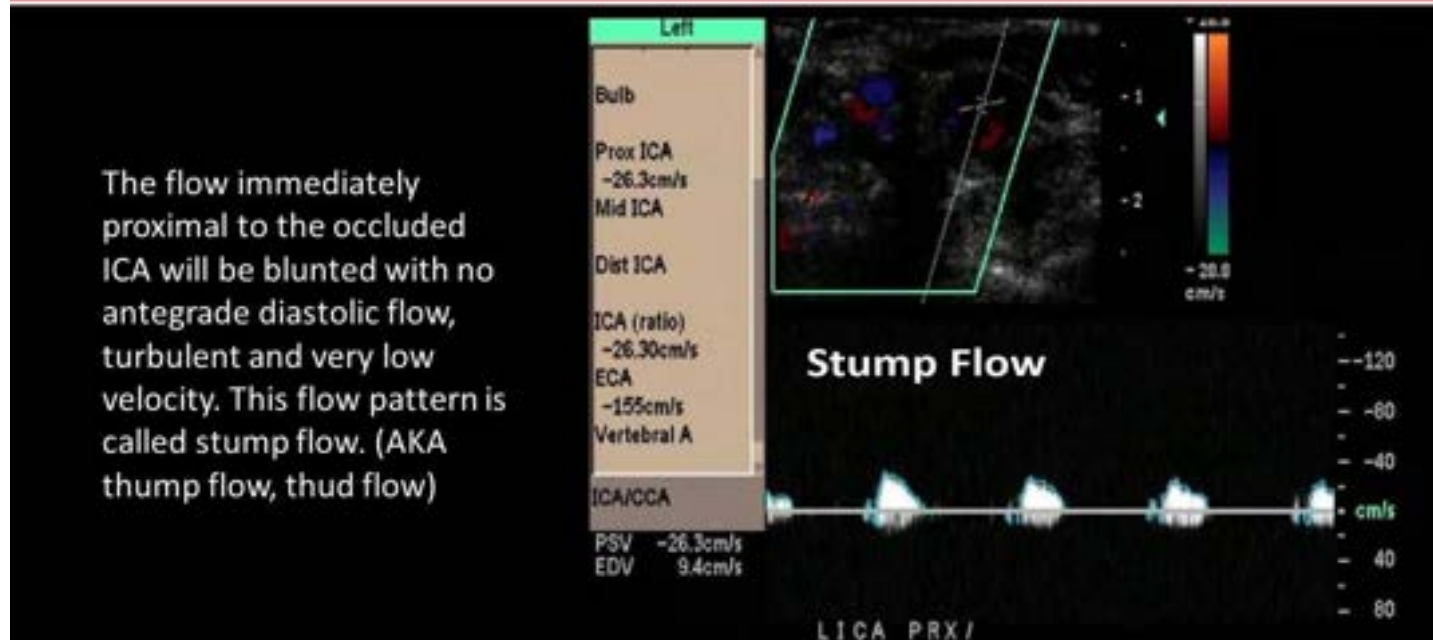


ICA Occlusion



An ICA occlusion is present distally in the carotid system. The proximal CCA demonstrates a high resistance waveform with increased velocity. The distal CCA demonstrates lower velocity, high resistance flow.

ICA Occlusion



Compensatory Flow:

- When there is a significant stenosis or an occlusion in one carotid system, the carotid system on the other side can aid in feeding both sides of the brain
- Flow velocities in the contralateral carotid system will be elevated
- For example, if there is a left CCA occlusion, the velocities in the right CCA, ECA and ICA can increase with compensatory flow
- If there is also a carotid stenosis on the right side, the stenosis can be overestimated using velocity criteria because the velocities are falsely elevated due to compensatory flow
- Signs that compensatory flow is causing an overestimation of stenosis:
 - 2D plaque assessment indicates less stenosis than the velocities obtained
 - The CCA velocities are elevated on the side with the stenosis and diminished on the side with the occlusion
 - The ICA/CCA ratio indicates a lower level of stenosis than the velocity criteria

Collateral Formation:

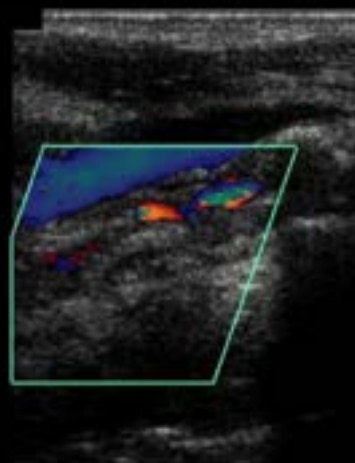
- Cross Over Collateralization
 - The information below refers to scanning through the transtemporal window

- Flow moves from the contralateral ICA into the contralateral ACA and then through anterior communicator into the ipsilateral ACA
- Flow toward the transducer in the ACA indicates ipsilateral ICA obstruction
- Increased peak velocity ($>150\text{cm/s}$) in the contralateral ACA
- EX: LT ICA Occlusion
 - flow from right ICA enters the circle of Willis and travels to the right ACA and then the anterior communicating artery to reach the left ACA
 - flow in the left ACA is normally moving away from the transducer, it will be moving toward the transducer in cross over collateralization
 - the right ACA can demonstrate increased flow velocity, as it compensates for the occlusion on the left side
- Right ICA occlusion = reversed right ACA and increased velocity in the left ACA; Left ICA occlusion = reversed left ACA and increased velocity in the right ACA
- Carotid compression test of the unaffected side causes a significant decrease or absent flow in the MCA on the side with the ICA occlusion
- ECA to ICA Collateralization
 - Flow from the superficial temporal/supraorbital artery to ophthalmic artery
 - Causes retrograde flow ipsilateral ophthalmic artery to fill ICA
 - Ophthalmic waveform will demonstrate decreased pulsatility and increased velocity as blood flows from a small artery into a larger branch (ICA)
- Posterior to Anterior Collateralization
 - Flow from vertebrobasilar system enters PCA with flow reversal into the posterior communicating artery to fill the ICA and branches distal to obstruction
 - Flow in PCA exceeds flow in MCA on the same side
 - Carotid compression test causes increased PCA flow on the affected side
 - The occipital artery is a branch of the ECA
 - Branches of the occipital artery meet branches of the vertebral artery
 - This is a possible collateral pathway for obstruction in the vertebral system or the ICA system on the same side.
- Carotid to Vertebral Collateralization - ICA flow enters posterior communicator and then the basilar/vertebral artery on the same side of the body

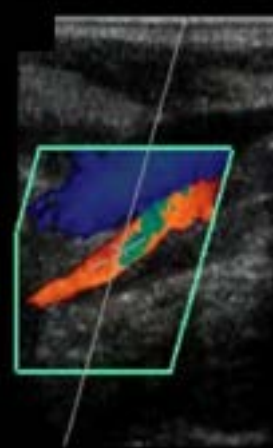
Carotid Compression Test:

- Performed by PHYSICIAN ONLY!!
- Used to evaluate the presence of collateral formation
- Compress unilateral CCA for approximately 5 minutes, if no symptoms occur, cerebral collaterals are present

ECA to ICA Collateralization

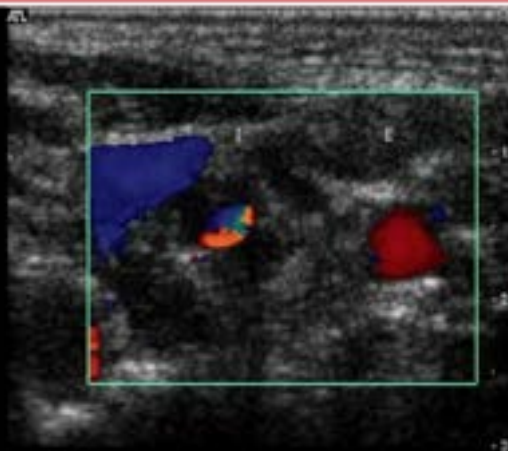


RT ICA P



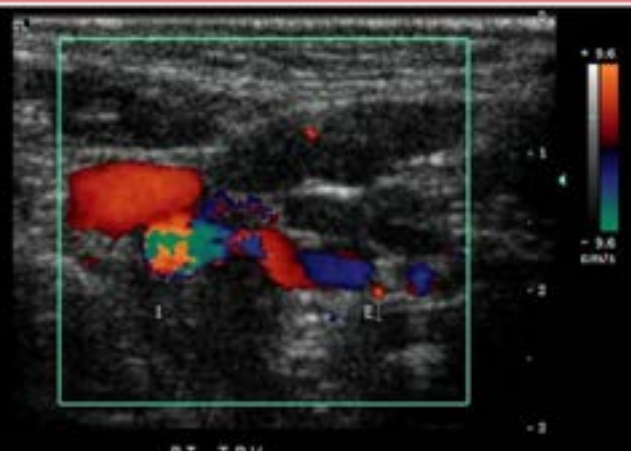
String flow identified at the Proximal ICA, yet the mid portion has significant flow present. This mid ICA flow must be coming in from another vessel. [Slide 1 of 3.](#)

ECA to ICA Collateralization



RT TRV

Proximal ICA with atherosclerosis causing significantly reduced flow.

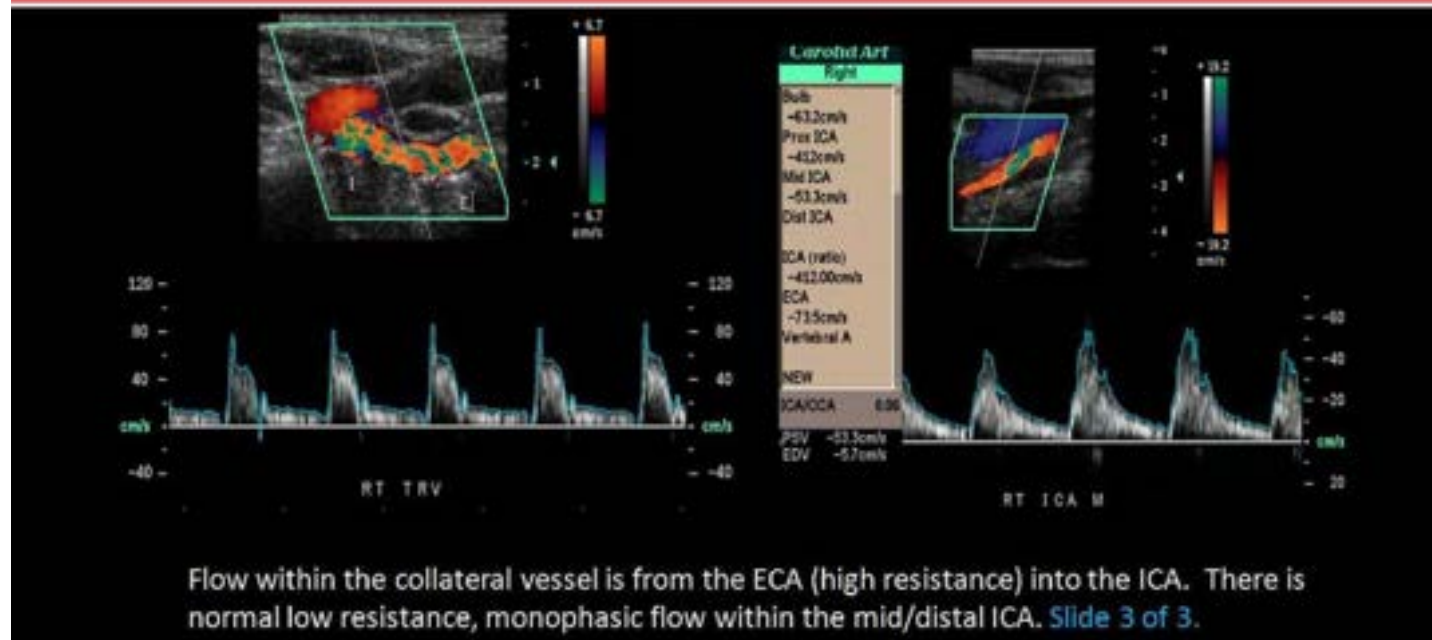


RT TRV

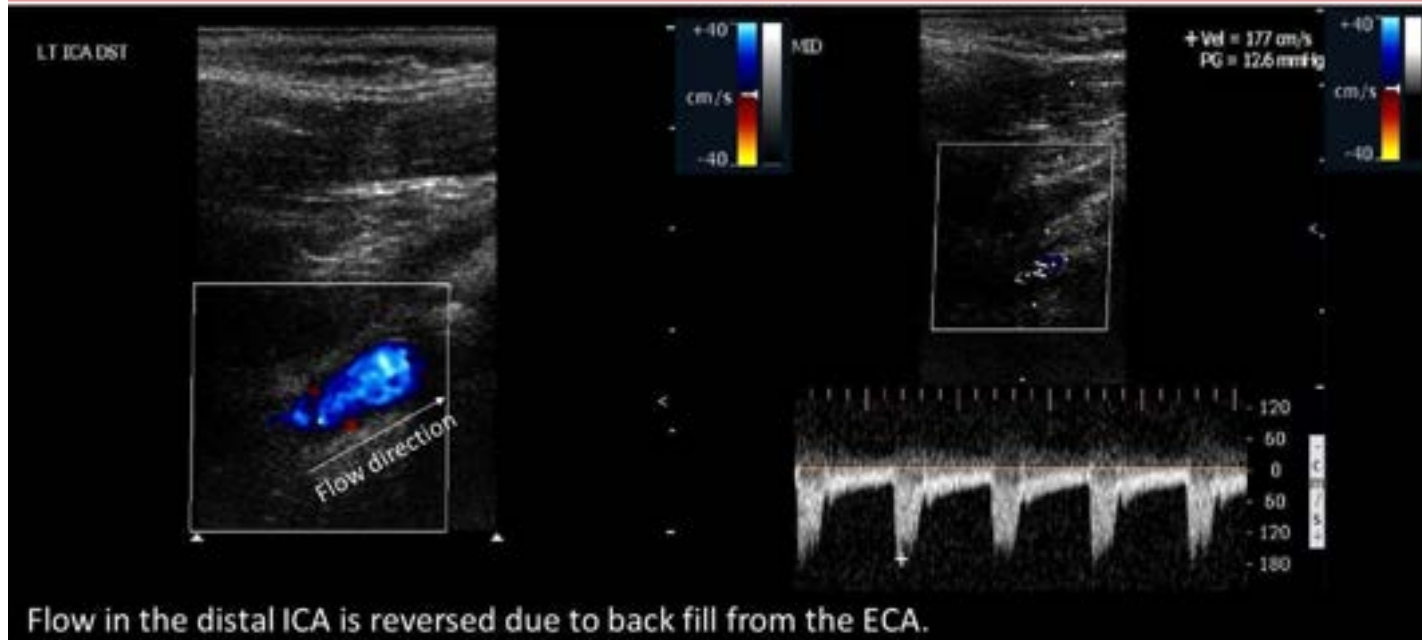
Mid ICA with collateral vessel from ECA filling the ICA lumen.

[Slide 2 of 3.](#)

ECA to ICA Collateralization



ECA to ICA Collateralization



Exam Limitations:

- Depth of vessel
- Path of vessel; high bifurcation
- Abnormal neck size
- Shadowing from atherosclerosis inhibits Doppler evaluation
- Edema of tissues
- Tremors
- Recent surgery
- Patient lack of cooperation, snoring
- Sclerosed neck tissue secondary to radiation treatments
- Tracheostomy

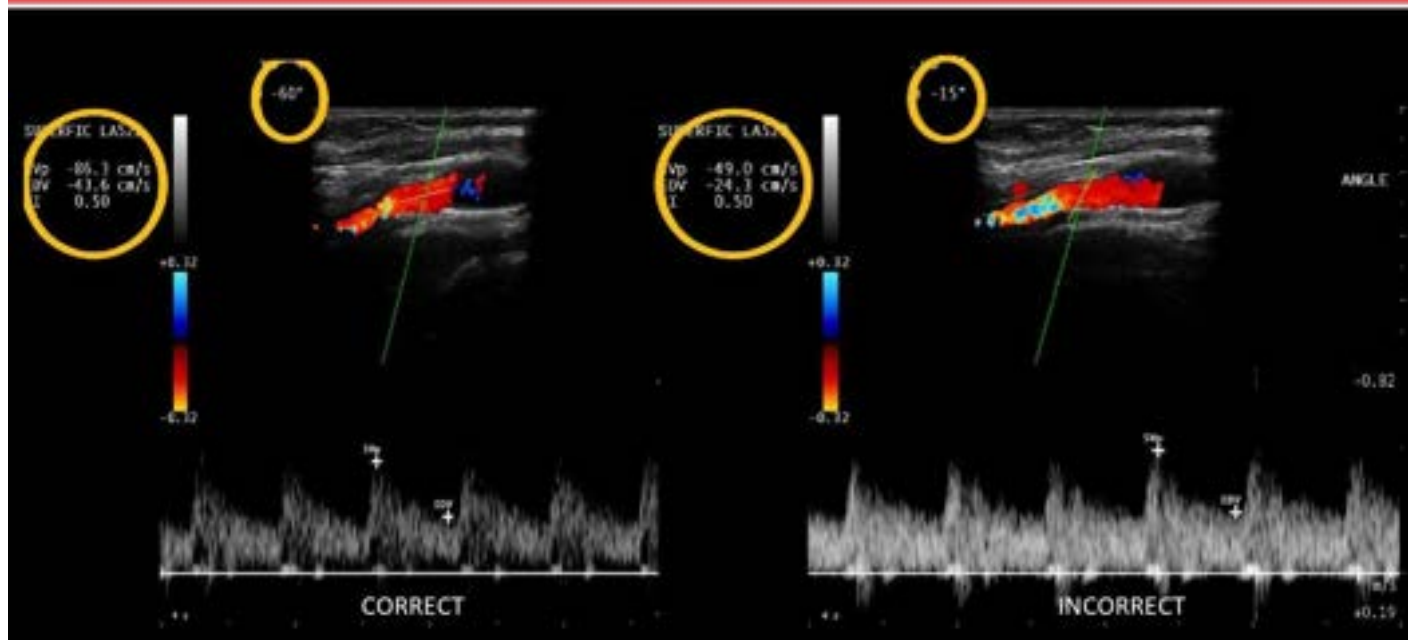
False Positives for Stenosis:

- Increased cardiac output
- Vessel tortuosity
- Inappropriate angle of insonation
- Compensatory flow - increased flow volume and velocity when opposite side is already occluded

False Negatives for Stenosis:

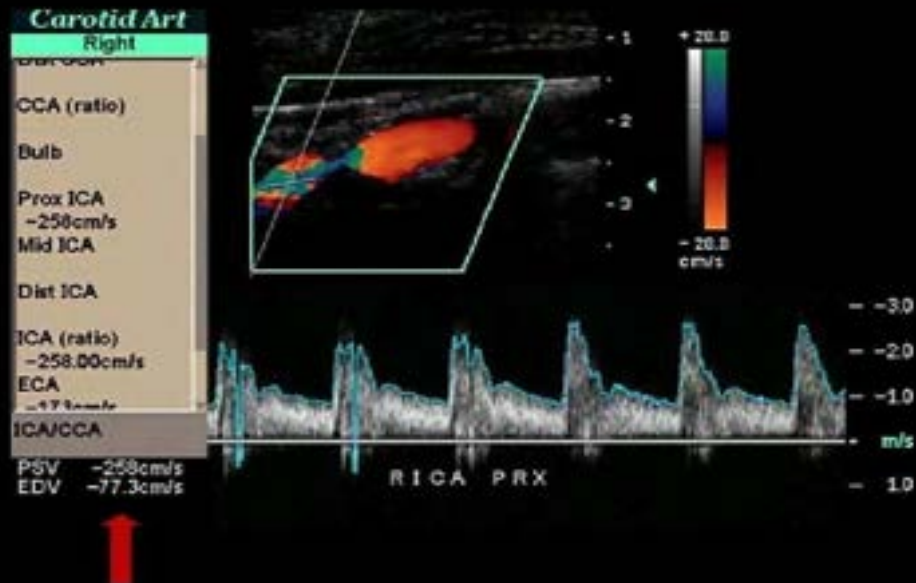
- Improper sample location, accelerated jet not detected (#1)
- Low cardiac output
- Stenosis is at the bulb (dilated area)
- Inappropriate angle of insonation
- Doppler angle over 60 degrees
- Incorrect Doppler cursor placement
- Collateral formation

Angle Correction

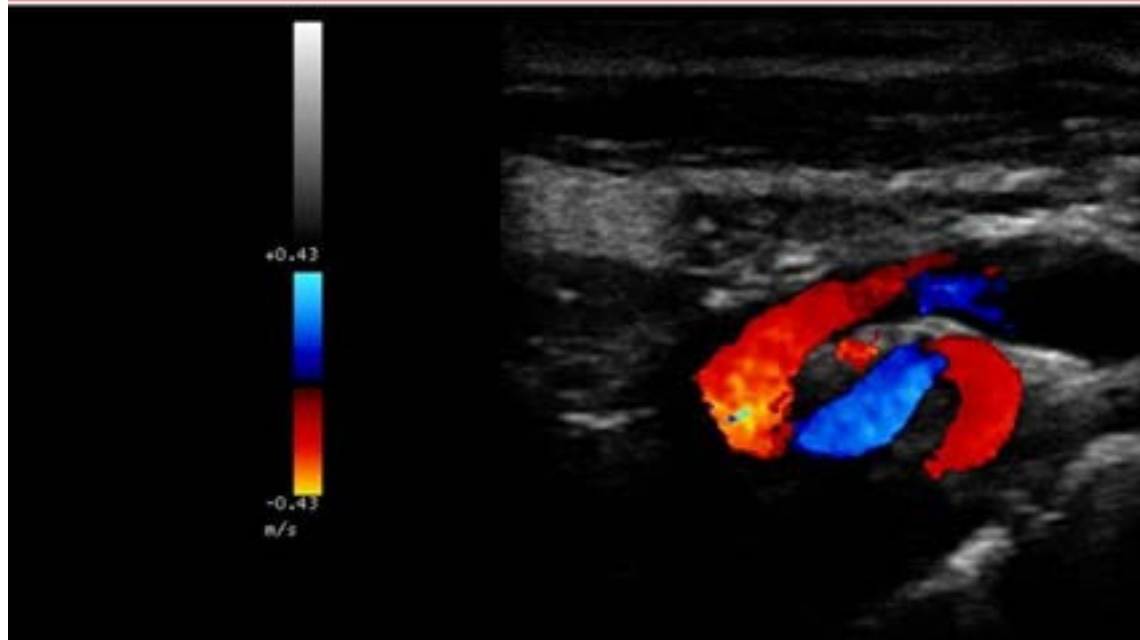


Improper Technique

The cursor angle should be aligned parallel to the flow. Using the incorrect angle can cause significant over- or underestimation of flow velocities and stenosis.



ICA Tortuosity

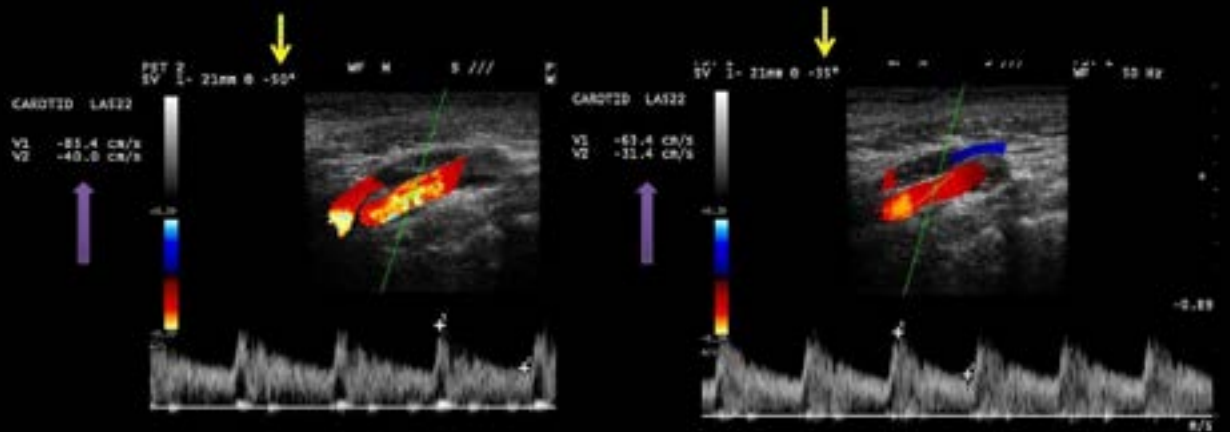


ICA Tortuosity



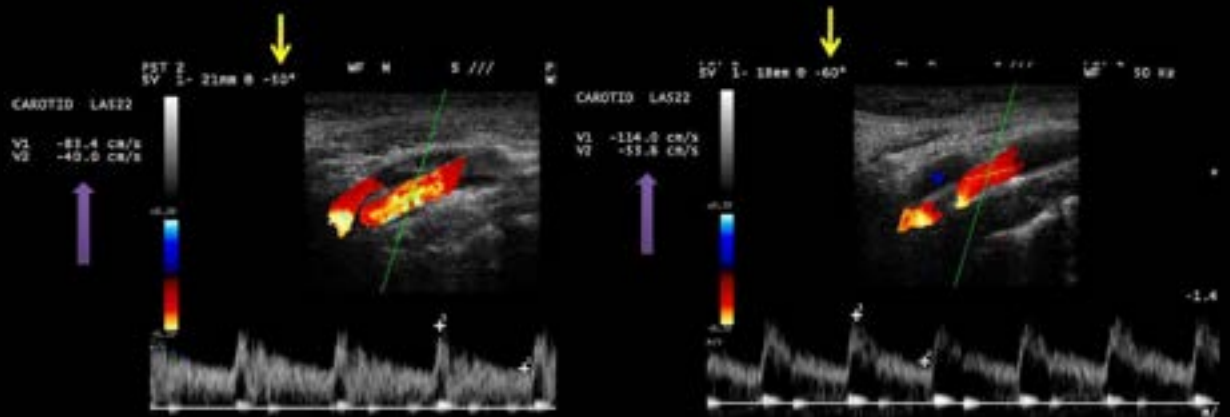
Vessel tortuosity will cause an increase in resistance and velocity.

Angle Correction Error



The cursor angle is "under-corrected" with an angle that is less than is necessary. This leads to underestimation of the velocity.

Angle Correction Error



The cursor angle is "over-corrected" with an angle greater than necessary. This leads to an overestimation of the velocity.

Treatment of Carotid Stenosis

- Lifestyle modification: diet, exercise, stop smoking
- Medication - aspirin is recommended for patients with mild to moderate stenosis because it reduces platelet aggregation in the vessels
- Invasive options: endarterectomy, angioplasty, atherectomy, stent

IMPORTANT DIFFERENCE BETWEEN 99% STENOSIS AND OCCLUSION, STENOSIS CAN BE TREATED!

Sterile Field Tips:

- All items and personnel in a sterile field must be sterile
- Sterile packages or fields are opened or created as close as possible to time of actual use
- Moist areas are not considered sterile
- Contaminated items must be removed immediately from the sterile field
- Gowns are considered sterile only in the front, from chest to waist and from the hands to slightly above the elbow
- Tables are considered sterile only at or above the level of the table
- Personnel with colds should avoid working while ill or apply a double mask
- Edges of sterile areas or fields (generally the outer inch) are not considered sterile
- When in doubt about sterility, discard the potentially contaminated item and begin again
- Items should not be used that have expired sterilization dates

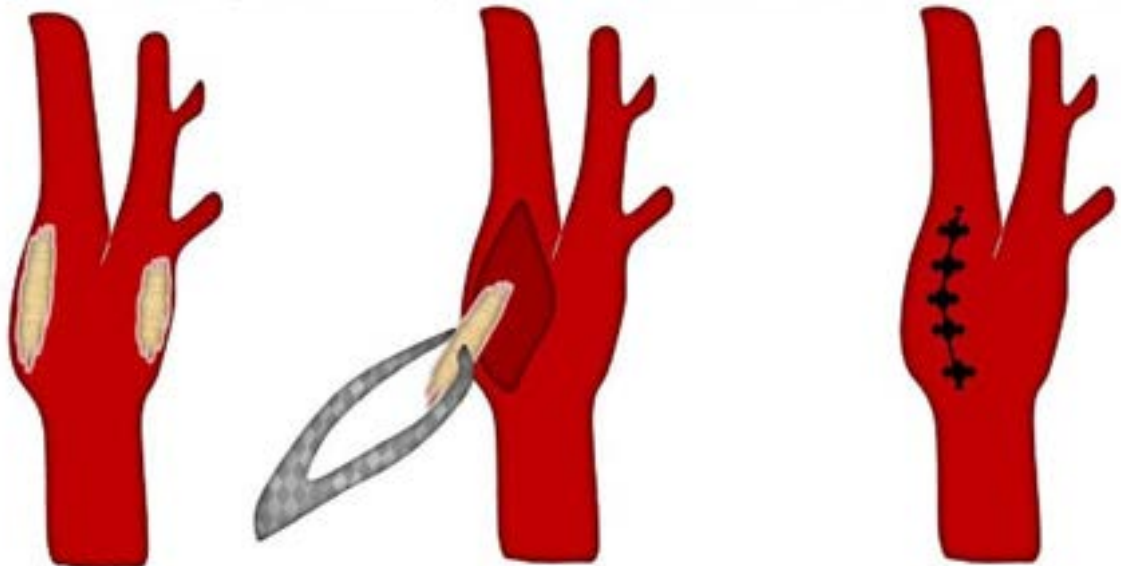
Interventional Procedures:

- Treatment recommended for ICA stenosis >70%
- Atherectomy
 - Uses a catheter with a rotating head and suction capabilities to break up the plaque and remove it from the artery
- Angioplasty
 - Uses a balloon tipped catheter to compress the atheroma and expand the vessel lumen
- Endarterectomy
 - The carotid artery is lacerated and the atheroma formation is scraped for removal
 - Complications include:
 - within the first month post-op, restenosis is attributed to suboptimal surgical techniques
 - intimal flap
 - dissection
 - shelf lesion - residual hyperechoic plaque in the lumen that has an abrupt edge
 - platelet aggregate - Hypoechoic/anechoic material adjacent to vessel wall that causes increased flow velocity
- Stent Placement
 - A mesh stent can be placed into the artery using a balloon catheter; usually made of metal alloys
 - Nitinol most common
 - Echogenic appearance with shadowing

- IMPORTANT: the Doppler cursor should be "walked" through the stent to locate the highest velocity
- Abnormal Findings:
 - Stent fracture - Irregular contour of stent with a visible abrupt edge; color and spectral Doppler turbulence noted at the area of the edge; strongly associated with calcification
 - Stent deformation - Wall of stent appears to protrude into vessel lumen; color-flow demonstrates a reduced lumen size; elevated PSV may be present depending on degree of deformation
 - Restenosis or Myointimal hyperplasia - homogeneous material lining stent wall
 - Thrombus formation, dissection and occlusion

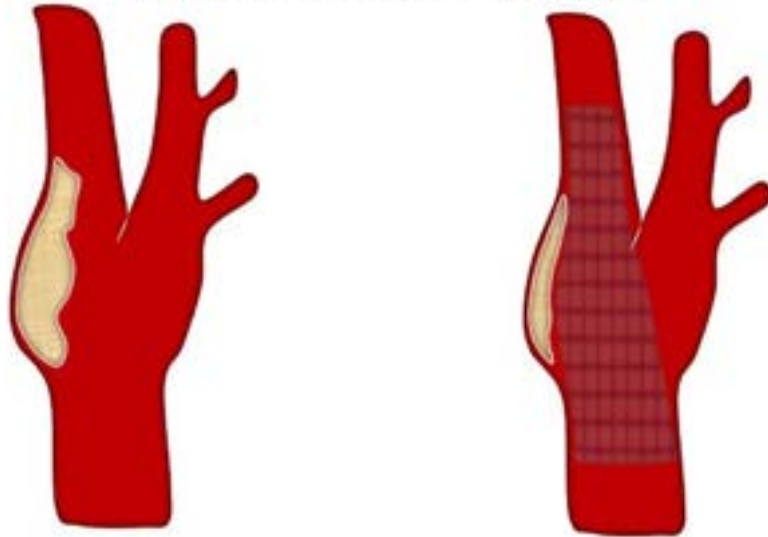
Endarterectomy

During an endarterectomy, the carotid artery is lacerated and the atheroma formation is scraped for removal.

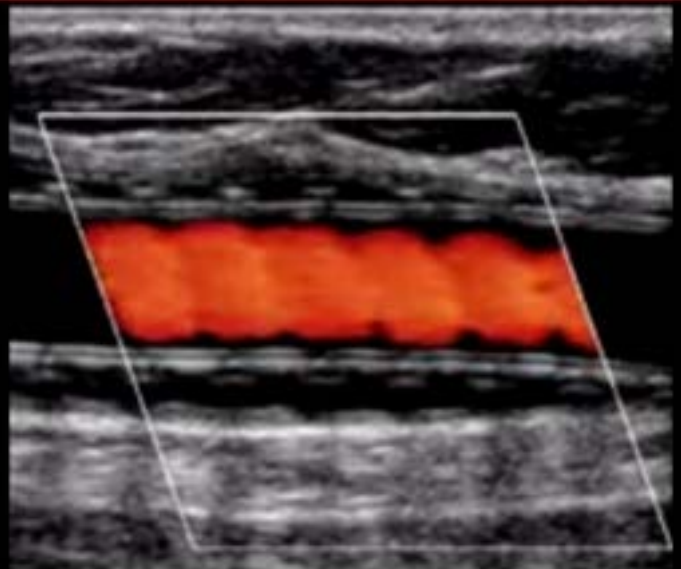
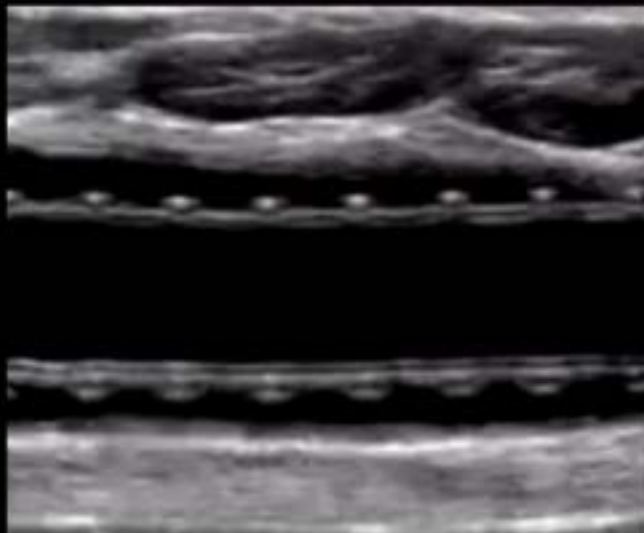


Carotid Stent

During stent placement, a catheter is inserted into the arterial system and advanced to the stenotic site. Once in the appropriate area the stent is expanded to increase lumen size.

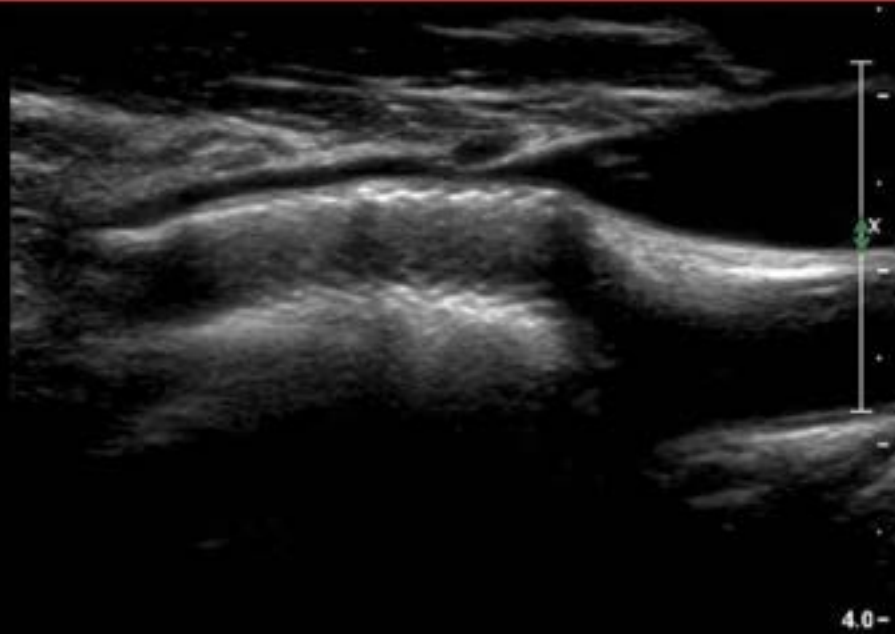


Carotid Stent



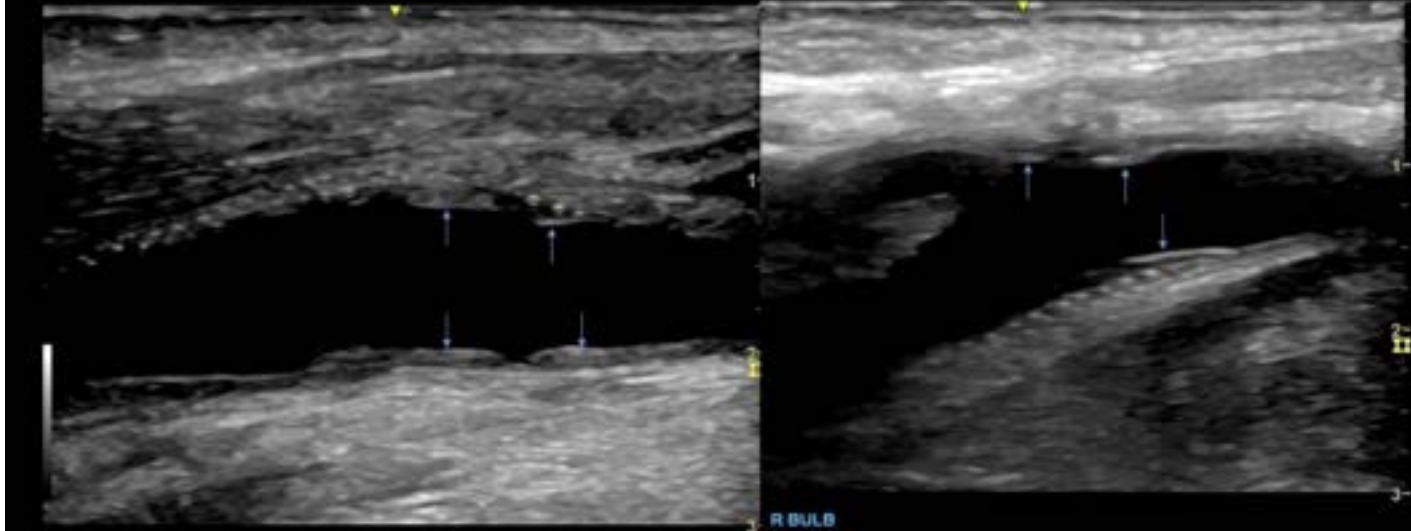
Note the echogenic material lining the lumen of the vessel. The reflective mesh material causes consistent "stripes" of posterior shadowing and drop out in the color display.

ICA Stent



Myointimal Hyperplasia

Myointimal hyperplasia refers to tissue overgrowth with healing after stent placement or endarterectomy. It can contribute to the recurrence of stenosis.



Cerebral Hyperperfusion or Reperfusion Syndrome:

- Rare, but serious, complication following revascularization
- Defined as a major increase in ipsilateral cerebral blood flow that is well above the metabolic demands of the brain tissue
- Causes hemorrhage
- Seen after endarterectomy or stent placement
- Presents as a triad of ipsilateral headache, contralateral neurological deficits, and seizure

Intraoperative Duplex Monitoring:

- Ultrasound performed on the vessel that is corrected prior to closing the surgical wound
- Provides objective functional assessment of the success of the vascular surgery
- Aids in modifying surgical procedure while in progress
- Provides baseline measurements for follow up
- Reduces the need for repeat surgery for same issue
- Requires sterile technique; sterile field, sterile sheath and sterile gel
- Performed on both lower extremity bypass grafts and carotid endarterectomy sites by the vascular surgeon
- Transducer placed within the sterile surgical wound directly on the vessel or graft
- The highest frequency linear transducer available should be used

- Technologist controls the ultrasound equipment from outside the sterile field
- Advantages of intraoperative US over intraoperative angiography include no vessel puncture required, real time flow velocity information, no radiation or contrast needed, more cost effective

Carotid Duplex Evaluation Post-Procedure:

- After an open surgical procedure
 - Endarterectomy or graft placement
 - If ultrasound performed soon after the procedure, artifacts produced by air in the patch or graft may inhibit the exam
 - To minimize artifact from air in the patch, use the most posterior approach possible to evaluate the bifurcation
 - Usually evaluated within first month following the procedure
 - Within the first month post-op, restenosis is attributed to suboptimal surgical techniques
 - Narrowing over the first 24 months after CEA is attributed to neointimal hyperplasia
 - After 2 years, stenosis occurs due to progression of atherosclerotic disease
 - Evaluate the vessels for residual plaque, neointimal hyperplasia, hematoma, pseudoaneurysm or leak
- After stent placement
 - Usually evaluated within first month following the procedure
 - Protocol must include evaluation of proximal native artery, proximal stent attachment site, proximal, mid and distal stent, distal stent attachment site and native artery distal to stent
 - Stent material can cause shadowing and color dropout
 - Serial studies are performed to assess the stent for changes
 - Restenosis is most common at the stent attachment sites

Velocity Criteria – Stented Carotid Artery

Doppler samples taken at the proximal attachment site, through the stent and at distal attachment site

% Stenosis	PSV	ICA/CCA Ratio
0-19%	<150cm/s	<2.15
20-49%	150-249cm/s	
50-79%	220-339cm/s	≥ 2.7
80-99%	≥ 340 cm/s	≥ 4.15

Lal BK1, Hobson RW 2nd, Tofighi B, Kapadia I, Cuadra S, Jamil Z. Duplex ultrasound velocity criteria for the stented carotid artery. 2008. J Vasc Surg. 47(1):63-73

Other Cerebrovascular Abnormalities

Vertebral Artery Stenosis/Occlusion:

- Difficult to diagnose with ultrasound due to portions of the artery are located within the transverse processes of the vertebrae
- Normal flow is low resistance, low to medium velocity
- Most common location of stenosis is at the vertebral origin
- Can lead to steal phenomenon

Diagnostic Criteria for Vertebral Artery Stenosis/Occlusion:

- Abnormal PSV $>100\text{cm/s}$ in most cases of stenosis
- Ratio of pre- and post-stenotic velocities >2
- Abnormal pulsatility (biphasic or triphasic)

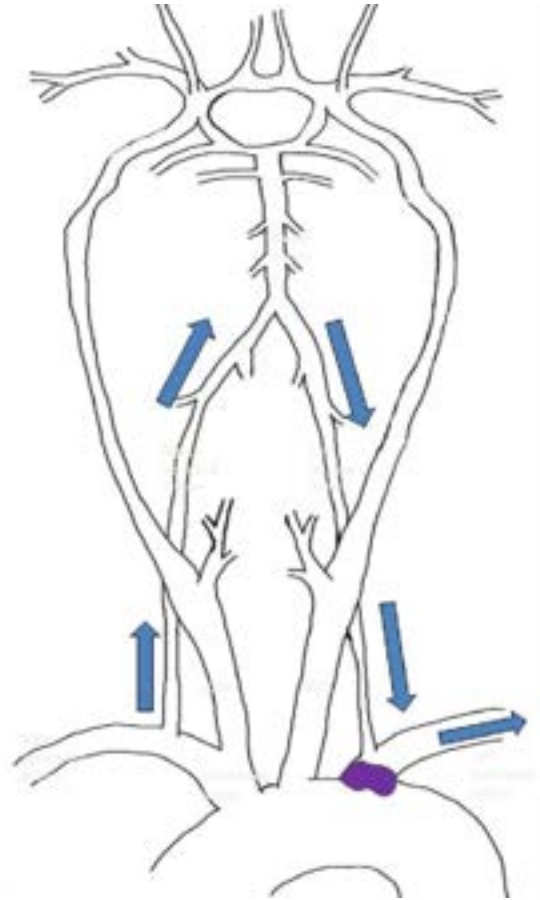
Subclavian Steal:

- Occlusion or stenosis of the proximal subclavian artery
- Most commonly occurs on the left side
- Distal subclavian artery fed by reversed flow through the ipsilateral vertebral artery
- Retrograde vertebral flow on affected side
- The contralateral vertebral artery usually increases in size and flow volume and velocity increases
- Symptoms include limb paralysis, vertigo, ataxia and syncope (vertebrobasilar symptoms)
- If a subclavian steal suspected, bilateral brachial pressures should be obtained; decreased BP 20mmHg or greater will be identified on affected side
- Anytime a subclavian steal is suspected the
- subclavian arteries should be evaluated by Doppler; monophasic subclavian flow detected on affected side
- If bidirectional or biphasic flow is detected in the vertebral artery, exercising the arm or applying a blood pressure cuff can be used to convert the flow to a complete steal
- Ask the patient to perform 40 flexion/extension maneuvers of the arm on the affected side and then re-evaluate the vertebral flow
- Inflate the cuff above the systolic pressure on the ipsilateral arm and maintain it for a 1-2 minutes; release the cuff and assess the flow in the abnormal vertebral artery; if a steal is present, the flow should change from biphasic to complete reversal of flow

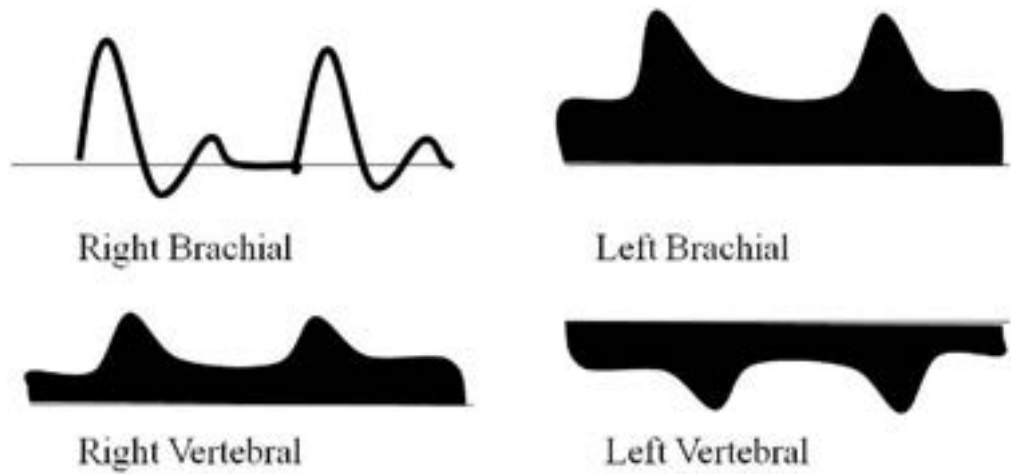
Subclavian Steal

85% involve
left vertebral artery

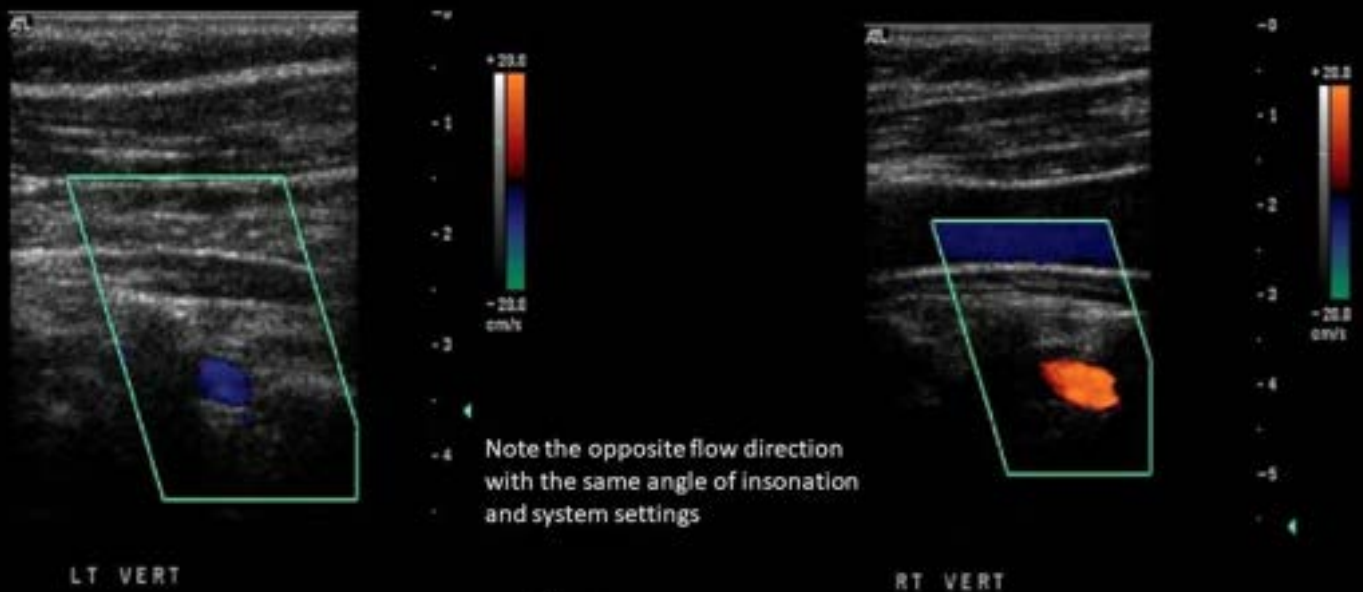
>20mmHg difference in
brachial pressures



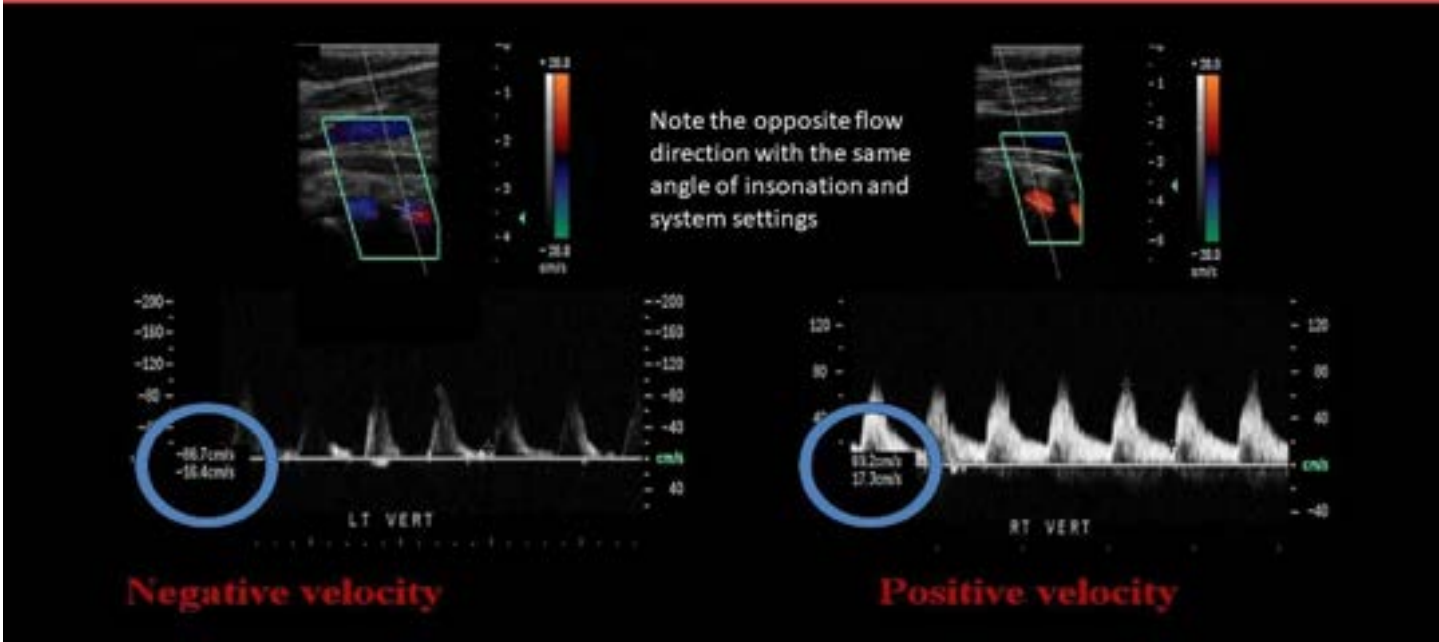
Subclavian Steal



Subclavian Steal

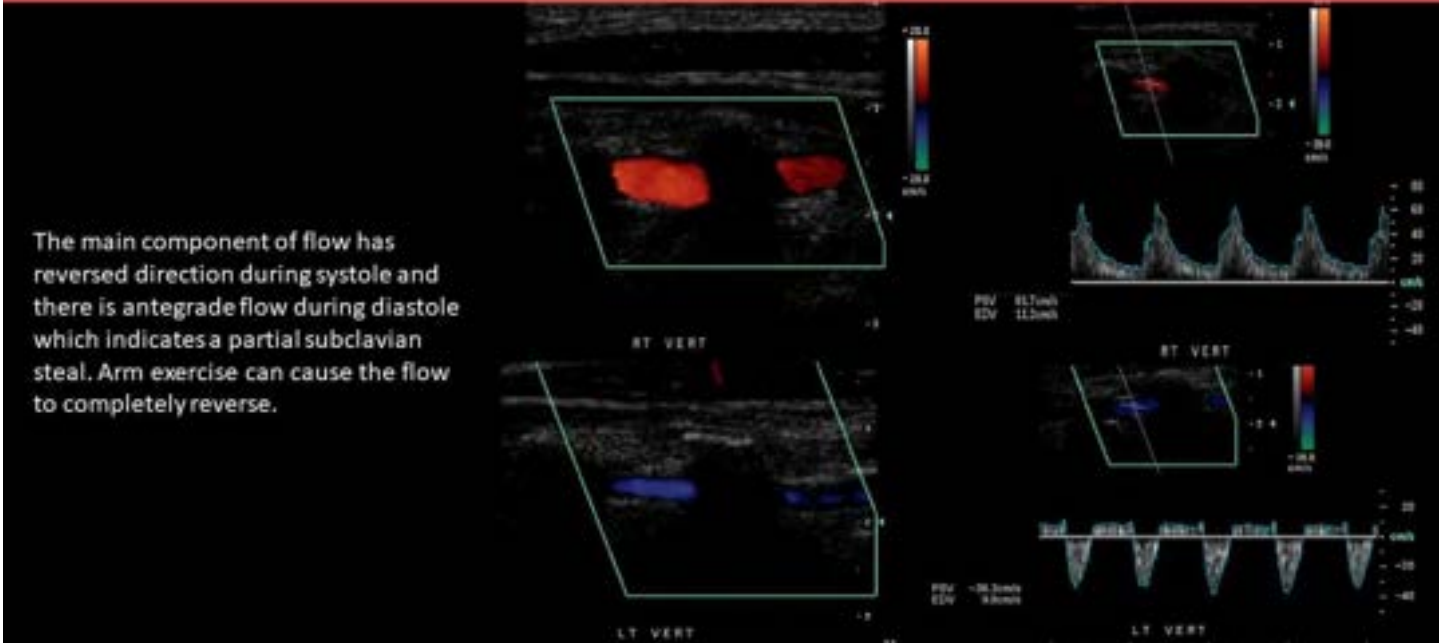


Subclavian Steal



Partial Subclavian Steal

The main component of flow has reversed direction during systole and there is antegrade flow during diastole which indicates a partial subclavian steal. Arm exercise can cause the flow to completely reverse.

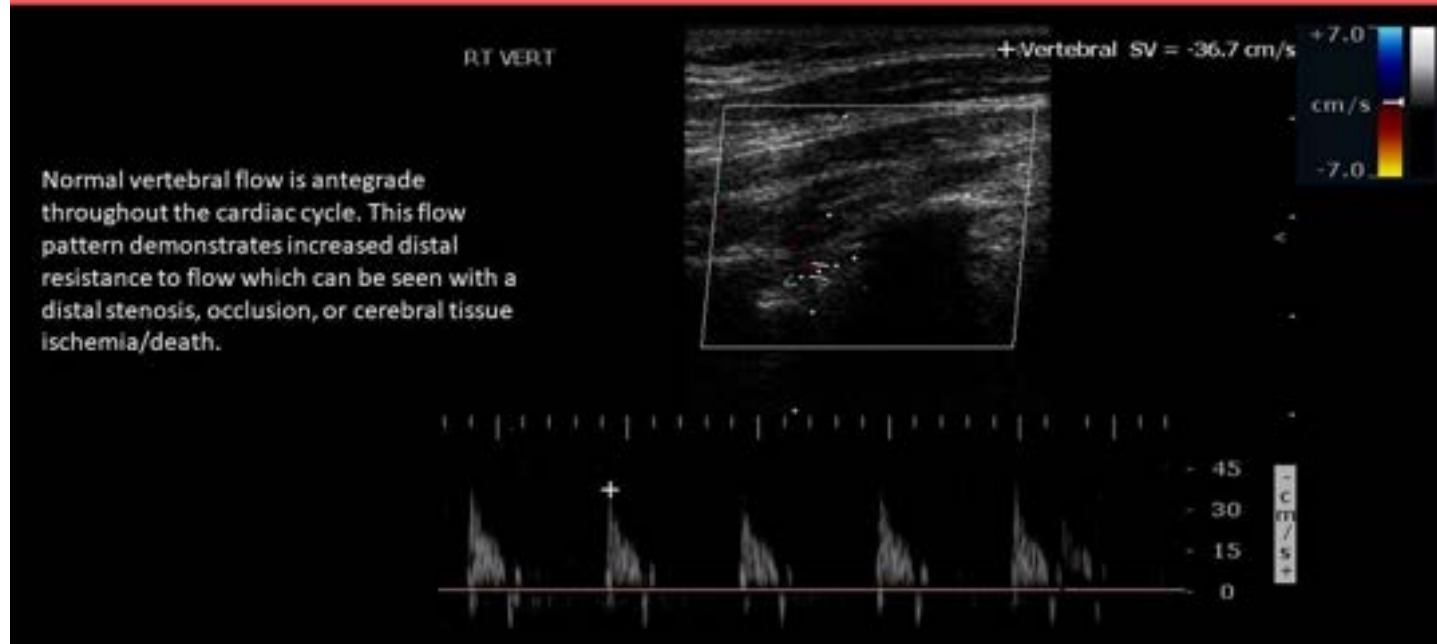


Left Subclavian Steal (Partial)



Courtesy of Dr. Taco Geertma
www.ultrasoundcases.info

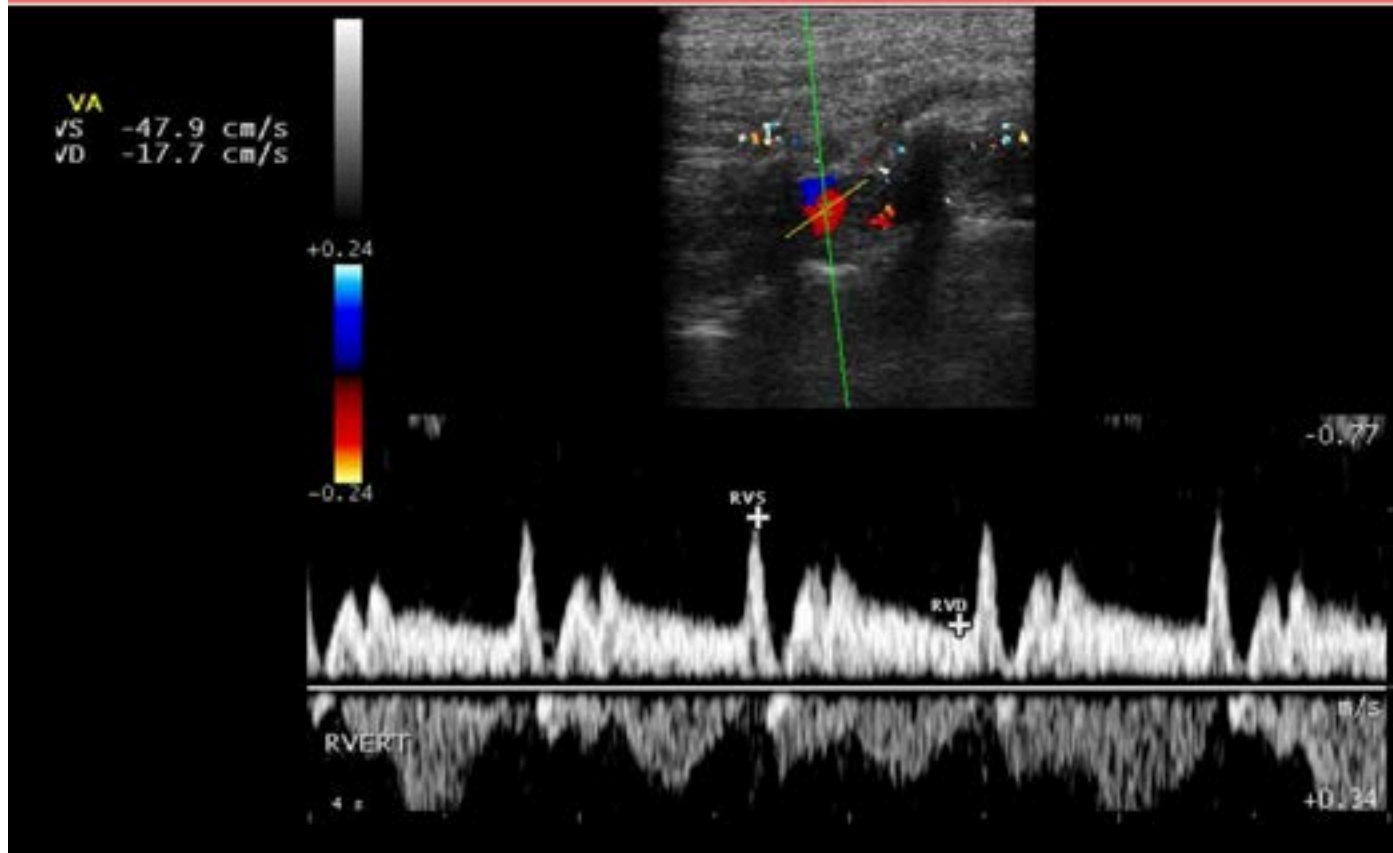
High Resistance Vertebral Artery Flow



Early Systolic Deceleration in the Vertebral Artery:

- The vertebral arteries originate from the subclavian arteries bilaterally
- Stenosis of the subclavian artery can cause changes in the vertebral flow
- Systolic deceleration is a sign of a significant stenosis in the ipsilateral side
- Systolic flow starts with a sharp peak followed by a sharp decrease in flow to the baseline
- Then flow resumes, increases again to decline through diastole
- This is caused by stenosis in the subclavian artery and less commonly the proximal vertebral artery
- This type of waveform is considered a sign of "pre-steal"
- Provocative maneuvers can be used to evaluate the ability to convert the waveform to a partial or complete steal
- Arm exercise followed by a repeat scan
- Pneumatic cuff inflation to a level of 20mmHg above systolic pressure on the affected arm for 2-3 minutes followed by rapid release and repeat scan

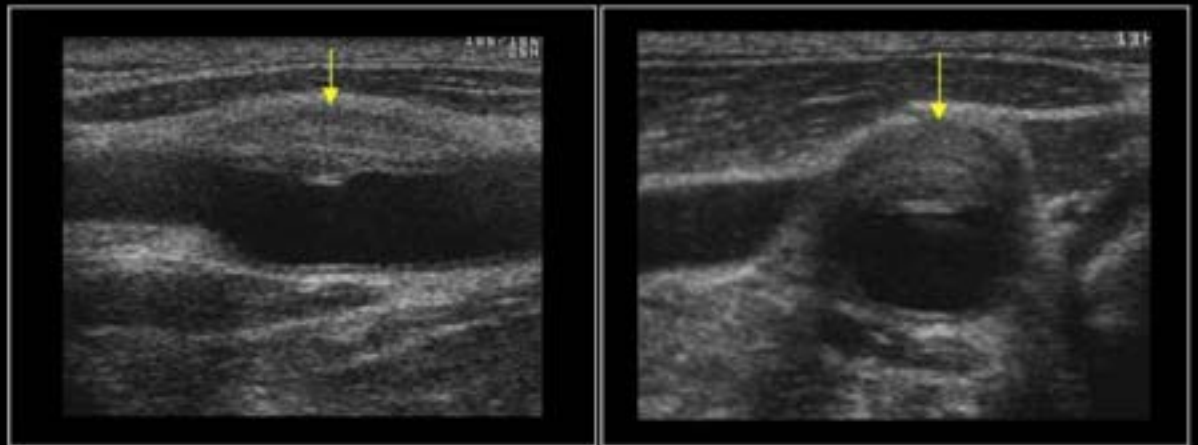
Early Systolic Deceleration



Arteritis:

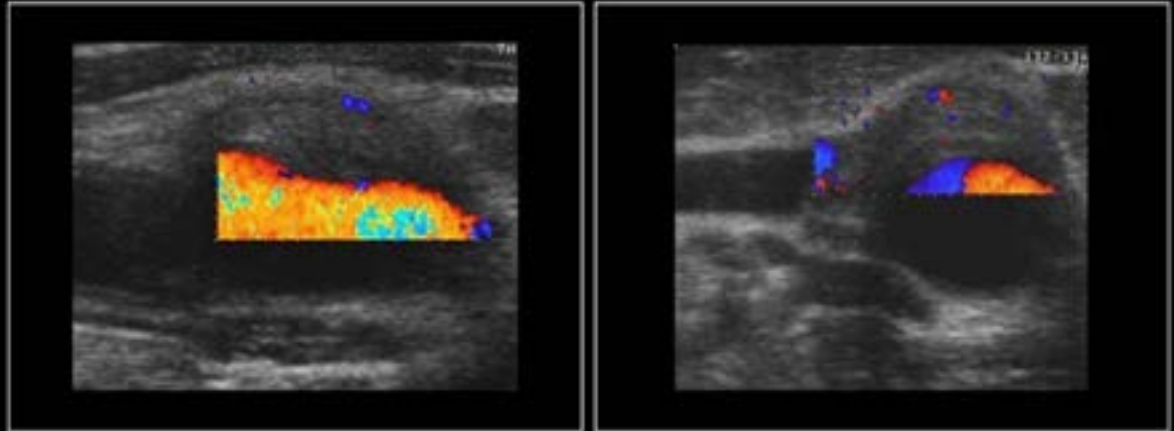
- Inflammation of the arterial wall
- Causes diffuse or focal thickening of the muscle layer
- May see increased vascularity in the wall of the inflamed segment
- Increased velocities with decreased lumen size
- Types of arteritis: Buerger arteritis, Takayasu arteritis, polyarteritis, temporal arteritis (Giant cell arteritis)

Arteritis



Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

Arteritis



Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

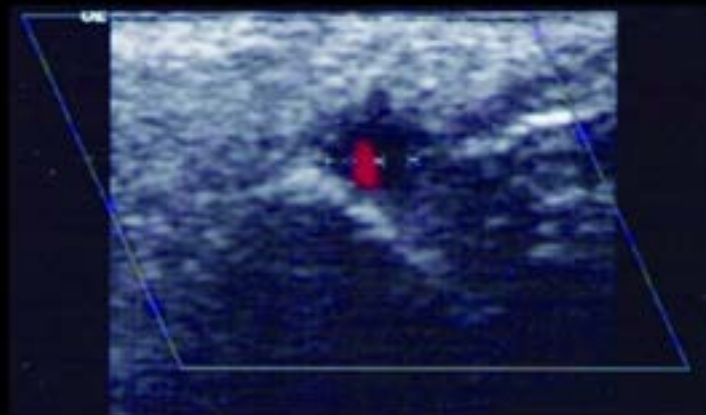
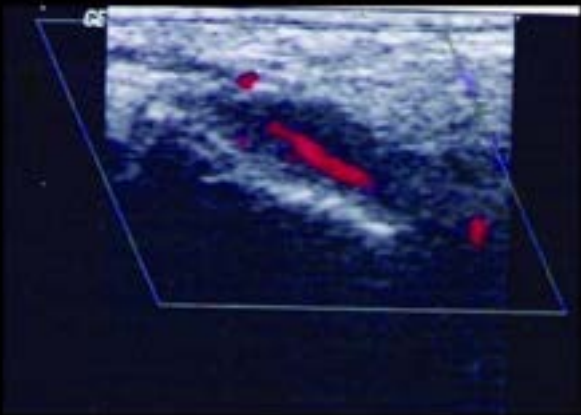
Temporal Arteritis:

- The superficial temporal artery courses anterior to the ear and can be palpated just above the level of the zygomatic arch
- Inflammation of the superficial temporal artery leading to flow reduction to frontal and parietal branches
- Stiffness in the neck, jaw claudication, headaches and sudden unilateral blindness may occur
- Blood tests performed:
 - To measure the erythrocyte sedimentation rate (how quickly the red blood cells fall to the bottom of the blood sample)
 - To measure C-reactive protein (CRP)
- Biopsy necessary for diagnosis
- AKA Giant cell arteritis
- High frequency linear transducer used for temporal artery evaluation
- 2D images demonstrate thickened arterial walls
- Halo Sign - a ring of thickened vessel wall tissue surrounding the vessel
- Normal flow pattern mimics the feeding vessel (ECA) waveform

- Diffuse vessel wall thickening and distal ischemia cause increased velocity and lower resistance flow
- Areas of greater wall thickening can cause a focal stenosis with increased resistance proximal to the stenosis and decreased resistance distal to the stenosis

Giant Cell Arteritis / Temporal Arteritis

Thickened arterial walls and increased resistance to flow are common signs of temporal arteritis.



Karahaliou M, Vaiopoulos G, Papaspyrou S, Kanakis MA, Revenas K, Sfikakis PP. Colour duplex sonography of temporal arteries before decision for biopsy: a prospective study in 55 patients with suspected giant cell arteritis. *Arthritis Res. Ther.* (2006)

Aneurysm:

- Rarely seen in the cervical carotid branches
- If found in the tiny intracranial vessels it is called a berry aneurysm
- Caused by trauma, infection, atherosclerosis or connective tissue disorder

Pseudoaneurysm:

- Blood escapes into surrounding tissues and becomes encapsulated
- Risk of hemorrhage/death; trauma, infection or true aneurysm
- Demonstrates a swirling flow pattern on color Doppler and a to and fro flow pattern on PW Doppler

AV Malformation and AV Fistula:

- An AVM is a congenital connection between the arterioles and venules (lack of capillary bed); usually has multiple connections and surrounding venous collaterals and dilated veins
- A Vein of Galen aneurysm is a severe form of intracranial AVM identified in fetal evaluations
- An AVF is a direct connection between the arterial and venous systems caused by trauma or vascular procedure; usually a single connection and NO surrounding collaterals due to acute cause
- Causes increased volume of flow in the venous system with pulsatility and turbulence
- Reduces resistance in the arterial inflow and increases diastolic flow
- Large or chronic arteriovenous connections can lead to heart failure

Dissection:

- Acute tear in the vessel intima
- If ICA affected, usually affects the first 2-4cm
- CCA dissection can be a continuation of an aortic arch dissection
- Creates false lumen
- Can occur due to trauma or spontaneously, without definitive cause
- Can lead to thrombosis, embolism, stenosis/occlusion, pseudoaneurysm and rupture
- Appears as an echogenic, mobile flap within vessel lumen that causes turbulence and flow reversal
- Associated with Marfan Syndrome and Ehler-Danlos syndrome (collagen vascular disorders)
- More common in the extracranial ICA than the intracranial ICA
- Considered a critical finding

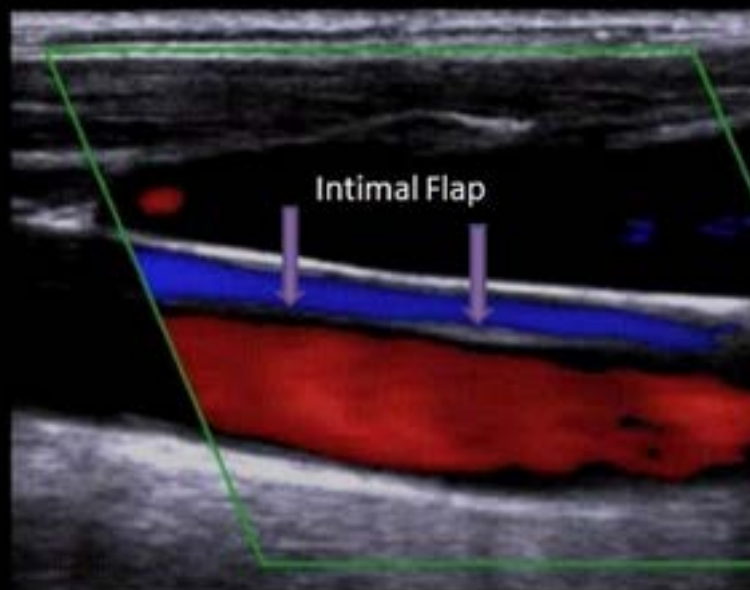
Diagnostic Criteria for Carotid Dissection:

- Evaluate the extent of the dissection and document the flow within the true and false lumen with color and PW Doppler
- Presence of a linear, echogenic flap in vessel lumen
- Two waveforms identified, one for true lumen and one for false lumen
- Microemboli in the ipsilateral MCA seen on TCD
- Anytime a carotid dissection is identified, the aortic arch should also be evaluated

Carotid Dissection



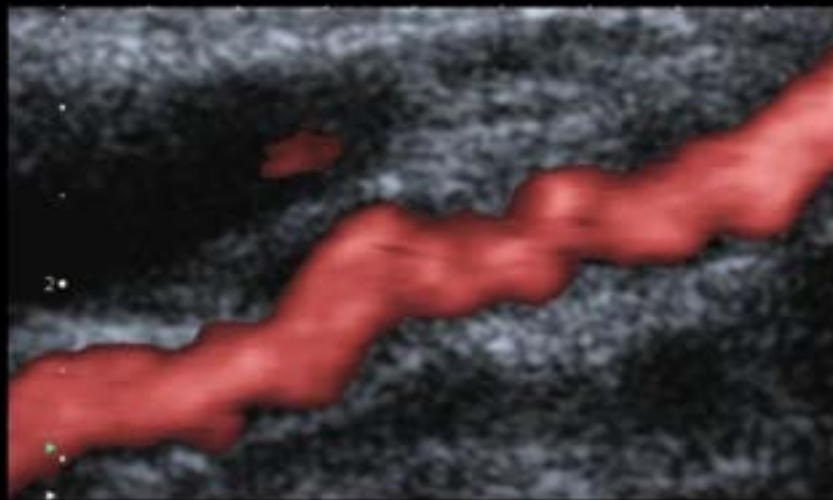
Carotid Dissection



Fibromuscular Dysplasia:

- Collagen growth in media layer of vessel (Atheroma formation occurs in the intimal layer)
- It is most commonly seen in the renal arteries
- Also seen in the carotid and mesenteric arteries
- Usually occurs in the mid or distal segment of the ICA, while atherosclerosis most commonly affects the proximal ICA
- Causing flow limiting lesions
- Rare but more common in young women
- String of pearls appearance on color Doppler
- Abnormal growth of media layer
- Does not shadow
- Causes stenosis/occlusion of vessel
- Best evaluated with contrast arteriography
- If FMD is identified in the carotid arteries, the renal and mesenteric arteries should also be evaluated for FMD

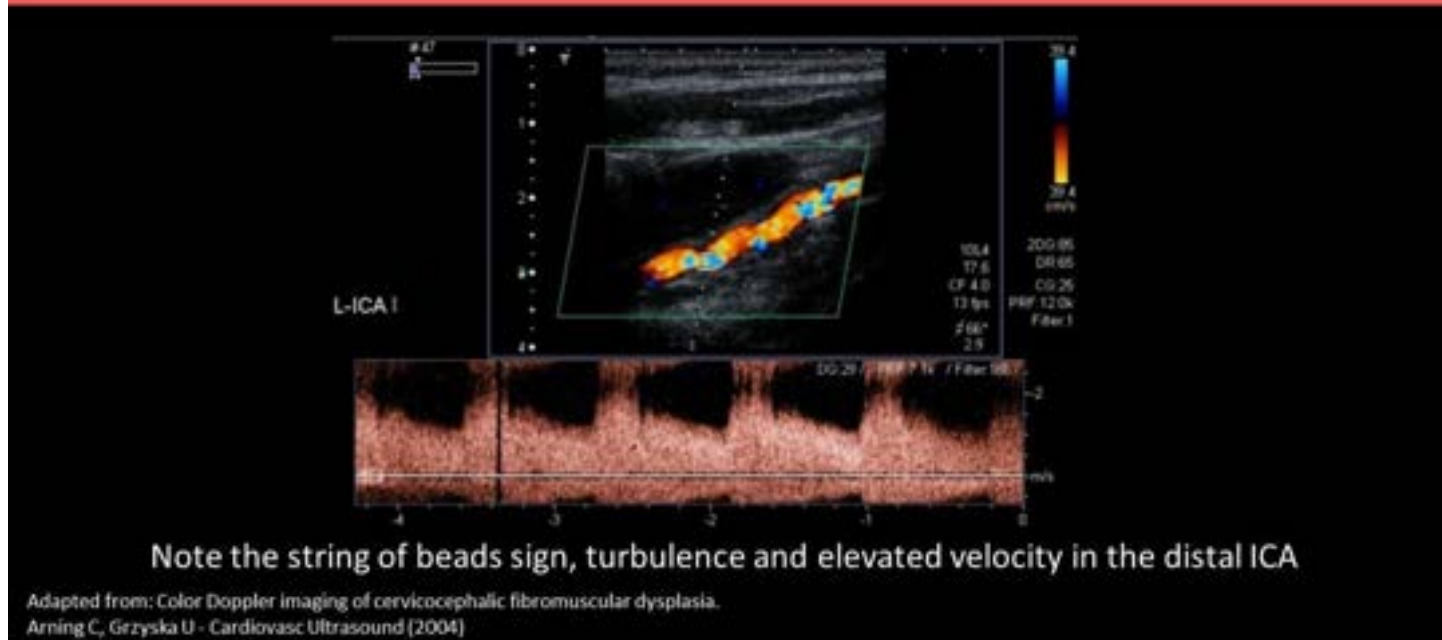
Fibromuscular Dysplasia



Note the string of beads sign in the mid/distal ICA

Adapted from: Color Doppler imaging of cervicocephalic fibromuscular dysplasia.
Arning C, Grzyska U - Cardiovasc Ultrasound (2004)

Fibromuscular Dysplasia



Carotid Body (Sinus):

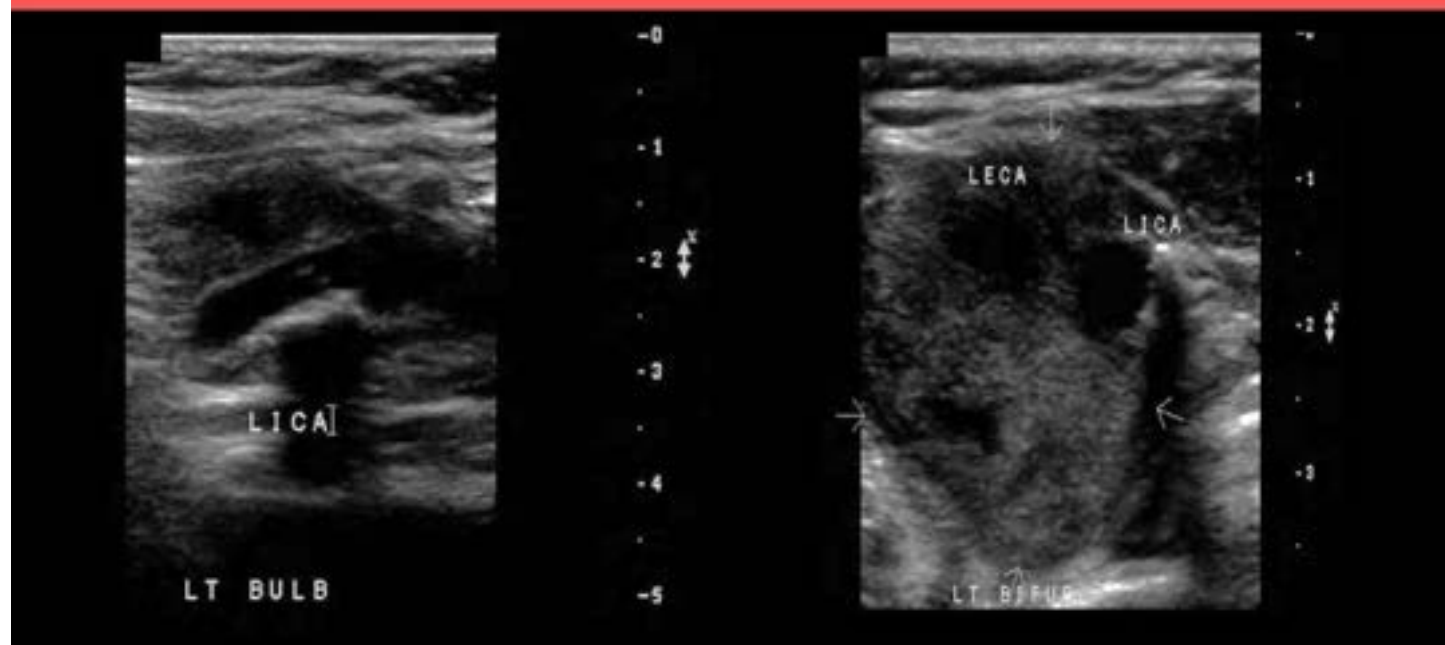
- The carotid body or carotid sinus is located at the bifurcation of the CCA, near the angle of the jaw
- The vessel walls in the dilated area above the CCA bifurcation contain numerous baroreceptors that monitor blood pressure
- Carotid sinus massage
 - Can be performed as a diagnostic or therapeutic procedure
 - The patient is placed on a tilt table and an EKG with blood pressure monitor is used to assess the patient during the procedure
 - The patient's head should be turned toward the contralateral side
 - A physician will massage the carotid artery at the level of the cricoid cartilage
 - Light pressure massage of the carotid sinus is performed for 5-10 seconds
 - Therapeutic
 - Used to interrupt paroxysmal atrial tachycardia or supraventricular tachycardia
 - Performed to reduce the heart rate
 - Causes a transient block in the conduction of the electrical signal in the heart
 - Diagnostic

- Used to evaluate a patient for hypersensitivity of the carotid sinus as the potential cause for the non-specific symptom of syncope
- If there is no stenosis identified in the extracranial arteries, this procedure may be performed
- An abnormal response will lead to hypotension (>50mmHg drop), bradycardia and feelings of dizziness or syncope
- Cannot be performed on patients that have a history of TIA, a bruit, or known stenosis

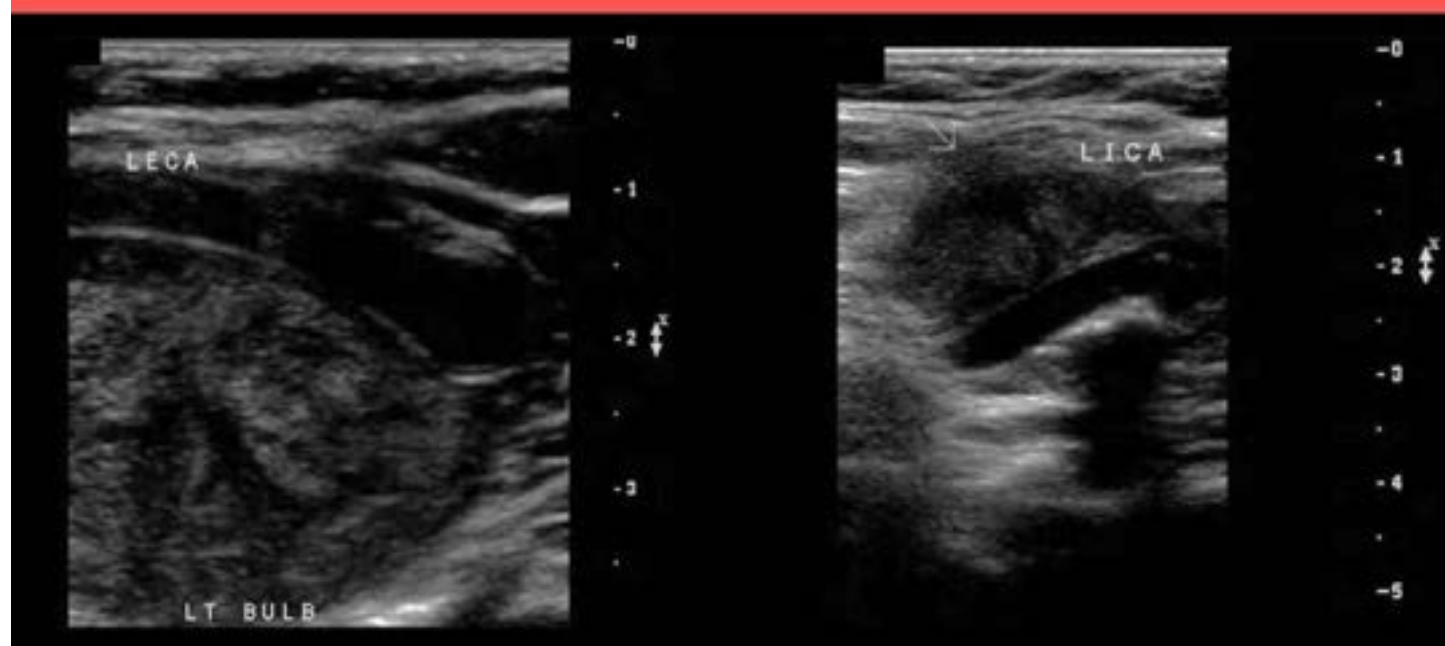
Carotid Body Tumor (CBT):

- Rare neoplasm composed of paraganglionic tissue
- More common in females
- Seen in people living at high altitudes; chronic hypoxia leads to an overactive and enlarged CBT
- Highly vascular structure fed by ECA
- Average measurements of normal carotid body = 5 x 3 x 2 mm
- Overgrowth of a normal structure that is responsible for detecting blood gases/pH
- Can compress ICA flow
- AKA glomus tumor, chemodectoma or paraganglioma
- Located at the bifurcation of the CCA
- Slow growth
- Lateral cervical mass that is pulsatile
- Pharyngeal bulge
- Dysphagia
- Syncope
- Hoarseness
- Dyspnea & cough
- Horner Syndrome

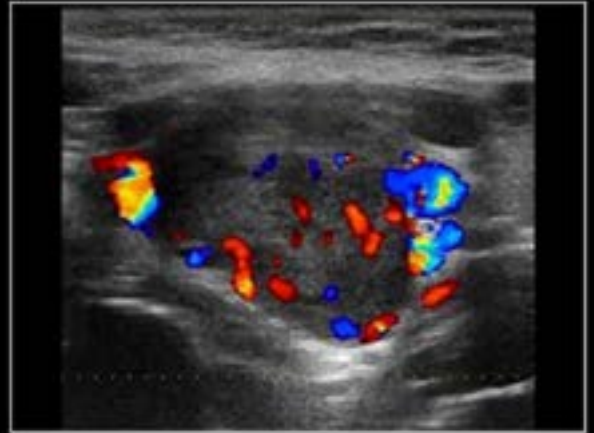
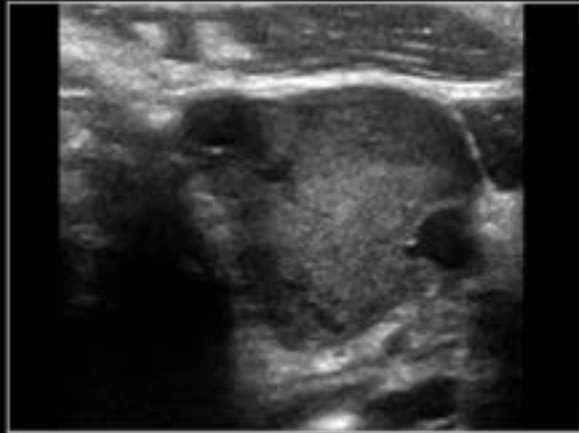
Carotid Body Tumor



Carotid Body Tumor

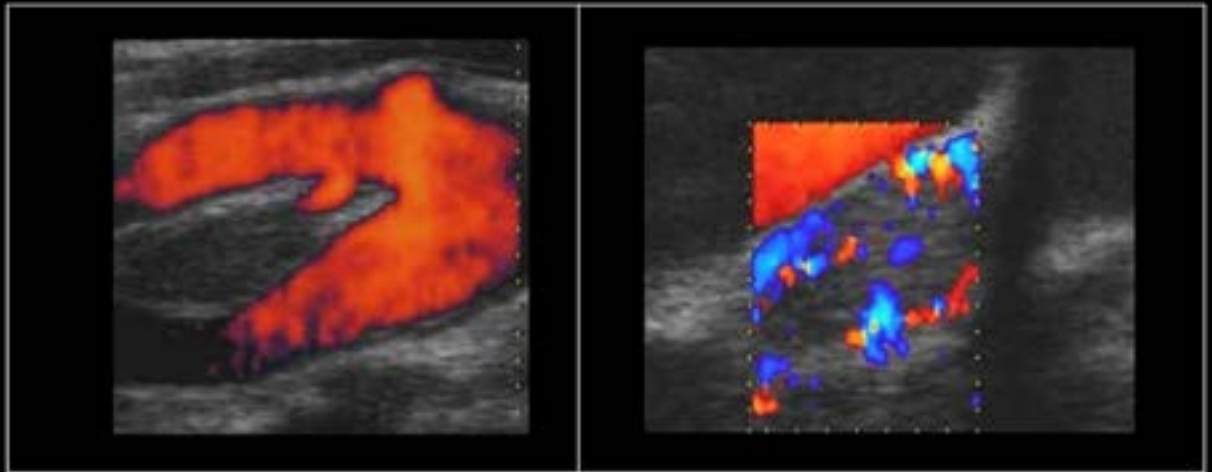


Carotid Body Tumor



Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

Carotid Body Tumor



Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

Thromboembolic Disease:

- Thrombus accumulation within the arterial system that produces embolisms that travel distally
- Associated with aneurysm, plaque formation, trauma, atrial fibrillation
- Symptoms vary with occlusion site of embolism
- Can be the cause of multiple TIAs

Neointimal Hyperplasia:

- AKA myointimal hyperplasia
- Intimal thickening post endarterectomy or stent placement; rapid thickening of the smooth muscle layer
- Flow limiting lesions
- Causes higher velocities throughout the ICA
- Leads to re-stenosis or occlusion usually within 2 years of the surgery

PHACE Syndrome:

- **P**osterior fossa malformations (cranial malformations such as Dandy Walker malformation), **H**emangiomas, **A**rterial anomalies, **C**oarctation of the aorta and other cardiac anomalies, **E**yes (ocular anomalies)
- Arterial stenosis or occlusion of cerebral arteries
- Absence or moderate-severe hypoplasia of the large cerebral arteries
- Aberrant origin or course of the large cerebral arteries
- Saccular aneurysms of any cerebral arteries

Eagle Syndrome:

- Elongation of the styloid process or a calcified stylohyoid ligament causes carotid compression
- Most commonly bilateral
- Symptoms include visual disturbances, syncope, carotid dissection
- Doppler ultrasound evaluation is obtained with the patient's head positioned to reproduce symptoms in order to demonstrate extrinsic compression and mechanical stenosis of the carotid artery

Cardiac Effects on Arterial Circulation:

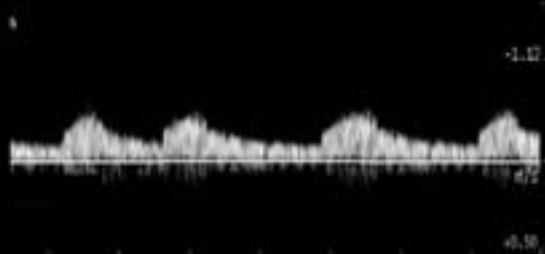
- The carotid arteries can demonstrate cardiac abnormalities by changes in their flow pattern
- BILATERAL FLOW CHANGES in the carotid arteries

Cardiac Output:

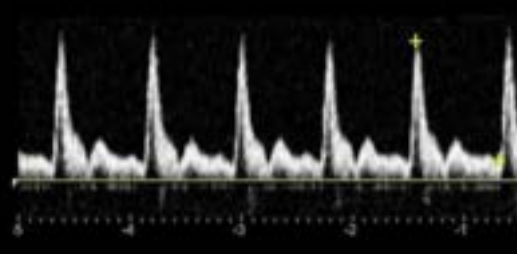
- Any change in cardiac output will be reflected in the vascular system
- High output can cause increased flow velocity in the arteries
- Low output can cause reduced resistance and flow velocity in the arterial system and pulsatile flow in the venous system

Cardiac Output

Coronary artery disease, dilated cardiomyopathy, and severe mitral stenosis are common causes of low cardiac output. Late peaking, lower velocity, lower resistance flow will be displayed in the carotid arteries.



Anemia, hyperthyroidism, and pregnancy are common causes of high cardiac output. Sharp peaking, high velocity, higher resistance flow will be displayed in the carotid arteries.



-
- heart valvular disease can cause abnormal pulsatile venous flow

Atrial Fibrillation:

- 80-90% of arterial emboli are from a cardiac source
- Atrial fibrillation can lead to clot formation in the left atrium
- If the clot breaks off, it can travel anywhere in the systemic arterial system
- If the foramen ovale is patent, the clot could also travel into the right heart and the pulmonary arteries

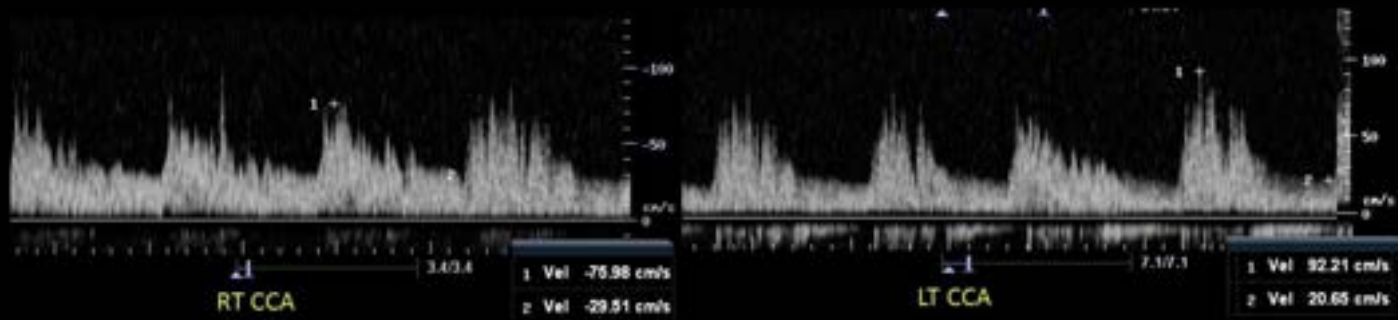
Aortic Stenosis:

- Usually caused by degenerative disease, but can be caused by congenital malformation of the aortic valve (bicuspid AV)
- Blood flow increases in velocity as it moves through the narrowed valve
- The flow into the ascending aorta will be turbulent
- Characteristics of post-stenotic flow can be transmitted to the carotid arteries: lower resistance, delayed upstroke, turbulence and reduced velocity

- Brachial pressures may also be lower than normal
- The ICA/CCA systolic and diastolic velocity ratios should be used to assess the patient for carotid stenosis

Aortic Valve Stenosis

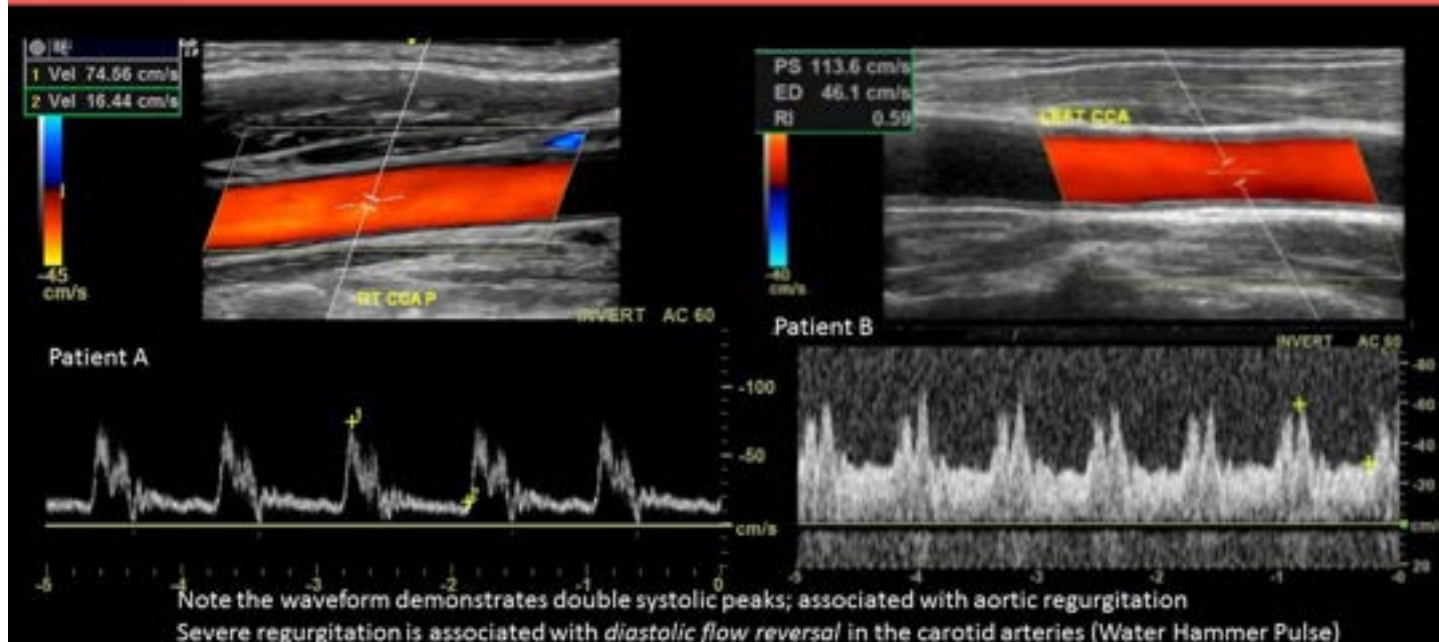
- Characteristics of post-stenotic flow can be transmitted to the carotid arteries: lower resistance, turbulence and reduced velocity
- The ICA/CCA SVR and DVR used to assess these patients for carotid stenosis



Aortic Regurgitation:

- Blood moves back through the aortic valve into the left ventricle during diastole
- Caused by diseased valve, congenital defects and chamber or aortic dilatation
- Significant regurgitation results in volume overload in the chamber receiving the retrograde blood flow
- With aortic regurgitation, volume overload occurs in the left ventricle
- Pulsus bisferiens can occur in the carotid arteries
- The waveforms will demonstrate double systolic peaks
- Severe regurgitation is associated with diastolic flow reversal in the carotid arteries (Water Hammer Pulse)

Pulsus Bisferiens



Mitral Stenosis:

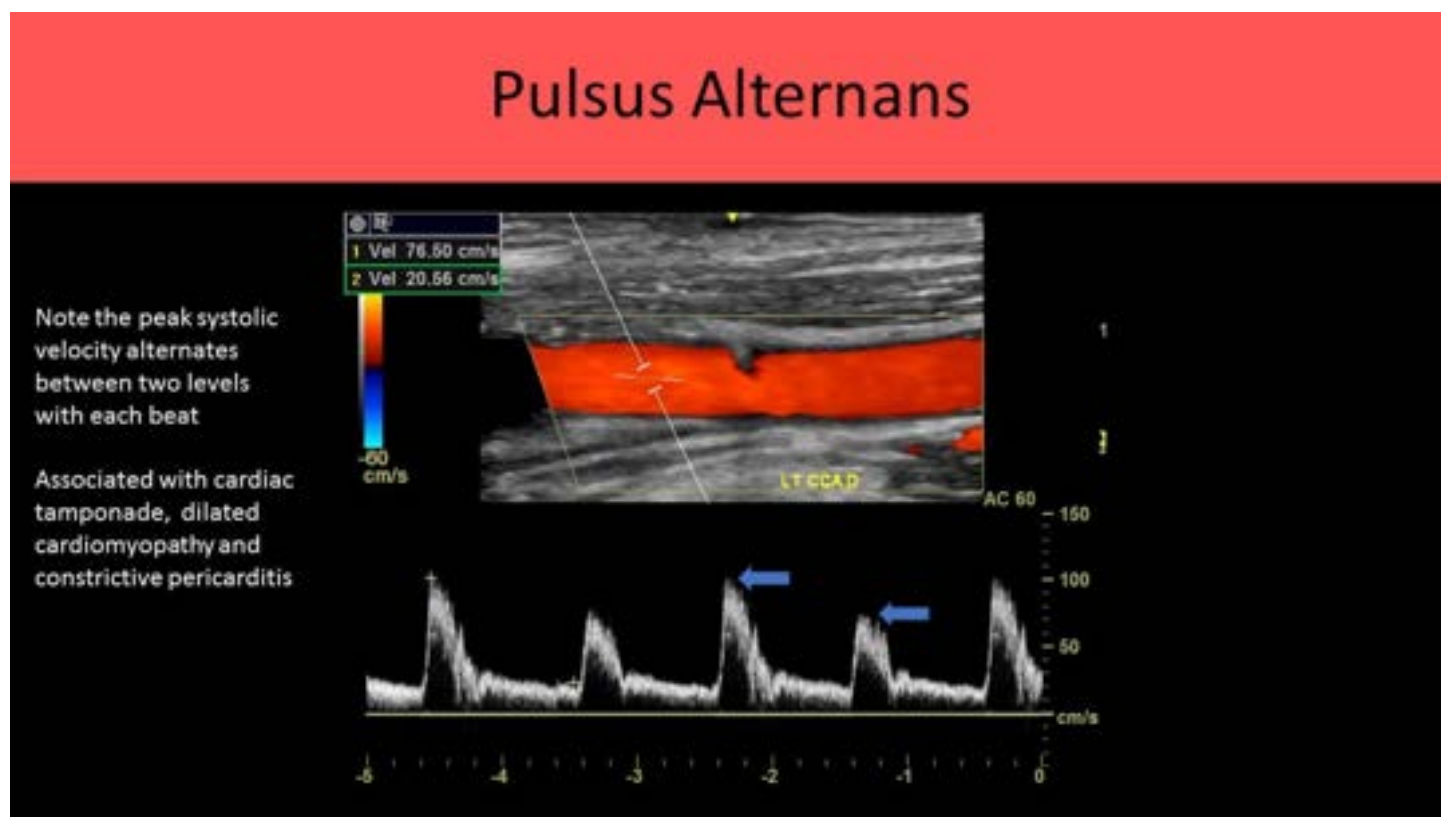
- Usually caused by degenerative disease
- Blood flow is reduced from the left atrium into the left ventricle
- Significant stenosis can cause left atrial pressure overload and reduced cardiac output
- The carotid arteries will demonstrate low resistance, low velocity flow with reduced cardiac output
- Bilateral tardus parvus waveforms in the carotid arteries can indicate severe mitral stenosis

Mitral Regurgitation:

- Blood moves back through the mitral valve into the left atrium during systole
- Caused by diseased valve, congenital defects and chamber dilatation
- Significant regurgitation results in volume overload in the chamber receiving the retrograde blood flow
- With mitral regurgitation, volume overload occurs in the left atrium
- Usually has NO effect on arterial or venous Doppler evaluations

Cardiac Tamponade:

- Critical finding
- Severe pericardial effusion within the fibrous pericardium causes extrinsic compression of the right heart chambers
- Respiratory variation in arterial flow velocities will occur
- Flow diminishes with inspiration and normalizes with expiration
- Another sign of cardiac tamponade is pulsus alternans
- Pulsus Alternans - the peak systolic velocity alternates between two levels with each beat
- Pulsus Alternans is also associated with dilated cardiomyopathy and constrictive pericarditis



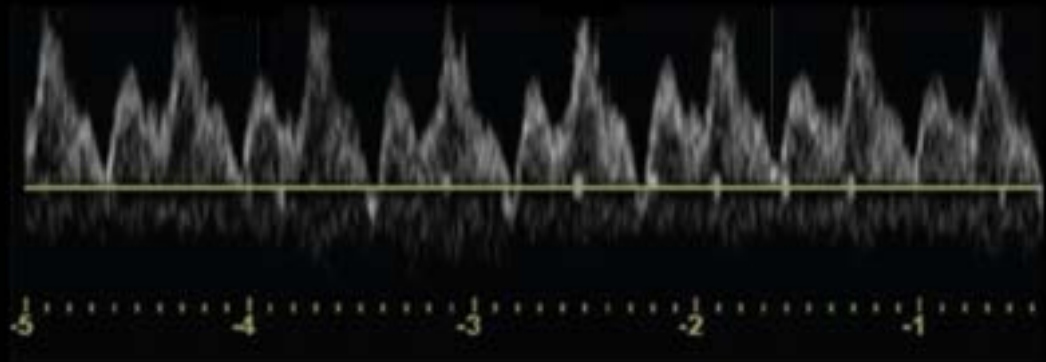
Intra-aortic Balloon Pump (IABP):

- Patients with severely reduced left ventricular function may require an assistive device
- The IABP is implanted in the patient's descending thoracic aorta and expanded and contracted to cause blood to flow throughout the body
- IABP assisted waveforms have 2 systolic peaks for each pulse
- The first peak indicates the left ventricular contraction
- The second peak indicates the balloon inflation

- EDV cannot be determined in the patients with an IABP because flow reversal is commonly seen after the balloon deflation
- The device can be temporarily disabled by a physician to allow the Sonographer to obtain representative waveforms for a more accurate assessment

Intra-aortic Balloon Pump

Carotid Waveform



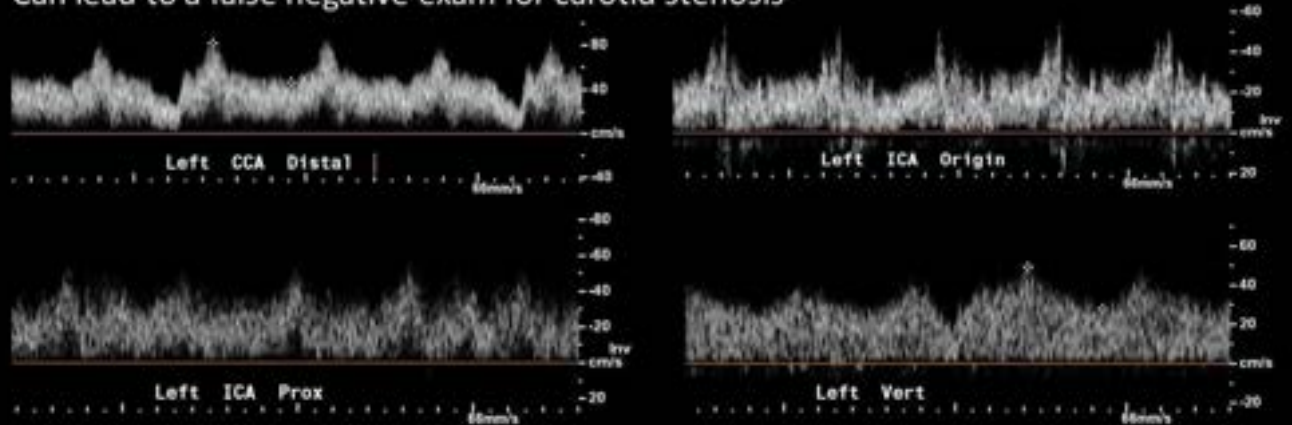
Adapted from: Ginat, D., Massey, H. T., Bhatt, S., & Dogra, V. S. (2011). Imaging of mechanical cardiac assist devices. *Journal of clinical imaging science*, 1, 21. doi:10.4103/2156-7514.80373 Copyright-open license

Left Ventricular Assist Device (LVAD):

- Used for patients with end stage heart failure
- Can prolong life for transplant patients and those not able to undergo transplant surgery
- The device pulls blood from the LV into a pump and the pump sends the blood into the aorta
- Complications include right heart dysfunction, thrombus formation, stroke, hemolysis
- Brachial pressure measurements must be recorded with each change in the pump settings
- Diminished peak systolic velocity, increased diastolic flow
- Mean flow velocity usually remains unchanged with changes to pump settings
- Can cause a false negative exam for carotid stenosis

LV Assist Device – Carotid Doppler

Diminished peak systolic velocity, increased diastolic flow
Mean flow velocity usually remains unchanged with changes to pump settings
Can lead to a false negative exam for carotid stenosis



Courtesy of Cleveland Clinic

Cerebrovascular “Most Commons”

The CCA bifurcation is most commonly assessed for the presence of a bruit

The CCA bifurcation is the most common site of plaque formation in the cerebrovascular system
2nd most common cerebrovascular site is at the origin of the CCA

In most patients the ECA is the anterior medial branch of CCA

Kinking of the ICA is the anatomic variant most commonly associated with symptomatic cerebral ischemia

Ischemic stroke is the most common type of stroke

Strokes are more common in African Americans and in men
Women die from strokes more often than men

Plaque in the carotid bifurcation is most common cause of a stroke, cardiac emboli #2

MCA is the most commonly occluded intracranial vessel with an acute stroke

Bilateral homonymous hemianopia is most commonly associated with an occipital infarct

Distal CCA IMT measurements are most commonly used to assess risk of cardiovascular events

Cerebrovascular “Most Commons”

Most common location of vertebral artery stenosis is at the origin

Subclavian steal most commonly occurs on the left side

Early systolic deceleration in the vertebral artery is most commonly caused by stenosis in the subclavian artery

Dissection is more common in the extracranial ICA than the intracranial ICA

FMD usually occurs in the mid or distal segment of the ICA, while atherosclerosis most commonly affects the proximal ICA

FMD is more common in young women

Carotid body tumor formation is more common in females

Eagle syndrome most commonly bilateral

Nitinol is the most common material used to make carotid stents

Upper Extremity Arterial Anatomy and Evaluation

Aortic Arch:

- First anatomic branches of the ascending aorta are right and left coronary arteries
- Normally arises from the right side of the heart and arches toward the left to course inferior through the chest and abdomen
- Three branches of aortic arch:
 - Innominate Artery - quickly branches into Rt CCA and Rt Subclavian, NO left innominate artery
 - Left CCA
 - Left Subclavian
- Bovine Arch
 - The most common anatomic variant of the arch
 - Only two direct branches of the arch
 - The innominate and left CCA originate as one short segment and then split
 - The second branch is the left subclavian artery
- Other variants in normal anatomy can cause complications
 - Double aortic arch - compresses the anterior aspect of the trachea and the posterior aspect of the esophagus
 - Pulmonary sling - compresses the anterior aspect of the trachea and the posterior aspect of the esophagus
 - Left aortic arch with aberrant right subclavian artery can cause compression of the esophagus
 - Right aortic arch with aberrant left subclavian artery can cause compression of the esophagus

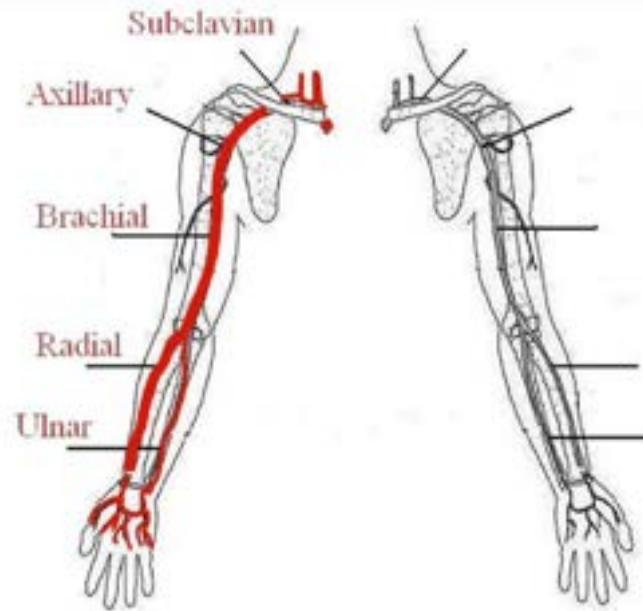
Upper Extremity Arteries:

******Remember the patient is in anatomic position, palms forward, thumbs are most lateral aspect of hands******

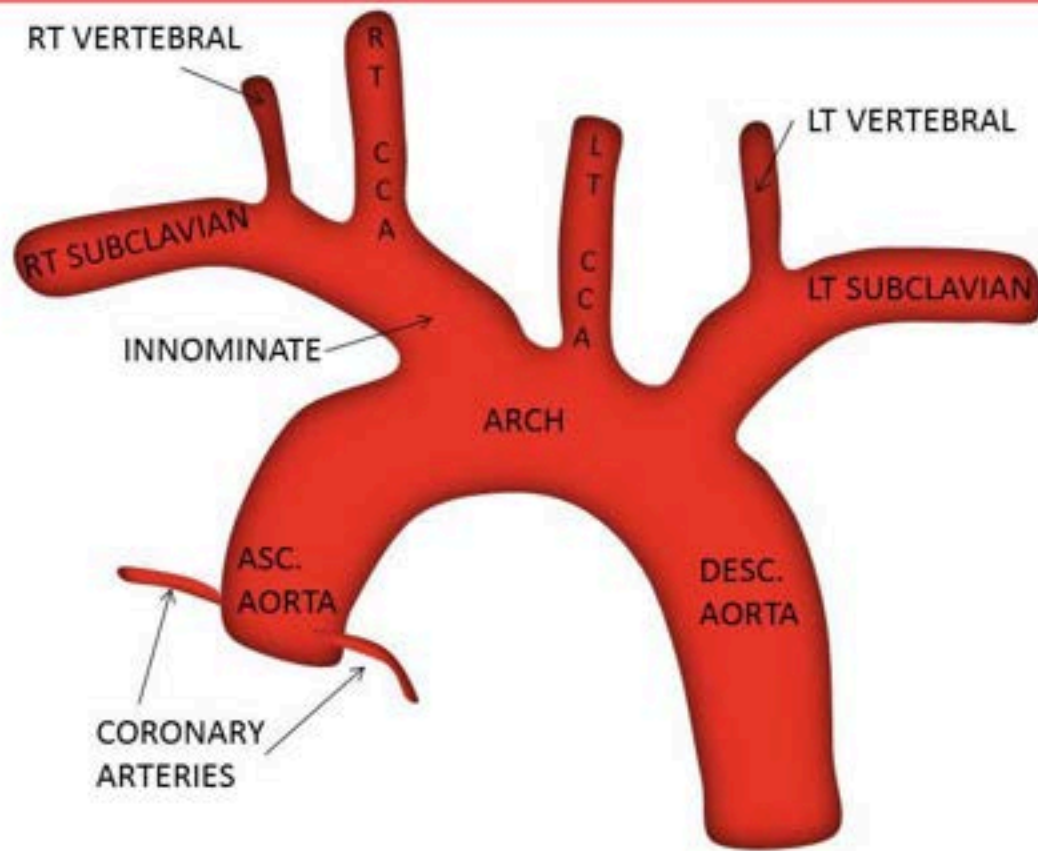
- **Innominate Artery:**
 - First major branch from the arch
 - Also called the brachiocephalic artery or trunk
 - Only ONE innominate artery, may be called the right innominate artery (NO left innominate artery)
 - Courses upwards and rightward, anterior to the trachea
 - Bifurcates posterior to the right sternoclavicular joint
 - The right CCA and right subclavian artery are the only two branches
- **Subclavian Artery:**
 - Right subclavian artery originates from the innominate artery
 - Aberrant right subclavian artery
 - In some patients the subclavian artery can be a direct branch of the aorta distal to the left subclavian origin
 - Called retroesophageal subclavian artery or aberrant subclavian artery
 - Usually originates from a dilated segment of the proximal descending aorta called the Kommerell diverticulum
 - Can cause dysphagia due to compression of the esophagus by the aberrant subclavian artery
 - Also related to Ortner syndrome - palsy of the recurrent laryngeal nerve
 - Left subclavian originates at the aortic arch (3rd branch)
 - Courses posterior to the anterior scalene muscle and clavicle
 - Courses anterior to the apex of the lung
 - Becomes the axillary artery at the level of the first rib
 - Branches from subclavian
 - Vertebral artery
 - Internal mammary artery (AKA internal thoracic artery)
 - Thyrocervical and Costocervical Trunks - can be differentiated from the vertebral arteries by their many branches and lower end-diastolic flow velocities
 - Triphasic waveform, high resistance
 - Flow velocity varies
 - Compare the velocities in both subclavian arteries; High peak systolic velocity can indicate stenosis in that artery
 - Biphasic or monophasic flow patterns associated with obstruction
- **Axillary Artery**
 - Portion of the artery in the axilla

- Becomes brachial artery just outside axilla
- Branches:
 - Superior artery
 - Thoracic artery
 - Thoracoacromial artery
 - Lateral thoracic artery
 - Subscapular artery
 - Anterior and posterior humeral artery
 - Thoracodorsal artery
- Brachial Artery
 - From distal axilla to antecubital fossa
 - Divides into radial and ulnar arteries just proximal to the antecubital fossa
- Radial Artery
 - Courses laterally on forearm
 - Branch joins the superficial palmar arch
 - Terminates at the deep palmar arch
- Ulnar Artery
 - Courses medially on forearm
 - Branch joins the deep palmar arch
 - Terminates at the superficial palmar arch
 - Supplies majority of the blood to the hand
- Digital Arteries
 - Originate from both palmar arches
 - Distribute blood to the fingers

Upper Extremity Anatomy



Aortic Arch



Physical Examination:

- **Palpation:**
 - Most arteries are palpated by placing two fingers over the artery
 - The radial and ulnar arteries are palpated to evaluate upper extremity pulses
 - If a hemodialysis graft is present, it should be assessed for the presence of a thrill (normal finding)
 - Loss of the thrill in a dialysis graft usually indicates occlusion
- **Auscultation:**
 - Commonly used to assess hemodialysis grafts
 - A graft stenosis can cause a bruit
 - A pseudoaneurysm will also cause a bruit
 - Water hammer pulse - associated with an occluded graft
- Evaluate color of the extremity when elevated and dependent

- Assess the skin and nails
- Test the capillary blush response
 - Press finger on extremity to create an area of pallor, time the delay until color returns
 - Used to assess ischemia in the distal extremity in patients with hemodialysis graft
 - Delayed capillary blush response indicates significant ischemic disease is present

UPPER EXTREMITY ARTERIAL EXAM TECHNIQUES:

- Exams are tailored for indication listed
- Raynaud disease-PPG before and after stimuli in affected area
- TOS-movements used to determine obstruction by anatomical structures
- Atherosclerosis-Doppler US and segmental pressures/PVR obtained

Upper Extremity Duplex Evaluation:

- 5-7MHz Linear probe used to evaluate the extremity vessels
- Patient supine with arm at 45-degree angle and externally rotated
- Most commonly performed to evaluate hemodialysis graft
- Atherosclerotic disease in the upper extremities is very uncommon; if present it is usually small vessel disease
- 2D images performed to evaluate atherosclerosis formation and vessel wall characteristics in the subclavian, axillary, brachial, radial and ulnar arteries
- Color and PW Doppler used to evaluate blood flow in the same vessels above
- The subclavian artery should be evaluated as close to its origin as possible (plaque formation is most common at the origin or bifurcation of a vessel)
- Velocity ratio can be used to assess disease; PSV at stenosis divided by the PSV proximal to the stenosis
- A ratio > 2 indicates hemodynamically significant disease in the vessel with the elevated velocity
- Segmental pressures and/or PVR are usually included in exam

Upper Extremity Arterial Duplex



Right

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Upper Extremity Arterial Duplex

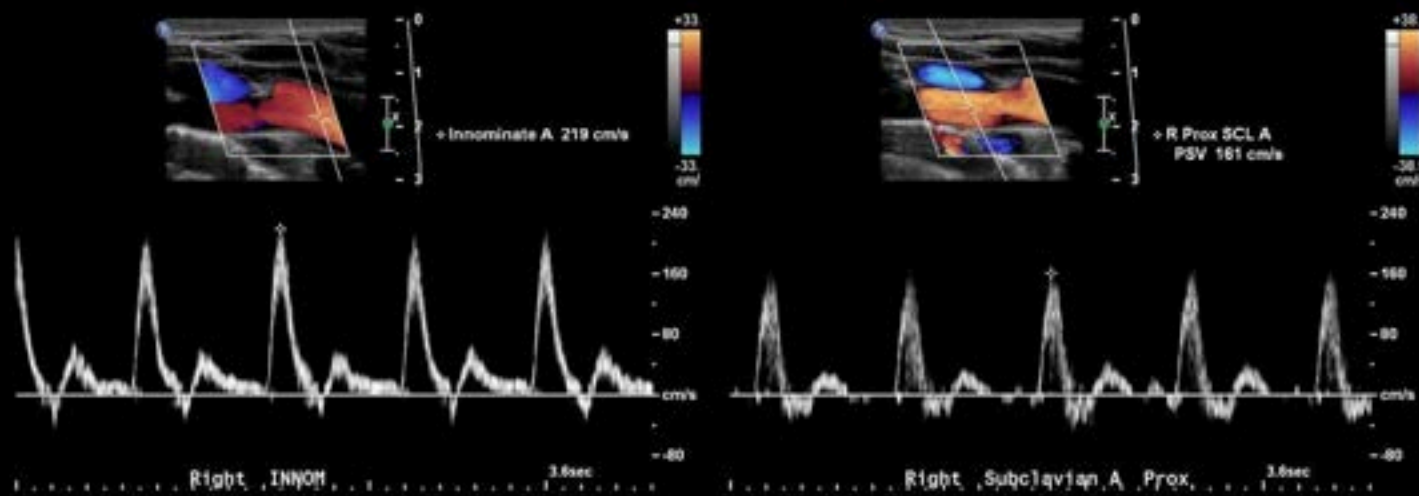


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Upper Extremity Arterial Duplex

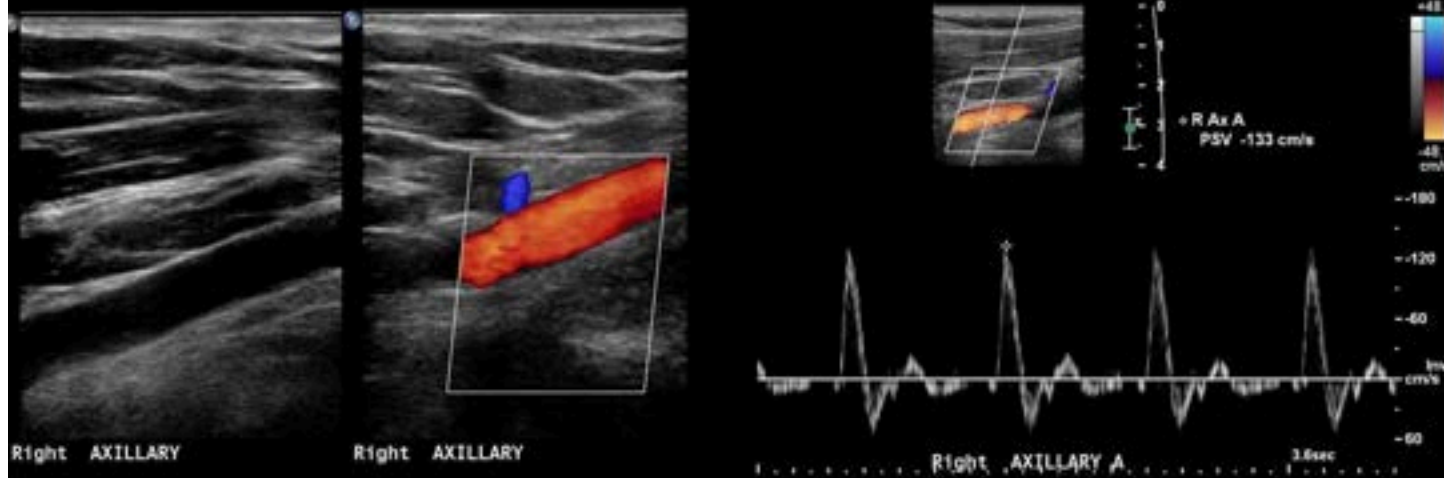


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Upper Extremity Arterial Duplex



Segmental Pressure Evaluation:

- Patient is examined in the supine position
- CW Doppler transducer usually used for the Doppler signal
- Cuffs placed snugly on upper arm, forearm and second digit
- Inflate the cuff to 20-30mmHg above the last audible signal
- Use the brachial artery to obtain the brachial pressure
- Use the radial artery to obtain the radial pressures in the forearm (and wrist if indicated by your lab)
- Use the ulnar artery to obtain the ulnar pressures in the forearm (and wrist if indicated by your lab)
- Finger pressures should be obtained with the patient sitting upright, hand placed on a positioning wedge or pillow for stability

- If a pressure measurement must be repeated, the cuff should be fully deflated for at least one minute before it is reinflated for the second time
- Record and compare pressure measurements
- Calculate the Wrist/Brachial Index using highest wrist pressure divided by the highest brachial pressure

Example:	RT	LT
Brachial	140	134
Radial	142	80
Ulnar	146	76
WBI	$146/140 = 1.04$	$80/140 = 0.57$

Disease suspected between LT Brachial & Forearm arteries. Because both forearm arteries are affected, the obstruction is most likely in the brachial artery segment that is distal to the brachial cuff

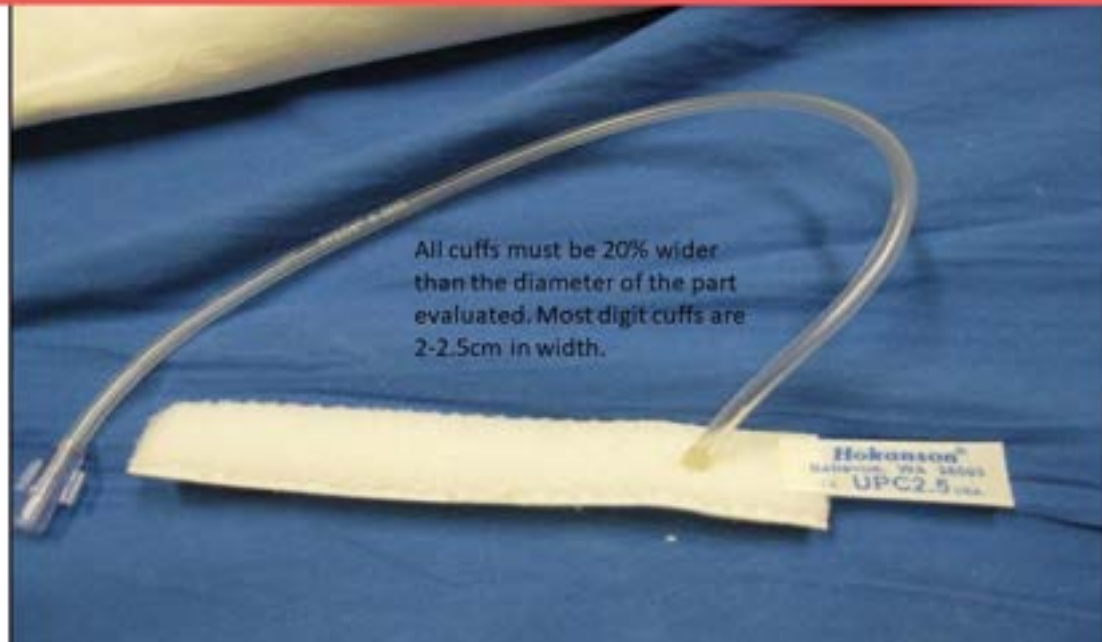
Upper Extremity Segmental Pressure Examination	
Normal	Abnormal
< 20mmHg difference between the same segments on both arms EX: RT Brachial 126mmHg LT Brachial 130mmHg	> 20mmHg difference between the same segments on both arms EX: RT Brachial 126mmHg LT Brachial 150mmHg
< 15mmHg difference between two consecutive segments of the same arm EX: RT Brachial 136mmHg RT Radial 130mmHg	> 15mmHg difference between two consecutive segments of the same arm EX: RT Brachial 136mmHg RT Radial 110mmHg
<10mmHg difference between the radial and ulnar arteries	> 10mmHg difference between the radial and ulnar arteries
	10-30mmHg difference between the radial and ulnar pressures indicates suspected disease
	> 30mmHg difference between radial and ulnar pressures confirms vessel with lower pressure is diseased
WBI ≥ 1.0	WBI < 1.0
	>0.15 difference in WBI on serial exams indicates disease progression

- Raynaud syndrome, atherosclerosis, Buerger disease and thoracic outlet syndrome all can lead to reduced digital pressures
- Finger/Brachial Index = Abnormal < 0.8
- Difference in pressure between digits > 15mmHg indicates reduced flow to the digit with the lower pressure
- Reduced single digit pressure with normal arm and wrist pressures indicates disease of the palmar arch or the digital artery with the lower pressure
 - When the arm pressures are normal but the digital pressure is abnormally reduced, there could be disease in the palmar arch or the single digit
 - To provide a complete evaluation, at least one more digit on the left hand should be evaluated
 - If the additional digit pressure is abnormal, disease in the palmar arch is the most likely cause for the reduced digit pressures
 - If the additional digit pressure is normal, the remaining digits should be tested to confirm that the reduced pressure is limited to the single digit and disease of that digital artery should be suspected
- Reduced pressure in the digits of one side of the hand indicates palmar arch disease
- Reduction in pressure of all the digits indicates disease in the radial and ulnar arteries or the palmar arch

Upper Extremity PVR



Digital Cuff



Pulse Volume Recording (PVR):

- Cuffs used to assess volume changes
- Most commonly used to assess the upper extremity for thoracic outlet syndrome
- Finger tracings should be obtained with the patient sitting upright, hand placed on a positioning wedge or pillow for stability
- Abnormal PVR tracings with stenosis or TOS - prolonged upstroke, rounded peak, loss of dicrotic notch, delayed downslope

Photoplethysmography(PPG):

- Finger tracings should be obtained with the patient sitting upright, hand placed on a positioning wedge or pillow for stability
- Sensors used to assess volume changes are placed on the fingers
- Commonly used to assess the upper extremity for thoracic outlet syndrome and Raynaud syndrome
 - Abnormal PPG tracings with stenosis or TOS - prolonged upstroke, rounded peak, loss of dicrotic notch, delayed downslope
 - Flatline tracing with occlusion with no collateral pathways
 - Raynaud syndrome will produce a tracing with a double peak with stimulation

- Manual compression of the forearm arteries can aid in determining structure/function of the palmar arch

Allen Test:

- Used to evaluate patency of the palmar arch for arterial harvesting of the radial artery
- Abnormal results can indicate arterial obstruction of the palmar arch or an incomplete palmar arch
- Radial artery compressed while fist clenched to induce pallor and then while maintaining radial compression the fist is relaxed to see if color returns (normal)
- If pallor remains, the ulnar artery is not patent and the radial artery cannot be harvested without causing damage to the hand circulation
- PPG sensors can be placed on the thumb or index finger to monitor flow changes during arterial compression
- Radial arterial compression should not cause any changes in the PPG tracing if the ulnar artery is patent
- If the ulnar artery is diseased, the PPG waveform amplitude will diminish or flatline completely
- Digital pressure cuffs could be used to assess blood pressure changes with radial artery compression
- Radial arterial compression should not cause any changes in the blood pressure readings if the ulnar artery is patent
- If the ulnar artery is diseased, the pressure readings will diminish significantly

Cold Sensitivity Testing:

- Used to assess patients with symptoms related to exposure to cold temperature
- PPG tracings obtained from the digits before and after immersion in ice water for up to 3 minutes
- Document patient signs and symptoms after stimuli used
- Obtain tracings just after immersion and after 5 minutes
- Immediately after immersion the amplitude of the waveform will decrease
- Abnormal if the amplitude of the waveform does not return to normal in five minutes

Radial Artery Mapping:

- Radial artery can be used for coronary bypass procedure
- Usually harvested from the non-dominant arm
- Diameter over 2mm can be used, but >2.5mm is preferred
- Preferred over GSV for coronary bypass because:

- Lumen size is suitable for procedure
 - Thicker wall layers
 - Easier to access for harvesting
- Contraindications include stenosis or occlusion of the radial artery and/or other arteries in the extremity, digital ischemia, incomplete palmar arch, and Raynaud syndrome
- Protocol
 - Evaluate the radial artery for stenosis
 - Measure the inner diameter of the proximal, mid and distal radial artery
 - Note the position of the brachial artery bifurcation related to the antecubital fossa (above or below)

Upper Extremity Arterial Abnormalities

Thoracic Outlet Syndrome (TOS):

- Muscular band, cervical rib or fibrous band crosses the brachial plexus in an abnormal location that causes compression with certain arm maneuvers
- Most patients asymptomatic
- May see dull aching pain, radiating from point of compression
- May see paresthesia/numbness, pain, increased symptoms with certain arm/neck positions or exercise
- May see extremity swelling with venous TOS
- Symptoms usually intermittent and vary with patient position

Types of TOS:

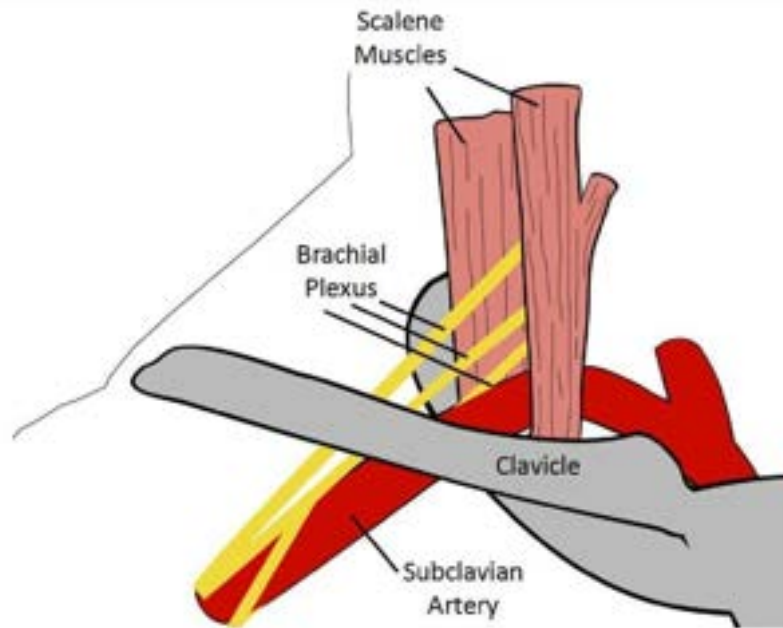
- Neurogenic TOS
 - Most common type (>90%)
 - Compression of the brachial plexus from cervical ribs, anterior scalene muscles and ligaments
 - Causes paresthesia, numbness, pain
 - Ultrasound evaluation will not identify the cause of the symptoms if only the nervous system is affected
- Arterial TOS
 - Least common (1%)
 - Subclavian artery compression from the head of the humerus
- Venous TOS
 - Subclavian vein compressed against the first rib and scalenus anticus muscle with abduction of the arm
 - Also called Paget-Schroetter syndrome or effort thrombosis

Exam Technique:

- Necessary to evaluate the patient first for atherosclerosis
- Establish normal flow patterns for the patient and then evaluate the changes in flow with changes in position
- Patient is seated with palms up and resting on the thighs
- Record the waveforms with the sweep speed reduced to better demonstrate changes with maneuvers
- The waveforms should be similar or larger than the resting position waveforms
- Reduction in amplitude of the waveform or "flatline" waveform indicates a positive exam
- PPG or PVR evaluation methods are preferred over Doppler (transducer can move when patient moves which negatively affects the evaluation)
- Perform PPG tracings of the 2nd digit (index finger) at rest and with maneuvers
- Perform PVR tracings with a brachial cuff at rest and with maneuvers
- Doppler evaluation - arterial
 - Evaluate the radial artery at the wrist
 - Obtain Doppler samples
 - Perform TOS maneuvers
 - Repeat Doppler samples
- Doppler evaluation - venous
 - Evaluate the distal subclavian vein
 - Establish baseline flow with Doppler
 - Perform TOS maneuvers

- Repeat Doppler samples

Normal Thoracic Outlet Anatomy



Tos

Hhs

uev cs

Takayasu

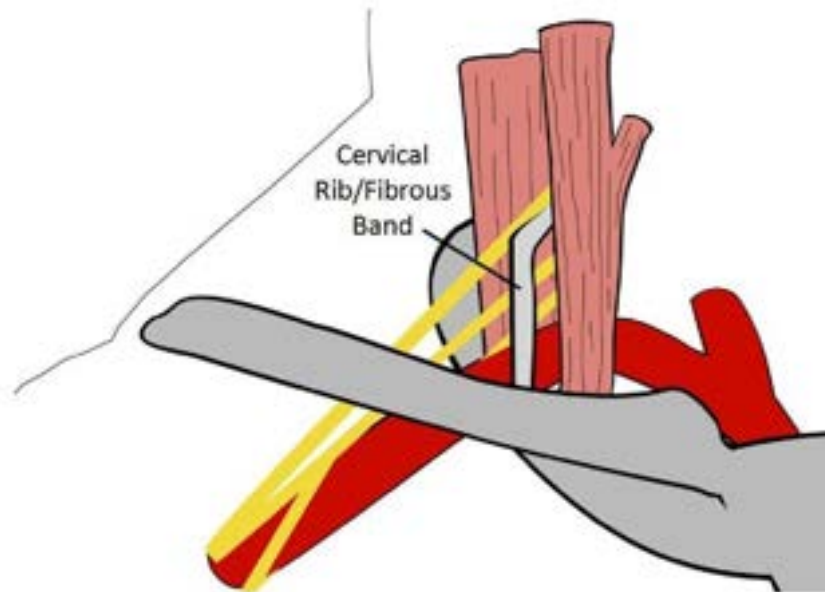
Raynaud

Atherosclerotic

Av malformation

Aneurysm

Abnormal Thoracic Outlet Anatomy



Neutral Position - TOS

Use this position to obtain baseline tracings before performing any maneuvers to initiate symptoms. Always record the positions that are used to evaluate the patient. When symptoms or waveform changes occur, note the patient position that caused the changes.



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90 Degree Abduction - TOS

Raise the arm up with hand at same level as the shoulder, palm up. Turn the head toward the affected arm and then away, while recording waveforms.



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180 Degree Abduction - TOS

Raise the arm straight up with the palm toward the head. Turn the head toward the affected arm and then away, while recording waveforms.



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Adson Maneuver - TOS

From neutral position, ask the patient to sit tall with their chest forward. Instruct them to take in a deep breath and hold it. Ask the patient to turn their head toward the affected side, extend the arm and raise it slightly. Finally, the patient should be evaluated with their head turned toward the unaffected side. Note symptoms and waveform changes that occur at any time during the maneuver.



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Costoclavicular Maneuver - TOS

From neutral position, push the elbows back, shoulders down and stick the chest out. Face forward with chin level. Turn the head toward the affected side and away, while recording waveforms.



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TOS Maneuvers:

- Inspiration/Expiration used with the change of position in some maneuvers
- 1. Resting neutral position, hand on lap
- 2. Arm at 90-degree angle to body
- 3. Arm at 180-degree angle to body
- 4. Adson's maneuver
- 5. Exaggerated military posture (costoclavicular)
- 6. Neutral position with head left/head right maneuver
- 7. Any position that the patient experiences symptoms

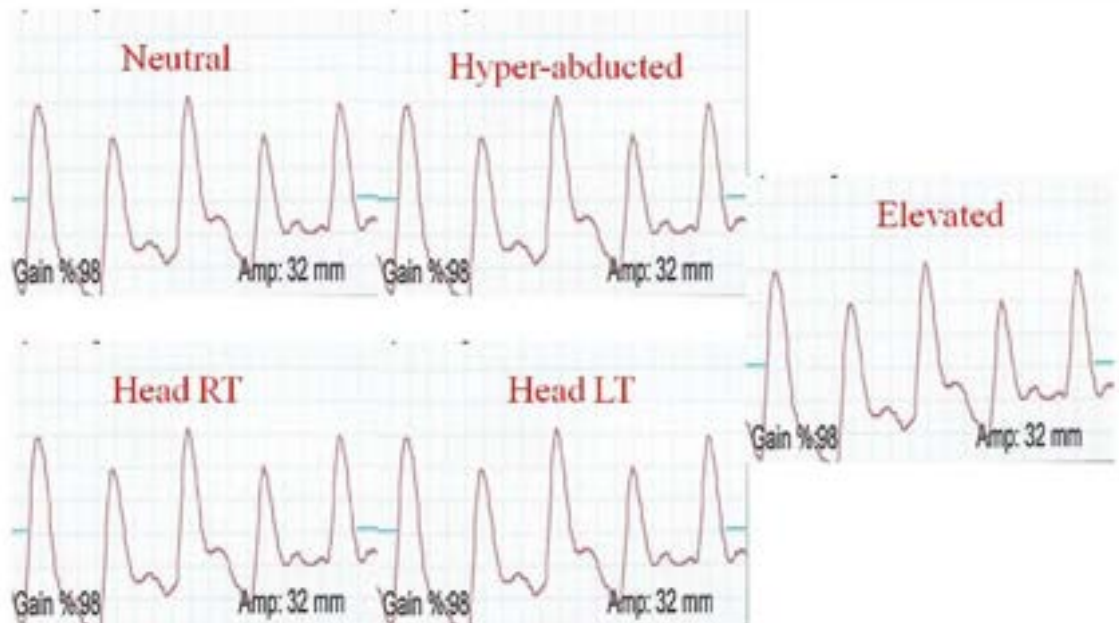
Adson's Maneuver:

- Palpate the radial pulse on the affected side with the elbow fully extended \Have the patient rotate their head to the side being tested and extend the neck
- Then abduct, extend, and laterally rotate the shoulder
- From this position, have the patient take a deep breath and hold
- Assess the pulse response
- A positive test is a decrease in pulse vigor from the starting position to the final position

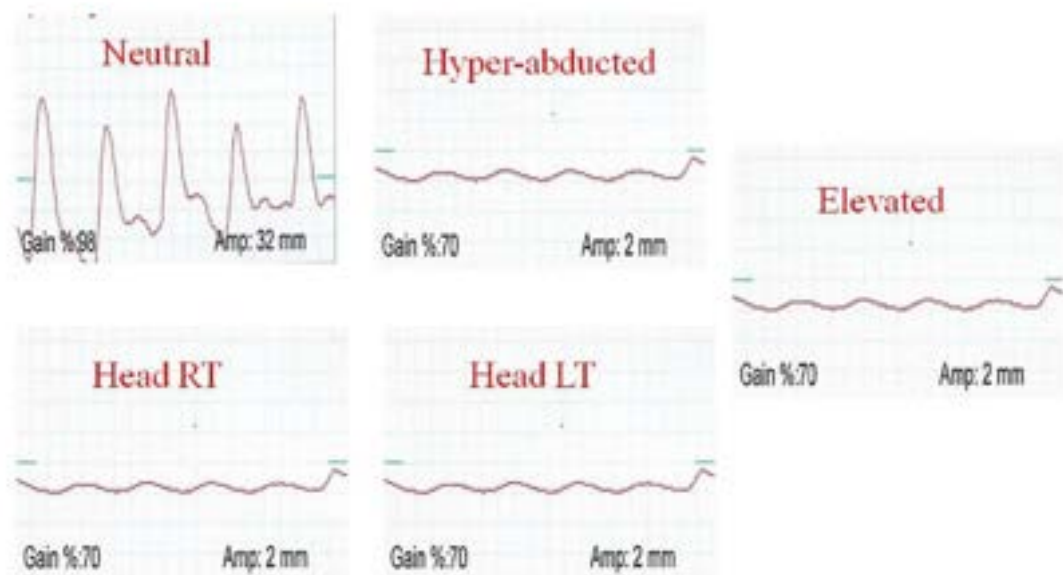
Abnormal Findings:

- PPG/PVR - significantly reduced amplitude or absent signal with patient maneuvers
- Arterial Doppler - significantly reduced or absent signal with patient maneuvers
- Venous Doppler - venous signal will become continuous or disappear
- When a change in the waveform is noted, ask the patient if they are experiencing symptoms
- Abnormal exam results + patient symptoms = positive exam for TOS

Normal PVR Exam – No TOS



Abnormal PVR Exam – Positive for TOS



Hypothenar Hammer Syndrome:

- Ulnar artery is commonly affected because it crosses over the hook of hamate bone
- Can be caused by repetitive compression or blunt trauma
- Associated with Raynaud syndrome
- Pain in palm of hand and with cold sensation
- Most commonly causes ischemia of fourth and fifth fingers
- Positive Tinel sign = distal paresthesia induced by compression of the affected portion of the entrapped ulnar nerve
- Positive Allen test
- Occlusion of the distal ulnar artery
- May also see irregular contour or aneurysm of the ulnar artery

UE Vascular Compression Syndromes:

- Variants in normal anatomy can cause complications
- Double aortic arch - compresses the anterior aspect of the trachea and the posterior aspect of the esophagus
- Pulmonary sling - compresses the anterior aspect of the trachea and the posterior aspect of the esophagus
- Left aortic arch with aberrant right subclavian artery can cause compression of the esophagus
- Right aortic arch with aberrant left subclavian artery can cause compression of the esophagus
- Innominate artery compression syndrome - tracheal compression by the innominate artery occurring in children; normal growth usually resolves the tracheal compression

Takayasu Arteritis:

- Presents as hypertrophic area(s) of inflamed tissue usually found in the aorta and its branches
- Associated with supralvalvular stenosis and aortic coarctation
- Commonly affects the subclavian and carotid arteries
- Can also affect the renal arteries
- Most commonly seen in young females

Raynaud Syndrome and Phenomenon:

- Vasospastic disorder that presents as a response to cold or emotional stimuli
- >70% occurs in females
- Can be associated with lupus, scleroderma, rheumatoid arthritis
- Primary
 - Called Raynaud Syndrome/Disease
 - Occurs in a NORMAL vascular system
 - Bilateral digital arterial spasm - INTERMITTENT
 - More common in women
 - Symptoms only occur when arterial spasm is present and are present for more than 2 years
- Secondary
 - Called Raynaud Phenomenon
 - Occurs in ABNORMAL obstructed system with vasoconstriction causing constant ischemia
 - May be unilateral
 - Pain is usually CONSTANT with significant arterial obstruction

- In contrast to Raynaud disease, Raynaud phenomenon presents concurrently with CREST syndrome, in which tissue necrosis results from spastic and occlusive disease
- CREST syndrome involves symptoms of generalized calcinosis, Raynaud phenomenon, esophageal dysfunction, scleroderma, and telangiectasia.
- Vasoconstriction of digital arteries occurs in response to stimuli
- Cold and emotional stimuli are common triggers
- Stimuli causes a series of color changes in the affected area; after the application of stimuli (cold) the area will turn WHITE due to ischemia caused by vasoconstriction, then BLUE due to cyanosis in the tissues and finally RED with vasodilation causing a localized "flushing" phenomenon
- More common in hands than feet
- Exam Protocol:
 - Rule out arterial stenosis and evaluate vascular response to stimuli
 - Patient evaluated in a sitting position
 - Evaluate the brachial, radial and ulnar arteries with Doppler and PVR equipment to rule out stenosis
 - Obtain the brachial pressures
 - Place the pressure cuff around the middle phalanx of the digit being evaluated and obtain pressure
 - Calculate the baseline Digit/Brachial Index using the highest brachial pressure
 - Place the PPG sensor on the pad of the digit being evaluated
 - PPG testing of digits at rest (evaluate for signs of proximal stenosis)
 - If the baseline PPG tracings are NORMAL and symptoms are caused by stimuli, cold sensitivity testing should be performed
 - Submerge hand in ice water up to 1-3 minutes (place hand in plastic bag)
 - Record submersion time
 - PPG testing immediately after stimuli removed
 - PPG testing 5 minutes after stimuli removed
 - Immediately after immersion the amplitude of the waveform will decrease
 - Abnormal response if the amplitude of the waveform does not return to normal in five minutes
 - Decreased flow with double peaked waveform unique characteristic of Raynaud syndrome
 - If the baseline PPG tracings are ABNORMAL, cold sensitivity testing should NOT be performed
 - Wrap the hands in a heating pad for 5 minutes
 - If the tracings return to normal flow patterns, secondary Raynaud Phenomenon is present

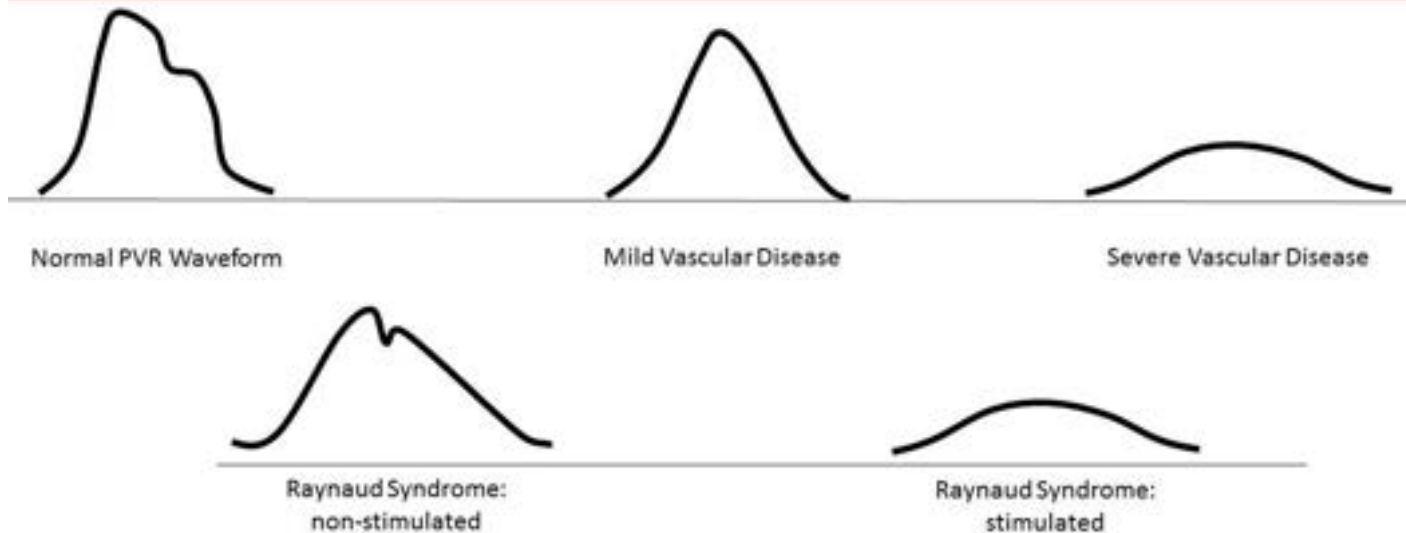
- If the tracings remain abnormal (like the baseline), small vessel occlusive disease should be suspected
- ***To reduce light from the exam room from causing artifact on the PPG tracing, cover the patient's hand with a towel or sheet

Raynaud Phenomenon



Adapted from: Voelter-Mahlknecht S, Rossbach B, Schleithoff C, Dransfeld CL, Letzel S, Mahlkecht U Sirtuin1 single nucleotide polymorphism (A2191G) is a diagnostic marker for vibration-induced white finger disease. Clin Epigenetics (2012)

PVR Waveforms



As disease progresses, PVR tracing amplitude decreases and the waveforms become elongated. Raynaud disease will demonstrate a double systolic peak due to increased vascular bed resistance.

Atherosclerotic Disease of the Upper Extremities:

- Uncommon finding
- Upper extremities seem to be somewhat "resistant" to atheroma formation
- Signs and symptoms are the same as PAD in the lower extremities
- Subclavian steal can be related to atheroma formation in the subclavian artery
- Normal Doppler waveforms demonstrate a high resistance, triphasic waveform with a spectral window

Collateral Formation:

- Subclavian obstruction - subclavian steal, abdominal and thoracic wall arteries
- Axillary obstruction - thoracic wall or shoulder
- Brachial obstruction - forearm, palmar arches with retrograde filling

AV Malformation or Fistula:

- AV malformation
 - A congenital malformation causing connection between the arterial and venous system

- It is usually an abnormal connection between arterioles and venules without a normal capillary bed between them
- Usually associated with multiple arterio-venous connections and dilated venous collaterals
- AV fistula
 - An acquired condition where arterial and venous flow are mixed
 - A traumatic or a surgically created AV fistula will usually only have a single direct connection between an artery and a vein and NO venous collaterals are present (acute disorder)
- Proximal arterial flow is low resistance with increased diastolic flow
- Venous outflow is pulsatile and turbulent
- Large or long-term AVM or AVF can lead to heart failure
- A centrally located AVF will usually lead to CHF, while a peripheral AVF will usually cause distal ischemia

Aneurysm:

- Uncommon finding in the upper extremity
- Subclavian aneurysms can be associated with distal embolization to the fingers
- Can be associated with chronic mechanical stressing of a vessel; secondary to TOS or hypothenar hammer syndrome

Upper Extremity Arterial “Most Commons”

Upper extremity arterial duplex exam is most commonly performed to evaluate hemodialysis graft

Pulse volume recording is most commonly used to assess the upper extremity for thoracic outlet syndrome

Photoplethysmography is commonly used to assess the upper extremity for thoracic outlet syndrome and Raynaud syndrome

Thoracic outlet syndrome is most commonly caused by compression of the brachial plexus nerves (Neurogenic TOS; non-vascular cause)

Hypothenar Hammer Syndrome commonly affects the ulnar artery because it crosses over the hook of hamate bone

Hypothenar Hammer Syndrome most commonly causes ischemia of fourth and fifth fingers (fed by ulnar artery)

Raynaud Syndrome is more common in women and in the hands (compared to the feet)

Upper Extremity Arterial “Most Commons”

Upper extremity arterial duplex exam is most commonly performed to evaluate hemodialysis graft

The venous anastomosis is the most common site of stenosis in a hemodialysis graft

Steal syndrome most commonly occurs in a radiocephalic AV fistula

The most common cause of hemodialysis graft failure is thrombosis

Hemodialysis Access Graft

Hemodialysis AV Fistula/Graft:

- Surgically created connection between artery and vein for hemodialysis
- Hemodialysis removes creatinine, urea and water from blood of patients in end-stage renal failure
- AV fistula is a direct connection between an artery and vein that is created to allow access point for dialysis port
- An AV graft uses a synthetic tube to connect an artery and vein to allow access point for dialysis port
- The closer the fistula to the heart, the greater the risk of developing heart failure
- This is why they are placed at the antecubital fossa and not the shoulder
- The vessels at the antecubital fossa offer the minimum sizes needed for the native vessels, while increasing the distance from the heart
- Pre-op assessment:
 - Needed to evaluate native vessels for selecting appropriate type of graft
 - Obtain bilateral brachial pressures - <20mmHg difference indicates normal findings
 - The brachial and radial arteries are evaluated for stenosis or variation in normal anatomy
 - Vessel diameters are measured from inner wall to inner wall
 - requires native vein >2.5mm for AVF
 - native vein >4mm for synthetic graft
 - native artery >2mm
 - A tourniquet can be placed on the upper arm to dilate the veins to assess maximum diameter
 - Veins should be evaluated to determine they have a straight course, are located within 1cm of skin surface and are free of obstruction
 - Requires palmar arch flow assessment (Allen Test) prior to bypass graft harvesting and insertion of a dialysis fistula
 - Distal radial artery should demonstrate >50cm/s peak flow velocity

- Reactive hyperemia can be performed to assess the feeding artery for appropriate increase in arterial diameter and vessel compliance for fistula placement and maturation
 - Clenching the ipsilateral fist during an upper extremity Doppler evaluation should increase distal resistance and pulsatility in the proximal arteries (high resistance, triphasic flow)
 - The clenching should be held for 2 minutes
 - Upon release of the clenched fist, the distal resistance drops significantly and flow increases to the hand (low resistance, monophasic flow)
 - The resistive index is measured and a value over 0.7 indicates the feeding artery will not work for a successful AVF creation

Indications for Post-Op AVF/AVG Evaluation:

- Dialysis equipment provides information regarding the functionality of the graft
 - Dynamic venous pressure (>200mmHg ABNL)
 - Access recirculation (>12% ABNL)
 - Urea reduction ratio (<60% ABNL)
- Palpable mass
- Significant hand pain
- Venous HTN
- Water hammer pulse - thumping pulse palpated with an acute occlusion of a graft; an occluded graft must be replaced, CANNOT be recanalized
- *Decreased or absent thrill*; Patent graft will produce a thrill - *not an indication of normal or abnormal, just patency*

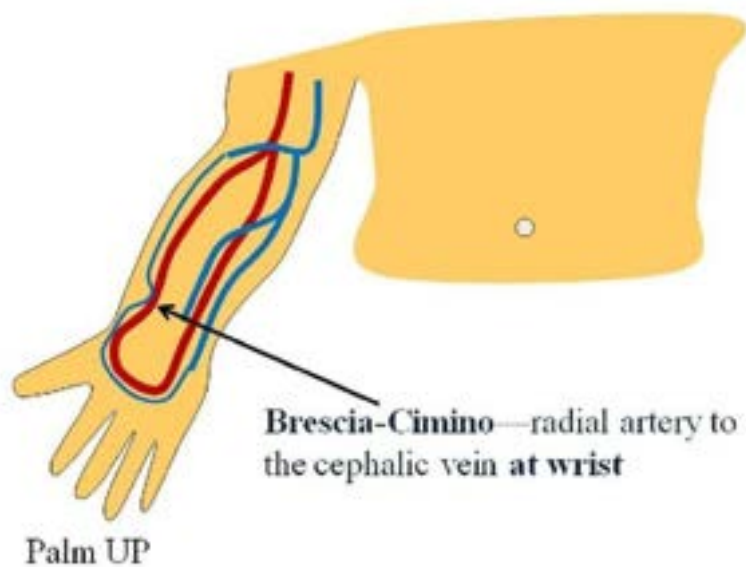
Types:

- Usually performed on the NON-dominant arm of the patient
 - Brescia-Cimino
 - Most common
 - Radial artery to the cephalic vein at wrist
 - "Snuffbox" fistula
 - The anatomical *snuffbox* is also called the radial fossa
 - Triangular depression on the lateral aspect of the dorsum of the hand
 - The radial artery, a branch of the radial nerve, and the cephalic vein are found in the snuffbox
 - Radial artery is connected to the cephalic vein at distal wrist

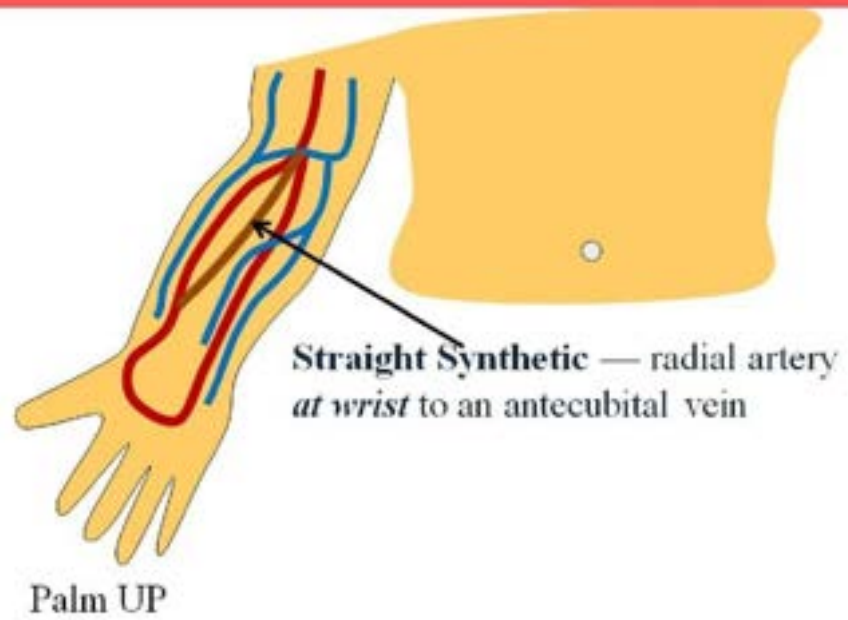
- Because the artery courses directly over the vein in this location, little vessel movement is required during surgery
 - Brachiocephalic - brachial artery and antecubital vein at elbow
 - Brachiobasilic - brachial artery and basilic vein at elbow
 - Radiobasilic - radial artery and basilic vein in the forearm
- Synthetic Graft
 - Used when native veins are inadequate or an AVF has failed
 - PTFE Gore-tex graft - made of polytetrafluoroethylene (PTFE), also known as Teflon
 - Dacron graft - made of synthetic polyester
 - Shorter duration of use and lower patency rates than AVF
 - Straight Synthetic Graft -most common; brachial artery to basilic vein in upper arm
 - Looped Synthetic Graft -most common; brachial artery to antecubital or cephalic vein at elbow; loop extends distally to wrist

*****Blood pressure cannot be taken on an arm with a dialysis graft/fistula present!!**

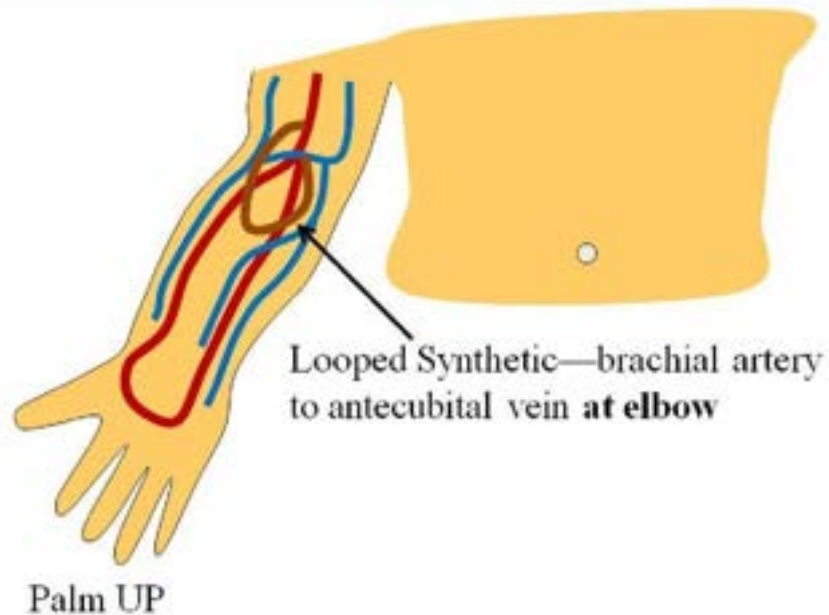
Hemodialysis Fistula



Hemodialysis Graft



Hemodialysis Graft



Exam Protocol:

- 7 - 12.5 MHz linear transducer
- Patient supine or seated with the arm extended and supported
- Arm externally rotated and placed at a 45 degree angle to the body
- Manually palpate the graft for the expected "thrill"
- 2D evaluation performed with high frequency transducer
- Color and PW Doppler evaluation performed with lower frequency transducer due to expected high velocity flow
- AV Fistula has a single anastomosis site; evaluate inflow artery proximal and distal to the fistula, fistula, outflow vein proximal and distal to the fistula
- AV Graft has two anastomosis sites; artery to graft and graft to vein
 - Inflow artery
 - Arterial anastomosis
 - Body of the graft in several locations
 - Venous anastomosis - #1 SITE OF STENOSIS!!!, more common in graft than AVF
 - Outflow vein - Also common site of stenosis
 - Evaluate for aneurysm, thrombus formation, perigraft fluid collection

- Measure the diameter at any area of suspected stenosis or aneurysm
- Use color Doppler to identify areas of turbulence, graft leak, stenosis
- Use PW Doppler to evaluate the native vessels, body of the graft and anastomosis sites

Duplex Evaluation:

- Venous flow proximal to an AV fistula or graft becomes pulsatile and turbulent due to the inflow of arterial flow distally
 - EX: radial artery to brachial vein graft will cause the subclavian vein to demonstrate high velocity, pulsatile flow
- Arterial flow proximal to AV fistula should be low resistance with increased diastolic flow
- Pressure will decrease in artery distal to fistula
 - EX: radial artery to brachial vein graft will cause the radial artery to demonstrate low resistance flow with increased diastolic flow proximal to the graft and immediately distal to the graft
- Normal flow peak systolic velocity within the graft should be 100 - 400cm/s
- End diastolic flow velocity should be 60 - 200 cm/s
- Abnormal findings:
 - Graft stenosis suspected with velocities greater than 400cm/s
 - Velocity ratio = PSV at stenosis / PSV artery proximal to stenosis
 - Ratio at a stenosis >2.0 is abnormal
 - Ratio at the anastomosis >3.0 abnormal
 - If the flow velocity doubles (or more) between two points in the graft, significant stenosis is suspected
 - Increase in flow velocity between two segments of 100% or more is considered abnormal
 - If the inflow artery demonstrates a triphasic waveform, graft occlusion is suspected
 - If the venous outflow vein demonstrates loss of spontaneous flow and respiratory phasicity, stenosis or obstruction of the vein is suspected
- Volume flow
 - Obtained in a straight vein segment, midgraft/fistula is preferred
 - Measure the diameter of the vein in the area of flow sampling
 - Measurement obtained by opening sample volume size to include all flow from anterior wall to posterior wall
 - Abnormal flow volume <500ml/min indicates stenosis
 - >1200ml/min indicates CHF

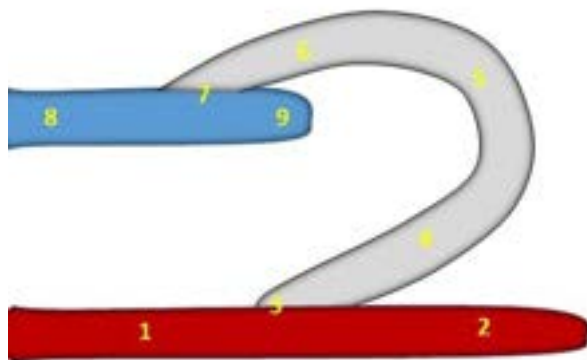
Most Common Sites of Dialysis Graft Stenosis:

- Venous anastomosis
- Outflow vein

***Due to arterial flow hitting venous wall, causes damage to lining resulting in hyperplasia
Complications:

- #1 cause of hemodialysis graft failure is thrombosis of the graft
- Stenosis/Occlusion
- Aneurysm of the graft
- Pseudoaneurysm caused by needle puncture for dialysis
- Can cause CHF due to increased flow in venous return to heart
- Portion of extremity distal to graft may suffer from ischemic symptoms due to steal syndrome
- Infection

Hemodialysis Graft Assessment



Sampling Sites for Doppler

1. Inflow artery proximal to graft
2. Inflow artery distal to graft
3. Proximal anastomosis
4. Proximal graft
5. Mid graft
6. Distal graft
7. Distal anastomosis
8. Outflow vein proximal
9. Outflow vein distal

Doppler Sample Location	Normal Findings
1. Inflow Artery Proximal To Graft	Low Resistance, Increased Diastolic Flow
2. Inflow Artery Distal To Graft	High Resistance, Reduced Diastolic Flow
3. Proximal Anastomosis	Turbulent Flow, Low Resistance
4. Proximal Graft	Turbulent Flow With 100-400cm/S PSV, <200 Cm/S EDV
5. Mid Graft	Similar Flow Velocity and Profile to Proximal Graft
6. Distal Graft	Similar Flow Velocity and Profile to Proximal and Mid Graft
7. Distal Anastomosis	Turbulent Flow, Low Resistance
8. Outflow Vein Proximal	Turbulent, Pulsatile, Elevated Velocity
9. Outflow Vein Distal	Slow moving or non-spontaneous flow; Can see retrograde flow toward hand due to high arterial pressure

AV Fistula/Graft Velocity Ratio

$$\frac{\text{PSV at stenosis}}{\text{PSV proximal to stenosis}}$$

Hemodialysis Graft Evaluation - ABNL

Graft lumen $< 2\text{mm}$; inner to inner dimension of TRUE lumen

Increased resistance in the inflow artery; biphasic or triphasic flow indicates significant abnormality

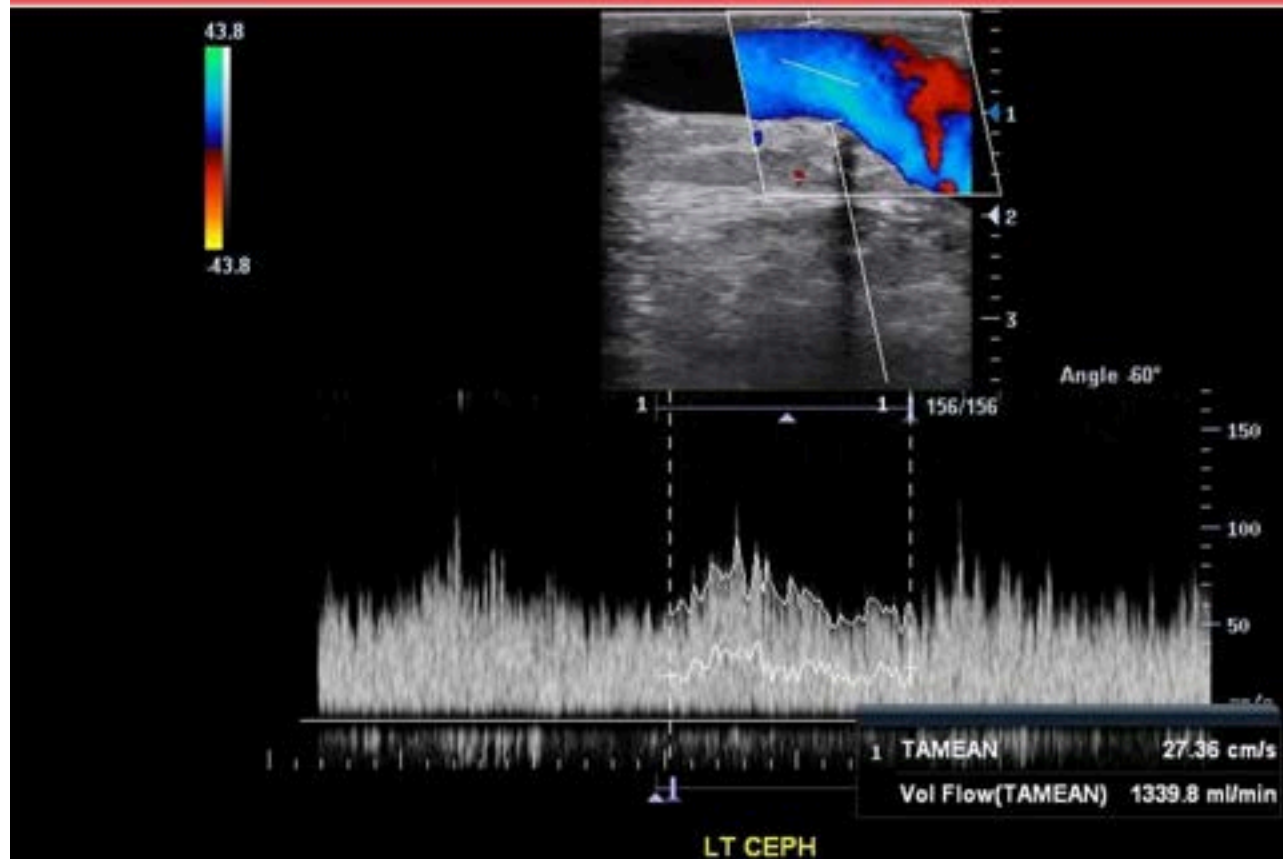
$< 100\text{cm/s}$ peak systolic velocity within the graft

$> 400\text{cm/s}$ peak systolic velocity within the graft

Velocity Ratio > 3.0 Graft / Inflow artery

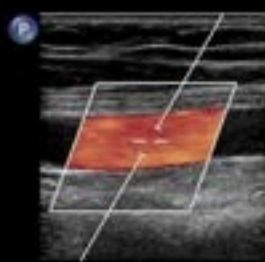
Velocity Ratio > 2.0 between two graft segments

Volume Flow Measurement



Hemodialysis Fistula – Arterial Inflow

Normal flow is turbulent and monophasic with moderate antegrade diastolic flow



PSV 151 cm/s
EDV 73.9 cm/s

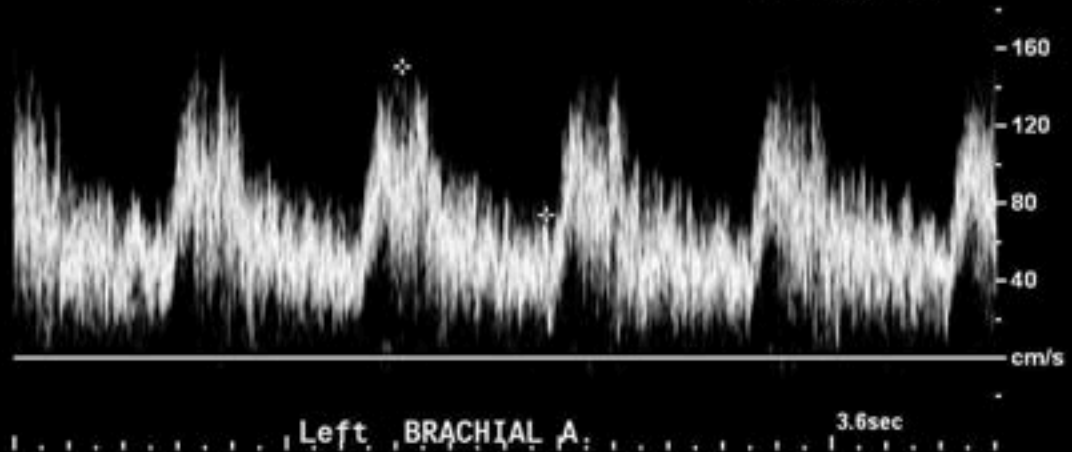
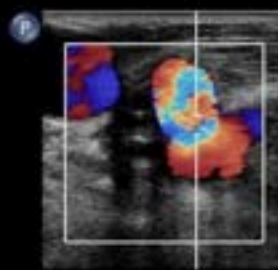


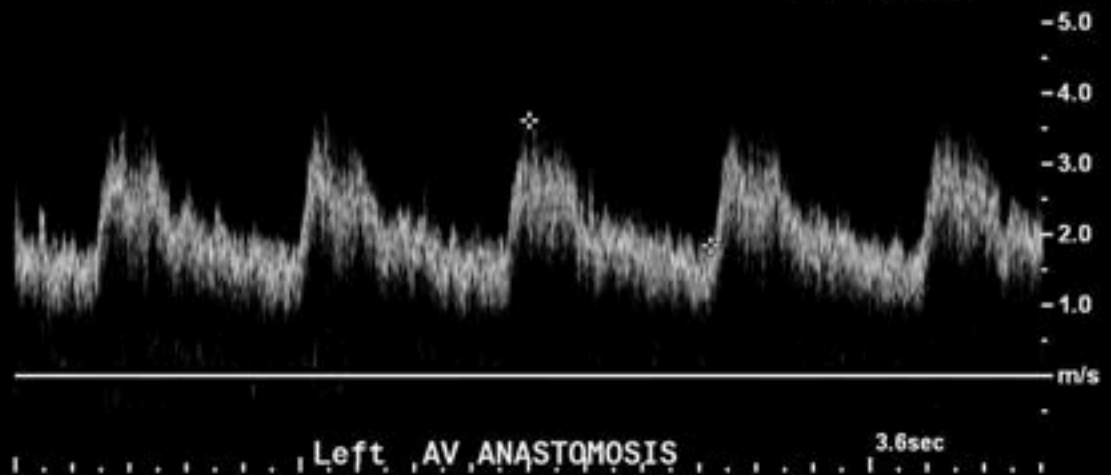
Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Hemodialysis Fistula - Anastomosis

Normal flow is turbulent and monophasic with moderate antegrade diastolic flow



PSV 361 cm/s
EDV 184 cm/s



Left AV ANASTOMOSIS

Image Courtesy of CLEMSON UNIVERSITY - CV IMAGING LEADERSHIP CONCENTRATION

Hemodialysis Fistula – Arterial Outflow

Normal flow is triphasic and moving away from the fistula (toward the hand)

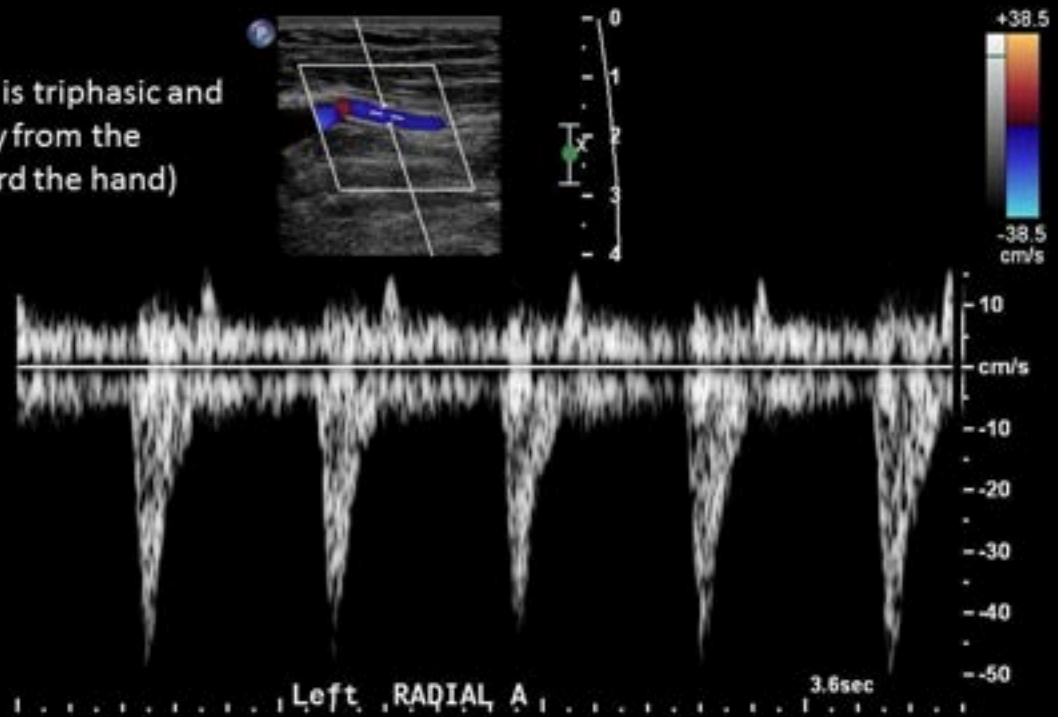


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Hemodialysis Fistula – Venous Outflow

Normal flow is turbulent and increased in velocity compared to a normal vein

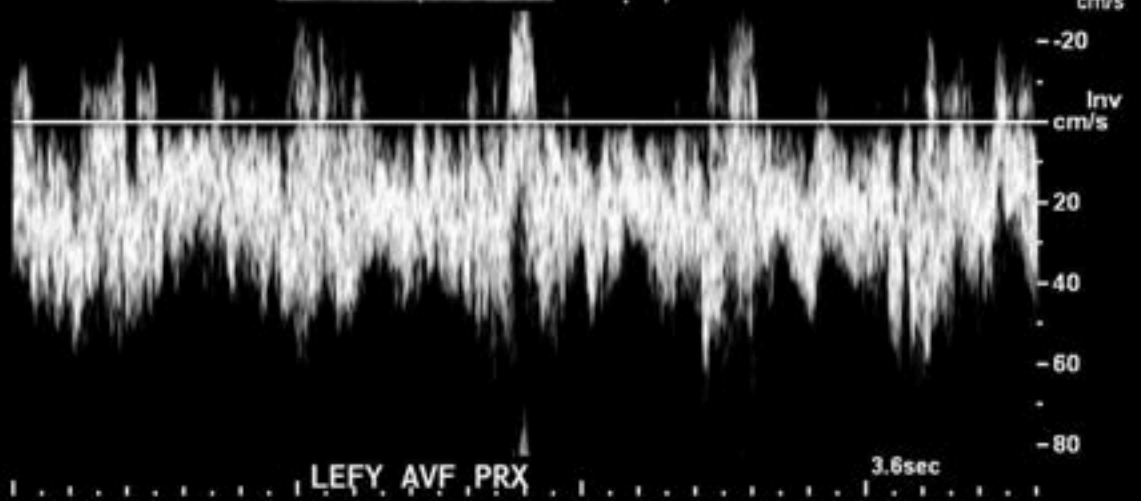
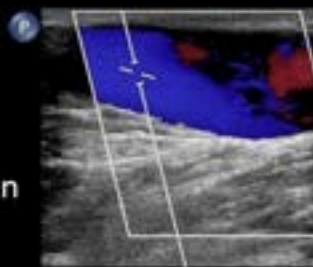
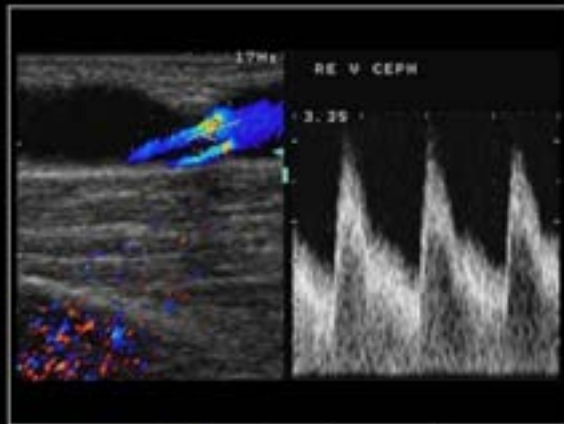
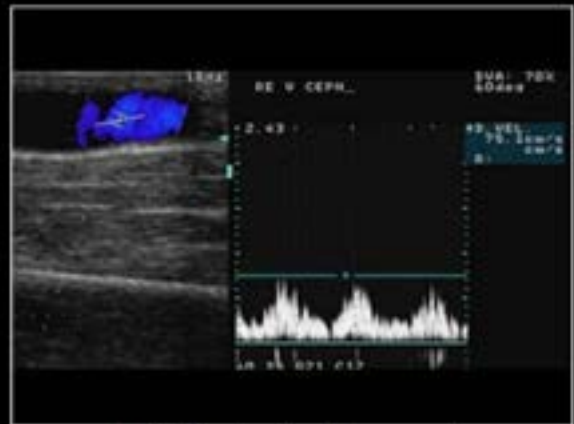


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Dialysis Fistula Stenosis



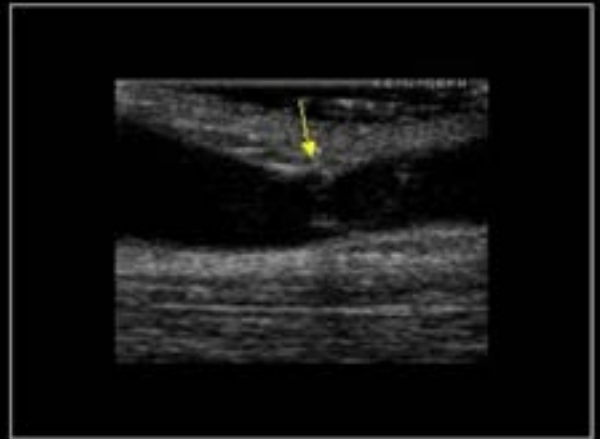
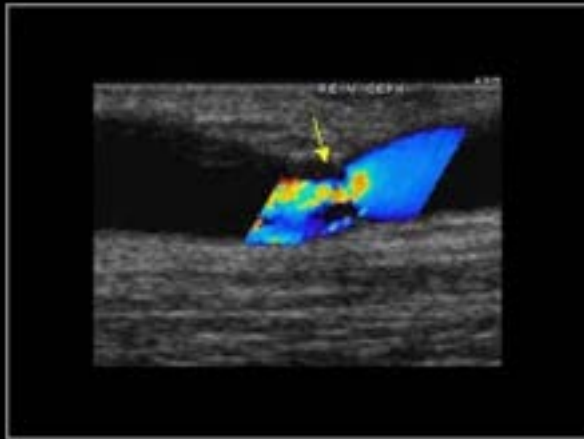
Cephalic vein graft stenosis



Cephalic vein distal to stenosis

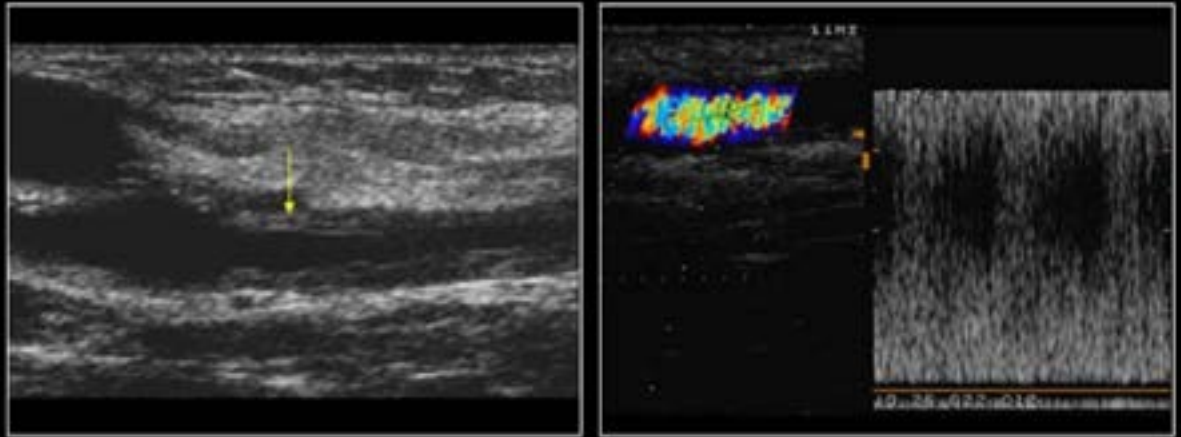
Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

Dialysis Fistula-Venous Stenosis



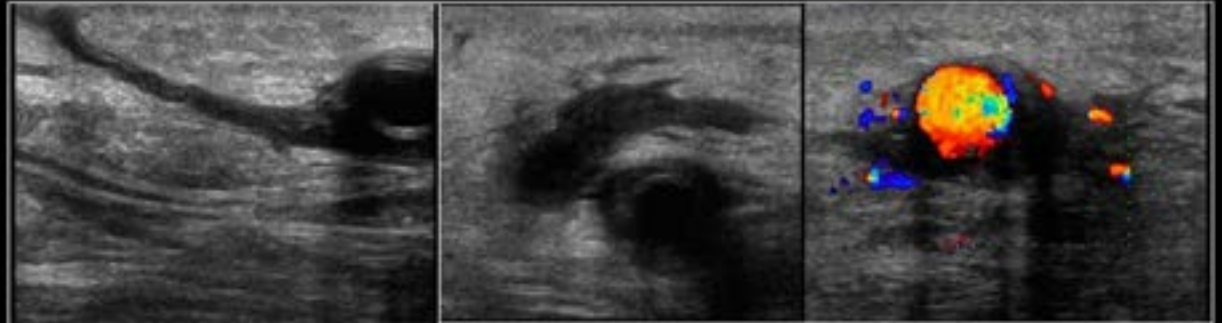
Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

Loop Dialysis Graft Stenosis



Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

Infected Dialysis Graft



Note the edema in the surrounding tissues.

Courtesy of Dr. Taco Geertsma
www.ultrasoundcases.info

Steal Syndrome:

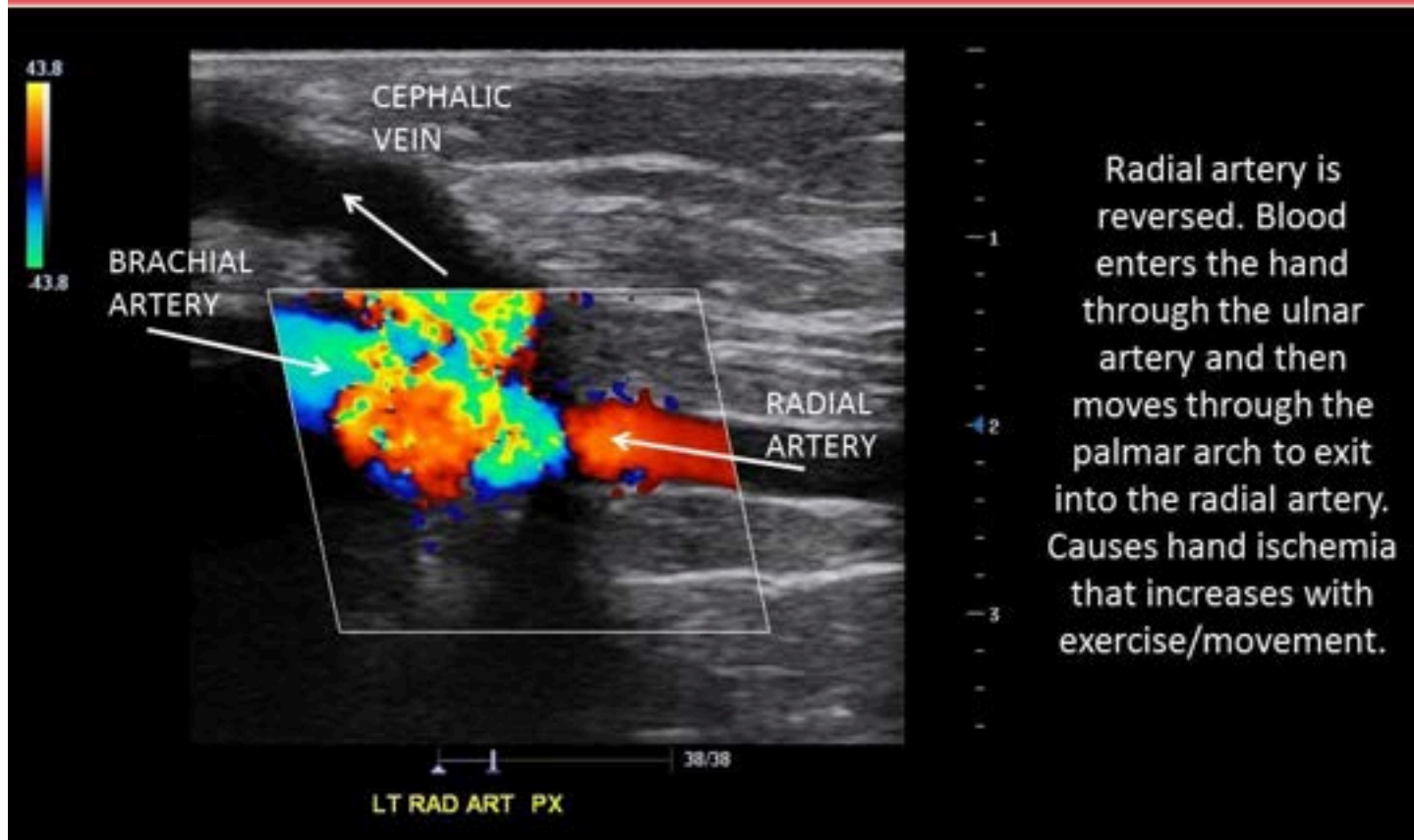
- Most commonly occurs in a radiocephalic fistula
- Ulnar artery will be antegrade and the radial artery will be retrograde
- Blood travels from the ulnar artery into the hand and moves through the palmar arch to exit the hand in the radial artery
- Caused by high volume flow in most cases, but can also occur with inflow stenosis
- Both cause flow to be "sucked up into" the radial artery and into the outflow vein
- PVR or PPG assessment of the affected digits should be compared to the unaffected digits of the opposite hand
- Symptoms include pain, polar sensation, paresthesia, Finger/Brachial Index <0.8 , cyanotic finger tips
- Diminished radial pulse on palpation
- Symptoms can increase with use of the arm (exercise)
- Duplex evaluation:
 - Inflow artery within the 2cm proximal to the fistula anastomosis site
 - Inflow artery within the first 2cm distal to the fistula anastomosis site

- This waveform is used to diagnose steal syndrome, if flow is retrograde in the distal radial (inflow) artery, then flow is moving from the ulnar artery through the palmar arch and cephalad toward the radial artery/fistula
 - Fistula at the confluence of the artery and vein
 - Outflow vein at multiple locations along the arm, above and below the fistula
- Must be treated by clearing the inflow stenosis or removing the high volume fistula completely in order to prevent permanent damage to the hand/fingers

Steal Syndrome



Steal Syndrome



Upper Extremity Venous Anatomy

Deep venous flow: Venules > deep digital veins > metacarpal veins > deep venous arches > radial/ulnar veins > brachial vein(s) > axillary vein > subclavian vein > innominate vein > SVC > right atrium

- Veins of the upper extremity originate at the confluence of the venules of deep digital veins of the fingers
- Metacarpal veins to deep venous arches which converge into the radial and ulnar veins at the wrist
- Calf and forearm veins are referred to as venae comitantes because 2 veins of the same name follow the same course as a single artery of the same name
- The superficial venous system of the arm is the primary drainage system of the extremity

*****CONSIDER ANATOMIC POSITION WHEN DISCUSSING VASCULAR ANATOMY OF THE ARMS, HANDS ROTATED WITH THUMBS POINTING AWAY FROM THE BODY AND PALMS FACE FRONT**

Radial Veins:

- Deep veins of the forearm
- Originate from the dorsal metacarpal veins
- Pair of veins that course adjacent to the radial artery along the lateral aspect of the forearm (anatomic position)
- Join with the paired ulnar veins to form the brachial vein(s) at the antecubital fossa; can be one or two brachial veins
- Normally demonstrate non-spontaneous flow but normal augmentation response

Ulnar Veins:

- Deep veins of the forearm
- Originate from the palmar arch
- Pair of veins that course adjacent to the ulnar artery along the medial aspect of the forearm (anatomic position)
- Join with the paired radial veins to form the brachial vein(s) at the antecubital fossa; can be one or two brachial veins
- Normally demonstrate non-spontaneous flow but normal augmentation response

Brachial Vein(s):

- May have 1 or 2 veins
- Deep vein(s) of the proximal arm
- Brachial vein(s) courses through anterior upper arm to meet the medial basilic vein at the axilla to become the axillary vein
- Spontaneous flow with respiratory phasicity

Axillary Vein:

- Deep vein
- Segment formed from the brachial/basilic junction
- Joins the cephalic vein to form the subclavian vein adjacent to lateral aspect of the clavicle
- Empties blood into the subclavian vein
- Spontaneous flow with respiratory phasicity

Subclavian Vein:

- Deep vein
- Formed by axillary and cephalic junction
- Located deep to the clavicle, usually courses somewhat parallel to clavicle
- Ends when internal jugular vein merges with it to form the innominate vein
- Spontaneous flow with respiratory phasicity AND pulsatile due to proximity to the heart
- Flow increases with inspiration and decreases with expiration

Brachiocephalic Vein:

- AKA innominate vein
- Internal jugular vein drains blood from the head/neck and joins the subclavian vein to form the innominate vein (brachiocephalic vein)
- Right and left brachiocephalic veins join to form the SVC
- Right brachiocephalic vein courses lateral and anterior to the right brachiocephalic artery
- Left brachiocephalic vein courses anterior to the left common carotid artery and right brachiocephalic artery to join the right brachiocephalic vein (NO left brachiocephalic artery)
- ***2 Innominate veins(Right and Left), 1 innominate artery (first branch of aortic arch)
- Right and left innominate veins merge to form the superior vena cava (SVC) which empties into the right atrium of the heart
- Spontaneous flow with respiratory phasicity AND pulsatile due to proximity to the heart
- Flow increases with inspiration and decreases with expiration

Internal Jugular Vein:

- It is formed by the union of inferior petrosal and sigmoid dural venous sinuses in the brain
- Receives tributaries from the face and neck and descends into the thorax, usually between the heads of the sternocleidomastoid muscle
- Merges with the subclavian vein to form the brachiocephalic vein
- The brachiocephalic veins (right and left) merge to form the superior vena cava
- The SVC and IVC both enter the right atrium of the heart

Basilic Vein:

- Superficial vein until it reaches the proximal arm and penetrates the deep fascia
- Considered superficial vein in the forearm and deep vein in the proximal arm due to its location in reference to the fascia
- Originates from the dorsal venous arch, medially in wrist, adjacent to ulnar bone, courses superiorly along the medial arm to join brachial vein in axilla
- Axillary vein formed by the junction of the basilic and brachial veins just distal to the axilla

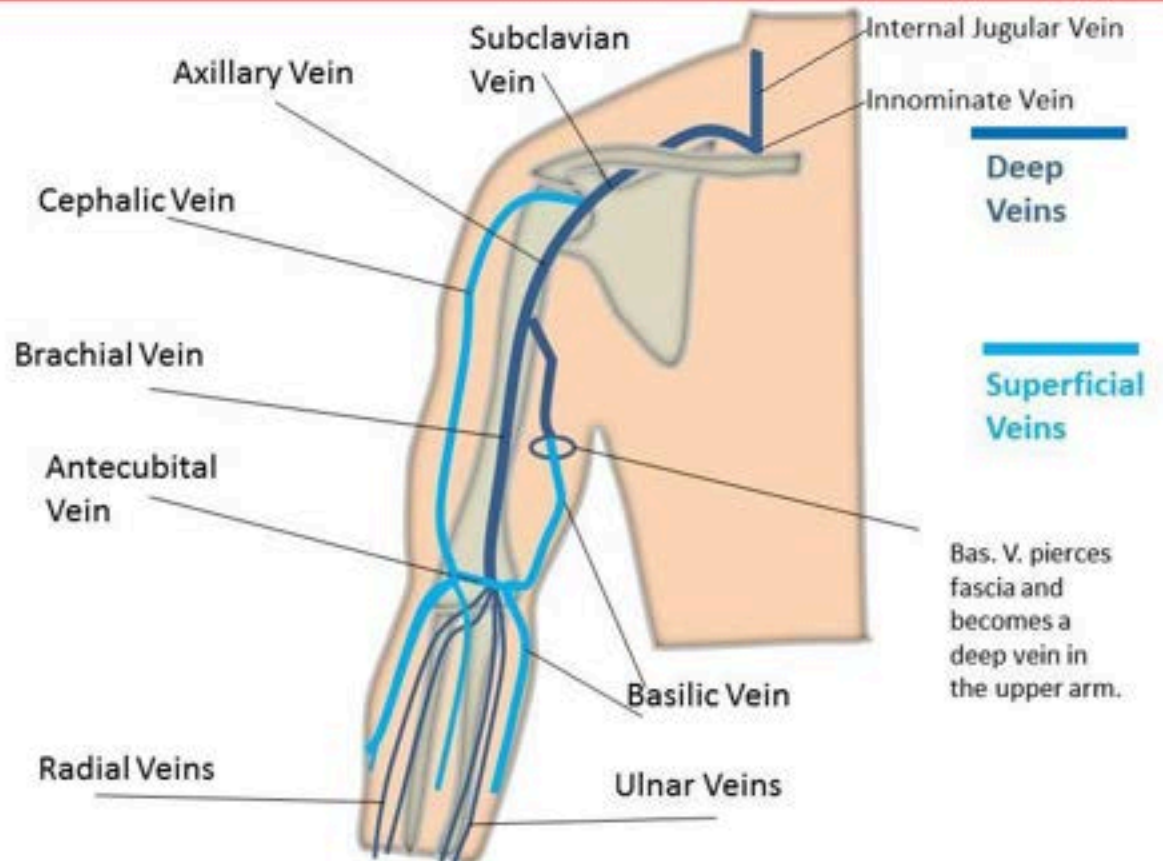
Cephalic Vein:

- Superficial vein
- Originates from the dorsal venous arch laterally in wrist, adjacent to radial bone
- Courses superiorly along the lateral aspect of the arm
- Joins axillary vein at shoulder to form the subclavian vein
- Most common upper extremity vein used for arterial bypass

Antecubital Vein:

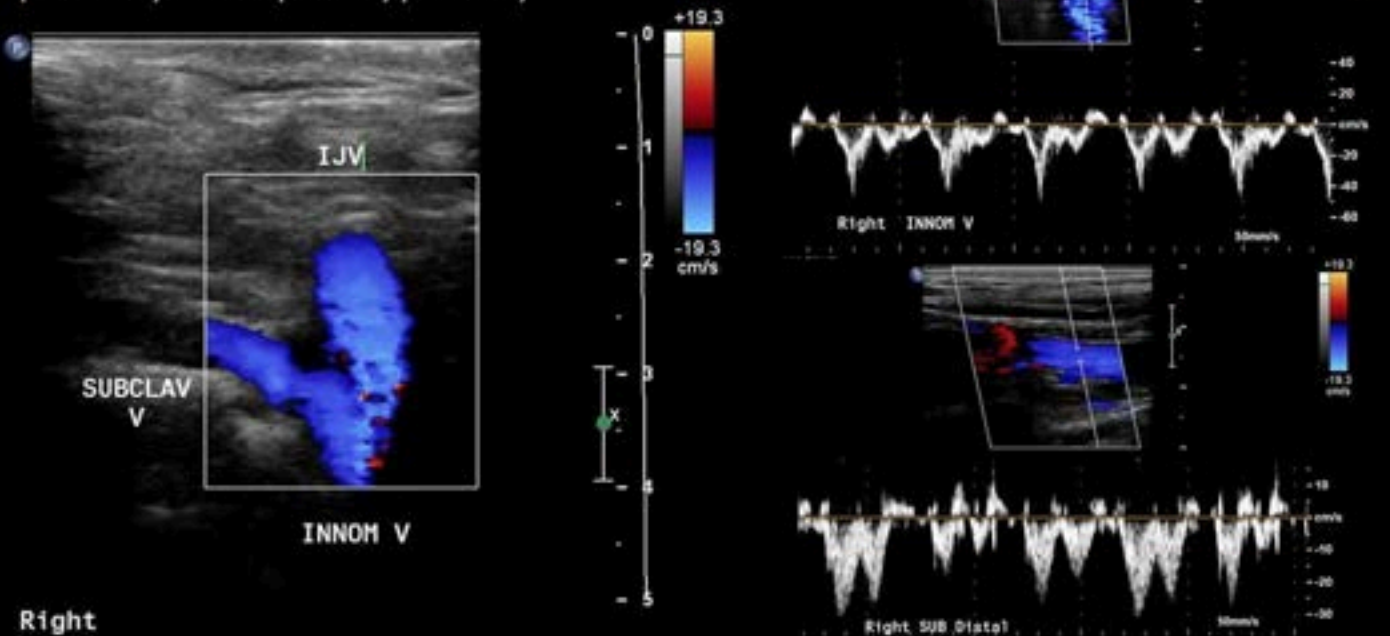
- AKA median cubital vein
- Superficial vein
- Connects cephalic and basilic veins in antecubital fossa
- Commonly used to draw blood

Upper Extremity Veins



Innominate and Subclavian Vein

The Innominate vein and Subclavian vein should normally demonstrate cardiac pulsatility and respiratory phasicity



Right

Image Courtesy of CLEMSON UNIVERSITY – CV IMAGING LEADERSHIP CONCENTRATION

Internal Jugular Vein

The Internal Jugular vein should normally demonstrate cardiac pulsatility and respiratory phasicity.

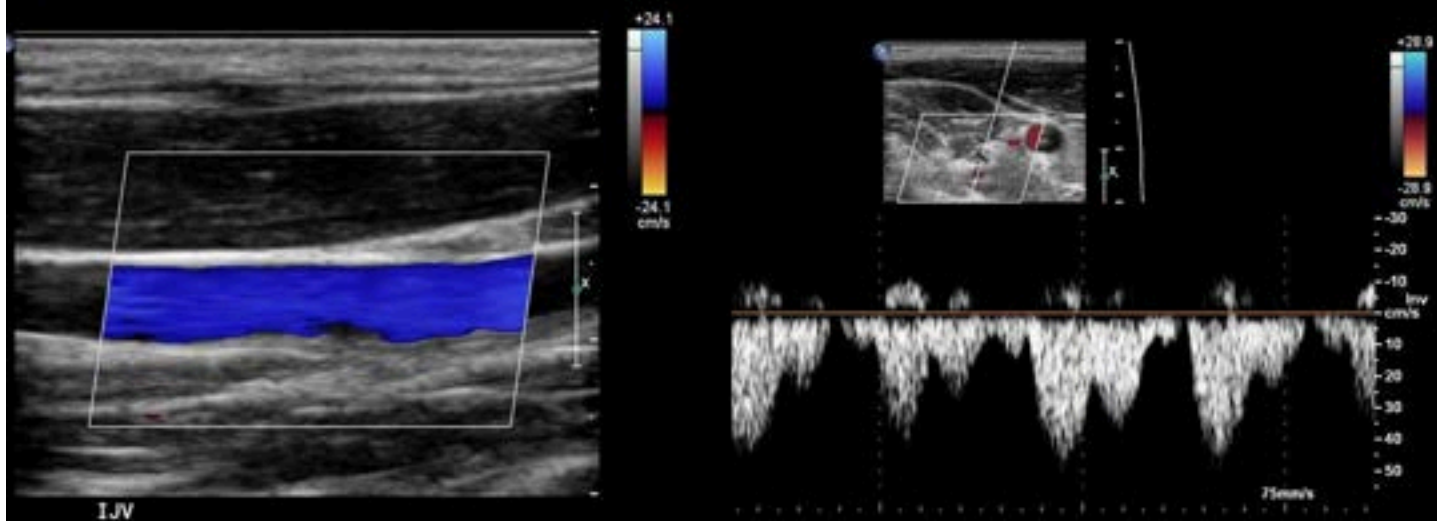


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Right Shoulder

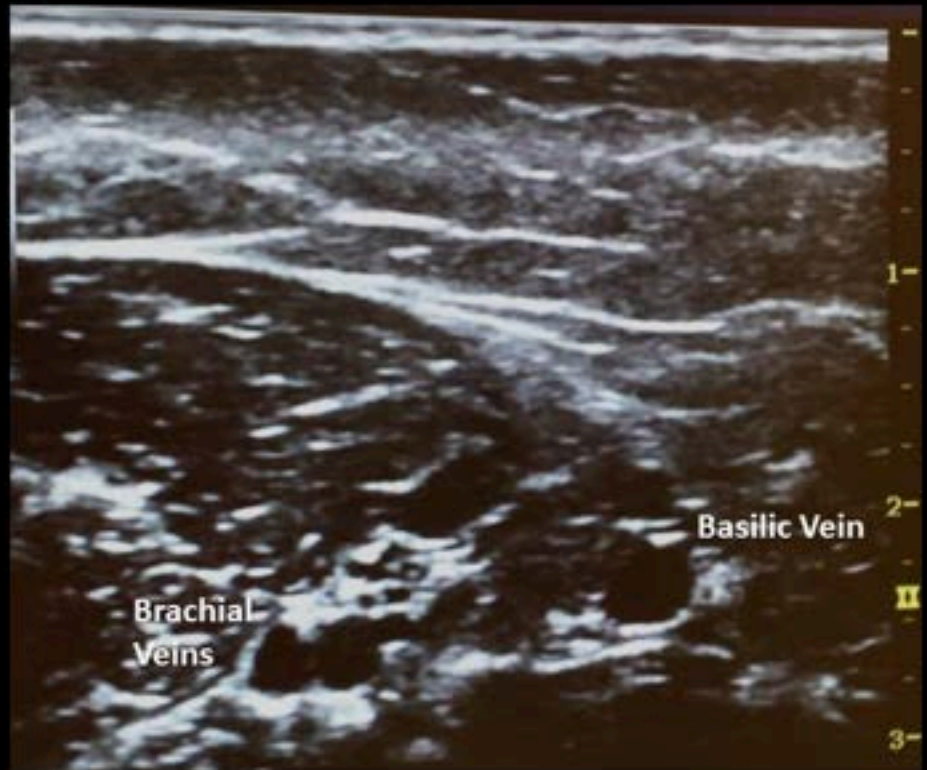


Junction of the Cephalic Vein with the Axillary Vein at the shoulder. The Subclavian Vein is formed at this junction.

Mid Right Arm

Keep in mind that the arm should be scanned in anatomical position.

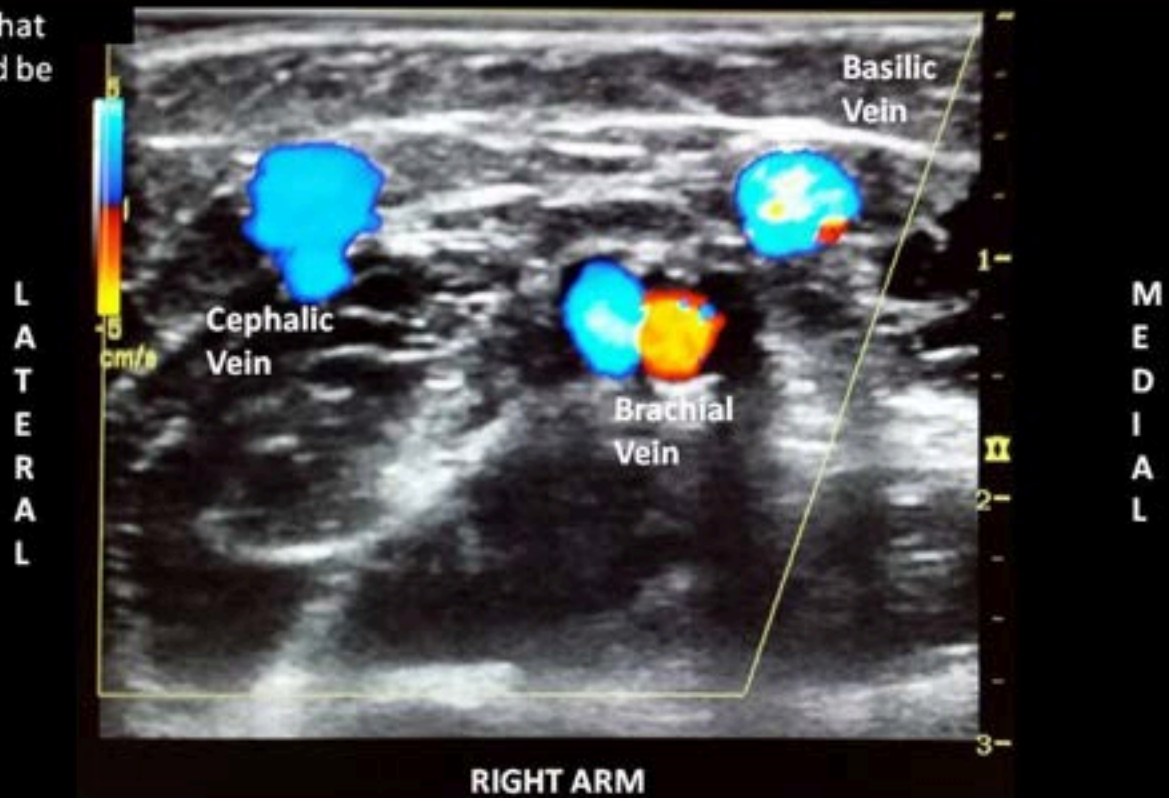
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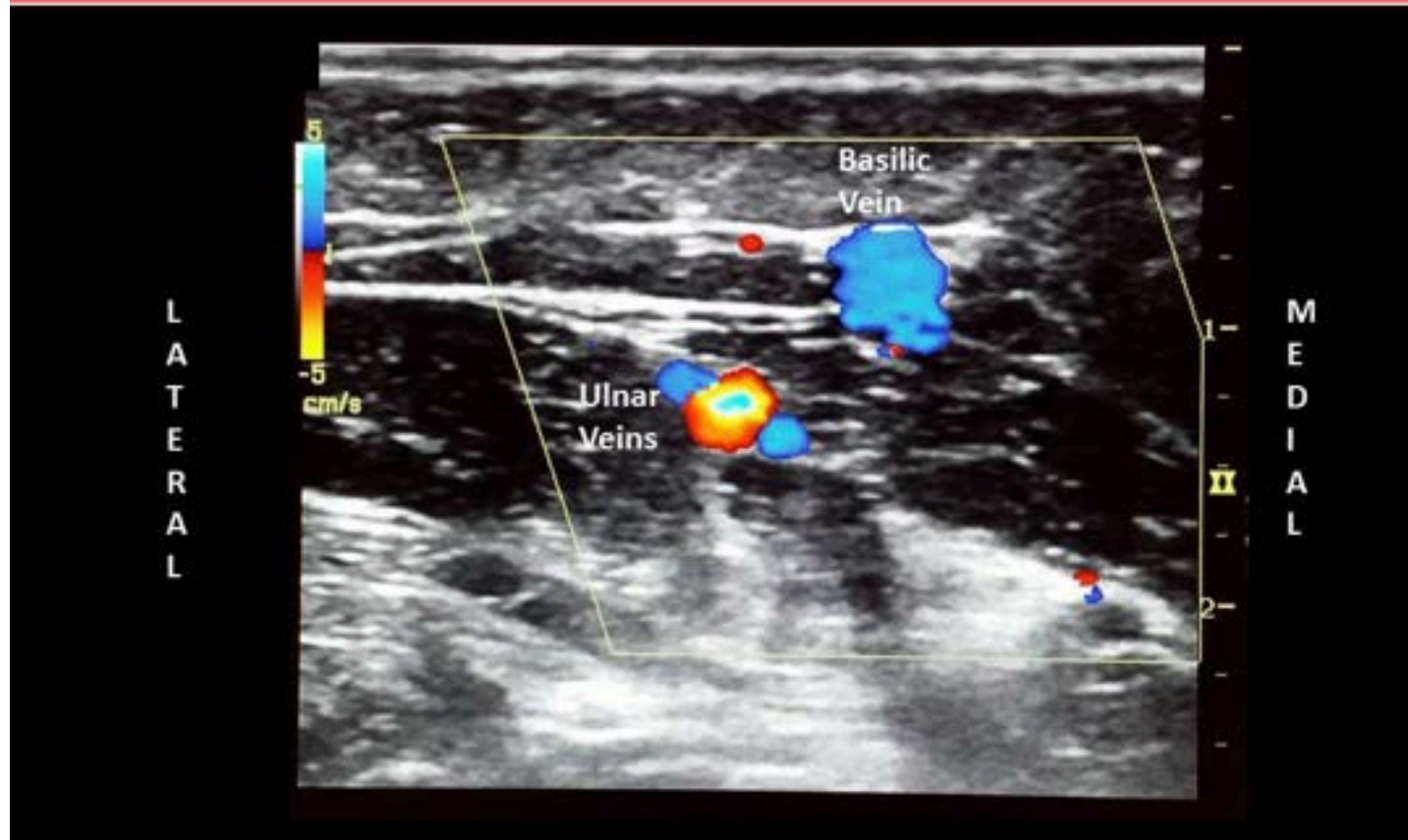
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Right Antecubital Fossa

Keep in mind that the arm should be scanned in anatomical position.



Right Distal Forearm



Superficial Arm Veins

The superficial course of the cephalic and basilic veins make them easy to compress even with light probe pressure.



